

SPACETIME PENROSE INEQUALITIES: ENCLOSING AREA, CHARGE, AND RIGIDITY

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ABSTRACT. We prove sharp spacetime Penrose inequalities for invariant ADM mass and minimum enclosing area in three and four spatial dimensions. Under their respective designated-end conventions, the neutral numerical results allow arbitrary second fundamental form, nonzero momentum, disconnected weakly future trapped boundary, and finitely many ends. Under the stated horizon hypotheses, equality for the neutral inequalities reconstructs the original exterior data as spacelike slices of Schwarzschild spacetime in three dimensions and Schwarzschild–Tangherlini spacetime in four; the four-dimensional equality theorem concerns a connected, one-ended exterior. We also prove a three-dimensional electric–magnetic charged upper-area inequality for one-ended data with nonzero momentum, with separate purely electric rest-frame corollaries allowing finitely many ends. Finally, for each fixed transverse-traceless seed and prescribed decaying solution branch, we prove a local Schwarzschild–anti-de Sitter inequality.

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1. INTRODUCTION

The Penrose inequality compares the mass seen at infinity with the size of a region hidden behind a trapped boundary. An initial data set consists of a Riemannian manifold (M, g) and a symmetric two-tensor K , representing the metric and second fundamental form of a spacelike hypersurface. For a hypersurface $S \subset M$ and a unit normal ν toward the chosen exterior, its future outward null expansion is

$$\theta_+(S) = H_S + \text{tr}_S K,$$

where H_S is positive on Euclidean spheres with outward normal. We call S weakly future outer trapped when $\theta_+(S) \leq 0$. A future marginally outer trapped surface (MOTS) satisfies $\theta_+ = 0$. The dominant energy condition requires the energy density of the constraint equations to be at least the norm of their momentum density. Precise normalizations and asymptotic assumptions are given separately for each spatial dimension below. The asymptotically flat results resolve the minimum-enclosing-area form of the spacetime Penrose conjecture in three and four spatial dimensions under those hypotheses.

The geometric quantity throughout the asymptotically flat results is an enclosing infimum. An enclosing cut separates the entire designated inner boundary from the chosen distant end. Its full area is counted, including portions coincident with the original boundary. Cuts may be

disconnected and need not themselves be trapped. This distinction matters: a trapped surface need not minimize area among enclosing surfaces. When several ends are present, the definition also specifies which of them lie on the excluded side. We give the exact admissible classes in each theorem, rather than identifying infima with different end constraints. In this overview, $A_{\min}$ denotes the enclosing infimum appropriate to the result being described; each part defines its precise cut class.

1.1. The results and their relation. For an asymptotically flat end, let (E, P) denote its ADM energy and momentum. Once $E > |P|$, its invariant mass is $m = \sqrt{E^2 - |P|^2}$. In three spatial dimensions, the basic numerical bound is

$$m \geq \sqrt{\frac{A_{\min}}{16\pi}}.$$

Here $A_{\min}$ denotes the relevant minimum enclosing area. The one-ended theorem is proved first under a timelike assumption on (E, P) . A positive conformal deformation then rules out a non-timelike ADM vector, and cutting off the unwanted distant ends gives the complete-data formulation. The latter permits a disconnected chosen exterior region as well as a disconnected trapped boundary. The precise statements are Theorem 2.2, Corollary 9.4, and Theorem 9.2.

Equality concerns the original data and requires additional geometric hypotheses. In Theorem 10.2, the chosen exterior is connected and one-ended, and its finite marginally trapped boundary is outer area-minimizing. Its outermostness condition excludes an enclosing weakly trapped cut that replaces any nonempty subset of the boundary components, even if other components remain coincident. Equality then forces a single spherical boundary component and a Schwarzschild spacetime realization of the entire exterior. The proof does not prescribe the data on the other side of that boundary.

For electric and magnetic fields, write Q_E, Q_B for their total charges, $Q = (Q_E^2 + Q_B^2)^{1/2}$, and $r = (A_{\min}/4\pi)^{1/2}$. Theorem 14.2 proves

$$m \geq Q, \quad r \leq m + \sqrt{m^2 - Q^2}$$

on a one-ended exterior. Its matter condition includes the electromagnetic momentum density. It allows arbitrary K and nonzero ADM momentum. The familiar lower bound $m \geq (r + Q^2/r)/2$ follows on the branch $r \geq Q$; it is not an equivalent formulation for every positive r . Purely electric rest-frame consequences, including a designated-end theorem with finitely many ends, follow with their own explicit cut convention. No charged equality classification is asserted.

In four spatial dimensions, an enclosing hypersurface has three-dimensional volume A_e . With $\omega_3 = 2\pi^2$, the theorem is

$$m_e \geq \frac{1}{2} \left(\frac{A_e}{\omega_3} \right)^{2/3}.$$

Theorem 22.2 allows arbitrary second fundamental form, finitely many asymptotically flat ends, and disconnected weakly future outer trapped boundary. Its hypotheses impose neither symmetry nor maximality. Theorem 22.4 completes this bound in the connected, one-ended horizon class: if the boundary is a future MOTS, outermost and outer area-minimizing, equality holds exactly for original exterior data realized by a smooth spacelike hypersurface in the regular Schwarzschild–Tangherlini extension. The embedding recovers both g and K , reaches spatial infinity, and attaches smoothly to a future-horizon section or the bifurcation sphere. Boosted ends and non-time-symmetric slices are included. These additional horizon assumptions belong only to the equality theorem; no disconnected equality or behind-horizon classification is asserted in four dimensions.

The last part treats a different asymptotic geometry. For cosmological constant -3 , fix a positive-mass Schwarzschild–anti-de Sitter background, a transverse-traceless tensor, and a decaying positive conformal solution branch generated by that tensor. Here “transverse-traceless” means divergence free and trace free in the background metric. Theorem 36.1 gives, on a sufficiently small interval depending on these fixed choices,

$$m_{\text{AH}} \geq \sqrt{\frac{A}{16\pi}} \left(1 + \frac{A}{4\pi}\right).$$

The quantity m_{AH} is the positive Lorentz norm of the hyperbolic mass covector, and A is the area of the prescribed marginally trapped boundary. The theorem establishes equality precisely for radial seeds at sufficiently small positive parameters. Identifying A with an enclosing infimum is a separate corollary with additional geometric hypotheses. The interval is not uniform over seeds or branches.

1.2. How the proofs fit together. The asymptotically flat numerical proofs have a common structure. First, the distant end is replaced and repaired so that its energy approaches the original invariant mass in a rest frame. The replacement preserves the dominant energy condition and controls every enclosing cut. Next, trapped-region boundaries and disjoint collars prepare a fixed exterior for a coupled elliptic deformation. The unknown graph function and conformal factor produce a Riemannian comparison metric with nonnegative scalar curvature, a controlled energy change, and an enclosing area no smaller than the required original infimum. The Riemannian Penrose inequality completes the comparison.

The three-dimensional elliptic construction is proved in full once. The charged argument uses this same system: it verifies its hypotheses, transports closed flux forms through the geometric changes, and proves a square estimate incorporating the electromagnetic momentum. In four dimensions the scalar identity and regularity estimates change, so we give the corresponding construction separately. Its final metric has zero second fundamental form. A minimizing enclosure therefore brings that actual metric directly within the Riemannian theorem’s hypotheses.

The equality proofs start from the unmodified exterior. Variations of the constraints and of the enclosing-area infimum give a normalized causal stationary field. A twist estimate makes its positive-norm region static, without assuming regularity of an interior zero set. A complete conformal double then forces Euclidean geometry. In three dimensions, the zero-mass argument uses the spin structure supplied by orientability, and the boundary identities give each component the total original area, forcing one component. In four dimensions, we prove the required compactification and reduce zero-mass rigidity to the smooth positive-mass theorem with arbitrary other ends. The final step in both cases reconstructs the original g, K and the smooth horizon attachment; it does not transfer equality through the numerical deformations. Figure 1 shows these two uses of the inequality.

Two ingredients keep the original-data argument independent of a chosen minimizing surface. Full obstacle perimeter includes every boundary-coincident piece. Compactness of all minimizing enclosures then gives the one-sided metric derivative as the minimum of their area derivatives. A positive measure on this minimizing space yields the variational identity; normal tests and the precise outermostness condition concentrate its area term on the original horizon. The causal-field estimates also separate a reusable finite-jet asymptotic argument from the charge normalization, so the converse can establish its asymptotic spacelike plane without assuming it.

The anti-de Sitter proof uses the Taylor coefficients of the given branch. Actual weighted remainder bounds connect those coefficients to the mass and area. The quadratic coefficient is a nonnegative quadratic form with a kernel of dimension four. A fourth-order argument treats its

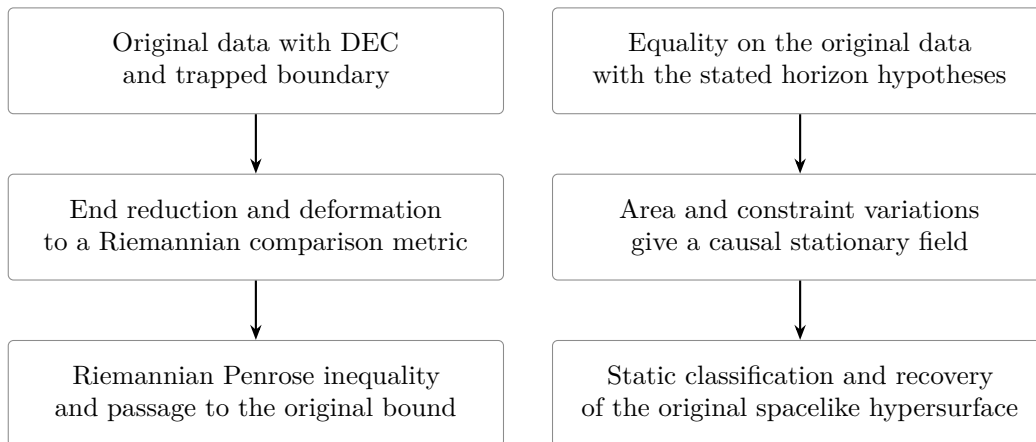


FIGURE 1. The numerical comparison and the equality argument. The bound proved on the left applies to the nearby data used in the variations on the right. Rigidity is obtained from the original data.

nonradial directions, while radial directions satisfy an exact mass conservation identity. This part is independent of the asymptotically flat elliptic construction.

1.3. Relation to earlier work. Penrose proposed the mass–horizon comparison in the context of gravitational collapse and cosmic censorship [45]. For time-symmetric data, Huisken and Ilmanen proved the connected-horizon inequality by weak inverse mean curvature flow, and Bray proved the total-area bound for possibly disconnected horizons by conformal flow [15, 31]. Bray and Lee extended the numerical Riemannian inequality to spatial dimensions below eight; spin enters their higher-dimensional equality statement [18]. Ben-Dov’s counterexample distinguishes apparent-horizon area from minimum enclosing area [14]. Carrasco and Mars constructed counterexamples for outermost generalized apparent horizons [20]. Bray and Khuri developed a generalized Jang reduction, including spherical existence results and conditional nonspherical constructions [16, 17].

Allen, Bryden, Kazaras, and Khuri obtained a universal suboptimal mass–area bound in three dimensions under their decay and horizon assumptions [4]. Khuri and Kunduri proved the sharp inequality for $SU(n+1)$ -invariant cohomogeneity-one data in spatial dimensions $2(n+1)$ [33]. The inequalities here have a different scope: the three- and four-dimensional results impose no such symmetry, and use their specified minimum enclosing quantities.

The stationary field arising from equality is related to the variational study of ADM mass. Hirsch and Huang proved a strong maximum principle for the norm of a causal Killing field and studied mass minimizers with fixed Bartnik boundary data [29]. Those boundary data and regularity hypotheses differ from the area variations used here. Once static vacuum geometry is obtained, the conformal-doubling method of Bunting and Masood-ul-Alam provides the classical model for the classification [19]. We prove the completeness and low-regularity zero-mass statement needed for our double. Higher-dimensional static-vacuum uniqueness was developed by Gibbons, Ida, and Shiromizu [27]. Here the static geometry and its complete base must first be recovered from the original initial data. The charged and anti-de Sitter parts give the comparisons with earlier results specific to those settings where their hypotheses can be stated precisely.

1.4. Reading conventions. The five parts are ordered by their mathematical dependencies. The three-dimensional numerical theorem precedes the global extension, rigidity, and charged application. The four-dimensional part states its own normalizations and contains both the numerical construction and original-data equality proof. The anti-de Sitter part is independent

and uses its own asymptotic geometry. The entry points are Sections 2, 10, 13, 22, and 36, respectively. Within each analytic construction the geometry and strictification are fixed before the deformation parameters. Uniform polynomial estimates and regularity estimates at a fixed parameter have distinct roles; the proofs specify the order of exhaustion and limits where each is used.

Part 1. Three-dimensional initial data

2. THREE-DIMENSIONAL EXTERIOR DATA AND THE MASS INEQUALITY

2.1. Initial data, normals, and enclosing area. We work in three spatial dimensions, with zero cosmological constant and $G = c = 1$. For a Riemannian metric g and a symmetric covariant two-tensor K , define the constraint densities by

$$(2.1) \quad 16\pi\mu = R_g + (\operatorname{tr}_g K)^2 - |K|_g^2, \quad 8\pi J = \operatorname{div}_g(K - (\operatorname{tr}_g K)g).$$

Here J is a covector field. The dominant energy condition is

$$(2.2) \quad \mu \geq |J|_g.$$

All initial data considered here are smooth, including at a compact boundary when one is present.

On an asymptotically flat end with Euclidean coordinates x and $r = |x|$, our basic assumptions are

$$(2.3) \quad g_{ij} - \delta_{ij} = O_2(r^{-q}), \quad K_{ij} = O_1(r^{-1-q}), \quad q > \tfrac{1}{2}.$$

The notation $O_j(r^{-a})$ includes the corresponding coordinate derivative bounds through order j . We assume that μ and $|J|_g$ are integrable and that the following ADM limits exist and are finite:

$$(2.4) \quad E = \frac{1}{16\pi} \lim_{r \rightarrow \infty} \int_{S_r} (\partial_j g_{ij} - \partial_i g_{jj}) n^i dA_\delta,$$

$$(2.5) \quad P_i = \frac{1}{8\pi} \lim_{r \rightarrow \infty} \int_{S_r} (K_{ij} - (\operatorname{tr}_g K)g_{ij}) n^j dA_\delta.$$

The normal and area form in these definitions are Euclidean. Whenever $E > |P|_\delta$, write

$$(2.6) \quad m = \sqrt{E^2 - |P|_\delta^2}.$$

For a two-sided surface Σ with specified unit normal ν , we use

$$H_\Sigma = \operatorname{div}_\Sigma \nu, \quad \theta_+(\Sigma) = H_\Sigma + \operatorname{tr}_\Sigma K.$$

Thus Euclidean spheres have positive mean curvature for the normal toward infinity. A future marginally outer trapped surface, abbreviated MOTS, satisfies $\theta_+ = 0$. A weakly future outer trapped surface satisfies $\theta_+ \leq 0$. Our spacetime sign convention is

$$(2.7) \quad K(Y, Z) = \mathbf{g}(\nabla_Y n, Z),$$

where n is the future unit timelike normal. Equivalently, in Gaussian normal coordinates the spatial metric has normal derivative $2K$.

Definition 2.1 (Exterior and enclosing area). An exterior is a connected orientable smooth three-manifold Ω with nonempty compact smooth boundary, complete as a metric space with its boundary included, and with exactly one end, which is asymptotically flat. An enclosing cut is a compact smooth embedded surface in $\bar{\Omega}$ separating the entire inner boundary from the sufficiently distant part of that end. Equivalently, it bounds the side containing the distant end; the other side is filled up to the inner obstacle. A cut may have several components and may coincide with the inner boundary. Its normal points toward the end. We put

$$a_g(\partial\Omega) = \inf\{| \Gamma |_g : \Gamma \text{ is an enclosing cut}\}.$$

When using perimeter compactness, we take the closure of this class among filled inner sets and include the area of any frontier coinciding with the obstacle. Smooth outward approximation gives the same infimum. Enclosing cuts and minimizing sets are taken with bounded complementary pockets filled. Auxiliary perimeter arguments may use competitors containing the inner obstacle before filling: filling a bounded pocket deletes its frontier, cannot increase perimeter, and leaves the enclosing infimum unchanged.

The definition permits coincidence because first variations of the minimum need not be realized by a cut lying strictly outside the obstacle. It permits disconnected cuts because neither the minimizing hull nor an intermediate trapped boundary need be connected. The perimeter formulation and its compactness properties are established in Section 3.

2.2. The one-ended exterior inequality.

Theorem 2.2 (Minimum-enclosing-area inequality). *Let (Ω, g, K) be an exterior as in Definition 2.1. Assume Equations (2.1)–(2.3), integrability of μ and $|J|_g$, and the finite ADM limits (2.4)–(2.5). Orient $\partial\Omega$ by the normal into Ω , and suppose $\theta_+(\partial\Omega) \leq 0$. If $E > |P|_\delta$, then*

$$(2.8) \quad \sqrt{E^2 - |P|_\delta^2} \geq \sqrt{\frac{a_g(\partial\Omega)}{16\pi}}.$$

No extension of the data across $\partial\Omega$ is required.

3. EXTERIOR APPROXIMATION AND REDUCTION TO A REST END

We reduce the invariant-mass inequality to an energy inequality on a controlled rest end. We replace the distant end, bend the replacement inside Schwarzschild spacetime, and repair its constraint deficit with a conformal change of vanishing mass cost. Throughout, (N, g, K) is a smooth exterior with one asymptotically flat end and compact smooth boundary B . The normal ν on B points into N . Completeness means completeness up to this boundary. We assume

$$(3.1) \quad g - \delta = O_2(r^{-q}), \quad K = O_1(r^{-1-q}), \quad q > \frac{1}{2},$$

and the integrability and dominant energy condition of Theorem 2.2. The symbol $a_g(B)$ denotes the filled enclosing perimeter infimum specified there. None of the results in this section requires the boundary to be connected or outermost.

3.1. A strict conformal direction.

Lemma 3.1 (Conformal constraint formulas). *For a smooth function f , set*

$$g_f = e^{4f}g, \quad K_f = e^{2f}K.$$

The corresponding constraint densities satisfy

$$(3.2) \quad 16\pi e^{4f}\mu_f = 16\pi\mu - 8\Delta_g f - 8|df|_g^2,$$

$$(3.3) \quad 8\pi J_f = e^{-2f}(8\pi J + 4K(\nabla f, \cdot)).$$

Here the second identity is an identity of covectors. Consequently,

$$(3.4) \quad \begin{aligned} 16\pi e^{4f}(\mu_f - |J_f|_{g_f}) &\geq 16\pi(\mu - |J|_g) \\ &\quad + 8(-\Delta_g f - |df|_g^2 - |K|_g|df|_g). \end{aligned}$$

On the fixed boundary,

$$(3.5) \quad \theta_{+,f} = e^{-2f}(\theta_+ + 4\partial_\nu f).$$

Equivalently, if $u = e^f > 0$, the last summand in Equation (3.4) is $8u^{-1}(-\Delta_g u - |K|_g|du|_g)$.

Proof. The scalar-curvature transformation in dimension three is $e^{4f} \text{Scal}_{g_f} = \text{Scal}_g - 8\Delta_g f - 8|df|^2$. Both quadratic terms in K scale by e^{-4f} , which proves Equation (3.2). Put $\pi = K - (\text{tr}_g K)g$. Then $\pi_f = e^{2f}\pi$. For a covariant symmetric tensor T , the connection difference for $\tilde{g} = \Omega^2 g$ gives, in dimension three,

$$\tilde{\nabla}^i(\Omega T)_{ij} = \Omega^{-1}(\nabla^i T_{ij} + 2T_{ij}\nabla^i \log \Omega - (\text{tr}_g T)\partial_j \log \Omega).$$

Take $\Omega = e^{2f}$ and use $\text{tr}_g \pi = -2\text{tr}_g K$. The terms containing $\text{tr}_g K$ cancel, leaving Equation (3.3). Taking its norm introduces another factor e^{-2f} . The triangle inequality, with the factor two between the constraint normalizations 16π and 8π , yields Equation (3.4). Finally, $\nu_f = e^{-2f}\nu$, $H_f = e^{-2f}(H + 4\partial_\nu f)$, and $\text{tr}_{B,g_f} K_f = e^{-2f} \text{tr}_{B,g} K$. This proves Equation (3.5). The last assertion follows from $\Delta \log u + |d \log u|^2 = u^{-1} \Delta u$. $\square$

Lemma 3.2 (Strict direction under the original decay). *Choose*

$$0 < \delta < \beta < \min(q, 1).$$

There are smooth positive functions w and ϕ on N , with $w = r^{-3-\delta}$ on the end, such that

$$(3.6) \quad \partial_\nu \phi = -1 \quad \text{on } B, \quad \phi = O_2(r^{-1}),$$

$$(3.7) \quad -\Delta_g \phi - |K|_g |d\phi|_g \geq w, \quad \frac{-\Delta_g \phi}{w} \longrightarrow 1.$$

The Euclidean normal flux

$$L_\phi = \lim_{r \rightarrow \infty} \int_{S_r} \partial_r \phi \, dA_\delta$$

is finite and negative. For every finite $s \geq 0$, set

$$(3.8) \quad u_s = 1 + s\phi, \quad g_s = u_s^4 g, \quad K_s = u_s^2 K.$$

These data are complete up to B and satisfy DEC, strictly for $s > 0$. If $\theta_+ \leq 0$ on B , their boundary expansion remains nonpositive and is strictly negative for $s > 0$. Their decay exponent can be taken to be $\min(q, 1) > 1/2$, their constraint densities are integrable, and

$$(3.9) \quad E_s = E + sA_\phi, \quad P_s = P, \quad A_\phi = -\frac{L_\phi}{2\pi} > 0.$$

Their full enclosing infima satisfy

$$(3.10) \quad a_g(B) \leq a_{g_s}(B) \leq (1 + s\|\phi\|_\infty)^4 a_g(B).$$

In particular $E_s \rightarrow E$ and $a_{g_s}(B) \rightarrow a_g(B)$ as $s \downarrow 0$. No sign assumption on the ADM vector is needed.

If K is compactly supported and the metric has symbol estimates of all orders with $g-\delta = O(r^{-1})$, the function can instead be chosen with $\phi = O_k(r^{-1})$ for every k , while retaining Equations (3.6) and (3.7).

Proof. Choose smooth $b \geq |K|_g$, positive everywhere and equal to $b_0 r^{-1-\beta}$ sufficiently far out, where $b_0 > 0$ is chosen large enough. This is possible because $\beta < q$. Choose a smooth positive regularizer ζ , equal to r^{-2} on the end, and extend w smoothly and positively to N . We solve

$$(3.11) \quad -\Delta_g \phi = b\sqrt{|d\phi|_g^2 + \zeta^2} + w, \quad \partial_\nu \phi = -1, \quad \phi \longrightarrow 0.$$

The regularizer makes this equation smooth even at critical points.

First truncate at a large coordinate sphere S_R and prescribe $\phi = 0$ there. The Neumann and Dirichlet conditions occur on disjoint boundary components. For $0 \leq t \leq 1$, replace b in Equation (3.11) by tb . At $t = 0$, the mixed Laplacian is invertible: its Dirichlet face gives the Poincare inequality, and the weak solution obtained from its coercive quadratic form is smooth by

boundary regularity. At any solution, the linearization is a Laplacian with a smooth drift of norm at most b and with the homogeneous mixed boundary conditions. The strong maximum principle and the Hopf boundary principle make its kernel zero. It is a Fredholm perturbation of the mixed Laplacian of index zero, and is therefore invertible. The boundary estimates used here are the ordinary elliptic estimates for Dirichlet and normal Neumann conditions: positive definiteness of g verifies their principal complementing condition, and the disjoint boundary components can be covered separately [1]. The maximum principles and local elliptic estimates below are used in their usual uniformly elliptic forms [28].

To close continuation at $t = 1$, first work on a fixed truncation. The elliptic $W^{2,p}$ estimate and gradient interpolation give

$$\|\phi\|_{W^{2,p}} \leq C_p(1 + \|\phi\|_{L^p} + \|d\phi\|_{L^p}) \leq \frac{1}{2}\|\phi\|_{W^{2,p}} + C'_p(1 + \|\phi\|_{L^p}).$$

The constants include the fixed boundary data and are uniform in t . If the supremum norms of solutions on this truncation were unbounded, divide them by their supremum norms. The preceding estimate with $p > 3$, compactness, and the equation give a nonzero limit v with homogeneous mixed data satisfying

$$-\Delta_g v = t_* b |dv|_g.$$

This can be written $\Delta_g v + A \cdot dv = 0$ with a bounded measurable drift: take $A = t_* b \nabla v / |dv|$ off the critical set and zero on it. The strong and Hopf maximum principles exclude a nonzero solution with the homogeneous mixed conditions. Thus the supremum norms are bounded. The $W^{2,p}$ estimate, Schauder estimates, and the smooth equation now give the bounds needed for closedness of continuation. Existence on each truncation follows.

Every such solution is nonnegative. A negative minimum cannot be interior because the right side of Equation (3.11) is positive. At an inner-boundary minimum, the outward-domain derivative would be negative by the Hopf principle, whereas the prescribed derivative in that direction is 1. The outer value is zero.

It remains to make these bounds independent of R . For fixed sufficiently large A and then sufficiently large r_0 , the positive radial function

$$v_0(r) = r^{-1}(1 - Ar^{-\delta})$$

satisfies, on $r \geq r_0$,

$$-\Delta_g v_0 - b |dv_0|_g = A\delta(1 + \delta)r^{-3-\delta} + O(r^{-3-q}) + O(r^{-3-\beta}) \geq c_0 w.$$

The constants are fixed independently of the truncation. Since $b\zeta = o(w)$, a sufficiently large multiple of v_0 is a supersolution of the regularized equation. The comparison principle gives

$$(3.12) \quad 0 \leq \phi(x) \leq C(1 + M_R)v_0(r), \quad M_R = \sup_{N \cap \{r \leq r_0\}} \phi, \quad r_0 \leq r \leq R.$$

The notation for the compact core includes all points outside the end chart. Were M_R unbounded along expanding truncations, the normalized solutions ϕ/M_R would have locally uniform $W^{2,p}$ bounds, including at B , by the preceding local estimates and Equation (3.12). A subsequence would converge locally in C^1 to a nonnegative function with maximum one on the core, homogeneous inner Neumann data, and a bounded-drift homogeneous equation. Equation (3.12) makes its value tend to zero at infinity. It consequently attains a positive maximum, contradicting the strong or Hopf principle for that homogeneous equation. This proves the uniform core bound. The contradiction uses the homogeneous bounded-drift equation.

Local compactness and diagonal extraction give a smooth solution of Equation (3.11), with $\phi = O(r^{-1})$. The claimed derivative count can be checked directly on annuli. Put

$$\phi_\rho(y) = \rho\phi(\rho y), \quad g_\rho(y) = g(\rho y), \quad 1 < |y| < 4.$$

On smaller fixed annuli the equation is

$$-\Delta_{g_\rho} \phi_\rho = \rho b(\rho y) \sqrt{|d\phi_\rho|_{g_\rho}^2 + (\rho^2 \zeta(\rho y))^2} + \rho^3 w(\rho y).$$

Its coefficients have uniform $C^{1,1}$ bounds from the two derivatives in Equation (3.1); the scaled drift is $O(\rho^{-\beta})$ and the scaled source is $O(\rho^{-\delta})$. The bounded zeroth-order norm, interior $W^{2,p}$ estimates, and gradient interpolation first give uniform $C^{1,\alpha}$ bounds for some $\alpha > 0$. The right side is then uniformly C^α . Schauder estimates give uniform $C^{2,\alpha}$ bounds on smaller annuli. This proves $\phi = O_2(r^{-1})$ without using a third derivative of g or a second derivative of K .

The right side of Equation (3.11) is $w + O(r^{-3-\beta})$. It is integrable and proves Equation (3.7). The divergence theorem, with outward-domain normal $-\nu$ on B , gives

$$(3.13) \quad \lim_{r \rightarrow \infty} \int_{S_r} \partial_{n_g} \phi \, dA_g = -|B|_g - \int_N (b \sqrt{|d\phi|_g^2 + \zeta^2} + w) \, dV_g.$$

The Euclidean flux differs by $O(r^{-q})$ and has the same finite limit.

For every fixed finite $s \geq 0$, the factor u_s in Equation (3.8) is bounded and at least one, so the transformed data are complete up to B . The u -form of Lemma 3.1 gives

$$16\pi u_s^4 (\mu_s - |J_s|_{g_s}) \geq 16\pi (\mu - |J|_g) + \frac{8s}{u_s} (-\Delta_g \phi - |K|_g |d\phi|_g) \geq \frac{8s}{u_s} w,$$

$$\theta_{+,s} = u_s^{-2} \left(\theta_+ - \frac{4s}{u_s} \right).$$

This proves the asserted signs for all finite s , without a smallness restriction.

Since $u_s - 1 = O_2(r^{-1})$, the transformed decay exponent is $\min(q, 1) > 1/2$. The exact energy transformation is

$$16\pi u_s^4 \mu_s = 16\pi \mu - 8s u_s^{-1} \Delta_g \phi.$$

Here $\Delta_g \phi = O(r^{-3-\delta})$, and the additional current term is bounded by a fixed- s constant times $|K|_g |d\phi|_g = O(r^{-3-q})$. Both are integrable in dimension three. The bounded positive conformal factor also preserves integrability in the transformed volume measure.

The leading metric change is $4s\phi\delta$, whose ADM energy flux is $-8s \int_{S_r} \partial_r \phi \, dA_\delta$ before division by 16π . All remaining metric flux terms are $O_s(r^{-q}) + O_s(r^{-1})$. Writing $\pi = K - (\text{tr}_g K)g$, one has $\pi_s = u_s^2 \pi$, so the change in its Euclidean momentum flux is $O_s(r^2 r^{-1} r^{-1-q}) = O_s(r^{-q})$. This proves Equation (3.9); Equation (3.13) gives $A_\phi > 0$. No $1/r$ coefficient expansion for ϕ is needed. Finally, the cut class is unchanged and every cut has area $\int_\Gamma u_s^4 \, dA_g$. Bounding u_s between 1 and $1 + s\|\phi\|_\infty$ and taking infima proves Equation (3.10), including coincident and disconnected cuts.

For the final assertion, take b nonnegative, compactly supported, and at least $|K|$; the proof is unchanged. The end equation is then simply $-\Delta_g \phi = w$. Once the two-derivative estimate has been established, successive annular elliptic estimates use the assumed symbol bounds for g and w to give $\phi = O_k(r^{-1})$ for every k . $\square$

3.2. Compactness of enclosing minimizers. The end replacement will change the metric on receding annuli. The next lemma keeps area-minimizing enclosures in a fixed compact set, where local metric convergence controls their areas.

Lemma 3.3 (Enclosing areas under receding changes). *Let g_j be smooth metrics on the fixed exterior N , converging to g locally uniformly up to B . Suppose that in a fixed chart outside a compact set they satisfy, with constants independent of j ,*

$$(3.14) \quad c\delta \leq g_j \leq C\delta, \quad r|\partial g_j|_\delta \leq C.$$

Suppose also that each g_j has an asymptotically flat end, possibly in a different chart, and admits a fixed compact enclosing competitor with uniformly bounded area. Then minimizing filled enclosures

exist, their boundaries lie in one fixed compact subset of N , and

$$a_{g_j}(B) \longrightarrow a_g(B).$$

The perimeter and smooth enclosing infima agree. Each minimizing boundary is $C^{1,1}$, and is smooth minimal away from contact with B .

Proof. For purposes of perimeter minimization only, extend the metrics through a collar of B and attach a fixed compact filling. The filling need not satisfy any constraint or curvature condition. Extend g_j so that the extensions converge uniformly on this fixed collar and filling. A filled enclosure is then a bounded set containing the prescribed inner obstacle.

For each fixed metric, sufficiently large spheres in its asymptotically flat chart have strictly positive outward mean curvature. Clipping a bounded enclosure at any such sphere cannot increase perimeter. To see this, integrate the divergence of the outward unit normal field of the sphere foliation over the part of the enclosure outside the sphere. Its divergence is positive. Its flux through the exterior boundary is at most the area of that boundary, whereas its flux on the cutting sphere is the negative of the new cutting area. Thus the discarded boundary area is at least the inserted area. Approximation gives the same statement for finite-perimeter sets. The direct method on the resulting bounded domain now gives a minimizer. The clipping observation makes this a minimizer against every bounded competitor, not merely those inside the chosen truncation.

Off the obstacle the boundary is locally perimeter minimizing. If it contains a point of radius r sufficiently large, the coordinate ball of radius $c_1 r$ about that point avoids the obstacle. After scaling this ball to unit size, Equation (3.14) gives uniform ellipticity and uniform first-derivative bounds for its area integrand. The local perimeter density estimate gives

$$\text{Area}_{g_j}(\partial F_j \cap B_{c_1 r}) \geq c_2 r^2.$$

Here $c_1, c_2 > 0$ are independent of j and r . One may obtain this estimate from the relative isoperimetric inequality and the comparison which removes, or fills, the set in concentric balls; integration of the resulting differential inequality gives the area lower bound. The fixed competitor bounds the left side above independently of j . Consequently the minimizing boundaries lie in a fixed compact set.

Perimeter compactness now gives a limiting filled enclosure. Uniform metric convergence on that compact set and lower semicontinuity imply

$$a_g(B) \leq \liminf_j a_{g_j}(B).$$

Conversely, using a fixed enclosure with g -area arbitrarily close to $a_g(B)$ gives the reverse upper-limit inequality.

The local obstacle regularity theorem for pure area minimization with a C^2 obstacle gives $C^{1,1}$ regularity, and smoothness off contact, in dimension three; its local estimates require only the appropriate bounded geometry of the smooth metric [31, Regularity Theorem 1.3]. This applies to the fixed smooth obstacle and our smooth one-sided metrics after the collar extension. Finally, push a minimizing boundary slightly away from its contact set by a smooth vector field agreeing there with the outward obstacle normal. In a small obstacle collar its flow increases the signed distance to the obstacle. The resulting C^1 embedded boundary is a positive distance from the obstacle. Smooth local graph approximation, within that distance, preserves enclosure and changes area by an arbitrarily small amount. This proves equality of the smooth and perimeter infima and also justifies using smooth fixed competitors in the upper-limit argument. $\square$

3.3. Schwarzschild reference charges. We next construct the vacuum end used in the replacement. Its charges must agree with those of the original data before we bend it to a rest slice.

Lemma 3.4 (A reference end with prescribed timelike charges). *Let $E > |P|$ and $m = (E^2 - |P|^2)^{1/2}$. There is a spacelike plane near spatial infinity of Schwarzschild spacetime of mass m whose induced data, in asymptotically Euclidean coordinates on the plane, have ADM charges (E, P) . The reference data satisfy*

$$g_* - \delta = O_k(r^{-1}), \quad K_* = O_k(r^{-2})$$

for every derivative order k .

Proof. In static isotropic coordinates (T, y) the Schwarzschild metric is

$$(3.15) \quad -\left(\frac{1 - m/(2r)}{1 + m/(2r)}\right)^2 dT^2 + \left(1 + \frac{m}{2r}\right)^4 \delta_{ij} dy^i dy^j, \quad r = |y|.$$

Its perturbation h from Minkowski space has symbol decay r^{-1} on every fixed spacelike cone. Constant Lorentz transformations preserve this decay. We verify the transformation of the ADM charges directly.

In inertial coordinates for the background Minkowski metric η , put

$$(3.16) \quad \begin{aligned} \mathcal{F}_{\ell ab} &= \partial_a h_{\ell b} - \partial_\ell h_{ab} \\ &+ \eta_{\ell b}(\partial^d h_{ad} - \partial_a \operatorname{tr}_\eta h) - \eta_{ab}(\partial^d h_{\ell d} - \partial_\ell \operatorname{tr}_\eta h). \end{aligned}$$

This tensor is antisymmetric in its first two slots, and differentiation of the displayed formula gives $\partial^\ell \mathcal{F}_{\ell ab} = 2G_{ab}^{\operatorname{lin}}$. The exact Einstein tensor vanishes; its terms beyond the linearization are $O(r^{-4})$. Thus this divergence is $O(r^{-4})$.

Use an inertial frame in which the plane under consideration is $x^0 = 0$. With $\eta = \operatorname{diag}(-1, 1, 1, 1)$, direct substitution in Equation (3.16) gives

$$\begin{aligned} \mathcal{F}_{i00} &= \partial_j h_{ij} - \partial_i h_{jj}, \\ \mathcal{F}_{i0k} &= 2(K_{ik}^{\operatorname{lin}} - \delta_{ik} \operatorname{tr} K^{\operatorname{lin}}) + \partial_k h_{0i} - \delta_{ik} \partial_j h_{0j}, \end{aligned}$$

where

$$K_{ik}^{\operatorname{lin}} = \frac{1}{2}(\partial_0 h_{ik} - \partial_i h_{0k} - \partial_k h_{0i}).$$

The convention is the normal rate of the spatial metric divided by two. For fixed k , the final vector has zero flux through every closed sphere. Indeed it equals

$$\partial_j(\delta_{jk} h_{0i} - \delta_{ik} h_{0j}),$$

the divergence of a tensor antisymmetric in i, j . This zero-flux identity is local to a neighborhood of the sphere: extend the potential smoothly inside before applying the divergence theorem, if necessary. The nonlinear errors in the induced metric and second fundamental form give vanishing flux errors. Therefore the limiting fluxes of $\mathcal{F}_{i0b}$ are $16\pi(E, P)$.

These fluxes are the integrals of the dual two-form of the first two slots of $\mathcal{F}$, with its translation slot held fixed. They therefore transform tensorially under a constant Lorentz change of frame. Independence of the limiting spacelike cut follows directly from Stokes' theorem: two large cuts on planes of slopes bounded strictly below one can be joined within a fixed spacelike cone by a three-dimensional cylinder of volume $O(R^3)$, all of whose points have radius comparable to R . Its divergence error is $O(R^{-4})$, so the flux difference is $O(R^{-1})$. Replacing the resulting ellipsoidal cuts by coordinate spheres on either plane gives the same estimate. The limiting translation charges consequently transform by the Lorentz representation. The static plane has charges

$(m, 0)$, as follows by inserting Equation (3.15) into the ADM integral. Its Lorentz orbit is precisely the set of future timelike vectors of norm m , proving the assertion.

More explicitly the plane can be written $T = v \cdot y$, $|v| < 1$. Writing $y = Ax$, where $A = (I - v \otimes v)^{-1/2}$, makes the induced Minkowski metric in the x coordinates equal to δ . The displayed decay then follows from the symbol estimates in Equation (3.15) and the formula for the second fundamental form. These coordinates will be used when identifying the reference end with the given end. $\square$

3.4. Moments and compact linear corrections. All operators in the next two lemmas are Euclidean. For symmetric covariant tensors define

$$(3.17) \quad \mathcal{L}h = \partial_i \partial_j h_{ij} - \Delta \operatorname{tr} h, \quad (\mathcal{D}k)_i = \partial_j (k_{ij} - (\operatorname{tr} k) \delta_{ij}).$$

The formal adjoint kernel of $\mathcal{L}$ consists of affine functions; the formal adjoint kernel of $\mathcal{D}$ consists of Euclidean Killing fields. Orthogonality to these kernels is the compatibility condition for the compact inverses below.

Lemma 3.5 (Compact scalar and symmetric divergence inverses). *Let $\mathcal{A} \subset \mathbb{R}^3$ be a fixed connected open annulus and let $Q \Subset \mathcal{A}$. Smooth sources supported in Q admit the following compact corrections, supported in a fixed larger compact subset of $\mathcal{A}$.*

- (i) *If $\int f = 0$ and $\int x^i f = 0$ for $i = 1, 2, 3$, there is a smooth symmetric tensor h with $\mathcal{L}h = f$ and $\|h\|_{W^{k+2,p}} \leq C_{k,p} \|f\|_{W^{k,p}}$.*
- (ii) *If $\int V \cdot Y = 0$ for every Euclidean Killing field Y , there is a smooth symmetric tensor k with $\mathcal{D}k = V$ and $\|k\|_{W^{k_0+1,p}} \leq C_{k_0,p} \|V\|_{W^{k_0,p}}$.*

The estimates hold for integers $k, k_0 \geq 0$ and $1 < p < \infty$.

Proof. We first record the ordinary divergence inverse being used. On a ball, the regularized Bogovskii operator solves $\operatorname{div} u = f$ for a mean-zero compactly supported source, preserves compact support, and gains one derivative in $W^{k,p}$; see [23, §3.1, Equation (3.13), Theorem 3.2, Corollary 3.4, and Remark 3.5]. To apply it on the annulus, cover Q by finitely many balls compactly contained in $\mathcal{A}$, adding finitely many overlapping balls so that the overlap graph is connected. Partition the source among the balls. On a spanning tree of the overlap graph, pass each piece's integral to its parent by subtracting and adding that integral times a fixed unit-integral bump in the overlap. Starting at the leaves leaves every piece with zero integral; the root does also because the total integral vanishes. The decomposition has bounded Sobolev norms because all cutoffs and bumps are fixed. The sum of the ball inverses gives a compact inverse on the annulus. For sources in a fixed compact set its output support lies in another fixed compact set. This construction may be repeated twice using slightly enlarged compact sets still contained in $\mathcal{A}$.

For the scalar assertion, solve $\partial_i u_i = f$ compactly. Integration by parts gives

$$\int u_i = - \int x^i f = 0.$$

Apply the divergence inverse to each component of u , obtaining $\partial_j A_{ij} = u_i$. Symmetrizing A does not change its double divergence. Denote this symmetric matrix by S and put

$$h = S - \frac{1}{2}(\operatorname{tr} S)\delta.$$

Since $h - (\operatorname{tr} h)\delta = S$, its linearized scalar curvature is f . The two applications of the divergence inverse give the two-derivative gain.

For the vector assertion, translation orthogonality gives $\int V_i = 0$. Solve $\partial_j B_{ij} = V_i$ component-wise. Orthogonality to rotations and integration by parts imply

$$0 = \int (x^j V_i - x^i V_j) = - \int (B_{ij} - B_{ji}).$$

Thus the skew part $S_{ij} = (B_{ij} - B_{ji})/2$ has zero integral. Apply the divergence inverse to its independent components to obtain a tensor D , skew in its first two indices, such that $\partial_k D_{ijk} = -S_{ij}$. Define

$$(3.18) \quad C_{ijk} = D_{ijk} - D_{ikj} - D_{jki}.$$

Index interchange shows both $C_{ijk} = -C_{ikj}$ and $(C_{ijk} - C_{jik})/2 = D_{ijk}$. Consequently

$$T_{ij} = B_{ij} + \partial_k C_{ijk}$$

is symmetric and satisfies $\partial_j T_{ij} = V_i$. The last statement uses the skew symmetry in j, k to cancel the double divergence of C . Set $k = T - (\text{tr } T)\delta/2$ to invert trace reversal. The first inverse gains one derivative, the second gains another, and the last derivative of C leaves a net gain of one. All constructions preserve smoothness and have the stated support and norm bounds. $\square$

Lemma 3.6 (Integrable sources give sublinear first moments). *For data satisfying Equation (3.1) and integrable $\mu, |J|$, put $h = g - \delta$ and $\pi = K - (\text{tr}_\delta K)\delta$. Then*

$$(3.19) \quad \begin{aligned} \mathcal{L}h &= 16\pi\mu + O(r^{-2-2q}), \\ \partial_j \pi_{ij} &= 8\pi J_i + O(r^{-2-2q}). \end{aligned}$$

In particular these Euclidean sources are integrable. Their scalar boundary fluxes paired with affine functions having zero constant term, and their vector boundary fluxes paired with rotational Killing fields, are $o(R)$ on coordinate spheres of radius R .

Proof. The scalar-curvature expansion has remainder bounded by

$$C(|h||\partial^2 h| + |\partial h|^2 + |K|^2),$$

and the replacement of the momentum constraint by Euclidean divergence has remainder bounded by $C(|h||\partial K| + |\partial h||K|)$. These are $O(r^{-2-2q})$, using exactly the derivatives in Equation (3.1). They are integrable because $2 + 2q > 3$.

For any integrable function F on the end and fixed $c > 1$,

$$(3.20) \quad \frac{1}{R} \int_{r_0 < |x| < cR} |x| |F(x)| \, dx \longrightarrow 0.$$

Indeed the contribution inside any fixed radius M tends to zero after division by R ; the remaining contribution is at most $c \int_{|x| > M} |F|$. Let $R \rightarrow \infty$ and then $M \rightarrow \infty$.

For an affine function f , the vector field

$$\mathcal{Q}_f(h)^i = f(\partial_j h_{ij} - \partial_i \text{tr } h) - (\partial_j f)h_{ij} + (\partial_i f) \text{tr } h$$

satisfies $\partial_i \mathcal{Q}_f(h)^i = f \mathcal{L}h$. For a Euclidean Killing field Y ,

$$\partial_j (\pi_{ij} Y^i) = Y^i \partial_j \pi_{ij},$$

because π is symmetric. Integrating from a fixed inner sphere and using Equation (3.20) proves the asserted $o(R)$ estimates. This argument does not assert the existence of a center-of-mass or angular-momentum limit and requires no parity condition. $\square$

Proposition 3.7 (Annular replacement with a vanishing deficit). *Suppose $E > |P|$, and let (g_*, K_*) be the reference end from Lemma 3.4, identified with the given end in the common asymptotically Euclidean coordinates. For all sufficiently large R there are smooth data $(\tilde{g}_R, \tilde{K}_R)$ equal to (g, K) on $r \leq R$ and to (g_*, K_*) on $r \geq 4R$ such that*

$$(3.21) \quad 16\pi(\tilde{\mu}_R - |\tilde{J}_R|_{\tilde{g}_R}) \geq -\eta_R R^{-3}, \quad \eta_R \longrightarrow 0.$$

The possible negative part is supported in $R < r < 4R$. The metrics there are uniformly Euclidean comparable and have uniformly bounded scaled first derivatives. The construction uses no derivatives of the original data beyond those in Equation (3.1) for its quantitative estimates.

Proof. Fix $\frac{1}{2} < \beta < \min(q, 1)$ and work on the annulus $\mathcal{A} = \{1 < |y| < 4\}$ with $x = Ry$. The scaled fields are

$$g_R(y) = g(Ry), \quad k_R(y) = RK(Ry),$$

and likewise for the reference data. Their constraint densities are the original densities multiplied by R^2 . If $h_R = g_R - \delta$, then

$$(3.22) \quad \|h_R\|_{C^2(\mathcal{A})} + \|k_R\|_{C^1(\mathcal{A})} + \|h_{*,R}\|_{C^2(\mathcal{A})} + \|k_{*,R}\|_{C^1(\mathcal{A})} \leq CR^{-\beta}.$$

Choose $\chi \in C^\infty(\mathcal{A})$ with $0 \leq \chi \leq 1$, equal to one near $|y| = 1$ and zero near $|y| = 4$, with all its derivatives supported in a fixed compact subannulus. Blend both fields by this cutoff.

Write the constraints in the unnormalized form

$$\mathcal{C}(g, k) = (\text{Scal}_g + (\text{tr}_g k)^2 - |k|_g^2, \text{div}_g(k - (\text{tr}_g k)g)).$$

Their Euclidean linear part is $(\mathcal{L}h, \mathcal{D}k)$. On fields satisfying Equation (3.22), the nonlinear remainder is bounded in supremum norm by $CR^{-2\beta}$. Thus the blended constraints equal the blended original constraint sources, plus a quadratic error and the compact commutators

$$\begin{aligned} f_R &= \mathcal{L}(\chi(h_R - h_{*,R})) - \chi\mathcal{L}(h_R - h_{*,R}), \\ V_R &= \mathcal{D}(\chi(k_R - k_{*,R})) - \chi\mathcal{D}(k_R - k_{*,R}). \end{aligned}$$

They have $C^{0,1}$ norms at most $CR^{-\beta}$. In detail the scalar commutator applied to a symmetric tensor h is

$$2\partial_i\chi(\partial_j h_{ij} - \partial_i \text{tr } h) + (\partial_i \partial_j \chi)h_{ij} - (\Delta\chi) \text{tr } h;$$

one derivative of this expression uses at most two derivatives of h . The vector commutator is $(\partial_j \chi)(k_{ij} - (\text{tr } k)\delta_{ij})$; one derivative uses at most one derivative of k . No derivative of the complete matter sources is needed.

Every affine moment of f_R and every Killing-field moment of V_R is $o(R^{-1})$. For the constant moments, integration by parts on $\mathcal{A}$ gives R^{-1} times the difference between the physical translation fluxes on its two boundary spheres, less the integrals of the blended linear sources. The flux difference tends to zero because the reference and original ADM vectors agree. Replacing the physical momentum integrand by its Euclidean trace reversal costs $O(R^{1-2q})$, which tends to zero. By Lemma 3.6, the physical linear-source integrals on this receding annulus tend to zero as well. For a linear scalar test or a rotational vector test the corresponding factor is R^{-2} . Its physical boundary fluxes are $o(R)$ by the same lemma, while the weighted source integrals are $o(R)$ by Equation (3.20). This proves the claim.

Remove these small moments by fixed smooth bumps. Explicitly, for a basis $p_1, \dots, p_4$ of affine functions and a fixed nonnegative bump ω positive on a ball inside the annulus, the Gram matrix

$$G_{ab} = \int \omega p_a p_b$$

is positive definite. Subtract $\omega \sum_{a,b} p_a (G^{-1})_{ab} \int p_b f_R$ from f_R . For the vector moments use in exactly the same way the Gram matrix $\int \omega Y_a \cdot Y_b$ of a basis of the six Euclidean Killing fields. It

too is positive definite, since a nonzero Killing field cannot vanish on an open ball. Both bump corrections have every fixed smooth norm $o(R^{-1})$.

Apply Lemma 3.5 with the negative of the moment-free commutators. Since their $W^{1,p}$ norms are $O(R^{-\beta})$, choosing $p > 3$ gives metric and tensor corrections of size $O(R^{-\beta})$ in $W^{3,p}$ and $W^{2,p}$, respectively. Sobolev embedding gives the needed $C^{2,\alpha}$ and $C^{1,\alpha}$ bounds. Their supports stay in a fixed compact subset of $\mathcal{A}$. Extend them by zero and undo the scaling. The metric remains positive definite for large R .

To check DEC, denote the original scaled sources by (A_R, B_R) , so $A_R \geq 2|B_R|_{g_R}$. The reference sources vanish. The corrected sources therefore have the form

$$(\chi A_R, \chi B_R) + o(R^{-1}) + O(R^{-2\beta})$$

in supremum norm. The norm of B_R changes by at most $CR^{-\beta}|B_R| \leq CR^{-2\beta}$ when the metric is changed, since the pointwise constraint formula gives $|B_R| \leq CR^{-\beta}$. All other nonlinear correction terms have the same quadratic bound. Because $2\beta > 1$, the scaled DEC deficit is $o(R^{-1})$. Multiplication by R^{-2} gives Equation (3.21). Outside the correction annulus the data are either the original DEC data or the exact vacuum reference data. Smoothness follows from smoothness of the original fields and of the compact inverse constructions, irrespective of the sizes of their unneeded higher derivatives. $\square$

3.5. Bending to a rest slice and repairing the deficit.

Lemma 3.8 (A vacuum bend with uniform geometry). *The data in Proposition 3.7 can be modified farther out, inside their exact Schwarzschild part, so that they are induced by a constant-static-time slice outside a compact set. This modification adds no constraint deficit. Write (g_R, K_R) for the resulting unscaled data. In the common end coordinates the metrics are uniformly Euclidean comparable, and throughout all transition annuli*

$$(3.23) \quad |\partial g_R|_\delta + |K_R|_\delta \leq C/r.$$

There is a smooth proper radius ρ_R on the end which equals the original coordinate radius on the gluing annulus and equals the rest isotropic radius beyond the transitions, with

$$(3.24) \quad cr \leq \rho_R \leq Cr, \quad |d\rho_R|_{g_R}^2 \geq c, \quad |\Delta_{g_R}\rho_R| \leq C/r.$$

All transition annuli lie between fixed multiples of R ; the constants can depend on the prescribed timelike ADM vector, but not on R .

Proof. Use the description $T = v \cdot y$, $y = Ax$, from Lemma 3.4. Choose $R_b = c_b R$ so large, with c_b fixed, that the region $|y| \geq R_b$ is entirely inside the exact reference part. Let ψ be a smooth function equal to one on $(-\infty, 0]$, zero on $[1, \infty)$, and taking values in $[0, 1]$. Replace the plane by the graph

$$(3.25) \quad T_R(y) = (v \cdot y)\psi\left(\frac{\log(|y|/R_b)}{L}\right).$$

Its Euclidean gradient satisfies

$$|dT_R|_\delta \leq |v|(1 + \|\psi'\|_\infty/L).$$

Choose a fixed L making the right side strictly smaller than one. The lapse and spatial coefficients in Equation (3.15) converge to their Minkowski values, so for all sufficiently large R this graph is uniformly spacelike. In fact for positive Schwarzschild mass the lapse is smaller than one and the spatial conformal factor exceeds one, which only improves this estimate on the static exterior.

The graph agrees with the original plane before R_b and with a constant-time slice after $e^L R_b$. Its first derivative is bounded, and its second and third derivatives are $O(r^{-1})$ and $O(r^{-2})$. The induced metric is uniformly comparable with δ ; differentiating its pullback formula gives the first

bound in Equation (3.23). The second fundamental form uses the graph's second derivatives and ambient first derivatives, divided by a uniformly positive spacelikeness factor, and gives the other bound. These estimates remain true in the x coordinates because A is a fixed invertible linear map. The constraints vanish on the entire bent part by the Gauss–Codazzi equations in exact vacuum Schwarzschild spacetime.

It remains to make the radius precise. On the bent region keep $\rho_R = |x| = |A^{-1}y|$. Beyond the bend, where the slice is already static, write $y = r\omega$ and put $a(\omega) = |A^{-1}\omega|$. Choose R_c a sufficiently large fixed multiple of $R_0 e^L$, and a cutoff χ_0 equal to one before zero and zero after one. In that static region set

$$\rho_R(r, \omega) = r \exp \left[\chi_0 \left(\frac{\log(r/R_c)}{L_0} \right) \log a(\omega) \right].$$

For fixed sufficiently large L_0 its radial derivative is bounded below by a positive constant: differentiating in r gives the positive factor ρ_R/r times $1 + L_0^{-1} \chi_0' \log a$, which is at least $1/2$. The function and its first two Euclidean derivatives have the bounds $\rho_R \asymp r$, $|d\rho_R| \leq C$, and $|\partial^2 \rho_R| \leq C/r$. Before this interpolation, the same bounds hold for $|A^{-1}y|$, and uniform metric comparability gives the gradient lower bound. Combining these facts with Equation (3.23) proves Equation (3.24). After the interpolation, $\rho_R = r$ is precisely the static isotropic radius. All radius ratios used in the construction are fixed, independently of R . $\square$

Lemma 3.9 (A radial DEC repair of vanishing mass cost). *Consider the data obtained from Proposition 3.7 and Lemma 3.8. There is a smooth positive conformal factor u_R , constant on the unchanged inner region and tending to one at infinity, such that $(u_R^4 g_R, u_R^2 K_R)$ satisfy DEC everywhere. Moreover*

$$(3.26) \quad \|u_R - 1\|_\infty = o(R^{-1}), \quad |du_R| = o(R^{-2}) \quad \text{on the transition region.}$$

The final end is exactly Schwarzschild with $K = 0$ and mass $m + o(1)$. The boundary expansion retains its original weak sign.

Proof. All geometry estimates below are uniform in R . Enlarge the fixed transition region slightly so that, in terms of $t = \rho_R/R$, it lies inside an interval (a, b) with $0 < a < b < \infty$. Arrange that $t \geq b$ is in the static spherical region, and that the support of the possible deficit lies in a fixed smaller interval $I \Subset (a, b)$. Increasing b if needed makes these assertions follow from Lemma 3.8. We take $a = 1$: the compact supports in the annular correction leave a fixed neighborhood of $t = 1$ free of deficit. The factor constructed below will consequently be constant throughout the original region $r \leq R$, and is extended by that constant over the compact core of N .

Let $\eta_R \rightarrow 0$ be as in Equation (3.21), replacing it by a positive majorant if necessary. We will choose $\varepsilon_R = C_0 \eta_R$ and construct

$$u_R = 1 + \frac{\varepsilon_R}{R} F_R(\rho_R/R),$$

where F_R and the relevant derivatives are bounded independently of R . Write $z_R = -F_R'$. On $[a, b+1]$, initially define z_R by

$$(3.27) \quad z_R' = C_1 z_R + \omega_1, \quad z_R(a) = 0,$$

where ω_1 is smooth, nonnegative, zero near a , and at least one on I . Take C_1 sufficiently large, depending only on the constants in Equations (3.23) and (3.24). The solution is nonnegative and has uniform bounds on this fixed interval. For a function with $F_R' = -z_R$ the chain rule gives

$$\begin{aligned} -\Delta_{g_R} u_R - |K_R|_{g_R} |du_R|_{g_R} &= \frac{\varepsilon_R}{R^3} \left[z_R' |d\rho_R|_{g_R}^2 + R z_R \Delta_{g_R} \rho_R - R z_R |K_R|_{g_R} |d\rho_R|_{g_R} \right] \\ &\geq \frac{\varepsilon_R}{R^3} (c_1 z_R' - C_2 z_R). \end{aligned}$$

Choose C_1 so that this is nonnegative and is at least $c_1 \varepsilon_R R^{-3}$ on I .

The preceding differential inequality need not be imposed on the spherical tail. There $K_R = 0$ and the Schwarzschild radial Laplace flux coefficient is $(r + m/2)^2$. Put $c_R = m/(2R)$ and, on the still spherical interval near b , define

$$A_R(t) = (t + c_R)^2 z_R(t).$$

It has positive derivative wherever $z_R > 0$. Continue it to a positive constant on $[b + 1, \infty)$ while keeping $A'_R \geq 0$. For an explicit smooth continuation, continue the solution of Equation (3.27) to $b + 1$, multiply the derivative of $(t + c_R)^2 z_R(t)$ by a nonnegative cutoff equal to one near b and zero near $b + 1$, and integrate with the initial value $A_R(b)$. Define $z_R = A_R(t)/(t + c_R)^2$ on this tail, and set it to zero for $t \leq a$. All joins are smooth, the functions are nonnegative, and their bounds on finite intervals are uniform in R . The constant tail value, denoted A_R^∞ , is bounded independently of R .

Now define

$$F_R(t) = \int_t^\infty z_R(s) \, ds.$$

It is nonnegative, uniformly bounded, and constant before a . The preceding construction gives a superharmonic u_R on the spherical tail, because its signed radial flux is $-\varepsilon_R A_R(t)$ and is nonincreasing. It also gives Equation (3.26). For large R , $1 \leq u_R \leq 2$. Increasing C_0 ensures

$$-\Delta_{g_R} u_R - |K_R| |du_R| \geq \frac{u_R}{8} [16\pi(|J_R|_{g_R} - \mu_R)]_+.$$

On the original inner region the conformal factor is constant and the original data satisfy DEC. Elsewhere this inequality and Lemma 3.1 prove DEC for the repaired data. Using u_R , rather than estimating $\log u_R$ separately, incorporates the gradient-square term exactly.

On the final end,

$$u_R = 1 + \frac{\varepsilon_R A_R^\infty}{r + m/2}.$$

Writing $a_R = \varepsilon_R A_R^\infty = o(1)$, one obtains the exact identity

$$(3.28) \quad u_R^4 \left(1 + \frac{m}{2r}\right)^4 \delta = \left(1 + \frac{m/2 + a_R}{r}\right)^4 \delta.$$

Thus the final mass is $m + 2a_R = m + o(1)$ and the momentum is zero. Multiplication of K_R by u_R^2 preserves its compact support. On the inner boundary, where u_R is constant, the expansion is simply multiplied by u_R^{-2} , so its weak sign is preserved. $\square$

3.6. The numerical reduction. The preceding construction has replaced the end while preserving DEC and the boundary expansion sign. A final strict conformal change puts the data in the class used for the elliptic deformation.

Proposition 3.10 (Reduction to strict data with a controlled rest end). *Let (N, g, K) satisfy the exterior hypotheses of Theorem 2.2, including $\theta_+ \leq 0$ on its smooth compact boundary, and let $m = (E^2 - |P|^2)^{1/2} > 0$. There is a sequence of smooth exterior data (g_j, K_j) on N with the following properties:*

- (i) $\mu_j > |J_j|_{g_j}$ everywhere and $\theta_{+,j} < 0$ on B . The constraint densities are integrable. On the end their DEC margin is bounded below by $c_j r^{-3-\delta}$ for some $c_j > 0$ and one fixed $0 < \delta < 1$.
- (ii) K_j is compactly supported, and in a rest chart $g_j - \delta = O_k(r^{-1})$ and $R_{g_j} = O_k(r^{-3-\delta})$ for every k . In particular $\text{Ric}_{g_j} = O(r^{-3})$ there. Their ADM momentum is zero.

- (iii) $(g_j, K_j) \rightarrow (g, K)$ smoothly on every fixed compact subset up to the boundary, their ADM energies satisfy $E_j \rightarrow m$, and $a_{g_j}(B) \rightarrow a_g(B)$.

Consequently it suffices to prove the energy/enclosing-area inequality for the class in (i)–(ii) in order to prove Theorem 2.2 in its stated class.

Proof. Apply Proposition 3.7, Lemma 3.8, and Lemma 3.9 along any sequence $R_j \rightarrow \infty$. The resulting DEC exteriors have compactly supported second fundamental form and an exact rest Schwarzschild end with energy tending to m . They agree with the original data on expanding compact subsets except for a constant conformal factor tending to one. Hence they converge smoothly on each fixed compact subset. Their boundary expansion is weakly negative.

Apply the last part of Lemma 3.2 separately to each of these fixed exteriors, using the same chosen $0 < \delta < 1$ and a compactly supported drift coefficient. Choose its positive parameter s in Equation (3.8) small enough that the energy change, $\|\log(1 + s\phi)\|_\infty$, and the supremum of its first derivative multiplied by the fixed original coordinate radius are at most $1/j$. Also require the change in the first j smooth norms on the first j members of a compact exhaustion to be at most $1/j$. All these quantities are finite for the fixed j -th exterior, so these choices are possible. No estimate uniform in the receding bending radius is required for this final strictification.

The resulting data satisfy (i). The strict conformal factor has symbol decay of all orders and does not enlarge the support of K . On the far Schwarzschild end the drift vanishes, so $-\Delta\phi = w = r^{-3-\delta}$. Since the rest Schwarzschild metric has zero scalar curvature and $K = 0$ there, the exact u -formula gives

$$R_{g_j} = 8s(1 + s\phi)^{-5}r^{-3-\delta}.$$

This proves the scalar-curvature bound in (ii). The Ricci bound follows from the two-derivative metric decay. The energy change tends to zero and the momentum change is zero. The prescribed smallness gives smooth local convergence, proving (ii) and the local convergence and charge assertions in (iii).

The metrics before strictification satisfy the uniform comparisons and scaled first-derivative bounds of Lemma 3.3, in the fixed original end chart: the reference plane, the bend, and the radius interpolation involve only fixed linear maps and fixed radius ratios. The radial repair preserves these bounds. The final strictification parameters were chosen to preserve them as well. That lemma therefore gives the convergence of enclosing infima.

If the reduced energy inequality is available, applying it to these data gives

$$E_j \geq \sqrt{a_{g_j}(B)/(16\pi)}.$$

Passing to the limit proves $m \geq \sqrt{a_g(B)/(16\pi)}$. Only the energies and enclosing infima enter this passage to the original data. $\square$

4. TRAPPED DOMAINS, COLLARS, AND WEAK SUPPORTS

We prepare the domain and boundary values for the elliptic deformation. The construction has three steps: choose maximal trapped regions, foliate their outer collars, and fix small expansion offsets. Throughout, the data satisfy Proposition 3.10. Write S_0 for their inner boundary. Thus S_0 is strictly future outer trapped, K has compact support, and the single end has the smooth $O(r^{-1})$ bounds obtained in that reduction. In particular, sufficiently large coordinate spheres have positive outward mean curvature. The function

$$(4.1) \quad \mathbf{m}(x) = 8\pi(\mu(x) - |J(x)|_g)$$

is positive, has a positive lower bound on every compact set, and has the positive $r^{-3-\delta}$ lower bound on the end supplied by the reduction, with $0 < \delta < 1$. Let A_* denote the reduced data's filled enclosing perimeter infimum.

4.1. The bounding-region convention. For a symmetric tensor Q and a two-sided surface Σ with a specified unit normal ν , put

$$(4.2) \quad \Theta_Q(\Sigma, \nu) = H_\Sigma(\nu) + \text{tr}_\Sigma Q.$$

For the boundary of a domain the specified normal always points *out of that domain*. Notice both elementary identities

$$(4.3) \quad \Theta_{Q-(c/2)g} = \Theta_Q - c, \quad \Theta_{-Q}(\Sigma, -\nu) = -\Theta_Q(\Sigma, \nu).$$

The factor $1/2$ in the first formula is the reciprocal of the surface dimension.

We use the following geometric theorem. Its boundary parts may be disconnected.

Theorem 4.1 (MOTS barriers and the total trapped region). *Let (N, g, Q) be a smooth compact three-dimensional initial data set with boundary $\Gamma_- \sqcup \Gamma_+$. Assume Γ_+ is nonempty and has positive expansion with its normal out of N . The inner part Γ_- may be empty; when it is present, assume its expansion is negative with its normal into N .*

Consider all smooth domains whose inner boundary is Γ_- and whose remaining boundary Σ lies in the interior of N and satisfies $\Theta_Q(\Sigma) \leq 0$. They avoid Γ_+ , and their normal on Σ points out of the domain. If their union is nonempty, its closure has a smooth compact embedded stable MOTS frontier in the interior, separating it from Γ_+ . This closure is itself the closure of such a domain and contains every domain in the defining class. If Γ_- is nonempty, the two strict barriers also give a nonempty smooth embedded enclosing MOTS.

The total-region assertion is [7, Definitions 7.1–7.2 and Theorem 7.3]; the barrier assertion is Theorem 3.1 there, with stability furnished by Theorem 4.1. The preliminary conventions in Section 2 of that paper allow an empty inner boundary. These results require no energy condition on Q . We therefore apply them to the shifted tensors in Equation (4.3), even though these tensors need not satisfy the dominant energy condition. A component of the complement of a smooth trapped domain which meets no designated outer boundary may be filled: its entire boundary is then deleted and the remaining boundary has the same expansion. The maximal region is therefore filled in this relative sense.

We use compact inner fillings only to express enclosing competitors. One may attach a smooth collar behind S_0 , extend the one-sided metric smoothly, and complete the inner side by a fixed compact filling. All competitors contain that filling and only their boundaries in the original exterior are counted. This introduces no geometric condition behind S_0 . For perimeter minimization with this smooth obstacle, the relevant regularity input is [31, Regularity Theorem 1.3(iii)]: the minimizing boundary is $C^{1,1}$ and is smooth away from contact. We retain the filled enclosing class and its smooth approximation from Section 3. The trapped regions below are not assumed to minimize perimeter.

4.2. Choosing the two maximal regions. Fix a large sphere S_{R_0} outside $\text{supp } K$ such that all coordinate spheres with radius at least R_0 have positive outward mean curvature. A compact smooth domain with $\Theta_{\pm K} \leq c \leq 0$ cannot reach this part of the end. Indeed, at a point where its boundary maximizes the coordinate radius, it lies inside the tangent coordinate sphere and has the same outward normal. The graph second-derivative comparison gives $H_{\partial D} \geq H_{S_r} > 0$ there; also $K = 0$ there. This contradicts its assumed expansion. The same argument applies to an enclosing domain with prescribed inner obstacle, since the maximum radius is attained on its free boundary. Consequently all the negative-threshold regions used below lie in one fixed

compact radial range. They may be constructed using any sufficiently distant sphere as the outer barrier, without dependence on that choice.

For

$$(4.4) \quad \max_{S_0} \Theta_K(S_0) < c < 0,$$

let $\mathcal{B}_c$ be the closed total trapped region for the condition $\Theta_K \leq c$, with S_0 as the required inner boundary and with its inner filling understood. It is nonempty: a sufficiently short outward collar of S_0 supplies a strict seed. Theorem 4.1, applied to $K - (c/2)g$, gives its smooth stable frontier with expansion exactly c . The regions $\mathcal{B}_c$ increase with c . Their complement toward the end has no bounded component, since such a component could be filled to enlarge the total trapped region.

We will choose $c_b < 0$ close to zero and set $\mathcal{B} = \mathcal{B}_{c_b}$. In the complement of $\mathcal{B}$, use the full black frontier and a distant coordinate sphere as *outer* barrier faces. With the normal out of this complement, the black frontier has expansion for $-K$ equal to $-c_b > 0$. For $c < 0$ close to zero, let $\mathcal{W}_c$ be the closed total trapped region of compact smooth domains in that complement with

$$(4.5) \quad \Theta_{-K} = H - \text{tr}_\Sigma K \leq c.$$

There is no required inner boundary in this application of Theorem 4.1. The shifted expansion of a reversed black face is $-c_b - c > 0$, and that of a distant sphere is also positive. Thus the theorem applies whenever the defining union is nonempty. When the union is empty we simply put $\mathcal{W}_c = \emptyset$. When nonempty, it has smooth stable frontier of white expansion c and lies a positive distance from all black faces. The family $\mathcal{W}_c$ is increasing, with the black region held fixed.

To control limits of the deformation, we choose thresholds at which these regions are right-continuous. The following lemma includes empty regions.

Lemma 4.2 (Right-continuous thresholds). *Let E_c , c in an interval, be an increasing family of closed subsets of a fixed compact metric space. Outside a countable set of parameters,*

$$(4.6) \quad E_{c_j} \longrightarrow E_c \quad \text{in Hausdorff distance whenever } c_j \downarrow c.$$

Here continuity at $E_c = \emptyset$ means that $E_{c'}$ is empty for every sufficiently close $c' > c$.

Proof. Choose a countable dense set $\{x_i\}$ and a number L larger than the diameter of the compact space. Set $d_c(x) = \min\{L, \text{dist}(x, E_c)\}$, with the value L when E_c is empty. For each i , $c \mapsto d_c(x_i)$ is a bounded nonincreasing real function, so its right discontinuities form a countable set. Avoid their countable union. Then $d_{c_j}(x_i) \rightarrow d_c(x_i)$ for all i whenever $c_j \downarrow c$. Every d_c is 1-Lipschitz. A finite net chosen from the dense set therefore upgrades this convergence to uniform convergence on the compact space.

For nonempty E_c , if points $y_j \in E_{c_j}$ stayed a fixed distance from E_c , uniform convergence evaluated at y_j would contradict $d_{c_j}(y_j) = 0$. The opposite Hausdorff inclusion follows from $E_c \subset E_{c_j}$. For empty E_c , uniform convergence to L is incompatible with any nonempty E_{c_j} , since its distance function has a zero. This proves the stated convention as well. $\square$

Choose c_b at such a right-continuity point of the black family, then choose $c_w < 0$ at such a point of the white family for this fixed black region. Both choices can be made arbitrarily close to zero. Write $\mathcal{W} = \mathcal{W}_{c_w}$, and let Ω be the closure of the component of the complement of $\mathcal{B} \cup \mathcal{W}$ which reaches the selected end. Its boundary B is a finite disjoint union of complete components of the two smooth frontiers. Write

$$(4.7) \quad B = B_b \sqcup B_w, \quad H = H_B(\nu), \quad P_B = \text{tr}_B K,$$

where ν points into Ω . Thus

$$(4.8) \quad H + P_B = c_b \quad \text{on } B_b, \quad H - P_B = c_w \quad \text{on } B_w.$$

The set B_w may be empty, and B_b need not contain every black frontier component: a white component may cut a black face off from the end. We retain exactly the boundary of the component that reaches infinity. The compact, disjoint frontiers create no intersection corners.

The excluded side of B contains S_0 and a collar of it. Hence every compact smooth enclosing cut T in Ω , with all its components retained, is an enclosing competitor for the original filled obstacle after adjoining the discarded regions to its inner side. It follows that

$$(4.9) \quad |T|_g \geq A_*.$$

The same conclusion holds for perimeter competitors. This is a statement about inclusion of competitor classes; no minimizing property of B is used.

4.3. Outward collars, including zero stability eigenvalue.

Lemma 4.3 (Collars of a maximal frontier). *Let Σ be a connected component of the frontier of a closed total trapped region at threshold c , for a fixed tensor Q . There is a smooth foliation Σ_s , $0 \leq s < s_0$, of an outward collar, with $\Sigma_0 = \Sigma$ and positive outward normal velocity, such that*

$$(4.10) \quad \Theta_Q(\Sigma_s) > c \quad (0 < s < s_0), \quad \Theta_Q(\Sigma_0) = c.$$

In particular its leaf parameter s is a smooth defining function and $|\nabla s|$ is bounded above and below by positive constants.

Proof. Use normal graphs $\Sigma(v)$ over Σ and pull their expansion back to Σ . The map

$$\Phi : C^{2,\alpha}(\Sigma) \longrightarrow C^{0,\alpha}(\Sigma), \quad \Phi(v) = \Theta_Q(\Sigma(v)) - c$$

is smooth near zero and has linearization $L = D\Phi(0)$, the MOTS stability operator for $Q - (c/2)g$. Its principal part is $-\Delta_\Sigma$. Stability gives a principal eigenvalue $\lambda \geq 0$ and a positive smooth eigenfunction φ . If $\lambda > 0$, take $v = s\varphi$. Smooth dependence gives $\Phi(s\varphi) = s\lambda\varphi + O(s^2)$ in C^0 , which is positive for all sufficiently small $s > 0$. The graph velocity is positive as well.

Suppose $\lambda = 0$. The kernel of L is spanned by φ , and the adjoint kernel is spanned by a positive smooth function φ^* . These are the principal-eigenfunction facts for a scalar elliptic operator on a connected closed surface. For example, simplicity of the kernel also follows by writing a kernel element as $v = \varphi w$ and applying the strong maximum principle to the resulting equation for w , whose zeroth-order term vanishes. Consider the augmented map

$$\mathcal{G}(v, a) = \left(\Phi(v) - a, \int_\Sigma v \, dA \right).$$

Its derivative at $(0, 0)$ is

$$(4.11) \quad (v, a) \longmapsto \left(Lv - a, \int_\Sigma v \, dA \right).$$

This map is an isomorphism. Indeed, to solve $Lv - a = f$ with $\int v = b$, the Fredholm compatibility condition uniquely fixes

$$a = -\frac{\int_\Sigma \varphi^* f \, dA}{\int_\Sigma \varphi^* \, dA}.$$

The equation for v then has a solution; adding a multiple of φ imposes its prescribed integral uniquely. Elliptic estimates and this one-dimensional normalization give a bounded inverse between the indicated Hölder spaces.

The implicit-function theorem consequently supplies smooth $v(s), a(s)$ with $\mathcal{G}(v(s), a(s)) = (0, s)$. Differentiating at zero and pairing with φ^* gives $a'(0) = 0$ and

$$v'(0) = \frac{\varphi}{\int_{\Sigma} \varphi \, dA} > 0.$$

After shortening the interval, all these graphs form an outward foliation and have constant expansion $c + a(s)$. If $a(s) \leq 0$ at any positive s in this interval, replace only this frontier component by $\Sigma(v(s))$. The collar is disjoint from all other components and designated barriers, so the new smooth bounding domain strictly contains the maximal region and its every free boundary component has expansion at most c . This contradicts maximality. Therefore $a(s) > 0$ for every such $s > 0$.

In either case positive graph velocity and compactness give the asserted bounds on $|\nabla s|$. Smooth elliptic bootstrapping of the graph equation gives smooth leaves in the zero-eigenvalue case. $\square$

Apply this lemma separately with $Q = K$ to black faces and $Q = -K$ to white faces. There are finitely many faces, so choose disjoint collars in Ω and shorten them to have a common positive parameter range when convenient. Write $\nu_s = \nabla s / |\nabla s|$ for their outward leaf normal; it agrees with ν at B .

4.4. From weak exterior supports to a smooth enclosure. The sublevel limits used below need not have smooth frontiers. We now show how an expansion bound on their exterior supports produces a smooth enclosure. The argument applies to arbitrary closed sets, without positive reach or a curvature bound on the supporting surfaces.

Definition 4.4. Let D be closed in a smooth three-manifold and let $x \in \partial D$. A smooth exterior support is a smooth function ϕ near x with $\phi(x) = 0$, $d\phi(x) \neq 0$, and $D \subset \{\phi \leq 0\}$ locally near x . Its normal is $n = \nabla \phi / |\nabla \phi|$. Define

$$(4.12) \quad \mathcal{E}_Q[\phi](x) = \frac{(g^{ij} - n^i n^j) \nabla_i \nabla_j \phi}{|\nabla \phi|} + \operatorname{tr} Q - Q(n, n).$$

We say D has exterior-support expansion at most c if every such support at every point of its frontier has $\mathcal{E}_Q[\phi] \leq c$.

For a smooth domain this is exactly the usual upper bound on its outward expansion: at contact an exterior supporting graph has mean curvature at most that of the enclosed smooth graph. The definition is unchanged by increasing smooth reparametrizations of ϕ , because the additional Hessian term is a multiple of $d\phi \otimes d\phi$ and has zero tangential trace.

Lemma 4.5 (Parallel neighborhoods and additive error). *Suppose the frontier of a closed set D lies in a fixed compact interior region of smooth (N, g, Q) and has exterior-support expansion at most c . Put $D_z = \{y : \operatorname{dist}(y, D) \leq z\}$. For sufficiently small $z > 0$, all exterior supports of D_z have expansion at most*

$$(4.13) \quad c + Cz.$$

The allowable radius and C depend only on the compact ambient geometry and Q , and not on the support, its second fundamental form, or the regularity of D .

Proof. Take a support ϕ to D_z at x' . There is a nearest point $x \in D$ and a length- z minimizing unit-speed geodesic $\gamma : [0, z] \rightarrow N$ from x to x' . For the radii under consideration the entire segment lies in a fixed smooth compact neighborhood and is shorter than its injectivity radius.

The ball $\overline{B}_z(x)$ is contained in D_z and touches its exterior support at x' . Its tangent plane there therefore agrees with that of the support and

$$(4.14) \quad \frac{\nabla \phi(x')}{|\nabla \phi(x')|} = \dot{\gamma}(z).$$

Let $P_t : T_x N \rightarrow T_{\gamma(t)} N$ denote parallel transport and put $P = P_z$.

For every $X \in T_x N$ solve the Jacobi boundary problem

$$(4.15) \quad J_X'' + \text{Rm}(J_X, \dot{\gamma})\dot{\gamma} = 0, \quad J_X(0) = X, \quad J_X(z) = PX.$$

Here primes denote covariant differentiation along γ . The short interval and absence of conjugate points make this problem uniquely solvable. Its estimates can be seen without division by an uncontrolled quantity. In a parallel frame write $j_X(t) = P_t^{-1} J_X(t)$ and $\mathcal{R}(t)$ for the curvature coefficient. The integral equation is

$$j_X(t) = X + tA_X - \int_0^t (t-s)\mathcal{R}(s)j_X(s) \, ds, \quad A_X = \frac{1}{z} \int_0^z (z-s)\mathcal{R}(s)j_X(s) \, ds.$$

It implies, by absorption for small z ,

$$(4.16) \quad \sup_{0 \leq t \leq z} |j_X(t)| \leq 2|X|, \quad |A_X| \leq Cz|X|.$$

Also $\langle J_X, \dot{\gamma} \rangle$ is affine in t and has equal endpoint values. Thus $\langle A_X, \dot{\gamma}(0) \rangle = 0$.

We may therefore prescribe the first jet of a smooth unit vector field V near x by

$$V(x) = \dot{\gamma}(0), \quad \nabla_X V(x) = A_X.$$

To realize it with uniform second-jet bounds, choose a smooth local orthonormal frame obtained by radial parallel transport from x . In that frame take the affine coefficient vector with the prescribed value and first derivatives, and normalize its length. The orthogonality just proved means normalization preserves its prescribed first derivatives. The frame has uniformly bounded derivatives on a fixed small coordinate ball, and Equation (4.16) therefore gives uniformly bounded second derivatives of V . This construction uses the radial direction and the ambient geometry, with no derivatives of the support ϕ .

Define the local endpoint map

$$F_z(y) = \exp_y(zV(y)).$$

Its differential at x is the terminal value of the Jacobi field with initial values $X, \nabla_X V(x)$. By Equation (4.15), it has the exact property

$$(4.17) \quad F_z(x) = x', \quad (dF_z)_x = P, \quad |(\nabla dF_z)_x| \leq Cz.$$

The last bound follows by differentiating the smooth map $(y, v, z) \mapsto \exp_y(zv)$ twice in y , including the first and second derivatives of V . At $z = 0$ the map is the identity and its covariant second derivative vanishes. Its z -derivative is uniformly bounded for the bounded jets just constructed, giving the displayed Cz bound. The exact middle identity, rather than a near-isometry estimate, is essential.

The map F_z takes D locally into D_z , since its defining geodesic from any $y \in D$ has length z . Consequently $\psi = \phi \circ F_z$ is an exterior support to D at x . Its derivative is $P^* d\phi$, so its gradient length is exactly $|\nabla \phi(x')|$ and, by Equation (4.14), its normal is $\dot{\gamma}(0)$. The covariant chain rule gives

$$\text{Hess } \psi(X, Y) = \text{Hess } \phi(PX, PY) + d\phi((\nabla dF_z)(X, Y)) \quad \text{at } x.$$

Trace over an orthonormal basis of $\dot{\gamma}(0)^\perp$ and divide by the common gradient length. The first term is exactly the corresponding normalized tangential trace at x' , and the remaining term has

absolute value at most Cz . Finally, the tensor trace changes by at most Cz , by the C^1 bound on Q and parallel transport. It follows that

$$|\mathcal{E}_Q[\psi](x) - \mathcal{E}_Q[\phi](x')| \leq Cz.$$

The assumed support inequality at x proves Equation (4.13). There is no term containing $z|\text{Hess } \phi|$. $\square$

Lemma 4.6 (A smooth enclosure from weak supports). *Let N be a smooth compact connected three-manifold with boundary $\Gamma_- \sqcup \Gamma_+$, where Γ_+ is nonempty and Γ_- may be empty. Let D be a nonempty closed subset which contains a relative collar of Γ_- when that boundary is prescribed, and stays a positive distance from Γ_+ . Suppose its interior frontier has exterior-support expansion at most c for a smooth tensor Q .*

For every $c' > c$ such that $\Theta_Q(\Gamma_+) > c'$ with its normal out of N , there is a smooth bounding domain E containing D in its relative interior, containing the required inner collar, and avoiding Γ_+ , whose free boundary satisfies

$$(4.18) \quad \Theta_Q(\partial E \setminus \Gamma_-) = c'.$$

Its boundary and the domain may be disconnected. In particular D is contained in the closed total trapped region at any such larger threshold. No regularity or positive-thickness assumption on D is required.

Proof. All distance neighborhoods in this proof are relative to N . Since D contains a collar of Γ_- and avoids Γ_+ , their new frontiers, at sufficiently small radii, lie in a compact subset of the interior. A minimizing segment ending on a new frontier has its nearest point on the interior frontier of D ; it cannot start on Γ_- because its covered collar has fixed positive width. Thus the proof of Lemma 4.5 applies unchanged. Choose $\beta > 0$ so small that its conclusion holds for $0 < z \leq \beta$, all these frontiers avoid the prescribed boundary, and

$$(4.19) \quad c + C\beta < c'.$$

We first construct a smooth inner enclosure without imposing any expansion bound on it. Smoothly approximate the continuous distance function $d_D = \text{dist}(\cdot, D)$ on N uniformly, with error less than $\beta/32$, using a finite coordinate cover, mollification, and a partition of unity. Choose a regular value in $(\beta/3, \beta/2)$ of the resulting smooth function. Its sublevel set U has smooth interior boundary and, for example, satisfies

$$(4.20) \quad D_{\beta/4} \subset U \subset D_{3\beta/4}.$$

The boundary Γ_- is contained in its relative interior and Γ_+ is excluded. If a component of $N \setminus \overline{U}$ meets no component of Γ_+ , fill it. This only deletes components of the smooth free boundary. After this filling the second inclusion in Equation (4.20) need not hold for its interior, but every remaining free boundary component still lies in the indicated thin neighborhood, and the first inclusion still holds.

Choose $\eta \in C^\infty(N)$, $0 \leq \eta \leq 1$, equal to one on that free boundary and supported in $\{d_D < \beta\}$. Its existence uses the positive separation between this compact smooth boundary and the closed set $\{d_D \geq \beta\}$; it requires no smoothness of d_D . For a sufficiently large finite constant A , the tensor

$$(4.21) \quad Q_A = Q - \frac{1}{2}(c' + A\eta)g$$

makes the free boundary of U a strictly negative inner barrier:

$$\Theta_{Q_A}(\partial U) = \Theta_Q(\partial U) - c' - A < 0.$$

All prescribed outer faces remain strictly positive for Q_A , because $\eta = 0$ there and their original expansion exceeds c' . Apply the barrier part of Theorem 4.1 on each component of $N \setminus \overline{U}$. Each

meets Γ_+ by the filling step and has nonempty inner boundary by connectedness of N and $U \neq \emptyset$. We obtain smooth MOTSs Σ bounding a domain E which contains $\bar{U}$ and avoids Γ_+ . They satisfy, with their normal out of E ,

$$(4.22) \quad \Theta_Q(\Sigma) = c' + A\eta \geq c'.$$

We claim Σ misses $\{d_D < \beta\}$. Otherwise set $z = \min_\Sigma d_D < \beta$. The first inclusion in Equation (4.20) gives $z \geq \beta/4 > 0$. Moreover D_z lies on the inner side of Σ . To see this directly, if $d_D(y) < z$, a shortest path from y to D of length less than z cannot cross Σ : such a crossing would have distance from D less than z . Since $D \subset E$, the whole path and y lie in E . Closure gives $D_z \subset \bar{E}$.

At a point attaining the minimum, a defining function for the smooth boundary Σ is therefore an exterior support to D_z , with the same outward normal as in Equation (4.22). The parallel support bound and Equation (4.19) imply

$$\Theta_Q(\Sigma) \leq c + Cz < c',$$

contradicting Equation (4.22). This excludes the entire modification range, so $\eta = 0$ on Σ and proves Equation (4.18). The contained $D_{\beta/4}$ gives strict enclosure of D . If desired, fill any remaining complementary components which meet no designated outer face; this deletes smooth boundary components and preserves all stated properties. $\square$

For an exterior, truncate by a distant sphere that is strict at the larger threshold. In the black construction, $\mathcal{B}$ already contains the original inner collar. In the white construction there is no required inner boundary, and the black faces belong to Γ_+ . The later applications therefore need compact avoidance of the foreign faces; they need no boundary regularity of the weak set. Empty weak sets can be omitted.

4.5. Small offsets and the prepared domain.

Proposition 4.7 (Prepared domain). *The thresholds and the domain above can be chosen with the following properties. The numbers $c_b, c_w < 0$ are right-continuity points of their respective full maximal-region families. The boundary of Ω consists of the faces in Equation (4.8), has disjoint smooth outward leaf collars as in Lemma 4.3, and preserves the enclosing-area lower bound in Equation (4.9).*

There are a smooth compactly supported function C on Ω , constants $C_b \geq 0$, $C_w \leq 0$, and positive constants k_B on its face collars such that

$$(4.23) \quad \begin{aligned} C_w \leq C \leq C_b, \quad C &= C_b - k_B s \quad \text{in a black collar near } B_b, \\ C &= C_w + k_B s \quad \text{in a white collar near } B_w. \end{aligned}$$

For an absent type, its corresponding constant is simply a bound. Set

$$(4.24) \quad b_- = c_b - C_b < 0, \quad b_+ = -c_w - C_w > 0.$$

There is a smooth positive function ρ , equal to a positive constant times r^{-4} on the far end and satisfying symbol derivative bounds there, such that, for every $h \in [b_-, b_+]$,

$$(4.25) \quad |(h + C) \operatorname{tr} K| + |\nabla C| + \rho \leq \frac{1}{2} \mathbf{m} \quad \text{everywhere on } \Omega.$$

The thresholds, collars, offset, and ρ are fixed before the deformation parameters are chosen. After fixing the thresholds and collars, the offset and its first derivatives can be made arbitrarily small.

Proof. The large-sphere comparison above confines all negative-threshold regions to a fixed compact set, independently of how close the thresholds are to zero. Choose a slightly larger compact set $\mathcal{K}$ containing this range and $\operatorname{supp} K$ in its interior. Let

$$m_0 = \min_{\mathcal{K}} \mathbf{m} > 0, \quad T_0 = \sup_{\mathcal{K}} |\operatorname{tr} K|.$$

First restrict both negative thresholds to an interval so close to zero that

$$\max\{|c_b|, |c_w|\} T_0 < m_0/8,$$

while retaining Equation (4.4). Lemma 4.2 allows the required successive choices in that interval. Construct the two regions, select Ω , and fix their disjoint collars inside $\mathcal{K}$ as above. All ensuing constants may depend on these fixed choices.

Here is an explicit offset construction. On each collar choose a smooth nonincreasing cutoff $\chi(s)$, equal to one near $s = 0$ and zero near the collar's far end. Choose $\kappa_B > 0$ so that $1 - \kappa_B s > 0$ throughout that collar. For a common amplitude $a > 0$, define the contribution of a black collar to be $a\chi(s)(1 - \kappa_B s)$, and that of a white collar to be $-a\chi(s)(1 - \kappa_B s)$. Extend by zero outside the disjoint collars and sum. This produces a smooth function on the manifold with boundary Ω , supported in $\mathcal{K}$, with $C_b = a$, $C_w = -a$, and $k_B = a\kappa_B$ on the respective collars. These remain valid bounds when a type is absent. Its C^1 norm tends to zero with a , even though the collar widths have already been fixed.

For $h \in [b_-, b_+]$ and $C \in [C_w, C_b]$,

$$|h + C| \leq \max\{|c_b|, |c_w|\} + C_b - C_w.$$

Choose a so small that $(C_b - C_w)T_0 + \|\nabla C\|_\infty < m_0/8$. The first two terms of Equation (4.25) are then at most $m_0/4$ on $\mathcal{K}$ and vanish outside $\mathcal{K}$. Finally take a fixed smooth positive weight equal to r^{-4} far out and multiply it by a sufficiently small positive constant. The end lower bound for $\mathfrak{m}$ and $\delta < 1$, together with compact positivity, allow the choice $\rho \leq \mathfrak{m}/4$ everywhere. This proves Equation (4.25). All remaining assertions were proved in constructing the regions and collars. $\square$

At the assigned face heights the quantity $h + C$ consequently has the useful exact values

$$(4.26) \quad \begin{aligned} h + C &= c_b = P_B + H && \text{on } B_b \text{ when } h = b_-, \\ h + C &= -c_w = P_B - H && \text{on } B_w \text{ when } h = b_+. \end{aligned}$$

These identities determine the deformation's boundary signs. Right-continuity and Lemma 4.6 will exclude limiting sublevel sets away from the corresponding full trapped regions.

5. THE ELLIPTIC DEFORMATION AND EXCLUSION OF THE FLOOR

We define the coupled system, derive its scalar-curvature identity, and choose a continuation path along which the conformal factor cannot touch its lower bound. We use the fixed data of Proposition 4.7. Thus Ω has one controlled asymptotically flat end, K is compactly supported, and its compact boundary $B = B_b \cup B_w$ consists of black and white faces. Either type may be absent. The normal ν points into Ω , and

$$H = \operatorname{div}_B \nu, \quad P_B = \operatorname{tr}_B K.$$

The numbers $b_- < 0 < b_+$, the smooth compactly supported offset C , and the positive function $\rho = O(r^{-4})$ are fixed as in that proposition. In particular,

$$(5.1) \quad b_- + C = P_B + H \text{ on } B_b, \quad b_+ + C = P_B - H \text{ on } B_w.$$

All compact sets and geometric constants in this section are fixed after this preparation. Extend the end radius to a smooth function on $\bar{\Omega}$ bounded below by one. Write Ω_R for the large spherical truncations and S_R for their outer boundary.

A *polynomial constant* is bounded by $C(1+n)^N$, with C, N depending on the prepared data and fixed auxiliary profiles, but not on R , the homotopy parameter, or the solution. Constants permitted to depend arbitrarily on fixed n will be identified explicitly. First fix $0 < \epsilon < 1$, then choose n sufficiently large, and finally require $R \geq R_{\min}(n, \epsilon)$. We always take $n \geq 4$ and $n > 2/\epsilon$.

5.1. Variables and equations. Choose a smooth nonincreasing $\vartheta : \mathbb{R} \rightarrow [0, 1]$ equal to one on $(-\infty, 0]$ and zero on $[1, \infty)$. Define

$$(5.2) \quad \begin{aligned} p(t) &= n\vartheta(nt), & l(t) &= \exp\left(\int_0^t p(s) \, ds\right), \\ L(t) &= l(t)e^{2t}, & p_L &= p + 2. \end{aligned}$$

Thus $l(0) = 1$, $l(t) = e^{nt}$ for $t \leq 0$, and l is constant for $t \geq 1/n$. Derivatives of p of every fixed order are polynomially bounded. Set

$$(5.3) \quad \ell = e^{-\epsilon n}, \quad \tau = \ell^{3/2}.$$

Whenever $t \geq -\epsilon$, one has $\ell \leq l(t) \leq e$.

For a function f , set

$$(5.4) \quad \begin{aligned} \sigma &= |\nabla f|, & D &= 1 + l^2 \sigma^2, & d &= D^{-1}, \\ a &= l\sigma/\sqrt{D}, & v &= a^2 = 1 - d, & w &= l\nabla f/\sqrt{D}, & \chi &= 4d/(4 + pv), \\ Z &= t + \frac{1}{8} \log D, & u &= L\sqrt{D}, & U &= l\sqrt{D}, & h &= \tau f. \end{aligned}$$

Derivatives and contractions refer to g unless indicated otherwise. At $\sigma > 0$, write $e = \nabla f/\sigma$ and $A_j = I + (j - 1)e \otimes e$. The matrices used below have the direction-free expressions

$$(5.5) \quad A_d = I - w \otimes w, \quad A_\chi = I - \frac{p + 4}{4 + pv} w \otimes w.$$

Vectors and covectors are identified with g .

The unknowns will be (f, Z) . For fixed ∇f , $\partial Z/\partial t = 1 + pv/4 > 0$, and its defining expression tends to $-\infty$ and $+\infty$ at the two ends of the t -axis. It therefore defines a unique smooth $t = t(Z, \nabla f)$ for all inputs, without a floor assumption. The formulas above are smooth at $\nabla f = 0$, and $0 < \chi \leq d \leq 1$ for finite inputs. Define

$$(5.6) \quad \bar{g} = g + l^2 \, df^2, \quad \hat{g} = e^{4t} \bar{g}, \quad H^f = \frac{l}{\sqrt{D}} \text{Hess } f, \quad S = K + H^f.$$

The original trace equation is

$$(5.7) \quad \text{tr}_{A_\chi} S = F, \quad F = h + C,$$

where $\text{tr}_A S$ is contraction with the contravariant matrix A . Along the continuation path we will specify modified right sides F . The full flux is

$$(5.8) \quad V = uA_\chi(4\nabla Z + K(w, \cdot)).$$

At S_R prescribe $f = Z = 0$. Initially the inner Dirichlet values are $h = b_-$ on B_b and $h = b_+$ on B_w .

5.2. The exact scalar-curvature identity. The next identity separates the original constraint densities, a nonnegative quadratic term, and a divergence. The second equation will prescribe that divergence so that the deformed metric has nonnegative scalar curvature.

Proposition 5.1 (Deformation identity). *Let f, t be smooth and $F = \text{tr}_{A_\chi}(K + H^f)$. At $\sigma > 0$ decompose S relative to $\mathbb{R}e \oplus e^\perp$:*

$$T = S|_{e^\perp \times e^\perp}, \quad M = S(e, \cdot)|_{e^\perp}, \quad q = S(e, e), \quad \text{tr } T = F - \chi q.$$

Then

$$(5.9) \quad \frac{1}{2} e^{4t} \text{Scal}_{\hat{g}} = 8\pi(\mu + J(w)) + \mathcal{T} + w(F) - F \text{tr } K - u^{-1} \text{div } V,$$

where

$$(5.10) \quad \begin{aligned} \mathcal{T} = & \frac{1}{2}|T^{\text{tf}}|^2 + (M + pa\nabla_{\perp}t) \cdot (M + (p+2)a\nabla_{\perp}t) \\ & + 4(p+1)|dt|_{A_d}^2 + (\chi - \chi^2/4)q^2 + 2(p+1)\chi qa \partial_e t \\ & - \chi Fq/2 + 3F^2/4. \end{aligned}$$

After substituting $F = \text{tr}_{A_\chi} S$, this expression extends smoothly across $\sigma = 0$.

Proof. Introduce an auxiliary coordinate s_0 and the stationary Lorentz metric

$$\mathbf{g} = -l^2(ds_0 - df)^2 + \bar{g} = -U^2 ds_0^2 + g(dx + \beta ds_0, dx + \beta ds_0), \quad \beta = l^2 \nabla f = Uw.$$

Its s_0 -slices have future normal $N_0 = (\partial_{s_0} - \beta)/U$ and second fundamental form

$$(5.11) \quad k = -\frac{\text{sym} \nabla \beta}{U} = -H^f - p(dt \otimes w^b + w^b \otimes dt).$$

Write $\theta = \text{tr} k$ and, for symmetric tensors, put

$$\mathcal{B}(A, B) = \langle A, B \rangle - (\text{tr} A)(\text{tr} B), \quad P_A = A - (\text{tr} A)g, \quad Q = K - k.$$

The normal expansion equation and contracted Gauss equation are

$$\begin{aligned} \text{Ric}_{\mathbf{g}}(N_0, N_0) &= -N_0(\theta) - |k|^2 + U^{-1}\Delta U, \\ \text{Scal}_{\mathbf{g}} &= \text{Scal}_g + 2\text{Ric}_{\mathbf{g}}(N_0, N_0) + |k|^2 - \theta^2. \end{aligned}$$

Consequently $\text{Scal}_{\mathbf{g}} = \text{Scal}_g + |k|^2 + \theta^2 + 2N_0(\theta) - 2U^{-1}\Delta U$. Stationarity gives $UN_0(\theta) = -\beta(\theta)$, and $\text{div} \beta = -U\theta$ follows from (5.11). Thus

$$(5.12) \quad U \text{Scal}_{\mathbf{g}} = U(\text{Scal}_g + \mathcal{B}(k, k)) - 2\text{div}(\nabla U + \theta\beta).$$

In the coordinates $s_0 - f$ the same metric is a static warped product. Its mixed Christoffel symbols are $\mathbf{\Gamma}_{is}^s = \partial_i \log l$ and $\mathbf{\Gamma}_{ss}^i = l\bar{\nabla}^i l$; contraction gives $\text{Scal}_{\mathbf{g}} = \text{Scal}_{\bar{g}} - 2l^{-1}\bar{\Delta}l$. Since $\bar{g}^{-1} = A_d$ and $dV_{\bar{g}} = (U/l)dV_g$,

$$(5.13) \quad \bar{\Delta}\phi = \frac{l}{U} \text{div} \left(\frac{U}{l} A_d \nabla \phi \right).$$

Direct differentiation of U and (5.11) gives

$$\nabla U - \frac{U}{l} A_d \nabla l = -k(\beta, \cdot).$$

In fact both sides equal

$$\frac{l^4}{U} \text{Hess} f(\nabla f, \cdot) + \frac{l^3}{U} [|\nabla f|^2 dl + (\nabla f)(l) df],$$

as follows by differentiating $U^2 = l^2 + l^4|\nabla f|^2$. Substitution in (5.12) yields

$$U \text{Scal}_{\bar{g}} = U(\text{Scal}_g + \mathcal{B}(k, k)) + 2\text{div}(P_k \beta).$$

The constraints read $16\pi\mu = \text{Scal}_g - \mathcal{B}(K, K)$ and $8\pi J = \text{div} P_K$. Using $P_K : \nabla \beta = -U\mathcal{B}(K, k)$, expansion therefore gives

$$(5.14) \quad U \text{Scal}_{\bar{g}} = U[16\pi(\mu + J(w)) + \mathcal{B}(Q, Q)] - 2\text{div}(P_Q \beta).$$

Indeed the derivative of P_K cancels its term in $\text{div}(P_K \beta)$, and $\mathcal{B}(Q, Q) + 2\mathcal{B}(K, k) = \mathcal{B}(K, K) + \mathcal{B}(k, k)$.

The three-dimensional conformal formula is

$$\frac{1}{2}e^{4t} \text{Scal}_{\bar{g}} = \frac{1}{2} \text{Scal}_{\bar{g}} - 4\bar{\Delta}t - 4|dt|_{A_d}^2.$$

For $Y = P_Q w$, replacing U by $u = e^{2t}U$ gives $-U^{-1} \operatorname{div}(UY) = -u^{-1} \operatorname{div}(uY) + 2Y(t)$. Since $u/(U/l) = L$, Equation (5.13) similarly gives

$$-4\bar{\Delta}t - 4|dt|_{A_d}^2 = -u^{-1} \operatorname{div}(4uA_d \nabla t) + 4(p+1)|dt|_{A_d}^2.$$

Before the flux modification, the scalar-curvature expression is

$$(5.15) \quad \begin{aligned} 8\pi(\mu + J(w)) + \frac{1}{2}\mathcal{B}(Q, Q) + 2P_Q(w, \nabla t) + 4(p+1)|dt|_{A_d}^2 \\ - u^{-1} \operatorname{div}(u(P_Q w + 4A_d \nabla t)). \end{aligned}$$

At $\sigma > 0$,

$$(5.16) \quad 4dZ = (4 + pv)dt + v d \log \sigma.$$

Write $x = \partial_e t$ and $y = \nabla_{\perp} t$. The blocks of Q are

$$\begin{aligned} Q_{\perp\perp} &= T, & Q_{e\perp} &= M + pay, & Q_{ee} &= q + 2pax, \\ \operatorname{tr} Q &= F + (1 - \chi)q + 2pax. \end{aligned}$$

Hence $(P_Q w)_{\perp} = a(M + pay)$ and $(P_Q w)_e = a(\chi q - F)$. Together with $aH^f(e, \cdot) = v d \log \sigma$, Equation (5.16) proves component by component

$$(5.17) \quad V/u = P_Q w + 4A_d \nabla t + wF.$$

Also $uw = e^{2t}l^2 \nabla f$, so

$$(5.18) \quad \operatorname{div}(uw)/u = F + (1 - \chi)q - \operatorname{tr} K + 2(p+1)ax.$$

Replacing the flux in (5.15) adds $w(F) + F \operatorname{div}(uw)/u$. The remaining quadratic expression is

$$(5.19) \quad \begin{aligned} \mathcal{T} &= \frac{1}{2}\mathcal{B}(Q, Q) + 2P_Q(w, \nabla t) + 4(p+1)|dt|_{A_d}^2 \\ &+ F \left[F + \frac{p+4}{4+pv} S(w, w) + 2(p+1)w(t) \right]. \end{aligned}$$

Here $(1 - \chi)q = (p+4)S(w, w)/(4+pv)$, which gives a smooth formula at $w = 0$.

For the explicit coefficients, substitute $|T|^2 = |T^{\text{tf}}|^2 + (F - \chi q)^2/2$ in $\mathcal{B}(Q, Q)/2$. The transverse terms become $(M + pay) \cdot (M + (p+2)ay)$. The coefficients of q^2, Fq, F^2 are respectively $\chi - \chi^2/4, -\chi/2, 3/4$. The qx coefficient is $2(p+1)\chi a$, the Fx terms cancel, and the remaining x^2 term is $4(p+1)dx^2$. These are the terms in (5.10). $\square$

Lemma 5.2 (Polynomial coercivity). *The expression $\mathcal{T}$ is nonnegative. For $0 \leq p \leq n$,*

$$(5.20) \quad |dt|_{A_d}^2 + |T|^2 + |M|^2 + dq^2 + F^2 \leq C(1+n)^N \mathcal{T}$$

with universal C, N . At $dt = 0$ there is the stronger comparison, uniform in p, d ,

$$(5.21) \quad c(|T^{\text{tf}}|^2 + |M|^2 + \chi q^2 + F^2) \leq \mathcal{T} \leq C(|T^{\text{tf}}|^2 + |M|^2 + \chi q^2 + F^2).$$

Proof. With $x = \partial_e t$ and $y = \nabla_{\perp} t$, exact completion gives

$$(5.22) \quad \begin{aligned} \mathcal{T} &= \frac{1}{2}|T^{\text{tf}}|^2 + |M + (p+1)ay|^2 + [4(p+1) - v]|y|^2 \\ &+ 4(p+1)d \left(x + \frac{\chi a q}{4d} \right)^2 \\ &+ \frac{3}{4} \left(F - \frac{\chi q}{3} \right)^2 + \frac{4d(9-d)}{3(4+pv)^2} q^2. \end{aligned}$$

Completing first in x leaves $3\chi q^2/(4+pv)$; completing next in F leaves $3\chi/(4+pv) - \chi^2/12 = 4d(9-d)/[3(4+pv)^2]$. The latter coefficient is at least $32d/[3(4+n)^2]$. It controls dq^2 polynomially. The F -square then controls F^2 , since $\chi^2 q^2 \leq dq^2$. The transverse gradient coefficient is at least three, and the shifted M -square controls M with an additional factor $(1+n)^2$. The squared

d -norm of the shift in the x -square is $\chi^2 v q^2 / (16d) \leq d q^2 / 16$, so it also controls dx^2 . Finally $|T|^2 = |T^{\text{tf}}|^2 + (F - \chi q)^2 / 2$. This proves (5.20). If $dt = 0$, the sole cross term is $-\chi F q / 2$. Using $|\chi F q| / 2 \leq \chi q^2 / 4 + \chi F^2 / 4$ and $0 < \chi \leq 1$ proves (5.21). $\square$

Lemma 5.3 (Boundary identity). *On a face where f is constant,*

$$(5.23) \quad V_\nu / u = d(H + 4\partial_\nu t) - H + w_\nu(F - P_B).$$

Its mean curvature in $\hat{g}$, with normal into Ω , is $e^{-2t}\sqrt{d}(H + 4\partial_\nu t)$.

Proof. The tangential Hessian is $\text{Hess } f|_{TB} = (\partial_\nu f) \Pi_g$. Thus $\text{tr}_\perp H^f = w_\nu H$ and $\chi q = F - P_B - w_\nu H$. The normal component of (5.8), using (5.16), is

$$V_\nu / u = 4d\partial_\nu t + \chi w_\nu q = 4d\partial_\nu t + w_\nu(F - P_B) - vH.$$

The $\bar{g}$ -normal is $\sqrt{d}\nu$. Its mean curvature is $\sqrt{d}H$, because the tangential metric is unchanged and its normal first variation is multiplied by $\sqrt{d}$. The conformal mean-curvature formula gives the asserted mean curvature for $\hat{g}$. $\square$

5.3. The source, boundary condition, and continuation path. We now prescribe the divergence and inner flux. For the original system, the source is chosen to give the deformed metric nonnegative scalar curvature. A penalty near the lower bound will exclude contact with that bound along the continuation path.

Choose $3/4 < \gamma < 1$ and fixed positive smooth weights

$$\rho_0 \asymp r^{-4}, \quad \rho_1 \asymp r^{-2\gamma},$$

with differentiated end bounds. Let $m : \mathbb{R} \rightarrow [0, 1]$ be smooth, equal to one on $(-\infty, 0]$ and zero on $[1, \infty)$, and set

$$m_0(t) = m(n(t + \epsilon)), \quad \mathcal{P} = C_n(\rho_0 + v\rho_1).$$

The positive δ_0 is fixed independently of n, R in Proposition 5.9; thereafter C_n is a sufficiently large polynomial constant. Define

$$(5.24) \quad \Xi = \delta_0 \mathcal{T} + \rho + \delta_0 \tau a \sigma - m_0(t) \mathcal{P}.$$

Choose a smooth face function $N_B > 1 + \sup_B |H|$. The original second equation and its inner condition are

$$(5.25) \quad \text{div } V = u\Xi, \quad V_\nu / u = -H + w_\nu(F - P_B) - N_B d.$$

Lemma 5.3 gives $H + 4\partial_\nu t = -N_B$. Only $-\epsilon \leq t \leq -\epsilon + 1/n$ is penalized in the admissible range. Since $n > 2/\epsilon$, this band has negative t , and

$$(5.26) \quad \ell \leq l \leq \epsilon \ell, \quad c\ell \leq L \leq C\ell \quad \text{on the admissible penalty band.}$$

The latter constants can be uniform for $0 < \epsilon < 1$.

Proposition 5.4 (Specified homotopy). *There is a continuous piecewise smooth family of systems indexed by $s \in [0, 4]$, beginning at (5.7), (5.25) and the original Dirichlet values, and ending at $f = 0$ and the scalar problem with zero source and zero inner conormal flux. Throughout its first three stages the source is (5.24), with $\mathcal{T}$ evaluated using the actual trace F . Each trace equation has right-side derivative at least one in h , with $x, Z, \nabla f$ fixed.*

Proof. Fix a smooth $j : \mathbb{R} \rightarrow [0, 1]$ which vanishes for $t \leq 0$ and equals one for $t \geq 1$. For $0 \leq \lambda \leq 1$ put

$$V_\lambda = uA_\chi(4\nabla Z + \lambda K(w, \cdot)), \quad \mathcal{B} = -H + w_\nu(F - P_B) - N_B d.$$

Before the last stage use the boundary condition

$$(5.27) \quad (V_\lambda)_\nu/u = [1 - (1 - \lambda)j(t)] \cdot [\mathcal{B} - (1 - \lambda)(A_\chi K(w, \cdot))_\nu].$$

Outer values remain $f = Z = 0$ throughout.

We specify two smooth collar modifications. Let $\nu_s = \nabla s/|\nabla s|$ be the outward unit normal to a collar leaf; the leaf coordinate s here is local and is separate from the homotopy parameter. Choose disjoint collar cutoffs equal to one at the faces. Take $\eta > 0$ so small that $b_- + 3\eta < b_+$. A black contribution to $D_0(x, w, h)$ is a negative constant times a collar cutoff, times a smooth directional cutoff supported where $w \cdot \nu_s < -1/2$ and equal to one at -1 , times a nonincreasing height cutoff equal to one below $b_- + \eta$ and zero above $b_- + 2\eta$. A white contribution has positive sign, its directional cutoff supported where $w \cdot \nu_s > 1/2$ and equal to one at 1 , and a nondecreasing height cutoff equal to zero below $b_+ - 2\eta$ and one above $b_+ - \eta$. Decrease η to separate these ranges. Their sum D_0 is smooth, bounded, supported in the collars, and nondecreasing in h . It is nonpositive for $h \leq b_- + \eta$, nonnegative for $h \geq b_+ - \eta$, and vanishes on black outward and white inward directions, for every length $0 \leq |w| \leq 1$.

Choose the magnitudes larger than $2 \sup_B |H| + 1$. Equation (5.1) then gives, for $F = h + C + D_0$ at the assigned boundary height $b = b_\pm$,

$$(5.28) \quad F(x, +\nu, b) \geq P_B + H, \quad F(x, -\nu, b) \leq P_B - H.$$

These are limiting directional evaluations at $a = 1, \chi = 0$. The compatible direction has equality; D_0 corrects the other one. Its derivatives on bounded height ranges are independent of n . Choose a smooth compactly supported extension $b(x)$, constant equal to $b_\pm$ on each smaller collar. On these collars let $D_1(x, w) = A_1 w \cdot \nu_s$, extended by a collar cutoff, with $A_1 > \sup_B |H| + 1$. Then $D_1(x, 0) = 0$ and $D_1(x, \pm\nu) = \pm A_1$ on B .

The four stages are the following.

- (i) For $0 \leq s \leq 1$, set $\lambda = 1 - s$, keep $F = h + C$ and $h|_B = b$, and use $\operatorname{div} V_\lambda = u\Xi$ and (5.27).
- (ii) For $1 \leq s \leq 2$, set $\lambda = 0$, $\alpha = s - 1$, retain $h|_B = b$, and replace the trace right side by $F = h + C + \alpha D_0(x, w, h)$. Keep the same scalar equation and boundary prescription.
- (iii) For $2 \leq s \leq 3$, set $\lambda = 0$, $\zeta = s - 2$, prescribe $h|_B = (1 - \zeta)b$, and use

$$(5.29) \quad F = h + (1 - \zeta)[C + D_0(x, w, h + \zeta b(x))] + \zeta[\operatorname{tr}_{A_\chi} K + D_1(x, w)].$$

Keep the same scalar equation and boundary prescription. At a face, $h + \zeta b = b$. The value at $w = \pm\nu$ is the convex combination of the preceding directional value and $P_B + D_1(x, \pm\nu)$, so (5.28) persists.

- (iv) For $3 \leq s \leq 4$, retain the trace problem at $\zeta = 1$ and multiply both Ξ and the right side of (5.27) by $\eta_s = 4 - s$.

The prescriptions agree at the common endpoints. Their h -derivatives are $1, 1 + \alpha \partial_h D_0$, or $1 + (1 - \zeta) \partial_h D_0$, all at least one.

At $\zeta = 1$ the trace equation is $\operatorname{tr}_{A_\chi} H^f = h + D_1(x, w)$ with zero Dirichlet values. At a positive interior maximum of f , one has $w = 0$, $A_\chi = I$, $D_1 = 0$, and $l\Delta f = \tau f > 0$, a contradiction. A negative minimum is excluded in the same way. Thus $f = 0$ and $t = Z$ in the last stage. At $s = 4$ the remaining equation is $\operatorname{div}(4L(Z)\nabla Z) = 0$, with zero inner conormal flux and $Z = 0$ on S_R . Multiplication by Z and integration give $Z = 0$. For the fixed-point construction in Section 7, the last map therefore has output identically zero. $\square$

5.4. Elementary height and end barriers.

Lemma 5.5 (Height control). *Every smooth trace solution in Proposition 5.4 satisfies $|h| \leq C_0$, independently of n, R , the homotopy parameter, and the input Z . Its actual right side satisfies $|F| \leq C_F$ with the same independence.*

Proof. The collar modifications vanish at $w = 0$. At an interior extremum the trace equation reads either $\text{tr } K + l\Delta f = h + C$ or $\text{tr } K + l\Delta f = h + (1 - \zeta)C + \zeta \text{tr } K$. The maximum and minimum inequalities bound h by fixed bounds for $|\text{tr } K| + |C|$. Boundary values range between $b_-, 0, b_+$. This proves the height bound without using the floor. Boundedness of D_0, D_1 and $|\text{tr}_{A_\chi} K| \leq 3|K|$ proves the assertion for F . $\square$

Lemma 5.6 (End height and outer boundary). *There are $r_0, C > 0$, independent of n, R and the homotopy parameter, such that for $R \geq 2r_0$,*

$$(5.30) \quad |h| \leq C(r^{-\gamma} - R^{-\gamma}) \quad (r_0 \leq r \leq R).$$

In particular,

$$(5.31) \quad |\nabla f| \leq CR^{-1-\gamma}/\tau \quad \text{on } S_R.$$

For fixed ϵ, n , a sufficiently large lower bound for R gives $t > -\epsilon$ on S_R .

Proof. Take r_0 outside the supports of K, C, D_0, D_1 . There $F = h$ in the first three stages, and $h = 0$ in the last one. For $\psi = r^{-\gamma}$, with axis $\nabla\psi/|\nabla\psi|$, the controlled end gives, uniformly for $0 < \chi \leq 1$,

$$\text{tr}_{A_\chi} \text{Hess } \psi = \gamma((\gamma + 1)\chi - 2)r^{-\gamma-2} + O(r^{-\gamma-3}) < 0$$

after enlarging r_0 . At a comparison contact, coefficients use the given Z and the common gradient of test and solution. The positive factor $l/(\tau\sqrt{D})$ preserves this sign. Height monotonicity therefore makes $C(\psi - R^{-\gamma})$ an upper barrier and its negative a lower barrier. Choose C to dominate C_0 on $r = r_0$ for every $R \geq 2r_0$. Differentiating at the common outer value gives (5.31); tangential derivatives there vanish.

At S_R , $Z = 0$. The monotonicity of its defining expression implies $t > -\epsilon$ if $\ell^2|\nabla f|^2 < e^{8\epsilon} - 1$. It suffices to require $C^2\ell^2\tau^{-2}R^{-2-2\gamma} < e^{8\epsilon} - 1$, which is a permissible lower bound depending on fixed n, ϵ . $\square$

5.5. Exclusion of a contact with the floor. We exclude a proposed floor contact using the equations at that point. These estimates require neither an upper bound for Z nor a bound for $|\nabla f|$.

Lemma 5.7 (Curvature at a minimum of t). *At an interior point where $dt = 0$ and $\text{Hess } t \geq 0$,*

$$(5.32) \quad \frac{1}{2}e^{4t} \text{Scal}_{\hat{g}} \leq \frac{1}{2} \text{Scal}_g - v \text{Ric}_g(e, e) + \mathcal{S}(H^f),$$

where, for a symmetric tensor A ,

$$(5.33) \quad \begin{aligned} \mathcal{S}(A) = & \frac{1}{4}(\text{tr}_\perp A)^2 - \frac{1}{2}|A_{\perp\perp}^{\text{tf}}|^2 \\ & + dA_{ee} \text{tr}_\perp A - d|A_{e\perp}|^2. \end{aligned}$$

There are universal $c_, \eta_* > 0$ and polynomial constants C_n such that, at $dt = 0$,*

$$(5.34) \quad \mathcal{T} - \mathcal{S}(H^f) \geq c_*\mathcal{T} - C_n(F^2 + |K|^2).$$

If $K = 0$, the more useful uniform estimate is

$$(5.35) \quad \mathcal{T} - \mathcal{S}(H^f) \geq c_*\mathcal{T} - C_nvF^2.$$

Proof. Consider the graph of f in the Riemannian product with warped fiber $(\Omega \times \mathbb{R}, g + l^2 db_0^2)$. Its induced metric is $\bar{g}$. At $dt = 0$ one also has $dl = 0$, and its second fundamental form, up to orientation, is H^f . Write $E = l^{-1}\partial_{b_0}$. The graph unit normal is $\sqrt{d}E - ae$. The ambient curvature formulas at this point are

$$\begin{aligned} \text{Scal}_{\text{amb}} &= \text{Scal}_g - 2\Delta l/l, \\ \text{Ric}_{\text{amb}}|_{T\Omega} &= \text{Ric}_g - \text{Hess } l/l, \quad \text{Ric}_{\text{amb}}(E, E) = -\Delta l/l, \end{aligned}$$

with zero mixed Ricci terms. Hence the ambient contribution to half the graph scalar curvature is

$$\frac{1}{2} \text{Scal}_g - v \text{Ric}_g(e, e) - v \text{tr}_{\perp} \text{Hess } l/l.$$

The last term is nonpositive: at $dt = 0$, $\text{Hess } l = lp \text{Hess } t$ and $p \geq 0$. The second-form contribution is $[(\text{tr}_{\bar{g}} H^f)^2 - |H^f|_{\bar{g}}^2]/2$. Since the inverse graph metric is A_d , its expansion is exactly (5.33). Finally the conformal contribution at this point is $-4\bar{\Delta}t \leq 0$. This proves (5.32).

For the algebraic estimates we must retain the degenerate normal coefficient. First evaluate $\mathcal{S}$ on $S = K + H^f$, whose trace in the transverse plane is $F - \chi q$. Subtracting (5.33) from (5.10) at $dt = 0$ gives

$$\begin{aligned} (5.36) \quad \mathcal{T} - \mathcal{S}(S) &= |T^{\text{tf}}|^2 + (1+d)|M|^2 \\ &\quad + \chi(1+d-\chi/2)q^2 - dFq + F^2/2. \end{aligned}$$

The normal coefficient is at least χ , while $|dFq| \leq \chi q^2/2 + d^2 F^2/(2\chi)$. Because

$$d^2/\chi = d(4+pv)/4 \leq 1+n/4,$$

Equation (5.21) implies $\mathcal{T} - \mathcal{S}(S) \geq c\mathcal{T} - C(1+n)F^2$ with universal $c > 0$.

To replace S by $H^f = S - K$, polarize (5.33). The difference is bounded by

$$C|K|(|T^{\text{tf}}| + |M| + |F| + d|q|) + C|K|^2.$$

Indeed the only extra normal product is $dq \text{tr}_{\perp} K$; all remaining products involve transverse entries, dM , or $\text{tr}_{\perp} S = F - \chi q$. By (5.21) and $d/\sqrt{\chi} \leq C\sqrt{1+n}$, this bound is at most $C\sqrt{1+n}|K|\sqrt{\mathcal{T}} + C|K|^2$. Young's inequality absorbs a fixed small fraction of $\mathcal{T}$ and costs only $C(1+n)|K|^2$. This proves (5.34).

If $(1+p)v \leq \eta_*$, then d and χ are within $C\eta_*$ of one. At $d = \chi = 1$ the normal block in (5.36) is $3q^2/2 - Fq + F^2/2$; its symmetric matrix has smallest eigenvalue $(2 - \sqrt{2})/2 > 0$. Choose $\eta_* > 0$ small enough that its coefficients remain within half this spectral gap. The transverse coefficients are already bounded below. For $K = 0$, Equation (5.21) therefore gives $\mathcal{T} - \mathcal{S}(H^f) \geq c_*\mathcal{T}$ in this regime. In the remaining regime $v > \eta_*/(1+p)$, use (5.34) with $K = 0$ and absorb $(1+n)F^2$ into $C(1+n)^2 v F^2$. This proves (5.35). All formulas extend to $\sigma = 0$ by their invariant expressions; there $d = \chi = 1$ and the full quadratic expressions are independent of the temporary choice of axis. $\square$

Lemma 5.8 (Homotopy errors at a floor contact). *There is a fixed compact set $\mathcal{K} \subset \bar{\Omega}$ containing the supports of K, C, b, D_0, D_1 such that, at an interior minimum $t = -\epsilon$ of a smooth solution in the first three stages,*

$$(5.37) \quad \text{div } V_{\lambda}/u \geq 2\delta_0 \mathcal{T} + \tau a \sigma - C(1+n)^N [\mathbf{1}_{\mathcal{K}} + v(\rho_0 + \rho_1)].$$

Here $\delta_0 > 0$ can be fixed sufficiently small independently of n, R and the homotopy parameter.

Proof. At this contact $dt = 0$ and $p = n$ is locally constant as a function of t near its value $-\epsilon$. Differentiating w in an orthonormal frame gives

$$(5.38) \quad \nabla_i w_j = H_{ij}^f - w_j H_{ik}^f w^k.$$

In the e -frame the normal vector slot is multiplied by d . Equations (5.21) and $d/\sqrt{\chi} \leq C\sqrt{1+n}$ therefore show

$$(5.39) \quad |\nabla w| \leq C(1+n)^N(|K| + \sqrt{\mathcal{T}}).$$

No bound for the uncontracted normal Hessian is asserted.

Each first-three-stage right side can be written as a smooth function $F(x, w, h, p)$. Its h -derivative is at least one. On the bounded height range supplied by Lemma 5.5, its explicit x - and w -derivatives are polynomially bounded and compactly supported, apart from the derivative of the term h . This follows directly from the fixed collar cutoffs and $A_\chi = I - (p+4)w \otimes w/(4+pv)$, whose denominator is at least four. For example its first w -derivatives on $|w| \leq 1$ are bounded by $C(1+n)^2$, and the other explicit modifications have bounds independent of n . Combining these bounds with (5.39) and Young's inequality therefore produces a fixed finite power of $1+n$. Since $\nabla p = 0$ at the contact and $w(h) = \tau a \sigma$, the chain rule and (5.39) give, for every fixed $\eta > 0$,

$$(5.40) \quad w(F) \geq \tau a \sigma - \eta \mathcal{T} - C_\eta(1+n)^N \mathbf{1}_\mathcal{K}.$$

The dependence $D_0(x, w, h + \zeta b(x))$ only adds the fixed compact derivative of b and does not change this conclusion.

The omitted flux is $(1-\lambda)uA_\chi K(w, \cdot)$. Let $B_K = A_\chi K(w, \cdot)$. Its derivative is bounded by a polynomial constant times $1 + \sqrt{\mathcal{T}}$ on $\mathcal{K}$, by (5.39); derivatives of p vanish. The weight term requires the contraction rather than a bound for $|\nabla \log u|$. At the contact,

$$\nabla \log u = H^f(w, \cdot), \quad \langle \nabla \log u, B_K \rangle = a^2(\chi H_{ee}^f K_{ee} + H_{e\perp}^f \cdot K_{e\perp}).$$

This involves only χH_{ee}^f and $H_{e\perp}^f$ and is controlled by (5.21). It follows that

$$(5.41) \quad |\operatorname{div}(uB_K)|/u \leq \eta \mathcal{T} + C_\eta(1+n)^N \mathbf{1}_\mathcal{K}.$$

Combine the scalar identity with (5.32). The difference $8\pi\mu - \operatorname{Scal}_g/2$ equals $((\operatorname{tr} K)^2 - |K|^2)/2$ and is compactly supported; so is J . The term $F \operatorname{tr} K$ is compactly bounded by Lemma 5.5. On $\mathcal{K}$, Equation (5.34) therefore contributes $c_* \mathcal{T} - C(1+n)^N$. Outside $\mathcal{K}$, $K = 0$, $F = h$, and (5.35) together with Lemma 5.6 gives $c_* \mathcal{T} - C(1+n)^N v r^{-2\gamma}$. The remaining Ricci term is bounded below by $-C\mathbf{1}_\mathcal{K} - C v r^{-3}$; enlarge $\mathcal{K}$ if necessary. Both end weights are bounded by a multiple of $\rho_0 + \rho_1$. Finally subtract the omitted flux and use (5.40)–(5.41), choosing their two η 's with sum at most $c_*/2$. Fix $0 < \delta_0 < \min(c_*/4, 1/4)$. Decreasing it slightly if necessary gives (5.37). $\square$

Proposition 5.9 (Floor exclusion). *Choose δ_0 as in Lemma 5.8. There is a polynomial choice $C_n = C(1+n)^N$ in the penalty such that, for all sufficiently large n and all $R \geq R_{\text{floor}}(n, \epsilon)$, no smooth solution anywhere along Proposition 5.4 can satisfy $t \geq -\epsilon$ and attain $t = -\epsilon$. In particular every such admissible solution has $t > -\epsilon$ on $\overline{\Omega_R}$.*

Proof. Choose R_{floor} to include the requirement of Lemma 5.6. A floor contact cannot occur on S_R . In the first three stages, an inner contact has $j(t) = 0$. Equation (5.27) then cancels the omitted flux on both sides and gives $V_\nu/u = \mathcal{B}$. Equation (5.23) forces $H + 4\partial_\nu t = -N_B$, so $\partial_\nu t < 0$. This contradicts the nonnegative derivative into Ω_R at a boundary minimum.

Consider an interior contact. Let $E_n = C(1+n)^N[\mathbf{1}_\mathcal{K} + v(\rho_0 + \rho_1)]$ denote the error in (5.37). Because ρ_0 has a positive minimum on $\mathcal{K}$, $\mathbf{1}_\mathcal{K} \leq C_\mathcal{K} \rho_0$. Also $v\rho_0 \leq \rho_0$, and ρ/ρ_0 is globally bounded. We can therefore choose a polynomial C_n so large that, pointwise for all $0 \leq v \leq 1$,

$$(5.42) \quad \mathcal{P} \geq E_n + 2\rho + \rho_0.$$

At the floor $m_0 = 1$, so the scalar equation and the lower estimate give

$$\begin{aligned} 0 &\geq \delta_0 \mathcal{T} + (1 - \delta_0) \tau a \sigma + \mathcal{P} - E_n - \rho \\ &\geq \rho + \rho_0 > 0, \end{aligned}$$

a contradiction. Enlarging C_n below preserves this inequality and preserves its polynomial dependence.

In the last stage $f = 0$, $t = Z$, $u = L$, and $v = 0$. At an interior minimum, $\mathcal{T} = (|K|^2 + (\text{tr } K)^2)/2$. Increase C_n polynomially if necessary so that $\delta_0 \mathcal{T} + \rho - C_n \rho_0 < 0$ at all such points. For $\eta_s > 0$, the scalar equation at a floor minimum has left side $4L\Delta t \geq 0$ and strictly negative right side. At an inner floor minimum its flux is $4\partial_\nu t = \eta_s(-H - N_B) < 0$, also impossible. If $\eta_s = 0$, multiplication of $\text{div}(4L(t)\nabla t) = 0$ by t , with zero outer value and zero inner flux, gives $t = 0$. Thus the last stage has no floor contact either. $\square$

We have excluded floor contact for every smooth admissible solution along the specified homotopy. Section 6 supplies the remaining a priori bounds, and Section 7 proves existence. The choice of $\mathcal{P}$ uses only polynomial losses in n ; it is independent of all fixed- n Schauder constants.

6. A PRIORI ESTIMATES FOR THE DEFORMATION

We bound the height, boundary slopes, and conformal variable before using classical regularity. Fix the prepared exterior of Proposition 4.7, its thresholds and collars, and $\epsilon > 0$. We use the variables and four stages of Proposition 5.4. A solution is smooth on a finite truncation and satisfies $t \geq -\epsilon$, so the estimates include putative floor-contact solutions. No Hessian bound or upper bound for Z is assumed.

A quantity denoted by Π_n is bounded by $C(1+n)^N$, with C, N independent of the truncation, the homotopy parameter, and n . It can increase at successive uses. The fixed data, collars, and ϵ can enter C, N . A constant denoted by $C(n)$ can depend arbitrarily on these fixed data and on n, ℓ, τ ; it is still independent of the truncation and the homotopy parameter. Constants explicitly described as fixed do not depend on n . Recall that

$$(6.1) \quad \ell = e^{-\epsilon n}, \quad \tau = \ell^{3/2}, \quad \ell \leq l \leq e, \quad 0 \leq p \leq n, \quad \frac{4d}{4+n} \leq \chi \leq d.$$

In particular $\Pi_n \tau / \ell \rightarrow 0$. This small factor will absorb only terms whose constants are polynomial.

6.1. Sublevels and exclusion of boundary layers. For notation in this section, let $\mathcal{B}_c$ be the full closed maximal black region at threshold c , and let $\mathcal{W}_c$ be the corresponding white region with $\mathcal{B}_{c_b}$ held fixed. These are the families used in Proposition 4.7.

Lemma 6.1 (Uniform separation from the wrong height). *In the first two homotopy stages, the following holds. If $Q \subset \bar{\Omega}$ is compact and disjoint from $\mathcal{B}_{c_b}$, there are $\eta_Q > 0$ and n_Q such that*

$$(6.2) \quad h \geq b_- + \eta_Q \quad \text{on } Q$$

for every $n \geq n_Q$, every sufficiently large allowed truncation, and every admissible or floor-contact solution in those stages. If Q is disjoint from $\mathcal{W}_{c_w}$, the analogous conclusion is $h \leq b_+ - \eta_Q$. The compact sets are relative to the full closure: they may contain faces of the opposite color.

Proof. We prove the lower assertion. It is enough to consider an arbitrary sequence $n_i \rightarrow \infty$, $R_i \rightarrow \infty$ and solutions in either of the first two stages. The elementary height and end estimates in Lemmas 5.5 and 5.6 give

$$(6.3) \quad |h_i| \leq C_0, \quad |h_i| \leq C(r^{-\gamma} - R_i^{-\gamma}) \quad \text{on the end}, \quad 3/4 < \gamma < 1.$$

Near each white face extend h_i to the other side by the constant b_+ . Extend the fixed geometric coefficients smoothly there. The extended functions are continuous. Define the lower relaxed limit

$$\underline{h}(x) = \lim_{j \rightarrow \infty} \inf \{h_i(y) : i \geq j, \text{ dist}(x, y) < 1/j\}.$$

It is lower semicontinuous, bounded, and satisfies the limiting end bound. No equicontinuity of h_i is asserted.

Choose $b_- < k' < \min\{0, b_- + \eta\}$, where η is the low-height sign range of the added term D_0 . Suppose that a smooth function ϕ touches $\underline{h}$ from below at x , with $\underline{h}(x) \leq k'$ and $d\phi(x) \neq 0$. Exclude for the moment the black faces. Subtracting a small fourth-order localization term from ϕ makes the contact strict without changing its two-jet. Minimization on a small closed ball then produces points $x_i \rightarrow x$ and constants $c_i \rightarrow 0$ such that $\phi + c_i$ touches h_i from below at x_i , and $h_i(x_i) \rightarrow \underline{h}(x)$. The boundary of this ball cannot contain x_i by strictness. Nor can x_i lie on a white face or on the extended side: their values there are $b_+ > k'$. Thus every sufficiently large i uses the interior trace equation.

At these contacts,

$$\nabla f_i = \tau_i^{-1} \nabla \phi, \quad d_i \leq \frac{\tau_i^2}{\ell_i^2 |d\phi(x_i)|^2} \rightarrow 0, \quad a_i \rightarrow 1, \quad \chi_i \rightarrow 0.$$

Here $a_i = l_i |\nabla f_i| / \sqrt{D_i}$ is the deformation variable. Positivity of A_{χ_i} and the Hessian inequality at a lower contact imply

$$\text{tr}_{A_{\chi_i}} K + \frac{l_i}{\tau_i \sqrt{D_i}} \text{tr}_{A_{\chi_i}} \text{Hess } \phi \leq F_i(x_i).$$

In the second stage $D_0 \leq 0$ at these values; in the first stage it is absent. Consequently $\limsup F_i(x_i) \leq k' + C_b$. Passing to the limit gives

$$(6.4) \quad \frac{\text{tr}_{(d\phi)^\perp} \text{Hess } \phi}{|d\phi|} + \text{tr}_{(d\phi)^\perp} K \leq k' + C_b.$$

This is the expansion of the level surface of ϕ , oriented toward increasing ϕ . The only Hessian used in this passage is the fixed test Hessian. Removing the localization proves the assertion for non-strict contacts as well.

We now transfer the test inequality to a closed sublevel set. Fix $b_- < k < k'$ and put $E_k = \{\underline{h} \leq k\}$. Set

$$(6.5) \quad \psi_j(s) = \frac{1}{1 + \exp[-j^2(s - k - j^{-1})]}, \quad v_j = \psi_j(\underline{h}).$$

The lower relaxed limit of v_j is the function v which equals zero on E_k and one on its complement. Indeed, at a point of E_k the fixed-point values $\psi_j(\underline{h}(x))$ tend to zero. If $\underline{h}(x) > k$, lower semicontinuity supplies a neighborhood where $\underline{h} > k + \delta$ for some $\delta > 0$, and v_j tends uniformly to one there.

Let φ be a nonzero-gradient lower test of v at a zero value. The same strict-contact localization gives lower tests $\varphi + c_j$ of v_j at $y_j \rightarrow x$, with $v_j(y_j) \rightarrow 0$. Since $\psi_j(k') \rightarrow 1$, one has $\underline{h}(y_j) < k'$ eventually. Locally about y_j , the test value is in $(0, 1)$, so $\psi_j^{-1}(\varphi + c_j)$ is a smooth lower test of $\underline{h}$. An increasing change of defining function does not change its oriented level expansion: if $\theta' > 0$, then

$$\text{Hess}(\theta \circ \varphi) = \theta' \text{Hess } \varphi + \theta'' d\varphi \otimes d\varphi, \quad \frac{\text{tr}_{(d\varphi)^\perp} \text{Hess}(\theta \circ \varphi)}{|d(\theta \circ \varphi)|} = \frac{\text{tr}_{(d\varphi)^\perp} \text{Hess } \varphi}{|d\varphi|}.$$

Equation (6.4) therefore holds for the test $\varphi + c_j$ and passes to φ . A smooth exterior support of E_k , with defining function negative on the containing inner side, is a lower test of v : shrink its neighborhood so that its positive values are less than one. Thus every such support has expansion at most $k' + C_b$.

This argument also rules out a hidden layer at a white face. The reversed white face is an exterior support of $E_k \cap \bar{\Omega}$ at any contact there, with expansion

$$-H_B + P_B = -c_w > 0.$$

For k' sufficiently close to b_- , this is strictly greater than $k' + C_b$. The high extension ensured that all preceding low-value tests came from the interior equation, so the support contradiction applies at the face itself. Hence E_k avoids every white face. The end estimate makes E_k compact.

Adjoin the entire closed region $\mathcal{B}_{c_b}$ and fill bounded complementary pockets. At a point on its smooth black frontier, every exterior support of the union also contains the black region locally. The tangency comparison therefore bounds its expansion by $c_b < k' + C_b$. At all other frontier points the preceding sublevel argument applies. Filling pockets only removes frontier pieces and preserves this property. The resulting compact closed filled set contains the required original inner barrier with a collar, because the black region already does. It avoids the outer barriers. Lemma 4.6 places it in a smooth black trapped region at any threshold $c'' > k' + C_b$ sufficiently close to c_b . It is consequently contained in $\mathcal{B}_{c''}$.

Finally let Q be as in the lemma. Hausdorff right continuity at c_b gives a $\delta > 0$ such that $\mathcal{B}_{c_b+\delta} \cap Q = \emptyset$; decrease δ so that the enlarged threshold is still negative and all barriers remain strict. Choose k, k', c'' with

$$b_- < k < k', \quad k' + C_b < c'' < c_b + \delta.$$

The construction implies $E_k \cap Q = \emptyset$ for every sequence under consideration. If the uniform estimate $h > k$ on Q failed, choose violating solutions with $n_i \rightarrow \infty$, $R_i \rightarrow \infty$ and $x_i \in Q$, $h_i(x_i) \leq k$. A subsequence has $x_i \rightarrow x \in Q$ and $\underline{h}(x) \leq k$, a contradiction. This gives (6.2), after decreasing its positive margin.

For the upper assertion apply the entire argument to $(-h, -K, -C)$, interchanging colors. The reversed black faces have strictly positive white expansion $-c_b$. Their exclusion and compactness give positive distance from the black region before applying the white version of Lemma 4.6. Adjoining the full white region is therefore legitimate in the fixed black complement. Empty white regions and missing face types cause no change: only an actually nonempty sublevel would supply a trapped seed and hence a contradiction. $\square$

For the truncation quantifier, start with any $R_{\min}(n) \rightarrow \infty$ that satisfies the outer estimates of Section 5, and enlarge it when necessary. Failure of any asserted estimate for arbitrarily large n and $R \geq R_{\min}(n)$ supplies exactly the sequence excluded above. Only finitely many compact sets will be needed below, so their thresholds can be chosen simultaneously.

6.2. Signed collar barriers.

Lemma 6.2 (Boundary slopes). *There are fixed constants $c, C_* > 0$ such that, for sufficiently large n and the allowed truncations, the first two stages satisfy*

$$(6.6) \quad \begin{aligned} c/\tau &\leq \partial_\nu f \leq C_*/\tau && \text{on black faces,} \\ c/\tau &\leq -\partial_\nu f \leq C_*/\tau && \text{on white faces.} \end{aligned}$$

During the third stage one still has $|\partial_\nu f| \leq C_/\tau$. On every fixed compact region needed for the final scalar-curvature estimate, the first two stages satisfy $b_- \leq h \leq b_+$.*

Proof. Write s for the smooth leaf parameter in an outward collar and $N = \nabla s/|ds|$. All geometric coefficients of this fixed collar are bounded, with $|ds|$ bounded above and below. At a comparison contact with $h_0 = b_- + \psi(s)$, $\psi' > 0$, the shared gradient gives

$$(6.7) \quad d \leq C \frac{\tau^2}{\ell^2(\psi')^2}, \quad 1 - a \leq d, \quad n\chi \geq c_1 d \quad (n \geq 4).$$

If the slopes are in a fixed positive interval, the first bound is $d \leq C\ell$. None of these comparisons uses an upper bound for Z .

The trace operator on this test is

$$(6.8) \quad \begin{aligned} & \operatorname{tr}_{A_\chi} \left(K + \frac{l}{\tau\sqrt{D}} \operatorname{Hess} h_0 \right) \\ &= P_s + aH_s + \chi K(N, N) + a\chi \left(\frac{\operatorname{Hess} s(N, N)}{|ds|} + |ds| \frac{\psi''}{\psi'} \right), \end{aligned}$$

where P_s is the tangential trace of K on the leaf. Thus it is the leaf expansion $H_s + P_s$, with an error of absolute value at most Cd , plus the displayed logarithmic-slope term. The same formula for a negative slope has a replaced by $-a$ in the Hessian terms. This is an evaluation at the contact variables; it does not differentiate the unknown t .

On a black collar $H_s + P_s \geq c_b$ and $C = C_b - k_B s$. Choose small positive slopes so that $\psi(0) = 0$ and $\psi(s) \leq k_B s/2$. At the outer end of a sufficiently short fixed collar, Lemma 6.1 lets this test lie below the solution. In the compatible direction the add-on D_0 vanishes. The residual of the trace equation on the test is therefore at least

$$(6.9) \quad k_B s/2 - Cd + a\chi |ds| \frac{\psi''}{\psi'}.$$

Choose a smooth nonnegative $\beta_n(s)$ equal to An for $0 \leq s \leq 1/n$, supported in $0 \leq s \leq 2/n$, and satisfying $\int \beta_n \leq 2A$. Define

$$\psi'(s) = m \exp \left(\int_0^s \beta_n(q) dq \right).$$

Choose the fixed A large enough that the final term of (6.9) dominates Cd on $s \leq 1/n$, using (6.7) and $a \geq 1/2$ there. On $s \geq 1/n$, the first term dominates $Cd \leq C\ell$ for large n ; the taper contributes with the good sign. Since the slope multiplier is at most e^{2A} , a fixed sufficiently small $m > 0$ enforces all the previous small-slope requirements. This constructs a strict lower barrier with slopes bounded above and below independently of n .

An upper barrier from b_- is constructed in the same collar with

$$\psi'(s) = M \exp \left(- \int_0^s \beta_n(q) dq \right).$$

The smooth leaf expansion differs from its boundary value by $O(s)$. Choose the fixed M so large that $\psi(s) \geq C_2 s$ dominates this variation, the offset variation, and the height bound at the outer collar end. The logarithmic-slope term is now negative. On $s \leq 1/n$ it dominates Cd ; on the rest of the collar the negative margin proportional to s does so. The upper barrier is strict at every possible positive contact. Its slopes lie in $[Me^{-2A}, M]$, an interval independent of n .

To check the comparison signs, at a negative interior minimum of $h - h_0$ the Hessian of h dominates that of h_0 , the gradients and Z agree, and hence so do t, l, A_χ . The right side is strictly increasing in the height. Thus a strict positive residual for a lower barrier contradicts the equation. At a positive maximum the inequalities reverse, proving the upper-barrier assertion. Taking the one-sided derivative at $s = 0$ proves the black part of (6.6). Replacing (h, K, C) by $(-h, -K, -C)$ proves the white part.

In the third stage the boundary height is the assigned interpolated constant, say b_ζ . Construct tests $b_\zeta + \psi(s)$ and $b_\zeta - \psi(s)$ with large positive ψ' and $(\log \psi')' = -\beta_n$. At $s = 0$, the two directional inequalities of Proposition 5.4 say precisely that the positive-slope test has nonpositive limiting residual and the negative-slope test nonnegative limiting residual. At every comparison contact, the fixed positive barrier slope and Equation (6.7) give $d \leq C\ell$, so $a \geq 1/2$ for large n . On $1/2 \leq |w| \leq 1$, the first w -derivatives of A_χ in Equation (5.5) are bounded independently of $0 \leq p \leq n$. The other collar terms have fixed smooth coefficients on the bounded height range. Thus, at fixed boundary height the coefficient and geometric errors are $O(s + d + \chi)$,

since $|w - (\pm N)| = 1 - a \leq d$ and the radial comparison segment stays in this range. Changing the test height supplies, by height monotonicity, a residual of magnitude at least $\psi(s)$ with the required sign. Large fixed slopes dominate the $O(s)$ error and the outer-end height bound. The logarithmic-slope term has the required sign for both tests, as follows from (6.8); it dominates the $O(d)$ error on the initial $1/n$ interval. The remaining margin dominates it beyond that interval. This proves the two-sided upper derivative bound, without using sublevel exclusion in the third stage.

Finally, on each own-color collar the lower barrier just constructed gives the required one-sided height bound. On the rest of a fixed compact region use Lemma 6.1; the opposite height bound on an own-color collar follows from that lemma as well, since the closed collar is disjoint from the other region. A finite cover proves $b_- \leq h \leq b_+$ on the compact regions in the statement. $\square$

6.3. A trace estimate with polynomial constants. The collar trace estimate must have polynomial constants: it will control the boundary flux both here and in the mass estimate. We therefore prove it before comparing with the ordinary product graph.

Lemma 6.3 (Weighted and unweighted collar traces). *In the first two stages let $\mathcal{C}$ be a fixed union of disjoint collars of the inner faces, shortened if necessary. For every nonnegative smooth function φ supported in those collars,*

$$(6.10) \quad \int_B L\varphi \, dA_g \leq \Pi_n \int_{\mathcal{C}} u[(1 + \sqrt{\mathcal{T}})\varphi + |d\varphi|_{A_\chi}] \, dV_g,$$

$$(6.11) \quad \int_B \varphi \, dA_g \leq \Pi_n \int_{\mathcal{C}} \sqrt{D}[(1 + \sqrt{\mathcal{T}})\varphi + |d\varphi|_{A_\chi}] \, dV_g.$$

The estimates also hold without a support condition on φ if the right sides include a fixed collar cutoff equal to one on B , with its derivative included in the coefficient of φ .

Proof. For $\sigma > 0$, write $e_f = \nabla f / \sigma$, the axis denoted by e in Section 5. Set

$$W = l\bar{\nabla}f = \frac{l\nabla f}{D} = \sqrt{d}ae_f.$$

Its norm in $\bar{g}$ is $a \leq 1$. For any covector ξ ,

$$(6.12) \quad |\xi(W)| \leq |\xi|_{A_d} \leq \sqrt{1 + n/4} |\xi|_{A_\chi}.$$

Direct differentiation, using $uW = Lw$, gives

$$(6.13) \quad \frac{\operatorname{div}(uW)}{u} = \sqrt{d} \left[\operatorname{tr}_{A_d} H^f + (dp + p_L)a\partial_{e_f}t \right],$$

$$(6.14) \quad \frac{\operatorname{div}(\sqrt{D}W)}{\sqrt{D}} = \sqrt{d} \left[\operatorname{tr}_{A_d} H^f + dp a\partial_{e_f}t \right].$$

For example, $\operatorname{div} w = \operatorname{tr}_{A_d} H^f + dp a\partial_{e_f}t$, while differentiating L adds $p_L a\partial_{e_f}t$ to the first formula. Coercivity in Lemma 5.2 bounds both right sides by $\Pi_n(1 + \sqrt{\mathcal{T}})$. More explicitly, writing $q = (K + H^f)_{e_f e_f}$, the normal Hessian term has coefficient $d^{3/2}$, so it is bounded by $\sqrt{d}|q| + C$. The other Hessian terms are transverse traces, and the gradient term is at most $(2n + 2)\sqrt{d}|\partial_{e_f}t|$. All are controlled by the stated coercivity estimate. This also establishes the formulas and bounds at zero gradient by their smooth direction-free extensions.

Let $\varepsilon_B = 1$ on a black collar and -1 on a white collar. The signed slopes give $w_\nu = \varepsilon_B a$ on B and $a \geq 1/2$ for large n . Thus

$$uW_\nu = \varepsilon_B La, \quad \sqrt{D}W_\nu = \varepsilon_B a.$$

The outward integration normal is $-\nu$. Apply the divergence theorem to $-\varepsilon_B u W \varphi$, with a fixed collar cutoff. Its boundary flux is $La\varphi$. Equations (6.12) and (6.13), and the fixed cutoff derivative, give (6.10). Replace u by $\sqrt{D}$ and use (6.14) for (6.11). No factor ℓ^{-1} was used. $\square$

6.4. High-level energy and the first upper bound. The first stage contains the drift in the divergence equation. We first bound this stage uniformly for $\lambda \in [0, 1]$. There is a number $M_0 = M_0(n) > 0$ such that

$$(6.15) \quad \Xi \geq \delta_0 \mathcal{T} + \rho + \frac{1}{2} \delta_0 \tau a \sigma \quad \text{on } \{Z > M_0\}.$$

Indeed, outside the support of m_0 this is immediate. On its support, $-\epsilon \leq t \leq -\epsilon + 1/n$, so l is comparable to ℓ . The identity $D = e^{8(Z-t)}$ shows that $a\sigma$ tends uniformly to infinity as $Z \rightarrow \infty$ in that band. The penalty $\mathcal{P}$ is bounded on the whole exterior by Π_n . It is therefore absorbed by $\delta_0 \tau a \sigma / 2$ above a fixed $M_0(n)$. This choice is independent of R and λ .

Choose a smooth nondecreasing k_0 equal to zero on $Z \leq M_0$, equal to one on $Z \geq M_0 + 1$, and with bounded derivative. Test $\text{div } V_\lambda = u\Xi$ by $k_0(Z)$. Its outer boundary term vanishes since $Z = 0$. The drift cross term is bounded by

$$uk'_0 |\langle dZ, K(w, \cdot) \rangle_{A_\chi}| \leq 2uk'_0 |dZ|_{A_\chi}^2 + Cuk'_0 |K|^2.$$

The last term has bounded integral: it is supported on a fixed compact set and on $M_0 \leq Z \leq M_0 + 1$, where the floor and $D = e^{8(Z-t)}$ bound u and σ by $C(n)$.

At a black face $F - P_B = H$, $w_\nu = a$; at a white face $F - P_B = -H$, $w_\nu = -a$. In both cases

$$-H + w_\nu(F - P_B) = (a - 1)H.$$

The additional omitted-drift term in the homotopy boundary condition has absolute value at most $C\chi \leq Cd$, since $A_\chi K(w, \cdot)$ has normal component $\chi w_\nu K(\nu, \nu)$. The remaining term is $-N_B d$, and the boundary multiplier lies between zero and one. Hence

$$(6.16) \quad |(V_\lambda)_\nu| \leq Cud = CL\sqrt{d} \leq C\frac{\tau}{\ell}L \quad \text{on } B,$$

where the last inequality uses the lower slope in (6.6). Integration now gives

$$(6.17) \quad \begin{aligned} & \int_{\Omega_R} uk_0 \left(\delta_0 \mathcal{T} + \rho + \frac{1}{2} \delta_0 \tau a \sigma \right) dV_g + 2 \int_{\Omega_R} uk'_0 |dZ|_{A_\chi}^2 dV_g \\ & \leq C(n) + C\frac{\tau}{\ell} \int_B Lk_0 dA_g. \end{aligned}$$

Apply (6.10) with the fixed collar cutoff. The boundary cost is at most

$$\Pi_n \frac{\tau}{\ell} \int_{\mathcal{C}} u \left[k_0(1 + \sqrt{\mathcal{T}}) + k'_0 |dZ|_{A_\chi} \right] dV_g.$$

Let $\rho_{\mathcal{C}} > 0$ be the minimum of ρ on the closed collars. Pointwise, $1 + \sqrt{\mathcal{T}} \leq C(\rho + \delta_0 \mathcal{T})$ there, with C fixed. Since $\Pi_n \tau / \ell \rightarrow 0$, the first summand is absorbed in the first integral of (6.17). Young's inequality bounds the derivative summand by

$$uk'_0 |dZ|_{A_\chi}^2 + C(\Pi_n \tau / \ell)^2 uk'_0.$$

The second term again has bounded compact integral because it lies in the transition strip of k_0 . We obtain

$$(6.18) \quad \int uk_0 (\mathcal{T} + \rho + \tau a \sigma) dV_g + \int uk'_0 |dZ|_{A_\chi}^2 dV_g \leq C(n).$$

The smallness requirement on n in this absorption is independent of M_0 ; only its resulting right-hand constant depends on M_0 .

The polynomial boundary absorption is complete. We may now use arbitrary fixed- n comparison constants. Write Γ for the ordinary product graph of f in $g + (db_0)^2$, and $d\Gamma = \sqrt{1 + \sigma^2} dV_g$.

The measures $d\Gamma$ and $\sqrt{D}dV_g$ are comparable by constants depending only on ℓ . On every fixed compact set Q ,

$$(6.19) \quad \int_{\Gamma \cap \pi^{-1}(Q)} e^{2Z} d\Gamma \leq C_Q(n),$$

where π is projection to the base. To check this without an upper bound for t , observe that

$$e^{2Z} = e^{2t} D^{1/4}, \quad u = l e^{2t} \sqrt{D}, \quad D^{1/4} \leq C(n)(1 + \tau a \sigma).$$

The last inequality follows by splitting $l\sigma \leq 1$ and $l\sigma > 1$, using $\ell \leq l \leq e$; on the second set $a \geq 1/\sqrt{2}$ and the right side grows linearly in σ , whereas $D^{1/4}$ grows at most as $C\sqrt{\sigma}$. Thus the high-level part of (6.19) follows from (6.18) and the positive minimum of ρ on Q . On $Z \leq M_0 + 1$, both σ and e^{2Z} are bounded by the floor identity, which controls the remaining compact integral.

6.5. Sobolev inequality and iteration on the ordinary graph. The comparisons in this subsection are uniform in the solution, R , and the homotopy parameter, but may depend arbitrarily on n . If $d_0 = (1 + \sigma^2)^{-1}$, then

$$|d\varphi|_{A_{d_0}} = |\nabla_\Gamma \varphi|, \quad \frac{d}{d_0} = \frac{1 + \sigma^2}{1 + l^2 \sigma^2}.$$

The last ratio is bounded above and below by constants depending only on ℓ ; Equation (6.1) therefore compares A_{d_0}, A_d, A_χ without any bound for Z .

Lemma 6.4 (Local graph Sobolev inequality). *For a smooth function ω supported over a fixed compact base patch, which may meet an inner face,*

$$(6.20) \quad \|\omega\|_{L^6(d\Gamma)}^2 \leq C(n) \int_\Gamma [|\nabla_\Gamma \omega|^2 + (1 + \mathcal{T})\omega^2] d\Gamma.$$

Proof. Extend the fixed base patch smoothly and embed a compact enlargement isometrically in Euclidean space by Nash's embedding theorem [41, Theorem 2]. Its product with the vertical line gives an isometric Euclidean immersion of the graph. The second fundamental form of the base embedding is bounded, so its contribution to the graph mean curvature vector is bounded. The scalar mean curvature in the product is

$$H_\Gamma = \frac{\sqrt{D}}{l\sqrt{1 + \sigma^2}} \left(\text{tr}_{e_f^\perp} H^f + d_0 H^f(e_f, e_f) \right).$$

The prefactor is bounded by $C(n)$, and $d_0/\sqrt{d} = \sqrt{1 + l^2 \sigma^2}/(1 + \sigma^2) \leq e$. Coercivity therefore gives $|\mathbf{H}_\Gamma| \leq C(n)(1 + \sqrt{\mathcal{T}})$. This estimate involves controlled quadratic energy, not a pointwise bound for the Hessian.

The ordinary Michael–Simon inequality [40], in the compact-support form stated in [47, Chapter 4, Section 5, Theorem 5.7], gives after the boundary extension described below, for nonnegative φ ,

$$\left(\int_\Gamma \varphi^{3/2} d\Gamma \right)^{2/3} \leq C \left[\int_\Gamma (|\nabla_\Gamma \varphi| + |\mathbf{H}_\Gamma| \varphi) d\Gamma + \int_{\partial\Gamma} \varphi dA \right].$$

One can obtain the boundary form directly from the compact-support version by smoothly extending each smooth graph across its boundary and cutting off on a collar tending to zero: the cutoff derivative integral converges to the boundary integral, and the added curvature integral tends to zero. This argument is applied to each smooth graph before the uniform estimate is taken. The only boundary in the support is an inner face, where f is constant and $dA = dA_g$. Equation (6.11) and the fixed- n graph comparisons bound its contribution by

$$C(n) \int_\Gamma [(1 + \sqrt{\mathcal{T}})\varphi + |\nabla_\Gamma \varphi|] d\Gamma.$$

Apply the resulting inequality to $\varphi = |\omega|^4$. Cauchy–Schwarz bounds its right side by

$$C(n)\|\omega\|_6^3 \left(\int_{\Gamma} [|\nabla_{\Gamma}\omega|^2 + (1 + \mathcal{T})\omega^2] d\Gamma \right)^{1/2}.$$

The left side is $\|\omega\|_6^4$; division, with the zero case separate, and squaring prove the assertion. $\square$

To turn the integral bound into an upper bound, let η be a nonnegative cutoff over a fixed compact patch. For k sufficiently large, independently of R and the solution, test the first-stage divergence equation by

$$q = \eta^2 e^{2kZ} / L.$$

The positive measure after cancellation of L is $\sqrt{D} dV_g$. The differentiated test is

$$dq = \frac{e^{2kZ}}{L} [2\eta d\eta + \eta^2 (2k dZ - p_L dt)].$$

Its principal exponential contribution is $8k\eta^2 e^{2kZ} |dZ|_{A_\chi}^2$. The source is bounded below by $\delta_0 \mathcal{T} - \Pi_n$ everywhere. Coercivity and $A_\chi \leq A_d$ give

$$4p_L |\langle dt, dZ \rangle_{A_\chi}| \leq \frac{1}{4} \delta_0 \mathcal{T} + C(n) |dZ|_{A_\chi}^2.$$

Choose $k \geq k_*(n)$ so that the second term is absorbed by the principal contribution. The drift terms, with $|K(w, \cdot)|_{A_\chi} \leq |K|$, obey

$$\begin{aligned} 2k |\langle K(w, \cdot), dZ \rangle_{A_\chi}| &\leq k |dZ|_{A_\chi}^2 + Ck |K|^2, \\ p_L |\langle K(w, \cdot), dt \rangle_{A_\chi}| &\leq \frac{1}{4} \delta_0 \mathcal{T} + C(n) |K|^2. \end{aligned}$$

The cutoff cross terms are handled by the same Cauchy inequalities, leaving $C(n)k e^{2kZ} (|d\eta|_g^2 + \eta^2)$. The penalty is bounded and contributes to this latter expression. Finally, Equation (6.16) before its last inequality gives $|q(V_\lambda)_\nu| \leq C\eta^2 e^{2kZ} \sqrt{d} \leq C\eta^2 e^{2kZ}$. Converting to the ordinary graph yields

$$\begin{aligned} &\int_{\Gamma} e^{2kZ} \eta^2 (\mathcal{T} + k |\nabla_{\Gamma} Z|^2) d\Gamma \\ (6.21) \quad &\leq C(n)k \int_{\Gamma} e^{2kZ} (\eta^2 + |d\eta|_g^2) d\Gamma + C(n) \int_B e^{2kZ} \eta^2 dA_g. \end{aligned}$$

Use the unweighted trace with $\varphi = e^{2kZ} \eta^2$. The boundary term is at most

$$C(n) \int_{\Gamma} e^{2kZ} \left[\eta^2 (1 + \sqrt{\mathcal{T}} + k |\nabla_{\Gamma} Z|) + \eta |d\eta|_g \right] d\Gamma.$$

Young's inequality absorbs its $\sqrt{\mathcal{T}}$ term into the $\mathcal{T}$ integral and its $k |\nabla_{\Gamma} Z|$ term into the $k |\nabla_{\Gamma} Z|^2$ integral. The latter absorption costs $C(n)k\eta^2 e^{2kZ}$, which has the allowed size. Hence

$$(6.22) \quad \int_{\Gamma} e^{2kZ} \eta^2 (\mathcal{T} + k |\nabla_{\Gamma} Z|^2) d\Gamma \leq C(n)k \int_{\Gamma} e^{2kZ} (\eta^2 + |d\eta|_g^2) d\Gamma,$$

with constants independent of $k \geq k_*$.

Apply Lemma 6.4 to $\omega = \eta e^{kZ}$ and use (6.22). For nested base patches $U_r \subset U_{r'}$ and a cutoff satisfying $|d\eta|_g \leq C/(r' - r)$, this gives

$$(6.23) \quad \|e^Z\|_{L^{6k}(\Gamma|_{U_r})} \leq \left[C(n)k^2 (1 + (r' - r)^{-2}) \right]^{1/(2k)} \|e^Z\|_{L^{2k}(\Gamma|_{U_{r'}})}.$$

Here $\Gamma|_U$ denotes the part of the graph projecting to U . Iterate with $k_j = 3^j k_*$ and radii decreasing geometrically from r' to r . The sums $\sum k_j^{-1}$ and $\sum j/k_j$ are finite. Taking the product in (6.23) therefore gives

$$(6.24) \quad S(r) := \sup_{U_r} e^Z \leq C(n)(r' - r)^{-A(n)} \|e^Z\|_{L^{2k_*}(\Gamma|_{U_{r'}})}.$$

Take all these patches inside a slightly larger fixed patch on which (6.19) holds. Increasing k_* to at least two, interpolation gives

$$\|e^Z\|_{2k_*} \leq S(r')^{1-1/k_*} \left(\int e^{2Z} d\Gamma \right)^{1/(2k_*)}.$$

Thus $S(r) \leq C(n)(r' - r)^{-A(n)} S(r')^{1-1/k_*}$. For every $\alpha > 0$, Young's inequality makes this

$$(6.25) \quad S(r) \leq \alpha S(r') + C(n, \alpha)(r' - r)^{-A(n)k_*}.$$

Choose radii increasing geometrically to a fixed outer radius r_1 , and then choose $\alpha < 2^{-A(n)k_*-1}$. Iterating (6.25) has a convergent geometric sum. Its final remainder $\alpha^j S(r_j)$ tends to zero, because each individual smooth solution is bounded on the closed outer patch. The resulting smaller-patch bound is independent of that individual supremum. A finite cover proves a uniform compact upper bound for Z , including at every inner face.

Beyond a fixed sphere containing the drift support, the equation is $\operatorname{div}(4uA_\chi \nabla Z) = u\Xi$, and $\Xi > 0$ above M_0 . At an interior maximum above M_0 its left side is nonpositive, because $dZ = 0$ and $\operatorname{Hess} Z \leq 0$. The outer boundary value is zero. The compact bound on that fixed sphere consequently extends the first-stage upper bound to the whole truncation.

6.6. The remaining stages and the bounded-variable conclusion. In the second and third stages the drift is zero. Their source has the same high- Z positivity as (6.15), with a uniform threshold for the bounded homotopy range. By Lemma 6.2, on an inner face

$$(6.26) \quad Z = t + \frac{1}{8} \log(1 + l^2 \sigma^2) \leq t + \frac{1}{8} \log(1 + e^2 C_*^2 / \tau^2).$$

Thus sufficiently high boundary Z forces $j(t) = 1$. The prescribed boundary flux is then zero, since $\lambda = 0$. At the face $A_\chi \nu = \chi \nu$, so this is exactly $\partial_\nu Z = 0$. The maximum principle excludes a high interior maximum, and the boundary point principle excludes a high maximum on a smooth inner face with this zero conormal derivative. Both principles are applied to the smooth individual solution on its finite domain; its coefficients are positive definite there. Locating the maximum requires no uniform ellipticity constant. The outer value is zero, so these two stages also have a uniform upper bound.

In the fourth stage $f = 0$ and $t = Z$. When the common source/flux scale is positive, the same maximum and boundary argument works with its positive high-level source. At scale zero the homogeneous mixed problem and the outer zero Dirichlet value give $Z = 0$. The upper bound is consequently uniform also as the scale tends to zero.

Proposition 6.5 (Bounds before regularity). *Fix the prepared data and $\epsilon > 0$. There are n_0 and $R_{\min}(n) \rightarrow \infty$ such that, for each fixed $n \geq n_0$ and every $R \geq R_{\min}(n)$, all smooth solutions of the four-stage homotopy with $t \geq -\epsilon$ satisfy*

$$(6.27) \quad |h| \leq C_0, \quad |f| \leq C_0/\tau, \quad -\epsilon \leq t \leq Z \leq C(n), \quad |\nabla f| \leq C(n).$$

The constants are uniform in R and in the homotopy parameter. They include putative floor-contact solutions, which are then excluded by Proposition 5.9. The fixed signed slopes, the third-stage upper slopes, and the polynomial trace estimates are those of Lemmas 6.2 and 6.3.

Proof. It remains to extract the bounds for the conformal variable and the gradient. Since $Z = t + \log D/8$ and $D \geq 1$, the floor gives $-\epsilon \leq t \leq Z$. The upper bound for Z gives

$$\sigma^2 = \frac{e^{8(Z-t)} - 1}{l^2} \leq \ell^{-2}(e^{8(C(n)+\epsilon)} - 1).$$

This proves the asserted fixed- n gradient bound. The height bounds are Lemma 5.5. The finite collection of sublevel and collar estimates and the polynomial absorption choose n_0 once. Enlarge $R_{\min}(n)$ to include the outer estimates and all fixed compact patches used above. The next section derives Hessian and continuity estimates from these bounds and uses them to prove existence. $\square$

7. REGULARITY AND EXISTENCE OF THE ELLIPTIC DEFORMATION

We now pass from the bounded variables in Proposition 6.5 to smooth solutions. The second equation contains Hessian squares, so classical estimates cannot yet be applied directly. We first control the gradient of the first unknown with a measurable axial coefficient. Its Hessian Morrey bound then gives continuity of the second unknown and permits classical regularity. Finally we define the scalar Dirichlet solution operator on all bounded input sets and apply degree to Proposition 5.4.

Throughout this section, ordinary regularity constants may depend on all fixed deformation parameters, including n and τ . No such constant will be used as a polynomial-in- n estimate. Bounds uniform in the outer truncation radius will be stated explicitly.

7.1. An estimate for a measurable axial coefficient. We use D for coordinate derivatives in the analytic lemmas. Balls are Euclidean coordinate balls, and a boundary ball is their intersection with one side of a smooth boundary. A family of boundary patches has bounded geometry here if normal-coordinate charts, their inverses, the metric and its inverse, and the derivatives through the orders used below have fixed bounds. We only apply the lemmas to smooth metrics and smooth boundaries. Their doubled metrics will be Lipschitz and smooth on each side of a single plane.

Lemma 7.1 (Axial gradient estimate). *Let U' be compactly contained in an interior or boundary coordinate patch U in dimension three. In a boundary patch assume that z is constant on the boundary face. Suppose z is smooth up to that face, $|\nabla z| \leq L$, and, almost everywhere,*

$$(7.1) \quad A_\chi : \text{Hess } z = G, \quad A_\chi = I - (1 - \chi)e \otimes e, \quad e = \frac{\nabla z}{|\nabla z|}, \quad 0 < \chi_0 \leq \chi \leq 1,$$

where $\|G\|_\infty \leq Q$. At zero gradients any measurable unit axis for which the equation holds is allowed. Then there are $\alpha \in (0, 1)$ and $C < \infty$, depending only on χ_0 , the patch geometry and the separation of U' from the other patch boundaries, such that

$$(7.2) \quad \|\nabla z\|_{C^{0,\alpha}(U')} \leq C(L + Q).$$

For all sufficiently small coordinate balls centered in U' , with intersection with the domain understood at a boundary, one also has

$$(7.3) \quad \int_{B_r \cap U} |\text{Hess } z|^2 dV_g \leq C(L + Q)^2 r^{1+2\alpha}.$$

In particular, the constants are independent of continuity moduli of χ and G .

Proof. The proof combines a Cordes estimate, a divergence identity for the gradient, and improvement of affine approximations. We first establish these estimates, then handle boundary reflection and the iteration.

A Cordes estimate above exponent two. For a Euclidean unit vector e ,

$$(7.4) \quad \|I - A_\chi\|_F = 1 - \chi \leq 1 - \chi_0.$$

For a compactly supported smooth function v , Plancherel's theorem gives $\|D^2v\|_2 = \|\Delta v\|_2$. The Calderón–Zygmund operator taking Δv to the full array D^2v is bounded on L^p , and interpolation makes its norm arbitrarily close to one as $p \rightarrow 2$. Choose once and for all

$$(7.5) \quad 2 < p_0 < 3, \quad C_{p_0}(1 - \chi_0/2) < 1.$$

Rescale a metric-normalized patch until the coordinate principal matrix differs from its Euclidean axial matrix by less than $\chi_0/2$ in Frobenius norm. Absorbing this difference in the Laplacian estimate, applying a cutoff, and using $\|Dv\|_p \leq \eta\|D^2v\|_p + C_\eta\|v\|_p$ gives

$$(7.6) \quad \|D^2v\|_{L^{p_0}(B_{1/2})} + \|Dv\|_{L^{p_0}(B_{1/2})} \leq C(\|v\|_{L^{p_0}(B_1)} + \|A_\chi : \text{Hess } v\|_{L^{p_0}(B_1)}).$$

The Christoffel terms are bounded first-order terms in this estimate. The same argument applies when $v = z - L_0$ and L_0 is coordinate affine: its extra right side is bounded by $C\|Dg\|_\infty|DL_0|$. The estimate uses bounded measurable principal coefficients only; the smoothness of the solutions allows all of its applications without an approximation issue. The linear estimates just used are the ordinary Laplacian estimates; see [28, Chapters 7 and 9].

The identity controlling a near-maximal gradient. Write $H = \text{Hess } z$ and $P = \nabla z$. Equation (7.1) gives the pointwise vector identity

$$(7.7) \quad A_\chi \nabla \frac{|P|^2}{2} - GP = (H - \Delta z g)P.$$

Indeed $HP = |P|H(e, \cdot)$ and $G = \Delta z - (1 - \chi)H(e, e)$, so the two axial terms cancel. The identity is also valid at $P = 0$. Taking the divergence of its right side, rather than differentiating χ , yields

$$(7.8) \quad \text{div} \left(A_\chi \nabla \frac{|P|^2}{2} - GP \right) = |H|^2 - (\Delta z)^2 + \text{Ric}(P, P).$$

Choose $a_0 > 0$ so small that $(1 + a_0)(1 - \chi_0)^2 < 1$. Young's inequality and the equation show

$$(7.9) \quad |H|^2 - (\Delta z)^2 \geq c_0|H|^2 - C_0G^2,$$

where $c_0 > 0$ depends only on χ_0 .

Consider solutions normalized by $|\nabla z| \leq 1$ on B_1 . After making the scale small, let the forcing and the rescaled geometric errors have size at most δ . The nonnegative function $s = 1 - |\nabla z|^2$ then satisfies

$$(7.10) \quad \text{div}(A_\chi \nabla s + 2G\nabla z) \leq -2c_0|\text{Hess } z|^2 + C\delta$$

in a weak sense. In the Euclidean case the last term can be replaced by $C\delta^2$. We allow $C\delta$ to include the geometric errors.

For the compactness step, suppose the normalized errors tend to zero in a sequence and $\inf_{B_{1/2}} s \rightarrow 0$. The inhomogeneous weak Harnack inequality, applied to Equation (7.10) after dropping its favorable Hessian term, gives $s \rightarrow 0$ in measure on $B_{1/2}$. The boundedness $0 \leq s \leq 1$ also gives convergence in every finite L^p norm there. To recover the Hessian information, take a cutoff η supported inside $B_{1/2}$ and use $\eta^2(b - s)_+$ in the weak supersolution inequality. Ellipticity and Young's inequality give

$$(7.11) \quad \int_{\{s < b\}} \eta^2 |Ds|^2 \leq Cb^2 \int |\eta|^2 + o(1)$$

for each fixed $b > 0$ as the errors vanish. The error fields in Equation (7.10) are bounded and tend to zero, so their products with Ds are absorbed in deriving this inequality. Equation (7.6) supplies a uniform L^2 bound for Ds . Its integral over $\{s \geq b\}$ tends to zero in L^1 by Cauchy's

inequality and convergence in measure. On $\{s < b\}$ use Equation (7.11), then let $b \downarrow 0$. Thus $Ds \rightarrow 0$ locally in L^1 . Testing Equation (7.8) with a nonnegative compact cutoff and using Equations (7.9) and (7.7) now gives

$$(7.12) \quad \int_{B_{1/4}} |\text{Hess } z|^2 \rightarrow 0.$$

Uniform Lipschitz compactness implies that, after subtracting a constant and taking a subsequence, z tends uniformly on a smaller ball to an affine function. Its slope has length one because $|\nabla z| \rightarrow 1$ in measure and the Hessians tend to zero.

Consequently, for any prescribed sufficiently small $b_* > 0$ there are fixed $r_* \in (0, 1/4)$, $k_* \in (r_*, 1)$ and $\delta_* > 0$ such that every normalized solution with errors at most δ_* has one of the following properties:

$$(7.13) \quad \begin{aligned} & \sup_{B_{r_*}} |\nabla z| \leq k_*; \quad \text{or} \\ & \sup_{B_{r_*}} |z - L_0| \leq b_* r_*, \quad \frac{1}{2} \leq |DL_0| \leq 2. \end{aligned}$$

If this conclusion failed, solutions whose gradient suprema approach one and whose errors approach zero would contradict Equation (7.12). A gradient-drop constant can always be weakened toward one to ensure $k_* > r_*$.

A fixed-axis equation. Once a nonzero affine slope appears, we compare with an equation whose axis is fixed. Fix a Euclidean unit vector e_0 . Suppose

$$(7.14) \quad \Delta_{e_0^\perp} Y_0 + \chi(x) \partial_{e_0 e_0} Y_0 = 0 \quad \text{in } B_{3/4}, \quad \|Y_0\|_{W^{2,2}(B_{3/4})} \leq C_1.$$

The derivative $v = \partial_{e_0} Y_0$ belongs to $W^{1,2}$ and, in distributions, satisfies

$$(7.15) \quad \Delta_{e_0^\perp} v + \partial_{e_0} (\chi \partial_{e_0} v) = 0.$$

This follows by differentiating the distribution $\chi \partial_{e_0} v$ as a whole. It does not assume a derivative of χ exists as a function. De Giorgi's estimate for this uniformly elliptic divergence equation, followed by Caccioppoli applied to $v - v(x_0)$, gives an exponent $\alpha_1 \in (0, 1)$ and

$$(7.16) \quad [v]_{C^{0,\alpha_1}(B_{1/2})} \leq C, \quad \int_{B_r(x_0)} |Dv|^2 \leq C r^{1+2\alpha_1}$$

for balls in a smaller fixed interior region. These are the scalar divergence estimates with ellipticity χ_0 ; see [28, Chapter 8].

All components of DY_0 , not only v , are controlled by this estimate. In fact,

$$(7.17) \quad \Delta Y_0 = f_0, \quad f_0 = (1 - \chi) \partial_{e_0} v, \quad \int_{B_r(x_0)} |f_0| \leq C r^{2+\alpha_1}.$$

The last inequality follows from Cauchy's inequality and Equation (7.16). Take the Newton potential of f_0 restricted to a slightly larger interior ball; its difference from Y_0 is harmonic in that ball. For two points at distance h , the contribution to the difference of gradient potentials from annuli of radius $r \leq 2h$ is bounded by $C \sum r^{\alpha_1} \leq Ch^{\alpha_1}$. On the remaining annuli the difference of gradient kernels is at most Chr^{-3} , so their contribution is

$$(7.18) \quad Ch \sum_{r \geq 2h} r^{\alpha_1-1} \leq Ch^{\alpha_1}.$$

The harmonic remainder has its usual interior derivative estimates. Therefore, decreasing the interior ball if needed,

$$(7.19) \quad \|Y_0\|_{C^{1,\alpha_1}(B_{1/4})} \leq C.$$

Improvement of a nonzero affine approximation. Suppose now

$$(7.20) \quad \|z - L_0\|_{L^\infty(B_1)} \leq b, \quad \frac{1}{4} \leq |DL_0| \leq 4, \quad |\nabla z| \leq 8.$$

Let the forcing and the first-derivative metric errors be at most $b\delta$. For $Y = (z - L_0)/b$, Equation (7.6) on fixed nested balls gives

$$(7.21) \quad \|Y\|_{W^{2,p_0}(B_{7/8})} \leq C.$$

In particular, this estimate does not require a bound for DY in the supremum norm. Since $Dz = DL_0 + bDY$, the variable axis converges in measure to $e_0 = DL_0/|DL_0|$ as $b \rightarrow 0$. More quantitatively, the coefficient error E caused by replacing this axis by e_0 is bounded and satisfies

$$(7.22) \quad \|E\|_{L^q(B_{7/8})} \leq Cb^\vartheta + Cb\delta, \quad q = \frac{2p_0}{p_0 - 2}, \quad \vartheta = \frac{p_0 - 2}{2} > 0.$$

Indeed the unit-direction map is Lipschitz where $|bDY| < |DL_0|/2$, and elsewhere its difference is bounded; the L^{p_0} bound in Equation (7.21) and interpolation give Equation (7.22). The same estimate includes the metric principal-part error. Hölder's inequality, with $1/2 = 1/q + 1/p_0$, shows that

$$(7.23) \quad \|[I - (1 - \chi)e_0 \otimes e_0] : D^2Y\|_{L^2(B_{3/4})} \leq C(b^\vartheta + \delta).$$

Solve for a zero-Dirichlet correction w on $B_{3/4}$ with the left side of Equation (7.23) as its source. This does not require a nondivergence Green function. The convex Dirichlet Hessian inequality $\|D^2w\|_2 \leq \|\Delta w\|_2$ and Equation (7.4) make the Laplace solution operator a contraction on $W^{2,2} \cap W_0^{1,2}$. Hence

$$(7.24) \quad \|w\|_{W^{2,2}(B_{3/4})} \leq C\chi_0^{-1}(b^\vartheta + \delta), \quad \|w\|_\infty \leq C(b^\vartheta + \delta).$$

The last implication uses $W^{2,2} \hookrightarrow C^{0,1/2}$ in dimension three. The convex Hessian inequality is the Miranda–Talenti estimate; see [49, Section 2, Theorem 3]. Now $Y_0 = Y - w$ satisfies Equation (7.14) with a uniform $W^{2,2}$ bound.

Choose $0 < \alpha_2 < \alpha_1$ and then a fixed $\lambda_0 \in (0, 1/4)$ such that $C\lambda_0^{1+\alpha_1} \leq \lambda_0^{1+\alpha_2}/4$ in Equation (7.19). By first taking δ small and then $b \leq b_*$ small, Equation (7.24) has supremum norm at most $\lambda_0^{1+\alpha_2}/4$. Taylor expansion of Y_0 at the center therefore gives an affine function L_1 with

$$(7.25) \quad \sup_{B_{\lambda_0}} |z - L_1| \leq b\lambda_0^{1+\alpha_2}, \quad |DL_1 - DL_0| \leq Cb.$$

All constants are uniform in the slope range in Equation (7.20).

Reflection at a constant Dirichlet face. Subtract the boundary value and use normal coordinates with the face given by $x_3 = 0$. Reflect the metric across this plane and reflect z oddly. Tangential derivatives of z vanish on the face, so the extension is C^1 . There is no distributional Hessian atom. The doubled metric is Lipschitz; the coefficient and lower-order estimates in Equation (7.6) hold with its essential first-derivative bound.

The flux in Equation (7.7) may have a normal jump. On either side of the face its normal trace is

$$(7.26) \quad ((\text{Hess } z - \Delta z g) \nabla z)_\nu = -(\text{tr}_T \text{Hess } z) \partial_\nu z.$$

Because the boundary value is constant, $\text{tr}_T \text{Hess } z$ is its mean-curvature coefficient times $\partial_\nu z$. The jump is therefore bounded by $C|\text{II}||\nabla z|^2$. At scale r it is a bounded plane density of size $O(r)$. Write the plane density and flux after multiplication by the reflected coordinate volume density, so that the following divergence is Euclidean. Such a signed density $a(x') \delta_{\{x_3=0\}}$ is the divergence of the bounded field $a(x') \mathbf{1}_{\{x_3>0\}} \partial_3$. Thus its negative part is another small divergence error in Equation (7.10). Smooth-side curvature errors are $O(r^2)$. Weak Harnack and the low-truncation

argument remain valid, and the flatness comparison uses the same Lipschitz metric estimates. This proves all the preceding alternatives for balls meeting the reflection plane as well.

Iteration and the Morrey consequence. Keep α_2, λ_0 from the affine improvement fixed. Choose b_* sufficiently small that $Cb_*/(1-\lambda_0^{\alpha_2}) < 1/4$, with C the slope-increment constant in Equation (7.25). Then choose r_*, k_*, δ_* in Equation (7.13) for this fixed b_* . Normalize initially by a fixed multiple of $L + Q$ and choose a uniformly small initial radius so that the metric and forcing errors satisfy all the preceding smallness conditions. If the first case of Equation (7.13) holds, rescale the ball by r_* and the gradient bound by k_* . The normalized forcing decreases by $r_*/k_* < 1$. Repeating this step either continues forever or reaches the second case. In the former case the gradient decays with exponent $\log k_*/\log r_*$. In the latter case apply Equation (7.25) successively, retaining exponent α_2 . At each affine step, rescale z by dividing by the spatial scale, subtracting only an additive constant; this preserves its gradient bound. Its slope changes have total size at most $Cb_*/(1-\lambda_0^{\alpha_2})$, so the slopes remain in $[1/4, 4]$. At the j th step the error size is $b_*\lambda_0^{j\alpha_2}$, while the normalized forcing and metric first-derivative errors have size $O(\lambda_0^j)$. Since $\alpha_2 < 1$, their ratio to the error size only decreases. This justifies every subsequent application of the improvement lemma. To combine the two branches, now choose

$$(7.27) \quad 0 < \alpha \leq \min \left\{ \alpha_2, \frac{\log k_*}{\log r_*} \right\}, \quad \alpha < 1.$$

At every center we have consequently obtained affine functions $L_{x,r}$ satisfying

$$(7.28) \quad \sup_{B_r(x)} |z - L_{x,r}| \leq C(L + Q)r^{1+\alpha}.$$

The slope differences at consecutive radii are bounded by $C(L + Q)r^\alpha$; comparing the affine approximations on overlapping balls gives Equation (7.2). This also identifies their limiting slope with Dz , since z is smooth. The reflected version gives the estimate up to a boundary.

Finally apply the L^2 version of Equation (7.6) to $z - L_{x,2r}$ after rescaling. The affine error contributes $C(L + Q)r^{1/2+\alpha}$ to the Hessian L^2 norm; the bounded forcing and affine Christoffel term contribute $C(L + Q)r^{3/2}$. Squaring and using $\alpha < 1$ proves Equation (7.3). $\square$

7.2. Bounded solutions with quadratic gradient growth. The Hessian Morrey bound controls the forcing in the second equation through the following oscillation estimate. A positive measure in the source bound also accommodates reflected boundary terms.

Lemma 7.2 (Natural-growth oscillation estimate). *On a three-dimensional ball suppose Z is bounded, $|Z| \leq M$, and is a smooth solution, or a reflected weak solution, of*

$$(7.29) \quad \operatorname{div}(aDZ + b) = e.$$

Assume a is measurable and symmetric, $\lambda I \leq a \leq \Lambda I$, $|b| \leq B$, and

$$(7.30) \quad |e| \leq C_0(1 + |DZ|^2) \, dx + \mu, \quad \mu(B_r(x)) \leq C_\mu r^{1+\beta}$$

for some $\beta > 0$ and every sufficiently small ball in the patch. Then Z has a local Hölder estimate whose constants depend only on the displayed bounds and the patch separation. The conclusion also holds up to a smooth boundary with bounded conormal flux or constant Dirichlet data, provided the corresponding reflected measure satisfies Equation (7.30).

Proof. Write $\mathcal{L} = -\operatorname{div}(aD)$ and, for $s \in \{-1, 1\}$, set $V_s = e^{skZ}$. The weak chain rule gives

$$(7.31) \quad \mathcal{L}V_s = \operatorname{div}(skV_sb) - skV_se - k^2V_s(aDZ \cdot DZ + b \cdot DZ).$$

Choose $k \geq \max\{1, 2C_0/\lambda\}$ and use $B|DZ| \leq (\lambda/2)|DZ|^2 + B^2/(2\lambda)$. Since $e^{-kM} \leq V_s \leq e^{kM}$, we obtain

$$(7.32) \quad \mathcal{L}V_s \leq \operatorname{div} F_s + \nu, \quad |F_s| \leq ke^{kM}B, \quad \nu \leq C \, dx + C\mu.$$

On a ball B_R compactly contained in the patch, solve $\mathcal{L}w_s = \operatorname{div} F_s + \nu$ with zero Dirichlet data. The bounded vector source has a solution of supremum norm at most CR : apply the L^p source estimate, $p > 3$, whose scaled factor is $R^{1-3/p}$, to $\|F_s\|_p \leq CR^{3/p}$. For the positive measure source use the Green bound $0 \leq \mathcal{G}(x, y) \leq C|x - y|^{-1}$ for uniformly elliptic measurable divergence coefficients [38, Section 7, Theorem (7.1) and Remark 2]. To use that interior comparison uniformly for all $x, y \in B_R$, extend the coefficients elliptically to B_{2R} . Domain monotonicity bounds the zero-Dirichlet Green function on B_R by the one on B_{2R} , where B_R is a fixed compact interior region after rescaling. This gives the stated uniform bound. Integration over concentric annuli gives

$$(7.33) \quad \sup_{x \in B_R} \int_{B_R} \mathcal{G}(x, y) d\nu(y) \leq C \int_0^{2R} \frac{\nu(B_t(x))}{t^2} dt + CR^{-1} \nu(B_{2R}(x)) \leq C(R^2 + R^\beta).$$

One may first approximate the measure by smooth positive densities. The uniform potential bound and the energy identity $\int a Dw \cdot Dw = \int w d\nu$ give a bounded sequence in $W_0^{1,2}$ and hence the required weak solution. Thus, with $\gamma = \min\{1, \beta\} > 0$,

$$(7.34) \quad \|w_s\|_\infty \leq \varepsilon_R, \quad \varepsilon_R = CR^\gamma.$$

Let $M_R = \sup_{B_R} Z$ and $m_R = \inf_{B_R} Z$. The functions

$$(7.35) \quad H_+ = e^{kM_R} - e^{kZ} + w_+ + \varepsilon_R, \quad H_- = e^{-km_R} - e^{-kZ} + w_- + \varepsilon_R$$

are nonnegative and satisfy $\mathcal{L}H_\pm \geq 0$. Weak Harnack controls their L^q means on $B_{R/2}$ by their infima on $B_{R/4}$. Each exponential deficit is comparable, with constants depending on kM , to $M_R - Z$ or $Z - m_R$, respectively. At least one of the sets

$$\{Z \leq (M_R + m_R)/2\} \cap B_{R/2}, \quad \{Z \geq (M_R + m_R)/2\} \cap B_{R/2}$$

has at least half the measure of $B_{R/2}$. Apply weak Harnack to the corresponding deficit. It follows that either the upper supremum decreases, or the lower infimum increases, by at least $c(M_R - m_R) - C\varepsilon_R$ on $B_{R/4}$. Consequently

$$(7.36) \quad \operatorname{osc}_{B_{R/4}} Z \leq (1 - c) \operatorname{osc}_{B_R} Z + CR^\gamma.$$

Iteration proves a Hölder estimate with any sufficiently small positive exponent below both γ and $-\log(1 - c)/\log 4$.

For the boundary estimate, flatten a face and put $S = \operatorname{diag}(1, 1, -1)$. An even extension of Z uses $a_- = Sa_+S$ and $b_- = Sb_+$. Its flux has a plane jump whose density is twice the prescribed normal flux, hence bounded. An odd extension after subtracting a constant Dirichlet value uses $a_- = Sa_+S$, $b_- = -Sb_+$, and the sign-reflected bulk source. Its normal flux is continuous. The transformed metrics and volume densities are absorbed into the displayed coefficients. A bounded plane density has ball mass $O(r^2)$, so it is admissible with exponent one. This proves the asserted boundary version. $\square$

Proposition 7.3 (Classical bounds for bounded-variable solutions). *Fix the deformation parameters and assume the bounds in Proposition 6.5. Every smooth solution in the homotopy of Proposition 5.4, with $t \geq -\epsilon$, has uniform local classical estimates of every finite order, up to the inner and outer faces. At fixed n the constants are independent of the homotopy parameter and of all sufficiently large outer radii R . The local patches can be chosen with a uniform positive radius throughout the end.*

Proof. Start with the bounds for f, Z, t, σ from Proposition 6.5. At fixed parameters they bound u, l, D above and away from zero, and give a uniform $\chi_0 > 0$. After dividing the scalar trace

equation by $l/\sqrt{D}$, its right side is

$$(7.37) \quad G_f = \frac{\sqrt{D}}{l} (F_s - \text{tr}_{A_\chi} K).$$

It is bounded throughout the homotopy, including its added collar terms. Lemma 7.1 therefore gives a uniform $C^{1,\alpha}$ estimate for f and

$$(7.38) \quad \int_{B_r} |\text{Hess } f|^2 \leq Cr^{1+2\alpha}$$

on the uniform patches, including boundary patches where f is constant.

After multiplying by the Riemannian volume density, the second equation has coordinate principal coefficients $a^{ij} = 4\sqrt{\det g} u A_\chi^{ij}$, a bounded flux vector b , and a source bounded in absolute value by

$$(7.39) \quad C(1 + |DZ|^2 + |D^2 f|^2).$$

Indeed t is a smooth function of x, Z, Df on the bounded variable range. Thus Dt is a bounded linear combination of $DZ, D^2 f$ and fixed smooth terms; the expression $\mathcal{T}$ is quadratic in these quantities. All other source terms have lower order. The same volume density is included in b, e ; it is smooth and bounded above and away from zero on the chosen patches, so it preserves these bounds and uniform ellipticity. The prescribed conormal flux at the inner boundary is bounded. At the outer face $Z = 0$. Apply Lemma 7.2, with $\mu = C|D^2 f|^2 dx$ and the bounded reflected plane density. Equation (7.38) supplies its measure growth hypothesis. We obtain a uniform positive Hölder exponent for Z .

Now Z and Df are Hölder, so all coefficients and the right side of the scalar f -equation are Hölder on their bounded variable range. Interior and constant-Dirichlet boundary Schauder estimates give $f \in C^{2,\theta}$ for a uniform $\theta > 0$. Expand the Z -equation in nondivergence form. Its principal coefficients are uniformly elliptic and $C^{0,\theta}$, while its right side is bounded by $C(1 + |DZ|^2)$. On an inner face Df is normal. Therefore A_χ has zero normal-tangential components there and normal eigenvalue χ . Its boundary equation solves for

$$(7.40) \quad \partial_\nu Z = \mathcal{H}_s(x, Z, f, Df),$$

where $\mathcal{H}_s$ is smooth on a fixed bounded variable range. Its derivatives with respect to x and its displayed scalar arguments are bounded at fixed parameters. The outer condition remains constant Dirichlet. These are different, disjoint smooth boundary components; there is no change of boundary condition along an edge or a corner.

It remains to absorb the quadratic gradient growth. Fix $p > 3$. The linear local $W^{2,p}$ estimate for the leading nondivergence operator with Equation (7.40) or outer Dirichlet data has the form, on fixed nested patches $U_x \Subset V_x$,

$$(7.41) \quad \|Z\|_{W^{2,p}(U_x)} \leq C \left(1 + \|DZ\|_{L^{2p}(V_x)}^2 + \|Z\|_{W^{1,p}(V_x)} \right).$$

Here the boundary norm is controlled by

$$(7.42) \quad \|\mathcal{H}_s(x, Z, f, Df)\|_{W^{1-1/p,p}(\partial V_x)} \leq C(1 + \|Z\|_{W^{1,p}(V_x^+)}),$$

using trace and composition after extending its smooth expression through the collar. The derivatives of f, Df entering this extension are already bounded. The leading matrix has a fixed Hölder modulus, so its small oscillation on sufficiently small patches permits the usual freezing and absorption in the linear estimate. The normal boundary operator is uniformly oblique and satisfies the complementing condition. These are the classical local elliptic boundary estimates [1, 2].

Let $[Z]_{C^{0,\theta}} \leq H_0$. On balls or half-balls of radius h , the scaled Gagliardo–Nirenberg inequality, applied after subtracting a constant, gives

$$(7.43) \quad \begin{aligned} \|DZ\|_{L^{2p}(B_h)}^2 &\leq C \operatorname{osc}_{B_h} Z \|D^2 Z\|_{L^p(B_h)} \\ &\quad + Ch^{3/p-2} (\operatorname{osc}_{B_h} Z)^2. \end{aligned}$$

One can use a bounded extension operator on a half-ball; its constants are uniform in the boundary charts. This is the interpolation inequality of [43, Lecture II, Equation (2.2)] with its lower-order term on a bounded patch. Cover V_x by radius- h patches of bounded overlap and take the ℓ^p sum of Equation (7.43). It follows that

$$(7.44) \quad \|DZ\|_{L^{2p}(V_x)}^2 \leq CH_0 h^\theta \|D^2 Z\|_{L^p(V_x^+)} + CH_0^2 h^{2\theta-2}.$$

Also, for any $\eta > 0$, $\|Z\|_{W^{1,p}(V_x)} \leq \eta \|D^2 Z\|_{L^p(V_x^+)} + C_\eta$. Each enlarged patch V_x^+ is covered by a bounded number of the patches U_y , independently of R . Set $S = \sup_y \|Z\|_{W^{2,p}(U_y)}$, finite for each smooth finite-domain solution. Equations (7.41) and (7.44) give

$$(7.45) \quad S \leq C(H_0 h^\theta + \eta)S + C(h, \eta).$$

Choose h and then η so the first coefficient is less than $1/2$. We obtain a uniform $W^{2,p}$ bound and hence a $C^{1,1-3/p}$ bound for Z .

The coefficients in both equations and in the normal boundary condition now have the regularity required for Schauder bootstrap. First the nondivergence Z -equation has Hölder source, giving $Z \in C^{2,\theta'}$; the scalar f -equation then gains another derivative. Repetition gives every finite classical derivative supported by the smooth data. The same procedure holds on the uniform unit patches of the asymptotic end and on the large outer spheres, whose normal-coordinate constants are uniform. This proves the asserted independence of R . The reduced background has end symbol bounds of every order by Proposition 3.10; the bootstrap here does not assume additional derivatives of the original asymptotic data. $\square$

7.3. The scalar Dirichlet solution operator. The compact map requires a scalar solve for every bounded input Z . Its definition must therefore work without a floor bound on the input.

Lemma 7.4 (Scalar solve for unrestricted bounded inputs). *Fix $n \geq 4$, $\tau > 0$, a finite smooth truncation Ω_R , and a homotopy parameter $s \in [0, 4]$. For every $Z \in C^{1,b}(\overline{\Omega_R})$, $0 < b < 1$, the scalar trace equation of Proposition 5.4, with its prescribed componentwise constant Dirichlet values for f , has a unique solution. It lies in $C^{3,b}$ and is smooth to the extent allowed by the input. The solution operator*

$$(7.46) \quad f = \mathcal{S}_s(Z)$$

is continuous in (s, Z) , including across the concatenation points of the homotopy. On bounded subsets of $C^{1,b}$ its $C^{3,b}$ norm is uniformly bounded at these fixed parameters. No lower bound for t is required.

Proof. The implicit definition of t has derivative $\partial Z / \partial t = 1 + pv/4 > 0$. Thus t and all the coefficients are smooth functions of x, Z, Df , also at $Df = 0$. There is no dependence of the principal coefficient on the height f itself. For bounded height the added collar terms in F_s are bounded, and its derivative with respect to $h = \tau f$ is at least one. At $Df = 0$ the equation and this height sign give upper and lower constant barriers. Hence every scalar solution satisfies

$$(7.47) \quad |h| \leq C_0,$$

where the constant can be chosen independently of Z and the homotopy parameter. We include the fixed boundary values in its choice.

We first derive the coefficient bounds at an unbounded slope. If $|Z| \leq M$ and $\sigma \rightarrow \infty$, the implicit equation forces $t \rightarrow -\infty$ uniformly in Z . There $p = n$ and $l = e^{nt}$, so its exact form is

$$(7.48) \quad e^{8Z} = e^{8t} + e^{(2n+8)t} \sigma^2.$$

Consequently

$$(7.49) \quad \begin{aligned} t &= -\frac{\log \sigma}{n+4} + O(1), & l &\asymp \sigma^{-n/(n+4)}, \\ d &\asymp \sigma^{-8/(n+4)}, & \chi &\asymp \sigma^{-8/(n+4)}, & \frac{\sqrt{D}}{l} &\asymp \sigma. \end{aligned}$$

The constants are uniform for $|Z| \leq M$ at fixed n . Since $n \geq 4$, the exponent $8/(n+4)$ is at most one. Combining the large-slope estimates with compactness of the remaining variable range yields

$$(7.50) \quad \chi \geq \frac{c}{1+\sigma}, \quad |G_f| \leq C(1+\sigma)$$

whenever Equation (7.47) holds.

We prove a global gradient estimate using this radial ellipticity. Let δ be smaller than a fixed collar radius, half the distance between distinct boundary components, and a normal injectivity radius of a smooth extension of the finite domain. On a boundary distance collar use

$$(7.51) \quad \psi(r) = k^{-1} \log(1 + Br), \quad \psi'' = -k(\psi')^2.$$

The upper and lower barriers are the assigned face value plus or minus ψ . On these tests the axis is normal to the distance leaves, so their principal trace is $\chi\psi'' + \psi'\Delta r$. At slopes at least one, Equation (7.50) gives

$$(7.52) \quad -k\chi(\psi')^2 \leq -\frac{1}{2}ck\psi'.$$

Choose k large enough to dominate all the bounded geometry terms and the linear growth in Equation (7.50). Next shrink δ so that $1/(2k\delta)$ exceeds any fixed large-slope threshold used here. Finally choose B large enough that $B\delta \geq 1$ and $\psi(\delta)$ exceeds twice the height oscillation. Then $\psi' \geq 1/(2k\delta)$ on this collar and the two barriers compare by the height monotonicity. This gives a uniform boundary gradient bound.

For the interior estimate, increase B if necessary and compare two points using

$$(7.53) \quad f(x) - f(y) - \psi(\text{dist}(x, y)), \quad 0 < \text{dist}(x, y) < \delta.$$

The distance is taken in the fixed smooth extension. The boundary barriers exclude a positive maximum with one endpoint on the boundary. The inequality $\psi(\delta) > 2 \text{osc } f$ excludes the other boundary of the two-point domain. At a positive interior maximum let $r = \text{dist}(x, y)$ and $q = \psi'(r)$. The two gradients have length q and are parallel transports along the connecting geodesic. Vary both endpoints simultaneously in parallel transverse directions. The second variation formula for the short geodesic yields

$$(7.54) \quad \text{tr}_{e_x^\perp} \text{Hess } f(x) - \text{tr}_{e_y^\perp} \text{Hess } f(y) \leq Cqr.$$

Varying each endpoint separately along that geodesic gives $f_{;e_x e_x}(x) \leq \psi''(r)$ and $f_{;e_y e_y}(y) \geq -\psi''(r)$. Subtracting the two scalar equations therefore gives

$$(7.55) \quad -2C(1+q) \leq G_f(x) - G_f(y) \leq Cqr + (\chi_x + \chi_y)\psi''(r).$$

By Equations (7.50) and (7.51), the last term is at most $-ckq$ at the large comparison slopes. Our choice of k gives a contradiction. Thus Equation (7.53) is nonpositive. Reversing x, y proves the same bound for the absolute difference and, on letting $y \rightarrow x$, a global Lipschitz bound for f . This argument used no continuity modulus for $\chi_x - \chi_y$; only the common transverse eigenvalues and the radial lower bound were needed.

For existence interpolate the normalized equation with $\Delta f = \tau f$, retaining the given boundary values. Explicitly, for $a \in [0, 1]$ take principal matrix $(1-a)I + aA_\chi$ and right side $(1-a)\tau f + aG_f$. The height sign, the linear gradient growth, and Equation (7.50) persist uniformly in a . The height and logarithmic gradient bounds just proved are therefore uniform. After these bounds the interpolating equations are uniformly elliptic. Their matrices still have the axial form, with axial eigenvalue $(1-a) + a\chi$, so Lemma 7.1 applies. Schauder estimates then give a uniform $C^{2,\theta}$ bound at fixed input Z . The coefficients and right side are now Lipschitz in position on bounded input sets, which permits their $C^{0,b}$ estimates and an upgrade to $C^{2,b}$. Differentiating once, or using the next Schauder estimate, yields $f \in C^{3,b}$ because $Z \in C^{1,b}$.

The scalar linearization has the form $a^{ij}D_{ij}\varphi + b^iD_i\varphi - c\varphi$ with $c \geq c_1\tau > 0$ on the bounded variable range. Indeed the principal matrix is independent of the height, while $\partial_f G_f \geq \tau\sqrt{D}/l > 0$. The Dirichlet maximum principle gives uniqueness and a zero kernel; the linear elliptic Dirichlet theory gives an inverse. The implicit-function theorem gives openness of the interpolation parameter set. The uniform classical bounds and compactness give closedness, starting from the invertible Laplace endpoint. This proves existence. The same maximum principle applied to the difference of two solutions proves uniqueness of the nonlinear solution. The implicit-function theorem, or compactness and uniqueness, gives continuous dependence on Z and on the homotopy parameter. At the finitely many concatenation points the scalar equations agree, so the same continuity holds there. $\square$

7.4. A compact equation and the order of parameter choices. Recall the parameter $s \in [0, 4]$ of Proposition 5.4: $s = 0$ is the desired system, $s = 1$ has zero drift, $s = 2$ has the full collar modification, and $s = 3$ has $f = 0$. On $[3, 4]$ the scalar source and inner flux are scaled to zero; at $s = 4$ the scalar equation still has the unique solution $f = 0$. We will define a compact map along this path and show that its fixed points remain in the admissible set.

Theorem 7.5 (Existence on finite truncations). *For the prepared domain of Proposition 4.7 and every sufficiently small fixed $\epsilon > 0$, there is an integer $n_0 \geq 4$ and, for each $n \geq n_0$, a radius $R_{\min}(n)$, with $R_{\min}(n) \rightarrow \infty$, such that the original deformation system has a smooth solution on every Ω_R with $R \geq R_{\min}(n)$. It satisfies $t > -\epsilon$ on the closure. At fixed n , these solutions have the local bounds of Proposition 7.3, uniformly as $R \rightarrow \infty$. In particular, any sequence $R_j \rightarrow \infty$ of such radii has a subsequence converging smoothly on compact subsets of $\overline{\Omega}$ to a solution of the original equations with $t \geq -\epsilon$.*

Proof. We first make the radius convention precise. Fix the prepared data, all thresholds and collars, and ϵ . Increase n_0 so that

$$(7.56) \quad n_0 \geq \max\{4, \lceil 2/\epsilon \rceil\}$$

and so that the floor test and all estimates of Proposition 6.5 apply for $n \geq n_0$. Let $R_{\text{ap}}(n)$ be a sufficient lower radius for those estimates, and let R_{geo} contain all compact boundary faces and supports used in the construction. The end barrier of that proposition gives a constant C_{out} , independent of n, R, s , such that every admissible or floor-contact homotopy solution has

$$(7.57) \quad |\nabla f| \leq C_{\text{out}}\tau^{-1}R^{-1-\gamma} \quad \text{on the outer sphere.}$$

The scalar last stage has $f = 0$ and satisfies the same bound. Choose, enlarging to a radius of a smooth truncation if necessary,

$$(7.58) \quad R_{\min}(n) = \max\left\{n, R_{\text{geo}}, R_{\text{ap}}(n), \left(\frac{2C_{\text{out}}\ell}{\tau\sqrt{e^{8\epsilon}-1}}\right)^{1/(1+\gamma)}\right\}.$$

At $Z = 0$, the value of its implicit expression at $t = -\epsilon$ is $-\epsilon + \frac{1}{8} \log(1 + \ell^2 \sigma^2)$. Equations (7.57) and (7.58) make this strictly negative. Strict monotonicity of that expression in t therefore excludes an outer floor contact. The term n in Equation (7.58) ensures $R_{\min}(n) \rightarrow \infty$.

The sublevel and collar estimates apply to arbitrary smooth admissible or floor-contact solution sequences with $n \rightarrow \infty$ and expanding outer cutoffs, uniformly over the compact ranges of the homotopy parameters. If the needed conclusion failed for arbitrarily large n with $R \geq R_{\min}(n)$, one could select such a violating sequence. The chosen radius convention would make it precisely an expanding sequence excluded by those proofs. Thus a single n_0 , followed by the radius choices above, suffices for every homotopy parameter. No upper bound for Z is inserted in the earlier contact estimates: there $\sigma \asymp \tau^{-1}$ and $l \geq \ell$ alone give $d \lesssim (\tau/\ell)^2$.

Fix now $n \geq n_0$ and $R \geq R_{\min}(n)$ for the rest of the degree argument. Let

$$(7.59) \quad X_b = \{Z \in C^{1,b}(\overline{\Omega_R}) : Z = 0 \text{ on the outer face}\}, \quad 0 < b < 1.$$

For an input $Z \in X_b$, first solve $f = \mathcal{S}_s(Z)$ by Lemma 7.4. Compute $t, u, A_\chi, \mathcal{T}$ and the full homotopy source from this input pair. Write B_s for the frozen drift summand in the flux, E_s for the full right side of its divergence equation, and q_s for the prescribed inner flux. Thus before the final scaling $B_s = \lambda_s u A_\chi K(w, \cdot)$ and $E_s = u \Xi$; on the final interval E_s and q_s have the simultaneous scalar factor prescribed in Proposition 5.4. Define $\mathcal{K}_s(Z) = \tilde{Z}$ by the linear mixed problem

$$(7.60) \quad \begin{cases} \operatorname{div}(4uA_\chi \nabla \tilde{Z} + B_s) = E_s & \text{in } \Omega_R, \\ (4uA_\chi \nabla \tilde{Z} + B_s)_\nu = q_s & \text{on } B, \\ \tilde{Z} = 0 & \text{on the outer face.} \end{cases}$$

The normal ν in the second line points into Ω_R ; the variational formulation uses its negative as the integration normal. This changes the sign of the boundary functional, not coercivity. On a bounded input set, the scalar gradient estimate bounds the coefficients above and away from zero. The nonempty outer Dirichlet face gives a Poincaré inequality, so the linear problem is uniquely solvable by its coercive bilinear form.

On bounded input sets in $C^{1,b}$, Lemma 7.4 bounds f in $C^{3,b}$. Hence the leading coefficient and drift in Equation (7.60) are $C^{1,b}$, the bulk source is $C^{0,b}$, and the prescribed boundary data are $C^{1,b}$. The derivatives of $t(x, Z, Df)$ involve only DZ and $D^2 f$, so this assertion includes the quadratic expression $\mathcal{T}$. Linear regularity gives a bounded output set in $C^{2,b}$. The inclusion $C^{2,b} \hookrightarrow C^{1,b}$ is compact on the finite domain. Continuous dependence of the scalar solve and the linear problem makes $(s, Z) \mapsto \mathcal{K}_s(Z)$ a continuous compact homotopy on bounded sets. These assertions hold also at the concatenation parameters because the defining equations coincide there. A fixed point solves the coupled system; repeated Schauder estimates make it smooth.

Admissibility is evaluated after the scalar solve. Define the open subset of the parameter–input product

$$(7.61) \quad \mathcal{O} = \{(s, Z) \in [0, 4] \times X_b : \min_{\Omega_R} t_s[Z] > -\epsilon, \|Z\|_{C^{1,b}} < M_1\}.$$

It is relatively open because the scalar solve and the implicit function defining t depend continuously on (s, Z) . Propositions 6.5 and 7.3 give a uniform bound in $C^{1,b}$ for all fixed points satisfying $t \geq -\epsilon$. If initially needed, use their higher classical bounds to reach the chosen exponent b . Choose M_1 larger than this bound. No fixed point lies on the norm boundary of $\mathcal{O}$. No fixed point lies on its floor boundary either: outer contacts were excluded above and all remaining contacts are excluded by Proposition 5.9.

To apply degree on the moving admissible set, we need compactness of its fixed points. The set of fixed points in $\overline{\mathcal{O}}$ is compact: their X_b norms are bounded, the parameter interval is

compact, and $Z = \mathcal{K}_s(Z)$ with a compact homotopy. Any limit has $t \geq -\epsilon$ by continuity; it is a smooth fixed point and cannot lie on the floor. Thus these fixed points are compactly separated from the excluded boundary. The excision form of Leray–Schauder homotopy invariance applies to the sections $\mathcal{O}_s$; equivalently, cover the compact fixed-point set by finitely many product neighborhoods contained in $\mathcal{O}$ and use ordinary homotopy invariance and excision on the resulting finite parameter subdivision. This is the compact perturbation-of-identity degree [35].

At $s = 4$, the scalar equation has the unique solution $f = 0$. The drift vanishes, and both E_4 and q_4 vanish for every input. Equation (7.60) then has output identically zero. Moreover, $t_4[0] = 0 > -\epsilon$, so $0 \in \mathcal{O}_4$ and

$$(7.62) \quad \deg(I - \mathcal{K}_4, \mathcal{O}_4, 0) = 1.$$

Homotopy invariance gives the same degree at $s = 0$. There is therefore a fixed point in $\mathcal{O}_0$, which is the desired smooth solution with $t > -\epsilon$.

The bounds of Proposition 7.3 are independent of R at fixed n . A diagonal subsequence on a compact exhaustion, using boundary charts at the fixed inner faces, converges smoothly to a solution on $\bar{\Omega}$. The floor inequality passes to $t \geq -\epsilon$. The next section establishes the end normalization, decay, and flux for these limits. $\square$

8. PASSAGE TO INFINITY AND THE MASS INEQUALITY

We complete the energy inequality by controlling the end flux of the deformation and applying the Riemannian Penrose theorem. Fix the reduced initial data supplied by Proposition 3.10, and the prepared exterior Ω of Proposition 4.7. Its boundary B separates the original inner obstacle from the end. Let A_* be the enclosing-area infimum of that original obstacle, measured in the reduced metric g . In particular every cut enclosing B has g -area at least A_* .

We use the notation of Section 5 and take the physical endpoint of the homotopy:

$$F = h + C, \quad h = \tau f, \quad \operatorname{div} V = u\Xi, \quad \Xi = \delta_0 \mathcal{T} + \rho + \delta_0 \tau a \sigma - m_0(t) \mathcal{P}.$$

Here $0 < \delta_0 < 1$, $\ell = e^{-\epsilon n}$, $\tau = \ell^{3/2}$, and all constants in the prepared data are fixed before n is taken large. Constants marked C_n in this section are bounded by $C(1+n)^C$, with fixed exponents. Other constants explicitly allowed to depend on fixed n need not have such a bound.

8.1. The limiting solution and its end.

Lemma 8.1 (End control). *Fix $\epsilon > 0$ and a sufficiently large $n \geq \max\{4, \lceil 2/\epsilon \rceil\}$. The truncation solutions of Theorem 7.5 have a subsequence converging smoothly on compact subsets of $\bar{\Omega}$ to a solution of the physical system with $t \geq -\epsilon$. On the asymptotically flat end, for some $c_n > 0$ and fixed- n constants $C_j(n)$,*

$$|\nabla^j f| \leq C_j(n) e^{-c_n r} \quad \text{for each fixed } j, \quad Z, t = O_2(r^{-1}).$$

The differences $t - Z$, $\bar{g} - g$, and their coordinate derivatives of any fixed order decay exponentially. The vector field V satisfies

$$(8.1) \quad V = 4\nabla_g t + O(r^{-3})$$

and has a finite outward flux limit.

Proof. The compact convergence, including at the fixed inner boundary, follows from Theorem 7.5 and its uniform fixed- n classical estimates. All equations and inner boundary conditions pass to this limit. To preserve the end normalization, we prove the decay estimates on the truncations before passing to infinity.

Outside a fixed large sphere, $K = C = 0$, and the first equation is

$$\mathrm{tr}_{A_\chi} \mathrm{Hess}_g f = \frac{\tau \sqrt{D}}{l} f.$$

At fixed n the preceding estimates make the left side uniformly elliptic, and its positive height coefficient is bounded below by a positive constant depending on n . It is bounded above as well. For sufficiently small $c_n > 0$, the functions $\exp[-c_n(r - r_0)]$ are positive supersolutions of the corresponding linear equation outside a sufficiently large r_0 : their radial second derivative has size c_n^2 , while their tangential second derivatives have size c_n/r , and the fixed positive height term dominates both. Multiplying by a constant bounds the solution on S_{r_0} ; the zero outer Dirichlet value is also bounded. The maximum principle applied to both signs of f gives the uniform exponential estimate. The fixed- n unit-patch estimates, followed by the local homogeneous linear estimates for this equation, give the same decay for derivatives. The corresponding boundary estimates apply on the large outer spheres. Iterating the smooth equations gives each required finite derivative order.

The relation $Z = t + \frac{1}{8} \log(1 + l^2 |\nabla f|^2)$ now shows that $t - Z$ is exponentially small, with derivatives. The graph errors have the same property. On the end the second equation becomes

$$(8.2) \quad \Delta_g Z + b_n(Z) |dZ|_g^2 = O(r^{-4}), \quad b_n(s) = p_L(s) - \delta_0(p_L(s) - 1).$$

Indeed $A_\chi = I$ and $u = L(Z)$ up to exponentially small errors, whereas $\mathcal{T} = 4(p_L(Z) - 1) |dZ|^2$ up to such errors. The ρ_0 penalty weight is $O(r^{-4})$; the term with ρ_1 is multiplied by the exponentially small v . All coefficients in this statement are bounded on the fixed- n solution range. Differentiably controlled exponential errors can absorb any fixed polynomial power of r .

Define an increasing smooth function on that range by

$$\Phi_n(0) = 0, \quad \Phi'_n(s) = \exp\left(\int_0^s b_n(z) dz\right).$$

Its derivative is bounded above and below by positive fixed- n constants. The chain rule in (8.2) gives

$$\Delta_g \Phi_n(Z) = O(r^{-4}).$$

The function is bounded and vanishes at the outer cutoff. To obtain a bound independent of that cutoff, choose $0 < \eta < 1$. For large r , a positive multiple of $r^{-1} - r^{-1-\eta}$ satisfies

$$-\Delta_g(r^{-1} - r^{-1-\eta}) \geq cr^{-3-\eta}.$$

Here the Euclidean leading term is $\eta(1 + \eta)r^{-3-\eta}$, and the metric error is $O(r^{-4})$. It dominates the right side after increasing the multiple and the fixed inner radius. Comparison for both signs yields $|\Phi_n(Z)| \leq C'(n)r^{-1}$, where $C'(n)$ is a fixed- n constant with no polynomial claim. Consequently $Z = O(r^{-1})$.

At sufficiently large radius this estimate and the exponential graph error give $t > -\epsilon + 1/n$; hence the floor penalty vanishes there. Rescale each annulus of radii comparable to r to unit size. The equation for $\Phi_n(Z)$ has a bounded rescaled source and uniformly controlled metric coefficients. Local $W^{2,p}$ estimates with $p > 3$ first give the gradient decay. Its source is then a smooth weight times a smooth function of the bounded variables, plus the already controlled exponential errors. Schauder estimates, or a further Sobolev bootstrap followed by them, give $Z = O_2(r^{-1})$. Inverting Φ_n preserves these bounds, and the same holds for t .

Since $L(0) = 1$, we have $u = 1 + O(r^{-1})$ and $A_\chi = I + O(e^{-c_n r})$. With $K = 0$ on the end, the definition of V gives (8.1). Moreover $\mathrm{div} V = u\Xi$ is integrable there: $\mathcal{T} = O(r^{-4})$, $\rho = O(r^{-4})$, the penalty has vanished, and $\tau a \sigma$ decays exponentially. The divergence theorem on annuli proves the existence of the outward flux limit. $\square$

Lemma 8.2 (Curvature, completeness, and ADM charge). *For the solution in Lemma 8.1,*

$$\hat{g} = e^{4t}(g + l^2 df^2)$$

is a smooth complete metric on $\bar{\Omega}$, with nonnegative scalar curvature and strictly negative mean curvature on B for the normal pointing into Ω . Its end is asymptotically flat with

$$\hat{g} - \delta = O_2(r^{-1}), \quad R_{\hat{g}} = O(r^{-3-\delta'}), \quad \delta' = \min\{\delta, 1\} > 0,$$

where the reduced end has $R_g = O(r^{-3-\delta})$. If

$$\mathfrak{F}_n = \lim_{r \rightarrow \infty} \int_{S_r} V_\nu dA_g,$$

then

$$(8.3) \quad E_{\hat{g}} = E - \frac{\mathfrak{F}_n}{8\pi}.$$

Proof. At the physical endpoint the curvature identity gives

$$\begin{aligned} \frac{1}{2}e^{4t}R_{\hat{g}} &= 8\pi(\mu + J(w)) + (1 - \delta_0)\mathcal{T} + (1 - \delta_0)\tau a\sigma \\ &\quad + w \cdot \nabla C - F \operatorname{tr} K - \rho + m_0(t)\mathcal{P}. \end{aligned}$$

The height bounds on the compact supports of K and C give $h \in [b_-, b_+]$. Since $|w| \leq 1$, the terms $|F \operatorname{tr} K| + |\nabla C| + \rho$ are dominated by $8\pi(\mu - |J|)$, by the choices in Proposition 4.7 and the choice of ρ . Outside those supports the same conclusion uses $8\pi(\mu - |J|) - \rho \geq 0$. All remaining terms are nonnegative. This proves $R_{\hat{g}} \geq 0$.

The boundary identity and the prescribed flux give $H + 4\partial_\nu t = -N_B < 0$. Since f is constant on each face, this is exactly the mean-curvature sign for $\hat{g}$. Also $\hat{g} \geq e^{-4\epsilon}g$, so any escaping curve has infinite length; the compact boundary is included. Smoothness and completeness therefore follow.

Lemma 8.1 gives the stated metric decay. The reduced construction supplies $R_g = O(r^{-3-\delta})$ with symbol estimates on the rest end. Since graph errors are exponentially small, the conformal formula gives

$$R_{\hat{g}} = e^{-4t}(R_g - 8\Delta_g t - 8|dt|_g^2) + O(e^{-c_n r}).$$

Equation (8.2), the vanishing far-end penalty, and the gradient estimate give $\Delta_g t = O(r^{-4})$. This proves the asserted scalar-curvature decay, including the pointwise falloff required by the AF convention of [15, Definition 21 and Theorem 19].

Exponential graph errors contribute no ADM flux. The conformal change and $t = O_2(r^{-1})$ give

$$E_{\hat{g}} = E - \frac{1}{2\pi} \lim_{r \rightarrow \infty} \int_{S_r} \partial_\nu t dA_g.$$

Replacing the metric normal and area form by their Euclidean versions costs $o(1)$, and the $O(r^{-3})$ error in (8.1) has vanishing sphere integral. This proves (8.3). $\square$

8.2. The boundary and floor losses. The mass formula reduces the remaining estimate to a lower bound for the flux. We absorb the inner boundary term into the positive bulk source and then estimate the penalty near the floor. Both steps use only polynomial constants.

Lemma 8.3 (Flux lower bound). *For fixed $\epsilon > 0$, the physical solutions satisfy*

$$\mathfrak{F}_n \geq -\varepsilon_n, \quad 0 \leq \varepsilon_n \leq C(1+n)^C(\ell + \ell^2/\tau) \rightarrow 0.$$

All constants in this estimate are independent of the fixed- n classical regularity constants.

Proof. The normal ν on B points into Ω , so the divergence theorem has the sign

$$(8.4) \quad \mathfrak{F}_n = \int_{\Omega} u \Xi \, dV_g + \int_B V_{\nu} \, dA_g.$$

Each integral is finite for a fixed n , by Lemma 8.1 and compact smoothness.

At a black face, $F - P_B = H$ and $w_{\nu} = a$. At a white face, $F - P_B = -H$ and $w_{\nu} = -a$. Thus at either type of face,

$$V_{\nu}/u = -(1-a)H - N_B d \geq -Cd.$$

The signed boundary gradient bounds give $\sigma \geq c/\tau$, so

$$\sqrt{d} \leq \frac{1}{l\sigma} \leq C\tau/\ell, \quad \int_B V_{\nu} \, dA_g \geq -C\frac{\tau}{\ell} \int_B L \, dA_g.$$

Use the polynomial collar trace estimate from Section 6 with the constant test function:

$$\int_B L \, dA_g \leq C(1+n)^C \int_{\mathcal{C}} u(1 + \sqrt{\mathcal{T}}) \, dV_g,$$

where $\mathcal{C}$ is a fixed compact union of collars. Its polynomial constant is independent of the upper Z bound. Since ρ has a fixed positive minimum on $\mathcal{C}$,

$$\int_{\mathcal{C}} u(1 + \sqrt{\mathcal{T}}) \, dV_g \leq C \int_{\Omega} u(\delta_0 \mathcal{T} + \rho) \, dV_g.$$

The factor $C(1+n)^C \tau/\ell = C(1+n)^C \ell^{1/2}$ tends to zero. For large n , the entire negative boundary term in (8.4) is therefore at most one quarter of $\int_{\Omega} u(\delta_0 \mathcal{T} + \rho) \, dV_g$.

It remains to estimate the floor penalty. On its support, $-\epsilon \leq t \leq -\epsilon + 1/n$, so l and L are comparable to ℓ , with constants independent of n . Directly from their definitions,

$$\begin{aligned} u &= L\sqrt{1 + l^2\sigma^2} \leq C(\ell + \ell^2\sigma), \\ uv &= \frac{Ll^2\sigma^2}{\sqrt{1 + l^2\sigma^2}} \leq C\ell^2\sigma, \\ ua\sigma &= Ll\sigma^2 \geq c\ell^2\sigma^2. \end{aligned}$$

Consequently

$$um_0\mathcal{P} \leq C_n[\ell\rho_0 + \ell^2\sigma(\rho_0 + \rho_1)].$$

Young's inequality, with the square term chosen to be a quarter of $\delta_0\tau ua\sigma$, yields

$$um_0\mathcal{P} \leq \frac{1}{4}\delta_0\tau ua\sigma + C_n \left[\ell\rho_0 + \frac{\ell^2}{\tau}(\rho_0^2 + \rho_1^2) \right].$$

Squaring the constant only changes the polynomial C_n . The integrals of ρ_0 , ρ_0^2 , and ρ_1^2 are finite: on the end their orders are respectively r^{-4} , r^{-8} , and $r^{-4\gamma}$, with $4\gamma > 3$. Integrating leaves at most $C(1+n)^C(\ell + \ell^2/\tau)$, after absorption.

Substitute both bounds in (8.4). Positive fractions of the three bulk terms $\delta_0\mathcal{T}$, ρ , $\delta_0\tau ua\sigma$ remain. Dropping them proves the claim. Finally $\ell^2/\tau = \ell^{1/2}$, so the error tends to zero exponentially against every polynomial loss. $\square$

8.3. The Riemannian horizon and the inequality. The deformed metric has nonnegative scalar curvature, strictly negative inner mean curvature toward the end, and the required asymptotic decay. We replace its inner boundary by an outer area-minimizing minimal enclosure, then compare its area with the original enclosing infimum.

Theorem 8.4 (Reduced energy inequality). *The reduced data satisfy*

$$E \geq \sqrt{\frac{A_*}{16\pi}}.$$

Consequently Theorem 2.2 holds in its entire stated exterior class.

Proof. For each fixed sufficiently large n , minimize $\hat{g}$ -area among filled cuts enclosing B , using a large outer sphere as an auxiliary barrier. The area-compactness and obstacle arguments of Section 3 apply to this smooth AF metric. A fixed enclosing competitor bounds the area, and local perimeter density estimates prevent minimizers from escaping to infinity. The positive mean curvature of sufficiently large outer spheres excludes contact there, as in Lemma 3.3. For the inner barrier, let U_s be a short outward normal collar of the filled obstacle and Y its outward unit foliation field. Strict negativity gives $\operatorname{div}_{\hat{g}} Y < 0$ throughout the collar. If a perimeter minimizer E omitted a positive-volume portion $D = U_s \setminus E$, Gauss–Green and $|Y|_{\hat{g}} \leq 1$ would give

$$P_{\hat{g}}(E \cup U_s) - P_{\hat{g}}(E) \leq \int_D \operatorname{div}_{\hat{g}} Y \, dV_{\hat{g}} < 0.$$

Thus E contains a whole inner collar; obstacle regularity upgrades this inclusion to its regular representative. Its free boundary is disjoint from the obstacle. The resulting cut Γ_n is compact, smooth, and minimal in dimension three; it may be disconnected. It is outer area-minimizing by its minimizing property. Equivalently one may first use the ordinary trapped-region theorem for $(\hat{g}, 0)$ and then take the outer minimizing enclosure. The boundary version of the Riemannian Penrose theorem applies to the exterior of Γ_n , by Lemma 8.2 and [15, Theorem 19].

For every two-plane the restriction of $\bar{g} = g + l^2 df^2$ dominates that of g . The floor therefore gives

$$|\Gamma_n|_{\hat{g}} \geq e^{-4\epsilon} |\Gamma_n|_g \geq e^{-4\epsilon} A_*.$$

The last inequality uses that B , and hence Γ_n , cuts off the original filled obstacle. Lemmas 8.2 and 8.3 now give

$$E + \frac{\varepsilon_n}{8\pi} \geq E_{\hat{g}} \geq \sqrt{\frac{|\Gamma_n|_{\hat{g}}}{16\pi}} \geq e^{-2\epsilon} \sqrt{\frac{A_*}{16\pi}}.$$

Let $n \rightarrow \infty$ with ϵ fixed, and then $\epsilon \downarrow 0$. This proves the reduced inequality.

Finally apply Proposition 3.10 to any exterior in Theorem 2.2. The reduced energies converge to its invariant ADM mass and the enclosing infima converge to its original enclosing infimum. Applying the reduced bound to each member of that sequence and passing to the limit gives Equation (2.8). $\square$

9. COMPLETE DATA AND MULTIPLE ENDS

9.1. The complete initial-data formulation. Let (M, g, K) be a connected orientable smooth initial data set without boundary, with g complete. Suppose that outside a compact set it has finitely many asymptotically flat ends. On each end assume

$$g - \delta = O_2(r^{-q}), \quad K = O_1(r^{-1-q}), \quad q > \frac{1}{2},$$

and finite ADM energy and momentum limits as in Equations (2.4) and (2.5). The exponents may differ between ends; their minimum is still larger than $1/2$. Assume that the constraint densities (2.1) satisfy $\mu \geq |J|_g$ and that μ and $|J|_g$ are integrable on M .

Definition 9.1 (Admissible boundary and minimum enclosing area). Fix an asymptotically flat end of M . An admissible boundary is a nonempty compact smooth embedded two-sided surface $S = \partial D^+$, possibly disconnected, where D^+ is a smooth open region containing the entire

sufficiently distant part of the chosen end. The region D^+ need not be connected and may contain other ends. For the normal ν pointing into D^+ , require

$$H_S + \text{tr}_S K \leq 0.$$

Define

$$(9.1) \quad A_{\min}(S) = \inf \left\{ \text{Area}_g(\partial D'^+) : \begin{array}{l} D'^+ \subset D^+, \quad D'^+ \text{ contains the chosen end at infinity,} \\ \partial D'^+ \text{ is a compact smooth embedded boundary} \end{array} \right\}.$$

Competitors may be disconnected or coincide with S . They need not be trapped or minimal. No condition on the inward null expansion is imposed.

Theorem 9.2 (Spacetime Penrose inequality for complete initial data). *For the complete initial data just specified, every chosen end and every admissible boundary satisfy*

$$(9.2) \quad E > |P|_\delta, \quad \sqrt{E^2 - |P|_\delta^2} \geq \sqrt{\frac{A_{\min}(S)}{16\pi}}.$$

Here (E, P) is the ADM vector of the chosen end. The assertion allows finitely many ends and disconnected admissible boundaries.

9.2. Removing the timelike hypothesis and recovering the global statement. The timelike exterior inequality already proved also rules out a non-timelike ADM vector for an exterior with weakly future outer trapped boundary. This observation then gives the complete initial-data formulation by restricting to a suitable one-ended exterior.

Lemma 9.3 (Positive enclosing area). *Every exterior in Definition 2.1 satisfies $0 < a_g(B) < \infty$, where B is its boundary.*

Proof. A sufficiently large coordinate sphere gives finiteness. For positivity, choose a small disk in one component of B and a smooth proper embedded ray from its center to the end, normal to B initially and straight in the asymptotic coordinates outside a compact set. The ray has a tubular neighborhood with disk cross-sections; its coordinate width can be fixed and positive on the straight tail. In product coordinates on this tube, pull back a smooth two-form compactly supported in the transverse disk and with integral one. Extend it by zero through the lateral boundary of the tube. The resulting two-form ω is smooth and closed on the exterior, including its boundary, and has bounded pointwise norm. The bound follows from compactness on the initial part and asymptotic flatness on the tail.

Orient the transverse disk so that the flux of ω through a large coordinate sphere is one. A smooth enclosing cut in the interior and a sufficiently large sphere bound a compact region disjoint from B . Stokes' theorem therefore gives $\int_\Gamma \omega = 1$. The same identity holds for coincident cuts by smooth outward approximation. Consequently

$$1 = \left| \int_\Gamma \omega \right| \leq \|\omega\|_{L^\infty(g)} \text{Area}_g(\Gamma).$$

Taking the infimum proves positivity. If the filled perimeter closure of the cut class is used, its infimum agrees with the smooth infimum by Definition 2.1 and the approximation established in the exterior reduction. $\square$

Corollary 9.4 (The exterior inequality without a timelike assumption). *Let (N, g, K) satisfy all the hypotheses of Theorem 2.2 except the assumption $E > |P|_\delta$. Then $E > |P|_\delta$ and the conclusion of that theorem holds. No extension of the data across the compact boundary B is required.*

Proof. Use the conformal family of Lemma 3.2, which requires no sign assumption on the ADM vector. For every finite $s \geq 0$, its data (g_s, K_s) satisfy all exterior hypotheses except possibly timelikeness, with

$$E_s = E + sA_\phi, \quad P_s = P, \quad A_\phi > 0.$$

Equation (3.10) and Lemma 9.3 give $a_{g_s}(B) \geq a_g(B) > 0$. Suppose that $E \leq |P|_\delta$ and put $s_0 = (|P|_\delta - E)/A_\phi \geq 0$. For every $s > s_0$, Equation (3.9) places (N, g_s, K_s) in the timelike class of the already proved Theorem 2.2. That theorem and the preceding area bound imply

$$\sqrt{(E + sA_\phi)^2 - |P|_\delta^2} \geq \sqrt{\frac{a_g(B)}{16\pi}} > 0.$$

Letting $s \downarrow s_0$ gives a contradiction. Therefore $E > |P|_\delta$, and applying Theorem 2.2 to the original data proves the claimed inequality. $\square$

Proof of Theorem 9.2. Let D_0 be the connected component of D^+ containing the distant chosen end. Its boundary is a union of components of S and is nonempty: otherwise D_0 would be both open and closed in the connected manifold M , hence equal to M , contrary to $S \neq \emptyset$. Put $X_0 = \overline{D_0}$, including its smooth compact boundary. Because S is compact, each sufficiently distant end of M lies entirely in D_0 or entirely outside it. Outside a compact set, X_0 is precisely the union of the original distant ends contained in D_0 . In particular, X_0 has finitely many ends, all asymptotically flat.

For every end of X_0 other than the chosen one, choose a sufficiently distant open coordinate tail, and let N be X_0 with all these open tails removed. These tails can be chosen disjoint and outside S . Removing them preserves connectedness: an excursion of a path into a removed tail can be replaced by a path on its connected coordinate sphere. The region N is a smooth manifold with nonempty compact boundary B , consisting of ∂D_0 and the added coordinate spheres, and its only end is the chosen one.

The region is closed in the complete manifold (M, g) and has smooth boundary. It is complete for its intrinsic distance with the boundary included: an intrinsic Cauchy sequence is ambient Cauchy, its ambient limit lies in N , and smooth boundary charts compare the intrinsic distance locally with Euclidean distance on a half-ball. The restricted constraints, integrability, and decay satisfy the exterior hypotheses. The chosen ADM vector is still (E, P) .

On the inherited part of B , the normal into N agrees with the normal into D^+ , so the assumed expansion is nonpositive. On an added sphere in another end, the normal into N points toward decreasing radius. Consequently

$$H_B = -\frac{2}{r} + O(r^{-1-q}), \quad \text{tr}_B K = O(r^{-1-q}),$$

and its future outward expansion is strictly negative when its radius is sufficiently large. Corollary 9.4 therefore applies to (N, g, K) and gives

$$(9.3) \quad E > |P|_\delta, \quad \sqrt{E^2 - |P|_\delta^2} \geq \sqrt{\frac{a_g(B)}{16\pi}}.$$

Finally,

$$(9.4) \quad a_g(B) \geq A_{\min}(S).$$

To see this, first take an enclosing cut Γ lying in the interior of N . The component on its distant chosen-end side is an open region $D'^+ \subset D_0 \subset D^+$, with compact smooth boundary consisting of components of Γ . It is a competitor in Equation (9.1); its area is at most that of Γ . If a cut coincides with any part of B , approximate it by smooth cuts pushed into the interior of N , with areas converging to its area. This also handles cuts coinciding with a newly introduced sphere.

Taking the infimum proves Equation (9.4). Other components of D^+ cause no difficulty, since the original competitor class permits the chosen-end component alone. Combining Equations (9.3) and (9.4) proves the theorem. $\square$

The conformal family above is used only to establish the numerical inequality and to exclude a non-timelike ADM vector. The equality argument is carried out on the original exterior under its separately stated hypotheses; no equality conclusion is passed through this family or through the truncation of other ends.

Part 2. Equality and Schwarzschild rigidity

10. EQUALITY AND THE NUMBER OF BOUNDARY COMPONENTS

We now classify equality on the original initial data. The numerical inequality has already been proved for a larger boundary class; that larger class is necessary for the variations below. We retain the constraint, ADM, and spacetime conventions of Equations (2.1)–(2.7). In particular J is a covector, the dominant energy condition is $\mu \geq |J|_g$, and $K = \frac{1}{2}\mathcal{L}_n g$ for the future normal. All data remain smooth; the asymptotic hypotheses require only the two metric derivatives and one K derivative in Equation (2.3), with $q > 1/2$.

The numerical input and the cut convention. Theorem 2.2 applies to a connected orientable smooth three-dimensional exterior, complete as a metric space with its nonempty compact smooth boundary included, and with exactly one AF end. For smooth data with $g - \delta = O_2(r^{-q})$, $K = O_1(r^{-1-q})$, $q > 1/2$, integrable μ and $|J|_g$, existing finite ADM limits, $\mu \geq |J|_g$, $E > |P|_\delta$, and $\theta_+(\partial\Omega) \leq 0$ for the normal into the exterior, it gives

$$\sqrt{E^2 - |P|_\delta^2} \geq \sqrt{\frac{a_g(\partial\Omega)}{16\pi}}.$$

It requires no fill-in and no outermostness, outer area minimization, or connectedness of the boundary. In the variational proof we apply this established theorem to nearby data on the same one-sided exterior.

The infimum is over the enclosing cuts of Definition 2.1: each cut separates the entire inner boundary from the distant part of the unique end, may be disconnected, and may coincide with the original boundary. The normal points toward that end, and coincident frontier area is counted. Enclosing cuts and minimizing sets are taken with bounded complementary pockets filled. During a perimeter argument we may enlarge the competitor class to all relatively compact finite-perimeter sets containing a fixed smooth inner obstacle. This class is closed under unions and intersections; its minimizers fill bounded complementary pockets, because filling such a pocket removes perimeter. Its infimum agrees with the smooth-cut infimum by the outward approximation proved in Section 11.1.

Definition 10.1 (Partial-coincidence outermostness). Let the boundary of a one-ended exterior be $S = \bigsqcup_{i=1}^\ell S_i$, with $1 \leq \ell < \infty$ and each S_i connected. The boundary is outermost in the partial-coincidence sense if no enclosing cut $\Gamma \neq S$ has all of the following properties: every connected component of Γ is either an entire original component S_i or is wholly contained in the open exterior; at least one component lies in the open exterior; and $\theta_+(\Gamma) \leq 0$ everywhere for the normal toward the end.

This condition includes comparisons that retain some original components and replace others by interior components. It will be used only after contact regularity has shown that the active minimizing cuts have exactly this form. We make no claim here for the weaker condition excluding only wholly interior enclosing cuts.

Theorem 10.2 (Finite-component exterior equality rigidity). *Let (M, g, K) be a smooth connected orientable complete initial data set without boundary, consisting of a compact core and finitely many asymptotically flat ends. On each end assume $g - \delta = O_2(r^{-q})$, $K = O_1(r^{-1-q})$, with $q > 1/2$, and existing finite ADM flux limits. Assume that μ and $|J|_g$ are integrable and that $\mu \geq |J|_g$ everywhere.*

Choose an end and a finite nonempty disjoint union

$$S = \bigsqcup_{i=1}^{\ell} S_i, \quad 1 \leq \ell < \infty,$$

of compact connected smooth embedded two-sided surfaces without boundary, separating that end from a complementary region. Let M_{ext} be the closure of the connected side toward the chosen end. Suppose that $\partial M_{\text{ext}} = S$, that M_{ext} has exactly this one end, and that, for the normal pointing into M_{ext} :

- (i) *every S_i is a future MOTS, so $H_{S_i} + \text{tr}_{S_i} K = 0$;*
- (ii) *S is outermost in the partial-coincidence sense of Definition 10.1;*
- (iii) *every enclosing cut in the closed exterior has area at least $A = |S|_g = \sum_{i=1}^{\ell} |S_i|_g > 0$.*

Let (E, P) be the ADM vector of the chosen end. If

$$(10.1) \quad E > |P|_{\delta}, \quad m = \sqrt{E^2 - |P|_{\delta}^2} = \sqrt{\frac{A}{16\pi}},$$

then $\ell = 1$ and S is diffeomorphic to a sphere. There is a proper smooth spacelike embedding of the entire original M_{ext} into the maximally extended Schwarzschild spacetime of mass m , inducing g and the given K for a consistent future unit normal. The chosen end approaches the corresponding spatial infinity. The image of S is a smooth cross-section of the corresponding future horizon or the bifurcation sphere. The embedding and its future normal are smooth up to S , and the image extends as a smooth spacelike hypersurface through that boundary.

Corollary 10.3 (Strictness for disconnected boundary). *Under the hypotheses of Theorem 10.2, retain $E > |P|_{\delta}$ but do not assume its equality in mass and area. If $\ell \geq 2$, then*

$$\sqrt{E^2 - |P|_{\delta}^2} > \sqrt{\frac{A}{16\pi}}.$$

Proof. The original exterior is complete as a metric space with boundary, as follows. An intrinsic Cauchy sequence is Cauchy in the complete ambient manifold, and its ambient limit remains in the closed exterior. A smooth interior or boundary half-ball chart then gives convergence in the intrinsic distance. Its compact boundary and unique end place it in Definition 2.1. Since S is an enclosing cut and every enclosing cut has area at least A , its enclosing infimum is A . Theorem 2.2 gives the non-strict inequality. Equality would imply $\ell = 1$ by Theorem 10.2, contradicting $\ell \geq 2$. $\square$

The proof of Theorem 10.2 occupies the next two sections. Each S_i is orientable because it is two-sided in the orientable manifold M , but no genus is prescribed. The proof first constructs a causal stationary field on the original exterior and concentrates its area multiplier on the entire boundary. A single complete conformal-double argument then gives positive intrinsic curvature and the same area A on every component, forcing $\ell = 1$. Schwarzschild reconstruction follows with connectedness established by the argument. The conclusion concerns only the chosen exterior; it allows a nonconstant time graph and nonzero momentum in the original asymptotic frame.

11. EQUALITY, FIRST VARIATIONS, AND A CAUSAL STATIONARY FIELD

We return to the original exterior in Theorem 10.2. Write its nonempty compact boundary as

$$S = \bigsqcup_{i=1}^{\ell} S_i, \quad 1 \leq \ell < \infty,$$

where each S_i is a closed connected orientable smooth surface. No genus is prescribed. Each component is a future MOTS, and S is outer area-minimizing. We use the precise outermostness condition of that theorem: there is no enclosing cut with at least one component in $\text{Int } M_{\text{ext}}$, every other component either in that interior or equal to an S_i , and $\theta_+ \leq 0$ on every component. Thus the condition permits comparison cuts that retain some original boundary components. All the arguments in this section take place on this original exterior. We use the exterior inequality of Theorem 2.2, already proved in the preceding sections, for nearby data on the same manifold with boundary. Those nearby data need not have an outermost or outer area-minimizing boundary. This distinction is essential for the variations below.

Write

$$A = |S|_g = 16\pi m^2, \quad b^0 = \frac{E}{m}, \quad b^i = -\frac{P_i}{m}, \quad c = \frac{1}{2m}, \quad \tau = \text{tr}_g K.$$

Thus $A = \sum_{i=1}^{\ell} |S_i|_g$ counts all boundary components. The vector (b^0, b^i) is future unit timelike. For a nearby ADM vector define the fixed linear functional

$$(11.1) \quad \mathcal{E} = b^0 E_{\text{new}} + b^i P_{\text{new},i}.$$

The reverse Lorentzian Cauchy–Schwarz inequality gives $\mathcal{E} \geq \sqrt{E_{\text{new}}^2 - |P_{\text{new}}|^2}$ when the nearby ADM vector is future timelike. At the original data both sides equal m and have the same first derivative. Moreover,

$$(11.2) \quad 16\pi \frac{d}{da} \sqrt{\frac{a}{16\pi}} \Big|_{a=A} = c.$$

We shall prove the following statement. The notation O_2 for the vector field uses the original end coordinates, component by component.

Proposition 11.1 (Causal adjoint at equality). *There are a function u and a vector field X , smooth on the one-sided manifold M_{ext} , with the following properties.*

(1) $u \geq |X|_g$ everywhere and $u > 0$ in $\text{Int } M_{\text{ext}}$. For every q' with $1/2 < q' < \min\{q, 1\}$,

$$(11.3) \quad u = b^0 + O_2(r^{-q'}), \quad X = b^i \partial_i + O_2(r^{-q'}).$$

(2) In the interior,

$$(11.4) \quad \text{sym } \nabla X = -uK,$$

$$(11.5) \quad \Delta u = -\text{div}(K(X, \cdot)^{\sharp}),$$

$$(11.6) \quad \text{Hess } u = u(\text{Ric}_g + \tau K - 2K^2) - \mathcal{L}_X K + 8\pi X_{(i} J_{j)}.$$

Here $(K^2)_{ij} = K_i^k K_{kj}$, and parentheses denote averaged symmetrization. These equations extend smoothly to S .

(3) On $\mathbb{R} \times \text{Int } M_{\text{ext}}$ the Lorentzian metric

$$(11.7) \quad \mathbf{g} = -u^2 dt^2 + g_{ij} (dx^i + X^i dt)(dx^j + X^j dt)$$

has stationary Killing field $\xi = \partial_t$ and induces (g, K) on $t = 0$, with future normal $n = u^{-1}(\partial_t - X)$ and the second fundamental form convention $K = \frac{1}{2}\mathcal{L}_n g$. Its Einstein

tensor satisfies

$$(11.8) \quad u\mathbf{G}_{ij} = -8\pi X_{(i}J_{j)}, \quad u\mu + J(X) = 0.$$

Set $N = u^2 - |X|^2$. Then $\mu N = 0$, and, throughout this stationary spacetime,

$$(11.9) \quad \mathbf{G} = \frac{8\pi\mu}{u^2} \xi^b \otimes \xi_b, \quad \text{Scal}_{\mathbf{g}} = 0, \quad \text{Ric}_{\mathbf{g}}(\xi, \cdot) = 0.$$

In particular, the spacetime is vacuum on $\{N > 0\}$.

(4) With ν pointing into the exterior,

$$(11.10) \quad X = u\nu, \quad \partial_\nu u + K(X, \nu) = \kappa, \quad \kappa = \frac{1}{4m} \quad \text{on } S.$$

For each $i = 1, \dots, \ell$, either $u > 0$ everywhere on S_i , or $u = 0$ everywhere on S_i . This alternative is made separately on each component; the constant κ is common to all components.

Equation (11.9) deliberately allows nonzero matter where $N = 0$. This section does not infer vacuum from the existence of a causal stationary field alone. The treatment of that null region belongs to Section 12.

11.1. The derivative of the enclosing area. We use filled enclosing sets, with coincidence with any component of S allowed. To make the perimeter convention explicit for arbitrary genus, attach a compact orientable handlebody to the inner side of each S_i , identifying its boundary smoothly with S_i . Every closed connected orientable surface bounds such a handlebody. The resulting manifold has no boundary and retains the original asymptotic end. Extend g smoothly and positively through the collars and over these caps, by extending its collar coefficients and then joining to a positive metric on each cap by a partition of unity. The finite union of the caps is a compact smooth obstacle with boundary S .

Minimize the full perimeter in this capped manifold among relatively compact finite-perimeter sets containing the obstacle almost everywhere. No condition on complementary pockets is imposed on these competitors, so this class is closed under unions and intersections. A minimizing set has no bounded complementary pocket: filling one removes its frontier and decreases perimeter. Full perimeter counts the frontier coinciding with S , in agreement with Definition 2.1. Every contributing boundary lies in the original closed exterior, so neither the choice of cap nor its interior metric changes its area. Only the metric is extended; the constraints and all variations below remain on M_{ext} . Let $\mathcal{T}$ be the collection of boundaries attaining the enclosing infimum for the original metric. Each boundary in $\mathcal{T}$ has area A ; S itself is one of them. Perimeters and tangent-plane measures count every boundary component.

Lemma 11.2 (Compact minimizing cuts and the area envelope). *Let g_s be a smooth path of metrics through g whose first and second parameter derivatives are uniformly bounded relative to g , and whose end geometry has uniformly bounded scaled first derivatives for s near zero. Put $h = \dot{g}_0$, and let $a(s)$ be its filled enclosing infimum. Then*

$$(11.11) \quad a'(0+) = \min_{T \in \mathcal{T}} a'_T(h), \quad a'_T(h) = \frac{1}{2} \int_T \text{tr}_T h \, dA_g.$$

The space $\mathcal{T}$ is compact for the convergence of its sets in local L^1 and of its unoriented tangent-plane area measures. The function $T \mapsto a'_T(h)$ is continuous for that topology. Off the obstacle, two cuts in $\mathcal{T}$ have the same tangent plane at each point where they meet.

Proof. A fixed enclosing competitor bounds all minimizing perimeters from above. A minimizing boundary that reaches arbitrarily large end radius has a point with a ball of radius comparable to that radius which is disjoint from the obstacle. The scaled metric bounds and the interior

perimeter density estimate give a lower area bound equal to a positive constant times the square of that radius. This contradicts the fixed upper bound. Thus all minimizing boundaries for the paths under consideration stay in a fixed compact set; this is also the compactness argument in Lemma 3.3. One may minimize on a large compact truncation, whose sphere is a strict mean-curvature barrier, and then remove the truncation. Perimeter compactness gives a limiting filled set. The fixed-obstacle condition is closed under this convergence, and lower semicontinuity gives a minimizing boundary.

The regularity input is the smooth-obstacle perimeter theorem: in ambient dimension three, a perimeter minimizer outside a C^2 obstacle is $C^{1,1}$ and is smooth minimal away from contact [31, Regularity Theorem 1.3(iii)]. These minimizers realize the smooth-cut infimum as well. Every smooth filled enclosing cut is an admissible finite-perimeter competitor. Conversely, flow a regular minimizing boundary for a small positive time by a smooth vector field pointing strictly outward along S and supported in its collars. The moved filled set contains the obstacle in its interior. A sufficiently close smooth C^1 approximation of that compact embedded boundary still encloses the obstacle, and its area tends to the original perimeter as the flow time and approximation error tend to zero. Filling any bounded complementary pockets only decreases area. The resulting smooth cuts prove equality of the two infima. This construction applies to each nearby metric as well.

The local obstacle estimates and the interior density estimates apply uniformly on the fixed compact region. For a convergent sequence of minimizing sets, lower semicontinuity and the upper bound from a fixed minimizing competitor give convergence of the perimeters. Express the normal vector measures in a fixed smooth local g -orthonormal frame. These are smooth matrix multiples of the distributional derivatives of the set indicators, so they converge weakly, and their Euclidean total variations are precisely the convergent g -perimeters. Strict convergence of these vector measures gives convergence of the integrals of every continuous function of position and unit normal; in particular it gives convergence for every continuous function of the unoriented tangent plane. This is the strict-measure continuity theorem, applied locally and then with a partition of unity [5]. The same argument works when the metrics vary: their uniform convergence converts convergence of the varying perimeters into strict convergence for the fixed metric.

For any rectifiable cut of bounded area the pointwise area-element expansion gives, uniformly over those cuts,

$$(11.12) \quad |T|_{g_s} = |T|_g + sa'_T(h) + O(s^2)|T|_g.$$

Testing a fixed member of $\mathcal{T}$ gives the upper bound in Equation (11.11). For the opposite bound, take minimizers T_s for g_s . Any sequence $s \downarrow 0$ has a subsequence converging as above to some $T \in \mathcal{T}$. Since $|T_s|_g \geq A$, Equation (11.12) gives

$$\frac{a(s) - A}{s} \geq a'_{T_s}(h) + O(s) \longrightarrow a'_T(h) \geq \min_{T' \in \mathcal{T}} a'_{T'}(h).$$

This proves the derivative formula and the asserted continuity.

Finally, if D_1, D_2 are two minimizing filled sets, then their union and intersection are admissible and perimeter submodularity gives

$$P(D_1 \cup D_2) + P(D_1 \cap D_2) \leq P(D_1) + P(D_2) = 2A.$$

Each term on the left is at least A , so both are minimizers. Different tangent planes at an intersection off the obstacle would give a corner in one of these two minimizing boundaries, contradicting their interior regularity. Thus the tangent planes agree there. $\square$

For later use, the common plane is a continuous field on the union of these cuts, locally away from S . Indeed, take points x_j on cuts T_j converging to a point x away from S . Compactness

gives a limiting cut. The interior density estimate puts x on that cut, and strict perimeter convergence to its smooth boundary gives local tangent convergence by the interior regularity theorem. The common plane property makes the limit independent of the selected cut.

11.2. A positive separator for the active constraints. Use the fixed unit ball bundle

$$\mathcal{B} = \{(x, v) : x \in M_{\text{ext}}, |v|_g \leq 1\}, \quad C(x, v) = 16\pi(\mu(x) + J_x(v)),$$

and its closed active subset $\mathcal{A} = \{C = 0\}$. For a varying metric transport v isometrically by the symmetric positive square root: if $g_s(\cdot, \cdot) = g(A_s \cdot, \cdot)$, put $v_s = A_s^{-1/2}v$. Thus

$$(11.13) \quad \dot{v}_0 = -\frac{1}{2}h^\sharp v, \quad |v_s|_{g_s} = |v|_g.$$

The linear variation C'_H below includes this argument variation.

Fix the strict conformal direction from Lemma 3.2, and write it as

$$Q = (4\phi g, 2\phi K).$$

Here $\partial_\nu \phi = -1$, $\phi = O_2(r^{-1})$ has a finite flux, and for some $0 < \delta < \min\{q, 1\}$ and a smooth positive weight $w = r^{-3-\delta}$ on the end,

$$(11.14) \quad -\Delta\phi - |K||\nabla\phi| \geq w, \quad \frac{-\Delta\phi}{w} \rightarrow 1.$$

Decrease δ if necessary in using that lemma. Let $\mathcal{V}$ be the real linear space generated by Q , all smooth compactly supported variations (h, p) up to S , and the following four smooth variations cut off to vanish on a fixed large compact set:

$$(11.15) \quad h_{ij}^{(0)} = \frac{\delta_{ij}}{r}, \quad p^{(0)} = 0; \quad h^{(a)} = 0, \quad p^{(a)} = B(e_a),$$

where, on the end,

$$(11.16) \quad B(p)_{ij} = r^{-2}(p_i n_j + p_j n_i - (\delta_{ij} - n_i n_j)p \cdot n), \quad n = x/r.$$

These tensors are Euclidean tracefree and divergence-free. The scalar-curvature linearization at the Euclidean metric annihilates δ/r outside the cutoff. Consequently the non-strict generators have

$$(11.17) \quad C'_H = O(r^{-3-q}) = o(w) \quad \text{on the end, uniformly for } |v| \leq 1.$$

The same estimate includes the frame term in Equation (11.13). The momentum prototypes have independent momentum fluxes and the scalar prototype has nonzero energy flux.

On an active ray the background conformal scaling term vanishes, so the exact conformal formulas give

$$(11.18) \quad C'_Q = -8\Delta\phi + 8K(v, \nabla\phi), \quad \frac{C'_Q}{w} \rightarrow 8, \quad -\theta'_Q = 4 \quad \text{on } S.$$

In particular, every $H \in \mathcal{V}$ has a well-defined coefficient $a(H)$ of Q , characterized equivalently by $C'_H/w \rightarrow 8a(H)$ on the end. This also shows that the coefficient is independent of its representation by generators.

Lemma 11.3 (Multiplier identity). *There are a probability measure ω on $\mathcal{T}$, a nonnegative finite measure η on S , a nonnegative locally finite measure Λ on $\mathcal{A}$, and a number $z \geq 0$ such that, for every $H \in \mathcal{V}$,*

$$(11.19) \quad 16\pi\mathcal{E}'(H) - c \int_{\mathcal{T}} a'_T(h) d\omega(T) = \langle C'_H, \Lambda \rangle - \int_S \theta'_H d\eta + za(H).$$

The pairing is well defined on $\mathcal{V}$. The last term vanishes on compact variations and on the four prototypes.

Proof. Adjoin an end point to $\mathcal{A}$, collapsing every ray that escapes spatial compact sets to that point; if $\mathcal{A}$ is compact, take an additional isolated end point. The bounded fibers make the result a compact space. Equations (11.17) and (11.18) show that C'_H/w extends continuously to it, with value $8a(H)$. Take the disjoint compact union of this space, S , and $\mathcal{T}$, and associate to H the continuous function whose values on the three pieces are

$$(11.20) \quad \frac{C'_H}{w}, \quad -\theta'_H, \quad ca'_T(h) - 16\pi\mathcal{E}'(H),$$

respectively. Smooth dependence of the boundary expansion and Lemma 11.2 give the remaining continuity.

We first prove that no image of this linear map is strictly positive at every point of the compact test space. Suppose otherwise, and write $H = aQ + H_0$. Positivity at the end point gives $a > 0$. Realize its tangent by the actual path

$$(11.21) \quad g_s = e^{4as\phi}(g + sh_0), \quad K_s = e^{2as\phi}(K + sp_0).$$

The metric remains positive and uniformly comparable to g for small $|s|$. The varied data retain the required asymptotic decay with exponent $\min\{q, 1\} > 1/2$, and their charge variations have finite limits. The conformal factor has its stated flux. Thus the path has differentiable ADM fluxes.

Here are the estimates which ensure nonlinear feasibility. On the end the tensors in H_0 are $O_2(r^{-1})$ and $O_1(r^{-2})$, and their far-end Euclidean linear constraints vanish. Terms in their linear constraint variation containing a background error cost $O(sr^{-3-q})$, while new quadratic terms cost $O(s^2r^{-4})$. Changing the orthonormal identification has the same bound: the background momentum is $O(r^{-2-q})$, and the metric variation is $O(r^{-1})$. Applying the exact conformal law to the intermediate data $(g + sh_0, K + sp_0)$, and keeping the nonnegative background term with its positive scaling factor, therefore gives, uniformly over the full unit ball of test vectors,

$$(11.22) \quad e^{4as\phi}C_s = C + s(8aw + o(w)) + O(s^2w) \quad \text{on the end.}$$

The $o(w)$ is uniform as $r \rightarrow \infty$ at fixed H , and the quadratic bound is uniform for small s . The gradient square in the conformal formula is $O(r^{-4}) = O(w)$ because $\delta < 1$. One may use the composition of the two natural isometries of the unit balls in deriving this estimate; it tests the same DEC. At active rays its derivative agrees with Equation (11.13), since the difference of two isometric identifications is tangent to the unit sphere and is annihilated by J at an active ray.

It follows from Equation (11.22) that the DEC holds on a fixed far end for all sufficiently small positive s . On the remaining compact ball bundle, the derivative is strictly positive on the closed active set. A neighborhood of that set has the same positive derivative with a fixed smaller margin. On its compact complement the original C has a positive minimum. Taylor's formula now gives the DEC throughout the compact region as well. The strict inequality $-\theta'_H > 0$ on compact S gives $\theta_{+,s} < 0$ for small positive s . The new sources are integrable: outside a compact set their changes, after the harmless positive rescaling of the original sources, have the integrable bounds just displayed. The ADM vector stays future timelike. Hence these are data to which Theorem 2.2 applies.

Finally, strict positivity on the compact cut space gives

$$c \min_{T \in \mathcal{T}} a'_T(h) - 16\pi\mathcal{E}'(H) > 0.$$

By Lemma 11.2 and Equation (11.2), this says that the right derivative of $16\pi(\mathcal{E}(s) - \sqrt{a(s)/(16\pi)})$ is strictly negative. The expression is zero at $s = 0$, whereas the supporting-energy inequality and Theorem 2.2 make it nonnegative for small positive s . This is a contradiction.

The image of the linear test map is therefore disjoint from the open cone of strictly positive continuous functions. The separating form of the Hahn–Banach theorem gives a nonzero continuous

linear functional, nonnegative on nonnegative functions, which annihilates the image. The Riesz representation theorem represents it by nonnegative finite measures on the three compact pieces. Its mass on the objective piece $\mathcal{T}$ cannot be zero: otherwise its value on the image of Q , which is strictly positive on both remaining pieces including their end point, would be positive. Normalize this objective mass to one. On the finite active rays divide the corresponding measure by w and call the result Λ . It is locally finite; the original finite measure makes its pairing with C'_H finite. Put the end-point mass, multiplied by eight, into z . Rearranging the annihilation identity gives Equation (11.19) with the stated signs. $\square$

Define the scalar and vector moments of Λ with the fixed volume-density convention by

$$(11.23) \quad \langle u, f \rangle = \int f(x) d\Lambda(x, v), \quad \langle X, \alpha \rangle = \int \alpha_x(v) d\Lambda(x, v).$$

Initially these are measures, including possible measures on S . Their positivity and the support of Λ give, distributionally,

$$(11.24) \quad u \geq |X|_g, \quad u\mu + J(X) = 0.$$

The first statement means domination of the total variation of the vector measure by the scalar measure; in particular it can be tested against arbitrary continuous unit covector fields.

11.3. Removing the interior area multipliers.

Lemma 11.4 (Normal slice tests). *For every smooth s compactly supported in $\text{Int } M_{\text{ext}}$ there are compact variations H_ϵ with metric component $h_\epsilon = 2sK$ such that*

$$(11.25) \quad C'_{H_\epsilon} \longrightarrow 0 \quad \text{uniformly on the active rays over compact sets.}$$

Their supports lie in one fixed compact set.

Proof. Choose a compact neighborhood of the support of s . In a Gaussian time collar put

$$\mathbf{g}_\epsilon = -dt^2 + g_\epsilon(t), \quad g_\epsilon(0) = g, \quad \partial_t g_\epsilon(0) = 2K.$$

Choose the second time derivative so that its spatial Einstein components at time zero are

$$(11.26) \quad (\mathbf{G}_\epsilon)_{ij} = 8\pi(D_\epsilon)_{ij}, \quad D_\epsilon = \frac{J \otimes J}{\mu + \epsilon}.$$

This is a free choice of jets, not a choice of evolution satisfying an energy condition. To see its solvability, the terms in the spatial Einstein tensor containing $\partial_t K$ are its trace reversal, with a fixed nonzero overall sign. On symmetric tensors in dimension three this map is invertible: its eigenvalue is one on the tracefree part and minus two on the pure-trace part. All remaining terms are determined by g, K and their spatial derivatives. Thus smooth second jets realizing Equation (11.26) exist; a quadratic extension with a time cutoff realizes them by a smooth Lorentzian metric for sufficiently small time. The constraints give at time zero

$$(11.27) \quad \mathbf{G}_\epsilon(n, n) = 8\pi\mu, \quad \mathbf{G}_\epsilon(n, e_i) = 8\pi J_i,$$

where $n = \partial_t$ and e_i is a spatial orthonormal frame.

The regularization has the precise compact convergence needed below. Define $D = J \otimes J/\mu$ where $\mu > 0$, and $D = 0$ at $\mu = 0$. The DEC implies

$$(11.28) \quad |D_\epsilon| \leq \mu, \quad |\nabla D_\epsilon| \leq 2|\nabla J| + |\nabla \mu|.$$

The second inequality follows by differentiating the quotient and using $|J| \leq \mu$ in each term. Smooth nonnegative μ has $\nabla \mu = 0$ at its zero set. There also $\nabla J = 0$: in a smooth frame $|J_i| \leq \mu$, so each directional derivative of J_i vanishes. It follows first that D is differentiable with zero derivative at that set, and then from Equation (11.28) that this derivative is continuous there. On a compact set, split into $\mu \geq \delta$ and $\mu < \delta$. On the first part the quotient and its derivatives

converge uniformly as $\epsilon \rightarrow 0$. On the second part the bound in Equation (11.28) is uniformly small as $\delta \downarrow 0$, by compactness and continuity at the zero set. Therefore

$$(11.29) \quad D_\epsilon \longrightarrow D \quad \text{in } C^1 \text{ on every compact set.}$$

Vary the slice by the graphs $t = as(x)$ at $a = 0$, using its induced metric and second fundamental form. Their variation is compactly supported where s and its derivatives are supported, and its metric component is $2sK$. We prove that its first DEC variation tends to zero at an active ray (x, v) .

If $\mu(x) > 0$, activity implies $\mu = |J|$, $|v| = 1$, and $v = -J^\sharp/\mu$ at that point. Write $\zeta = n + v$. On the portion of the initial slice where $\mu > 0$ the limiting tensor prescribed above is

$$(11.30) \quad \mathbf{G}_0 = \frac{8\pi}{\mu} \alpha \otimes \alpha, \quad \alpha(n) = \mu, \quad \alpha(e_i) = J_i.$$

The covector α is causal and is nonnegative on future causal vectors. At the active point, ζ is null and $\alpha(\zeta) = 0$. Parallel transport ζ along any spatial curve in the initial slice, using the spacetime connection. It remains future null, so its contraction with α is a nonnegative function with a minimum zero at x . Thus

$$(\nabla_i \alpha)(\zeta) = 0, \quad (\nabla_i \mathbf{G}_0)(e_i, \zeta) = 0 \quad \text{at } x.$$

The spatial spacetime connection at time zero depends only on g, K , not on ϵ . Equation (11.29) therefore also gives convergence of these spatial covariant derivatives. Contracted Bianchi, applied before taking the limit, states in an orthonormal frame that

$$(11.31) \quad (\nabla_n \mathbf{G}_\epsilon)(n, \zeta) = \sum_i (\nabla_{e_i} \mathbf{G}_\epsilon)(e_i, \zeta).$$

Only the normal derivative of the constraint components is used here; Bianchi expresses it through the spatial first derivatives controlled above. Its limiting value is zero.

The derivative of the contraction defining the DEC on the graph also differentiates its two vector arguments. The term varying the first argument vanishes because $\mathbf{G}_0(\cdot, \zeta) = 0$. For the second argument, its isometrically transported spatial part still has length one, so the varied ζ stays null. At the active point α is proportional to the covector obtained by lowering ζ ; hence $\alpha(\zeta) = 0$. This kills the second argument term. Together with Equation (11.31) it proves the desired limit at such a ray.

If $\mu(x) = 0$, then $J = 0$ and their spatial first derivatives vanish. The limiting tensor and its spatial first derivatives are zero. Equation (11.31) now kills the required normal derivative, and both argument terms vanish. This applies to all active v , including $|v| < 1$. Finally, all the expressions just used involve only fixed compactly bounded jets, D_ϵ , and its spatial first derivatives. The compact C^1 convergence makes the convergence uniform, including near $\mu = 0$. This proves Equation (11.25). $\square$

Lemma 11.5 (Concentration of the area multiplier). *The measure ω in Lemma 11.3 is concentrated on the single cut S .*

Proof. Use the variations of Lemma 11.4 in Equation (11.19). Their ADM and boundary variations vanish. Their supports lie in a fixed compact set, on which Λ is finite, so uniform convergence permits passage to the limit in its pairing. The metric variation is $2sK$, independent of the regularization. It follows that

$$(11.32) \quad \int_{\mathcal{T}} \int_T s \operatorname{tr}_T K \, dA_g \, d\omega(T) = 0 \quad \text{for every } s \in C_c^\infty(\operatorname{Int} M_{\text{ext}}).$$

Consider the positive aggregate area measure $\beta = \int_{\mathcal{T}} dA_T \, d\omega(T)$ off S . By Lemma 11.2 and the observation following its proof, the function $F(x) = \operatorname{tr}_{\Pi(x)} K(x)$, where $\Pi(x)$ is the common

tangent plane, is continuous on the union of the cuts, locally away from S . In particular it is measurable there. Equation (11.32) says the signed Radon measure $F\beta$ is zero. Thus $F = 0$ β -almost everywhere. Fubini's theorem on a countable compact exhaustion shows that for ω -almost every cut its tangential trace of K is zero at area-almost every point off S . Each such cut is smooth minimal there, so continuity improves this to

$$H_T = 0, \quad \text{tr}_T K = 0 \quad \text{on } T \setminus S.$$

There can be no cancellation in this argument between cuts with different tangent planes: the common-plane property is exactly what made F a single function of position.

Fix one of these cuts. Near a point of contact with S , the $C^{1,1}$ obstacle graph and the smooth graph of S have the same first derivatives on their coincidence set. Their second derivatives agree almost everywhere on that set. One way to see the latter assertion is to apply the fact that a Lipschitz function vanishing on a measurable set has derivative zero almost everywhere on that set, first to the graph difference and then to each of its first derivatives. The outward orientations agree at contact. Since S is a MOTS, the cut consequently satisfies $H_T + \text{tr}_T K = 0$ almost everywhere across contact, as well as on its complement. In a graph chart this is a uniformly elliptic quasilinear equation for a $C^{1,1}$ function. Its coefficients are Hölder in position and first derivatives. Interior elliptic regularity followed by differentiation therefore makes the graph smooth. Thus the whole cut is a smooth MOTS.

Suppose the cut touches a component S_i . At each contact point, the two smooth graph equations have the same orientation and one graph lies on one side of the other. Their difference satisfies a linear uniformly elliptic equation with bounded coefficients. The strong comparison principle implies local coincidence. The contact set $T \cap S_i$ is therefore open in S_i ; it is also closed by compactness. Connectedness of S_i gives $S_i \subset T$. Local coincidence and compactness then make S_i a connected component of T . Consequently, every component of T either equals one of the original S_i or is disjoint from S and lies entirely in $\text{Int } M_{\text{ext}}$. This argument does not use the genus of S_i .

The cut is smooth and encloses the entire obstacle. Its filled set has no bounded complementary pocket, since filling such a pocket would remove positive perimeter and contradict minimality. If T had any interior component, it would thus be an enclosing smooth MOTS with at least one interior component and with every remaining component either interior or equal to an original S_i . This is exactly a cut excluded by the partial-coincidence outermostness hypothesis, even if it retains some original boundary components. Hence T has no interior component. There is a subset $I \subset \{1, \dots, \ell\}$ such that

$$T = \bigsqcup_{i \in I} S_i, \quad \sum_{i \in I} |S_i|_g = |T|_g = A = \sum_{i=1}^{\ell} |S_i|_g.$$

Each $|S_i|_g$ is strictly positive. The equality of these finite sums therefore forces $I = \{1, \dots, \ell\}$, so $T = S$. It follows that ω -almost every cut is S , proving the lemma. $\square$

11.4. The interior adjoint equations. We now know that the area term in Equation (11.19) is exactly $ca'_S(h)$. In particular, arbitrary variations compactly supported in the open exterior have zero total multiplier pairing. We first use this fact before making any regularity assumption on the moments.

Lemma 11.6 (Interior regularity and the full metric equation). *The moments (u, X) are smooth in the open exterior and satisfy Equations (11.4)–(11.6) and $u\mu + J(X) = 0$ there. Their full first jet satisfies a homogeneous linear first-order system with smooth background coefficients.*

Proof. For a compact variation $p = \dot{K}$ with $h = 0$, direct differentiation and integration of the momentum divergence give the coefficient

$$2[u(\tau g - K) - \text{sym} \nabla X + (\text{div} X)g]$$

against p . It is zero distributionally. Taking its trace gives $\text{div} X = -u\tau$, and substituting back proves Equation (11.4).

Next use the compact simultaneous conformal variation $(h, p) = (4\psi g, 2\psi K)$. Its background scaling term is $-64\pi\psi(u\mu + J(X))$, which vanishes by Equation (11.24). The remaining pairing is

$$-8\langle u, \Delta\psi \rangle + 8\langle X, K(\nabla\psi, \cdot) \rangle = 0.$$

This is Equation (11.5) in distributions. Taking a divergence of the symmetric-gradient equation and using $\text{div} X = -u\tau$ also gives

$$(11.33) \quad \Delta X_j + \text{Ric}_{jk} X^k = (\tau\delta_j^i - 2K_j^i)\nabla_i u + u(\nabla_j \tau - 2\nabla^i K_{ij}).$$

Together with the lapse equation this is a second-order elliptic system with scalar Laplacian principal part and smooth lower-order couplings. Locally the measure moments belong to some negative Sobolev space. The local Laplace estimate applied to this system improves their Sobolev order by one, since the right sides have at most one derivative of the unknowns. Iterating with nested compact cutoffs puts them in every local Sobolev space. Sobolev embedding then gives smoothness. The distributional inequalities and complementarity now hold pointwise in the interior.

We give the full metric calculation, including the dependence of the momentum test vectors on the metric. Put $P = K - \tau g$. For compactly supported variations the constraint pairing is the first variation of

$$(11.34) \quad I(g, K) = \int \left[u(R + \tau^2 - |K|^2) + 2X^i \nabla^j P_{ij} \right] dV_g,$$

together with the frame correction

$$(11.35) \quad -8\pi \int h(X, J^\sharp) dV_g.$$

The integration may be localized to a fixed region containing the variation. Changing its volume form instead of retaining the original fixed measure adds one half of its background integrand times $\text{tr} h$. That integrand is $16\pi(u\mu + J(X)) = 0$, so this replacement does not change the pairing. Equation (11.35) follows directly from Equation (11.13) and has the displayed negative sign.

Integrating the shift term once by parts makes it $-\int P^{ij}(\mathcal{L}_X g)_{ij} dV_g$ up to a boundary term unaffected by these variations. In taking a pure metric variation, u , contravariant X , and covariant K are fixed. The elementary variation formulas used are

$$\begin{aligned} \dot{R} &= -\text{Ric}^{ij} h_{ij} + \nabla^i \nabla^j h_{ij} - \Delta(\text{tr} h), \\ \dot{\tau} &= -K^{ij} h_{ij}, \quad (|K|^2)^\cdot = -2(K^2)^{ij} h_{ij}, \quad (dV)^\cdot = \frac{1}{2} \text{tr} h dV, \\ (P^{ij})^\cdot &= -h^i_a K^{aj} - h^j_a K^{ia} + (K^{ab} h_{ab}) g^{ij} + \tau h^{ij}, \\ (\mathcal{L}_X g)^\cdot &= \mathcal{L}_X h. \end{aligned}$$

For example, integrating the last term by parts and combining it with the variation of P^{ij} gives the following coefficient from the entire shift integral:

$$(\mathcal{L}_X K)_{ij} - (\text{div} X) K_{ij} - [X(\tau) + K^{ab} \nabla_a X_b] g_{ij}.$$

The scalar-curvature terms contribute $\text{Hess } u - (\Delta u)g - u \text{ Ric}$. Consequently the vanishing metric coefficient is the explicit equation

$$(11.36) \quad \begin{aligned} 0 = & \nabla_i \nabla_j u - (\Delta u)g_{ij} - u \text{ Ric}_{ij} + 2u(K^2 - \tau K)_{ij} \\ & + \frac{u}{2}(R + \tau^2 - |K|^2)g_{ij} + (\mathcal{L}_X K)_{ij} - (\text{div } X)K_{ij} \\ & - [X(\tau) + K^{ab}\nabla_a X_b]g_{ij} - 8\pi X_{(i}J_{j)}. \end{aligned}$$

This calculation can equally be read distributionally, since each coefficient is linear in the moments and their derivatives.

For completeness, its simplification uses no vacuum assumption. Equation (11.4) gives

$$\text{div } X = -u\tau, \quad K^{ab}\nabla_a X_b = -u|K|^2, \quad \text{tr}(\mathcal{L}_X K) = X(\tau) - 2u|K|^2.$$

Taking the trace of Equation (11.36) and using $16\pi\mu = R + \tau^2 - |K|^2$ and $J(X) = -u\mu$ yields

$$(11.37) \quad \Delta u + X(\tau) = \frac{u}{2}(R + \tau^2 + |K|^2).$$

Substitution of this identity into Equation (11.36) gives exactly Equation (11.6).

These equations have the asserted finite-type property. The right side of Equation (11.6) is linear in $u, X, \nabla X$, with smooth background coefficients, since

$$(\mathcal{L}_X K)_{ij} = X^k \nabla_k K_{ij} + K_{kj} \nabla_i X^k + K_{ik} \nabla_j X^k.$$

The second derivatives of X are also fixed by first jets. To make this explicit, put $B_{ij} = \nabla_{(i} X_{j)} = -uK_{ij}$. Commuting covariant derivatives gives

$$(11.38) \quad \begin{aligned} \nabla_i \nabla_j X_k = & \nabla_i B_{jk} + \nabla_j B_{ik} - \nabla_k B_{ij} \\ & + \frac{1}{2}([\nabla_i, \nabla_j]X_k - [\nabla_i, \nabla_k]X_j - [\nabla_j, \nabla_k]X_i). \end{aligned}$$

The commutators are curvature times X , and differentiating $B = -uK$ uses only $u, \nabla u$ and the background first derivative of K . Thus every derivative of $(u, X, \nabla u, \nabla X)$ is a homogeneous linear expression in that same first jet. This proves the final assertion. $\square$

11.5. Boundary continuation and the ADM normalization.

Lemma 11.7 (Continuation, boundary atoms, and the end constants). *The smooth interior fields extend smoothly to the one-sided boundary. The original moment measures have no boundary-supported part. They satisfy Equation (11.3), and $u > 0$ in the interior.*

Proof. In a smooth normal collar (s, y) of S , all background coefficients in the first-jet system of Lemma 11.6 are smooth up to $s = 0$. Along a normal line the system is an ordinary linear differential equation for the full first jet, with bounded coefficients. Solve it down to $s = 0$ from a fixed interior collar surface. Smooth dependence on the initial data and on y supplies a smooth extension, which agrees with the original solution by uniqueness along each normal line. In particular all its one-sided derivatives have the values supplied by the system. The same system also proves that an interior first jet which vanishes at one point vanishes everywhere on the connected interior, by continuation along piecewise smooth paths.

We next establish the asymptotic form without having prescribed growth to the measures. Let Y denote all coordinate components of (u, X) . The original $O_2(r^{-q})$, $O_1(r^{-1-q})$ bounds, the constraints, and Equations (11.6) and (11.38) give

$$(11.39) \quad |\partial^2 Y| \leq Cr^{-1-q'}|\partial Y| + Cr^{-2-q'}|Y|, \quad \frac{1}{2} < q' < \min\{q, 1\}.$$

Only the original derivative bounds occur here: the curvature and $\partial K, J$ are $O(r^{-2-q'})$, and the coefficients of first jets are $O(r^{-1-q'})$.

Fix a large sphere. Radial integration of Equation (11.39), with the supremum of $|Y|/r + |\partial Y|$ up to radius r , bounds this supremum by its initial value plus its integral against $Cr^{-1-q'}$ dr . Gronwall's inequality therefore gives, uniformly in angle,

$$|Y| = O(r), \quad |\partial Y| = O(1), \quad |\partial^2 Y| = O(r^{-1-q'}).$$

Each coordinate gradient has a limit along a radial ray, with error $O(r^{-q'})$. Two points on a sphere can be joined on that sphere by a path of length $O(r)$; the last Hessian estimate shows that their gradient limits agree. Thus there is one constant matrix L such that

$$\partial Y = L + O(r^{-q'}), \quad Y = Lx + O(r^{1-q'}).$$

The scalar field u is nonnegative on every large sphere, so its linear coefficient is zero. Then $u = O(r^{1-q'})$, and $|X| \leq u$ eliminates every linear coefficient of X as well. Now Equation (11.39) improves to $|\partial^2 Y| = O(r^{-1-2q'})$. Since the gradients have limit zero, radial integration gives $|\partial Y| = O(r^{-2q'})$. As $2q' > 1$, the values have limits along radial rays. Their differences on a large sphere are $O(r^{1-2q'})$, so there is a single constant limit Y_∞ . With Y bounded, one more use of Equation (11.39) gives

$$(11.40) \quad Y = Y_\infty + O(r^{-q'}), \quad \partial Y = O(r^{-1-q'}), \quad \partial^2 Y = O(r^{-2-q'}).$$

At this point the original measure moments could still have a component supported on S . Subtract the integration by parts of the smooth interior fields from the compact identity (11.19). Test with $p = 0$ and a metric variation whose value and full first jet vanish on S . The area and expansion variations and all the smooth integrated boundary terms then vanish. The remaining scalar-curvature variation on S contains

$$-\partial_\nu^2(\text{tr}_S h).$$

This is an arbitrary smooth function on S : in collar coordinates one may take the tangential trace to be a prescribed multiple of s^2 times a cutoff. All momentum and frame terms involve only the value or first jet of h and hence vanish on S . It follows that the scalar moment has no boundary measure. Domination $|X| \leq u$ as measures eliminates a vector boundary measure also.

We can therefore integrate the smooth adjoint by parts all the way to S and to a large coordinate sphere. Use a prototype from Equation (11.15), with its cutoff chosen to vanish near S . Its area and expansion variations vanish, and $a(H) = 0$. Its constraint pairing is integrable by Equation (11.17) and Equation (11.40). Interior terms vanish by the adjoint equations. If the constants in the latter equation are (u_∞, X_∞) , its outer boundary term is

$$(11.41) \quad \int_{S_R} [u_\infty(\partial_j h_{ij} - \partial_i h_{jj}) + 2X_\infty^j(p_{ij} - (\text{tr}_\delta p)\delta_{ij})] n^i dA_\delta + o(1).$$

For clarity, every omitted term vanishes under the stated decay: the derivatives of u, X are $O(r^{-1-q'})$ and multiply $h = O(r^{-1})$; their nonconstant values multiply $\partial h, p = O(r^{-2})$; and the metric-variation momentum terms contain $Kh = O(r^{-2-q'})$. Multiplication by sphere area still leaves $O(r^{-q'})$. Thus the limit of Equation (11.41) is $16\pi(u_\infty E' + X_\infty^i P'_i)$.

The scalar prototype has $E' = 1/2$ and $P' = 0$; for the momentum prototype in direction p , $E' = 0$ and $P' = 2p/3$. The latter follows by integrating $B(p)_{ij}n^j = r^{-2}(p_i + (p \cdot n)n_i)$. Equation (11.19) for these four independent fluxes consequently gives

$$u_\infty = b^0, \quad X_\infty^i = b^i.$$

This proves Equation (11.3) and in particular shows that the adjoint is nontrivial.

If u vanished at an interior point, smooth nonnegativity would give $\nabla u = 0$ there. The inequality $|X| \leq u$ gives $X = 0$; since u has zero differential, it also forces $\nabla X = 0$ at that point.

The whole first jet would be zero. Its continuation property would make (u, X) identically zero, contradicting $u_\infty = b^0 > 0$. Thus $u > 0$ in the interior. $\square$

11.6. The stationary Einstein tensor. We can now form the nondegenerate Lorentzian metric (11.7) on the open exterior. All its coefficients are independent of t . Its future unit normal to a constant-time slice is $n = u^{-1}(\partial_t - X)$, and

$$\frac{1}{2}\mathcal{L}_n g = -\frac{1}{2u}\mathcal{L}_X g = K$$

by Equation (11.4). Thus this construction recovers the original second fundamental form, including its sign.

Here is an explicit curvature verification of the stationary equation; in particular the modified metric adjoint is not being identified with vacuum Killing initial data by assertion. The Gauss–Codazzi formulas and the stationary normal-derivative formula for K , with the convention just specified, give

$$(11.42) \quad \text{Ric}_{\mathbf{g},ij} = \text{Ric}_{g,ij} + \tau K_{ij} - 2(K^2)_{ij} - u^{-1}[(\mathcal{L}_X K)_{ij} + (\text{Hess } u)_{ij}],$$

$$(11.43) \quad \text{Scal}_{\mathbf{g}} = R + \tau^2 + |K|^2 - 2u^{-1}(\Delta u + X(\tau)).$$

These identities can also be checked from the lapse-shift metric without an evolution assumption: its scalar decomposition is

$$u \text{Scal}_{\mathbf{g}} = u(R + |K|^2 - \tau^2) - 2 \text{div}(\nabla u + \tau X),$$

and $\text{div } X = -u\tau$ changes this into Equation (11.43). The spatial Ricci formula is its tensor counterpart, with normal change $u^{-1}(-\mathcal{L}_X K)$ and lapse acceleration $u^{-1} \text{Hess } u$. Equivalently, variation of the reduced scalar action $\int u(R + |K|^2 - \tau^2) dV$ at fixed lapse and contravariant shift has coefficient $-u\mathbf{G}_{ij}$. This agrees with the explicit coefficient (11.36) before its frame correction.

Equation (11.37) in Equation (11.43) gives $\text{Scal}_{\mathbf{g}} = 0$. Equation (11.6) in Equation (11.42) now gives

$$u\mathbf{G}_{ij} = u \text{Ric}_{\mathbf{g},ij} = -8\pi X_{(i} J_{j)}.$$

The normal and mixed Einstein components are the constraints, $\mathbf{G}(n, n) = 8\pi\mu$ and $\mathbf{G}(n, e_i) = 8\pi J_i$. This proves Equation (11.8).

It remains to draw exactly the correct consequence of causality. At every interior point,

$$(11.44) \quad 0 = u\mu + J(X) \geq u\mu - |J||X| \geq u(\mu - |J|) \geq 0.$$

If $u > |X|$, these equalities force $\mu = |J| = 0$. If $\mu > 0$, they force

$$(11.45) \quad |X| = u, \quad |J| = \mu, \quad J = -\frac{\mu}{u} X^\flat.$$

When $\mu = 0$, the DEC gives $J = 0$ and all Einstein components already vanish. When $\mu > 0$, use the orthonormal coframe with $\theta^0(n) = 1$ and $\theta^i(e_j) = \delta_j^i$. The constraints and the spatial Einstein equation give

$$\mathbf{G} = 8\pi\mu \left(\theta^0 - \frac{X_i}{u} \theta^i \right)^2 = \frac{8\pi\mu}{u^2} \xi^\flat \otimes \xi^\flat.$$

The covector in parentheses is null and annihilates $\xi = un + X$. This proves the tensor identity in Equation (11.9) everywhere, with $\mu N = 0$. Its trace is zero and its contraction with ξ is zero, so $\text{Ric}_{\mathbf{g}}(\xi, \cdot) = 0$. All these identities hold on every stationary time translate. No strict timelikeness has been inferred on the internal null set.

11.7. Free boundary variations. We finish the proof of Proposition 11.1 by determining the boundary data. By Lemma 11.7, all moments now have smooth volume densities up to S , with no constraint measure on S itself. The integration-boundary normal is $-\nu$.

Take an arbitrary compact K variation p , allowing arbitrary boundary values, with $h = 0$. Integration of the momentum term has boundary contribution

$$-2 \int_S [p(X, \nu) - (\text{tr } p)X_\nu] \, dA.$$

The boundary expansion variation is $\text{tr}_S p$, and the area variation vanishes. Separating the mixed and tangential-trace components of p in Equation (11.19) gives

$$(11.46) \quad X_T = 0, \quad d\eta = 2X_\nu \, dA.$$

Indeed, the remaining boundary expression is

$$-2 \int_S p(X_T, \nu) \, dA + \int_S (\text{tr}_S p)(2X_\nu \, dA - d\eta),$$

and these components of p are independent. This also identifies η as a smooth density.

Now use the compact conformal variation $(4\psi g, 2\psi K)$, allowing arbitrary value and normal derivative of ψ on S . Its area derivative is $4 \int_S \psi \, dA$, and its expansion derivative is $4\partial_\nu \psi$, since $\theta_+ = 0$. Integrating the lapse equation by parts with normal $-\nu$ yields

$$(11.47) \quad \begin{aligned} -4c \int_S \psi \, dA &= 8 \int_S [u\partial_\nu \psi - (\partial_\nu u + K(X, \nu))\psi] \, dA \\ &\quad - 4 \int_S \partial_\nu \psi \, d\eta. \end{aligned}$$

The independently prescribed normal derivative gives $d\eta = 2u \, dA$. Comparing with Equation (11.46) yields $X = u\nu$. The independently prescribed value then gives

$$\partial_\nu u + K(X, \nu) = \frac{c}{2} = \frac{1}{4m}.$$

This proves Equation (11.10) with its normal orientation and numerical normalization. The boundary values and normal derivatives used in these tests can be supported near any one S_i . The same total-area coefficient c therefore gives the same constant $\kappa = c/2$ on every boundary component.

Let L be the MOTS stability operator for outward normal graph variation at S : a graph with initial speed $s\nu$ has expansion derivative Ls . Its principal part is $-\Delta_S$ and its coefficients are smooth. Extend an arbitrary smooth $s\nu$ to a compact vector field Y on the one-sided exterior and test with

$$h = \mathcal{L}_Y g, \quad p = \mathcal{L}_Y K.$$

Only its one-sided jets at S are needed; an actual diffeomorphism of the manifold with boundary is not required to be a member of the variation space. In the open exterior these are spatial diffeomorphism variations. Their DEC derivative vanishes at every active ray: under diffeomorphism transport, the derivative is a spatial derivative of a nonnegative scalar at its zero; the difference from the isometric vector transport is tangent to the unit sphere, and J annihilates that difference at an active ray. At $\mu = 0$ the latter statement also holds since $J = 0$. There is no boundary constraint measure left to test.

The ADM variation vanishes, the area variation is $\int_S Hs \, dA$, and the expansion variation is Ls . Using $d\eta = 2u \, dA$ in Equation (11.19) gives

$$c \int_S Hs \, dA = 2 \int_S uLs \, dA \quad \text{for every } s \in C^\infty(S).$$

Consequently

$$(11.48) \quad 2L^*u = cH \quad \text{on } S.$$

Outer area minimization implies $H \geq 0$: the first area variation of every nonnegative outward speed is nonnegative. Thus the nonnegative smooth $u|_S$ satisfies $L^*u \geq 0$. The strong minimum principle applies with no sign assumption on the original zeroth-order coefficient: subtract a sufficiently large constant from the operator with positive Laplacian principal part, using $u \geq 0$, to obtain the usual nonpositive zeroth-order coefficient. Apply this principle on each connected S_i . It follows that either $u > 0$ everywhere on S_i , or u is identically zero on S_i [28, Chapter 3]. The argument is componentwise and does not require the same alternative on distinct components.

All assertions of Proposition 11.1 have now been proved. On any component S_i where $u = 0$, one also has $X = 0$ and $\nabla X = 0$: tangential derivatives of $X = 0$ vanish, and Equation (11.4) supplies the remaining normal derivatives there. On any component S_i where $u > 0$, the same equation and Equation (11.10) give

$$\partial_\nu N = 2u(\partial_\nu u + K(X, \nu)) = 2u\kappa \quad \text{on } S_i.$$

These two local boundary behaviors apply independently at each component and will be used in the static classification.

12. STATIC CLASSIFICATION AND THE ORIGINAL HYPERSURFACE

We now work entirely with the original equality data. The preceding variational argument supplies a causal stationary field; it does not initially supply a vacuum exterior or exclude interior points where that field is null. We first prove staticity where the stationary field is timelike, construct the complete static base, and use its conformal double to exclude an interior null set. We then reconstruct the original hypersurface, including its boundary. The conformal step follows the method of Bunting and Masood-ul-Alam [19]. We give the completeness and rigidity arguments needed here, without assuming a domain-of-communications theorem or regularity of an interior Killing horizon.

Proposition 12.1 (Static classification and hypersurface reconstruction). *Let the original exterior satisfy the hypotheses of Theorem 10.2, and let u, X be the fields supplied by Proposition 11.1. Then the number of boundary components is $\ell = 1$, and that component is diffeomorphic to S^2 . Moreover,*

$$N := u^2 - |X|_g^2 > 0 \quad \text{on } \text{int } M_{\text{ext}}.$$

The metric and function

$$h = g + N^{-1}X^\flat \otimes X^\flat, \quad \lambda = \sqrt{N}$$

form the Schwarzschild static base of mass m , with its horizon boundary attached in the regular base coordinates described below. There is a proper smooth spacelike embedding of the original M_{ext} into maximally extended Schwarzschild spacetime of mass m whose induced metric and future second fundamental form are g, K . Its boundary is a cross-section of the future horizon if $u|_S > 0$, and the bifurcation sphere if $u|_S = 0$. The embedding extends as a smooth spacelike hypersurface through S , and its chosen end approaches the corresponding spatial infinity.

We record precisely the outputs of Proposition 11.1 that will be used. Put $\Omega = \text{int } M_{\text{ext}}$. The fields u, X are smooth up to the one-sided boundary, $u > 0$ on Ω , and $u \geq |X|_g$. For any fixed

$$\frac{1}{2} < q_0 < \min\{q, 1\},$$

their end asymptotes have an $O_2(r^{-q_0})$ remainder and satisfy

$$(12.1) \quad (u, X) \longrightarrow (b^0, b) = \left(\frac{E}{m}, -\frac{P}{m} \right), \quad (b^0)^2 - |b|^2 = 1, \quad b^0 > 0.$$

On $\mathbb{R} \times \Omega$, the Lorentzian metric

$$(12.2) \quad \mathbf{g} = -u^2 dt^2 + g_{ij}(dx^i + X^i dt)(dx^j + X^j dt)$$

has Killing field $\xi = \partial_t$, satisfies $\text{Ric}_{\mathbf{g}}(\xi, \cdot) = 0$, and is vacuum wherever $N > 0$. In addition,

$$(12.3) \quad \text{sym } \nabla X = -uK, \quad X|_S = u\nu, \quad \partial_\nu u + K(X, \nu) = \kappa := \frac{1}{4m}.$$

For each connected component S_i , either $u > 0$ everywhere on S_i or $u = 0$ everywhere on S_i . The alternative may initially depend on i ; the same positive constant κ occurs on all components. These statements include the null-dust alternative before vacuum has been proved: only the contraction $\text{Ric}_{\mathbf{g}}(\xi, \cdot) = 0$ will be used at points where $N = 0$.

12.1. The Killing norm and the twist. Let $\mathcal{P} = \{N > 0\} \subset \Omega$. On this open set define

$$(12.4) \quad \begin{aligned} h &= g + N^{-1}X^\flat \otimes X^\flat, & C &= h^{-1} = g^{-1} - u^{-2}X \otimes X, \\ \lambda &= \sqrt{N}, & A &= N^{-1}X^\flat, & F &= dA. \end{aligned}$$

Thus

$$(12.5) \quad \mathbf{g} = -N(dt - A)^2 + h, \quad dV_h = \frac{u}{\sqrt{N}} dV_g.$$

The tensor C extends smoothly and nonnegatively across the interior zero set of N , because $u > 0$ there.

Lemma 12.2 (The original boundary has a timelike collar). *There are pairwise disjoint collars of the finitely many components $S_1, \dots, S_\ell$, each contained in $\mathcal{P}$ away from its boundary. In the formulas for an individual collar, write S for its boundary component. If $u|_S > 0$, then*

$$(12.6) \quad dN|_S = 2\kappa X^\flat|_S = 2u\kappa\nu^\flat.$$

If $u|_S = 0$, and $s \geq 0$ is the original g -normal distance from S , there are smooth a, Y such that

$$(12.7) \quad u = sa, \quad X = s^2 Y, \quad a|_S = \kappa, \quad N = s^2(a^2 - s^2|Y|_g^2).$$

The AF end is also contained in $\mathcal{P}$.

Proof. Fix one component and use the local notation S of the statement. On S , $X = u\nu$, hence $N = 0$ and its tangential derivatives vanish. The normal-normal component of the KID equation gives

$$\langle \nabla_\nu X, \nu \rangle = -uK(\nu, \nu).$$

Consequently

$$\partial_\nu N = 2u\partial_\nu u - 2u\langle \nabla_\nu X, \nu \rangle = 2u(\partial_\nu u + uK(\nu, \nu)) = 2u\kappa.$$

This proves Equation (12.6), including positivity on an exterior collar when $u|_S > 0$.

If $u = 0$ on S , then $X = 0$ there and all its tangential covariant derivatives vanish. The normal-normal and normal-tangential components of $\text{sym } \nabla X = 0$ imply $\nabla_\nu X = 0$ on S . Smooth division by s and then s^2 yields Equation (12.7); the boundary equation gives $a|_S = \kappa > 0$. Compactness of each S_i makes its collar assertion uniform. Since there are finitely many disjoint compact components, shrink their collars to be pairwise disjoint and choose a common positive collar width below all of these finitely many widths. Thus every original boundary component has a timelike collar, including when the boundary lapse alternatives differ between components. Finally, Equation (12.1) gives $N \rightarrow 1$. $\square$

Lemma 12.3 (Norm and twist identities). *Write $\mathcal{G}_{\alpha\beta} = \nabla_\alpha^\mathbf{g} \xi_\beta$, with full tensor contraction used in $|\mathcal{G}|_\mathbf{g}^2$. On Ω ,*

$$(12.8) \quad \frac{1}{u} \operatorname{div}_g(uC \, dN) = -2|\mathcal{G}|_\mathbf{g}^2.$$

On $\mathcal{P}$, the antisymmetric contravariant tensor

$$Q^{ij} = uNC^{ik}C^{j\ell}F_{k\ell}$$

satisfies

$$(12.9) \quad \nabla_i Q^{ij} = 0.$$

Proof. The Killing equation makes $\mathcal{G}$ antisymmetric. The differentiated Killing equation and $\operatorname{Ric}_\mathbf{g}(\xi, \cdot) = 0$ imply $\square_\mathbf{g} \xi = 0$. Since $\mathbf{g}(\xi, \xi) = -N$, differentiation twice gives $\square_\mathbf{g} N = -2|\mathcal{G}|_\mathbf{g}^2$. For a t -independent scalar, the inverse metric in Equation (12.2) has spatial block C and volume density $u\sqrt{\det g}$. Its wave operator is therefore $u^{-1} \operatorname{div}_g(uC \, d\cdot)$, proving Equation (12.8).

For completeness, put $\theta = dt - A$. The Killing one-form is $\xi^\flat = -\lambda^2 \theta$, and

$$\xi^\flat \wedge d\xi^\flat = -\lambda^4 \theta \wedge F.$$

The Killing identity

$$d *_\mathbf{g} (\xi^\flat \wedge d\xi^\flat) = 2 *_\mathbf{g} (\xi^\flat \wedge \operatorname{Ric}_\mathbf{g}(\xi, \cdot))$$

has zero right side. Taking the Hodge star in the orthonormal time form $\lambda\theta$ shows that its left side, up to an irrelevant overall orientation sign, is $d(\lambda^3 *_h F)$. It follows that

$$\operatorname{div}_h(\lambda^3 C^{ik} C^{j\ell} F_{k\ell}) = 0.$$

Using $\lambda^3 dV_h = uN dV_g$ gives Equation (12.9). For an antisymmetric two-tensor the extra connection term on the free index vanishes, so this change of volume density gives exactly the stated g -divergence. $\square$

Lemma 12.4 (Logarithmic control at an arbitrary null set). *Near the interior zero set of N , put $B = |X|_g$, $e = X/B$, and $T = e^\perp$. For every compactly supported smooth cutoff η in a sufficiently small neighborhood of that set,*

$$(12.10) \quad \sup_{\varepsilon > 0} \int u\eta^2 \frac{|dN|_g^2}{(N + \varepsilon)^2} dV_g < \infty.$$

In particular,

$$(12.11) \quad \int_{\mathcal{P}} \eta^2 |d_T \log N|^2 dV_g < \infty.$$

No regularity or measure assumption on $\{N = 0\}$ is required.

Proof. On each such compact neighborhood, u and B are bounded below. Indeed $u > 0$ in Ω and $B = u$ on $\{N = 0\}$. The tensor C acts as the identity on T and has eigenvalue N/u^2 on e .

We first justify a useful estimate for the smooth nonnegative function N :

$$(12.12) \quad |dN|_g^2 \leq CN$$

on a compact set after slightly enlarging its neighborhood. Choose a uniform normal-coordinate radius and a constant L bounding the Hessian and large enough that $|dN|/L$ is smaller than that radius. Taylor's inequality on a geodesic in direction $-\nabla N/|\nabla N|$, at length $|dN|/L$, gives $0 \leq N - |dN|^2/(2L)$. This proves Equation (12.12).

Let $n = (\partial_t - X)/u$ be the future unit normal to the stationary slices, and complete e by an orthonormal pair e_1, e_2 in T . Since $\xi = un + Be$,

$$\mathcal{G}_{in} = -\frac{N_i/2 + B\mathcal{G}_{ie}}{u}.$$

Expanding all temporal and spatial terms in the full squared norm gives

$$(12.13) \quad \begin{aligned} -2|\mathcal{G}|_{\mathbf{g}}^2 &= \frac{|\mathrm{d}N|_g^2}{u^2} + \frac{4B}{u^2} \sum_{a=1}^2 N_a \mathcal{G}_{ae} \\ &\quad - \frac{4N}{u^2} \sum_{a=1}^2 \mathcal{G}_{ae}^2 - 4\mathcal{G}_{12}^2. \end{aligned}$$

The fields and their derivatives are bounded on the compact neighborhood. Equations (12.8), (12.12), and (12.13) therefore imply

$$(12.14) \quad \frac{1}{u} \operatorname{div}_g(uC \, \mathrm{d}N) \leq C_0(N + |\mathrm{d}_T N|).$$

Multiply by $u\eta^2/(N + \varepsilon)$ and integrate by parts. With E_ε denoting the integral in Equation (12.10), the result is

$$E_\varepsilon - 2 \int \frac{u\eta C(\mathrm{d}\eta, \mathrm{d}N)}{N + \varepsilon} \mathrm{d}V_g \leq C_0 \int u\eta^2 \mathrm{d}V_g + C_0 \int \frac{u\eta^2 |\mathrm{d}_T N|}{N + \varepsilon} \mathrm{d}V_g.$$

Cauchy's inequality bounds the cutoff term by $E_\varepsilon/4 + C \int u |\mathrm{d}\eta|_C^2 \mathrm{d}V_g$. Since $|\mathrm{d}_T N| \leq |\mathrm{d}N|_C$, the last term is bounded by $E_\varepsilon/4 + C \int u\eta^2 \mathrm{d}V_g$. This proves Equation (12.10). Fatou's lemma on the positive set gives Equation (12.11), using the local lower bound for u . $\square$

Lemma 12.5 (Vanishing twist). *The two-form F vanishes on all of $\mathcal{P}$.*

Proof. First note the exact contraction identity with its free index lowered using g :

$$(12.15) \quad g_{ik} Q^{kj} A_j = \frac{1}{u} \left(X^j (\mathrm{d}X^b)_{ij} - \frac{B^2}{N} (\mathrm{d}_T N)_i \right).$$

It follows by inserting

$$F = N^{-1} \mathrm{d}X^b - N^{-2} \mathrm{d}N \wedge X^b \quad \text{and} \quad CX = \frac{N}{u^2} X$$

into $g_{ik} Q^{kj} A_j$. Both terms on the right of Equation (12.15) are orthogonal to X . At $X = 0$ its second numerator is interpreted as $B^2 \mathrm{d}N - \mathrm{d}N(X)X^b$, so no choice of e is needed there.

The end asymptotes give a constant one-form A_∞ with

$$A - A_\infty = O_1(r^{-q_0}), \quad F = O(r^{-1-q_0}).$$

Choose a smooth function equal to a linear primitive of A_∞ near infinity and zero outside a larger end neighborhood. Its differential α is closed, equals A_∞ near infinity, and vanishes near S and all possible interior zeros of N . These zeros are contained in a fixed compact subset of Ω , by Lemma 12.2. Put $\beta = A - \alpha$, so $\mathrm{d}\beta = F$.

For each component choose a collar cutoff χ_i that is zero near S_i and one outside its collar. Take $\chi = \chi_{\text{end}} \chi_{\text{null}} \prod_{i=1}^\ell \chi_i$, initially supported away from S , infinity, and $\{N = 0\}$. The boundary collars are pairwise disjoint; on the support of $\mathrm{d}\chi_i$ all other boundary factors equal one. Testing Equation (12.9) by $\chi\beta_j$ gives

$$(12.16) \quad \frac{1}{2} \int_{\mathcal{P}} \chi u N C^{ik} C^{j\ell} F_{ij} F_{k\ell} \mathrm{d}V_g = - \int_{\mathcal{P}} Q^{ij} (\partial_i \chi) \beta_j \mathrm{d}V_g.$$

The integrand on the left is nonnegative.

We remove the interior cutoff first, with the other two cutoffs fixed. Let it transition from zero to one on $\varepsilon < N < 2\varepsilon$, with derivative bounded by C/ε . Near that band $\beta = A$. Equation (12.15), its orthogonality to X , and $N \asymp \varepsilon$ bound its contribution by

$$C \int_{\{\varepsilon < N < 2\varepsilon\}} (1 + |d_T \log N|^2) dV_g$$

on a fixed compact set. This tends to zero by Lemma 12.4. Indeed the bands tend pointwise to the empty set within $\mathcal{P}$, and their indicators are dominated by an integrable function there. This reasoning remains valid when $\{N = 0\}$ has positive measure.

Next remove the boundary cutoffs, keeping the end cutoff fixed. In the collar of S_i let s be the original normal distance and let χ_i transition on $s \in (\delta, 2\delta)$ with derivative bounded by C_i/δ . For this local computation write $S = S_i$. If $u|_S > 0$, then $N \asymp s$, $X = u\nu + O(s)$, and $d_T N = O(s)$. The flux in Equation (12.15) is bounded. Its orthogonality to X makes its g -normal component $O(s)$, so multiplication by the cutoff derivative $O(\delta^{-1})$ and integration over the collar gives a term tending to zero. If $u|_S = 0$, Equation (12.7) gives

$$u \asymp s, \quad X = O(s^2), \quad dX^b = O(s), \quad N \asymp s^2, \quad dN = O(s).$$

The entire flux is then $O(s^2)$, with the same conclusion. Each boundary error therefore tends to zero independently of which lapse alternative holds on that component. The derivative of the finite product is the finite sum of these collar contributions; letting $\delta \downarrow 0$ makes their sum tend to zero.

Finally use a radial cutoff on $R < r < 2R$. Here $Q = O(r^{-1-q_0})$ and $\beta = O(r^{-q_0})$; its contribution is $O(R^{1-2q_0})$, which tends to zero. Fatou's lemma in Equation (12.16) now makes its nonnegative integrand vanish everywhere on $\mathcal{P}$. Since $uN > 0$ and C is positive definite there, $F = 0$. $\square$

12.2. The complete static base. Let $\mathcal{E}$ be the component of $\mathcal{P}$ containing the connected far AF end. We first derive the static equations on every component of $\mathcal{P}$, then prove that there is only one component. Since $F = 0$, the form A is locally exact. Equation (12.5) can locally be written as $-\lambda^2 dT^2 + h$. The Christoffel symbols involving T are $\Gamma_{Ti}^T = \lambda_i/\lambda$ and $\Gamma_{TT}^i = \lambda h^{ij} \lambda_j$. The vacuum equations consequently give, on all of $\mathcal{P}$,

$$(12.17) \quad \lambda \operatorname{Ric}_h = \operatorname{Hess}_h \lambda, \quad \Delta_h \lambda = 0, \quad \operatorname{Scal}_h = 0.$$

These equations are global even though no global primitive of A has yet been constructed.

Lemma 12.6 (Connected positive set). *The set $\mathcal{P}$ equals its infinity component $\mathcal{E}$. In particular, every original boundary collar belongs to $\mathcal{E}$.*

Proof. The unique end of M_{ext} lies in $\mathcal{P}$ outside a compact set. Any component V of $\mathcal{P}$ other than $\mathcal{E}$ therefore has compact closure in the original closed exterior M_{ext} . Its frontier there lies in $\{N = 0\} \cup S$: an interior frontier point with $N > 0$ has a connected positive neighborhood meeting V and hence belongs to V , a contradiction. The continuous function $\lambda = \sqrt{N}$ is zero on that frontier. The frontier is nonempty; otherwise V would be open and closed in the connected open exterior Ω , which also contains the far end in $\mathcal{E}$.

Choose any point of V , where $\lambda > 0$. Compactness of the closure and the zero frontier values imply that the positive maximum of λ is attained at an interior point of V . The strong maximum principle for the smooth locally elliptic equation $\Delta_h \lambda = 0$ makes λ constant on V , contradicting its zero frontier values. Thus no such component exists. Every boundary collar from Lemma 12.2 is a subset of $\mathcal{P} = \mathcal{E}$, as claimed. $\square$

Lemma 12.7 (Completion and the lapse range). *The function λ satisfies $0 < \lambda < 1$ on $\mathcal{E}$. Approach to an interior zero of N has infinite h -length. Attaching all components of S gives a regular base boundary on which*

$$(12.18) \quad h|_{TS} = g|_{TS}, \quad \lambda = 0, \quad \partial_{\nu_h} \lambda = \kappa, \quad \Pi_h = 0.$$

At a component S_i with $u|_{S_i} > 0$, the base metric admits a smooth reflection with odd lapse. At a component with $u|_{S_i} = 0$, it admits a C^2 reflection with odd lapse. These componentwise attachments form a complete base with compact boundary, and its two-copy double is a connected orientable complete manifold.

Proof. Near an interior zero set, B, u have positive lower bounds. Contracting $F = 0$ with X in Equation (12.15) gives

$$|d_T \log N| \leq B^{-2} |X| |dX^\flat| \leq C.$$

Together with Equation (12.12), this yields

$$(12.19) \quad |d \log N|_h^2 = |d_T \log N|^2 + \frac{|d_e N|^2}{u^2 N} \leq C$$

on compact neighborhoods of the interior zero set. A finite-length path approaching that set would make $\log N \rightarrow -\infty$, contradicting this bound.

Every possible limit of $\mathcal{E}$ in the compact part of the original closed exterior either belongs to $\mathcal{E}$, lies in $\{N = 0\}$, or belongs to S . At the latter two types of points $\lambda \rightarrow 0$, while $\lambda \rightarrow 1$ at the AF end. If $\lambda > 1$ somewhere, a positive superlevel set bounded away from one is contained in a compact subset of $\mathcal{E}$. The harmonic function would attain an interior maximum, forcing it constant, contrary to its limit at infinity. Thus $\lambda \leq 1$. If equality holds at one interior point, the strong maximum principle gives $\lambda \equiv 1$ on $\mathcal{E}$. That component then has no interior boundary, since N remains one at any such boundary point. It is open and closed in connected Ω , hence equals Ω , contradicting the timelike collar where $\lambda \rightarrow 0$ at S . Therefore $0 < \lambda < 1$.

For boundary regularity fix a component S_i and write S for it through the following local calculation. Suppose first that $u|_S > 0$. By Equation (12.6), use (N, y^1, y^2) as original smooth collar coordinates. Smooth division gives

$$X^\flat = a(N, y) dN + N \beta_A(N, y) dy^A, \quad a(0, y) = \frac{1}{2\kappa}.$$

Writing $N = \lambda^2$, the coefficients of h are

$$(12.20) \quad \begin{aligned} h_{\lambda\lambda} &= 4\lambda^2 g_{NN} + 4a^2, \\ h_{\lambda A} &= 2\lambda(g_{NA} + a\beta_A), \\ h_{AB} &= g_{AB} + \lambda^2 \beta_A \beta_B, \end{aligned}$$

with all right-side coefficients evaluated at (λ^2, y) . They extend smoothly to negative λ . The map $(\lambda, y) \mapsto (-\lambda, y)$ is an isometry: the diagonal blocks are even and the mixed block is odd. At zero, $h_{\lambda\lambda} = \kappa^{-2}$, $h_{\lambda A} = 0$, and $h_{AB} = g_{AB}|_S$. The fixed hypersurface of this reflection is totally geodesic, and all assertions in Equation (12.18) follow.

If $u|_S = 0$, put $D = a^2 - s^2|Y|^2$ in Equation (12.7). Then

$$h = g + s^2 D^{-1} Y^\flat \otimes Y^\flat, \quad \lambda = s\sqrt{D}, \quad D|_S = \kappa^2.$$

These are smooth in the original one-sided collar. The static equations extend by continuity to give $\text{Hess}_h \lambda = 0$ at S . Its tangential components are $\kappa \Pi_h$, so $\Pi_h = 0$. In h -Gaussian coordinates $\rho \geq 0$, write $h = d\rho^2 + h_\rho$. Then $\partial_\rho h_\rho|_0 = 0$, and $\partial_\rho^2 \lambda|_0 = 0$. Even reflection of h_ρ and odd reflection of λ are therefore C^2 .

These constructions apply on the finitely many disjoint collars, with no compatibility condition between different lapse alternatives. They attach the entire original S , by Lemma 12.6. Let $\bar{\mathcal{E}}$ denote this attached base and let $j : \bar{\mathcal{E}} \rightarrow M_{\text{ext}}$ be its identity map on the interior and on boundary points. It is a homeomorphism onto the original exterior with any interior zero set removed, but need not be a boundary diffeomorphism: on a positive-lapse collar the original coordinate is $N = \lambda^2$. The identity $h = g + N^{-1}X^\flat \otimes X^\flat$ gives $h \geq j^*g$ in the interior and, by continuity in each regular collar, on the attached boundary as a nonnegative quadratic-form inequality.

It remains to check completeness without classifying the missing sets. The original exterior is complete as a metric space with boundary: a finite-length Cauchy path is Cauchy in the complete ambient initial-data manifold and its limit remains in the closed exterior. The inequality $h \geq j^*g$ makes the image of a finite h -length escaping path have such an original limit. An interior limit with $N > 0$ is an ordinary smooth base limit. An interior zero is impossible by Equation (12.19). A limit in S is accounted for by the regular collar just constructed. The AF end has infinite distance. These possibilities exhaust finite-length escapes, proving completeness of $\bar{\mathcal{E}}$ with its boundary included.

Take two copies of this base and identify corresponding points of each S_i by the collar reflections just constructed. The resulting double is connected because the base is connected and its boundary is nonempty. Give the second copy the opposite orientation; the identifications yield an orientable manifold without boundary. There is no assertion that this double is simply connected. The reflected metric h_d and odd reflected lapse are C^2 across every join, and smooth at the positive-lapse joins. The static scalar and harmonic equations hold continuously across the joins, since the indicated second jets match.

For completeness, fold the double onto $\bar{\mathcal{E}}$ by identifying the two copies. The fold is distance nonincreasing, because lengths on either side are the base lengths. A Cauchy sequence in the double projects to a Cauchy sequence in the complete base. The base is a complete locally compact length space, hence is proper, so the projected sequence lies in a compact base set. The preimage of that set is a quotient of two compact copies and is compact. The original sequence consequently has a convergent subsequence and, being Cauchy, converges. This proves completeness of the double. $\square$

12.3. Static asymptotics and the compactified end. After a constant linear change of the original end coordinates, $h = \delta + O_2(r^{-q_0})$ and $\lambda = 1 + O_2(r^{-q_0})$. Indeed its original constant metric is $I + b \otimes b$, by Equation (12.1). We prove the improvement required for conformal doubling explicitly; the harmonic-coordinate mechanism agrees with Bartnik [11, Theorem 3.1, Proposition 3.3, and Theorem 4.3].

Lemma 12.8 (Harmonic-coordinate expansion). *There are asymptotically Cartesian coordinates on the end and a number $0 < \epsilon < 1$ such that, with $\gamma = \lambda^2 h$ and $U = \log \lambda$,*

$$(12.21) \quad \gamma_{ij} = \delta_{ij} + O_k(r^{-1-\epsilon}), \quad U = -\frac{M}{r} + \frac{b_1 \cdot x}{r^3} + O_k(r^{-2-\epsilon})$$

for every fixed finite k . Here $M > 0$; the vector b_1 is unrelated to the shift asymptote b . Only the original two metric derivatives and the stated adjoint decay are needed to start this improvement.

Proof. The three-dimensional conformal formulas applied to Equation (12.17) give

$$(12.22) \quad \text{Ric}_\gamma = 2dU \otimes dU, \quad \Delta_\gamma U = 0.$$

Initially $\gamma - \delta, U = O_2(r^{-q_0})$. We give the coordinate construction with this derivative count.

Move to a sufficiently distant end and extend γ to a metric on $\mathbb{R}^3$, equal to δ inside a large ball, using a fixed-ratio cutoff annulus. In the resulting coordinates write

$$\Delta_\gamma = (\delta^{ab} + a^{ab})\partial_{ab} + c^a\partial_a.$$

The coefficients satisfy $a = O_2(r^{-q_0})$, $c = O_1(r^{-1-q_0})$, and their scaled Hölder norms through the orders needed here are small after increasing the cutoff radius. The bounded second derivatives in the hypothesis give scaled Lipschitz bounds on first derivatives, so they suffice for any fixed Hölder exponent less than one at this stage.

Put $s_0 = 1 - q_0 \in (0, 1/2)$. For a source with weighted scaled $C^{0,\alpha}$ size $O(r^{s_0-2})$, the Euclidean inverse normalized to vanish at zero is

$$(12.23) \quad Pf(x) = -\frac{1}{4\pi} \int_{\mathbb{R}^3} \left(\frac{1}{|x-y|} - \frac{1}{|y|} \right) f(y) dy.$$

It maps that weighted norm boundedly to scaled $C^{2,\alpha}$ size $O(r^{s_0})$. To verify the zeroth-order bound, split at $|y| = 2|x|$. On the far part, the kernel difference is bounded by $C|x||y|^{-2}$, and its integral is $O(|x|^{s_0})$ because $s_0 < 1$. On the near part the separate kernels have the same bound because $s_0 > 0$. Interior Poisson estimates on rescaled annuli give the derivative and Hölder bounds. The ordinary local estimates also apply on the fixed inner ball.

The equations

$$v^i = P(-c^i - a^{ab}\partial_{ab}v^i - c^a\partial_av^i)$$

are contractions in that weighted space. Indeed the operator norm of the last two terms is bounded by a constant times the small scaled norms of a and rc . They produce harmonic coordinates $x^i + v^i$ with $v = O_{C^{2,\alpha}}(r^{1-q_0})$ and $Dv = O(r^{-q_0})$. These functions are coordinates sufficiently far out.

There is a derivative issue in changing metric coordinates which must be addressed before using a classical Ricci equation. The original assumptions bound a, c , respectively, in scaled $W^{2,\infty}$ and $W^{1,\infty}$. Differentiate the equation for v once and apply local $W^{2,p}$ estimates to Dv on rescaled annuli. All differentiated coefficients and forcing are controlled by these bounds and the preceding $C^{2,\alpha}$ estimate. For every finite p this gives

$$v = O_{W^{3,p}}(r^{1-q_0}).$$

Consequently the transformed $\gamma - \delta, U$ have scaled $W^{2,p}$ size $O(r^{-q_0})$. Take $p > 3$; Morrey embedding gives scaled $C^{1,\alpha}$ control for some $\alpha > 0$. In the harmonic chart Equation (12.22) has the elliptic form

$$-\frac{1}{2}\gamma^{ab}\partial_{ab}\gamma_{ij} + Q_{ij}(\gamma^{-1}, D\gamma) = 2U_iU_j, \quad \gamma^{ab}\partial_{ab}U = 0,$$

where Q_{ij} is quadratic in $D\gamma$. Schauder estimates first give scaled $C^{2,\alpha}$ control; differentiating this elliptic system then gives $\gamma - \delta, U = O_k(r^{-q_0})$ for each finite k . No higher original-coordinate derivative bound has been assumed.

We use two elementary exterior Poisson estimates. If $w = o(1)$, $\Delta_\delta w = f$, and $f = O_k(r^{-2-t})$, $1 < t < 2$, then

$$(12.24) \quad w = \frac{a}{r} + O_k(r^{-t}).$$

To prove this, cut w off on a fixed inner annulus and extend it by zero; this changes f only on a compact set. The extended solution equals the Newton potential of its Laplacian, since their difference is an entire decaying harmonic function. The coefficient is $a = -(4\pi)^{-1} \int f$. For $|y| < r/2$, subtracting $1/r$ from the kernel costs

$$Cr^{-2} \int_{|y| < r/2} |y||f(y)| dy = O(r^{-t}).$$

The remaining region, including the integrable kernel singularity, has the same bound. Rescaled interior estimates prove the derivative version. If instead $f = O_k(r^{-4-\epsilon})$, $0 < \epsilon < 1$, its first moments converge and the same reasoning gives

$$(12.25) \quad w = \frac{a}{r} + \frac{d \cdot x}{r^3} + O_k(r^{-2-\epsilon}).$$

Here subtract $1/r + (x \cdot y)/r^3$ in the near region. The remainder costs $Cr^{-3} \int_{|y| < r/2} |y|^2 |f(y)| dy = O(r^{-2-\epsilon})$; the far region has that order as well.

Set $\epsilon = 2q_0 - 1 \in (0, 1)$. The harmonic-coordinate equations and initial symbol bounds yield

$$\Delta_\delta(\gamma_{ij} - \delta_{ij}), \quad \Delta_\delta U = O_k(r^{-2-2q_0}).$$

Equation (12.24) gives

$$\gamma_{ij} = \delta_{ij} + \frac{A_{ij}}{r} + O_k(r^{-1-\epsilon}), \quad U = \frac{a}{r} + O_k(r^{-1-\epsilon}).$$

The harmonic-coordinate condition is $\partial_j(\sqrt{\det \gamma} \gamma^{ij}) = 0$. Expanding it at the displayed order gives

$$\left(A_{ij} - \frac{1}{2}(\text{tr } A)\delta_{ij}\right) \frac{x^j}{r^3} = O(r^{-2-\epsilon}).$$

Taking a radial limit shows $A = \frac{1}{2}(\text{tr } A)I$. Its trace in dimension three is zero, hence $A = 0$. Since $D^2U = O_k(r^{-3})$, the scalar equation now improves to

$$\Delta_\delta U = (\delta^{ab} - \gamma^{ab})\partial_{ab}U = O_k(r^{-4-\epsilon}).$$

Equation (12.25) proves Equation (12.21), initially with a real M .

Its positivity uses the lapse range from Lemma 12.7. The function $-U > 0$ is γ -harmonic. For $0 < \eta < 1$ and a fixed positive C , the function

$$w = r^{-1} + Cr^{-1-\eta}$$

is positive and strictly γ -subharmonic sufficiently far out: $\Delta_\gamma r^{-1} = O(r^{-4-\epsilon})$, while the leading Laplacian of $r^{-1-\eta}$ is $\eta(1+\eta)r^{-3-\eta}$. Choose $c > 0$ so small that $-U \geq cw$ on a large inner sphere. The difference tends to zero and is superharmonic, so the exterior minimum principle gives $-U \geq cw$ throughout that end. Taking the radial limit proves $M \geq c > 0$. $\square$

Lemma 12.9 (The two conformal metrics). *Define*

$$h_\pm = \left(\frac{1 \pm \lambda}{2}\right)^4 h \quad \text{on } \mathcal{E}.$$

Both are scalar flat. The plus end satisfies $h_+ = \delta + O_k(r^{-1-\epsilon})$, hence has zero ADM mass. The minus end compactifies by one point to a nondegenerate $C^{1,\epsilon} \cap W_{\text{loc}}^{2,p}$ metric for some $p > 3$, with scalar curvature zero as a distribution at that point.

Proof. Scalar flatness follows from Equation (12.17) and the three-dimensional conformal scalar-curvature formula, since $1 \pm \lambda$ are harmonic. In terms of the preceding variables,

$$(12.26) \quad h_+ = \cosh^4(U/2)\gamma, \quad h_- = \sinh^4(U/2)\gamma.$$

The first factor is $1 + O_k(r^{-2})$, proving the assertion for h_+ . Its ADM flux is $O(r^{-\epsilon})$, so its mass is zero.

For the minus end invert $x = y/|y|^2$, write $\rho = |y|$, and put $R(y) = I - 2y \otimes y/\rho^2$. The inversion differential is $\rho^{-2}R(y)$, with $R(y)^T R(y) = I$. Equation (12.21) gives

$$I^*\gamma = \rho^{-4}(I + O_k(\rho^{1+\epsilon})), \quad U(I(y)) = -M\rho + (b_1 \cdot y)\rho + O_k(\rho^{2+\epsilon}).$$

Derivatives of the angular matrix $R(y)$ cost the corresponding powers of ρ^{-1} , so the displayed metric estimate retains the symbol bounds. Dividing the expansion of $\sinh(U/2)$ by ρ , and using $\epsilon < 1$, gives

$$(12.27) \quad I^* h_- = \left(\frac{M}{2}\right)^4 \left[\left(1 - \frac{4b_1 \cdot y}{M}\right) I + O_k(\rho^{1+\epsilon}) \right].$$

In particular, its only first-order term is linear and scalar.

For the remainder $E(y)$, the bounds

$$|E| \leq C\rho^{1+\epsilon}, \quad |DE| \leq C\rho^\epsilon, \quad |D^2E| \leq C\rho^{\epsilon-1}$$

show that $E(0) = DE(0) = 0$ extends it as $C^{1,\epsilon}$. To see the Hölder estimate between two points, use the gradient bound when their separation is comparable to their radii; otherwise integrate the Hessian along the short segment in a common annulus. Moreover choose

$$3 < p < \frac{3}{1-\epsilon}.$$

Then $D^2E \in L^p$. Integration by parts across small coordinate spheres has an error bounded by a constant times their area, because the first derivatives are bounded. It follows that the punctured classical derivatives are the weak derivatives of the extension. The Christoffel symbols are continuous with weak derivatives in L^p , so scalar curvature is its usual L^p expression in $\partial\Gamma$ and Γ . It vanishes off the point and therefore also as a distribution on the whole ball. There is no point curvature term. $\square$

12.4. Zero-mass rigidity with arbitrary remaining ends. The completed conformal manifold can still have ends corresponding to an interior zero set. We need a rigidity statement that permits these ends, and also the regularity in Lemma 12.9. The smooth arbitrary-end positive mass theorem is established in [21, Theorem B]. In our scalar-flat situation the following direct spinor proof also handles the compactified point. It uses Witten's identity [53]; no asymptotic condition is imposed on any other end.

Lemma 12.10 (Scalar-flat rigidity at a faster-decaying end). *Let $(W, \mathbf{g})$ be a connected orientable complete three-dimensional Riemannian manifold without boundary. Assume that its metric is locally $C^{1,\alpha} \cap W^{2,p}$, for some $\alpha > 0, p > 3$, is scalar flat as a distribution, and has an end on which it is smooth with*

$$\mathbf{g} = \delta + O_2(r^{-1-\epsilon}), \quad \epsilon > 0.$$

The other ends can be arbitrary. Then $(W, \mathbf{g})$ is isometric to Euclidean space. The asserted regularity is sufficient at a compactified point and at a reflected hypersurface.

Proof. An orientable three-manifold is spin. Choose a spin structure and use a smooth background metric to identify its spinor bundle with that for $\mathbf{g}$. The positive square-root change of orthonormal frames has the stated metric regularity. The spin connection consequently has continuous locally bounded coefficients, with weak first derivatives in L^p_{loc} . Write ∇^S for that connection, D for the Dirac operator, and c for Clifford multiplication.

The weak Lichnerowicz formula has the following integrated form for every compactly supported H^1 spinor ϕ :

$$(12.28) \quad \|D\phi\|_2^2 = \|\nabla^S \phi\|_2^2.$$

Here and below norms and integrals are for $\mathbf{g}$. To check the formula at the indicated regularity, first use a smooth compactly supported spinor. Approximate the metric on its support in C^1 and $W^{2,p}$ by smooth metrics, using the same background-bundle identification. The smooth Lichnerowicz formula passes to the limit: connection coefficients converge uniformly and scalar curvatures converge in L^p . The limiting scalar term is zero by the assumed distributional scalar

flatness and the L^p curvature expression. Density then proves Equation (12.28) for compact H^1 spinors. Thus no integration surface is retained around the compactified point.

We next establish the coercivity needed to complete compact spinors. For a compactly supported scalar function f on the AE end, one-dimensional integration by parts along coordinate rays gives

$$(12.29) \quad \int_{r_0}^{\infty} f^2 \, dr \leq 4 \int_{r_0}^{\infty} r^2 |\partial_r f|^2 \, dr.$$

The inner boundary term has the favorable sign. Apply this to $f = |\phi|$ and integrate over angles. Metric comparability and $|\mathrm{d}|\phi|| \leq |\nabla^S \phi|$ give

$$(12.30) \quad \int_{\text{AE}} r^{-2} |\phi|^2 \, \mathrm{d}V_{\mathfrak{g}} \leq C \|\nabla^S \phi\|_2^2.$$

For any fixed compact $K \subset W$, connect it to a fixed end annulus by a relatively compact connected domain. Poincaré's inequality with the norm on that annulus retained gives

$$\|\phi\|_{L^2(K)} \leq C_K \left(\|\mathrm{d}|\phi|\|_{L^2(\text{domain})} + \|\phi\|_{L^2(\text{annulus})} \right).$$

For example, this version follows from the ordinary Poincaré inequality after subtracting the mean; the observed annulus controls that mean because it has positive measure. Equation (12.30) controls its last term. We have proved

$$(12.31) \quad \|\phi\|_{L^2(K)} \leq C_K \|\nabla^S \phi\|_2.$$

Connectedness is the only global input in this propagation.

Let $\mathcal{H}$ be the completion of compactly supported smooth spinors in the norm $\|\nabla^S \phi\|_2$. Equation (12.31) embeds $\mathcal{H}$ in L_{loc}^2 , and the locally bounded connection then embeds it in H_{loc}^1 . Both $\nabla^S : \mathcal{H} \rightarrow L^2$ and $D : \mathcal{H} \rightarrow L^2$ are well-defined, and Equation (12.28) makes the latter an isometry. Its range is therefore closed. Equation (12.30) also passes to this completion.

Choose an arbitrary constant spinor z in the ordinary orthonormalized coordinate frame on the distinguished end. Let ψ_0 equal z near infinity and vanish before a slightly smaller end neighborhood. Because the connection there is $O(r^{-2-\epsilon})$, $\nabla^S \psi_0, D\psi_0 \in L^2$. Orthogonally project $D\psi_0$ onto the closed subspace $D\mathcal{H}$ and choose $\eta \in \mathcal{H}$ such that

$$D\eta = -\text{proj}_{D\mathcal{H}}(D\psi_0).$$

Set $\psi = \psi_0 + \eta$, $w = D\psi$. Then $w \in L^2$ and

$$\int \langle w, D\phi \rangle \, \mathrm{d}V_{\mathfrak{g}} = 0 \quad \text{for every compact smooth } \phi.$$

Thus $Dw = 0$ weakly. Local elliptic regularity for a first-order elliptic operator with locally Lipschitz principal coefficients and bounded lower-order coefficients gives $w \in H_{\text{loc}}^1$. This regularity can also be seen by freezing the principal coefficients on small coordinate balls: the constant-coefficient Dirac estimate controls one derivative in L^2 , the oscillation is absorbed, and commutators of mollification are bounded by the local Lipschitz norm. Apply the estimate to regularized w and pass to a weak limit. The metric in the lemma has more than this coefficient regularity.

Completeness supplies compactly supported Lipschitz distance cutoffs χ_R , equal to one on the radius- R ball and with $|\mathrm{d}\chi_R| \leq C/R$. Indeed a complete locally compact Riemannian length space is proper, and a piecewise linear cutoff of its distance has these properties. Equation (12.28) applies to $\chi_R w$, and the weak equation gives

$$\|\nabla^S(\chi_R w)\|_2^2 = \|D(\chi_R w)\|_2^2 = \|c(\mathrm{d}\chi_R)w\|_2^2 \leq \frac{C}{R^2} \|w\|_2^2.$$

On every fixed compact set the cutoff eventually equals one. It follows that $\nabla^S w = 0$. A parallel spinor has constant length on the connected manifold; the distinguished end has infinite volume, so $w \in L^2$ forces $w = 0$. No condition at another end entered this argument.

It remains to show that ψ , rather than just $D\psi$, is parallel. Define the continuous sesquilinear form

$$\mathcal{B}(a, b) = \int \left(\langle \nabla^S a, \nabla^S b \rangle - \langle Da, Db \rangle \right) dV_g$$

on fields for which the indicated derivatives are in L^2 . For compact ϕ , the polarized Lichnerowicz identity, with one slot compactly supported, gives $\mathcal{B}(\psi_0, \phi) = 0$. Approximate η in $\mathcal{H}$ by compact spinors. Continuity and Equation (12.28) imply

$$\mathcal{B}(\psi_0, \eta) = 0, \quad \mathcal{B}(\eta, \eta) = 0, \quad \mathcal{B}(\psi, \psi) = \mathcal{B}(\psi_0, \psi_0).$$

The last expression is zero. To see this directly, integrate the Lichnerowicz identity for ψ_0 up to a large end sphere. There is no contribution at another end because ψ_0 vanishes there. On that sphere $\psi_0 = O(1)$ and $\nabla^S \psi_0, D\psi_0 = O(r^{-2-\epsilon})$; the boundary integral is $O(r^{-\epsilon})$ and tends to zero. This is also the zero ADM boundary term of the spin proof. Since $D\psi = w = 0$, we conclude that $\|\nabla^S \psi\|_2^2 = 0$.

For $z \neq 0$ the resulting parallel spinor cannot be zero. Otherwise $\eta = -\psi_0$, contradicting Equation (12.30), because a nonzero constant spinor has infinite r^{-2} -weighted L^2 norm on the end. The construction is linear in z , so it produces a full complex basis of parallel spinors at each point. Their parallel equations first imply $C^{1,\beta}$ regularity locally for some $\beta > 0$: from $H^1 \subset L^6$ and the bounded connection one obtains $W^{1,6}$, hence local Hölder continuity. The connection coefficients are locally Hölder by the metric regularity, so the parallel equations make the first derivatives locally Hölder as well. Their curvature equations hold weakly. The spin representation is faithful on the Lie algebra $\mathfrak{spin}(3)$, so a full parallel spinor frame makes the Riemann curvature zero. Equivalently, its spinor bilinears give a global parallel orthonormal tangent frame. The dual coframe is closed by torsion-freeness and integrates locally to coordinates in which the metric is exactly Euclidean. This also removes any apparent metric singularity at the compactified point.

Finally, completeness makes the universal flat cover Euclidean $\mathbb{R}^3$. The global parallel frame makes its deck transformations translations. A nontrivial discrete translation group has rank at least one, and a quotient by it has volume growth at most quadratic. The distinguished AE end gives a cubic lower bound: radial paths in that end put a Euclidean annulus of radius comparable to R inside a metric ball of radius CR . The two volume bounds are incompatible. The deck group is therefore trivial, proving the lemma. $\square$

Lemma 12.11 (The infinity component has no interior missing boundary). *The component $\mathcal{E}$ is all of Ω . In particular $N > 0$ everywhere in the open exterior and the stationary metric is vacuum there.*

Proof. The complete base of Lemma 12.7 reaches every component of S . Take its complete two-copy double, use h_+ on the plus copy and h_- on the minus copy, and identify corresponding boundary points. The signed reflected lapse expresses the joined metric as

$$q = \left(\frac{1 + \lambda_{\text{signed}}}{2} \right)^4 h_d.$$

It is smooth at each positive-lapse join and C^2 at each zero-lapse join. The matched second jets make its scalar curvature zero across every join, as well as on the two open sides. Compactify the unique AF end of the minus copy by Lemma 12.9.

We verify completeness of this conformal manifold without placing any restriction on the number or regularity of the possible internal-null ends. On the plus copy the factor $(1 + \lambda)/2$ is at least $1/2$. Fix a sufficiently large original AF tail. On its complement the continuous

function $\sqrt{N}$ is defined on a compact subset of the original closed exterior, equals zero on S and the internal zero set, and is strictly less than one elsewhere by Lemma 12.7. Its maximum on this compact set is therefore some $a_0 < 1$. Thus on the entire corresponding minus remainder $(1 - \lambda)/2 \geq (1 - a_0)/2 > 0$ uniformly. On the double away from the minus AF tail the conformal metric is consequently bounded below by a fixed positive multiple of the complete doubled base metric. All joins are already attached. A finite-length escape can therefore only run down the minus AF tail, and its proved nondegenerate one-point compactification supplies its limit. More explicitly, truncate that tail at a large sphere. The other side of the sphere is complete with compact boundary by the uniform lower bound, and the compactified tail is a compact metric region with that same boundary; their gluing is complete.

The conformal manifold is connected, orientable, and without boundary. It has local $C^{1,\alpha} \cap W^{2,p}$ regularity for some $\alpha > 0$ and $p > 3$: the finitely many joins are at least C^2 , and the sole compactification point has the regularity established in Lemma 12.9. Its scalar curvature is zero distributionally, including at that point, and its distinguished plus end has the fast decay required by Lemma 12.10. That lemma makes this entire manifold Euclidean space.

Choose a sufficiently large sphere in the distinguished plus end. Under the Euclidean isometry it is a compact embedded sphere. Its end tail is the unbounded component of the complement: paths from large end radii to that sphere have lengths tending to infinity, and the tail has precisely that sphere as frontier. The other component is bounded and has compact closure. A sequence in $\mathcal{E}$ approaching an interior point with $N = 0$ lies outside the chosen end tail and escapes every compact set of the complete conformal manifold. Indeed it cannot converge at an ordinary plus point by continuity of N , cannot converge on any joined component because every original boundary collar has no interior zero, and cannot converge at a point in the open minus copy or its compactified end. Such a sequence is impossible in the compact Euclidean complement. Thus $\mathcal{E}$ has no interior boundary in Ω .

It is a nonempty open and closed subset of connected Ω , and hence equals Ω . Vacuum now follows from Proposition 11.1, since $N > 0$ everywhere in Ω . $\square$

Lemma 12.12 (Equality forces a single spherical boundary component). *In the Euclidean conformal double, every original component S_i is a round sphere of radius $a = (8\kappa)^{-1}$ and has original area $|S_i|_g = 16\pi m^2 = |S|_g$. Consequently $\ell = 1$.*

Proof. The completed conformal double in the proof of Lemma 12.11 is Euclidean. In particular its plus metric $q = h_+$ is flat up to every original boundary component. Put $\phi = (1 + \lambda)/2$. On each S_i , Lemma 12.7 gives

$$\phi = \frac{1}{2}, \quad \partial_{\nu_h} \log \phi = \kappa, \quad \Pi_h = 0, \quad h|_{TS_i} = g|_{TS_i}.$$

The conformal second-fundamental-form formula, with normal pointing into the plus side, is therefore

$$\Pi_q = \phi^2 (\Pi_h + 2\partial_{\nu_h} \log \phi h|_{TS_i}) = 8\kappa q|_{TS_i}, \quad q|_{TS_i} = \frac{1}{16} g|_{TS_i}.$$

Let F_i be the Euclidean position vector on S_i and ν_i its unit normal into the plus side. Its shape operator is $8\kappa I$, so the tangential differential of $F_i - a\nu_i$ is zero for $a = (8\kappa)^{-1}$. Connectedness makes $F_i - a\nu_i$ a constant point c_i . Thus the image lies on the sphere of radius a centered at c_i , with ν_i its outward radial normal. The image is open in that sphere by local invertibility and closed by compactness, hence is the whole sphere. The Euclidean embedding therefore identifies S_i diffeomorphically with it. Since $q|_{TS_i} = g|_{TS_i}/16$, its area satisfies

$$(12.32) \quad |S_i|_g = 16|S_i|_q = 64\pi a^2 = \frac{\pi}{\kappa^2} = 16\pi m^2 = |S|_g.$$

Summing over the finite family yields $|S|_g = \sum_{i=1}^{\ell} |S_i|_g = \ell |S|_g$. The boundary is nonempty and $|S|_g > 0$, hence $\ell = 1$. In particular no connectedness or spherical topology assumption was needed before this step. $\square$

Lemma 12.13 (Identification of the base and its mass). *The regular completion of (Ω, h, λ) is the Schwarzschild static exterior*

$$(12.33) \quad h = \left(1 + \frac{a}{\rho}\right)^4 \delta, \quad \lambda = \frac{1 - a/\rho}{1 + a/\rho}, \quad \rho \geq a,$$

where $a = m/2$. The completion is homeomorphic to the original M_{ext} , and S is identified diffeomorphically with its horizon sphere in the regular base boundary structure.

Proof. Lemma 12.12 identifies the boundary in the Euclidean conformal double as one round sphere of radius $a = (8\kappa)^{-1}$. The plus side is its outside: its normal is the outward radial normal, its entire boundary is this sphere, and it contains the distinguished unbounded end. Center Euclidean coordinates at this sphere and write ρ for their radius.

In the resulting Euclidean coordinates, write $\psi = 2/(1 + \lambda)$, so $h = \psi^4 \delta$. The scalar-flat conformal equation gives $\Delta_\delta \psi = 0$ outside the sphere, with $\psi = 2$ on it and $\psi \rightarrow 1$ at infinity. The function $1 + a/\rho$ has the same data. Their difference vanishes by the exterior maximum principle. This proves Equation (12.33). Its Schwarzschild mass is $2a = (4\kappa)^{-1} = m$, since $\kappa = 1/(4m)$. In particular $a = m/2$.

The regular base collar has the same underlying boundary topology as the original collar, even when its coordinate is $\lambda = \sqrt{N}$ instead of the original defining function N . Together with the unchanged interior this identifies its completion homeomorphically with M_{ext} . An interior base isometry extends uniquely to the metric completions. On the regular boundary structures it is smooth: its boundary restriction preserves induced lengths, hence is a smooth boundary isometry, and normal geodesic coordinates extend it smoothly to a collar. Thus the completion has the claimed global topology, and $\Omega \cong (a, \infty) \times S^2$ is simply connected. $\square$

12.5. The time graph and the Kruskal attachment.

Proof of Proposition 12.1. Lemma 12.12 proves $\ell = 1$ and the spherical topology of S . Lemma 12.11 proves $N > 0$ on Ω , and Lemma 12.13 gives the global static base with mass m . For the reconstruction write $M = m$, so $\kappa = 1/(4M)$. Simple connectivity and Lemma 12.5 give a global smooth function H with

$$(12.34) \quad dH = -A.$$

In static Schwarzschild coordinates use time $T = t + H(x)$. Equations (12.4) and (12.34) give the exact identity

$$(12.35) \quad -N dT^2 + h = -u^2 dt^2 + g_{ij} (dx^i + X^i dt)(dx^j + X^j dt).$$

The graph $T = H(x)$, corresponding to $t = 0$, therefore induces the original metric g . The stationary future unit normal is $n = (\partial_t - X)/u$. With the second-fundamental-form convention of the theorem,

$$(12.36) \quad K_{\text{graph}} = \frac{1}{2u} (\partial_t g - \mathcal{L}_X g) = -\frac{1}{2u} \mathcal{L}_X g = K.$$

This calculation recovers the original K , including a non-time-symmetric slice.

We next establish smoothness in the original boundary structure. If $u|_S = 0$, Equation (12.7) gives

$$A = \frac{Y^\flat}{a^2 - s^2 |Y|_g^2}.$$

This is smooth in the original collar. The primitive H therefore extends smoothly and finitely to S . For instance, fix its values on one interior collar section and integrate its smooth normal derivative to the boundary. Tangential derivatives of this extension agree with those prescribed by $dH = -A$, by continuity from the interior.

If $u|_S > 0$, the numerator $X^\flat - (2\kappa)^{-1} dN$ vanishes as a one-form on S , by Equation (12.6). Smooth division by the defining function N gives

$$(12.37) \quad A = \frac{1}{2\kappa} d \log N + \beta, \quad H = -\frac{1}{2\kappa} \log N + B,$$

where β and B are smooth in the original collar. The smoothness of B follows by applying the same normal-integration argument to its differential $-\beta$. In this case $H \rightarrow +\infty$ at the boundary, so static time itself is not a regular attachment coordinate.

Let r be the Schwarzschild area radius. On the whole exterior

$$N = 1 - \frac{2M}{r}, \quad r = \frac{2M}{1 - N}.$$

Normalize the tortoise coordinate and Kruskal coordinates by

$$(12.38) \quad \begin{aligned} r_* &= r + 2M \log \left(\frac{r}{2M} - 1 \right), \\ U &= -e^{-\kappa(T-r_*)}, \quad V = e^{\kappa(T+r_*)}. \end{aligned}$$

On the right exterior $U < 0, V > 0$, and the Schwarzschild metric extends as

$$g_M = -\frac{32M^3}{r} e^{-r/(2M)} dU dV + r^2 d\omega^2.$$

An exact useful form of Equation (12.38) is

$$(12.39) \quad U = -Q(N)\sqrt{N}e^{-\kappa T}, \quad V = Q(N)\sqrt{N}e^{\kappa T}, \quad Q(N) = \frac{\exp(1/[2(1-N)])}{\sqrt{1-N}}.$$

The function Q is smooth and positive near zero, with $Q(0) = \sqrt{e}$.

If $u|_S = 0$, both H and $\sqrt{N} = s\sqrt{a^2 - s^2|Y|^2}$ are smooth. Substitution into Equation (12.39) gives $U = V = 0$ on S and

$$\partial_s U|_S = -\sqrt{e} \kappa e^{-\kappa H|_S} < 0, \quad \partial_s V|_S = \sqrt{e} \kappa e^{\kappa H|_S} > 0.$$

This is a smooth attachment at the bifurcation sphere. If $u|_S > 0$, Equation (12.37) instead gives

$$(12.40) \quad U = -Q(N)Ne^{-\kappa B}, \quad V = Q(N)e^{\kappa B}.$$

These are smooth in the original collar; U vanishes simply and $V|_S = \sqrt{e} e^{\kappa B|_S} > 0$. Thus the attachment is on the future horizon, with each generator met once after the angular identification.

The angular regularity requires a further check when $u|_S > 0$, because the regular base coordinate was λ , not the original N . On the smooth reflected collar in Equation (12.20), normal projection onto its fixed boundary is invariant under $\lambda \mapsto -\lambda$. This follows from uniqueness of the normal geodesics and reflection invariance of the metric. Schwarzschild angular coordinates are constant along these normal geodesics, and their boundary values are the smooth round-sphere identification from Lemma 12.13. They are therefore smooth even functions of λ . A smooth even function, with smooth tangential parameters, is a smooth function of $N = \lambda^2$ for $N \geq 0$: all odd Taylor coefficients vanish, and repeated application of $(2\lambda)^{-1}\partial_\lambda$ to the Taylor remainder gives continuous derivatives of every order at zero. This proves the required angular smoothness in the original collar. In the zero boundary-lapse case the original collar is already a smooth base collar, so its completed base isometry supplies the angular extension directly.

We have constructed a smooth map $\iota : M_{\text{ext}} \rightarrow \text{Schwarzschild}$. Equation (12.35) extends to S by continuity, hence $\iota^* \mathbf{g}_M = g$ there as well. The positive definiteness of g proves that $d\iota$ is injective at every boundary point. Interior injectivity follows from the global base isometry; boundary injectivity follows from the sphere identification. The boundary image, at $r = 2M$, is disjoint from the open image, where $r > 2M$. The map is proper: a compact subset of Schwarzschild has bounded area radius, and its inverse image is closed in the compact base region corresponding to $[2M, R] \times S^2$. Thus the injective immersion is an embedding.

The future normal extends smoothly even where $u = 0$. To avoid taking a limit of the expression involving u^{-1} , choose the smooth future timelike Kruskal vector $Z = \partial_U + \partial_V$ near the boundary. Let $Z^\top$ be its tangential projection along the embedded spacelike hypersurface and set

$$\hat{n} = \frac{Z - Z^\top}{\sqrt{-\mathbf{g}_M(Z - Z^\top, Z - Z^\top)}}.$$

The denominator is positive, and this is a smooth future unit normal. On the open exterior it agrees with the normal used in Equation (12.36). Its second fundamental form is smooth and equals K in the interior, so equality extends to S .

There is also a smooth spacelike extension of the image through its boundary. In the positive boundary-lapse case, use (U, ω) as hypersurface coordinates; its normal derivative is nonzero by Equation (12.40). The remaining null coordinate is a smooth graph function and can be extended across $U = 0$. In the zero boundary-lapse case use (χ, ω) , where $\chi = (V - U)/2$; the displayed normal derivatives give $\partial_s \chi > 0$. Again extend the remaining graph function across zero. Smooth one-sided graph functions extend across a smooth boundary, and compactness of S , together with openness of the spacelike condition, preserves spacelikeness on a sufficiently short collar. This extension makes no assertion about the original data behind S .

Finally, the exact inverse-metric identity gives

$$(12.41) \quad |dH|_h^2 = \left| \frac{X^b}{N} \right|_C^2 = \frac{|X|_g^2}{u^2 N} \rightarrow \frac{|b|^2}{(b^0)^2} < 1.$$

Choose a constant $a_0 < 1$ larger than the limiting slope norm. Along radial rays in the Schwarzschild base, uniformly in angle,

$$|H(r, \omega)| \leq C + a_0 \int_R^r \frac{d\rho}{\sqrt{1 - 2M/\rho}} = a_0 r + O(\log r).$$

Since $r_* = r + O(\log r)$, it follows that

$$H - r_* \rightarrow -\infty, \quad H + r_* \rightarrow +\infty.$$

This is approach to the chosen Schwarzschild spatial infinity. It allows a nonzero asymptotic tilt and does not impose $P = 0$ or $K = 0$. All conclusions of Proposition 12.1 follow. $\square$

Propositions 11.1 and 12.1 prove Theorem 10.2.

Part 3. Charge

13. THE CHARGED UPPER-AREA INEQUALITY

We now extend the neutral exterior inequality to divergence-free electric and magnetic fields with neutral exterior matter satisfying the physical dominant energy condition. In the presence of charge, the numerical statement is an upper bound for the area radius,

$$(13.1) \quad r \leq m + \sqrt{m^2 - Q^2}, \quad m \geq Q,$$

where Q is the magnitude of the total charge and m is the invariant ADM mass. For disconnected horizons, it is essential to retain this upper-bound formulation: the familiar expression $(r + Q^2/r)/2$ is not increasing throughout the positive r -axis. Khuri–Weinstein–Yamada developed a charged conformal-flow approach for multiple horizon components [34]; the distinction between the two area branches is already visible in the counterexamples of Weinstein–Yamada [52].

For non-time-symmetric data, Disconzi–Khuri developed electric graph transport preserving charge and divergence, and a reduction conditional on a coupled Jang–inverse-mean-curvature-flow system [26]. We combine the neutral construction above with the charged numerical theorem of the companion paper [44, Theorem 2.3]. Proposition 14.4 transfers the one-sided exterior inequality to the charged notation; Proposition 17.3 transfers the elliptic construction and its estimates. Their proofs identify the precise neutral results and verify the normalizations. The charged preparation satisfies the elliptic hypotheses by Lemma 17.2.

Theorem 14.2 also excludes a non-timelike ADM vector for a nonempty smooth compact obstacle. Physical matter DEC implies total DEC, so the endpoint corollary already proved for neutral exteriors applies; Lemma 15.2 verifies its hypotheses here.

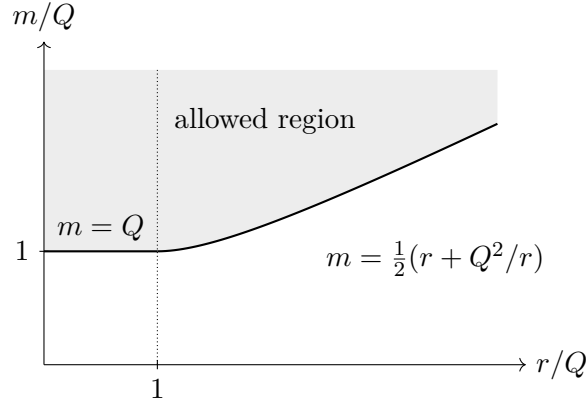


FIGURE 2. The charged upper-area inequality for $Q > 0$. For $r \leq Q$, it requires only $m \geq Q$. For $r \geq Q$, its boundary is $m = (r + Q^2/r)/2$. The mass-lower-bound expression must therefore be restricted to this latter branch.

Two ingredients carry the charge through the neutral construction. First, we represent the fields by closed flux two-forms and keep those forms through an annular replacement, a change to a rest end, and conformal repairs. This produces strict data whose rest energies approach the original invariant mass and whose enclosing areas approach the original infimum. The quantitative repair uses only the stated weak asymptotic derivative bounds on the original data. Second, after aligning the total charge by a constant electromagnetic rotation, a pointwise square completion controls the graph norm of one closed two-form. The other rotated form has zero total charge. This single-form comparison accommodates the electromagnetic momentum density and makes the original neutral elliptic system sufficient.

The proof proceeds through four geometric stages. Section 15 constructs strict charged rest data. Section 16 prepares their exterior boundary with a positive matter margin. Sections 17–19 use the specified neutral system to produce Riemannian comparison metrics with the same total charge, no smaller enclosing area, and a controlled energy. Section 20 derives their Riemannian endpoint from the companion charged numerical theorem and removes the parameters in their prescribed order.

14. EXTERIORS, FLUX FORMS, AND CONFORMAL CHANGES

Throughout this part, the spatial dimension is three, $G = c = 1$, and the cosmological constant is zero. Data are smooth up to every specified boundary. A subscript on a norm specifies its metric when needed. For a two-sided surface, the designated unit normal ν points into the exterior toward the chosen end, and

$$H = \operatorname{div}_\Sigma \nu, \quad \theta_+ = H + \operatorname{tr}_\Sigma K.$$

Thus outward Euclidean spheres have positive mean curvature. The spacetime convention is $K(Y, Z) = \mathbf{g}(\nabla_Y n, Z)$ for the future unit normal n .

Definition 14.1 (Charged exterior). A charged exterior consists of a connected orientable smooth three-manifold N with nonempty compact smooth boundary S , a Riemannian metric g complete with S included, a symmetric covariant tensor K , and vector fields $\mathcal{E}, \mathcal{B}$. There is exactly one end. In coordinates x on that end, with $r_x = |x|$, for some $q > 1/2$,

$$(14.1) \quad g_{ij} - \delta_{ij} = O_2(r_x^{-q}), \quad K_{ij} = O_1(r_x^{-1-q}), \quad \mathcal{E}^i, \mathcal{B}^i = O_1(r_x^{-2}).$$

The notation $O_k(r^{-b})$ bounds coordinate derivatives of order $j \leq k$ by Cr^{-b-j} . Define the total constraint densities by

$$(14.2) \quad 16\pi\mu = R_g + (\operatorname{tr}_g K)^2 - |K|_g^2, \quad 8\pi J_i = \nabla^j (K_{ij} - (\operatorname{tr}_g K)g_{ij}).$$

The functions μ and $|J|_g$ are integrable. The matter densities are

$$(14.3) \quad \mu_m = \mu - \frac{|\mathcal{E}|_g^2 + |\mathcal{B}|_g^2}{8\pi}, \quad J_m = J - \frac{(\mathcal{E} \times \mathcal{B})^b}{4\pi},$$

where the cross product uses the orientation and metric. They satisfy

$$(14.4) \quad \mu_m \geq |J_m|_g, \quad \operatorname{div}_g \mathcal{E} = \operatorname{div}_g \mathcal{B} = 0.$$

The boundary satisfies $\theta_+(S) \leq 0$. We define ADM energy and momentum by the following finite limits [10]:

$$(14.5) \quad E = \frac{1}{16\pi} \lim_{R \rightarrow \infty} \int_{|x|=R} (\partial_j g_{ij} - \partial_i g_{jj}) n_\delta^i dA_\delta,$$

$$(14.6) \quad P_i = \frac{1}{8\pi} \lim_{R \rightarrow \infty} \int_{|x|=R} (K_{ij} - (\operatorname{tr}_g K)g_{ij}) n_\delta^j dA_\delta,$$

$$(14.7) \quad Q_E = \frac{1}{4\pi} \lim_{R \rightarrow \infty} \int_{|x|=R} g(\mathcal{E}, \nu_R) dA_g, \quad Q_B = \frac{1}{4\pi} \lim_{R \rightarrow \infty} \int_{|x|=R} g(\mathcal{B}, \nu_R) dA_g.$$

In the first two formulas the normal and area element are Euclidean; in the last two, ν_R is the outward g -unit normal. No extension of the physical data across S is required.

14.1. Filled enclosing cuts. A smooth enclosing cut Γ is a compact embedded surface separating all of S from the distant end. Its inner side contains the obstacle, and bounded complementary pockets are filled. Components of Γ may coincide with components of S . The frontier so obtained, including any coincident part, is counted in

$$(14.8) \quad a_g(S) = \inf_\Gamma \operatorname{Area}_g(\Gamma).$$

All cuts are oriented toward the end. They need not be trapped or minimal, and an individual component need not separate S from infinity by itself.

14.2. The charged bound.

Theorem 14.2. *Let $(N, g, K, \mathcal{E}, \mathcal{B})$ be a charged exterior in the sense of Definition 14.1, with inner boundary S , ADM energy and momentum (E, P) , and total charges (Q_E, Q_B) . Then $E > |P|$. Setting*

$$m = \sqrt{E^2 - |P|^2}, \quad Q = \sqrt{Q_E^2 + Q_B^2}, \quad r = \sqrt{a_g(S)/(4\pi)},$$

we have

$$m \geq Q, \quad r \leq m + \sqrt{m^2 - Q^2}.$$

Here $a_g(S)$ is the minimum enclosing area, defined as an infimum over all smooth filled enclosing cuts. Neither the inner boundary nor a competing cut is required to be connected. The boundary condition is only $H + \text{tr}_S K \leq 0$, with the normal pointing toward infinity. The electromagnetic fields are divergence free, and the remaining matter satisfies the dominant energy condition. The statement places no maximality, symmetry, or zero-momentum restriction on the initial data.

Lemma 14.3. *For every exterior as above, $0 < a_g(S) < \infty$. If the infimum is instead defined using filled finite-perimeter inner sets that retain the obstacle and count its coincident frontier, its value is unchanged.*

Proof. After forgetting K and the fields, Definition 14.1 gives an exterior in Definition 2.1, with the same boundary S . Lemma 9.3 therefore gives $0 < a_g(S) < \infty$.

For the perimeter assertion, apply Lemma 3.3 to the constant sequence $g_j = g$. Local convergence is immediate. The falloff in Equation (14.1) gives uniform end bounds $c\delta \leq g \leq C\delta$ and $r_x|\partial g| \leq C$, and a fixed large coordinate sphere is a compact enclosing competitor of finite area. The lemma consequently identifies the smooth and perimeter infima. Its competitors are the same bounded filled inner sets used here: they retain the entire obstacle, bounded complementary pockets are filled, and the full frontier, including contact with S , is counted. Its outward smoothing preserves enclosure and proves the claimed equality of infima. The auxiliary filling in that perimeter argument requires no extension of the physical data or energy condition. $\square$

14.3. The neutral exterior inequality in this notation. We first record the form of Theorem 2.2 used below. The transfer retains arbitrary compact boundary topology and a potentially disconnected boundary.

Proposition 14.4 (Neutral exterior inequality). *Let (N, g, K) be a smooth connected orientable one-ended exterior, complete with its nonempty compact smooth boundary S included. Suppose the metric and tensor satisfy the first two bounds in (14.1), the densities (14.2) are integrable as above, the ADM limits (14.5)–(14.6) are finite, and $\mu \geq |J|_g$ throughout N . If $H + \text{tr}_S K \leq 0$ and $E > |P|$, then*

$$\sqrt{E^2 - |P|^2} \geq \sqrt{a_g(S)/(16\pi)}.$$

No complete fill-in across S is part of the hypothesis.

Proof. Apply Theorem 2.2 with its exterior $\Omega = N$. Definition 14.1, after the field requirements are omitted, uses the same smoothness, orientation, one-endedness and completeness convention as Definition 2.1. Equations (14.2), (14.5) and (14.6) are exactly its constraint and ADM conventions. The decay, integrability, weak expansion and future timelike hypotheses in the proposition are precisely those of that theorem. The class of filled enclosing cuts is also identical: all boundary components are enclosed, bounded pockets are filled, and coincident obstacle frontier is counted. Hence its enclosing infimum is $a_g(S)$, giving the displayed inequality. The theorem is one-sided and requires no extension across S . $\square$

14.4. Flux forms and a conformal inequality. Put $X_1 = \mathcal{E}$, $X_2 = \mathcal{B}$, and

$$(14.9) \quad \alpha_i = \iota_{X_i} dV_g.$$

The divergence equations are $d\alpha_i = 0$. Their fluxes across every filled enclosing cut are $4\pi Q_E$ and $4\pi Q_B$. In every change of metric below, retaining a flux form means defining its new vector field by contraction with the new volume form. This preserves closedness and all fluxes without choosing a vector potential.

For $|Y|_g \leq 1$, define

$$(14.10) \quad I_g(Y) = |X_1|_g^2 + |X_2|_g^2 + 2\langle X_1 \times X_2, Y \rangle_g,$$

$$(14.11) \quad D_g(Y) = 8\pi(\mu + J(Y)) - I_g(Y).$$

Then

$$(14.12) \quad I_g(Y) \geq 0, \quad \inf_{|Y|_g \leq 1} D_g(Y) = 8\pi(\mu_m - |J_m|_g) =: M_0.$$

The first assertion uses $2|X_1 \times X_2| \leq |X_1|^2 + |X_2|^2$. The second follows by minimizing the linear term $8\pi J_m(Y)$ over the unit ball. Both I_g and D_g are invariant under a constant $SO(2)$ rotation of the pair of forms. The same elementary inequality and the triangle inequality imply

$$(14.13) \quad 8\pi(\mu - |J|_g) \geq M_0.$$

Lemma 14.5 (Conformal charged comparison). *Let $u \geq 1$ be smooth, and set $g' = u^4 g$, $K' = u^2 K$, retaining the two flux forms. For every $|Y|_g \leq 1$,*

$$(14.14) \quad u^4 D_{g'}(u^{-2}Y) \geq D_g(Y) + 4u^{-1}(-\Delta_g u - |K|_g |du|_g).$$

At a fixed oriented surface,

$$(14.15) \quad \theta'_+ = u^{-2}(\theta_+ + 4\partial_\nu \log u).$$

Here $\Delta_g = \operatorname{div}_g \nabla$.

Proof. The conformal scalar-curvature formula and the scaling of the tensor terms in (14.2) give

$$16\pi u^4 \mu' = 16\pi \mu - 8u^{-1} \Delta_g u.$$

For $f = \log u$, the difference of the two connections is

$$\nabla'_A B - \nabla_A B = 2(df(A)B + df(B)A - g(A, B)\nabla f).$$

Taking the divergence of the rescaled trace reversal $u^2(K - (\operatorname{tr}_g K)g)$ therefore yields

$$8\pi u^2 J' = 8\pi J + 4K(\nabla \log u, \cdot).$$

The new field vectors are $X'_i = u^{-6}X_i$, so their components in orthonormal frames scale by u^{-4} and $I_{g'}(u^{-2}Y) = u^{-8}I_g(Y)$. Consequently

$$u^4 D_{g'}(u^{-2}Y) = D_g(Y) + (1 - u^{-4})I_g(Y) + 4u^{-1}(-\Delta_g u + K(\nabla u, Y)).$$

Dropping the nonnegative second term and estimating the last tensor contraction proves (14.14). Finally, $H' = u^{-2}(H + 4\partial_\nu \log u)$ and $\operatorname{tr}_{\Sigma}^{g'} K' = u^{-2} \operatorname{tr}_{\Sigma}^g K$, which give (14.15). $\square$

15. CHARGE-PRESERVING REDUCTION TO STRICT REST DATA

Throughout this section the closed forms α_1, α_2 are fixed. For each new metric they define new electromagnetic vector fields by contraction with its volume form. Thus every change below preserves both Maxwell constraints and both signed charges. We first apply the neutral endpoint corollary to establish timelikeness of the original data.

Proposition 15.1 (Strict rest reduction). *For a charged exterior as in Definition 14.1, $E > |P|$ and $m = (E^2 - |P|^2)^{1/2} > 0$. There are smooth data (g_j, K_j) on the same exterior, with the same flux forms, satisfying the following properties.*

- (1) *The physical matter condition is strict everywhere, the original boundary S has $\theta_{+,j} < 0$, and the constraint densities remain integrable. For one fixed $0 < \delta < 1$ the matter margin satisfies*

$$M_{0,j} = 8\pi(\mu_{m,j} - |J_{m,j}|_{g_j}) \geq c_j r_y^{-3-\delta} \quad \text{far out}, \quad c_j > 0.$$

- (2) *In a rest chart y , K_j is compactly supported and*

$$g_j - \delta_{\text{eucl}} = O_k(r_y^{-1}), \quad R_{g_j} = O_k(r_y^{-3-\delta}) \quad \text{for every } k.$$

The fields have the required $O_1(r_y^{-2})$ decay. In particular the ADM momentum is zero and $\text{Ric}_{g_j} = O(r_y^{-3})$.

- (3) *Smooth convergence holds on every compact set, including at S , and*

$$(15.1) \quad E_j \longrightarrow m, \quad a_{g_j}(S) \longrightarrow a_g(S), \quad (Q_{E,j}, Q_{B,j}) = (Q_E, Q_B).$$

Moreover $g_j \geq (1 - \varepsilon_j)g$ globally for some $\varepsilon_j \downarrow 0$.

The proof has four steps: establish timelikeness from the neutral endpoint corollary; replace the distant end by a vacuum reference with matching ADM vector; bend and repair while retaining the flux forms; and make the inequalities strict by a separate elliptic construction. All quantitative estimates for the replacement use exactly the original O_2 metric and O_1 tensor assumptions.

15.1. Timelikeness.

Lemma 15.2. *The original ADM vector satisfies $E > |P|$.*

Proof. The physical matter condition and Equation (14.13) give

$$\mu - |J|_g \geq \mu_m - |J_m|_g \geq 0.$$

These are the actual total constraint densities of (g, K) . Definition 14.1 supplies smoothness, orientation, one-endedness, completeness with the nonempty compact boundary included, the required metric and tensor falloff, integrable total densities, finite ADM limits, and weak future trapping with the normal toward the end. The total constraint and ADM conventions are exactly those of Equations (2.1), (2.4) and (2.5). All hypotheses of Corollary 9.4 are therefore satisfied. Applying it to the unchanged data (N, g, K) yields $E > |P|$ for their original ADM vector. $\square$

15.2. A matching vacuum reference and annular replacement.

Lemma 15.3 (Boosted vacuum reference). *There is an auxiliary Schwarzschild end of mass m whose induced data (g_*, K_*) , in asymptotically Euclidean coordinates x , have ADM vector (E, P) and satisfy $g_* - \delta_{\text{eucl}} = O_k(r_x^{-1})$, $K_* = O_k(r_x^{-2})$ for every k .*

Proof. Lemma 15.2 gives $E > |P|$. The constraint and ADM conventions in Equations (14.2), (14.5), and (14.6) agree with those in Part I. Lemma 3.4 therefore gives the required reference end, including the stated differentiated decay and the future second-form convention.

For the later bend, retain its static isotropic coordinates:

$$(15.2) \quad \mathbf{g} = - \left(\frac{1 - m/(2r_y)}{1 + m/(2r_y)} \right)^2 dT^2 + \left(1 + \frac{m}{2r_y} \right)^4 |dy|^2.$$

The reference plane is $T = v \cdot y$, with $|v| < 1$ chosen to give (E, P) . Its asymptotically Euclidean coordinates are $y = Ax$, where $A = (I - vv^\top)^{-1/2}$, since $A^\top(I - vv^\top)A = I$. $\square$

For the remaining construction, fix $1/2 < \beta < \min(q, 1)$.

Lemma 15.4 (Annular replacement). *For sufficiently large R there are smooth data equal to (g, K) on $r_x \leq R$ and to (g_*, K_*) on $r_x \geq 4R$, with possible neutral DEC deficit bounded by $\eta_R R^{-3}$ in between, where $\eta_R \rightarrow 0$. The estimates use no derivatives beyond the assumed two metric derivatives and one tensor derivative.*

Proof. The original data satisfy neutral DEC by Equation (14.13). Definition 14.1 gives the smooth one-ended exterior, completeness with its compact boundary included, decay $g - \delta = O_2(r_x^{-q})$ and $K = O_1(r_x^{-1-q})$ with $q > 1/2$, integrable total constraint densities, and finite ADM limits. The total constraint and ADM normalizations agree with those used in Proposition 3.7. The reference in Lemma 15.3 has matching charges in the common x coordinates, and $E > |P|$ by Lemma 15.2. That proposition applies and gives

$$16\pi(\tilde{\mu}_R - |\tilde{J}_R|_{\tilde{g}_R}) \geq -\eta_R R^{-3}, \quad \eta_R \rightarrow 0,$$

with the stated exact agreement on both sides of the annulus and with quantitative estimates using only the original two metric derivatives and one tensor derivative. Absorbing the fixed normalization into η_R gives the assertion. The scaled construction in Equation (3.22) and the compact corrections in that proposition's proof also give

$$\|\tilde{g}_R(R \cdot) - I\|_{C^2(\{1 < |z| < 4\})} + \|R\tilde{K}_R(R \cdot)\|_{C^1(\{1 < |z| < 4\})} \leq CR^{-\beta}.$$

The two closed flux forms are retained throughout; their contribution to the physical matter deficit is estimated next. $\square$

15.3. Bending and repairing the physical matter condition. Beyond the annular replacement, use the exact Schwarzschild part and replace the plane by the graph

$$(15.3) \quad T_R(y) = (v \cdot y)\psi\left(\frac{\log(r_y/R_b)}{L_0}\right).$$

Here ψ is smooth, is one before zero and zero after one, and takes values in $[0, 1]$. The fixed multiple R_b/R is large enough that the bend begins beyond $r_x = 4R$. First fix L_0 so large that $|v|(1 + \|\psi'\|_\infty/L_0) < 1$. Then the graph is uniformly spacelike: in Equation (15.2) the spatial factor is greater than one and the lapse is less than one. Its constraints vanish by Gauss–Codazzi with the prescribed future normal.

Write (g_R, K_R) for the unscaled replacement and bend. Constants in the rest of this subsection may depend on the fixed original data and L_0 , but not on R . The graph formula and its first three derivatives give uniform metric comparability, $|\partial g_R| + |K_R| \leq C/r_y$ in the transition, and

$$(15.4) \quad c \leq |dr_y|_{g_R} \leq C, \quad |\Delta_{g_R} r_y| \leq C/r_y.$$

The same bounds hold on the gluing annulus, since $y = Ax$ is a fixed linear map. Outside a fixed multiple of R , the data are horizontal Schwarzschild and $K_R = 0$.

Keep the original closed forms in these coordinates. Their norms are $O(r_y^{-2})$, so their contribution to the negative part of $\inf_{|Y| \leq 1} D_{g_R}(Y)$ is $O(r_y^{-4})$. Choose fixed $0 < a < \xi_0 < b$ so that

the data are original for $r_y/R \leq \xi_0$ and horizontal Schwarzschild for $r_y/R \geq b$. Lemma 15.4 and vacuum of the reference and bend imply

$$(15.5) \quad [-\inf_{|Y|_{g_R} \leq 1} D_{g_R}(Y)]_+ \leq \begin{cases} 0, & r_y/R \leq \xi_0, \\ \eta_R R^{-3}, & \xi_0 \leq r_y/R \leq b, \\ Cr_y^{-4}, & r_y/R \geq b, \end{cases} \quad \eta_R \longrightarrow 0.$$

Thus this intermediate data set is not assumed to satisfy charged DEC.

Lemma 15.5 (Charged repair with vanishing energy cost). *There are smooth $u_R \geq 1$, constant on the unchanged core, for which $(u_R^4 g_R, u_R^2 K_R)$ with the fixed forms satisfy the physical matter condition. They retain weak trapping, have zero ADM momentum, and their energies tend to m . Moreover $\|u_R - 1\|_\infty = o(R^{-1})$, and the final metric has all-order $O(r_y^{-1})$ asymptotics and scalar curvature $R_{u_R^4 g_R} = O_k(r_y^{-4})$.*

Proof. Set $\xi = r_y/R$ and seek

$$u_R = 1 + \frac{e_R}{R} F_R(\xi), \quad F_R(\xi) = \int_\xi^\infty z_R(t) dt, \quad z_R \geq 0,$$

where $e_R \rightarrow 0$ will be chosen below. On $[a, b+1]$ initially solve $z'_R = C_0 z_R + \omega$, $z_R(a) = 0$, where $\omega \geq 0$ is smooth, zero near a , and at least one on $[\xi_0, b+1]$; extend $z_R = 0$ below a . Equation (15.4) yields

$$(15.6) \quad \begin{aligned} -\Delta_{g_R} u_R - |K_R|_{g_R} |du_R|_{g_R} &= \frac{e_R}{R^3} \{z'_R |dr_y|^2 + Rz_R \Delta r_y - Rz_R |K_R| |dr_y|\} \\ &\geq \frac{e_R}{R^3} (cz'_R - Cz_R). \end{aligned}$$

Choose C_0 large. This is nonnegative on $[a, b]$ and at least $c'e_R R^{-3}$ on $[\xi_0, b]$.

On the horizontal region put $A_R(t) = (t + m/(2R))^2 z_R(t)$. Its derivative from the preceding prescription is positive near b . Continue that derivative smoothly by a convex blend with t^{-2} over $[b, b+1]$, and integrate from $A_R(b)$. Thus $A'_R \geq c''t^{-2}$ for $t \geq b$ and $A'_R = t^{-2}$ eventually. Defining $z_R = A_R(t)/(t + m/(2R))^2$ there makes every join smooth. For fixed L_0 , $A_R^\infty = \lim_{t \rightarrow \infty} A_R(t)$ and F_R are bounded uniformly in R . The exact radial Laplacian on the horizontal slice is

$$(15.7) \quad -\Delta_{g_R} u_R = \frac{e_R}{R^3} \left(1 + \frac{m}{2R\xi}\right)^{-6} \xi^{-2} A'_R(\xi) \geq c''' e_R R^{-3} \xi^{-4}.$$

Choose, for example, $e_R = C_1(\eta_R + R^{-1})$, with a sufficiently large fixed C_1 . For large R , $1 \leq u_R \leq 2$. Equations (15.6)–(15.7) then dominate $u_R/4$ times the deficit in Equation (15.5). Equation (14.14) proves the physical matter condition. The factor is constant near S , so Equation (14.15) preserves its weak expansion.

At infinity, for each fixed R ,

$$u_R = 1 + \frac{e_R A_R^\infty}{r_y} + O_k(r_y^{-2}) \quad \text{for every } k.$$

The ADM energy is therefore $m + 2e_R A_R^\infty \rightarrow m$ and the momentum vanishes because K_R has compact support. The differentiated radial formula gives scalar curvature $O_k(r_y^{-4})$. This also verifies integrability of the repaired constraints. Completeness follows from the comparability of the intermediate metric and $u_R \geq 1$. $\square$

15.4. Strictification by an exterior elliptic problem.

Lemma 15.6 (A strict conformal direction). *Fix one repaired exterior (g_0, K_0) from Lemma 15.5, and fix $0 < \delta < 1$. There exist smooth positive w_0, ϕ_0 such that*

$$(15.8) \quad \partial_\nu \phi_0 = -1 \text{ on } S, \quad -\Delta_{g_0} \phi_0 - |K_0|_{g_0} |d\phi_0|_{g_0} \geq w_0,$$

where $w_0 = r_y^{-3-\delta}$ and $-\Delta_{g_0} \phi_0 = w_0$ far out, and $\phi_0 = O_k(r_y^{-1})$ for every k . Its gradient has finite flux at infinity. All constants in this lemma may depend on the fixed repaired exterior.

Proof. Choose a smooth nonnegative compactly supported majorant $b_0 \geq |K_0|_{g_0}$ and a smooth positive extension w_0 of the specified tail. We solve the regularized equation

$$(15.9) \quad -\Delta_{g_0} \phi = b_0 \sqrt{1 + |d\phi|_{g_0}^2} + w_0, \quad \partial_\nu \phi = -1, \quad \phi \rightarrow 0.$$

First truncate at $r_y = T$ and impose outer Dirichlet value zero. Multiply b_0 by $t \in [0, 1]$ for continuation. At $t = 0$ the mixed Laplacian is invertible, by the Dirichlet Poincaré inequality and elliptic regularity. Every linearization is a uniformly elliptic Laplacian with drift of norm at most b_0 , with homogeneous Neumann data on S and Dirichlet data on the outer sphere. These faces are disjoint. The maximum and Hopf principles give zero kernel; as a compact lower-order perturbation of the mixed Laplacian, it is Fredholm of index zero and is invertible.

We supply the estimates needed for closedness. On this fixed truncation, the $W^{2,p}$ estimate and gradient interpolation give, uniformly in t ,

$$\|\phi\|_{W^{2,p}} \leq C(1 + \|\phi\|_{L^p} + \|d\phi\|_{L^p}) \leq \frac{1}{2}\|\phi\|_{W^{2,p}} + C'(1 + \|\phi\|_\infty).$$

If the supremum norms were unbounded, divide by them and take $p > 3$. A subsequence converges in C^1 to a nonzero solution v with homogeneous mixed data and $-\Delta_{g_0} v = t_* b_0 |dv|$. Indeed the normalized right side is $tb_0 \sqrt{M^{-2} + |dv|^2} + w_0/M$ when $M = \|\phi\|_\infty$. The limit is a homogeneous bounded-drift equation, taking drift $t_* b_0 \nabla v / |dv|$ off the critical set and zero on it. The strong and Hopf maximum principles exclude such a nonzero v . The resulting supremum and $W^{2,p}$ bounds, followed by Schauder estimates for the smooth regularized equation, close continuation. Thus every sufficiently large truncation has a smooth solution.

All solutions are nonnegative: a negative minimum cannot be interior since the source is positive, or on S since the outward-domain derivative there is $-\partial_\nu \phi = 1$, contradicting the Hopf sign at a minimum. Fix a core containing $\text{supp } b_0$. On its exterior $f = r_y^{-1} - r_y^{-1-\delta}$ is positive and

$$-\Delta_{g_0} f = \delta(1 + \delta)r_y^{-3-\delta} + O(r_y^{-4}) \geq cw_0$$

after enlarging the fixed core. Comparison with a multiple of f gives, independently of T ,

$$(15.10) \quad 0 \leq \phi \leq C(1 + M_T)r_y^{-1} \quad \text{outside the core}, \quad M_T = \sup_{\text{core}} \phi.$$

If $M_T \rightarrow \infty$ along an exhaustion, divide by M_T . Local $W^{2,p}$ estimates, including the fixed inner Neumann boundary, and Equation (15.10) give a limit $v \geq 0$ with maximum one on the core, homogeneous inner Neumann condition, $-\Delta v = b_0 |dv|$, and $v \rightarrow 0$ at infinity. Its positive global maximum is attained, contradicting the strong or Hopf principle. Hence the core bounds are uniform in T . Diagonal compactness and bootstrap give a smooth global solution of Equation (15.9), positive also on S by the same minimum argument.

On the end its equation is the linear Poisson equation $-\Delta_{g_0} \phi_0 = w_0$. On a scaled annulus the function $\rho \phi_0(\rho \cdot)$ is uniformly bounded; the scaled metric has all-order bounds and the scaled source is $O(\rho^{-\delta})$ with all its derivatives. Interior Poisson estimates of each order therefore give $\phi_0 = O_k(r_y^{-1})$. The full right side of Equation (15.9) is integrable. The divergence theorem then gives a finite metric gradient flux, and its difference from the Euclidean flux is $O(r_y^{-1})$. This proves every assertion. $\square$

For $u = 1 + s\phi_0$, $s > 0$, use $(u^4 g_0, u^2 K_0)$ and the fixed forms. Equations (14.14) and (15.8) imply

$$(15.11) \quad M_{0,s} \geq 4su^{-5}w_0 > 0, \quad \theta_{+,s} = u^{-2}(\theta_{+,0} - 4s/u) < 0 \quad \text{on } S.$$

Compact support of K_0 persists. The conformal scalar formula gives the asserted all-order $r_y^{-3-\delta}$ scalar decay, and thus integrability. The ADM energy change is

$$(15.12) \quad E_s - E_0 = -\frac{s}{2\pi} \lim_{r \rightarrow \infty} \int_{r_y=r} \partial_{\nu_r} \phi_0 dA_{g_0} = O(s).$$

The metric and Euclidean fluxes agree in this formula by the estimates just proved. For the fixed repaired datum the finite coefficient and any finite collection of compact smooth norms can be made arbitrarily small by choosing s small; no bound uniform in the bending radius is needed.

15.5. Global area control and the diagonal limit.

Completion of the proof of Proposition 15.1. It remains to check area and organize the parameters. On the vacuum plane and bend, write $dT_R = \psi a + b$, where $a = v \cdot dy$ and $|b| \leq CL_0^{-1}|dy|$. Since $0 \leq \psi \leq 1$, $dT_R^2 \leq a^2 + CL_0^{-1}|dy|^2$ as quadratic forms. The spatial factor and lapse in Equation (15.2) consequently give

$$(15.13) \quad g_R \geq |dy|^2 - dT_R^2 \geq |dy|^2 - (v \cdot dy)^2 - \frac{C}{L_0}|dy|^2 \geq \left(1 - \frac{C'}{L_0}\right)|dx|^2.$$

The constants here depend on the fixed boost, not on R or large L_0 . In the gluing annulus $g_R = I + O(R^{-\beta})$, whereas the original metric is uniformly $I + o(1)$ throughout the receding end. On the unchanged core the metrics agree. Both conformal changes increase the metric. Thus first choosing $L_0 \rightarrow \infty$ and then $R \rightarrow \infty$ sufficiently fast gives the global comparison $g_j \geq (1 - \varepsilon_j)g$ with $\varepsilon_j \rightarrow 0$.

For each fixed L_0 the energy of the repaired data tends to m by Lemma 15.5. Select R to make this energy error, the constant core conformal change, and the preceding comparison error arbitrarily small. Finally choose the strictification parameter s_j to make Equation (15.12) tend to zero and to control the first j smooth norms on the first j sets of a compact exhaustion. This gives local smooth convergence and all the analytic assertions of the proposition. Each metric is complete up to S , by the global positive comparison and smoothness. The fixed linear relation $y = Ax$ and the all-order metric asymptotics preserve the $O_1(r_y^{-2})$ field bounds, since the forms themselves are fixed.

For every smooth filled enclosing cut Γ , the quadratic-form comparison gives

$$\text{Area}_{g_j}(\Gamma) \geq (1 - \varepsilon_j) \text{Area}_g(\Gamma).$$

Hence $\liminf_j a_{g_j}(S) \geq a_g(S)$. Conversely fix any such cut with area within $\zeta > 0$ of $a_g(S)$. Local convergence on this compact cut gives $\limsup_j a_{g_j}(S) \leq \text{Area}_g(\Gamma) \leq a_g(S) + \zeta$; let $\zeta \downarrow 0$. This proves the area limit directly for the smooth-cut infimum, including obstacle-coincident components. Finally each enclosing cut has the same flux of each fixed closed form by Stokes' Theorem, so both charges are preserved exactly. $\square$

The proposition constructs charged initial data on the original exterior and recovers its invariant mass; it does not assert that the original data admit a spacetime embedding or a global Lorentz change of asymptotic slice.

16. GEOMETRIC PREPARATION WITH THE PHYSICAL MATTER MARGIN

Fix a strict rest datum from Proposition 15.1, denoted by $(N, g, K, \alpha_1, \alpha_2)$, with obstacle S_0 , energy $E_* > 0$, and $A_* = a_g(S_0)$. Retain its geometry and end estimates; in particular, K has compact support and $\theta_+(S_0) < 0$. Its physical and neutral margins satisfy

$$(16.1) \quad M_0 = 8\pi(\mu_m - |J_m|_g) > 0, \quad 8\pi(\mu - |J|_g) \geq M_0, \quad M_0 \geq c_* r_y^{-3-\delta_*} \quad \text{far out,}$$

where $c_* > 0$ and $0 < \delta_* < 1$ are fixed for this datum. Every choice and constant in this section concerns this fixed datum.

16.1. The bounding regions and their orientations. For a symmetric tensor T write $\Theta_T(\Sigma, \nu) = H_\Sigma(\nu) + \text{tr}_\Sigma T$. The geometric input is the following form of the three-dimensional barrier and total trapped-region theorems [7, Section 2, Theorems 3.1 and 4.1, Definitions 7.1–7.2, and Theorem 7.3].

Theorem 16.1 (Barrier and total trapped-region theorem). *Let (D, g, T) be smooth compact connected three-dimensional data with boundary $\Gamma_- \sqcup \Gamma_+$. Assume $\Gamma_+ \neq \emptyset$ and $\Theta_T(\Gamma_+) > 0$ with its normal out of D . The inner boundary may be empty; when present it satisfies $\Theta_T(\Gamma_-) < 0$ with its normal into D . All boundary parts may be disconnected.*

Consider all smooth bounding domains containing a relative collar of Γ_- , avoiding Γ_+ , and having interior free boundary with $\Theta_T \leq 0$ for the normal out of the bounding domain. If the union is nonempty, its closure is a maximal closed bounding region whose interior frontier is smooth, compact, embedded, stable, and satisfies $\Theta_T = 0$. It is itself the closure of an admissible domain. A nonempty strict inner barrier ensures a nonempty union and an enclosing stable MOTS. No energy condition or constraint equation is required for T .

Filling a complementary component incident on no designated outer boundary deletes whole free-boundary components and preserves the expansion of those retained. This filling convention is understood in Theorem 16.1 and throughout the construction. Two identities fix the shifted tensors and all orientations:

$$(16.2) \quad \Theta_{T-(c/2)g} = \Theta_T - c, \quad \Theta_{-T}(\Sigma, -\nu) = -\Theta_T(\Sigma, \nu).$$

Choose R_0 beyond $\text{supp } K$ so that every coordinate sphere of radius at least R_0 has positive outward mean curvature. Every compact bounding domain with $\Theta_{\pm K} \leq c \leq 0$ is confined inside this fixed radial range. Indeed, at a free-boundary point of maximum radius beyond R_0 , the boundary lies inside the tangent coordinate sphere with the same outward normal. The local graph comparison gives $H_{\partial D} \geq H_{S_r} > 0$, whereas $K = 0$. This contradicts the expansion bound. Consequently all families below can be constructed on a single compact truncation, independently of the sufficiently distant outer sphere.

For $\max_{S_0} \Theta_K(S_0) < c < 0$, let $\mathcal{B}_c$ be the *full* closed maximal region for $\Theta_K \leq c$, with S_0 as required inner boundary. Applying Theorem 16.1 to $K - (c/2)g$ is legitimate: S_0 is strictly negative and the distant sphere strictly positive for the shifted expansion. Thus $\mathcal{B}_c$ is nonempty, contains a collar of S_0 , and has smooth stable frontier at expansion c . These regions increase with c .

Fix such a threshold c_b and put $\mathcal{B} = \mathcal{B}_{c_b}$. In its filled complement, designate both the full black frontier and a distant sphere as *outer* barriers. There is no required inner boundary. For $c < 0$, let $\mathcal{W}_c$ be the full closed maximal region of compact bounding domains satisfying $\Theta_{-K} \leq c$ there. On the reversed black frontier the expansion of $-K - (c/2)g$ equals $-c_b - c > 0$; it is positive on the distant sphere as well. Theorem 16.1 therefore applies whenever the union is nonempty. Its smooth stable frontier has $\Theta_{-K} = c$ and positive separation from the black frontier. If the union is empty, set $\mathcal{W}_c = \emptyset$. The white family increases with c , always with the black region held fixed.

We choose the two thresholds successively at points of Hausdorff right continuity of these full families. Here is the elementary selection argument, including emptiness. For any increasing closed family E_c in a compact metric space, choose a cap L larger than its diameter and put $d_c(x) = \min\{L, \text{dist}(x, E_c)\}$, taking $d_c \equiv L$ if $E_c = \emptyset$. At every point of a countable dense set, this is a bounded nonincreasing function of c and has at most countably many right discontinuities. Avoid their union. The common 1-Lipschitz bound and a finite-net argument then give uniform convergence $d_{c_j} \rightarrow d_c$ whenever $c_j \downarrow c$. For nonempty E_c this gives Hausdorff convergence, since $E_c \subset E_{c_j}$. For empty E_c , uniform convergence to L forces every sufficiently close $E_{c'}$, $c' > c$, to be empty: otherwise its distance function has a zero. Thus c_b , and then c_w for this fixed $\mathcal{B}$, can be chosen arbitrarily close to zero with the stated continuity, including the empty-white case.

Put $\mathcal{W} = \mathcal{W}_{c_w}$ and let X be the closure of the component toward infinity of $N \setminus (\mathcal{B} \cup \mathcal{W})$. Its boundary is the entire adjacent frontier, a finite disjoint union $B = B_b \sqcup B_w$ of complete black and white face components. It is nonempty because the excluded side contains S_0 and its collar. Both full frontiers are compact and disjoint, so selection creates no corners. With ν pointing into X , set $H = H_B(\nu)$ and $P_B = \text{tr}_B K$. Then

$$(16.3) \quad H + P_B = c_b \quad \text{on } B_b, \quad H - P_B = c_w \quad \text{on } B_w.$$

Either face type may be absent from B ; for example a white face can separate a black face from infinity. The full regions used to choose the thresholds are retained in the preparation even when some of their frontier components are not adjacent to X .

The restricted data on X are smooth and complete up to B , with the original single end. Every enclosing cut Γ for B is also a cut for S_0 after the discarded side is adjoined to its inner region and bounded pockets are filled. Therefore

$$(16.4) \quad \text{Area}_g(\Gamma) \geq A_*, \quad \frac{1}{4\pi} \int_{\Gamma} \alpha_i = \frac{1}{4\pi} \int_B \alpha_i = Q_i \quad (i = 1, 2), \quad (Q_1, Q_2) = (Q_E, Q_B).$$

The flux equalities follow from closedness and Stokes' Theorem between the cut and a distant sphere, all normals pointing toward the end. These assertions concern the whole frontier and do not require any individual component to separate S_0 from infinity.

16.2. Stable collars and a stronger offset bound.

Lemma 16.2 (Outward leaf collars). *Each connected face has a smooth outward collar with leaf parameter $s \geq 0$, $s = 0$ on the face, and $|\nabla s|$ bounded above and below by positive constants. Its leaves satisfy $\Theta_K > c_b$ for black faces and $\Theta_{-K} > c_w$ for white faces whenever $s > 0$. The collars may be chosen disjoint in X .*

Proof. Let Σ be a connected component of a full maximal frontier at threshold c , for $T = K$ or $-K$, with its outward normal. For normal graphs define $\Phi(v) = \Theta_T(\Sigma(v)) - c$ as a map $C^{2,\alpha}(\Sigma) \rightarrow C^{0,\alpha}(\Sigma)$. Its linearization $\mathcal{L}$ is the stability operator of $T - (c/2)g$. It has principal eigenvalue $\lambda \geq 0$ and a positive eigenfunction φ . If $\lambda > 0$, the graphs $v = s\varphi$ have $\Phi(s\varphi) = s\lambda\varphi + O(s^2) > 0$ for small $s > 0$.

If $\lambda = 0$, the kernels of $\mathcal{L}$ and its adjoint are one-dimensional, with positive generators φ and φ^* . The bordered linear map

$$(16.5) \quad (v, a) \mapsto \left(\mathcal{L}v - a, \int_{\Sigma} v dA_g \right)$$

is invertible. In fact, solving $\mathcal{L}v - a = f$ first determines $a = -\int_{\Sigma} \varphi^* f dA_g / \int_{\Sigma} \varphi^* dA_g$ by Fredholm compatibility, and adding a unique multiple of φ fixes the prescribed integral of v . Elliptic estimates give a bounded inverse. The implicit-function theorem applied to $(\Phi(v) - a, \int v) = (0, s)$ now yields constant-expansion leaves with $a'(0) = 0$ and $v'(0) = \varphi / \int_{\Sigma} \varphi dA_g > 0$. They foliate a short outward collar. If $a(s) \leq 0$ for any positive small s , adjoining this collar and replacing just this

whole frontier component by its new leaf would strictly enlarge the full maximal bounding region while preserving expansion $\leq c$ on every free component. The collar avoids all other faces and designated barriers, so this is an admissible competitor, contradicting maximality. Thus $a(s) > 0$.

Smooth elliptic regularity gives smooth leaves; positive graph velocity gives the two bounds on $|\nabla s|$. There are finitely many faces. Shortening the collars of those adjacent to X makes them disjoint and contained in X . $\square$

Proposition 16.3 (Prepared exterior with physical margin). *The thresholds and exterior above can be chosen with their stated full-region right continuity and collars, and with the following additional data. There are constants $C_b \geq 0$, $C_w \leq 0$ and a smooth compactly supported function C satisfying*

$$(16.6) \quad \begin{aligned} C_w \leq C \leq C_b, \quad C &= C_b - k_B s \quad \text{near each black face,} \\ C &= C_w + k_B s \quad \text{near each white face,} \end{aligned}$$

where $k_B > 0$ is constant on each individual collar. Set

$$(16.7) \quad b_- = c_b - C_b < 0, \quad b_+ = -c_w - C_w > 0.$$

There is a smooth positive weight ρ , equal to a positive constant times r_y^{-4} on the end with differentiated symbol bounds, such that, for every real $h \in [b_-, b_+]$ and every $x \in X$,

$$(16.8) \quad |(h + C) \operatorname{tr}_g K| + |\nabla C|_g + \rho \leq \frac{1}{2} M_0(x).$$

The thresholds, full regions, exterior, collars, offset, assigned heights, and weight are all fixed before any elliptic deformation parameter. After fixing the thresholds and collars, the offset can have arbitrarily small C^1 norm. The constants of an absent face type remain valid bounds. Equation (16.4) holds.

Proof. Take a fixed compact neighborhood $\mathcal{K}$ containing the uniform confinement range and $\operatorname{supp} K$ in its interior, and put $\eta_* = \min_{\mathcal{K}} M_0 > 0$ and $T_* = \sup_{\mathcal{K}} |\operatorname{tr}_g K|$. Restrict both negative thresholds in advance to an interval so close to zero that $\max\{|c_b|, |c_w|\} T_* < \eta_*/8$, while maintaining the strict black inner-barrier inequality. The countable-exception argument permits the successive right-continuous choices there. Construct the regions and fix their disjoint collars inside $\mathcal{K}$.

On each collar choose a smooth cutoff $\zeta_B(s)$ in $[0, 1]$, equal to one near zero and zero near the far end. Fix $\kappa_B > 0$ with $1 - \kappa_B s > 0$ throughout the collar. For a common amplitude $a_* > 0$ define the black contribution to C as $a_* \zeta_B(s)(1 - \kappa_B s)$ and the white contribution as its negative. Extend by zero and sum on the disjoint collars. Then $C_b = a_*$, $C_w = -a_*$, $k_B = a_* \kappa_B$, and $\|C\|_{C^1} = O(a_*)$ with constants depending only on the fixed collars and profiles. For every assigned height,

$$|h + C| \leq \max\{|c_b|, |c_w|\} + 2a_*.$$

Choose a_* small enough that $2a_* T_* + \|\nabla C\|_\infty < \eta_*/8$. The first two terms of Equation (16.8) are then at most $\eta_*/4 \leq M_0/4$ on $\mathcal{K}$, and vanish outside it. Finally scale a fixed smooth positive weight, equal to r_y^{-4} far out, so that $\rho \leq M_0/4$ everywhere. This is possible by compact positivity and $r_y^{-4}/r_y^{-3-\delta_*} = r_y^{-1+\delta_*} \rightarrow 0$. This proves the asserted uniform inequality for all h and the parameter order. $\square$

Since $M_0 \leq 8\pi(\mu - |J|_g)$, Proposition 16.3 satisfies the exact neutral smallness hypothesis of Proposition 4.7. The proof above establishes the stronger charged choice directly. At the assigned face heights it also gives the exact boundary identities

$$(16.9) \quad h + C = c_b = P_B + H \quad (B_b), \quad h + C = -c_w = P_B - H \quad (B_w),$$

which will determine the signs in the neutral elliptic boundary conditions.

17. TRANSFER OF THE NEUTRAL ELLIPTIC CONSTRUCTION

The charged construction uses the neutral system with the total tensors (g, K) . We verify its prepared class and identify the equations, normalizations and estimates needed from Theorem 7.5, Lemmas 6.2 and 8.1, with the parameter choices in Proposition 5.9. The transfer proposition below is a consequence of those proved results; it introduces no new solvability hypothesis.

17.1. The class to which the input applies. Fix one strict rest datum supplied by Proposition 15.1, with original obstacle S_0 , energy $E_* > 0$, and $A_* = a_g(S_0)$. We record the complete preparation relevant to the neutral construction, so that its hypotheses are not replaced by a solvability assumption about arbitrary charged data.

Definition 17.1 (Prepared neutral class). The underlying reduced datum is smooth, orientable, connected, complete up to its nonempty compact smooth boundary S_0 , and has exactly one asymptotically flat end. Its boundary satisfies $H_{S_0} + \text{tr}_{S_0} K < 0$, with normal toward that end. The total constraint densities satisfy

$$\mu, |J|_g \in L^1(dV_g), \quad M_N := 8\pi(\mu - |J|_g) > 0, \quad M_N \geq cr_y^{-3-\delta} \quad \text{far out}, \quad 0 < \delta < 1,$$

for a fixed $c > 0$. The tensor K is compactly supported. In a rest chart y , $g - \delta_{\text{eucl}} = O_j(r_y^{-1})$ for every fixed j , and hence $\text{Ric}_g = O(r_y^{-3})$; the ADM energy is finite and the momentum is zero. Our reduced data also have $R_g = O_j(r_y^{-3-\delta})$ for every fixed j .

The prepared exterior X is the closure of the component toward infinity remaining after the full black region and the subsequent full white region in Proposition 16.3 are removed. More precisely, choose $c_b < 0$ with $c_b > \max_{S_0}(H + \text{tr}_{S_0} K)$ as a Hausdorff right-continuity point of the full maximal family with $H + \text{tr}_\Sigma K \leq c_b$ and the required inner collar of S_0 . With that black region fixed, choose $c_w < 0$ as a right-continuity point for the full white family with $H - \text{tr}_\Sigma K \leq c_w$, no required inner boundary, and the black faces and a distant sphere as outer barriers. If the selected region is empty, right continuity includes emptiness immediately to its right. The adjacent frontiers give a smooth compact boundary $B = B_b \sqcup B_w \neq \emptyset$, with

$$(17.1) \quad H + P_B = c_b \quad \text{on } B_b, \quad H - P_B = c_w \quad \text{on } B_w, \quad P_B = \text{tr}_B K,$$

where ν points into X and $H = \text{div}_B \nu$. The faces have disjoint smooth outward leaf collars, with leaf coordinate $s \geq 0$, $|ds| > 0$, and the expansion of the corresponding color strictly greater than its threshold for $s > 0$. Every cut enclosing B has g -area at least A_* .

The fixed offset and weights satisfy

$$\begin{aligned} C &\in C_c^\infty(\overline{X}), \quad C_w \leq C \leq C_b, \quad C_b \geq 0, \quad C_w \leq 0, \\ C &= C_b - k_B s \text{ near a black face}, \quad C = C_w + k_B s \text{ near a white face}, \quad k_B > 0, \\ b_- &= c_b - C_b < 0, \quad b_+ = -c_w - C_w > 0. \end{aligned}$$

Here k_B may depend on the face. Either color may be absent; its constant C_b or C_w then remains only a bound, and its face conditions are omitted. The function ρ is smooth and positive, equals a positive constant times r_y^{-4} sufficiently far out, and has symbol derivative bounds there. For every $h \in [b_-, b_+]$, the exact neutral smallness condition is

$$(17.2) \quad |(h + C) \text{tr} K| + |\nabla C| + \rho \leq \frac{1}{2} M_N \quad \text{throughout } X.$$

All thresholds, full regions, collars, offsets, and these weights are fixed before any deformation parameter is chosen. This is the preparation in Proposition 4.7 on the reduced class of Proposition 3.10.

Here “reduced class” means the properties in Proposition 3.10(i)–(ii), which explicitly singles out these properties as the class for its subsequent argument. The preparation uses these properties of the datum and the small offset choices of Section 4 and Proposition 4.7.

Lemma 17.2 (Verification for the charged preparation). *The outputs of Propositions 15.1 and 16.3 belong to Definition 17.1.*

Proof. Proposition 15.1 supplies the reduced geometry, integrability and end estimates; Proposition 16.3 supplies the full-region preparation. Equation (14.13) gives $M_N \geq M_0$, so the strict physical margin and its end lower bound imply their neutral counterparts. The stronger bound (16.8) implies (17.2). These are the actual total constraint densities of (g, K) , as required by the neutral construction. $\square$

17.2. Profiles, quadratic form, and boundary value problem. All derivatives and contractions below use g ; vectors and covectors are identified by g . Extend r_y smoothly over X , bounded below by one, and let X_R be a large spherical truncation, with outer face S_R . First fix a sufficiently small $0 < \epsilon < 1$, then a large integer $n \geq 4$ with $n > 2/\epsilon$, and then a sufficiently large R . These are the standing parameter conventions of Section 5. Choose a smooth nonincreasing $\vartheta : \mathbb{R} \rightarrow [0, 1]$, equal to one on $(-\infty, 0]$ and zero on $[1, \infty)$, and set

$$(17.3) \quad \ell = e^{-\epsilon n}, \quad \tau_0 = \ell^{3/2}, \quad p(t) = n\vartheta(nt), \quad l(t) = \exp\left(\int_0^t p(s) ds\right), \quad L(t) = e^{2t}l(t).$$

The symbol τ_0 is unrelated to $\tau = \text{tr } K$. In particular $l(t) = e^{nt}$ for $t \leq 0$, l is constant for $t \geq 1/n$, and $\ell \leq l \leq e$ when $t \geq -\epsilon$. Define

$$(17.4) \quad \begin{aligned} h &= \tau_0 f, \quad \sigma = |\nabla f|, \quad D = 1 + l^2 \sigma^2, \quad d = D^{-1}, \quad w = \frac{l \nabla f}{\sqrt{D}}, \quad a = |w|, \quad v = a^2 = 1 - d, \\ \chi &= \frac{4d}{4 + pv}, \quad A_d = I - w \otimes w, \quad A_\chi = I - \frac{p + 4}{4 + pv} w \otimes w, \\ Z &= t + \frac{1}{8} \log D, \quad u = L\sqrt{D}, \quad H^f = \frac{l}{\sqrt{D}} \text{Hess } f, \quad S_f = K + H^f, \\ F &= \text{tr}_{A_\chi} S_f, \quad V = u A_\chi (4\nabla Z + K(w, \cdot)). \end{aligned}$$

Matrix expressions are in a g -orthonormal frame, and $\text{tr}_A S = A^{ij} S_{ij}$. The unknowns are (f, Z) : for each fixed ∇f , the displayed expression for Z is strictly increasing and onto as a function of t , since $\partial_t Z = 1 + pv/4 > 0$. Thus it determines t smoothly without assuming a floor. In particular $0 < \chi \leq d \leq 1$.

For $\sigma > 0$, put $e = \nabla f / \sigma$ and decompose

$$T = S_f|_{e^\perp \times e^\perp}, \quad M = S_f(e, \cdot)|_{e^\perp}, \quad k = S_f(e, e), \quad y_t = (\nabla t)_\perp, \quad x_t = \partial_e t.$$

These T, M, k are tensor blocks; M is not a manifold or an ADM mass. We have $\text{tr } T = F - \chi k$, and T^{tf} means the trace-free part in the two-dimensional plane $e^\perp$. Set

$$(17.5) \quad \begin{aligned} \mathcal{T} &= \frac{1}{2} |T^{\text{tf}}|^2 + |M + (p + 1)ay_t|^2 + [4(p + 1) - v]|y_t|^2 \\ &+ 4(p + 1)d \left(x_t + \frac{\chi ak}{4d}\right)^2 + \frac{3}{4} \left(F - \frac{\chi k}{3}\right)^2 + \frac{4d(9 - d)}{3(4 + pv)^2} k^2. \end{aligned}$$

This is the exact square completion in Section 5. It extends smoothly across $\sigma = 0$; its direction-independent value there is

$$\mathcal{T} = \frac{1}{2} (|S_f|^2 + (\text{tr } S_f)^2) + 4(p + 1)|\nabla t|^2.$$

The invariant expression and smooth extension are proved in Lemma 18.1, and polynomial coercivity is proved in Lemma 18.2.

Fix $3/4 < \gamma < 1$ and positive smooth weights with $\rho_0 \asymp r_y^{-4}$, $\rho_1 \asymp r_y^{-2\gamma}$ and symbol derivative bounds on the end. In particular $\rho_0, \rho_0^2, \rho_1^2$ are integrable; integrability of ρ_1 itself is not required. For a smooth $m_c : \mathbb{R} \rightarrow [0, 1]$, equal to one on $(-\infty, 0]$ and zero on $[1, \infty)$, put

$$m_0(t) = m_c(n(t + \epsilon)), \quad \mathcal{P} = C_n(\rho_0 + v\rho_1).$$

Choose $N_B \in C^\infty(B)$ with $N_B > 1 + \sup_B |H|$. Use the choices furnished by Lemma 5.8 and Proposition 5.9: first a sufficiently small $0 < \delta_0 < 1$, independent of n, R , then a sufficiently large polynomial penalty $C_n = C_*(1 + n)^{N_*}$. The fixed constants may depend on the prepared data and auxiliary profiles. All auxiliary collar cutoffs, face extensions, and continuation choices are those furnished in that construction. Its physical endpoint is exactly

$$(17.6) \quad \begin{cases} F = h + C & \text{in } X_R, \\ \operatorname{div} V = u(\delta_0 \mathcal{T} + \rho + \delta_0 \tau_0 a \sigma - m_0(t) \mathcal{P}) & \text{in } X_R, \\ h = b_- \text{ on } B_b, \quad h = b_+ \text{ on } B_w, \\ V_\nu / u = -H + w_\nu(F - P_B) - N_B d & \text{on } B, \\ f = Z = 0 & \text{on } S_R. \end{cases}$$

Here $V_\nu = g(V, \nu)$ and $w_\nu = g(w, \nu)$; the inner normal is the negative of the integration-outward normal of X_R . At the assigned face heights, $F = P_B + H$ on black faces and $F = P_B - H$ on white faces. On the admissible penalty band one has

$$(17.7) \quad -\epsilon \leq t \leq -\epsilon + 1/n < 0, \quad \ell \leq l \leq e\ell, \quad c\ell \leq L \leq C\ell.$$

The last constants can be uniform for $0 < \epsilon < 1$.

17.3. The transferred existence and estimates.

Proposition 17.3 (Neutral elliptic construction for the prepared class). *For the prepared class in Definition 17.1, with the profiles, parameters, penalty choices, and physical endpoint in Equations (17.3)–(17.6), the following statements hold.*

- (1) Existence with the floor bound. *With δ_0, C_n chosen as in Proposition 5.9, for every sufficiently small fixed $\epsilon > 0$, Theorem 7.5 supplies n_0 , enlarged to include $n \geq 4$ and $n > 2/\epsilon$, and a sufficient radius $R_{\min}(n, \epsilon) \rightarrow \infty$ as $n \rightarrow \infty$, such that Equation (17.6) has a smooth solution on every X_R with $n \geq n_0$ and $R \geq R_{\min}(n, \epsilon)$. It satisfies $t > -\epsilon$ on $\overline{X_R}$. The radius includes the geometric supports, all a priori and floor thresholds, and the outer-boundary floor restriction in the proof of that theorem. No polynomial bound for this radius is asserted.*
- (2) Physical-endpoint height and slope bounds. *For smooth finite-truncation solutions with $t \geq -\epsilon$, for these sufficiently large n, R , Lemma 6.2 at the physical endpoint gives fixed $c, C_* > 0$ such that*

$$(17.8) \quad \frac{c}{\tau_0} \leq \partial_\nu f \leq \frac{C_*}{\tau_0} \quad (B_b), \quad \frac{c}{\tau_0} \leq -\partial_\nu f \leq \frac{C_*}{\tau_0} \quad (B_w).$$

These constants are independent of n, R after fixing the prepared data and ϵ . Also $b_- \leq h \leq b_+$ on every fixed compact region needed for the scalar-curvature argument, in particular on the supports of K, C . A condition on an absent face color is vacuous. No global height bound is being added to this statement.

- (3) Exhaustion and end normalization. *At fixed ϵ, n , the solutions have local smooth estimates of every fixed order, uniformly as $R \rightarrow \infty$, including up to the fixed inner boundary. Every*

sequence of allowed radii tending to infinity has a subsequence converging smoothly on compact subsets of $\bar{X}$ to a solution of Equation (17.6) with the outer boundary condition omitted and $t \geq -\epsilon$. The preceding compact height and face estimates pass to this limit. For sufficiently large $n \geq \max\{4, \lceil 2/\epsilon \rceil\}$, choose an exhausting sequence and the limiting solution supplied by Lemma 8.1. For some $c_n > 0$ and constants $C_j(n)$, this solution also satisfies

$$(17.9) \quad |\nabla^j f| \leq C_j(n) e^{-c_n r_y} \quad \text{for each fixed } j, \quad t, Z = O_2(r_y^{-1}).$$

Writing $\bar{g} = g + l^2 df^2$, both $t - Z$ and $\bar{g} - g$, together with their coordinate derivatives of every fixed order, decay exponentially. Moreover

$$(17.10) \quad V = 4\nabla_g t + O(r_y^{-3}), \quad \lim_{s \rightarrow \infty} \int_{r_y=s} g(V, \nu_s) dA_g \quad \text{exists and is finite,}$$

where ν_s points toward infinity.

Proof. We first identify the input class. The reduced properties in Definition 17.1 are the properties (i)–(ii) of Proposition 3.10; that proposition explicitly designates them as the class for the subsequent construction. In particular, provenance as a particular member of its approximation sequence is not an additional hypothesis. The full black and white regions, their right-continuity conventions including emptiness, the adjacent complete boundary components, and the disjoint outward collars are exactly those of Proposition 4.7. Equation (17.2) is its offset inequality with $M_N = 8\pi(\mu - |J|)$. Thus its preparation, including the possibility of either missing face color, applies to every datum in the stated class.

Next identify the system. The neutral construction uses the capillarity parameter denoted here by $\tau_0 = \ell^{3/2}$, to distinguish it from $\text{tr } K$. Its functions l, L , logarithmic profile derivative $p = l'/l$, graph quantities D, d, w, a, v , matrices A_d, A_χ , and unknowns (f, Z) are exactly Equations (17.3)–(17.4). The notation $p_L = L'/L$ in the neutral construction therefore means $p_L = p + 2$. Its tensor S and axial component q are denoted here by S_f and k , respectively. Its quadratic form is Equation (17.5), with no change of sign in the momentum constraint. The physical endpoint of Proposition 5.4 is its first continuation stage at $s = 0$, with drift coefficient $\lambda = 1$: its trace equation is $F = \tau_0 f + C$, and its divergence source is $\delta_0 \mathcal{T} + \rho + \delta_0 \tau_0 a \sigma - m_0(t) \mathcal{P}$. These are the two interior equations of Equation (17.6). The inner normal in both formulations points toward the end. The componentwise assigned heights and the identities $F = P_B + H$ on black faces and $F = P_B - H$ on white faces give exactly the same inner flux condition; the outer values are $f = Z = 0$ in both systems. All auxiliary continuation data are chosen by that construction after the fixed preparation, as stipulated above.

Lemma 5.8 and Proposition 5.9 choose a fixed $\delta_0 > 0$ and a polynomial C_n . Theorem 7.5 then supplies item 1, including its sufficiently large truncation radius and the strict floor; enlarging the lower bounds on n and R imposes the additional displayed numerical restrictions. At the physical endpoint, Lemma 6.2 gives item 2. Its slope constants are independent of n, R , its height interval is asserted on the fixed compact regions used in the curvature argument, and its face conditions are componentwise. No boundary connectedness is required.

For fixed ϵ, n , Proposition 7.3 and the exhaustion following Theorem 7.5 give the local smooth estimates and subsequential limit in item 3, including every fixed inner face. These estimates allow arbitrary fixed- n constants and do not change the polynomial penalty already chosen. Lemma 8.1 supplies the stated normalized end solution: the exponential decay of f and the graph errors, the $O_2(r_y^{-1})$ bounds for t, Z , and the finite flux with $V = 4\nabla_g t + O(r_y^{-3})$. These are exactly Equations (17.9)–(17.10). All three conclusions follow from the proved neutral construction. $\square$

The constants in the local regularity and end estimates may depend arbitrarily on the fixed n (and on the prepared datum and ϵ); neither they nor c_n^{-1} are claimed polynomial. The penalty

C_n , by contrast, is polynomial in n , as are the explicit algebraic losses proved in Lemma 18.2. No uniqueness or convergence as $n \rightarrow \infty$ is part of Proposition 17.3.

The neutral results used in the transfer have no Penrose equality premise. Proposition 17.3 uses their original physical endpoint, corresponding to the first continuation stage with its drift coefficient equal to one. It introduces neither a charged source term nor a solver for electromagnetic-subtracted constraints. Lemma 17.2 verifies its hypotheses for each fixed charged preparation. Henceforth we use its global limiting solution, always taking $R \rightarrow \infty$ first at fixed ϵ, n . The later numerical argument takes $n \rightarrow \infty$, then $\epsilon \downarrow 0$, and only then removes the strict rest-data approximation.

18. SCALAR CURVATURE AND ELECTROMAGNETIC COMPARISON

Fix the prepared datum, then ϵ and n , and take the smooth limiting solution supplied by Proposition 17.3. Throughout this section, derivatives, tensor contractions and cross products without a metric subscript are taken with respect to g . Put

$$(18.1) \quad \bar{g} = g + l^2 df \otimes df, \quad \hat{g} = e^{4t} \bar{g}, \quad \tilde{g} = e^{4\epsilon} \hat{g}.$$

The scalar and boundary identities of the neutral construction apply without changing the total constraint densities. We first identify their notation here, then use the physical matter margin to retain the total charge in the comparison metric.

18.1. The scalar identity.

Lemma 18.1 (Scalar identity). *With the definitions of Section 17, including the quadratic form in Equation (17.5), one has*

$$(18.2) \quad \frac{1}{2} e^{4t} R_{\hat{g}} = 8\pi(\mu + J(w)) + \mathcal{T} + w(F) - F \operatorname{tr} K - u^{-1} \operatorname{div} V.$$

Here μ, J are the total constraint densities of (g, K) . The quadratic form $\mathcal{T}$ extends smoothly through $\nabla f = 0$ and is nonnegative everywhere.

Proof. Apply Proposition 5.1. The variables $l, L, p, D, d, w, a, v, A_d, A_\chi, Z, u, H^f, F, V$ in Equations (17.3)–(17.4) are those of that proposition. Its tensor S is our S_f , its axial component q is our k , and its transverse and axial derivatives of t are y_t, x_t . Its logarithmic derivative $p_L = L'/L$ is $p + 2$. In particular the graph and conformal metrics, momentum convention, and full flux agree. The identity holds for arbitrary smooth f, t ; no elliptic equation or constraint beyond the definitions of μ, J is assumed in its proof.

Equation (5.22) is exactly Equation (17.5) under this correspondence. It proves nonnegativity, since $p \geq 0$ and $0 < d \leq 1$. The invariant formula (5.19) proves smoothness at $\nabla f = 0$, where its value is

$$\mathcal{T} = \frac{1}{2}(|S_f|^2 + (\operatorname{tr} S_f)^2) + 4(p+1)|\nabla t|^2.$$

Thus the shared identity gives (18.2), with the momentum term $+8\pi J(w)$. $\square$

Lemma 18.2 (Weighted coercivity). *At every point, with any axial direction at zero slope,*

$$(18.3) \quad |T|^2 + |M|^2 + dk^2 + F^2 + |y_t|^2 + dx_t^2 \leq 13(1+n)^2 \mathcal{T}.$$

The constant is independent of the solution, n , the truncation radius, and the size of $|\nabla f|$.

Proof. Use the square completion (5.22), identified above with (17.5). Its last coefficient is at least $2d/[3(1+p)^2]$ and at least $2\chi^2/3$. Its transverse-gradient coefficient is at least $3(p+1)$.

Comparing these terms and the shifted squares gives

$$\begin{aligned} dk^2 &\leq \frac{3}{2}(1+p)^2\mathcal{T}, & \chi^2 k^2 &\leq \frac{3}{2}\mathcal{T}, & |y_t|^2 &\leq \frac{\mathcal{T}}{3(p+1)}, \\ F^2 &\leq 3\mathcal{T}, & |M|^2 &\leq \frac{8}{3}(p+1)\mathcal{T}, & |T|^2 &\leq 4\mathcal{T}. \end{aligned}$$

Here the shifted F and M estimates use $|A+B|^2 \leq 2|A|^2 + 2|B|^2$; for T write $\text{tr } T = (F - \chi k/3) - 2\chi k/3$. The axial-gradient square gives

$$dx_t^2 \leq \frac{\mathcal{T}}{2(p+1)} + \frac{\chi^2 vk^2}{8d} \leq \left(\frac{1}{2(p+1)} + \frac{1}{6} \right) \mathcal{T} \leq \frac{2}{3}\mathcal{T},$$

because the ratio of its second coefficient to the last square coefficient is $3v/[2(9-d)] \leq 1/6$. Summing and using $0 \leq p \leq n$ proves (18.3). At zero slope the same estimates hold for every axial decomposition of the direction-independent value of $\mathcal{T}$. $\square$

18.2. The inner boundary.

Lemma 18.3 (Boundary identity and sign). *On every face of B , with ν pointing into X and $P_B = \text{tr}_B K$,*

$$(18.4) \quad \frac{V_\nu}{u} = d(H + 4\partial_\nu t) - H + w_\nu(F - P_B).$$

For a solution of Equation (17.6), the corresponding mean curvatures satisfy

$$(18.5) \quad H_{\hat{g}} = -e^{-2t}\sqrt{d}N_B < 0, \quad H_{\tilde{g}} = e^{-2\epsilon}H_{\hat{g}} < 0.$$

Both mean curvatures use the normal toward the designated end.

Proof. The assigned height is constant on each face, and the normal into X is the normal used in Lemma 5.3. Its flux, trace F , and tangential trace P_B agree with ours by the correspondence in Lemma 18.1. Applying Lemma 5.3 gives (18.4) and

$$H_{\hat{g}} = e^{-2t}\sqrt{d}(H + 4\partial_\nu t).$$

The boundary equation in (17.6) now implies $d(H + 4\partial_\nu t) = -N_B d$. Since $d > 0$, substitution gives (18.5); the last equality is the mean-curvature scaling under $\tilde{g} = e^{4\epsilon}\hat{g}$. $\square$

18.3. Retaining both total charges.

Proposition 18.4 (Charged comparison). *For every limiting solution furnished by Proposition 17.3, there are smooth closed two-forms $\tilde{\alpha}_1, \tilde{\alpha}_2$ on X whose end fluxes are $4\pi Q_E, 4\pi Q_B$, respectively, such that the fields $\tilde{\mathcal{E}} = \tilde{X}_1, \tilde{\mathcal{B}} = \tilde{X}_2$ defined by $\iota_{\tilde{X}_i} dV_{\tilde{g}} = \tilde{\alpha}_i$ satisfy*

$$(18.6) \quad R_{\tilde{g}} \geq 2(|\tilde{X}_1|_{\tilde{g}}^2 + |\tilde{X}_2|_{\tilde{g}}^2), \quad \text{div}_{\tilde{g}} \tilde{X}_i = 0.$$

Moreover $\tilde{g} \geq g$ as quadratic forms, X is complete up to B for $\tilde{g}$, and $H_{\tilde{g}} < 0$ on B .

Proof. Substituting Equation (17.6) into Equation (18.2), and using $w(h) = \tau_0 a \sigma$, gives

$$(18.7) \quad \begin{aligned} \frac{1}{2}e^{4t}R_{\hat{g}} &= 8\pi(\mu + J(w)) + (1 - \delta_0)\mathcal{T} + (1 - \delta_0)\tau_0 a \sigma \\ &+ w(C) - F \text{tr } K - \rho + m_0(t)\mathcal{P}. \end{aligned}$$

The physical matter condition says

$$8\pi(\mu + J(w)) - I_g(w) = 8\pi(\mu_m + J_m(w)) \geq M_0,$$

since $|w| < 1$. The compact height bound in Proposition 17.3 and Equation (16.8) imply $|w(C)| + |F \operatorname{tr} K| + \rho \leq M_0/2$; outside the supports of K and C only the already controlled term ρ remains. All other terms in Equation (18.7) are nonnegative. Consequently

$$(18.8) \quad \frac{1}{2}e^{4t}R_{\hat{g}} \geq I_g(w).$$

For $Q > 0$, take the constant rotation

$$O = \frac{1}{Q} \begin{pmatrix} Q_E & Q_B \\ -Q_B & Q_E \end{pmatrix} \in SO(2), \quad \begin{pmatrix} X_e \\ X_o \end{pmatrix} = O \begin{pmatrix} \mathcal{E} \\ \mathcal{B} \end{pmatrix};$$

for $Q = 0$, use $O = I$. Rotate the flux forms by the same matrix. Their charges become $(Q, 0)$, while I_g is unchanged: the sum of squared norms is invariant and $X_e \times X_o = \mathcal{E} \times \mathcal{B}$. Let $\alpha = \iota_{X_e} dV_g$. Completing one vector square gives

$$(18.9) \quad \begin{aligned} I_g(w) &= |X_o + w \times X_e|^2 + |X_e|^2 - |w \times X_e|^2 \\ &\geq |X_e|^2 - |w \times X_e|^2 = |\alpha|_g^2. \end{aligned}$$

The last equality is a two-form norm identity. In a positively oriented g -orthonormal frame with first vector parallel to w , the inverse graph metric has eigenvalues $(d, 1, 1)$. Writing $X_e = (x_1, x_2, x_3)$ gives

$$\alpha = x_1 e^2 \wedge e^3 + x_2 e^3 \wedge e^1 + x_3 e^1 \wedge e^2, \quad |\alpha|_g^2 = x_1^2 + d(x_2^2 + x_3^2) = |X_e|^2 - |w \times X_e|^2.$$

At $w = 0$ this formula is independent of the frame. This is the electric graph transport of [26, Section 2 and Appendix A], expressed through its flux form.

Retain the rotated pair $(\alpha, 0)$ and define the new pair by the inverse rotation

$$(18.10) \quad \begin{pmatrix} \tilde{\alpha}_1 \\ \tilde{\alpha}_2 \end{pmatrix} = O^T \begin{pmatrix} \alpha \\ 0 \end{pmatrix}.$$

This discards a form of zero *total* charge. Since O is constant, the new forms are closed, have separately the original total charges, and satisfy $|\tilde{\alpha}_1|_h^2 + |\tilde{\alpha}_2|_h^2 = |\alpha|_h^2$ for every metric h . No assertion about the individual component charges is needed. Flux conservation across every enclosing cut follows from Stokes' Theorem.

Finally set $s_0 = t + \epsilon \geq 0$, using the floor in Proposition 17.3. Scalar curvature under constant scaling and the conformal norm of a two-form give

$$R_{\tilde{g}} = e^{-4\epsilon} R_{\hat{g}}, \quad |\alpha|_{\tilde{g}}^2 = e^{-8s_0} |\alpha|_g^2.$$

Equations (18.8) and (18.9) therefore imply

$$R_{\tilde{g}} \geq 2e^{-4s_0} |\alpha|_g^2 \geq 2e^{-8s_0} |\alpha|_{\tilde{g}}^2 = 2 \sum_{i=1}^2 |\tilde{X}_i|_{\tilde{g}}^2.$$

The last equality uses the isometry $X \mapsto \iota_X dV_{\tilde{g}}$ between vectors and two-forms in dimension three. Closedness gives the divergence equations because $d(\iota_X dV_{\tilde{g}}) = (\operatorname{div}_{\tilde{g}} X) dV_{\tilde{g}}$. Also $\tilde{g} = e^{4s_0}(g + l^2 df \otimes df) \geq g$. A $\tilde{g}$ -Cauchy sequence is therefore g -Cauchy; its limit lies in X with B included, and local smooth comparison of the two metrics gives convergence for $\tilde{g}$ as well. Completeness follows, and the boundary sign is Lemma 18.3. $\square$

The asymptotic estimates, scalar-curvature integrability and ADM energy control needed for the final Riemannian comparison are established in Proposition 19.1.

19. ENERGY AND BOUNDARY FLUX

Fix a prepared strict rest datum, including its face collars, offsets and weights. All limits in this section hold for that datum and for a fixed sufficiently small $\epsilon > 0$, with $n \rightarrow \infty$ only after the fixed- n exhaustion in Proposition 17.3. Constants may depend on the datum and on ϵ , but not on n , unless expressly stated otherwise.

Proposition 19.1 (Energy of the comparison metric). *Let (f, t) be the global solutions of Equation (17.6) supplied by Proposition 17.3, and let $\tilde{g}_n = e^{4\epsilon} \hat{g}_n$ and the comparison fields be as in Proposition 18.4. In the normalized end coordinates $z = e^{2\epsilon} y$, these data satisfy*

$$\tilde{g}_n - \delta = O_2(|z|^{-1}), \quad \tilde{X}_{1,n}, \tilde{X}_{2,n} = O_1(|z|^{-2}), \quad R_{\tilde{g}_n} \in L^1(X, dV_{\tilde{g}_n}).$$

Their ADM energies are finite and obey

$$(19.1) \quad \tilde{E}_n \leq e^{2\epsilon} (E_* + \eta_n), \quad \eta_n \geq 0, \quad \eta_n \rightarrow 0.$$

No uniform bound in n on the solutions' classical regularity constants is needed for this conclusion.

Proof. The charged preparation belongs to the reduced neutral class by Lemma 17.2, and Proposition 17.3 supplies its exact physical endpoint after exhaustion at fixed ϵ, n . We may therefore apply the neutral end and flux results. We verify the relevant hypotheses before carrying out the additional scaling and field comparison.

The end and its ADM flux. The prepared tensor K is compactly supported, the end satisfies $g - \delta = O_j(r_y^{-1})$ and $R_g = O_j(r_y^{-3-\delta})$ for every fixed j , and the neutral offset inequality holds. The compact height bound is used only on the supports of K, C . The profiles, quadratic form and flux are exactly those identified in Lemma 18.1, with the capillarity parameter $\tau_0 = \ell^{3/2}$ denoted by τ in the neutral argument. Lemmas 8.1 and 8.2 consequently give

$$\hat{g}_n - \delta = O_2(r_y^{-1}), \quad R_{\hat{g}_n} \in L^1(X, dV_{\hat{g}_n}), \quad V = 4\nabla_g t + O(r_y^{-3}),$$

and the finite flux identity

$$(19.2) \quad E_{\hat{g}_n} = E_* - \frac{\mathfrak{F}_n}{8\pi}, \quad \mathfrak{F}_n = \lim_{s \rightarrow \infty} \int_{r_y=s} V_{\nu_s} dA_g.$$

Here ν_s points toward infinity. These statements require only fixed- n end estimates. In particular the exponential graph error has zero ADM contribution, and the leading conformal change $4t\delta$ has energy flux $-8\partial_t t$, as in the proof of Lemma 8.2.

The lower flux bound. Equation (17.6) has the source $\delta_0 \mathcal{T} + \rho + \delta_0 \tau_0 a \sigma - m_0(t) \mathcal{P}$ and the boundary flux used in Lemma 8.3. The two face identities are $F - P_B = H$, $w_\nu = a$ on black faces and $F - P_B = -H$, $w_\nu = -a$ on white faces. The signed slope estimates (17.8) hold on every existing face; no condition is needed on an absent color. The weight ρ has a fixed positive minimum on the compact collars, the penalty C_n is polynomial, and the weights $\rho_0, \rho_0^2, \rho_1^2$ are integrable because $\gamma > 3/4$. These are precisely the boundary, bulk and penalty hypotheses of that lemma. It gives

$$(19.3) \quad \mathfrak{F}_n \geq -\zeta_n, \quad 0 \leq \zeta_n \leq C(1+n)^N \left(\ell + \frac{\ell^2}{\tau_0} \right) = C(1+n)^N (\ell + \ell^{1/2}) \rightarrow 0.$$

Its proof absorbs the boundary contribution by the polynomial collar trace before estimating the floor penalty. Thus the constants in this bound do not involve the possibly much larger fixed- n classical regularity constants.

Scaling and electromagnetic fields. The constant rescaling $\tilde{g}_n = e^{4\epsilon} \hat{g}_n$ is asymptotically Euclidean in coordinates $z = e^{2\epsilon} y$, and

$$(19.4) \quad \tilde{E}_n = e^{2\epsilon} E_{\hat{g}_n}.$$

Indeed the coordinate derivative and sphere area factors in the ADM integral are $e^{-2\epsilon}$ and $e^{4\epsilon}$, respectively. This also gives the asserted metric decay and preserves scalar-curvature integrability.

The original flux forms have components $O_1(r_y^{-2})$ on this end. The forms in (18.10) are constant linear combinations of them and have the same decay. Define their vector fields by $\iota_{\tilde{X}_{i,n}} dV_{\tilde{g}_n} = \tilde{\alpha}_i$. The metric coefficients are constant at infinity up to $O_2(r_y^{-1})$, so this definition followed by the fixed coordinate dilation gives $\tilde{X}_{i,n} = O_1(|z|^{-2})$. Their integrals are unchanged by this coordinate change, so both total charges remain Q_E, Q_B . Proposition 18.4 supplies the charged scalar-curvature inequality for these very fields and metric.

Combining (19.2)–(19.4) proves (19.1) with $\eta_n = \zeta_n/(8\pi)$. Throughout this argument the datum and ϵ are fixed and the exhaustion precedes $n \rightarrow \infty$. $\square$

20. MINIMAL ENCLOSURE AND THE FINAL LIMITS

We now complete the proof of Theorem 14.2. All geometric data and preparation choices are fixed until the last step.

Lemma 20.1. *Let $(X, \tilde{g})$ be a comparison exterior furnished by the preceding sections. It has a smooth compact nonempty minimal boundary enclosure $\mathcal{H}$ whose exterior toward infinity is one ended and has no larger enclosing compact minimal boundary. Moreover,*

$$\text{Area}_{\tilde{g}}(\mathcal{H}) \geq A_*,$$

and restricting the comparison fields to that exterior preserves the two total charges.

Proof. The inner boundary B has strictly negative mean curvature for the normal into X . Sufficiently distant coordinate spheres have strictly positive outward mean curvature. Apply the trapped-region theorem of Andersson–Metzger with second tensor zero, required inner boundary B , and a large outer sphere [7, Theorem 7.3]. A strict inner collar makes the weakly trapped union nonempty. Its full marginal frontier is smooth, compact, and minimal. Take the component of its exterior incident on the distant sphere, together with the whole adjacent frontier, and fill bounded pockets on the inner side.

This construction is independent of sufficiently large truncations. Indeed, a weakly trapped bounding surface cannot have a largest-radius point in the region foliated by strictly mean-convex coordinate spheres: the outward mean-curvature comparison at that point is positive. Thus the full trapped union is confined to a fixed compact region. The frontier separates B from infinity and is nonempty.

If a larger compact minimal enclosure existed, its inner side, including the required collar, would be a permissible weakly trapped set, contrary to maximality. This also covers partial coincidence: components already present may be retained while other components are replaced. The comparison is of entire bounding regions. At a touching point of two minimal frontiers ordered from outside, the maximum principle identifies the coincident component.

The end-connected exterior with this boundary is complete up to the boundary. Restoring the inner regions discarded during preparation makes $\mathcal{H}$ a smooth cut enclosing S_0 . Therefore

$$\text{Area}_{\tilde{g}}(\mathcal{H}) \geq \text{Area}_g(\mathcal{H}) \geq a_g(S_0) = A_*.$$

The first inequality uses $\tilde{g} \geq g$. The closed comparison flux forms have the same integrals over $\mathcal{H}$ and a distant sphere, by Stokes' Theorem. Hence both total charges are preserved. $\square$

We use the following Riemannian consequence of the charged spacetime inequality [44, Theorem 2.3]. Let a smooth connected oriented three-dimensional exterior be complete with its nonempty compact outermost full minimal boundary included, have one strongly asymptotically flat end, and satisfy

$$R \geq 2(|\mathcal{E}|^2 + |\mathcal{B}|^2), \quad \text{div } \mathcal{E} = \text{div } \mathcal{B} = 0.$$

Assume metric decay $O_2(r^{-1})$, field decay $O_1(r^{-2})$, and integrable scalar curvature. Then, for ADM mass $\mathcal{M}$, total charge magnitude $\mathcal{Q}$, and full boundary area $4\pi r_H^2$,

$$(20.1) \quad \mathcal{M} \geq \mathcal{Q}, \quad r_H \leq \mathcal{M} + \sqrt{\mathcal{M}^2 - \mathcal{Q}^2}.$$

The boundary may be disconnected, and no area-charge restriction is imposed.

Here are the two reductions needed to apply the cited theorem. If $\mathcal{Q} > 0$, put

$$X = \frac{\mathcal{Q}_E \mathcal{E} + \mathcal{Q}_B \mathcal{B}}{\mathcal{Q}}, \quad \mathcal{E}' = \frac{\mathcal{Q}_E}{\mathcal{Q}} X, \quad \mathcal{B}' = \frac{\mathcal{Q}_B}{\mathcal{Q}} X.$$

These fields are divergence free, have the same two total charges and falloffs, and obey

$$\mathcal{E}' \times \mathcal{B}' = 0, \quad |\mathcal{E}'|^2 + |\mathcal{B}'|^2 = |X|^2 \leq |\mathcal{E}|^2 + |\mathcal{B}|^2.$$

If $\mathcal{Q} = 0$, take both new fields zero. With second tensor zero the resulting spacetime data have $J_m = 0$ and $8\pi\mu_m = R/2 - (|\mathcal{E}'|^2 + |\mathcal{B}'|^2) \geq 0$, so they satisfy the matter DEC of the cited theorem. Their boundary has zero future expansion.

Secondly, the full enclosing infimum is the full area of an outermost minimal boundary. To see this, if a full cut had smaller area, minimize perimeter among its full enclosing regions containing the original inner obstacle. Large mean-convex end spheres confine this problem; BV compactness and lower semicontinuity give a minimizer. Its free frontier is smooth and minimal in dimension three. At contact with the smooth minimal obstacle, the minimizing-hull regularity and the strong comparison principle give smooth coincidence; whole coincident components are retained. Filling bounded complementary pockets leaves the full frontier incident on the end. Since its area is strictly smaller, it cannot be just the original boundary. It is therefore a strictly larger full minimal enclosure, contrary to outermostness. This proves the area assertion, with every component counted. Theorem 2.3 of [44] now gives Equation (20.1), including positivity of the mass and both area branches. Only its numerical conclusion is used.

Proof of Theorem 14.2. Lemma 15.2 gives a future timelike ADM vector, and Proposition 15.1 supplies strict charged rest data with energy E_* and enclosing infimum A_* . Fix one such datum with positive energy and prepare its exterior by Proposition 16.3. For each sufficiently small fixed $\epsilon > 0$ and sufficiently large n , Proposition 17.3 gives a smooth limiting solution after exhausting the truncations at that fixed n .

The comparison construction gives the scalar inequality required in (20.1), closed field forms with the original total charges, and a metric no smaller than the prepared metric. Proposition 19.1 supplies strong asymptotic flatness, integrable scalar curvature, and an energy bound

$$\tilde{E}_n \leq e^{2\epsilon}(E_* + \eta_n), \quad \eta_n \rightarrow 0$$

at fixed data and ϵ . Lemma 20.1 supplies the appropriate outermost minimal boundary. All hypotheses of the Riemannian theorem are therefore satisfied on its exterior. In particular,

$$(20.2) \quad Q \leq \tilde{E}_n, \quad \sqrt{A_*/(4\pi)} \leq \tilde{E}_n + \sqrt{\tilde{E}_n^2 - Q^2}.$$

Let $F_Q(x) = x + \sqrt{x^2 - Q^2}$ for $x \geq Q$. This function is continuous and nondecreasing on its entire domain, including $x = Q$. By (20.2), the upper energy bound $e^{2\epsilon}(E_* + \eta_n)$ also lies in that domain. Replacing $\tilde{E}_n$ by this bound in (20.2), then sending $n \rightarrow \infty$, gives

$$e^{2\epsilon}E_* \geq Q, \quad \sqrt{A_*/(4\pi)} \leq F_Q(e^{2\epsilon}E_*).$$

Letting $\epsilon \downarrow 0$ yields

$$(20.3) \quad E_* \geq Q, \quad \sqrt{A_*/(4\pi)} \leq E_* + \sqrt{E_*^2 - Q^2}.$$

Only numerical upper bounds have been passed to the limit; no limiting comparison metric as $n \rightarrow \infty$ is needed.

Finally remove the strict rest-data approximation. Proposition 15.1 keeps Q_E, Q_B fixed, makes $E_* \rightarrow m = \sqrt{E^2 - |P|^2}$, and gives $A_* \rightarrow a_g(S)$. Continuity in (20.3) proves the theorem. This order of limits requires no constants uniform across the strict-data sequence. $\square$

Remark 20.2 (The area branches). For $0 < r \leq Q$, the conclusion follows from $m \geq Q$. For $r > Q$, it is equivalent to

$$m \geq \frac{1}{2} \left(r + \frac{Q^2}{r} \right).$$

The proof uses the upper-area formulation throughout, so disconnected boundaries of either type are covered. When $Q = 0$, it reduces to the neutral bound for $a_g(S)$; Corollary 9.4, applied as in Lemma 15.2, includes the non-timelike endpoint exclusion.

21. THE ELECTRIC REST-FRAME INEQUALITY AND FINITELY MANY ENDS

The one-ended charged theorem also gives the mass lower bound on the closed area-charge branch. We first record its purely electric rest-frame consequence. We then treat additional asymptotically flat ends by removing their distant tails and regarding the resulting spheres as further inner boundary components. This second argument uses only the electric rest-frame consequence.

Corollary 21.1 (Electric rest-frame inequality). *Let $(\Omega^3, g, K, \mathcal{E})$ be smooth and orientable, with Ω connected and complete with its nonempty compact smooth boundary S included, and with exactly one asymptotically flat end. The boundary may have finitely many components. Define*

$$16\pi\mu = R_g + (\text{tr}_g K)^2 - |K|_g^2, \quad 8\pi J = \text{div}_g(K - (\text{tr}_g K)g).$$

Assume that $\mu, |J|_g$ are integrable, that the ADM limits exist, and that, for some $q > 1/2$, the end satisfies

$$g - \delta = O_2(r^{-q}), \quad K = O_1(r^{-1-q}), \quad \mathcal{E} = O_1(r^{-2}).$$

There is no magnetic field. Suppose throughout Ω that

$$(21.1) \quad \text{div}_g \mathcal{E} = 0, \quad \mu - \frac{|\mathcal{E}|_g^2}{8\pi} \geq |J|_g,$$

and on S , with normal into Ω , that $H + \text{tr}_S K \leq 0$. Let $a = a_g(S)$ be the minimum enclosing area over all smooth filled cuts of the entire boundary, and put

$$r_a = \sqrt{\frac{a}{4\pi}}, \quad Q = \frac{1}{4\pi} \int_S \langle \mathcal{E}, \nu \rangle_g dA_g.$$

If the ADM momentum vanishes and $r_a \geq |Q|$, then the ADM energy m is positive and

$$(21.2) \quad m \geq \frac{1}{2} \left(r_a + \frac{Q^2}{r_a} \right).$$

Proof. The electromagnetic momentum is zero when the magnetic field is zero. Thus (21.1) is precisely the non-electromagnetic matter dominant energy condition in Theorem 14.2. Closedness of $\iota_{\mathcal{E}} dV_g$ identifies the displayed boundary charge with the charge at infinity. All other one-ended hypotheses are unchanged. That theorem gives $m > 0$, $m \geq |Q|$, and

$$r_a \leq m + \sqrt{m^2 - Q^2}.$$

The enclosing area is positive. Moreover, $m - \sqrt{m^2 - Q^2} \leq |Q| \leq r_a$. Consequently

$$(r_a - m + \sqrt{m^2 - Q^2})(r_a - m - \sqrt{m^2 - Q^2}) \leq 0.$$

Expanding and dividing by $2r_a > 0$ proves (21.2). This calculation includes $r_a = |Q|$; no limiting argument or equality classification is needed. $\square$

Corollary 21.2 (Outer area-minimizing boundary). *Under the hypotheses of Corollary 21.1, suppose that every filled enclosing cut of the whole boundary has area at least $A = \text{Area}_g(S)$. If $A \geq 4\pi Q^2$, then*

$$m \geq \sqrt{\frac{A}{16\pi}} + Q^2 \sqrt{\frac{\pi}{A}}.$$

In particular this applies to maximal purely electric source-free electrovacuum data with such a boundary and zero ADM momentum.

Proof. The boundary itself is an admissible cut, so the area assumption gives $a_g(S) = A$. Apply Corollary 21.1. For the stated electrovacuum specialization, $\text{tr}_g K = 0$, $\text{div}_g K = 0$, and $R_g = |K|_g^2 + 2|\mathcal{E}|_g^2$ imply $\mu = |\mathcal{E}|_g^2/(8\pi)$ and $J = 0$, hence (21.1). $\square$

21.1. The area associated with a designated end. Let Ω instead have finitely many asymptotically flat ends $\mathcal{E}_0, \dots, \mathcal{E}_k$, with $\mathcal{E}_0$ designated, and compact smooth boundary S . We allow $S = \emptyset$ only when $k \geq 1$. A designated-end cut is the full compact smooth manifold boundary $\Gamma = \partial Y$ of a smooth closed codimension-zero exterior $Y \subset \Omega$. Its intrinsic interior $W = \text{int } Y$ must be connected, lie in $\text{int } \Omega$, contain a distant tail of $\mathcal{E}_0$, and exclude distant tails of every other end and the entire S . Here ∂Y means the boundary of Y as a manifold, not its relative topological boundary in Ω ; in particular, every portion coinciding with S is counted. Bounded complementary pockets are assigned to the excluded side. The cut need not be connected, and its individual components need not separately separate Ω .

Define

$$(21.3) \quad a_0 = \inf_{\Gamma} \text{Area}_g(\Gamma), \quad r_0 = \sqrt{\frac{a_0}{4\pi}},$$

where the infimum is over these full designated-end cuts. This is a single enclosing-area infimum, not the sum of separate end or boundary component infima. A distant sphere in $\mathcal{E}_0$ is a competitor, so $a_0 < \infty$.

Theorem 21.3 (Electric rest-frame extension to finitely many ends). *Let $(\Omega^3, g, K, \mathcal{E})$ be smooth, connected and orientable, complete with its compact smooth boundary S included, and have the finitely many asymptotically flat ends just described. Assume (21.1) everywhere, integrable $\mu, |J|_g$, and $H + \text{tr}_S K \leq 0$ with the normal into Ω on every component of S . Assume the end falloff of Corollary 21.1 on each end, with a common $q > 1/2$, and finite ADM limits at the designated end. There is no magnetic field. Let*

$$Q_0 = \frac{1}{4\pi} \int_{S_R^0} \langle \mathcal{E}, \nu_R \rangle_g dA_g$$

be its conserved outward charge. Suppose its ADM momentum is zero and that $0 < a_0$ and $r_0 \geq |Q_0|$, with a_0 as in (21.3). Then its ADM energy m_0 is positive and

$$(21.4) \quad m_0 \geq \frac{1}{2} \left(r_0 + \frac{Q_0^2}{r_0} \right).$$

No momentum condition is imposed at the other ends.

Proof. For $k = 0$, this is Corollary 21.1. Suppose $k \geq 1$. In each unwanted end $\mathcal{E}_j$, choose a coordinate sphere $S_{R_j}^j$ sufficiently far out, and remove its open outside tail. Denote the retained manifold by $\Omega_{\mathbf{R}}$. The radii can be chosen beyond a connected compact core: fix paths in Ω connecting the finitely many inner end collars and the components of S , and put these paths inside that core before increasing the radii. Each retained end annulus is connected and attached to the core. Thus $\Omega_{\mathbf{R}}$ is connected, has exactly the designated end, and has nonempty compact smooth boundary

$$S_{\mathbf{R}} = S \sqcup \bigsqcup_{j=1}^k S_{R_j}^j.$$

It is a closed subset of the original complete manifold. A Cauchy sequence for its intrinsic length metric therefore has a limit in that subset; smooth boundary half-ball charts give convergence in the intrinsic metric as well. This proves completeness with its boundary included.

On an added sphere, the normal pointing into the retained exterior is the negative of the normal pointing toward that unwanted infinity. The asymptotic estimates give

$$H = -\frac{2}{R_j} + O(R_j^{-1-q}), \quad \text{tr}_{S_{R_j}^j} K = O(R_j^{-1-q}).$$

Hence its future outward expansion, in the retained-exterior orientation, is strictly negative for large R_j . The original boundary retains its weakly trapped sign. All fields are merely restricted, so the charged DEC and divergence equation are unchanged. No extension of the fields over a cap is involved. The designated end and its ADM energy and momentum are also unchanged.

Put $\mathcal{F} = \iota_{\mathcal{E}} dV_g$. The equation $d\mathcal{F} = 0$ and Stokes' theorem on the region between $S_{\mathbf{R}}$ and a distant sphere of the designated end give

$$(21.5) \quad \frac{1}{4\pi} \int_{S_{\mathbf{R}}} \mathcal{F} = Q_0,$$

where every inner component is oriented into the retained exterior. In particular, if Q_j is the outward charge of the j th unwanted end and $Q(S)$ is the charge of the original boundary with normal into Ω , then $Q_0 = Q(S) - \sum_{j=1}^k Q_j$. The charge in (21.4) is therefore the designated end charge, not generally $Q(S)$.

Let $a_{\mathbf{R}} = a_g(S_{\mathbf{R}})$ be the minimum enclosing area in the retained one-ended manifold. Every smooth filled cut of $S_{\mathbf{R}}$ also gives a designated-end cut in Ω : keep its chosen-end exterior, and assign phase zero to all removed tails. This introduces no new frontier because the original cut already encloses every added inner sphere. If the cut coincides with any added sphere, that sphere is the same smooth surface in the original end and its full area remains counted. The same observation applies to coincident portions of S . Selecting the component toward the designated end and filling bounded pockets keeps the whole-boundary convention and can only remove boundary components. Consequently

$$(21.6) \quad a_{\mathbf{R}} \geq a_0, \quad r_{\mathbf{R}} := \sqrt{\frac{a_{\mathbf{R}}}{4\pi}} \geq r_0 \geq |Q_0|.$$

This argument compares the smooth full-cut classes directly; it does not require attainment of either infimum or a limiting statement as $\mathbf{R} \rightarrow \infty$.

All hypotheses of Corollary 21.1 now hold on $\Omega_{\mathbf{R}}$, with charge Q_0 by (21.5). Hence

$$m_0 \geq \frac{1}{2} \left(r_{\mathbf{R}} + \frac{Q_0^2}{r_{\mathbf{R}}} \right).$$

The function $r \mapsto (r + Q_0^2/r)/2$ is nondecreasing for $r \geq |Q_0|$ (and $r > 0$). Combining this fact with (21.6) proves (21.4), including the endpoint $r_0 = |Q_0|$. $\square$

The finite-end theorem is explicitly a purely electric result in the rest frame of the designated end. The use of total full-cut area is essential: taking separate component areas, or applying a sum of end-energy estimates, would not give the comparison (21.6).

Part 4. Four spatial dimensions

22. THE INVARIANT PENROSE INEQUALITY IN FOUR SPATIAL DIMENSIONS

We prove a lower bound for the ADM rest mass in terms of the least three-volume needed to separate a chosen asymptotically flat end from the compact weakly trapped boundary and the other ends. The second fundamental form is arbitrary. The final comparison uses the Riemannian Penrose inequality in dimension four after a graph and conformal deformation; the required enclosure is constructed below.

22.1. Initial data and enclosing cuts. Four is the spatial dimension; the associated spacetime dimension is five, and a hypersurface area is a three-dimensional Riemannian volume. Set

$$\omega = \omega_3 = |S^3| = 2\pi^2, \quad \tau = \text{tr}_g K, \quad \mu = \frac{1}{2}(R_g + \tau^2 - |K|_g^2), \quad J_i = \nabla^j (K_{ij} - \tau g_{ij}).$$

Here $R_g = \text{Scal}_g$ denotes scalar curvature and ∇ is the Levi-Civita connection of g . The cosmological constant is zero. We use $\Delta_g = \text{div}_g \text{grad}_g$. For a hypersurface with specified unit normal ν , write

$$H = \text{div}_{\tan} \nu, \quad \text{tr}_{\tan} K = (g^{ij} - \nu^i \nu^j) K_{ij}, \quad \Theta_K = H + \text{tr}_{\tan} K.$$

At an inner boundary the normal points into the exterior. In an auxiliary trapped-region construction it points out of the bounded region.

The notation $O_j(r^{-a})$ includes coordinate derivatives of order $k \leq j$, bounded by $O(r^{-a-k})$. On an asymptotically flat end the ADM normalization is

$$(22.1) \quad E = \frac{1}{6\omega} \lim_{R \rightarrow \infty} \int_{|x|=R} (\partial_j g_{ij} - \partial_i g_{jj}) n_\delta^i \, dA_\delta,$$

$$(22.2) \quad P_i = \frac{1}{3\omega} \lim_{R \rightarrow \infty} \int_{|x|=R} (K_{ij} - \tau g_{ij}) n_\delta^j \, dA_\delta.$$

Here n_δ and dA_δ are Euclidean. Whenever a charge is used below, its stated limit is required to exist.

Definition 22.1 (Enclosing cuts). Let M have nonempty compact boundary S and finitely many asymptotically flat ends, and fix an end e . An admissible outer domain D_e is a connected smooth codimension-zero submanifold with boundary, closed as a subset of M , such that its manifold interior lies in $\text{int } M$, it contains the entire sufficiently distant part of e , and it contains no sufficiently distant part of any other end. Its entire intrinsic manifold boundary $\Gamma = \partial_{\text{man}} D_e$ is required to be compact. The enclosing volume is

$$A_e(S) = \inf_{D_e} |\partial_{\text{man}} D_e|_g.$$

All frontier coincident with S is counted. The cut may be disconnected. In the one-ended case $D_e = M$ is allowed and has intrinsic boundary S . We then also write A_* for this infimum.

Only these smooth cuts are used. Their infimum need not be attained. The definition treats the entire original boundary and all other ends as obstacles to the selected outer domain.

Theorem 22.2. *Let (M^4, g, K) be smooth and connected, with M orientable, K an arbitrary smooth symmetric covariant two-tensor, and $S = \partial M$ nonempty, compact, and smooth. Suppose*

g is complete as a metric space with S included. Outside a compact set let M have finitely many, and at least one, ends diffeomorphic to the complement of a closed ball in $\mathbb{R}^4$. Assume

$$\mu \geq |J|_g, \quad \mu, |J|_g \in L^1(M, dV_g),$$

and, on every end for a common $q > 1$,

$$g - \delta = O_2(r^{-q}), \quad K = O_1(r^{-1-q}),$$

with finite energy and momentum limits (22.1)–(22.2). On every component of S assume $\Theta_K \leq 0$, with normal into M .

Then for each end e ,

$$(22.3) \quad E_e \geq \sqrt{|P_e|_\delta^2 + \frac{1}{4} \left(\frac{A_e(S)}{\omega} \right)^{4/3}}.$$

In particular $E_e > |P_e|_\delta$, and the invariant mass satisfies

$$(22.4) \quad \sqrt{E_e^2 - |P_e|_\delta^2} \geq \frac{1}{2} \left(\frac{A_e(S)}{\omega} \right)^{2/3}.$$

The strict timelikeness is a conclusion: Lemma 24.1 establishes $A_e(S) > 0$, and the proof includes the non-timelike case. There is no extra decay assumption on $\tau = \text{tr}_g K$. All original boundary components use the same future trapping convention.

The coefficient is sharp. On the time-symmetric Schwarzschild–Tangherlini exterior [50],

$$g_m = \left(1 + \frac{m}{2r^2}\right)^2 \delta, \quad K = 0, \quad r \geq \sqrt{m/2},$$

the ADM energy is m , the momentum is zero, and the boundary has three-volume $\omega(2m)^{3/2}$. The areal radius $r + m/(2r)$ is nondecreasing on this exterior. Radial projection to its boundary is therefore nonexpanding on three-dimensional tangent volumes; its degree on an enclosing cut is one. The minimum enclosing volume is consequently the boundary volume, and (22.3) is an equality for this family. The equality statement below concerns the original exterior data and has additional horizon hypotheses; these are not assumptions of the numerical inequality.

22.2. Equality for a connected exterior. We also write $\theta_+ = \Theta_K$ for the future null expansion.

Definition 22.3 (The connected outermost horizon class). Let (Ω^4, g, K) satisfy the initial-data hypotheses of Theorem 22.2, with exactly one end and connected boundary S . We additionally require:

- (i) S is a future marginally outer trapped surface (MOTS): $\theta_+(S) = 0$ for the normal into Ω .
- (ii) S is outermost toward the end: no smooth compact embedded two-sided enclosing hypersurface in $\text{int } \Omega$, possibly disconnected and oriented toward the end, has $\theta_+ \leq 0$ everywhere.
- (iii) S is outer area-minimizing: every cut in Definition 22.1 has area at least $A := |S|_g > 0$. Consequently $A_e(S) = A$.

For a geometric mass $M_0 > 0$, the Schwarzschild–Tangherlini metric [50] in its static exterior is

$$(22.5) \quad \bar{g}_{M_0} = - \left(1 - \frac{2M_0}{r^2}\right) dt^2 + \left(1 - \frac{2M_0}{r^2}\right)^{-1} dr^2 + r^2 g_{S^3}, \quad r > \sqrt{2M_0},$$

where g_{S^3} is the unit round metric. “Regular maximal extension” refers to the usual extension through the future and past horizons, including their bifurcation sphere. The apparent singularity in Equation (22.5) at the horizon is a coordinate singularity.

Theorem 22.4 (Original-data exterior rigidity). *Let (Ω, g, K) belong to Definition 22.3, and put $m = \sqrt{E^2 - |P|_\delta^2}$. Then*

$$(22.6) \quad m = \frac{1}{2} \left(\frac{A}{\omega} \right)^{2/3}$$

if and only if the original initial data admit a global smooth spacelike embedding, including S , into the regular maximal extension of $\bar{g}_m$. The image lies in the closure of the corresponding exterior, the chosen end approaches its spatial infinity, and S maps to a smooth section of its future horizon or to its bifurcation sphere. The pullback of the spacetime metric is g ; for a consistent future unit normal n , the second fundamental form is precisely the original tensor K , with the convention

$$(22.7) \quad K(U, V) = \bar{g}_m(\bar{\nabla}_U n, V).$$

Both the embedding and the normal extend smoothly to S in horizon-regular coordinates.

Equivalently, every such Schwarzschild–Tangherlini hypersurface in $\bar{g}_{M_0}$ satisfying Definition 22.3 and the prescribed asymptotic hypotheses has invariant ADM mass M_0 , horizon area $\omega(2M_0)^{3/2}$, and equality in Theorem 22.2.

No spherical topology, simple connectivity, stationary development, or vacuum condition is assumed in Theorem 22.4. The conclusion permits non-time-symmetric and asymptotically boosted hypersurfaces. It asserts rigidity only of the exterior in Definition 22.3 and makes no assertion about data behind S . A graph in the singular static time coordinate is not required at the horizon.

22.3. Proof strategy and parameter order. The proof first replaces a timelike asymptotic end by a rest end, while preserving the dominant energy condition and comparing every enclosing cut. Strictification then gives positive energy slack and strict boundary trapping. Two threshold frontiers provide boundary collars and a small interval for the prescribed graph trace.

On the resulting exterior we choose a positive smooth coefficient function $l : \mathbb{R} \rightarrow (0, \infty)$, solve for real functions f, Z , define t from them, and construct

$$\hat{g} = e^{2t}(g + l(t)^2 df \otimes df).$$

The scalar identity retains the unrestricted term $-F\tau$, where F is the prescribed scalar trace in the graph equation. A polynomial penalty controls it in the first-floor estimate; the later height bounds let strict DEC slack absorb it in the final scalar comparison. The coupled existence proof uses a scalar principal matrix with three transverse eigenvalues equal to one. Its measurable axial coefficient is handled by the regularity theorem proved in Section 28.

The deformation produces nonnegative scalar curvature, negative boundary mean curvature, and a vanishing upper bound for the energy change. A separate perimeter minimization supplies an outer-minimizing minimal enclosure, so Bray–Lee’s Riemannian theorem applies. The resulting energy estimate transfers back to the original invariant mass. A positive conformal shell excludes null and spacelike ADM vectors, and truncating the other ends completes the endwise statement.

Section 24 proves the end reduction; Section 25 constructs the threshold exterior and collars. Sections 26 and 27 establish the equations, floor exclusion and global bounds. Sections 28 and 29 prove regularity, existence and the final Riemannian comparison. The shared enclosure results in Section 23 supply both the minimal boundary for that comparison and the one-sided area derivative used at equality.

The equality proof then returns to the original data. Sections 30 and 31 turn the numerical inequality for nearby DEC and trapped data into a causal stationary field. Those nearby data need not preserve outermostness or outer area minimization. Sections 32 and 33 remove twist, establish

completeness and classify the static base. Section 34 recovers the original metric and second fundamental form in regular horizon coordinates and checks the converse, including boosted ends.

Remark 22.5 (Order of constants and limits). Fix the geometry, strictification and $\epsilon > 0$ first. The integer n is then sufficiently large. A bound Π_n is polynomial in n , uniformly in the outer truncation radius, homotopy parameter and solution; fixed- n regularity constants may depend arbitrarily on n . Exhaust the outer radius at fixed n , then send $n \rightarrow \infty$ using only the polynomial bounds. Finally send $\epsilon \downarrow 0$, remove strictification at each fixed repaired end, and send the end-replacement radius to infinity.

23. ENCLOSING AREA, OBSTACLE MINIMIZERS, AND METRIC VARIATION

The cuts in Definition 22.1 count the entire intrinsic boundary, including every portion coincident with the original boundary. We identify their area infimum with a full-perimeter minimum. The resulting enclosure will be used in the numerical inequality. Its variation formula, which retains all tied minimizers, will be used in the equality argument.

Throughout this section, (X, g) is a smooth connected orientable four-dimensional Riemannian manifold with nonempty compact smooth boundary B . It is complete with B included and has exactly one end, which is asymptotically Euclidean. In coordinates on that end we assume $g - \delta = O_2(r^{-q})$ for some $q > 1$. No scalar-curvature assumption is needed in this section. Write

$$a_g(B) = \inf_{\Gamma} \text{Area}_g(\Gamma),$$

where Γ ranges over the full smooth enclosing cuts of Definition 22.1.

23.1. A perimeter problem that counts the original boundary. Choose a smooth manifold $\tilde{X}$ without boundary by attaching an inner side along B , and extend g smoothly through B . For example, one may use the topological double, with a smooth metric extension in a collar and an arbitrary smooth metric farther inside. Let O be the closed inner side, so that $\partial O = B$ and $\tilde{X} \setminus O = \text{int } X$. Neither completeness nor a curvature condition on the extension will be used. We continue to write g for this auxiliary extension.

Let $\mathcal{A}$ consist of sets E of locally finite perimeter in $\tilde{X}$ such that $O \subset E$ up to a null set and $E \setminus O$ is contained, up to a null set, in some compact subset of X . Sets are identified if their indicator functions agree almost everywhere. The perimeter is always

$$(23.1) \quad P_g(E) = |D\mathbf{1}_E|_g(\tilde{X}),$$

computed in the extended manifold. Thus $P_g(O) = \text{Area}_g(B)$, and perimeter on B is retained. Although O itself need not have finite volume, the competitors differ from it only in a compact exterior region, so the compactness arguments below are local BV arguments on a fixed bounded region. We use the standard perimeter compactness, Gauss–Green, and submodularity facts for BV sets [5, 39].

If D is a smooth exterior domain from Definition 22.1, its filled side is

$$E_D = O \cup (X \setminus \text{int } D).$$

Here $\text{int } D$ denotes the intrinsic manifold interior, which lies in $\text{int } X$. The distant end is absent from E_D , and

$$(23.2) \quad E_D \in \mathcal{A}, \quad P_g(E_D) = \text{Area}_g(\partial_{\text{man}} D).$$

The identity includes any frontier on B . In particular $D = X$ corresponds to $E_D = O$, with perimeter $\text{Area}_g(B)$.

Proposition 23.1 (Full-perimeter minimizing enclosures). *The minimum of P_g over $\mathcal{A}$ exists and equals $a_g(B) > 0$. Every minimizer has a regular representative whose frontier T is a nonempty compact embedded $C^{1,1}$ hypersurface, smooth and minimal on $T \setminus B$. Its graph functions belong locally to $W^{2,\infty}$, and hence to $W^{2,2}$, including at contact with B . Its outward coorientation points from the filled set toward a connected exterior containing the asymptotic end. All minimizing frontiers lie in a fixed compact subset of X .*

Every such frontier is a limit in C^1 , and in area, of admissible smooth enclosing cuts lying in $\text{int } X$. If in addition $H_B < 0$ for the normal into X , every minimizer contains a fixed collar of B on the exterior side. In this case T is itself a smooth minimal enclosing cut, and its closed exterior is smooth, connected, complete with its nonempty boundary included, and one-ended. The boundary of this exterior is outer area-minimizing there, with every boundary component counted.

Proof. A common outer truncation. On the coordinate end, let $Y = \nabla r / |\nabla r|_g$ be the unit normal pointing toward increasing radius. The differentiated asymptotic decay gives

$$(23.3) \quad \text{div}_g Y = \frac{3}{r} + O(r^{-1-q}) > 0 \quad (r \geq R_0)$$

after increasing R_0 . For a competitor E and almost every $R > R_0$, put $E_R = E \setminus \{r > R\}$. Gauss–Green on $V_R = E \cap \{r > R\}$ gives

$$\int_{V_R} \text{div}_g Y \, dV_g = \int_{\partial^* E \cap \{r > R\}} g(Y, \nu_E) \, dA_g - \int_{\{r=R\}} \mathbf{1}_E \, dA_g.$$

The trace on the slicing sphere is understood at a good slicing radius; ν_E is the measure-theoretic outward normal of the filled set. Consequently

$$(23.4) \quad P_g(E_R) - P_g(E) \leq - \int_{V_R} \text{div}_g Y \, dV_g \leq 0,$$

and the inequality is strict if V_R has positive volume.

Fix $R_1 > R_0$. For each competitor separately, choose a good radius in (R_0, R_1) and use Equation (23.4). The infimum therefore equals the infimum in the class with $E \setminus O$ confined to the same fixed truncation $\{r \leq R_1\}$ together with the compact core. BV compactness on a slightly larger compact region and lower semicontinuity give a minimizer. The constraints of containing O and being empty in the prescribed exterior tail are closed under this convergence. The competitor O gives a finite upper bound.

The resulting set is also a global minimizer in $\mathcal{A}$. Applied to any global minimizer, Equation (23.4) forbids positive filled volume beyond a good radius above R_0 . Choosing such a radius below a fixed number R_2 with $R_0 < R_2 < R_1$ shows that all minimizers have their filled exterior parts in the same compact truncation at R_2 . This also keeps their frontiers away from the artificial boundary used in the compactness argument. There is no requirement that one slicing radius be good for every BV set.

Regularity at the obstacle. A global minimizer is locally perimeter-minimizing among competitors that contain the smooth open obstacle O° . The relevant form of the Riemannian obstacle regularity theorem is the following: in a smooth Riemannian manifold of dimension less than eight, a local perimeter minimizer constrained to contain an open set with C^2 boundary has a $C^{1,1}$ frontier and is smooth away from contact. This is precisely Regularity Theorem 1.3(iii) of Huisken–Ilmanen [31, pp. 368–369], applied with zero bulk term. Its hypotheses are local, so the arbitrary geometry of the attached inner side and its possible noncompactness are irrelevant. Here the ambient dimension is four. The result describes the entire frontier of the regular representative, not just the reduced boundary outside an unaccounted exceptional set.

For clarity, the obstacle also gives the usual two-sided almost-minimality estimate needed for density and compactness estimates. Choose a smooth compactly supported vector field Z on $\tilde{X}$

with $|Z|_g \leq 1$ and $Z = \nu_O$ on B . Such a field is obtained from a tubular neighborhood of the compact smooth boundary. For a local replacement F , with $F \triangle E$ compactly supported, $F \cup O$ is admissible. Perimeter submodularity and Gauss–Green imply

$$(23.5) \quad \begin{aligned} P_g(E) &\leq P_g(F \cup O) \leq P_g(F) + P_g(O) - P_g(F \cap O), \\ P_g(O) - P_g(F \cap O) &\leq \int_{O \setminus F} \operatorname{div}_g Z \, dV_g \leq C \operatorname{Vol}_g(E \triangle F). \end{aligned}$$

Thus local replacements satisfy the standard perimeter almost-minimality inequality. In particular, the usual perimeter and volume density estimates apply, including at contact [39]. The $C^{1,1}$ conclusion above directly gives the asserted Sobolev regularity in graph charts. Off the obstacle, arbitrary local replacements are allowed; first variation gives zero mean curvature and interior regularity gives smoothness.

Take the representative that is the closure of its measure-theoretic interior. Its frontier is the compact $C^{1,1}$ hypersurface just described, and the filled and exterior sides occupy the two local sides of every frontier chart. This choice discards all irrelevant null-set slits. The frontier is nonempty: the filled set contains the open inner side, whereas its complement contains an exterior tail. On the connected extended manifold a BV indicator with zero perimeter is constant almost everywhere, which is impossible here. The nonempty regular frontier consequently has positive area.

Connectedness of the exterior. The complement of the regular filled set has a component containing the entire sufficiently distant end. Any other component is relatively compact, because the frontier is compact and X has one end. If such a component existed, adjoining it to the filled set would remove its nonempty boundary and introduce no new perimeter. More explicitly, near each frontier point there is only one exterior side; the boundary of a complement component is therefore a union of connected components of the compact embedded frontier. Filling a bounded component removes a positive amount of full perimeter, including any portion on B . This contradicts minimality. Thus the exterior is connected.

Equality with the smooth-cut infimum. Equation (23.2) gives $\min_{\mathcal{A}} P_g \leq a_g(B)$. To prove the reverse inequality, let T be a minimizing frontier with its outward coorientation. Choose a smooth compactly supported vector field V with $g(V, \nu_T) \geq c_0 > 0$ on T ; approximate its continuous unit normal in local charts and use a partition of unity. If Φ_t is its flow, compactness and transversality give a C^1 flow collar

$$\Psi : T \times (-\delta, \delta) \longrightarrow \tilde{X}, \quad \Psi(p, t) = \Phi_t(p),$$

in which the filled side is $\{t \leq 0\}$. For sufficiently small $\varepsilon > 0$ the displaced filled set satisfies

$$E_\varepsilon := \Phi_\varepsilon(E) = E \cup \Psi(T \times (0, \varepsilon]).$$

This inclusion is global: a point can change its membership only when its flow line crosses T , and every such short crossing is outward in the chosen collar. Thus $O \subset E \subset \operatorname{int} E_\varepsilon$. The displaced frontier $T_\varepsilon = \partial E_\varepsilon$ is disjoint from O and lies in $\operatorname{int} X$. Its distance from the compact original boundary is positive.

Choose a smaller cooriented collar of T_ε with compact closure disjoint from O , and a C^1 defining function there whose derivative along V is positive. Smooth this defining function in C^1 . A sufficiently close approximation still has positive V -derivative, and its zero set meets each collar fiber exactly once. It is consequently a smooth hypersurface given by a small graph over T_ε along those fibers. The graph displacement extends to a collar isotopy: in collar coordinates use $(p, t) \mapsto (p, t + \tau \chi(t) w(p))$, where w is the small graph height, χ equals one near zero and vanishes near the collar ends, and $0 \leq \tau \leq 1$. Smallness of w makes the derivative in t positive. This isotopy is the identity on O and in the distant end. It therefore preserves enclosure and

connectedness of the exterior. Its final smooth frontier bounds a connected smooth exterior domain, closed in X , and is exactly that domain's full intrinsic boundary.

The resulting cuts are admissible. Their areas tend first to the area of T_ε by C^1 approximation and then to $\text{Area}_g(T)$ as $\varepsilon \downarrow 0$. Hence $a_g(B) \leq P_g(E)$, proving equality of the two minimization problems and the claimed approximation. This also proves that the minimum is independent of the chosen inner extension.

A strict inner barrier. Suppose $H_B < 0$ for the normal into X . Compactness of B gives an exterior signed-distance collar $0 \leq t \leq \varepsilon$ whose unit normal $Y = \partial_t$ satisfies

$$\text{div}_g Y \leq -\beta < 0$$

for fixed $\varepsilon, \beta > 0$. For almost every $s \in (0, \varepsilon)$, let $O_s = O \cup \{0 < t < s\}$ and $W_s = O_s \setminus E$. Extend the collar field smoothly across B when applying Gauss–Green. The boundary flux identity on W_s gives

$$\int_{W_s} \text{div}_g Y \, dV_g = \int_{\{t=s\}} \mathbf{1}_{E^c} \, dA_g - \int_{\partial^* E \cap \{0 \leq t < s\}} g(Y, \nu_E) \, dA_g.$$

The second integral includes the frontier at $t = 0$. Therefore

$$(23.6) \quad P_g(E \cup O_s) - P_g(E) \leq \int_{W_s} \text{div}_g Y \, dV_g \leq -\beta \text{Vol}_g(W_s).$$

Minimality forces $\text{Vol}_g(W_s) = 0$. For each minimizer choose a good $s \in (\varepsilon/2, \varepsilon)$. Its regular representative then contains the whole open collar $0 \leq t < \varepsilon/2$ as interior points in $\tilde{X}$. Thus the same collar is included by all minimizers, despite allowing their good slicing levels to differ.

The frontier is now disjoint from B , hence is smooth and minimal. Its closed exterior D is a smooth submanifold with boundary T , closed as a subset of X , with connected interior and the same distant end. It is complete in its intrinsic length metric. Indeed an intrinsic Cauchy sequence is Cauchy for the original metric distance and converges to a point of the closed set D by completeness of X . Interior charts and smooth boundary half-ball charts identify the two local notions of convergence, with locally comparable path lengths, so the sequence converges also intrinsically in D .

Finally, a smooth enclosing cut in D determines a filled competitor containing the minimizing set E , hence also containing O . Its full area is at least $P_g(E) = \text{Area}_g(T)$ by global minimality. This is the required outer area-minimizing property, with all components and any frontier on T retained. $\square$

23.2. Compactness of all minimizers and the area derivative. For a regular minimizing frontier $T = \partial E$, define its unoriented tangent-plane area measure on the Grassmann bundle of three-planes by

$$(23.7) \quad \int \Phi \, dV_T = \int_T \Phi(x, T_x T) \, dA_g(x) \quad (\Phi \text{ continuous on the Grassmann bundle}).$$

Here V_T is a measure, not a volume form. Let $\mathcal{T}$ be the set of all minimizing filled sets, identified almost everywhere and recorded together with their regular frontiers. We give $\mathcal{T}$ the topology of local L^1 convergence of indicators and weak convergence of the measures in Equation (23.7). All variable parts and all these measures have support in one compact region by Proposition 23.1.

Proposition 23.2 (Compact minimizing space and right area derivative). *The space $\mathcal{T}$ is compact and metrizable. If h is a smooth symmetric two-tensor, the function*

$$(23.8) \quad a'_T(h) = \frac{1}{2} \int_T \text{tr}_{T_x T} h \, dA_g$$

is continuous on $\mathcal{T}$.

More generally, let g_s , $|s| < s_0$, be a smooth path of smooth metrics on X , with $g_0 = g$, uniformly comparable to g , and with uniformly bounded first and second parameter derivatives relative to g . Assume that their large coordinate spheres have strictly positive outward mean curvature beyond one common radius. Write $a(s)$ for their smooth-cut infimum and $h = \partial_s g_s|_{s=0}$. Then

$$(23.9) \quad a'(0+) = \min_{T \in \mathcal{T}} a'_T(h).$$

In particular the minimum on the right is attained. If $s_j \rightarrow 0$ and T_j is any minimizing frontier for g_{s_j} , a subsequence of its filled sets converges to a member of $\mathcal{T}$, and its g -tangent-plane area measures converge to the corresponding measure.

Proof. The proof of the outer barrier in Proposition 23.1 is uniform for this family: all minimizing frontiers lie in one compact set $L \subset X$. The metrics can be extended smoothly through the fixed boundary in a common collar. On that collar and on L , their smooth dependence on s supplies uniform local bounds and convergence to the fixed extended metric. The regularity and density results used above therefore remain available with uniform local bounds where needed.

First consider a sequence of minimizers for g . BV compactness gives $\mathbf{1}_{E_j} \rightarrow \mathbf{1}_E$ locally in L^1 along a subsequence, with E satisfying the same obstacle and exterior-tail constraints. Lower semicontinuity and global minimality imply

$$a_g(B) \leq P_g(E) \leq \liminf_j P_g(E_j) = a_g(B).$$

Thus E is another minimizer, and the perimeters converge as well.

This strict BV convergence also gives the claimed tangent-plane convergence, including mass carried by the original boundary. We give the localization argument to keep track of the metric norm. Set $\sigma_j = |D\mathbf{1}_{E_j}|_g$ and $\sigma = |D\mathbf{1}_E|_g$, as scalar perimeter measures on $\tilde{X}$. Every subsequential weak limit of σ_j dominates σ : this follows from lower semicontinuity of perimeter weighted by an arbitrary nonnegative continuous function. Their total masses converge and their supports lie in one compact set, so this domination must be equality. Hence σ_j converges weakly to σ . This conclusion uses only weighted lower semicontinuity, before any continuity theorem for polar directions is applied.

Now form the vector-valued measures $\mu_j = \nu_{E_j} \sigma_j$ and $\mu = \nu_E \sigma$. Gauss–Green and local L^1 convergence give $\mu_j \rightarrow \mu$. Choose finitely many smooth coordinate charts covering the common compact support and a nonnegative smooth partition of unity $\{\rho_\alpha\}$ subordinate to them. In each chart choose a smooth matrix $A_\alpha(x)$ such that $|A_\alpha(x)v|_\delta = |v|_g$. The coordinate vector measures

$$\lambda_{\alpha,j} = \rho_\alpha A_\alpha \mu_j \quad \text{satisfy} \quad |\lambda_{\alpha,j}|_\delta = \rho_\alpha \sigma_j.$$

They converge weakly, and their Euclidean total variation masses converge by the already established weak convergence of σ_j . The vector-measure continuity theorem therefore applies in each chart [48, Theorem 1.3]. Use its bounded continuous polar integrand

$$(x, z) \mapsto \Phi(x, (A_\alpha(x)^{-1}z)^\perp), \quad |z|_\delta = 1,$$

and sum over the partition. This proves weak convergence of V_{T_j} to V_T . Charts crossing B are ordinary interior charts of $\tilde{X}$, so no contact mass is removed in this localization. The compact base region and its Grassmann bundle are metrizable, so the indicated L^1 and weak-measure topology is metrizable. The subsequence argument proves compactness. Taking $\Phi(x, \Pi) = \frac{1}{2} \text{tr}_\Pi h$ proves the asserted continuity.

For the varying metrics, uniform comparison with $g_s \rightarrow g$ on L first gives

$$(23.10) \quad a(s) \longrightarrow a(0), \quad P_g(E_s) \longrightarrow a(0)$$

for any choice of a minimizer E_s for g_s . Indeed testing on a fixed g -minimizer gives the upper bound for $a(s)$; comparison of perimeters gives the lower bound, since $P_g(E_s) \geq a(0)$. Consequently every BV limit of such E_s is a g -minimizer and the same strict-convergence argument gives tangent-plane convergence. This proves the last assertion of the proposition.

It remains to calculate the derivative. The area element of a fixed three-plane has the uniform Taylor expansion

$$(23.11) \quad dA_{g_s}|_{\Pi} = \left(1 + \frac{s}{2} \operatorname{tr}_{\Pi} h + O(s^2)\right) dA_g|_{\Pi}.$$

The remainder is uniform over $x \in L$ and all three-planes Π , because of the parameter-derivative bounds and uniform metric positivity. Integrating over the fixed reduced boundary of a competitor gives

$$P_{g_s}(E) = P_g(E) + s \frac{1}{2} \int_{\partial^* E} \operatorname{tr}_{T_x \partial^* E} h dA_g + O(s^2) P_g(E).$$

For each $T \in \mathcal{T}$, using its fixed filled set as a competitor therefore yields

$$\limsup_{s \downarrow 0} \frac{a(s) - a(0)}{s} \leq a'_T(h).$$

The continuous function in Equation (23.8) has a minimum on $\mathcal{T}$, so this gives the required upper bound by that minimum.

Conversely, choose a minimizer E_s for each $s > 0$. Its g -perimeter is at least $a(0)$ and is uniformly bounded by Equation (23.10). Equation (23.11) then gives

$$\frac{a(s) - a(0)}{s} \geq \frac{1}{2} \int_{\partial E_s} \operatorname{tr}_{T_x \partial E_s} h dA_g - O(s).$$

Along every sequence $s \downarrow 0$, the compactness just proved gives a subsequence whose integral converges to $a'_{T_\infty}(h)$ for some $T_\infty \in \mathcal{T}$. This is at least the minimum in Equation (23.9). The lower and upper bounds agree, proving existence of the right derivative and the formula. No uniqueness or smooth selection of the minimizing frontier has been assumed. $\square$

Proposition 23.3 (A common tangent plane for tied minimizers). *Let*

$$\mathcal{L} = \bigcup_{T \in \mathcal{T}} (T \setminus B) \subset \operatorname{int} X.$$

If $x \in \mathcal{L}$ belongs to two minimizing frontiers, those frontiers have the same tangent plane at x . Consequently $x \mapsto \Pi_x$, where $\Pi_x = T_x T$ for any minimizing frontier through x , is a well-defined continuous three-plane field on $\mathcal{L}$ with its relative topology. In particular, for every smooth symmetric two-tensor K , the function $x \mapsto \operatorname{tr}_{\Pi_x} K$ is single valued and continuous there.

Proof. Let E and F be minimizing filled sets. Their union and intersection are admissible, and perimeter submodularity gives

$$P_g(E \cup F) + P_g(E \cap F) \leq P_g(E) + P_g(F) = 2a_g(B).$$

Each term on the left is at least $a_g(B)$, so both sets are also minimizers. If their original frontiers met with different tangent planes at a point outside B , the union and intersection would have a corner there: in smooth local graph coordinates, their boundary is the minimum or maximum of two functions with distinct first derivatives. That contradicts the C^1 regularity of the minimizing frontiers. Hence the tangent planes agree.

We also need continuity as the minimizing frontier varies. Suppose $x_j \in T_j \setminus B$ and $x_j \rightarrow x \in \mathcal{L}$. Proposition 23.2 gives, after passage to a subsequence, a minimizing limit T_∞ with convergence of indicators and tangent-plane area measures. Choose a small ball about x with closure disjoint

from B . Uniform local perimeter density bounds at x_j imply that every smaller ball about x carries positive limiting perimeter, so $x \in T_\infty$.

On this ball every compactly supported perimeter replacement is admissible, since the ball misses the obstacle. Thus each frontier is an ordinary local area-minimizing boundary. Its associated integral varifold is stationary for the fixed metric, has density one, and has no boundary in the ball. The limiting frontier T_∞ is smooth and has multiplicity one.

These facts supply more than strict BV convergence alone. Choose a sufficiently small continuity radius about x . Smoothness of T_∞ makes its normalized mass and its height and tilt excesses as close as desired to those of the single plane $T_x T_\infty$. Varifold convergence transfers these bounds to T_j for large j ; moving the centers to x_j changes the enclosing radii by a quantity tending to zero. The rescaled smooth-metric errors also tend to zero as the chosen radius decreases. Stationarity gives the required mean-curvature bound in the Riemannian small-excess regularity theorem. Equivalently, after a smooth local isometric embedding the Euclidean generalized mean curvature is bounded by the fixed ambient second fundamental form, whose rescaled contribution tends to zero.

The small-excess graphical regularity theorem consequently describes the entire support in a smaller ball as one graph, with uniform $C^{1,\alpha}$ bounds and tangent deviation tending to zero with these inputs [3, 39]; see [47, Chapter 5, §5, Theorem 5.2, p. 121]. It cannot leave an additional small sheet or spike outside that description. Oppositely cooriented sheets also contribute positively to varifold mass and are covered by the same conclusion. Express the graphs over the fixed plane $T_x T_\infty$ on a still smaller disk. Their uniform $C^{1,\alpha}$ bounds give subsequential $C^{1,\beta}$ convergence for $0 < \beta < \alpha$, and varifold convergence identifies every such graphical limit with the corresponding portion of T_∞ . Thus the whole graph sequence converges in C^1 there, which in particular gives

$$T_{x_j} T_j \longrightarrow T_x T_\infty.$$

The already proved common-plane assertion identifies the plane on the right with Π_x . Every convergent subsequence has this same plane limit; compactness of the Grassmannian then proves continuity for the whole sequence. The assertion about $\text{tr}_{\Pi_x} K$ follows at once. $\square$

24. REDUCTION TO A STRICT EXTERIOR WITH A REST END

This section reduces the invariant-mass assertion to an energy assertion for one-ended data with compactly supported second fundamental form. The replacement takes place arbitrarily far along the selected end. It preserves the dominant energy condition after a correction of vanishing energy cost. A proper map controls all enclosing cuts, including cuts that enter the replacement region.

The endpoint used in this section is Theorem 29.6, proved below: for the strict one-ended data described in Lemma 24.7,

$$(24.1) \quad E_g \geq \frac{1}{2} \left(\frac{A_*(g)}{\omega} \right)^{2/3}.$$

Here and throughout this section, $A_*(g)$ is the smooth-cut infimum of Definition 22.1, with the entire intrinsic frontier counted. The reduction to (24.1) does not use maximality.

24.1. The exact cut class and removal of other ends.

Lemma 24.1 (Cut reduction and positivity). *Under the hypotheses of Theorem 22.2, the quantity $A_e(S)$ is finite and strictly positive for every end e . Truncating all other ends at sufficiently large coordinate spheres gives a connected, complete one-ended exterior M_T with compact smooth*

boundary S_T . All its boundary components have future nonpositive expansion with the normal into M_T . Its selected-end charges are the original charges, and

$$(24.2) \quad A_*(M_T, g) \geq A_e(S), \quad \lim_{T \rightarrow \infty} A_*(M_T, g) = A_e(S).$$

The common parameter T in the limit can be replaced by any exhaustion in which every unwanted truncation radius tends to infinity.

Proof. A sufficiently distant sphere in the chosen end bounds an admissible outer domain, so the infimum is finite. To obtain a positive lower bound, choose an embedded arc from an open disk in S to the chosen end, and continue it along a straight coordinate ray. The compact part of the arc can be chosen embedded and disjoint from the boundary except at its initial point. A boundary collar and the tubular neighborhood theorem give a proper tube with coordinates

$$[0, \infty) \times B^3 \rightarrow M.$$

Its initial section lies in S ; its far part is a Euclidean tube of fixed transverse coordinate size. Choose a smooth nonnegative function η compactly supported in B^3 , with integral one, and define in this tube

$$\alpha = \eta(z) \, dz^1 \wedge dz^2 \wedge dz^3.$$

Extend α by zero across the lateral tube boundary. It is a smooth closed three-form on M . Its comass, meaning the supremum of its absolute value on unit simple three-vectors, has a finite bound C_α : the initial part is compact, and the far metric is uniformly comparable with the Euclidean metric. A sufficiently distant selected-end sphere has α -integral one, after fixing orientation.

Let D_e be any admissible outer domain with cut Γ . Cut off its selected-end tail at a sphere beyond Γ . The resulting domain is compact, since D_e contains no distant part of another end. Its boundary is the distant sphere together with the *entire* intrinsic boundary Γ , with the boundary orientations. Stokes' theorem gives

$$\left| \int_\Gamma \alpha \right| = 1, \quad |\Gamma|_g \geq C_\alpha^{-1}.$$

This proof applies to disconnected cuts and to frontier on S . In particular, it proves positivity before any end is truncated.

On an unwanted end, the normal of the truncation sphere that points into the retained manifold is the inward radial normal. The given decay therefore gives

$$H = -3T^{-1} + O(T^{-1-q}), \quad \text{tr}_{S_T} K = O(T^{-1-q}).$$

The new boundary has strictly negative future expansion for large T . The other boundary components are the original ones. Removing the open tails leaves a connected manifold: a path entering a removed tail can have that portion replaced by a path along its connected spherical cross-section. The retained manifold is closed in M . It is complete for its intrinsic metric as well. Indeed, an intrinsic Cauchy sequence is ambient Cauchy and has a limit in the retained set; near a smooth boundary or an interior point, the intrinsic and ambient local distances are comparable in a half-ball or ball chart.

Every admissible outer domain in M_T , viewed as a submanifold of M , is an admissible outer domain for e . A part of its frontier on a truncation sphere becomes a hypersurface in the interior of M , and remains part of its intrinsic boundary. This proves the first inequality in (24.2). Conversely, a fixed original admissible domain has compact frontier and contains no tail of any unwanted end. It is consequently contained in M_T for all sufficiently large truncation radii and is then an admissible domain there. Applying this observation to cuts with area within any prescribed positive error of $A_e(S)$ proves the limit. The selected-end data have not been changed, so neither selected-end charge changes. $\square$

We may thus work on one-ended data until the final transfer. The boundary is always included. No finite-perimeter relaxation or minimizing cut is needed in this section.

24.2. Conformal changes and an energy-raising shell.

Lemma 24.2 (Conformal constraint and boundary formulas). *Let $u > 0$ be smooth, and set $g_u = u^2 g$, $K_u = uK$. In dimension four, as an identity of scalar functions and an identity of covectors, respectively,*

$$(24.3) \quad u^2 \mu_u = \mu - 3u^{-1} \Delta_g u,$$

$$(24.4) \quad uJ_u = J + 3K(\text{grad}_g \log u, \cdot).$$

Consequently,

$$(24.5) \quad u^2(\mu_u - |J_u|_{g_u}) \geq \mu - |J|_g + 3u^{-1}(-\Delta_g u - |K|_g |du|_g).$$

For any consistently oriented hypersurface,

$$(24.6) \quad \Theta_{K_u} = u^{-1}(\Theta_K + 3\partial_\nu \log u).$$

Proof. Put $\varphi = \log u$. The scalar curvature formula and the tensor scalings are

$$R_{g_u} = u^{-2}(R_g - 6\Delta_g \varphi - 6|d\varphi|_g^2), \quad \tau_u = u^{-1}\tau, \quad |K_u|_{g_u}^2 = u^{-2}|K|_g^2.$$

Since $\Delta_g \varphi + |d\varphi|_g^2 = u^{-1} \Delta_g u$, these give (24.3). Write $P = K - \tau g$, so that $P_u = uP$. The connection difference is

$$\hat{\Gamma}_{ij}^a - \Gamma_{ij}^a = \delta_i^a \varphi_j + \delta_j^a \varphi_i - g_{ij} \varphi^a.$$

Contracting the covariant derivative of uP gives

$$uJ_{u,i} = J_i + 3P_{ij} \varphi^j - (\text{tr}_g P) \varphi_i = J_i + 3K_{ij} \varphi^j,$$

because $\text{tr}_g P = -3\tau$. This displays the cancellation of all trace-gradient terms. The covector norm scales by u^{-1} , whence

$$u^2 |J_u|_{g_u} = |J + 3K(\text{grad} \log u, \cdot)|_g.$$

The triangle inequality proves (24.5). Finally $\nu_u = u^{-1}\nu$, $H_{g_u} = u^{-1}(H_g + 3\partial_\nu \log u)$, and $\text{tr}_{\text{tan}, g_u} K_u = u^{-1} \text{tr}_{\text{tan}, g} K$, proving (24.6). $\square$

Lemma 24.3 (Positive shell). *For any one-ended data in Theorem 22.2 and every $\lambda > 0$, there is a conformal change preserving all its hypotheses, with*

$$E_\lambda = E + 2\lambda, \quad P_\lambda = P, \quad A_*(g_\lambda) \geq A_*(g).$$

The conformal factor is constant near the compact boundary and is at least one everywhere.

Proof. Choose $0 < \xi < \min(q, 1)$. For $T(r) = r^{-2} - r^{-2-\xi}$, at sufficiently large radius,

$$T' < 0, \quad -\Delta_g T - |K|_g |dT|_g = \xi(2 + \xi)r^{-4-\xi} + O(r^{-4-q}) > 0.$$

Take a nondecreasing smooth switch β from zero to one in this region, constant on neighborhoods of its endpoints, and set

$$G(r) = - \int_r^\infty \beta(s) T'(s) \, ds.$$

Extend G constantly throughout the interior. Then $G \geq 0$, it equals T sufficiently far out, and

$$-\Delta_g G - |K|_g |dG|_g = \beta(-\Delta_g T - |K|_g |dT|_g) - \beta' T' |dr|_g^2 \geq 0.$$

Use $u = 1 + \lambda G$ in Lemma 24.2. The dominant energy condition is preserved, as is the boundary expansion sign. The new densities are integrable: outside a compact set the additional terms in (24.3)–(24.4) are bounded by constants, depending on λ , times $r^{-4-\xi} + r^{-4-q}$. The new falloff exponent can be taken to be $\min(q, 2) > 1$.

For completeness, the leading change in the metric flux numerator is $-6\partial_i(\lambda G)$. Products involving $g - \delta$, $u - 1$, or their first derivatives have vanishing limiting flux. Hence (22.1) gives

$$E_\lambda - E = -\frac{\lambda}{\omega} \lim_{r \rightarrow \infty} \int_{S_r} \partial_r G \, dA_\delta = 2\lambda.$$

The transformed momentum tensor is exactly $u(K - \tau g)$. Its additional flux is $O(r^{-q})$, so $P_\lambda = P$ by (22.2). Since $u \geq 1$, every cut has at least its original area, proving the cut comparison. $\square$

The shell will be used only after proving the inequality for $E > |P|$. It then excludes all remaining ADM vectors; no preliminary strict timelikeness hypothesis will remain.

24.3. A reference end with prescribed timelike charges.

Lemma 24.4 (Explicit reference-plane charges). *For any $E > |P|$, put $m = (E^2 - |P|^2)^{1/2}$. A spacelike plane in the Schwarzschild–Tangherlini spacetime of mass m , sufficiently far from its central region, has induced vacuum data (g_*, K_*) with charges exactly (E, P) , and*

$$g_* - \delta = O_k(r^{-2}), \quad K_* = O_k(r^{-3}) \quad \text{for every integer } k \geq 0.$$

Here K_* is one half the metric rate in the future normal direction.

Proof. The exterior isotropic form of the static vacuum metric [50] is

$$(24.7) \quad \mathbf{g} = - \left(\frac{1 - m/(2|y|^2)}{1 + m/(2|y|^2)} \right)^2 db^2 + \left(1 + \frac{m}{2|y|^2} \right)^2 \sum_{i=1}^4 (dy^i)^2.$$

For $P = 0$ use $b = 0$. For the computation in general, rotate the first axis into the desired momentum direction, take $0 \leq v < 1$, and write $\gamma = (1 - v^2)^{-1/2}$. Introduce inertial coordinates by

$$b = \gamma(t + vx_1), \quad y_1 = \gamma(x_1 + vt), \quad y_A = x_A \quad (A = 2, 3, 4).$$

The plane $t = 0$ is $b = vy_1$. Its induced Minkowski metric is δ ; for sufficiently large radius it is spacelike for $\mathbf{g}$ as well. Let $s^2 = \gamma^2 x_1^2 + |x_\perp|^2$. The leading spacetime perturbation is $m|y|^{-2}(2db^2 + \sum_i (dy^i)^2)$. On the plane its nonzero components in the new coordinates are

$$(24.8) \quad \begin{aligned} h_{00} &= m(3\gamma^2 - 1)s^{-2}, & h_{01} &= 3m\gamma^2 v s^{-2}, \\ h_{11} &= m(3\gamma^2 - 2)s^{-2}, & h_{AB} &= m s^{-2} \delta_{AB}, \end{aligned}$$

with remainders $O_k(r^{-4})$. It follows by differentiating the spatial components that

$$(24.9) \quad \partial_j(g_*)_{ij} - \partial_i(g_*)_{jj} = 6m\gamma^2 x_i s^{-4} + O(r^{-5}).$$

To verify the momentum sign and normalization directly, the future second form, to the order that contributes to the flux, is

$$(K_*)_{ij} = \frac{1}{2}(\partial_t h_{ij} - \partial_i h_{0j} - \partial_j h_{0i}) + O(r^{-5}).$$

In addition to the spatial derivatives of s^{-2} , use $\partial_t(s^{-2}) = -2\gamma^2 v x_1 s^{-4}$ at $t = 0$. One obtains

$$K_* = \frac{m\gamma^2 v}{s^4} \begin{pmatrix} (3\gamma^2 + 2)x_1 & 3x_\perp^\top \\ 3x_\perp & -x_1 I_3 \end{pmatrix} + O(r^{-5}),$$

and therefore

$$(24.10) \quad K_* - (\text{tr}_{g_*} K_*)g_* = \frac{3m\gamma^2 v}{s^4} \begin{pmatrix} x_1 & x_\perp^\top \\ x_\perp & -\gamma^2 x_1 I_3 \end{pmatrix} + O(r^{-5}).$$

All discarded terms have vanishing sphere flux.

For $n \in S^3$, let $D(n) = \gamma^2 n_1^2 + |n_\perp|^2$. The ellipsoid $\gamma^2 x_1^2 + |x_\perp|^2 \leq 1$ has volume $\omega/(4\gamma)$ by a linear change of variables. Polar integration gives the same volume as $\frac{1}{4} \int_{S^3} D(n)^{-2} dA$. Thus

$$(24.11) \quad \int_{S^3} D(n)^{-2} dA = \omega/\gamma.$$

Equations (22.1), (24.9), and (24.11) give $E_* = m\gamma$. The first row of (24.10) gives $P_{*1} = m\gamma v$ with the denominator 3ω in (22.2). Each transverse momentum integrand is a constant multiple of $n_1 n_A D(n)^{-2}$, and integrates to zero by reflection. Choose $v = |P|/E$ and the rotation specified above. The constraints are vacuum because these are induced hypersurface data in the vacuum metric (24.7). The displayed expansions also give the asserted differentiated decay. $\square$

24.4. Compact inverses for the linear constraints. On Euclidean four-space write

$$(24.12) \quad \mathcal{L}b = \partial_i \partial_j (b_{ij} - (\text{tr}_\delta b) \delta_{ij}), \quad (\mathcal{D}k)_i = \partial_j (k_{ij} - (\text{tr}_\delta k) \delta_{ij}).$$

These are the linearizations at $(\delta, 0)$ of 2μ and J . The next lemma identifies every compatibility condition needed for a compact correction.

Lemma 24.5 (Compact linear constraint corrections). *Fix a compact subannulus U_0 of a bounded connected Euclidean annulus U . If f and F are smooth and supported in U_0 , and*

$$(24.13) \quad \begin{aligned} \int f \phi \, dz &= 0 \quad \text{for every affine } \phi, \\ \int F_i Y^i \, dz &= 0 \quad \text{for every Euclidean Killing field } Y, \end{aligned}$$

there are symmetric smooth tensors b, k , supported in a fixed compact subset of U , with $\mathcal{L}b = f$, $\mathcal{D}k = F$. For every integer $j \geq 0$ and every $1 < p_1 < \infty$, they can be chosen linearly in the sources so that

$$(24.14) \quad \|b\|_{W^{j+2, p_1}} \leq C \|f\|_{W^{j, p_1}}, \quad \|k\|_{W^{j+1, p_1}} \leq C \|F\|_{W^{j, p_1}}.$$

The constant depends only on U_0, U, j, p_1 and fixed localization choices. The moment conditions are also necessary for compactly supported solutions.

Proof. We first describe the scalar divergence inverse that will be used a finite number of times. On a ball the regularized Bogovskiĭ operator sends compactly supported smooth mean-zero data to compactly supported smooth vector fields with the prescribed divergence, gaining one W^{j, p_1} derivative. These are precisely the ball support and Sobolev properties of the regularized operator [23].

To use it inside U , cover a connected compact enlargement of the source support by finitely many balls with closures in U . Choose a subordinate smooth partition and a spanning tree in the intersection graph. For each non-root ball choose a smooth bump of integral one in its overlap with its parent. Starting at the leaves, subtract from the piece in that ball its integral times the overlap bump, and add that multiple of the bump to its parent's piece. The corrected piece has mean zero and remains compactly supported in its ball. At the root the remaining integral is zero because the original source has integral zero. Solve each piece in its ball and sum the resulting vector fields. All transfers and estimates involve a fixed finite family of bumps, so this gives a linear inverse with one derivative gain and support in a fixed compact subset of U . Choose successively larger compact subsets in advance when several inverses will be used. This permits every later inverse to act on the support produced by the preceding one, without approaching the annular boundary.

For the scalar constraint, solve $\partial_i v_i = f$. Compact support and the linear moments give

$$\int v_i \, dz = - \int z_i f \, dz = 0.$$

Solve in turn $\partial_j B_{ij} = v_i$, component by component. The symmetric tensor $S = (B + B^\top)/2$ still satisfies $\partial_i \partial_j S_{ij} = f$, since the double divergence of a skew tensor vanishes. The trace reversal in dimension four is inverted by

$$(24.15) \quad b = S - \frac{1}{3}(\text{tr}_\delta S)\delta.$$

It gives $b - (\text{tr}_\delta b)\delta = S$, and hence $\mathcal{L}b = f$. The two divergence inverses give the first estimate in (24.14).

For the vector constraint, the translational moments first allow a matrix B with $\partial_j B_{ij} = F_i$. Testing against the rotational fields gives

$$\int (B_{ij} - B_{ji}) \, dz = 0 \quad \text{for every } i, j.$$

Apply the divergence inverse to obtain tensors $D_{ijk} = -D_{jik}$ such that

$$\partial_k D_{ijk} = -\frac{1}{2}(B_{ij} - B_{ji}).$$

Set

$$C_{ijk} = D_{ijk} - D_{ikj} - D_{jki}.$$

Using only $D_{ijk} = -D_{jik}$ gives the two identities

$$C_{ijk} = -C_{ikj}, \quad \frac{1}{2}(C_{ijk} - C_{jik}) = D_{ijk}.$$

Consequently $S_{ij} = B_{ij} + \partial_k C_{ijk}$ is symmetric and $\partial_j S_{ij} = F_i$: its skew part cancels, while the added row divergence vanishes by the first identity. Apply (24.15) to S to obtain k . Here B has one derivative more than F , D has two, and the final differentiation loses one, proving the second estimate. All operations preserve the stated support and smoothness.

Finally, integration by parts pairs $\mathcal{L}b$ with $\text{Hess } \phi - (\Delta \phi)\delta$, which vanishes for affine ϕ . It pairs $\mathcal{D}k$ with the symmetric gradient of a Killing field, which also vanishes after trace reversal. This proves necessity of (24.13). $\square$

24.5. Replacement, vacuum bending, and repair.

Proposition 24.6 (Rest-end replacement). *Let (M, g, K) satisfy the one-ended hypotheses of Theorem 22.2, and suppose $E > |P|$. Put $m = (E^2 - |P|^2)^{1/2}$. There is a sequence of smooth data (g_R, K_R) on the same manifold, complete with the same compact boundary, satisfying the dominant energy condition and the same future boundary inequality, such that:*

- (i) K_R is compactly supported, and the ultimate end is an exact static Schwarzschild–Tangherlini spatial end;
- (ii) its energy is $m_R = m + o(1)$, its momentum is zero, and both constraint densities are integrable;
- (iii) $A_*(g_R) \geq (1 - o(1))A_*(g)$.

Every error here tends to zero as $R \rightarrow \infty$ for the fixed original data and fixed timelike vector. No uniformity as $E - |P|$ tends to zero is asserted or needed.

Proof. Use Lemma 24.4 for a reference end with exactly the original charges. We first join the data to this plane on $R < r < 4R$, controlling the full pointwise constraint deficit.

The scaled commutators. Fix $1 < p < \min(2, q)$, write $x = Rz$, and on $1 < |z| < 4$ use the scaled component fields

$$g(Rz), \quad k(z) = RK(Rz),$$

and their reference counterparts. This corresponds to rescaling lengths by R^{-1} , so both constraint densities scale by R^2 . The metric deviations and tensors have size $O(R^{-p})$ in C^2 and C^1 , respectively. Expanding the constraint map about $(\delta, 0)$ gives its linear part $(\mathcal{L}, \mathcal{D})$ from (24.12),

with remainder bounded in C^0 by CR^{-2p} . This follows directly from the scalar curvature formula: the nonlinear metric terms are bounded by $C(|b|D^2b + |Db|^2)$; the other scalar terms are quadratic in k . The momentum remainder is bounded by $C(|b|Dk + |Db||k|)$.

Choose a fixed smooth radial ψ , equal to one near $|z| = 1$ and zero near $|z| = 4$, with $0 \leq \psi \leq 1$. Blend the metric components and tensors by ψ . If b is the difference of the two scaled metric perturbations and k the difference of the two scaled tensors, the errors of the linear constraints relative to the blended sources are $f_R = [\mathcal{L}, \psi]b$ and $F_R = [\mathcal{D}, \psi]k$. For $T_{ij} = b_{ij} - (\text{tr}_\delta b)\delta_{ij}$,

$$(24.16) \quad [\mathcal{L}, \psi]b = (\partial_i \partial_j \psi)T_{ij} + 2(\partial_i \psi)\partial_j T_{ij},$$

$$(24.17) \quad ([\mathcal{D}, \psi]k)_i = (\partial_j \psi)(k_{ij} - (\text{tr}_\delta k)\delta_{ij}).$$

These expressions are supported in a fixed compact subannulus, and their C^1 norms are $O(R^{-p})$. In particular, the first bound uses only two metric derivatives, and the second only one tensor derivative.

All compatibility moments are small. We prove the sharper bound $o(R^{-2})$ for every moment in (24.13). Work momentarily in the physical coordinates, with

$$h = g - g_*, \quad T = K - (\text{tr}_\delta K)\delta - K_* + (\text{tr}_\delta K_*)\delta, \quad f = \mathcal{L}h, \quad F_i = \partial_j T_{ij}.$$

Both f and F belong to L^1 on the end. Indeed, the difference between the physical constraints and their Euclidean linear parts is $O(r^{-2-2\min(q,2)})$, which is integrable in dimension four. The physical densities are integrable by hypothesis and by reference vacuum, and Euclidean and physical measures and norms are uniformly comparable there.

An elementary consequence of integrability is

$$(24.18) \quad \int_{r_0 < r < 4R} r(|f| + |F|) \, dx = o(R).$$

To see this, split at a fixed radius L . The integral up to L , divided by R , tends to zero. The remaining integral divided by R is at most four times the L^1 tail beyond L , which can be made arbitrarily small. A finite first moment is not required.

For an affine function ϕ , define the flux primitive

$$Q_{\phi,i} = \phi(\partial_j h_{ij} - \partial_i h_{jj}) - (\partial_j \phi)h_{ij} + (\partial_i \phi)h_{jj}.$$

Its divergence is ϕf . For a Killing field Y , symmetry of T gives $\partial_j(Y^i T_{ij}) = Y^i F_i$. Let $\mathfrak{F}_\phi(r)$ and $\mathfrak{G}_Y(r)$ be the corresponding outward Euclidean sphere fluxes. For constant tests, matching the ADM charges gives

$$\mathfrak{F}_1(r) = o(1), \quad \mathfrak{G}_{e_i}(r) = o(1).$$

In the momentum flux the replacement of $\text{tr}_g K g$ by $(\text{tr}_\delta K)\delta$ costs at most $O(r^{2-2\min(q,2)}) = o(1)$. For homogeneous linear scalar tests and rotational fields, integration from a fixed sphere and (24.18) instead give

$$\mathfrak{F}_\phi(r) = o(r), \quad \mathfrak{G}_Y(r) = o(r).$$

Put $\psi_R(x) = \psi(x/R)$, and let $A_R = \{R < r < 4R\}$. Integration by parts, with $\psi_R = 1$ near the inner sphere and zero near the outer sphere, gives the exact formulas

$$(24.19) \quad \int_{A_R} \phi[\mathcal{L}, \psi_R]h \, dx = -\mathfrak{F}_\phi(R) - \int_{A_R} \psi_R \phi f \, dx,$$

$$(24.20) \quad \int_{A_R} Y^i([\mathcal{D}, \psi_R](K - K_*))_i \, dx = -\mathfrak{G}_Y(R) - \int_{A_R} \psi_R Y^i F_i \, dx.$$

In scaled coordinates, constant moments have the factor R^{-2} : the source scales by R^2 and volume by R^{-4} . A linear test in z contributes the additional factor R^{-1} . Thus (24.19)–(24.20) and the

preceding estimates show

$$(24.21) \quad \int f_R \phi \, dz = o(R^{-2}), \quad \int (F_R)_i Y^i \, dz = o(R^{-2})$$

for each element of a fixed basis of the affine functions or Killing fields.

Correction and the physical deficit. Choose a nonnegative smooth bump ζ , supported in a fixed ball within the annulus and positive on a smaller ball. For a basis of the affine tests, its Gram matrix $\int \zeta \phi_a \phi_b$ is positive definite. For a basis of Killing fields, the matrix $\int \zeta Y_a \cdot Y_b$ is likewise positive definite: a nonzero affine Killing field cannot vanish on an open ball. Subtract the corresponding bump combinations from f_R, F_R to erase their moments. By (24.21), the coefficients, and every fixed smooth norm of these bump corrections, are $o(R^{-2})$.

Apply Lemma 24.5 to the negatives of the moment-free commutators and add the resulting tensors to the blend. With $j = 1$ and $p_1 > 4$, Sobolev embedding and (24.14) bound the metric correction in C^2 and the tensor correction in C^1 by CR^{-p} . Their supports stay in a fixed enlarged compact subannulus. The new metric is positive for sufficiently large R and agrees exactly with the original data inside R and the plane outside $4R$.

Let $(\tilde{g}_R, \tilde{K}_R)$ denote these physical data. Their scaled constraints equal ψ times the original scaled constraints, plus an error $o(R^{-2}) + O(R^{-2p})$; the reference constraints vanish. The change in the momentum norm costs another $O(R^{-2p})$, because the metric change is $O(R^{-p})$ and the scaled momentum density is $O(R^{-p})$. Since $\psi \geq 0$, the original dominant energy condition implies, after undoing the scaling,

$$(24.22) \quad \tilde{\mu}_R - |\tilde{J}_R|_{\tilde{g}_R} \geq -\eta_R R^{-4}, \quad \eta_R \rightarrow 0.$$

Indeed the remaining physical error is $R^{-2}[o(R^{-2}) + O(R^{-2p})] = o(R^{-4})$, since $p > 1$. The possible deficit is confined to a fixed closed subannulus on the scale R . Replace η_R by a positive majorant tending to zero if necessary.

Bending the plane to a static slice. Only the exact reference region is used for this step. In its spatial coordinates y , write the plane as $b = v \cdot y$. Choose a smooth switch ψ_1 from one to zero on $[0, 1]$, constant outside that interval, and replace the plane farther out by the graph

$$(24.23) \quad b = (v \cdot y) \psi_1 \left(L^{-1} \log(|y|/R_b) \right).$$

Here R_b is a sufficiently large fixed multiple of R , so the graph initially lies wholly beyond the gluing annulus. For fixed $|v| < 1$, choose the fixed L so large that

$$|\text{grad}_y b| \leq |v|(1 + \|\psi'_1\|_\infty/L) < 1.$$

The Minkowski graph is uniformly spacelike. Since the perturbation (24.7) is $O(r^{-2})$, it remains uniformly spacelike in that metric for large R . It agrees with the plane on an open inner region and with $b = 0$ on an open outer region. These induced data are vacuum everywhere in the bend, so it adds no deficit.

In the fixed x chart, use $y = Ax$ where $A = (I - v \otimes v)^{-1/2}$, with the chosen rotation. The derivatives of (24.23) give $D^2 b = O(r^{-1})$, and the induced data, now denoted (g_R^0, K_R^0) , satisfy on all transition regions

$$(24.24) \quad c\delta \leq g_R^0 \leq C\delta, \quad |\partial g_R^0| + |K_R^0| \leq C/r.$$

Here $c, C > 0$ may depend on the fixed boost and switches, but not on large R . Outside a fixed multiple of R , the data are static in the y chart and have tensor zero.

For the repair choose a smooth radius $\mathfrak{r}$, equal to $|x|$ through the gluing annulus and through the bend, and equal to $|y|$ farther out in the static region. It can be chosen with

$$(24.25) \quad \mathfrak{r} \asymp r, \quad c \leq |\text{dt}|_{g_R^0} \leq C, \quad |\Delta_{g_R^0} \mathfrak{r}| \leq C/r.$$

Here is an explicit way to make the interpolation. In the static region put $a(n) = |A^{-1}n|$, $n = y/|y|$, and interpolate

$$\log \mathfrak{r} = \log |y| + (1 - \chi(\log(|y|/R_c))) \log a(n).$$

Take R_c beyond the bend and make χ change from zero to one over a sufficiently long fixed logarithmic interval. Its slope can be made so small that the derivative of $\log \mathfrak{r}$ with respect to $\log |y|$ lies between $1/2$ and $3/2$. Angular derivatives are bounded, second spatial derivatives are $O(r^{-1})$, and (24.24) proves (24.25). All transition intervals are contained between fixed multiples of R .

A conformal repair with vanishing mass cost. Write $s = \mathfrak{r}/R$. We construct a nonnegative function $F_R(s)$, constant on the inside, and use

$$u_R = 1 + \varepsilon_R R^{-2} F_R(s), \quad \varepsilon_R = C_0 \eta_R.$$

Choose a fixed interval $[a, b_1]$ in the s variable that starts before the possible deficit, contains all transition regions, and ends where the radius is $|y|$ and the data are static. This interval lies in the end for all large R . Put $z_R = -F'_R$ and prescribe

$$(24.26) \quad z_R(a) = 0, \quad z'_R = C_1 z_R + \omega_1,$$

where $\omega_1 \geq 0$ is smooth, vanishes on a neighborhood of a , and is at least one on a neighborhood of the possible deficit interval. Choose it to vanish near b_1 . The solution is nonnegative and its norms on this fixed interval are bounded independently of large R .

Differentiating u_R and using (24.24) and (24.25) gives fixed constants $c', C' > 0$ such that

$$(24.27) \quad -\Delta_{g_R^0} u_R - |K_R^0| |du_R|_{g_R^0} \geq \varepsilon_R R^{-4} (c' z'_R - C' z_R) \quad \text{on } a \leq s \leq b_1.$$

In fact the first positive term is $\varepsilon_R R^{-4} z'_R |d\mathfrak{r}|^2$; the other terms are $\varepsilon_R R^{-3} z_R \Delta \mathfrak{r}$ and the negative tensor term, both bounded below by $-C \varepsilon_R R^{-4} z_R$ here. Choose $C_1 > C'/c'$. Equation (24.26) then makes (24.27) nonnegative everywhere on this interval and at least $c' \varepsilon_R R^{-4}$ on the deficit region.

On the static region put $c_m = m/2$ and continue z_R so that

$$q_R(s) = s^3 \left(1 + \frac{c_m}{R^2 s^2} \right)^2 z_R(s)$$

is nondecreasing and becomes a positive constant $q_{R,\infty}$. This can be done smoothly with a uniform bound on $q_{R,\infty}$. Indeed, immediately past b_1 , continue (24.26); both its derivative and that of the positive prefactor make q'_R nonnegative for large R . Multiply this nonnegative derivative by a smooth switch that equals one initially and zero after one more fixed interval, and integrate. This preserves all matching derivatives and keeps q_R nondecreasing. The radial Laplacian of the static metric then shows $-\Delta_{g_R^0} u_R \geq 0$ there, since its radial divergence is proportional to $-q'_R$; also $K_R^0 = 0$.

Set $F_R(s) = \int_s^\infty z_R(t) dt$, extending it constantly before a . Its norm is bounded uniformly: the finite intervals have bounded length and data, while on the tail $z_R = O(s^{-3})$. Thus $1 \leq u_R \leq 2$ for large R . Choose the fixed C_0 large enough that $3u_R^{-1} c' \varepsilon_R \geq \eta_R$. Lemma 24.2, (24.22), and (24.27) now prove the dominant energy condition everywhere for

$$g_R = u_R^2 g_R^0, \quad K_R = u_R K_R^0.$$

The repair is constant in a neighborhood of the original boundary, and therefore preserves its expansion sign.

On the ultimate tail the integral for F_R is explicit:

$$u_R = 1 + \frac{a_R}{|y|^2 + c_m}, \quad a_R = \frac{1}{2} \varepsilon_R q_{R,\infty} = O(\varepsilon_R) = o(1).$$

Consequently

$$(24.28) \quad g_R = \left(1 + \frac{c_m + a_R}{|y|^2}\right)^2 \delta, \quad K_R = 0 \quad \text{on that tail.}$$

Its energy is $m + 2a_R = m + o(1)$, with zero momentum, by the direct static instance of Lemma 24.4. The densities vanish on this tail and are smooth on the remaining compact part, so they are integrable. The data are complete with boundary: a fixed compact core is smooth, and the end has a uniform positive metric lower bound for each R .

Comparison of every enclosing cut. The final metric dominates the original g for $r \leq R$, because the intermediate data there are the original ones and $u_R \geq 1$. For $r \geq R$, (24.24) and the static tail give $g_R \geq c\delta$ in the original x chart with a fixed $c > 0$. Choose $0 < \kappa < \min(1, \sqrt{c}/2)$. Let $L_R = R^{1/2}$, and choose a smooth strictly increasing radial map ρ_R equal to r for $r \leq L_R$, with derivative decreasing from one to κ on $[L_R, 2L_R]$, and derivative κ afterward. It defines a diffeomorphism

$$\Phi_R(x) = \rho_R(|x|) \frac{x}{|x|}$$

on the end, extended by the identity on the core. It is proper because $\rho_R(r) \rightarrow \infty$, preserves the selected end, and fixes the boundary. Its radial and tangential Euclidean dilations are ρ'_R and ρ_R/r . They are at most one everywhere, and for $r \geq R$ are at most $\kappa + O(R^{-1/2})$.

On the nonidentity region with $r \leq R$, both r and $\rho_R(r)$ tend uniformly to infinity; the original metric is therefore $(1 + o(1))\delta$ at both points. The dilation bound and $g_R \geq g$ show $\Phi_R^* g \leq (1 + o(1))g_R$ there. For $r \geq R$, the choice of κ , the same original-end decay, and $g_R \geq c\delta$ give this inequality as well. It is immediate on the identity core. We have thus proved the global tensor comparison

$$(24.29) \quad \Phi_R^* g \leq (1 + \epsilon_R)g_R, \quad \epsilon_R \rightarrow 0.$$

An admissible outer domain for g_R maps under this proper diffeomorphism to an admissible original outer domain, with all coincident frontier retained. Its three-dimensional area is at most $(1 + \epsilon_R)^{3/2}$ times the original cut's g_R -area. Taking infima gives

$$A_*(g) \leq (1 + \epsilon_R)^{3/2} A_*(g_R),$$

which proves the last assertion without a compactness assumption on minimizing sequences. $\square$

24.6. Strictification with a controlled asymptotic tail.

Lemma 24.7 (Strictification). *Let (M, g, K) be smooth one-ended data with compact boundary, complete with the boundary included, satisfying the dominant energy condition and $\Theta_K \leq 0$. Suppose K is compactly supported and the ultimate end is an exact static Schwarzschild–Tangherlini spatial end. Fix $0 < \delta_1 < 1$, and extend its radial coordinate to a positive smooth function r on M . There are smooth conformal data $(g_s, K_s) = (e^{2s\phi}g, e^{s\phi}K)$, for all sufficiently small $s > 0$, such that*

$$(24.30) \quad \Theta_{K_s} < 0, \quad \mu_s - |J_s|_{g_s} \geq c_s r^{-4-\delta_1}, \quad c_s > 0.$$

They are complete, have compactly supported tensor, integrable constraint densities, finite energy, and on the end

$$(24.31) \quad g_s - \delta = O_k(r^{-2}) \quad (k \geq 0), \quad R_{g_s} = O(r^{-4-\delta_1}).$$

Moreover $E_{g_s} \rightarrow E_g$ and $A_(g_s) \rightarrow A_*(g)$ as $s \downarrow 0$.*

Proof. Choose a smooth compactly supported $b_0 \geq |K|_g$, and solve

$$(24.32) \quad \begin{aligned} -\Delta_g \phi &= b_0 \sqrt{|\mathrm{d}\phi|_g^2 + r^{-6}} + r^{-4-\delta_1}, \\ \partial_\nu \phi &= -1 \quad \text{on } \partial M, \quad \phi \longrightarrow 0 \quad \text{at infinity.} \end{aligned}$$

The boundary normal here points into M . We give the solvability argument, including the estimates needed for the condition at infinity.

First truncate at an outer sphere S_T in the static region and prescribe $\phi = 0$ there. Multiply b_0 by a parameter $t_0 \in [0, 1]$. At zero the problem is the ordinary mixed Poisson problem, with a nonempty Dirichlet face and disjoint smooth Neumann faces. At any solution, the linearization for homogeneous variations is

$$v \longmapsto -\Delta_g v - t_0 b_0 \frac{\langle \mathrm{d}\phi, \mathrm{d}v \rangle_g}{\sqrt{|\mathrm{d}\phi|_g^2 + r^{-6}}}, \quad \partial_\nu v = 0 \text{ on } \partial M, \quad v = 0 \text{ on } S_T.$$

The drift has magnitude at most b_0 . The maximum principle and boundary point principle give a zero kernel: a nonzero positive maximum or negative minimum cannot occur in the interior or on a homogeneous Neumann face, and the Dirichlet values are zero. The mixed Laplacian has Fredholm index zero; adding the smooth first-order term preserves that index. Hence the linearization is invertible in the usual mixed Hölder spaces. The linear estimates and maximum principles used here are the ordinary scalar elliptic ones; see [28].

Fix a truncation and $p_1 > 4$. The mixed Laplace estimate, first-derivative interpolation, and the elementary bound

$$\sqrt{|\mathrm{d}\phi|^2 + r^{-6}} \leq |\mathrm{d}\phi| + r^{-3}$$

give

$$(24.33) \quad \|\phi\|_{W^{2,p_1}(M_T)} \leq C_T(1 + \|\phi\|_{L^\infty(M_T)}),$$

uniformly in t_0 . Nonnegativity follows from the minimum principle: the equation has strictly positive right side; at an inner boundary minimum the derivative into the domain cannot equal -1 , and the outer boundary value is zero.

If the suprema on a fixed truncation were unbounded, divide a sequence by its supremum. By (24.33) and compact Sobolev embedding, a subsequence converges in $C^{1,\alpha}$ for some $\alpha > 0$ to a nonnegative function v with supremum one, homogeneous mixed data, and

$$-\Delta_g v = t_* b_0 |\mathrm{d}v|_g.$$

The latter equation is a linear equation with bounded measurable drift along v : set the drift to $t_* b_0 \operatorname{grad} v / |\mathrm{d}v|$ where the gradient is nonzero, and to zero elsewhere. The strong maximum and boundary point principles again force $v = 0$, a contradiction. Thus the suprema are bounded. The nonlinearity in (24.32) is smooth on a fixed truncation, since r^{-6} has a positive minimum there. Sobolev embedding, Schauder estimates, and iteration now give all required classical bounds. Invertibility gives openness and these bounds give closedness of the parameter set. A solution therefore exists at $t_0 = 1$ on every sufficiently large truncation.

We next obtain bounds independent of T . Fix r_0 beyond the support of b_0 and in the static region. For large fixed r_0 , the positive function

$$W(r) = r^{-2}(1 - r^{-\delta_1})$$

satisfies $-\Delta_g W \geq c r^{-4-\delta_1}$ there, for some $c > 0$. Its Euclidean leading coefficient is $\delta_1(2 + \delta_1)$, while the static metric error is $O(r^{-6})$, dominated since $\delta_1 < 1$. Comparison with a sufficiently large multiple of $(1 + \sup_{r \leq r_0} \phi)W$ gives

$$(24.34) \quad 0 \leq \phi \leq C(1 + \sup_{r \leq r_0} \phi)r^{-2} \quad (r_0 \leq r \leq T),$$

with C independent of T . The comparison has the required ordering on S_{r_0} by the chosen multiple and on S_T because the solution vanishes there.

If the core suprema diverged along a sequence $T \rightarrow \infty$, normalize by those suprema. The local version of (24.33), including the fixed inner boundary, gives a subsequential limit with core supremum one. It solves the same homogeneous bounded-drift equation as above, has homogeneous inner Neumann data, and tends to zero at infinity by (24.34). Its positive global maximum is attained in a compact set. The strong maximum and boundary point principles exclude such a maximum. This contradiction gives uniform core bounds. Local compactness and exhaustion now solve (24.32) on M , with $\phi \geq 0$.

On the static tail the equation is simply $-\Delta_g \phi = r^{-4-\delta_1}$. Bound (24.34) and rescaled Poisson estimates on annuli imply

$$(24.35) \quad \phi = O_k(r^{-2}) \quad \text{for every } k \geq 0.$$

The right side and static coefficients have differentiated decay of all orders, so these estimates follow successively from the rescaled W^{2,p_1} and Schauder estimates. The limit

$$L_\phi = \lim_{r \rightarrow \infty} \int_{S_r} \partial_r \phi \, dA_\delta$$

exists: integration of $\Delta_g \phi$, which is integrable, gives the limit with physical normal and area; (24.35) and $g - \delta = O_1(r^{-2})$ make their difference tend to zero.

Now put $u = e^{s\phi}$. Lemma 24.2 gives

$$\begin{aligned} u^2(\mu_s - |J_s|_{g_s}) &\geq \mu - |J| + 3s(-\Delta_g \phi - |K| |\mathrm{d}\phi| - s |\mathrm{d}\phi|^2) \\ &\geq 3s(r^{-4-\delta_1} - s |\mathrm{d}\phi|^2). \end{aligned}$$

The function $|\mathrm{d}\phi|^2 r^{4+\delta_1}$ is bounded: it is smooth on the compact part and is $O(r^{-2+\delta_1})$ on the end. For all sufficiently small $s > 0$, the last expression is at least $\frac{3}{2}sr^{-4-\delta_1}$. Boundedness of ϕ then proves the gap in (24.30). The boundary formula is

$$\Theta_{K_s} = e^{-s\phi}(\Theta_K - 3s) < 0.$$

Equation (24.35) proves the metric falloff, and the scalar conformal formula on the static tail gives $R_{g_s} = O(r^{-4-\delta_1})$. There $K_s = 0$, so both density integrability and (24.31) follow. Completeness follows from the global comparison $g_s \geq g$.

Finally the same flux calculation as in Lemma 24.3 gives the exact energy change

$$E_{g_s} - E_g = -\frac{s}{\omega} L_\phi.$$

The remainder from the exponential and metric products has zero limiting flux by (24.35). Thus the energy is finite and converges to E_g . Pointwise, $g \leq g_s \leq e^{2s\|\phi\|_\infty} g$, so

$$A_*(g) \leq A_*(g_s) \leq e^{3s\|\phi\|_\infty} A_*(g).$$

This proves the asserted convergence of the exact cut infima. $\square$

24.7. Transfer of the prepared energy estimate.

Proposition 24.8 (Closure of the end reduction). *The prepared energy estimate (24.1) implies Theorem 22.2.*

Proof. First take one-ended original data with $E > |P|$. For each sufficiently large fixed R , Proposition 24.6 supplies a repaired rest end (g_R, K_R) of energy $m_R \rightarrow m$, where $m = (E^2 - |P|^2)^{1/2}$. Apply Lemma 24.7 to this fixed pair. Its output has all the hypotheses of Theorem 29.6: a smooth connected orientable complete one-ended exterior with nonempty compact smooth

boundary, compactly supported K , strictly negative future boundary expansion, a positive gap bounded below by a multiple of $r^{-4-\delta_1}$, and the end conditions (24.31). That theorem gives

$$E_{g_{R,s}} \geq \frac{1}{2} \left(\frac{A_*(g_{R,s})}{\omega} \right)^{2/3}.$$

Let $s \downarrow 0$ at fixed R . The strictification limits yield

$$m_R \geq \frac{1}{2} \left(\frac{A_*(g_R)}{\omega} \right)^{2/3}.$$

Now let $R \rightarrow \infty$ and use the proper-map comparison in Proposition 24.6. The result is

$$(24.36) \quad \sqrt{E^2 - |P|^2} \geq \frac{1}{2} \left(\frac{A_*(g)}{\omega} \right)^{2/3} \quad \text{when } E > |P|.$$

Suppose next that the original one-ended charges satisfy $E \leq |P|$. Choose any sequence $a_j > 0$ decreasing to zero and put $\lambda_j = (|P| + a_j - E)/2 > 0$. By Lemma 24.3, the conformally changed data have energy $|P| + a_j$, momentum P , and cut infimum at least the original positive $A_*(g)$. They are timelike, so (24.36) applied separately to each gives

$$\sqrt{(|P| + a_j)^2 - |P|^2} \geq \frac{1}{2} \left(\frac{A_*(g)}{\omega} \right)^{2/3} > 0.$$

The left side tends to zero, a contradiction. Thus all original one-ended data in this class have $E > |P|$, and (24.36) holds for all of them. The argument includes the cases of zero or negative energy and the null case. It takes no limit of the replacement geometries as the boost degenerates.

For finite-ended original data, apply the complete one-ended conclusion to a sufficiently distant truncation of every other end from Lemma 24.1. The charges at the selected end are unchanged, and its one-ended cut infimum is at least $A_e(S)$. Therefore

$$E_e \geq \sqrt{|P_e|^2 + \frac{1}{4} \left(\frac{A_e(S)}{\omega} \right)^{4/3}}.$$

This is Theorem 22.2 in the stated normalization. The only remaining task is to prove (24.1); the following sections do so by constructing a Riemannian metric and applying the minimal-enclosure comparison in Section 29. $\square$

25. THRESHOLD EXTERIORS AND GEOMETRIC BARRIERS

We work with the prepared one-ended data furnished by Lemma 24.7. Thus the original boundary has strictly negative future expansion, K has compact support, and, for a smooth positive extension r of the end radius,

$$(25.1) \quad \mu - |J|_g \geq c_0 r^{-4-\delta_1}, \quad c_0 > 0, \quad 0 < \delta_1 < 1.$$

The enclosing infimum for this prepared metric is denoted by A_* , with the intrinsic-boundary convention of Definition 22.1. This section constructs an exterior on which both signs of a small prescribed trace can be controlled. The original trapping convention is always the future convention $H + \text{tr}_S K < 0$. The tensor $-K$ below is used only to construct additional interior barriers.

25.1. Total regions inside a compact barrier. For a smooth symmetric tensor Q and an oriented hypersurface Σ , write

$$\Theta_Q(\Sigma) = H_\Sigma + \text{tr}_\Sigma Q.$$

When Σ bounds a compact region, its normal points out of that region. A smooth compact region may have several components. It includes its entire manifold boundary and is the closure of its interior. For a compact manifold X with boundary and a number c , define

$$(25.2) \quad T_c(X, Q) = \overline{\bigcup \{E : E \Subset \text{int } X \text{ is a smooth compact region, } \Theta_Q(\partial E) \leq c\}}.$$

The empty union has empty closure. All closures in this definition are taken in X .

Lemma 25.1 (Compact total region). *Let (X^4, g) be smooth, compact, connected, and orientable, with nonempty smooth boundary, and let Q be smooth. Suppose that $\Theta_Q(\partial X) > c$, with the normal pointing out of X on every boundary component. Then $T_c(X, Q)$ is either empty or a smooth compact region contained in $\text{int } X$. In the latter case its frontier has expansion c and is stable for outward variations. It contains every region in (25.2) and is itself one of those regions. In particular, no component of its complement whose closure is contained in $\text{int } X$ can be a bounded hole.*

Proof. Replace Q by $Q_c = Q - (c/3)g$. Since the hypersurfaces have dimension three,

$$(25.3) \quad \Theta_{Q_c}(\Sigma) = \Theta_Q(\Sigma) - c$$

for every oriented hypersurface. It suffices to prove the result for $c = 0$, and we use Q_c throughout this proof.

The global input is the smooth-frontier part of [9, Theorem 4.6]: for smooth complete asymptotically flat data in spatial dimensions $2 \leq n \leq 7$, a nonempty total trapped region has smooth embedded outermost stable MOTS frontier. No dominant energy condition is required for this conclusion. We explain how to meet its boundaryless asymptotically flat hypotheses while preserving exactly the compact problem.

First attach an outward cobordism from ∂X to one three-sphere. Each component of ∂X is a closed orientable three-manifold. Relative one-handles join these components, and the smooth surgery theorem for orientable three-manifolds gives a framed link surgery from the resulting connected three-manifold to S^3 [51, Theorem 6, p. 522]. Its trace consists of relative two-handles: reversing a four-dimensional two-handle trace again uses two-handles. Thus the cobordism has a handle decomposition relative to ∂X with indices at most two. Attach $S^3 \times [0, \infty)$ at its other end, obtaining a boundaryless manifold $\tilde{X}$. The handle Morse function provides a proper smooth height q on the extension, with $q = 0$ on ∂X , increasing in an outward collar, with only critical points of index at most two, and equal to a radial coordinate sufficiently far out. Extend q a short distance into X as a collar coordinate with negative values.

Extend g smoothly across ∂X . We may choose its values near every critical point of q so that

$$(25.4) \quad \text{tr}_V \text{Hess}_g q > 0 \quad \text{for every three-dimensional subspace } V \subset T_x \tilde{X}$$

at that critical point, and hence throughout a smaller neighborhood. Indeed, at an index-two point the eigenvalues of the Hessian can be arranged as $-a, -b, A, B$, with $a, b > 0$ and $\min(A, B) > a + b$; the sum of the three smallest is positive. At indices zero or one the same choice is easier. This is achieved by rescaling the positive and negative coordinate directions in the metric in a Morse chart. The neighborhoods are disjoint and away from the prescribed original data, so these local metrics extend to a smooth positive metric on the whole extension. Choose that metric Euclidean on the remote end.

At a regular point of a level of q , its mean curvature toward increasing q is

$$H_{\{q=\text{constant}\}} = \frac{\text{tr}_{(\text{grad } q)^\perp} \text{Hess } q}{|\text{d}q|}.$$

It is positive near the critical points by (25.4). Choose a smooth preliminary tensor extension agreeing with Q_c in the prescribed boundary collar. Near each critical point there is $a > 0$ such that $\text{tr}_V \text{Hess } q \geq a$ on every unit three-plane. After shrinking its neighborhood, $H \geq a/|\text{d}q|$ dominates the bounded tangential trace of the preliminary tensor. On the remaining compact regular part the level mean curvature is bounded below. Retain the tensor near the original boundary, and add a sufficiently large nonnegative multiple of the metric on the remaining compact part of the extension. The tangential trace of this addition is three times its coefficient. We can therefore ensure strictly positive expansion on every regular level of q . The given strict boundary inequality permits the matching in a collar of ∂X , including a collar on the original side. On the Euclidean remote end take the tensor to be zero; its round levels already have positive mean curvature. The transition to zero can be made inside that Euclidean region with a nonnegative metric multiple. We have obtained smooth complete one-ended data without boundary, exactly Euclidean with zero tensor sufficiently far out.

Every compact smooth weakly trapped region in this extension stays a positive distance inside X . To see this, extend q to a smooth function on the interior core with values below those of a fixed smaller collar. If a region enters that collar or the attached extension, the maximum of q on it occurs at a point in that part. A regular maximum cannot lie in the interior of the region. At a regular boundary maximum the region is on the inner side of a positively expanding level, with the same outward normal at contact. Tangential Hessian comparison gives $\Theta_{Q_c}(\partial E) \geq \Theta_{Q_c}(\{q = q_{\max}\}) > 0$, a contradiction. At a critical maximum in the interior, a direction of positive Hessian contradicts maximality. At a critical boundary maximum, the inward open half-space of a smooth domain contains a vector v with $\text{Hess } q(v, v) > 0$: choose the sign of a positive Hessian direction to point inward, and perturb it inward if it is tangent. A curve entering the region with this initial velocity has $q(\gamma(t)) = q(\gamma(0)) + t^2 \text{Hess } q(v, v)/2 + o(t^2)$, again a contradiction. Thus one fixed original-side collar is avoided by every such region.

The global total region consequently coincides with (25.2); all its candidate regions lie in the same compact subset of $\text{int } X$. The cited theorem gives its smooth stable frontier. With the closed representative used here it is a smooth compact region, its frontier has $\Theta_{Q_c} = 0$, and it contains every candidate. This proves the assertion at threshold c by (25.3). If a complementary component has closure in $\text{int } X$, filling it deletes whole smooth boundary components and changes no remaining outward expansion. The resulting larger region would still be a candidate in (25.2), contradicting maximality. $\square$

The extension in this proof is auxiliary. Neither a constraint inequality nor a topological restriction is imposed on it. We will also apply the lemma after compactly supported changes of Q that leave the strict outer barriers unchanged.

25.2. Threshold continuity and collars.

Lemma 25.2 (Continuity thresholds). *Suppose that the strict barrier condition in Lemma 25.1 holds for every c in an open interval I . The closed regions $T_c(X, Q)$ increase with c . Except at a countable set of thresholds, they are right-continuous in Hausdorff distance. At an empty right-continuity threshold the regions are empty for all sufficiently close larger thresholds.*

Proof. Inclusion of the defining classes proves monotonicity. Let L exceed the diameter of X , and associate to each threshold the function

$$d_c(x) = \begin{cases} \text{dist}(x, T_c(X, Q)), & T_c(X, Q) \neq \emptyset, \\ L, & T_c(X, Q) = \emptyset. \end{cases}$$

These functions are uniformly bounded, are 1-Lipschitz in x , and are nonincreasing in c . Fix a countable dense set $\{x_j\}$ in X . For each j the monotone real function $c \mapsto d_c(x_j)$ has at most countably many discontinuities. Away from the union of these exceptional sets, $d_{c_i}(x_j) \rightarrow d_c(x_j)$ whenever $c_i \downarrow c$. A finite net in X and the common Lipschitz bound promote convergence on the dense set to uniform convergence on X . For two nonempty compact sets the uniform distance between their distance functions equals their Hausdorff distance. If T_c is empty, uniform convergence to the constant $L > \text{diam } X$ is impossible along nonempty T_{c_i} , whose distance functions vanish somewhere. Thus eventual emptiness also follows. In particular every nonempty interval contains an admissible right-continuity threshold. $\square$

Lemma 25.3 (Outward leaf collars). *Let $T_c(X, Q)$ be nonempty, and let Σ be one connected component of its frontier. On the side exterior to this region, Σ has a smooth foliated collar, parametrized by $s \in [0, s_0]$, with $|ds| > 0$. With normal $N = \text{grad } s / |ds|$, every leaf satisfies*

$$(25.5) \quad \Theta_Q(\Sigma_s) \geq c.$$

The collars of distinct components can be chosen disjoint.

Proof. Write $\Theta(v)$ for the expansion of the outward normal graph of a small function v over Σ , pulled back to Σ . If ν is the outward normal and II its second fundamental form, the normal variation formulas give its linearization

$$Lv = -\Delta_\Sigma v + 2Q(\nu, \text{grad}_\Sigma v) + (\text{tr}_\Sigma \nabla_\nu Q - |\text{II}|^2 - \text{Ric}(\nu, \nu))v.$$

Thus L is a smooth scalar elliptic operator with principal part $-\Delta_\Sigma$. Stability gives a nonnegative principal eigenvalue λ_1 , with a positive eigenfunction ϕ . The principal eigenvalue of the adjoint is the same and has a positive eigenfunction ϕ^* ; these spectral facts apply on a compact connected manifold of any dimension [8, Lemma 4.1]. The shift in (25.3) has zero derivative under surface variations, so this is also the stability operator for expansion $\Theta_Q = c$.

If $\lambda_1 > 0$, the graphs $v = s\phi$ have $\Theta(v) = c + s\lambda_1\phi + O(s^2) > c$ for all sufficiently small $s > 0$. Their positive normal speed makes them a collar. It remains to treat $\lambda_1 = 0$.

Fix $0 < \alpha < 1$. The bounded linear map

$$(25.6) \quad C^{2,\alpha}(\Sigma) \times \mathbb{R} \longrightarrow C^\alpha(\Sigma) \times \mathbb{R}, \quad (v, a) \longmapsto \left(Lv - a, \int_\Sigma v \, dA \right)$$

is an isomorphism. Indeed, for prescribed (f, b) , pairing $Lv - a = f$ with ϕ^* determines

$$a = -\frac{\int_\Sigma \phi^* f \, dA}{\int_\Sigma \phi^* \, dA}.$$

The resulting right side $f + a$ is orthogonal to the adjoint kernel, so the Fredholm alternative solves $Lv = f + a$. The kernel is the span of ϕ ; adding its unique multiple that gives integral b proves existence and uniqueness in (25.6). Elliptic estimates give a bounded inverse.

The implicit-function theorem applied to

$$(v, a) \longmapsto \left(\Theta(v) - c - a, \int_\Sigma v \, dA \right)$$

therefore produces smooth graphs v_s of constant expansion $c + a_s$ with $v_0 = 0$, $a_0 = 0$, and $\int_{\Sigma} v_s \, dA = s$. Differentiation at zero and pairing with ϕ^* give

$$a'_0 = 0, \quad v'_0 = \frac{\phi}{\int_{\Sigma} \phi \, dA} > 0.$$

After shortening the parameter interval, the graphs have positive normal speed and lie strictly outside T_c for $s > 0$. If $a_s \leq 0$ for any such s , replace Σ by this outward graph and leave the other boundary components unchanged. The resulting smooth compact region would have expansion at most c and would strictly enlarge T_c . This contradicts Lemma 25.1. Thus $a_s > 0$ for every sufficiently small $s > 0$, proving (25.5) also in the degenerate case. Smooth elliptic regularity gives smooth leaves and parameter dependence. Compactness of the frontier gives finitely many components; sufficiently short collars are mutually disjoint. $\square$

25.3. Exterior supports and smooth enclosing regions. A *smooth exterior support* to a closed set D at $x \in \partial D$ is a smooth regular level $\{\psi = 0\}$ through x , defined in a neighborhood of x , such that D is locally contained in $\{\psi \leq 0\}$. Its outward normal is $\text{grad } \psi / |\text{d}\psi|$. Saying that D has expansion at most k in exterior supports means that every such support has $\Theta_Q \leq k$ at contact. This definition makes no regularity assumption on D .

The next argument is useful because the curvatures of these supports need not be bounded. Ordinary approximate transport of tangent metrics would produce an uncontrolled error multiplying their second fundamental forms. We use a map whose differential is an exact isometry at the contact point.

Lemma 25.4 (Exact support transport). *Let g, Q be smooth on a neighborhood of a compact set, and let D be a compact closed subset of its interior. Suppose that D has expansion at most k in exterior supports. There are $z_0 > 0$ and $C < \infty$ such that, for $0 < z < z_0$, every exterior support to*

$$D_z = \{y : \text{dist}_g(y, D) \leq z\}$$

has expansion at most $k + Cz$. The constant C depends only on bounds for the smooth background geometry and Q in the fixed compact neighborhood, and not on the support or its curvature. The allowable z_0 is also smaller than the distance required to keep these neighborhoods in the interior.

Proof. The assertion is immediate if D is empty. Let a support $\{\psi = 0\}$ touch D_z at x' , and choose $x \in D$ with $\text{dist}(x, x') = z$. This nearest point lies in ∂D , since an interior point could be moved a short distance toward x' to decrease the distance. Choose z_0 below a uniform injectivity radius. Let $\gamma : [0, z] \rightarrow X$ be the unit-speed minimizing segment from x to x' , put $u_0 = \dot{\gamma}(0)$, and denote parallel transport along γ by P_t . The ball of radius z centered at x lies in D_z . Consequently the support normal at x' is $P_z u_0$.

For each $X \in T_x X$, solve the Jacobi boundary problem

$$(25.7) \quad D_t^2 J_X + R(J_X, \dot{\gamma})\dot{\gamma} = 0, \quad J_X(0) = X, \quad J_X(z) = P_z X.$$

Here R denotes the Riemann curvature tensor. This problem is uniquely solvable for uniformly small z . To see the estimates explicitly, write $J_X(t) = P_t(X + W_X(t))$. The equation for W_X has zero endpoint values and is $W_X'' = -\mathcal{R}_t(X + W_X)$, with bounded curvature endomorphism $\mathcal{R}_t$. The Dirichlet inverse of the second derivative on $[0, z]$ has supremum-norm bound Cz^2 . Absorption for small z , followed by the differentiated one-dimensional Green formula, gives

$$(25.8) \quad \sup_{[0, z]} |W_X| \leq Cz^2 |X|, \quad |D_t J_X(0)| \leq Cz |X|.$$

The longitudinal component of J_X is affine with equal endpoint values. Hence the linear map $A_z X = D_t J_X(0)$ satisfies

$$\langle A_z X, u_0 \rangle = 0, \quad |A_z| \leq Cz.$$

These are precisely the first-jet compatibility conditions for a unit vector field U with $U(x) = u_0$ and $\nabla U(x) = A_z$.

For completeness, use normal coordinates y^i at x and extend vectors by parallel transport along the radial geodesics of those coordinates. Writing $E_v(y)$ for the resulting extension of $v \in T_x X$, set

$$V(y) = E_{u_0}(y) + \sum_i y^i E_{A_z e_i}(y), \quad U(y) = \frac{V(y)}{|V(y)|}.$$

On a uniformly small neighborhood V stays away from zero. At x this field has the required value and first derivative, since $A_z X \perp u_0$. It has uniformly bounded first and second derivatives by smoothness and compactness of the background. No vanishing second covariant derivative is required.

Define the local map

$$F_z(y) = \exp_y(zU(y)).$$

The geodesic variation determining $dF_z(x)X$ has initial Jacobi data $X, A_z X$, so (25.7) gives the exact identities

$$(25.9) \quad F_z(x) = x', \quad dF_z(x) = P_z, \quad |\nabla dF_z(x)| \leq Cz.$$

To verify the last estimate, the smooth map $E(y, v) = \exp_y(v)$ satisfies $E(y, 0) = y$. Differentiating $E(y, zU(y))$ twice in any fixed coordinate chart, with the uniform C^2 bound on U , gives $DF_z = I + O(z)$ and $D^2 F_z = O(z)$. In the covariant second derivative the domain and target connection terms cancel at $z = 0$; their remaining difference is $O(z)$. This proves the estimate uniformly in x, x' and u_0 .

For $y \in D$ sufficiently close to x , the curve defining $F_z(y)$ has length z , and therefore $F_z(y) \in D_z$. Thus $\psi \circ F_z$ is an exterior support to D at x . Its gradient is nonzero by the exact isometry in (25.9), and its normal is u_0 . For an orthonormal basis e_1, e_2, e_3 of $u_0^\perp$, the Hessian chain rule is

$$\text{Hess}(\psi \circ F_z)(e_a, e_a) = \text{Hess} \psi(P_z e_a, P_z e_a) + d\psi((\nabla dF_z)(e_a, e_a)).$$

Since $|d(\psi \circ F_z)|_x = |d\psi|_{x'}$, taking the tangential trace and dividing by this common norm yields

$$(25.10) \quad H_{\psi \circ F_z}(x) - H_\psi(x') = \sum_{a=1}^3 \langle P_z u_0, (\nabla dF_z)(e_a, e_a) \rangle = O(z).$$

There is no error involving $\text{Hess} \psi$ in this formula. Smoothness of Q similarly gives

$$\left| \text{tr}_{u_0^\perp} Q(x) - \text{tr}_{P_z(u_0^\perp)} Q(x') \right| \leq Cz.$$

Apply the assumed support inequality at x , then combine these two bounds. The expansion of the original support at x' is at most $k + Cz$, as claimed. $\square$

Lemma 25.5 (Smooth support enclosure). *Assume the hypotheses of Lemma 25.1 at threshold c' . Let $D \subseteq \text{int } X$ be a compact closed set whose every smooth exterior support has $\Theta_Q \leq k$, where $k < c'$. Then*

$$D \subset T_{c'}(X, Q).$$

In particular, the right side is nonempty whenever D is nonempty.

Proof. Assume $D \neq \emptyset$. Choose $\beta > 0$ small enough that D_β lies in $\text{int } X$, Lemma 25.4 applies for $0 < z \leq \beta$, and

$$(25.11) \quad k + C\beta < c'.$$

Smooth approximation of the distance function, followed by a regular-value choice, gives a smooth compact region E with

$$D_{\beta/4} \subset \text{int } E \subset E \subset \text{int } D_{3\beta/4}.$$

For example, approximate the distance uniformly to accuracy $\beta/16$ and choose a regular level between $7\beta/16$ and $9\beta/16$.

Choose a smooth function $a \geq 0$, supported where $\text{dist}(\cdot, D) < \beta$, that is a sufficiently large constant near ∂E . For $Q' = Q - ag$ we then have $\Theta_{Q'}(\partial E) = \Theta_Q(\partial E) - 3a < c'$. The boundary barriers of X are unchanged. Lemma 25.1 applied to Q' produces a nonempty smooth total region T' containing E . Its frontier is nonempty because T' is compactly contained in the interior of the connected manifold X with nonempty boundary. That outward frontier satisfies

$$(25.12) \quad \Theta_Q(\partial T') = c' + 3a \geq c'.$$

Since D has an open neighborhood in $E \subset T'$, the number $z = \text{dist}(D, \partial T')$ is positive and attained.

Suppose $z < \beta$. Then $D_z \subset T'$. Indeed, a minimizing segment of length at most z from D to a point outside T' would first cross $\partial T'$ at distance strictly less than z . Such segments stay in X by our choice of β . At a pair realizing the minimum distance, $\partial T'$ is therefore an exterior support to D_z . Its expansion is at most $k + Cz < c'$ by Lemma 25.4 and (25.11), contradicting (25.12). Consequently $z \geq \beta$.

The frontier of T' is outside the support of a , so its expansion for the original tensor is exactly c' . Thus T' itself is an admissible region in (25.2) for (X, Q, c') . Since it contains D , the defining maximality gives $D \subset T' \subset T_{c'}(X, Q)$. $\square$

25.4. The two colors and the trace allowance.

Proposition 25.6 (Geometric preparation). *For the prepared data satisfying (25.1), there are a closed connected exterior $\Omega \subset M$ with one end and nonempty compact smooth boundary B , two negative numbers c_b, c_w , and a decomposition of B into black and white components with the following properties. With the normal ν into Ω and $P_B = \text{tr}_B K$,*

$$(25.13) \quad H + P_B = c_b \quad \text{on black components}, \quad H - P_B = c_w \quad \text{on white components}.$$

The components come from closed total-region families $\mathcal{D}_c$ and $\mathcal{W}_c$ at right-continuity thresholds c_b and c_w , respectively. The white family is formed with the black region fixed, and is allowed to be empty. Every face has a disjoint smooth collar in Ω , with parameter $s \geq 0$, $|ds| > 0$, and the corresponding expansion toward increasing s is at least c_b or c_w .

There are $a_0 > 0$, a compactly supported smooth function C , and a positive smooth weight ρ , equal to a positive constant times $r^{-4-\delta_1}$ on the end, such that $|C| \leq a_0$ and, on smaller collars,

$$(25.14) \quad C = a_0 - \kappa_B s \quad \text{on black collars}, \quad C = -a_0 + \kappa_B s \quad \text{on white collars}, \quad \kappa_B > 0.$$

Set

$$(25.15) \quad b_- = c_b - a_0 < 0 < -c_w + a_0 = b_+.$$

Every real-valued function h taking values in $[b_-, b_+]$ on $\text{supp } K \cap \Omega$ satisfies the following inequality on Ω :

$$(25.16) \quad |(h + C)\tau| + |dC|_g + \rho \leq \mu - |J|_g, \quad \tau = \text{tr}_g K.$$

There is a smooth compactly supported extension b on Ω of the boundary values b_- on black faces and b_+ on white faces, constant on smaller collars. Finally, every smooth enclosing cut of B in Ω has g -volume at least A_ .*

Proof. Choose a distant sphere S_{R_0} beyond $\text{supp } K$, sufficiently large that it and all farther coordinate spheres have positive mean curvature. Their expansions for both K and $-K$ are positive. Let X_0 be the original compact core between S and S_{R_0} . It is connected: paths using the remote end can be replaced at its connected spherical cross section. Attach a smooth compact filling F behind the entire original boundary. Such a filling can be constructed by taking an

oppositely oriented copy of X_0 and capping its spherical boundary by a four-ball. Its remaining boundary identifies with S . Extend g and K smoothly across S into this filling, keeping the metric positive. Extension of the collar jets followed by a partition of unity does this; no constraint condition is required in F . The resulting compact manifold X has only the connected outer boundary S_{R_0} .

For later parameter choices, fix now

$$\sigma_* = \min_{X_0}(\mu - |J|_g) > 0, \quad T_* = \max_{X_0} |\tau|.$$

Choose $\gamma > 0$ so small that

$$(25.17) \quad \gamma T_* \leq \sigma_*/8$$

and the original boundary has expansion less than -2γ . By strictness and continuity, F together with a fixed short exterior collar of S is a smooth compact region whose outward boundary has expansion less than $-\gamma$. For every $c \in (-\gamma, 0)$ form

$$\mathcal{D}_c = T_c(X, K).$$

Here K denotes its smooth extension into F . Lemma 25.1 applies and $\mathcal{D}_c$ contains that entire filled neighborhood. Its smooth frontier therefore lies strictly in the original data. Its complement in X is connected. Indeed the collar of the connected outer sphere belongs to a unique complementary component; every other component would be an interior hole and could be filled, contrary to maximality.

Choose $c_b \in (-\gamma, 0)$ at a right-continuity threshold by Lemma 25.2. Write Y for the closed complement of $\text{int } \mathcal{D}_{c_b}$ in X . This is a smooth compact connected manifold with the outer sphere and the black frontier as its boundary. For the white tensor $-K$, all these boundary components have strictly positive outward expansion. On a black component the outward normal of Y is the reverse of the outward normal of $\mathcal{D}_{c_b}$, so the expansion is

$$(25.18) \quad -H - \text{tr}_{\partial \mathcal{D}_{c_b}} K = -c_b > 0.$$

This changes both the normal and the tensor in an auxiliary problem; the original boundary still has its prescribed future sign.

For $c \in (-\gamma, 0)$ define the white family

$$\mathcal{W}_c = T_c(Y, -K).$$

It may be empty. Lemmas 25.1 and 25.2 give a right-continuity threshold $c_w \in (-\gamma, 0)$. Every nonempty white region lies strictly inside Y , so $\mathcal{D}_{c_b}$ and $\mathcal{W}_{c_w}$ are disjoint closed sets, a positive distance apart. Notice that the black threshold and manifold Y are fixed before making this second threshold choice.

In the filled one-ended manifold take the component of the open complement of $\mathcal{D}_{c_b} \cup \mathcal{W}_{c_w}$ that reaches the end, and let Ω be its closure. Its boundary is the union of those entire frontier components adjoining this exterior. To justify the word “entire,” a smooth connected frontier component has a connected thin collar on each side; the component of its exterior side is therefore constant along the frontier. Both frontiers are compact and smooth with finitely many components, and they are disjoint. Thus Ω is smooth, closed, and connected, and its boundary B is a union of complete smooth components. It has one end, the original end, and lies in the original manifold because the black region contains the filling and a collar of S . Its boundary is nonempty: the component reaching infinity is separated from that nonempty filled region. The inherited metric is complete up to B ; boundary charts give local completeness at finite limit points, and completeness of the original end prevents escape in finite distance.

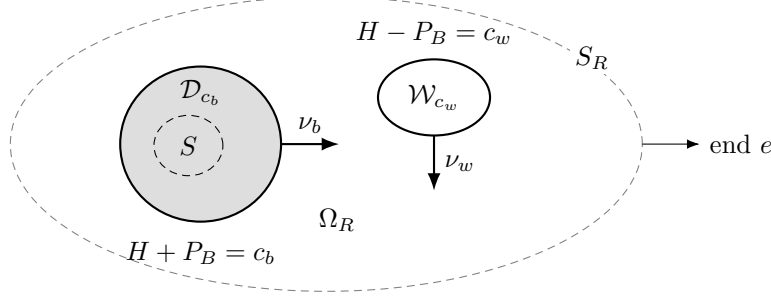


FIGURE 3. A schematic of the auxiliary exterior; one frontier component of each kind is shown. The black region encloses the original boundary. Both normals point into the retained exterior, but its black and white faces have prescribed expansion for K and $-K$, respectively. The drawing represents incidence and orientations, not the topology or dimension of the hypersurfaces.

Call its retained $\mathcal{D}_{c_b}$ faces black and its retained $\mathcal{W}_{c_w}$ faces white. The normals into Ω are the outward normals of these regions. Their threshold equations give (25.13). Figure 3 illustrates these orientations; Ω_R there denotes the part of Ω inside a distant coordinate sphere S_R .

Apply Lemma 25.3 to their respective total regions and shorten the collars so they lie on the Ω side and are mutually disjoint. This gives all the asserted leaf inequalities. The support enclosure Lemma 25.5 is available for the full families $\mathcal{D}_c$ and $\mathcal{W}_c$ on X and Y , respectively; in the latter case the reversed black barriers remain fixed.

All these collars lie in the fixed original core X_0 , so the lower bound σ_* was fixed before either threshold or any collar width. For each collar choose a smooth profile $q_B(s) \in [0, 1]$, supported in that collar, with $q_B(s) = 1 - \alpha_B s$ near its face and $\alpha_B > 0$. Such a profile is obtained by multiplying this linear function by a nonnegative cutoff supported before the linear function can become negative. With disjoint supports, define

$$C = a_0 \sum_B \varepsilon_B q_B(s), \quad \varepsilon_B = \begin{cases} 1, & B \text{ black,} \\ -1, & B \text{ white.} \end{cases}$$

There is a finite constant M_* , now depending on the chosen collars and profiles, such that $|C| \leq a_0$ and $|\mathrm{d}C| \leq a_0 M_*$. Formula (25.14) holds with $\kappa_B = a_0 \alpha_B > 0$. Only at this point choose $a_0 > 0$ small enough that

$$(25.19) \quad a_0(2T_* + M_*) \leq \sigma_*/8.$$

For $b_- \leq h \leq b_+$ we have $|h + C| \leq \gamma + 2a_0$, and hence on $X_0 \cap \Omega$

$$(25.20) \quad |(h + C)\tau| + |\mathrm{d}C| \leq \gamma T_* + a_0(2T_* + M_*) \leq \sigma_*/4.$$

At points outside $\mathrm{supp} K$ the trace term vanishes, so this estimate requires no restriction there on the value of h .

Finally choose a constant $\rho_* > 0$ so small that

$$\rho_* \leq c_0/4, \quad \rho_* \max_{X_0} r^{-4-\delta_1} \leq \sigma_*/4, \quad \rho = \rho_* r^{-4-\delta_1}.$$

On $X_0 \cap \Omega$, (25.20) plus this choice bounds the left side of (25.16) by $\sigma_*/2$. Outside X_0 , $K = C = 0$, and (25.1) gives the required inequality directly. This proves (25.16). This order of choices uses no lower bound on collar widths as c_b, c_w approach zero: the thresholds are chosen first, the collars next, and their amplitude last. Ordinary collar cutoffs also give the stated compactly supported smooth extension of the constants b_- and b_+ .

If D_e is an admissible closed connected outer domain for a cut in Ω , it is also closed in M , because Ω is closed in M . It is a smooth codimension-zero submanifold of M , its interior lies in $\text{int } M$, and it contains the distant original end. Its entire intrinsic boundary, including every part coincident with B , is therefore an admissible original enclosing cut. Its area is at least A_* . This proves the final assertion without assuming the existence of a minimizing cut. $\square$

26. THE SCALAR SYSTEM AND EXCLUSION OF THE FIRST FLOOR

Fix the prepared exterior and the geometric choices of Proposition 25.6. All contractions and derivatives in this section use its background metric g , unless another metric is indicated. In particular, $\tau = \text{tr}_g K$ is unrestricted. The boundary normal ν points into Ω . Let Ω_R denote the truncation at a large coordinate sphere S_R , so its ordinary outward normal at its inner boundary B is $-\nu$.

The purpose of the system is to produce a metric $\hat{g} = e^{2t}(g + l(t)^2 df^2)$ with nonnegative scalar curvature and negative boundary mean curvature. The parameter η_n below makes the prescribed trace depend strictly increasingly on f . A separate penalty prevents t from reaching $-\epsilon$. This section proves that prevention using a coarse height bound. The narrower height interval needed to use (25.16) will be proved subsequently.

26.1. Variables and a scalar-curvature identity. Fix $0 < \epsilon < 1$ and a smooth nonincreasing function $\vartheta : \mathbb{R} \rightarrow [0, 1]$ equal to one on $(-\infty, 0]$ and to zero on $[1, \infty)$. For every integer $n \geq 4$ define

$$(26.1) \quad \begin{aligned} p(t) &= n\vartheta(nt), & l(t) &= \exp\left(\int_0^t p(s) \, ds\right), \\ L_0(t) &= e^{2t}l(t), & \ell &= e^{-\epsilon n}, & \eta_n &= \ell^{3/2}. \end{aligned}$$

Thus $0 \leq p \leq n$, $l(t) = e^{nt}$ for $t \leq 0$, and l is constant with value between one and e for $t \geq 1/n$. In particular $l(-\epsilon) = \ell$. The unknowns are real functions f, Z . Define t implicitly, and then all the remaining variables, by

$$(26.2) \quad \begin{aligned} \sigma &= |\text{grad } f|, & D &= 1 + l(t)^2 \sigma^2, & d &= D^{-1}, & Z &= t + \frac{1}{6} \log D, & h &= \eta_n f, \\ w &= \frac{l \text{grad } f}{\sqrt{D}}, & a &= |w|, & v &= a^2 = 1 - d, & \chi &= \frac{3d}{3 + pv}, \\ A_d &= \text{Id} - w \otimes w, & A_\chi &= \text{Id} - \frac{3+p}{3+pv} w \otimes w, & u &= L_0 \sqrt{D}. \end{aligned}$$

We identify vectors and covectors using g . A symmetric endomorphism also denotes the corresponding contravariant tensor, and $|\xi|_A^2 = A(\xi, \xi)$ for a covector ξ . For fixed df , the derivative of the defining expression for Z with respect to t is $1 + pv/3 \geq 1$. That expression tends to $-\infty$ and $+\infty$ at the two ends of the real line. It therefore defines a unique smooth $t = t(x', Z, df)$, with x' the base point, for all inputs, including inputs below the proposed floor. No gradient bound is needed for this definition. The eigenvalues of A_d and A_χ on the line spanned by $\text{grad } f$ are d and χ ; their other three eigenvalues are one. Consequently

$$(26.3) \quad 0 < \frac{3d}{3+n} \leq \chi \leq d \leq 1.$$

Set

$$(26.4) \quad \begin{aligned} \bar{g} &= g + l^2 df^2, & \hat{g} &= e^{2t} \bar{g}, & H^f &= \frac{l}{\sqrt{D}} \text{Hess}_g f, & S_f &= K + H^f, \\ F &= \text{tr}_{A_\chi} S_f, & V &= u A_\chi (3 \text{grad } Z + K(w, \cdot)). \end{aligned}$$

Where $\sigma > 0$, let $e = \text{grad } f / \sigma$. Decompose S_f into its transverse block $T = S_f|_{e^\perp}$, mixed block $M = S_f(e, \cdot)|_{e^\perp}$, and axial entry $j = S_f(e, e)$; write $T^0 = T - \frac{1}{3}(\text{tr } T)g|_{e^\perp}$. Thus $\text{tr } T = F - \chi j$. Also write $x = e(t)$ and $y = (dt)|_{e^\perp}$. The symbol M in this block notation is a covector, not the original manifold.

Proposition 26.1 (Scalar identity). *For arbitrary smooth f, Z and arbitrary symmetric K ,*

$$(26.5) \quad \frac{1}{2}e^{2t} \text{Scal}_{\hat{g}} = \mu + J(w) + \mathcal{T} + w(F) - F\tau - u^{-1} \text{div}_g V,$$

where the smooth scalar $\mathcal{T}$ is given, on $\{df \neq 0\}$, by

$$(26.6) \quad \begin{aligned} \mathcal{T} = & \frac{1}{2}|T^0|^2 + |M + (p+1)ay|^2 - v|y|^2 + 3(p+1)(|y|^2 + dx^2) \\ & - \frac{1}{3}(F - \chi j)^2 + \chi j^2 - \chi Fj + F^2 + 2(p+1)\chi jax. \end{aligned}$$

The expression extends smoothly across $\{df = 0\}$, where any unit axis may be used in the block expression.

Proof. We derive the identity in a stationary Lorentzian metric, keeping its normal second form distinct from the prescribed tensor K . On the product with coordinate s' use

$$\mathbf{g} = -l^2(ds' - df)^2 + \bar{g}.$$

Its constant- s' slices have metric g . Their lapse, shift, and normal second form, with the convention that the second form is half the normal metric rate, are

$$(26.7) \quad \begin{aligned} U &= l\sqrt{D}, \quad \beta = l^2 \text{grad } f = Uw, \\ k &= -\frac{\text{sym } \nabla \beta}{U} = -H^f - p(dt \otimes w + w \otimes dt). \end{aligned}$$

Here sym includes the factor $1/2$. Put $\theta = \text{tr}_g k$, and let N_0 be the future unit normal. The contracted Gauss and normal expansion identities are

$$\text{Scal}_g = \mathbf{R} + 2\text{Ric}(N_0, N_0) + |k|^2 - \theta^2, \quad \text{Ric}(N_0, N_0) = -N_0\theta - |k|^2 + U^{-1}\Delta_g U.$$

Stationarity gives $UN_0\theta = -\beta(\theta)$ and $\text{div } \beta = -U\theta$. If

$$\mathbf{B}(A, B) = A : B - (\text{tr}_g A)(\text{tr}_g B), \quad P_A = A - (\text{tr}_g A)g,$$

the preceding two formulas consequently yield

$$(26.8) \quad U\mathbf{R} = U(\text{Scal}_g + \mathbf{B}(k, k)) - 2\text{div}(\text{grad } U + \theta\beta).$$

In the static coordinate $s' - f$ one instead has

$$\mathbf{R} = \text{Scal}_{\bar{g}} - 2l^{-1}\Delta_{\bar{g}}l, \quad \Delta_{\bar{g}}\phi = \frac{l}{U} \text{div} \left(\frac{U}{l} A_d \text{grad } \phi \right).$$

Direct differentiation of $U = l\sqrt{D}$ gives

$$\text{grad } U - \frac{U}{l} A_d \text{grad } l = -k(\beta, \cdot).$$

Moreover, using $J = \text{div } P_K$, symmetry of P_K , and $\text{sym } \nabla \beta = -Uk$ gives

$$\text{div}(P_K\beta) = U\{J(w) - \mathbf{B}(K, k)\}.$$

Substitute these identities into (26.8), use $\mu = (\text{Scal}_g - \mathbf{B}(K, K))/2$, and put $Q = K - k$. The result is

$$(26.9) \quad \frac{1}{2} \text{Scal}_{\bar{g}} = \mu + J(w) + \frac{1}{2}\mathbf{B}(Q, Q) - U^{-1} \text{div}(UP_Q w).$$

For example, the divergence arising directly from (26.8) is $U^{-1} \operatorname{div}(UP_k w)$; adding $J(w)$ combines it with $-U^{-1} \operatorname{div}(UP_K w)$ to give the last term in (26.9). This also fixes its sign.

The four-dimensional conformal formula is

$$\frac{1}{2}e^{2t} \operatorname{Scal}_{\hat{g}} = \frac{1}{2} \operatorname{Scal}_{\bar{g}} - 3\Delta_{\bar{g}} t - 3|\operatorname{dt}|_{A_d}^2.$$

Since $\Delta_{\bar{g}} t = U^{-1} \operatorname{div}(UA_d \operatorname{grad} t) - p|\operatorname{dt}|_{A_d}^2$ and $u = e^{2t}U$, changing the divergence weight from U to u gives

$$(26.10) \quad \begin{aligned} \frac{1}{2}e^{2t} \operatorname{Scal}_{\hat{g}} &= \mu + J(w) + \frac{1}{2}\mathbf{B}(Q, Q) + 2P_Q(w, \operatorname{grad} t) + 3(p+1)|\operatorname{dt}|_{A_d}^2 \\ &\quad - u^{-1} \operatorname{div}(u(P_Q w + 3A_d \operatorname{grad} t)). \end{aligned}$$

The following differential identities follow from (26.2):

$$(26.11) \quad \begin{aligned} 3dZ &= (3 + pv)dt + H^f(w, \cdot), \\ \operatorname{div} w &= \operatorname{tr}_{A_d} H^f + p dw(t), \\ d \log u &= (p + 2 + pv)dt + H^f(w, \cdot). \end{aligned}$$

They imply

$$(26.12) \quad \begin{aligned} \frac{V}{u} &= P_Q w + 3A_d \operatorname{grad} t + Fw, \\ u^{-1} \operatorname{div}(uw) &= \operatorname{tr}_g H^f + 2(p+1)w(t) \\ &= F + (1 - \chi)j - \tau + 2(p+1)w(t). \end{aligned}$$

Adding uFw inside the divergence in (26.10) therefore leaves $w(F) + Fu^{-1} \operatorname{div}(uw)$. Its trace contribution is exactly $-F\tau$, while the remaining quadratic expression is

$$(26.13) \quad \begin{aligned} \mathcal{T} &= \frac{1}{2}\mathbf{B}(Q, Q) + 2P_Q(w, \operatorname{grad} t) + 3(p+1)|\operatorname{dt}|_{A_d}^2 \\ &\quad + F \left(F + \frac{3+p}{3+pv} S_f(w, w) + 2(p+1)w(t) \right). \end{aligned}$$

This is smooth even at $w = 0$, since $(1 - \chi)j = (3 + p)S_f(w, w)/(3 + pv)$. For the block expansion, Q has blocks T , $M + pay$, $j + 2pax$, and $\operatorname{tr} T = F - \chi j$. Expanding (26.13) gives (26.6). At $w = 0$ it reduces to

$$\mathcal{T} = \frac{1}{2}(|S_f|^2 + (\operatorname{tr}_g S_f)^2) + 3(p+1)|\operatorname{dt}|^2,$$

which is independent of the auxiliary axis. This proves all assertions. $\square$

26.2. Coercivity and the boundary identity. Throughout the proof, Π_n denotes a bound of the form $C(1+n)^N$ for a finite nonnegative integer N . Its coefficient C and exponent N may depend on the fixed prepared data, ϵ , the chosen collars and smooth switches, and any explicitly fixed small absorption constant. They never depend on R , a homotopy parameter, or a solution. Different occurrences may denote larger such bounds. Constants denoted by $C(n)$ in later fixed- n analytic estimates have no polynomial restriction and will not enter a limit requiring polynomial control.

Lemma 26.2 (Polynomial coercivity). *For all inputs in (26.2),*

$$(26.14) \quad |T|^2 + |M|^2 + dj^2 + F^2 + |\operatorname{dt}|_{A_d}^2 \leq \Pi_n \mathcal{T}.$$

At a point where $\operatorname{dt} = 0$, there are absolute constants $c, C > 0$ such that

$$(26.15) \quad \begin{aligned} c(|T^0|^2 + |M|^2 + \chi j^2 + F^2) &\leq \mathcal{T} \\ &\leq C(|T^0|^2 + |M|^2 + \chi j^2 + F^2). \end{aligned}$$

Proof. Completing squares in (26.6) gives the exact identity

$$(26.16) \quad \begin{aligned} \mathcal{T} = & \frac{1}{2}|T^0|^2 + |M + (p+1)ay|^2 + (3(p+1) - v)|y|^2 \\ & + 3(p+1)d \left(x + \frac{\chi aj}{3d} \right)^2 + \frac{2}{3} \left(F - \frac{\chi j}{4} \right)^2 + \frac{d(48-3d)}{8(3+pv)^2} j^2. \end{aligned}$$

For the final coefficient, the identity used in this completion is

$$\chi - \frac{3}{8}\chi^2 - \frac{(p+1)\chi^2 v}{3d} = \frac{d(48-3d)}{8(3+pv)^2}.$$

The coefficient of $|y|^2$ is at least two, and the last coefficient is at least $45d/[8(3+n)^2]$. Thus $dj^2 \leq C(3+n)^2 \mathcal{T}$. The inequalities $\chi^2 \leq d^2 \leq d$ then recover F^2 from its shifted square and dx^2 from its shifted square, at polynomial cost. Recovering M costs at most $C(1+n)^2|y|^2$, and recovering T uses $\text{tr } T = F - \chi j$. This proves (26.14); these observations, for instance, allow one common bound $C(1+n)^2$ in that inequality.

When $dt = 0$, complete only the (F, j) terms to obtain

$$(26.17) \quad \mathcal{T} = \frac{1}{2}|T^0|^2 + |M|^2 + \frac{2}{3} \left(F - \frac{\chi j}{4} \right)^2 + \chi \left(1 - \frac{3\chi}{8} \right) j^2.$$

Since $5/8 \leq 1 - 3\chi/8 \leq 1$ and $\chi^2 \leq \chi$, this gives (26.15) with absolute constants. $\square$

Lemma 26.3 (Boundary conversion). *On any smooth face on which f is constant, let $P_B = \text{tr}_B K$, $w_\nu = \langle w, \nu \rangle$, and let H be its background mean curvature for the normal into Ω . Then*

$$(26.18) \quad \frac{V_\nu}{u} = d(H + 3\partial_\nu t) - H + w_\nu(F - P_B), \quad \hat{H} = e^{-t}\sqrt{d}(H + 3\partial_\nu t).$$

Proof. With $\Pi_B(X, Y) = g(\nabla_X \nu, Y)$ for tangent vectors, one has tangentially $\text{Hess } f = (\partial_\nu f)\Pi_B$, so $\text{tr } T = P_B + w_\nu H$ and $\chi j = F - P_B - w_\nu H$. Use this relation in the first identity of (26.12); as w is normal on the face, it gives

$$V_\nu/u = 3d\partial_\nu t + w_\nu(F - P_B - w_\nu H),$$

which is the first assertion because $w_\nu^2 = 1 - d$. The induced metric of $\bar{g}$ on the face equals $g|_B$, and its normal in base coordinates is $\sqrt{d}\nu$. In the tangential first variation, $l^2 df^2$ contributes zero since the tangential derivatives of f vanish there. Hence $H_{\bar{g}} = \sqrt{d}H$. Applying the boundary conformal formula with $\nu_{\bar{g}} = \sqrt{d}\nu$ proves the second assertion. Both assertions also hold when $df = 0$. $\square$

26.3. The equations and their four-stage homotopy. Extend the end radius to a positive smooth function $r \geq 1$ on Ω and set

$$(26.19) \quad \begin{aligned} \rho_0 = r^{-4-\delta_1}, \quad \rho_1 = r^{-3}, \quad m_0(t) = \vartheta(n(t+\epsilon)), \quad \mathcal{P} = C_n(\rho_0 + v\rho_1), \\ \Xi = \delta_0 \mathcal{T} + \rho + \delta_0 \eta_n a\sigma - m_0(t)\mathcal{P}. \end{aligned}$$

The constant $\delta_0 > 0$ and the polynomial C_n will be fixed in Proposition 26.7. The smooth positive weight ρ comes from (25.16); in particular $\rho \leq C\rho_0$. The combination $a\sigma = l|df|^2/\sqrt{D}$ is smooth also at $df = 0$, so all terms of Ξ are smooth functions of the indicated jets. Choose a constant $N_B > 1 + \sup_B |H|$, and put

$$\mathcal{B} = -H + w_\nu(F - P_B) - N_B d.$$

The desired system is

$$(26.20) \quad \begin{cases} F = h + C, & \text{in } \Omega_R, \\ \operatorname{div} V = u\Xi, & \text{in } \Omega_R, \\ h = b, \quad V_\nu/u = \mathcal{B}, & \text{on } B, \\ f = Z = 0, & \text{on } S_R. \end{cases}$$

In every boundary flux expression the symbol F is evaluated using the prescribed trace right side. Thus the boundary condition contains no second derivative of f . For a solution of (26.20), (26.18) immediately gives $\hat{H} = -e^{-t}\sqrt{d}N_B < 0$.

We specify the entire homotopy because its boundary modifications and strictly increasing height term will be used in the a priori estimates. The stages run from the desired system to the terminal problem with solution $f = Z = 0$; Section 29 transfers existence back along this homotopy. Take a smooth $j_0 : \mathbb{R} \rightarrow [0, 1]$ which is zero for $t \leq 0$ and one for $t \geq 1$. For $0 \leq \lambda \leq 1$ put

$$V_\lambda = uA_\chi(3 \operatorname{grad} Z + \lambda K(w, \cdot)).$$

In stages 1–3 use

$$(26.21) \quad \begin{cases} \operatorname{div} V_\lambda = u\Xi & \text{in } \Omega_R, \\ (V_\lambda)_\nu/u = [1 - (1 - \lambda)j_0(t)][\mathcal{B} - (1 - \lambda)(A_\chi K(w, \cdot))_\nu] & \text{on } B. \end{cases}$$

Keep the outer data $f = Z = 0$ throughout.

Here are explicit smooth choices of the auxiliary trace terms. Fix $0 < 4\xi_1 < b_+ - b_-$ and smooth height switches q_-, q_+ such that

$$\begin{aligned} q_-(z) &= 1 & (z \leq b_- + \xi_1), & \quad q_-(z) = 0 & (z \geq b_- + 2\xi_1), \\ q_+(z) &= 0 & (z \leq b_+ - 2\xi_1), & \quad q_+(z) = 1 & (z \geq b_+ - \xi_1), \end{aligned}$$

where q_- is nonincreasing and q_+ is nondecreasing. Choose a smooth direction switch q_d with values in $[0, 1]$, equal to one on $(-\infty, -3/4]$ and zero on $[-1/2, \infty)$. On each collar write $N = \operatorname{grad} s/|ds|$, so $N = \nu$ at the face, and choose a smooth cutoff ω_B supported there and equal to one near the face. The collars are disjoint. With fixed constants

$$A_0 > 2 \sup_B |H| + 1, \quad A_1 > \sup_B |H| + 1,$$

define, extending by zero outside the collars,

$$(26.22) \quad \begin{aligned} D_0(x', w, z) &= -A_0\omega_B(x')q_d(w \cdot N)q_-(z) && \text{on black collars,} \\ D_0(x', w, z) &= A_0\omega_B(x')q_d(-w \cdot N)q_+(z) && \text{on white collars,} \\ D_1(x', w) &= A_1\omega_B(x')w \cdot N && \text{on every collar.} \end{aligned}$$

Both terms vanish at $w = 0$, are bounded for $|w| \leq 1$, and have bounded derivatives of every order on bounded height ranges. These bounds are independent of n . In addition $\partial_z D_0 \geq 0$.

The successive trace prescriptions and inner Dirichlet data are as follows; the indicated parameter moves in the displayed direction.

Stage 1: Decrease λ from one to zero, keeping $F = h + C$ and $h|_B = b$.

Stage 2: Keep $\lambda = 0$, increase α_0 from zero to one, and use $F = h + C + \alpha_0 D_0(x', w, h)$ and $h|_B = b$.

Stage 3: Keep $\lambda = 0$, increase ζ from zero to one, and use

$$(26.23) \quad \begin{aligned} F &= h + (1 - \zeta)[C + D_0(x', w, h + \zeta b(x'))] \\ &\quad + \zeta[\operatorname{tr}_{A_\chi} K + D_1(x', w)], \quad h|_B = (1 - \zeta)b. \end{aligned}$$

Stage 4: Retain the terminal prescription $F = h + \text{tr}_{A_\chi} K + D_1$ and $h|_B = 0$. Keep $\lambda = 0$ and multiply both right sides of (26.21) by a common parameter γ , decreasing from one to zero.

The endpoints agree, so this is a continuous homotopy. All trace right sides have derivative at least one with respect to h , with (x', w, t) fixed. This is also the relevant derivative when comparing scalar solutions at a shared gradient and shared Z , since t is independent of the value of f .

Two consequences of these choices will be useful. First, at a stage 3 face, evaluate the prescribed right side at $h = (1 - \zeta)b$ and at the limiting values $w = \pm\nu$, $d = \chi = 0$. It obeys

$$(26.24) \quad F(\nu) \geq P_B + H, \quad F(-\nu) \leq P_B - H.$$

Indeed, on a black face $b + C = P_B + H$, $D_0(\nu, b) = 0$, and $D_0(-\nu, b) = -A_0$; on a white face $b + C = P_B - H$, $D_0(-\nu, b) = 0$, and $D_0(\nu, b) = A_0$. The terminal expressions are $P_B \pm A_1$. Each inequality follows for both endpoints from the choices of A_0, A_1 , and hence for their convex combination.

Second, throughout stage 4 the scalar solution is $f = 0$, for every trial Z . Its equation is $\text{tr}_{A_\chi} H^f = \eta_n f + D_1$, with zero boundary values. At a positive interior maximum $w = 0$, $D_1 = 0$, $A_\chi = \text{Id}$, and $l\Delta f = \eta_n f > 0$, a contradiction; a negative minimum is excluded in the same way. Consequently in this stage

$$(26.25) \quad f = 0, \quad t = Z, \quad \mathcal{T} = \mathcal{T}_K + 3(p+1)|dt|^2, \quad \mathcal{T}_K = \frac{1}{2}(|K|^2 + \tau^2),$$

and its two remaining equations are exactly

$$\text{div}(3u \text{grad } t) = \gamma u \Xi, \quad 3\partial_\nu t = \gamma[1 - j_0(t)](-H - N_B).$$

26.4. Coarse heights and an outer boundary estimate.

Lemma 26.4 (Height bound before the floor). *There exist fixed r_0 and $C_* > 0$, independent of n , R , and all homotopy parameters, with the following property. Every smooth solution of the trace equation in any stage, with its prescribed Dirichlet data, on Ω_R for $R \geq 2r_0$ satisfies*

$$(26.26) \quad |h| + |F| \leq C_*, \quad |h| \leq C_*(r^{-3/2} - R^{-3/2}) \quad (r_0 \leq r \leq R).$$

Here Z may be arbitrary; the divergence equation and the condition $t \geq -\epsilon$ are not assumed. In particular, with the prescribed outer value $Z = 0$,

$$(26.27) \quad |\text{grad } f| \leq C_* \eta_n^{-1} R^{-5/2} \quad \text{on } S_R.$$

For every fixed n there is $R_{\min}(n) \geq 2r_0$, chosen to tend to infinity with n , such that every such solution for $R \geq R_{\min}(n)$ has

$$(26.28) \quad t > -\epsilon \quad \text{on } S_R.$$

Proof. At an interior extremum of h , $w = 0$ and both auxiliary terms vanish. The trace equation becomes

$$l\Delta f = h + C - \tau \quad \text{in stages 1 and 2,} \quad l\Delta f = h + (1 - \zeta)(C - \tau) \quad \text{in stage 3.}$$

Since $l > 0$, comparison at a maximum and a minimum bounds $|h|$ by $\max(\sup_B |b|, \|C - \tau\|_\infty)$. Stage 4 has $h = 0$. All prescribed trace right sides then bound $|F|$ by a fixed constant, because D_0, D_1 are bounded and $\|A_\chi\| \leq 1$.

Choose r_0 beyond the supports of K, C, b, D_0, D_1 . At a comparison with a radial function, the axis of A_χ is $\text{grad}_g r / |dr|_g$. Uniformly for $0 < \chi \leq 1$, the prepared end expansion gives

$$(26.29) \quad \text{tr}_{A_\chi} \text{Hess}_g r^{-3/2} = \frac{3}{2} \left(-3 + \frac{5}{2}\chi \right) r^{-7/2} + O(r^{-11/2}) < 0$$

after increasing r_0 . The leading coefficient is at most $-3/4$, so the choice is independent of n . Take C_* large enough that $C_*(r_0^{-3/2} - R^{-3/2})$ dominates the global height bound for every $R \geq 2r_0$. On this annulus the scalar equation is

$$\frac{l}{\eta_n \sqrt{D}} \operatorname{tr}_{A_\chi} \operatorname{Hess}_g h = h.$$

The functions $\pm C_*(r^{-3/2} - R^{-3/2})$ are respectively upper and lower barriers by (26.29). At a putative first comparison point the gradients coincide, so the same positive trace matrix and the same positive prefactor occur in both expressions. The Hessian order and the proper height sign give a contradiction. This proves the end estimate. Differentiating the comparison at the outer boundary, where $h = 0$ tangentially, gives (26.27), absorbing the uniform end normal comparison into C_* .

Finally let $G(t) = t + \frac{1}{6} \log(1 + l(t)^2 \sigma^2)$ at an outer point. It is strictly increasing and $G(t) = Z = 0$. Thus $t > -\epsilon$ if

$$G(-\epsilon) = -\epsilon + \frac{1}{6} \log(1 + \ell^2 \sigma^2) < 0.$$

By (26.27), it suffices to choose $R_{\min}(n)$ so large that $C_*^2 \ell^2 \eta_n^{-2} R_{\min}(n)^{-5} < e^{6\epsilon} - 1$. This uses only fixed- n enlargement of the outer radius. $\square$

26.5. The stationary comparison at a minimum of t . The scalar identity is useful at a floor contact because a second, Riemannian Gauss calculation gives an upper bound for the same scalar curvature. Define, for a symmetric tensor A and a chosen unit axis e ,

$$(26.30) \quad \mathcal{S}(A) = \frac{1}{3}(\operatorname{tr}_\perp A)^2 - \frac{1}{2}|A_\perp^0|^2 + dA_{ee} \operatorname{tr}_\perp A - d|A_{e\perp}|^2.$$

Lemma 26.5 (Stationary Gauss comparison). *At an interior local minimum of t ,*

$$(26.31) \quad \frac{1}{2}e^{2t} \operatorname{Scal}_{\widehat{g}} \leq \frac{1}{2} \operatorname{Scal}_g - v \operatorname{Ric}_g(e, e) + \mathcal{S}(H^f).$$

At every point where $dt = 0$ there are an absolute $c_2 > 0$ and a polynomial bound Π_n such that

$$(26.32) \quad \mathcal{T} - \mathcal{S}(H^f) \geq c_2 \mathcal{T} - \Pi_n(vF^2 + |K|^2).$$

These conclusions extend to $df = 0$ with any unit axis.

Proof. Use the Riemannian warped product $g + l^2(db')^2$ and its graph $b' = f$. At $dt = 0$ one has $dl = 0$, so the graph second form is H^f , up to its immaterial overall sign. Its inverse metric is A_d . The contracted second-form contribution is

$$\frac{1}{2}\{(\operatorname{tr}_{A_d} H^f)^2 - |H^f|_{A_d \otimes A_d}^2\} = \mathcal{S}(H^f),$$

as is seen by writing $A_d = \operatorname{diag}(1, 1, 1, d)$. The ambient scalar curvature and Ricci blocks are

$$\operatorname{Scal}_g - 2\Delta l/l, \quad \operatorname{Ric}_g - \operatorname{Hess} l/l, \quad -\Delta l/l,$$

for the scalar, horizontal block, and unit vertical direction; the mixed block is zero. The graph unit normal has horizontal part $-w$ and vertical component $\sqrt{d}$. Therefore the ambient scalar minus twice its normal Ricci, divided by two, is

$$\frac{1}{2} \operatorname{Scal}_g - v \operatorname{Ric}_g(e, e) - v \operatorname{tr}_\perp(\operatorname{Hess} l/l).$$

At a stationary point $\text{Hess } l/l = p \text{Hess } t$, and the conformal term is $-3 \text{tr}_{A_d} \text{Hess } t$. Thus we have, more precisely, the equality

$$(26.33) \quad \begin{aligned} \frac{1}{2} e^{2t} \text{Scal}_{\widehat{g}} &= \frac{1}{2} \text{Scal}_g - v \text{Ric}_g(e, e) + \mathcal{S}(H^f) \\ &\quad - (3 + pv) \text{tr}_{\perp} \text{Hess}_g t - 3d \text{Hess}_g t(e, e). \end{aligned}$$

At a minimum the last two terms are nonpositive, proving (26.31).

For (26.32), first subtract $\mathcal{S}(S_f)$. An expansion using $\text{tr } T = F - \chi j$ gives exactly

$$(26.34) \quad \begin{aligned} \mathcal{T} - \mathcal{S}(S_f) &= |T^0|^2 + (1 + d)|M|^2 + \chi(1 + d - 2\chi/3)j^2 \\ &\quad + (\chi/3 - d)Fj + F^2/3. \end{aligned}$$

At $d = \chi = 1$ the (F, j) quadratic is $F^2/3 - 2Fj/3 + 4j^2/3$, whose matrix has determinant $1/3$ and positive trace. It is uniformly positive in a fixed neighborhood of this point. Since

$$1 - d = v, \quad 1 - \chi = \frac{(3 + p)v}{3 + pv},$$

there is an absolute $\kappa_0 > 0$ such that this neighborhood contains all inputs with $(1 + p)v \leq \kappa_0$. In that region (26.34) controls an absolute fraction of $|T^0|^2 + |M|^2 + \chi j^2 + F^2$.

In the remaining region, use $\chi(1 + d - 2\chi/3) \geq \chi$ and $|\chi/3 - d| \leq d$ to absorb the mixed term into, for example, $\chi j^2/4$ and a multiple of $(d^2/\chi)F^2$. The only ratio required is

$$(26.35) \quad \frac{d^2}{\chi} = \frac{d(3 + pv)}{3} \leq \frac{3 + n}{3}, \quad v > \frac{\kappa_0}{1 + p} \geq \frac{\kappa_0}{1 + n}.$$

Keeping also a fixed fraction of F^2 at the cost of another multiple of F^2 , this proves a lower bound by an absolute fraction of the stationary norm minus $C(1 + n)^2 v F^2$. Equation (26.15) converts the norm to $\mathcal{T}$.

It remains to replace S_f by $S_f - K$. Polarizing (26.30) bounds the difference by

$$C(|T| + d|j| + d|M|)|K| + C|K|^2.$$

The stationary norm controls $|T|$ and $|M|$, while $d|j| \leq (d^2/\chi)^{1/2}(\chi j^2)^{1/2}$. By (26.35) and Young's inequality, for every fixed $\alpha > 0$ this is at most $\alpha \mathcal{T} + C_\alpha(1 + n)|K|^2$. Choose α to retain half the already obtained positive coefficient of $\mathcal{T}$. This proves (26.32) with absolute $c_2 > 0$. $\square$

Lemma 26.6 (Errors from the trace switches and drift). *Let $\mathcal{K}$ be a fixed compact enlargement of the supports of K, C, b, D_0, D_1 . For every fixed $\delta_2 > 0$, every smooth trace solution in stages 1–3 satisfies, at a point where $dt = 0$,*

$$(26.36) \quad \begin{aligned} w(F) &\geq \eta_n a \sigma - \delta_2 \mathcal{T} - \Pi_n \mathbf{1}_{\mathcal{K}}, \\ u^{-1} |\text{div}(u A_\chi K(w, \cdot))| &\leq \delta_2 \mathcal{T} + \Pi_n \mathbf{1}_{\mathcal{K}}. \end{aligned}$$

The polynomial may depend on δ_2 and the fixed geometric choices, but no negative power of ℓ or η_n is required.

Proof. At $dt = 0$, differentiation gives

$$\nabla_i w = A_d H_i^f, \quad d \log u = H^f(w, \cdot), \quad dp = p'(t) dt = 0.$$

These formulas and (26.15) show

$$(26.37) \quad |\nabla w| + |d \log u|_{A_\chi} + |\nabla A_\chi| \leq \Pi_n(\sqrt{\mathcal{T}} + |K|).$$

For clarity, the axial entry in ∇w is dH_{ee}^f , and its control uses $d^2/\chi \leq (3 + n)/3$. In $|d \log u|_{A_\chi}$ the axial entry has the factor $\sqrt{\chi}$. Differentiating $A_\chi = \text{Id} - (3 + p)w \otimes w/(3 + pv)$ uses only the

preceding ∇w and coefficients bounded by powers of $1 + n$, since $3 + pv \geq 3$. This proves (26.37) without dividing by l or η_n .

Write the relevant prescribed trace as $\Phi(x', w, t, h)$. Its height derivative is at least one, so the chain rule contains the favorable term

$$\Phi_h w(h) \geq w(h) = \eta_n a \sigma.$$

All other terms are supported in $\mathcal{K}$. Lemma 26.4 bounds their height arguments in a fixed range; derivatives of D_0, D_1 and b are consequently bounded. Derivatives of $\text{tr}_{A_\chi} K$ use ∇K and (26.37). The t derivative has zero contribution at the point in question. Thus the remaining absolute error is bounded by $\Pi_n(1 + \sqrt{\mathcal{T}})\mathbf{1}_{\mathcal{K}}$. Young's inequality proves the first assertion.

Expanding the divergence in the second assertion differentiates A_χ, K , and w , and adds $\langle K(w, \cdot), d \log u \rangle_{A_\chi}$. The latter is bounded by $C|K| |d \log u|_{A_\chi}$. Equation (26.37), compact support, and Young's inequality give the second assertion with the same allowed dependencies. $\square$

26.6. Exclusion of first contact with the floor.

Proposition 26.7 (First-floor exclusion). *There are a fixed $\delta_0 > 0$ and a positive polynomial C_n in (26.19) such that, for every $n \geq 4$ and every $R \geq R_{\min}(n)$, no smooth solution at any stage of the homotopy satisfies*

$$\min_{\Omega_R} t = -\epsilon.$$

The choices depend only on the prepared data, ϵ , and the fixed switches and collars. They do not use a height bound sharper than Lemma 26.4.

Proof. Suppose first that a floor minimum is interior and the parameter is in one of stages 1–3. At that point $dt = 0$ and $\text{Hess } t \geq 0$. Subtract (26.31) from (26.5) and use $V_\lambda = V - (1 - \lambda)uA_\chi K(w, \cdot)$. This gives

$$\begin{aligned} u^{-1} \text{div } V_\lambda &\geq \mu - \frac{1}{2} \text{Scal}_g + J(w) + v \text{Ric}_g(e, e) + \mathcal{T} - \mathcal{S}(H^f) + w(F) - F\tau \\ &\quad - (1 - \lambda)u^{-1} \text{div}(uA_\chi K(w, \cdot)). \end{aligned}$$

Choose $\delta_2 > 0$ in Lemma 26.6 so small that the two losses $\delta_2 \mathcal{T}$ leave at least $c_2 \mathcal{T}/2$ in (26.32). The remaining tensor expressions $\mu - \text{Scal}_g/2 = (\tau^2 - |K|^2)/2$, J , and $F\tau$ are compactly supported and bounded by fixed constants, using Lemma 26.4. On the complement of $\mathcal{K}$, $F = h$ and $F^2 \leq Cr^{-3}$; also $|\text{Ric}_g| \leq Cr^{-4}$. On $\mathcal{K}$ the positive weight ρ_0 has a fixed positive minimum. Hence

$$\mathbf{1}_{\mathcal{K}} + vF^2 + v|\text{Ric}_g| \leq C(\rho_0 + v\rho_1).$$

Combining these bounds yields an absolute $c_3 > 0$ and a polynomial $Q_n \geq 0$ such that, at the proposed minimum,

$$(26.38) \quad u^{-1} \text{div } V_\lambda \geq c_3 \mathcal{T} + \eta_n a \sigma - Q_n(\rho_0 + v\rho_1).$$

Fix $0 < \delta_0 < \min(c_3/2, 1/4)$. Since $t = -\epsilon$ gives $m_0(t) = 1$, choose C_n to exceed $Q_n + 1 + \sup_\Omega(\rho/\rho_0)$. The right side of (26.38) then strictly exceeds $\delta_0 \mathcal{T} + \rho + \delta_0 \eta_n a \sigma - C_n(\rho_0 + v\rho_1) = \Xi$, contradicting (26.21). This choice is polynomial in n .

At an interior minimum in stage 4, (26.25) gives $f = 0$, $t = Z$, and $v = 0$. If $\gamma > 0$, its equation at the stationary point is

$$3\Delta t = \gamma(\delta_0 \mathcal{T}_K + \rho - C_n \rho_0).$$

Enlarge C_n , still polynomially, so that $C_n > 1 + \sup_\Omega(\delta_0 \mathcal{T}_K + \rho)/\rho_0$. The right side is strictly negative, contradicting $\Delta t \geq 0$. This additional bound is finite because K is compactly supported.

Consider next an inner boundary minimum in stages 1–3. There $j_0(-\epsilon) = 0$, so the same drift correction appears on both sides of (26.21). Cancelling it gives $V_\nu/u = \mathcal{B}$. Equation (26.18) and $d > 0$ imply

$$3\partial_\nu t = -H - N_B < 0.$$

This contradicts the nonnegative derivative into Ω_R at a boundary minimum. In stage 4 with $\gamma > 0$, the corresponding exact formula from (26.25) is $3\partial_\nu t = \gamma(-H - N_B) < 0$, giving the same contradiction. The outer boundary is excluded by (26.28).

Finally, at $\gamma = 0$, $f = 0$ and the remaining problem is $\operatorname{div}(3u \operatorname{grad} t) = 0$ with homogeneous inner conormal data and $t = 0$ on S_R . Testing by t gives $\int_{\Omega_R} 3u |dt|^2 = 0$. Connectedness and the nonempty Dirichlet boundary imply $t = 0$. Thus no stage admits floor equality. $\square$

The order of choices is now fixed: choose the background geometric data and ϵ , then δ_0 and the polynomial penalty C_n . At each n , enlarge the outer radius as in Lemma 26.4. Subsequent estimates may increase the lower bound on n or $R_{\min}(n)$, but none will replace C_n by an arbitrary fixed- n constant. In particular, the first-floor argument has paid for $F\tau$ using a bounded compact error and has not used the eventual small-height absorption in (25.16).

27. HEIGHT SEPARATION AND GLOBAL BOUNDS

Throughout this section the prepared geometry, its thresholds and collars, and $\epsilon \in (0, 1)$ are fixed. We consider arbitrary smooth solutions of the four stages of the homotopy in Section 26, on Ω_R , satisfying $t \geq -\epsilon$. The admissible radii satisfy $R \geq R_{\min}(n)$, where $R_{\min}(n) \rightarrow \infty$ and is large enough for the outer-boundary conclusion of Proposition 26.7. Every assertion of uniformity includes the homotopy parameter and R . As in Remark 22.5, Π_n denotes a polynomial bound in n , whereas $C(n)$ can depend arbitrarily on the fixed integer n . We shall keep this distinction through the boundary-flux estimate.

27.1. The limiting heights and the two total regions. The large factor l/η_n forces a nonconstant test level of h to approach a hypersurface whose expansion is controlled by the trace equation. A sublevel of the relaxed limiting height need not have a regular boundary. We first establish the expansion inequality for every smooth exterior support, including at a plateau and at a face of the opposite color.

Proposition 27.1 (Separation from the two total regions). *Let $A \subset \overline{\Omega}$ be compact. In the first two homotopy stages the following statements hold.*

- (i) *If $A \cap \mathcal{D}_{c_b} = \emptyset$, there exist $\xi_A > 0$ and n_A such that $h \geq b_- + \xi_A$ on A for every $n \geq n_A$.*
- (ii) *If $A \cap \mathcal{W}_{c_w} = \emptyset$, there exist $\xi_A > 0$ and n_A such that $h \leq b_+ - \xi_A$ on A for every $n \geq n_A$.*

Here the total regions are the full closed regions of Proposition 25.6, including components whose frontiers do not border Ω .

Proof. We give the lower-height argument first. Consider any sequence $n_i \rightarrow \infty$ of the indicated solutions; its radii tend to infinity. Extend h_i by the constant b_+ into a small collar across each white face. The functions on these enlarged neighborhoods are continuous. Define their lower relaxed limit by

$$\underline{h}(x) = \liminf_{\substack{i \rightarrow \infty \\ y \rightarrow x}} h_i(y).$$

It is lower semicontinuous and locally bounded by Lemma 26.4. Fix levels

$$(27.1) \quad b_- < k < k' < \min\{0, b_- + \xi_1\}, \quad k' + a_0 < 0,$$

where ξ_1 is the height-cutoff constant in D_0 . We initially work away from black faces.

Suppose a smooth ϕ touches $\underline{h}$ from below at x , with $\underline{h}(x) < k'$ and $d\phi(x) \neq 0$. Subtracting a small fourth-order function of distance makes the contact strict on a small closed neighborhood without changing its second jet. The definition of the relaxed limit and minimization on this neighborhood give, after passing to a subsequence, points $x_i \rightarrow x$ and constants c_i such that $\phi + c_i$ touches h_i from below at x_i , and $h_i(x_i) \rightarrow \underline{h}(x)$. These contacts are interior contacts of the original exterior: their heights are eventually strictly below $k' < b_+$, so they cannot occur on a white face or in its constant extension. This remains true when x itself is on a white face.

At contact, $dh_i = d\phi$ and $\text{Hess } h_i \geq \text{Hess } \phi$. The positive trace matrix therefore gives

$$(27.2) \quad \text{tr}_{A_{\chi_i}} K + \frac{1}{\sqrt{(\eta_{n_i}/l_i)^2 + |d\phi|^2}} \text{tr}_{A_{\chi_i}} \text{Hess } \phi \leq h_i + C + \alpha_i D_0, \quad 0 \leq \alpha_i \leq 1.$$

Here $\alpha_i = 0$ in the first stage. The cutoff construction gives $D_0 \leq 0$ at these low heights, everywhere: its white contribution vanishes there, and its black contribution is nonpositive. Moreover

$$\frac{l_i}{\eta_{n_i}} \geq \frac{\ell_i}{\eta_{n_i}} = e^{\epsilon n_i/2} \rightarrow \infty.$$

Consequently $|w_i| \rightarrow 1$, $d_i \rightarrow 0$, $\chi_i \rightarrow 0$, and A_{χ_i} tends to the orthogonal projection onto $(\text{grad } \phi)^\perp$. Passing to the limit in (27.2) proves

$$(27.3) \quad \Theta_K[\phi] := \text{tr}_{(\text{grad } \phi)^\perp} K + \frac{\text{tr}_{(\text{grad } \phi)^\perp} \text{Hess } \phi}{|d\phi|} \leq k' + a_0.$$

The normal here points toward increasing ϕ .

We next transfer this test inequality to the closed set $D_k = \{\underline{h} \leq k\}$. This step does not assume that k is a regular value. For integers $j \geq 1$, put

$$S_j(z) = (1 + e^{-j^2(z-k-j^{-1})})^{-1}.$$

The lower relaxed limit of $S_j(\underline{h})$ is the function I_k equal to zero on D_k and one off D_k . To verify this, at a point of D_k use the constant sequence of points and $S_j(\underline{h}(x)) \leq S_j(k) \rightarrow 0$. Off D_k , lower semicontinuity gives a neighborhood on which $\underline{h} \geq k + \delta$ for some $\delta > 0$, and hence uniform convergence to one there.

A smooth exterior support of D_k , with defining function $\phi = 0$ and D_k on the side $\phi \leq 0$, gives a lower test for I_k : shrink the neighborhood and multiply ϕ by a positive constant so that $|\phi| < 1/2$. Strict localization and the same minimization argument give lower tests $\phi + a_j$ for $S_j(\underline{h})$ at points $x_j \rightarrow x$, with contact values $v_j \rightarrow 0$. Each v_j lies in $(0, 1)$, so locally $S_j^{-1}(\phi + a_j)$ is a smooth lower test for $\underline{h}$. Its contact height satisfies

$$S_j^{-1}(v_j) = k + j^{-1} + j^{-2} \log \frac{v_j}{1 - v_j} \leq k + j^{-1} < k'$$

for all large j . Its gradient is nonzero. Composition with a smooth increasing function preserves the oriented level expansion: the additional Hessian term is a multiple of $d\phi \otimes d\phi$, whose tangential trace is zero. Thus (27.3) applies to these inverse tests. Passing to the limit shows that every smooth exterior support of D_k , away from black faces, has expansion at most $k' + a_0$.

This argument also rules out a boundary layer at a white face. The constant extension need not make $\underline{h}$ continuous there. Nevertheless all approximating low contacts used above are interior, so the support inequality holds at a limiting white-face contact. The reversed white face would itself be an exterior support of D_k with expansion

$$-H + P_B = -c_w > 0,$$

contradicting $k' + a_0 < 0$. Hence D_k misses every white face.

Adjoin the full black total region and consider $D_k \cup \mathcal{D}_{c_b}$ in the filled black barrier manifold. At a frontier point belonging to $\partial \mathcal{D}_{c_b}$, an exterior support of this union also supports the smooth region $\mathcal{D}_{c_b}$, so smooth tangency bounds its expansion by c_b . At other frontier points the bound is $k' + a_0 > c_b$. The coarse end-height decay makes the additional sublevel compact. It lies strictly inside the distant barrier sphere: otherwise a sphere at its greatest radius is an exterior support with positive expansion, a contradiction. There are no remaining contacts with the boundary of the filled barrier manifold. Lemma 25.5 therefore gives

$$(27.4) \quad D_k \cup \mathcal{D}_{c_b} \subset \mathcal{D}_{c'} \quad \text{whenever } k' + a_0 < c' < 0$$

in the chosen threshold range.

If A is disjoint from $\mathcal{D}_{c_b}$, its distance from that closed region is positive. Right continuity first permits $c' > c_b$ close enough that $A \cap \mathcal{D}_{c'} = \emptyset$. We then choose k, k' as in (27.1), close enough to b_- that $k' + a_0 < c'$. It follows that $A \cap D_k = \emptyset$. If a uniform eventual lower bound $h \geq k$ on A failed, a sequence of violating points in the compact set A would give a point of $A \cap D_k$ in its lower relaxed limit. This contradiction proves the first claim, with, for example, $\xi_A = k - b_-$.

For the second claim apply the same proof to $(-h, -K, -C)$. The low-height modification is now $-D_0$ and again has the required nonpositive sign. Extend across black faces by the high constant $-b_-$. At a limiting contact, the reversed black face has expansion $-c_b > 0$ for $-K$, and is excluded before using the white barrier manifold. Adjoining $\mathcal{W}_{c_w}$ and applying Lemma 25.5 now places the sublevel in $\mathcal{W}_{c'}$. Right continuity at c_w finishes the proof. This also covers an empty chosen white region: the distance-cap formulation of right continuity makes the nearby regions empty if necessary, and a nonempty enclosed sublevel is then impossible. $\square$

27.2. Collar comparisons and boundary slopes. We shorten the disjoint collars to $0 \leq s \leq s_0$ so that the offset is exactly linear there. Their outer leaves are a positive distance from the corresponding closed total region. Proposition 27.1 therefore provides a strict height gap on those leaves in the first two stages. All these leaves and the compact sets needed below are fixed before increasing n .

Lemma 27.2 (Collar slopes). *For all sufficiently large n , there are constants $0 < c_4 < C_4$, independent of n, R and the homotopy parameter, such that in the first two stages*

$$(27.5) \quad \frac{c_4}{\eta_n} \leq \varepsilon_B \partial_\nu f \leq \frac{C_4}{\eta_n}, \quad \varepsilon_B = \begin{cases} 1 & \text{on black faces,} \\ -1 & \text{on white faces.} \end{cases}$$

On these collars $h \geq b_-$ on black faces and their exterior collars, and $h \leq b_+$ on white faces and their exterior collars. In stage three,

$$(27.6) \quad |\partial_\nu f| \leq C_4/\eta_n.$$

In stage four $f = 0$.

Proof. Write $N = \text{grad } s / |\text{ds}|$ and let P_s, H_s be the tangential trace of K and the mean curvature of the collar leaf, with normal N . For a test height $h_* = b_* + \psi(s)$ of positive slope, comparison with a solution uses the test gradient and the solution's value of Z . With the resulting implicit t , direct contraction gives

$$(27.7) \quad \text{tr}_{A_\chi}(K + H^{f*}) = P_s + aH_s + \chi K(N, N) + a\chi \left(\frac{\text{Hess } s(N, N)}{|\text{ds}|} + |\text{ds}| \frac{\psi''}{\psi'} \right).$$

Indeed $f_* = h_*/\eta_n$ and $\text{Hess } f_* = (\psi' \text{Hess } s + \psi'' \text{ds}^2)/\eta_n$. The factor $l\psi' / (\eta_n \sqrt{D})$ equals $a/|\text{ds}|$, and $\text{tr}_{N^\perp} \text{Hess } s = |\text{ds}| H_s$. For a test $b_* - \psi(s)$ with $\psi' > 0$, all terms with prefactor a in (27.7) change sign.

If the slope is between two fixed positive constants, the floor and the fixed collar geometry imply

$$(27.8) \quad d \leq C(\eta_n/\ell)^2 = Ce^{-\epsilon n}, \quad 1 - a = \frac{d}{1+a} \leq d, \quad n\chi = \frac{3n}{3+pv}d \geq d,$$

and $a \geq 1/2$ for large n . Choose a smooth nonnegative cutoff ϑ_2 , equal to one on $[0, 1]$ and zero on $[2, \infty)$, and set

$$b_n(s) = A_2 n \vartheta_2(ns), \quad \int_0^{s_0} b_n(s) \, ds \leq 2A_2.$$

Thus slopes whose logarithmic derivative is $\pm b_n$ remain within a fixed multiplicative factor e^{2A_2} of their initial slopes.

On a black collar a lower barrier is $b_- + \psi_-(s)$, with $\psi_-(0) = 0$ and $\psi_-''/\psi_-' = b_n$. First choose A_2 large, depending only on bounded collar geometry. Then choose the slope multiplier sufficiently small that $\psi_-' \leq \kappa_B/2$ throughout the collar and that its value at s_0 is below the solution by the strict outer-leaf gap. The direction of its gradient is compatible with the face, so $D_0 = 0$ at a contact. Since $P_s + H_s \geq c_b$ and $C = a_0 - \kappa_B s$, the left side of the trace equation minus its prescribed right side at this test height is at least

$$(27.9) \quad \frac{\kappa_B s}{2} - Cd + a\chi |ds| b_n.$$

The constant multiplying d depends on collar geometry, not on the size of the chosen slope multiplier. On $s \leq 1/n$, the last term dominates Cd by (27.8) and the choice of A_2 . On $s \geq 1/n$, the first term dominates the exponentially small Cd for sufficiently large n . The residual is therefore strictly positive.

For an upper barrier use $b_- + \psi_+(s)$ with $\psi_+''/\psi_+' = -b_n$. Choose its minimum slope sufficiently large to dominate the coarse height at s_0 and all fixed linear spatial variations. Smoothness and $P_0 + H_0 = c_b$ give $|P_s + H_s - c_b| \leq Cs$. The residual is consequently bounded above by

$$(27.10) \quad -cs + Cd - a\chi |ds| b_n$$

for a fixed $c > 0$. It is strictly negative by the same two ranges of s . In particular, the favorable linear term in these comparisons comes from the height and the offset; we have not assumed a positive derivative of the leaf expansion at $s = 0$.

At any hypothetical crossing extremum, the test and the solution have the same gradient and the same Z , hence the same t and positive trace matrix. The prescribed right side is increasing in h with derivative at least one. The strict residual signs, Hessian comparison, and the endpoint inequalities exclude such a crossing. Differentiating the resulting barriers at $s = 0$, and using upper and lower positive bounds for $|ds|$, proves (27.5) on black faces. Sign reversal of h, K, C proves the white assertion. The compatible-direction modification again vanishes there.

In stage three let $b_* = (1 - \zeta)b$ be the assigned face height. Use $b_* + \psi(s)$ and $b_* - \psi(s)$, with $\psi' > 0$, $\psi''/\psi' = -b_n$, and with a large fixed minimum slope. For the limiting coefficients $a = 1$, $\chi = 0$ at $s = 0$ and height b_* , the directional inequalities in the homotopy construction are

$$F_{\text{presc}}(\nu) \geq P_B + H, \quad F_{\text{presc}}(-\nu) \leq P_B - H.$$

Their residuals have precisely the signs needed for the upper and lower tests, respectively. At positive s and finite d , smooth spatial variation and $1 - a \leq d$, $\chi \leq d$ cost at most $C(s + d)$. Increasing or decreasing the height by $\psi(s)$ improves the respective residual by at least $\psi(s)$. Choose the minimum slope to dominate the Cs error and the coarse height at the outer collar leaf. The logarithmic curvature contribution in (27.7) has the favorable sign for both tests and dominates Cd on $s \leq 1/n$; on $s \geq 1/n$ the linear height term dominates it. The same comparison proves (27.6). All constants are uniform in $\zeta \in [0, 1]$ by smoothness of the prescribed convex

combination. The assertion for stage four is the scalar maximum-principle conclusion already proved in the homotopy construction. $\square$

Combining the own-color collar comparison with Proposition 27.1 on the compact complement of smaller own-color collars yields

$$(27.11) \quad b_- \leq h \leq b_+ \quad \text{on } \overline{\Omega} \cap \text{supp } K \quad \text{in the first two stages, for all sufficiently large } n.$$

The finitely many compact sets used here are fixed first. Thus (27.11) has exactly the range required in (25.16); it was not used to establish the floor.

27.3. Polynomial weighted traces and flux. The slope sign supplies a trace inequality with a small coefficient in the boundary flux. Its proof uses weighted divergences, so it does not require an upper bound for Z or $|df|$. All measures in the next lemma are background g measures.

Lemma 27.3 (Weighted trace inequalities). *Let $\mathcal{C}$ be the union of fixed collars, slightly enlarged if necessary. In stages one and two, for every nonnegative smooth φ and every sufficiently large n ,*

$$(27.12) \quad \int_B L_0 \varphi \, dA_g \leq \Pi_n \int_{\mathcal{C}} u((1 + \sqrt{\mathcal{T}})\varphi + |d\varphi|_{A_\chi}) \, dV_g,$$

$$(27.13) \quad \int_B \varphi \, dA_g \leq \Pi_n \int_{\mathcal{C}} \sqrt{D}((1 + \sqrt{\mathcal{T}})\varphi + |d\varphi|_{A_\chi}) \, dV_g.$$

On B , the corresponding homotopy flux satisfies

$$(27.14) \quad |(V_\lambda)_\nu| \leq C u d \leq C \frac{\eta_n}{\ell} L_0.$$

The constants C and the coefficients of Π_n are independent of n, R and the solution.

Proof. Set $W = \sqrt{d}w$. For every covector ξ ,

$$(27.15) \quad |\xi(W)| \leq |\xi|_{A_d} \leq \sqrt{(n+3)/3} |\xi|_{A_\chi}, \quad uW = L_0 w, \quad \sqrt{D}W = w.$$

The first assertion follows along the axis from $|\xi(W)|^2 = dv\xi(e)^2 \leq d\xi(e)^2$ and is immediate transversely.

For completeness, write $Q_f = \text{tr}_{A_d} H^f$. Differentiating w and D gives

$$\begin{aligned} \nabla_i w_j &= H_{ij}^f - H^f(w, \partial_i)w_j + p d t_i w_j, \\ \text{div } w &= Q_f + p d w(t), \end{aligned} \quad Q_f = F + (d - \chi)j - \text{tr}_{A_d} K.$$

The two weighted divergences are exactly

$$(27.16) \quad D^{-1/2} \text{div}(\sqrt{D}W) = \sqrt{d}(Q_f + p d w(t)),$$

$$(27.17) \quad u^{-1} \text{div}(uW) = \sqrt{d}(Q_f + (p + 2 + p d)w(t)).$$

Indeed the second formula follows from $uW = L_0 w$ and $d \log L_0 = (p + 2) dt$. Since $\sqrt{d}|w(t)| \leq |dt|_{A_d}$ and $0 \leq d - \chi \leq d$, coercivity (26.14) bounds both right sides by $\Pi_n(1 + \sqrt{\mathcal{T}})$ on the fixed collars. No factor l^{-1} or η_n^{-1} occurs.

On a face $\varepsilon_B w_\nu = a \geq 1/2$ by (27.5). Integrate the divergence of $\varepsilon_B \zeta u W \varphi$ separately on each collar, where $\zeta = 1$ at B and $\zeta = 0$ near its outer leaf. The outward normal of Ω_R at B is $-\nu$, so the boundary term is $-\int_B L_0 a \varphi$. Taking absolute values of the remaining terms and using (27.15)–(27.17) proves (27.12). Replacing u by $\sqrt{D}$ proves (27.13). The derivatives of ζ cost only a fixed constant.

On the boundary the compatible slope makes $D_0 = 0$, and the prescribed heights and offsets give exactly $F - P_B = \varepsilon_B H$. Moreover $w_\nu = \varepsilon_B a$ and $(A_\chi K(w, \cdot))_\nu = \varepsilon_B a \chi K(\nu, \nu)$. Writing $q_\lambda = 1 - (1 - \lambda)j_0(t) \in [0, 1]$, the flux formula (26.21) is therefore

$$\frac{(V_\lambda)_\nu}{u} = q_\lambda \left[-\frac{d}{1+a} H - N_B d - (1 - \lambda) \varepsilon_B a \chi K(\nu, \nu) \right].$$

This is bounded in absolute value by Cd since $\chi \leq d$. Finally $ud = L_0 \sqrt{d}$ and $\sqrt{d} \leq C\eta_n/\ell$ at B by (27.5). This proves (27.14). $\square$

27.4. A high-level integral and graph Sobolev inequality. From this point, constants used to obtain a bounded range may depend arbitrarily on fixed n . The only large- n absorption needed first uses the polynomial constants just proved:

$$(27.18) \quad \Pi_n \frac{\eta_n}{\ell} = \Pi_n e^{-\epsilon n/2} \longrightarrow 0.$$

There exists $Z_0(n) > 0$, independent of the solution and R , such that

$$(27.19) \quad \Xi \geq \delta_0 \mathcal{T} + \rho + \frac{1}{2} \delta_0 \eta_n a \sigma \quad \text{on } \{Z > Z_0(n)\}, \quad \Xi \geq \delta_0 \mathcal{T} - C(n) \quad \text{everywhere.}$$

To see the first assertion, only the support of m_0 matters. There $-\epsilon \leq t \leq -\epsilon + 1/n$, whereas

$$D = e^{6(Z-t)}, \quad \eta_n a \sigma = \frac{\eta_n}{l} \left(\sqrt{D} - \frac{1}{\sqrt{D}} \right).$$

At fixed n this last expression tends uniformly to infinity with Z , and dominates the bounded penalty $\mathcal{P}$. The global lower bound follows from the bounded weights defining $\mathcal{P}$.

Lemma 27.4 (High-level integral). *In stage one, choose a smooth nondecreasing k_0 which is zero on $(-\infty, Z_0]$ and one on $[Z_0 + 1, \infty)$. For all sufficiently large fixed n ,*

$$(27.20) \quad \int_{\Omega_R} u k_0(Z) (\mathcal{T} + \rho + \eta_n a \sigma) \, dV_g \leq C(n).$$

Proof. The test $k_0(Z)$ vanishes on S_R . Integration by parts gives

$$\int u k_0 \Xi + 3 \int u k'_0 |dZ|_{A_\chi}^2 = - \int_B k_0 (V_\lambda)_\nu - \lambda \int u k'_0 \langle K(w, \cdot), dZ \rangle_{A_\chi}.$$

The last integral is absorbed into part of the gradient term plus $C(n)$. Indeed it is supported in a fixed compact set and in the strip $Z_0 \leq Z \leq Z_0 + 1$, where

$$(27.21) \quad u = l e^{3Z-t} \leq e^{1+3(Z_0+1)+\epsilon}.$$

Thus the residual strip integral has bounded background volume and a bounded weight, without any preliminary gradient estimate.

By Lemma 27.3, the absolute boundary contribution is at most

$$\Pi_n \frac{\eta_n}{\ell} \int_C u ((1 + \sqrt{\mathcal{T}}) k_0 + k'_0 |dZ|_{A_\chi}).$$

The fixed collars have $\min_C \rho > 0$. Consequently $1 + \sqrt{\mathcal{T}} \leq C(\rho + \delta_0 \mathcal{T})$ there. Choose n large using (27.18) to absorb the nondifferentiated term into the positive source in (27.19). Young's inequality absorbs the differentiated term into another part of $\int u k'_0 |dZ|_{A_\chi}^2$, with a bounded remainder by (27.21). This proves (27.20). $\square$

Let Γ_f be the ordinary product graph of f in $(\Omega_R \times \mathbb{R}, g + dz^2)$. Put $E_f = 1 + \sigma^2$ and $G_f = \text{Id} - (\sigma^2/E_f)e \otimes e$, the inverse graph metric on base covectors. The following comparisons hold without a bound for σ :

$$(27.22) \quad \begin{aligned} e^{-1}\sqrt{D} \, dV_g &\leq dV_{\Gamma_f} = \sqrt{E_f} \, dV_g \leq \ell^{-1}\sqrt{D} \, dV_g, \\ \ell^2 A_\chi &\leq G_f \leq \frac{e^2(n+3)}{3} A_\chi. \end{aligned}$$

They follow from $\ell \leq l \leq e$, hence $\ell^2 \leq D/E_f \leq e^2$, and $3d/(n+3) \leq \chi \leq d$. Thus graph measure and gradient norms are comparable to $\sqrt{D} \, dV_g$ and the A_χ norm, with constants depending only on fixed n .

On every fixed compact base set Q , (27.20) implies

$$(27.23) \quad \int_{\Gamma_f|_Q} e^{2Z} \, dV_{\Gamma_f} \leq C(n, Q).$$

In fact, for $x = \sqrt{D} \geq 1$, the elementary estimate $x^{2/3} \leq C(n)(1 + (\eta_n/l)(x - x^{-1}))$ gives

$$D^{1/3} \leq C(n)(1 + \eta_n a \sigma), \quad e^{2Z} \sqrt{D} = \frac{u}{l} D^{1/3}.$$

On $\{Z \geq Z_0 + 1\}$ use (27.20) and the positive minimum of ρ on Q . On its complement the floor bounds D and e^{2Z} , and the fixed base volume suffices.

Lemma 27.5 (Sobolev inequality on the graph). *Fix a compact base patch and a slightly larger compact neighborhood. For every smooth φ supported over the patch, allowed to meet B , a stage-one solution satisfies*

$$(27.24) \quad \left(\int_{\Gamma_f} |\varphi|^4 \, dV_{\Gamma_f} \right)^{1/2} \leq C(n) \int_{\Gamma_f} (|\nabla_{\Gamma_f} \varphi|^2 + (1 + \mathcal{T})\varphi^2) \, dV_{\Gamma_f}.$$

The constant depends on the fixed neighborhoods, and is independent of the graph and the outer radius.

Proof. Extend the compact base neighborhood smoothly to a compact Riemannian manifold and fix a smooth Euclidean isometric embedding by Nash's theorem [41]. Its product with a line embeds the product graph isometrically. The scalar mean curvature of the product graph is, up to its normal sign,

$$H_{\Gamma_f} = \frac{\sqrt{D}}{l\sqrt{E_f}} \left[\text{tr } T - \text{tr}_{e^\perp} K + E_f^{-1}(j - K(e, e)) \right].$$

The prefactor is at most ℓ^{-1} and $E_f^{-1} \leq e^2 d$. Coercivity therefore bounds its magnitude by $C(n)(1 + \sqrt{\mathcal{T}})$. The additional mean-curvature vector of the fixed embedding has magnitude at most four times the bound for its second fundamental form. Thus the Euclidean mean-curvature vector has the same bound, independently of the graph slope.

The four-dimensional Michael–Simon L^1 Sobolev inequality [40, 47], applied to $|\varphi|^3$, gives

$$\left(\int_{\Gamma_f} |\varphi|^4 \right)^{3/4} \leq C(n) \int_{\Gamma_f} (|\varphi|^2 |\nabla_{\Gamma_f} \varphi| + (1 + \sqrt{\mathcal{T}})|\varphi|^3) + C \int_{\partial\Gamma_f} |\varphi|^3.$$

We have retained the boundary term. Its form follows, for each individual smooth graph, by applying the boundaryless inequality to cutoffs tending to one up to the face: the integral of their normal derivatives tends to the face integral. Artificial patch edges do not contribute because φ vanishes there. Along B , f is constant on each face, so the graph face measure is exactly dA_g .

Apply (27.13) to $|\varphi|^3$ and use (27.22). This bounds the face contribution by the same two interior expressions. If $I = \int_{\Gamma_f} |\varphi|^4$, Cauchy–Schwarz now gives

$$I^{3/4} \leq C(n) I^{1/2} \left(\int_{\Gamma_f} (|\nabla_{\Gamma_f} \varphi|^2 + (1 + \mathcal{T}) \varphi^2) \right)^{1/2}.$$

For $I > 0$ divide by $I^{1/2}$ and square. The case $I = 0$ is immediate, proving (27.24). $\square$

27.5. Iteration and a bounded range for all stages.

Proposition 27.6 (Global bounded range). *For every sufficiently large fixed n there is a finite $C(n)$ such that every above-floor smooth solution in any of the four homotopy stages satisfies*

$$(27.25) \quad -\epsilon \leq t \leq Z \leq C(n), \quad |f| + |\text{grad } f| \leq C(n) \quad \text{on } \bar{\Omega}_R.$$

The constant is uniform in the allowed radii and homotopy parameters.

Proof. We first treat stage one, where the compactly supported drift can prevent a direct maximum argument. Let ψ be a smooth base cutoff supported in a fixed compact patch, permitted to meet B . For $k \geq k_*(n) \geq 1$, test the divergence equation with $\psi^2 e^{2kZ}/L_0$. We claim

$$(27.26) \quad \int_{\Gamma_f} e^{2kZ} \psi^2 (\mathcal{T} + k |\nabla_{\Gamma_f} Z|^2) \leq C(n) k \int_{\Gamma_f} e^{2kZ} (\psi^2 + |\text{d}\psi|_g^2),$$

where $C(n)$ is independent of k .

Here are the error estimates establishing the claim. Let $\text{d}\mu_D = \sqrt{D} \text{d}V_g$, and write $\kappa = K(w, \cdot)$ in this computation. Since $\text{d} \log L_0 = (p+2) \text{d}t$, the exact tested identity is

$$\begin{aligned} & \int e^{2kZ} \psi^2 (\Xi + 6k |\text{d}Z|_{A_\chi}^2) \text{d}\mu_D \\ &= - \int_B \frac{\psi^2 e^{2kZ}}{L_0} (V_\lambda)_\nu \text{d}A_g - 2k\lambda \int e^{2kZ} \psi^2 \langle \kappa, \text{d}Z \rangle_{A_\chi} \text{d}\mu_D \\ & \quad - 2 \int e^{2kZ} \psi \langle 3 \text{d}Z + \lambda \kappa, \text{d}\psi \rangle_{A_\chi} \text{d}\mu_D + \int e^{2kZ} \psi^2 (p+2) \langle 3 \text{d}Z + \lambda \kappa, \text{d}t \rangle_{A_\chi} \text{d}\mu_D. \end{aligned}$$

The global lower bound in (27.19) supplies $\delta_0 \mathcal{T}$ and costs only $C(n) \psi^2$ on the right. Coercivity bounds the final integrand, by Young’s inequality, by

$$\frac{1}{8} \delta_0 \psi^2 \mathcal{T} + C(n) \psi^2 |\text{d}Z|_{A_\chi}^2 + C(n) \psi^2.$$

Choose $k_*(n)$ sufficiently large to absorb this fixed multiple of the gradient square. The drift term carrying $2k$ is bounded by $k \psi^2 |\text{d}Z|_{A_\chi}^2 + C(n) k \psi^2$. The cutoff terms cost another fixed fraction of the k -weighted gradient square plus $C(n) k (\psi^2 + |\text{d}\psi|_g^2)$.

For the boundary term, (27.14) gives $|(V_\lambda)_\nu|/L_0 \leq C\sqrt{d} \leq C$. Apply (27.13) to $\psi^2 e^{2kZ}$. The resulting integrand, apart from its common factor e^{2kZ} , is bounded by

$$C(n) \left[(1 + \sqrt{\mathcal{T}}) \psi^2 + 2|\psi| |\text{d}\psi|_{A_\chi} + 2k \psi^2 |\text{d}Z|_{A_\chi} \right].$$

For any fixed coefficient $b = C(n)$ the estimates

$$b\sqrt{\mathcal{T}} \leq \frac{1}{8} \delta_0 \mathcal{T} + 2b^2/\delta_0, \quad 2bk |\text{d}Z|_{A_\chi} \leq k |\text{d}Z|_{A_\chi}^2 + b^2 k$$

show that the boundary cost is absorbed with the claimed remainder linear in k . Leaving fixed positive fractions of the coercive terms and using (27.22) proves (27.26).

Set $U = e^Z$. Applying (27.24) to ψU^k and using (27.26) yields

$$(27.27) \quad \left(\int_{\Gamma_f} \psi^4 U^{4k} \right)^{1/2} \leq C(n) k^2 \int_{\Gamma_f} U^{2k} (\psi^2 + |\text{d}\psi|_g^2).$$

Choose nested base patches $Q_r \Subset Q_s$ in a fixed coordinate neighborhood, or corresponding half patches at B , with $0 < s - r \leq 1$. Cutoffs have $|\mathrm{d}\psi| \leq C/(s - r)$. Iterate (27.27) with $k_j = 2^j k_*$ and geometric gaps. The factor at the j th step is at most

$$[C(n)k_j^2(1 + \mathrm{gap}_j^{-2})]^{1/(2k_j)}.$$

Its product is finite, since $\sum_j (j + 1)/k_j < \infty$. Tracking the total gap exponent gives

$$(27.28) \quad \sup_{Q_r} U \leq C(n)(s - r)^{-2/k_*} \left(\int_{\Gamma_f|Q_s} U^{2k_*} \, \mathrm{d}V_{\Gamma_f} \right)^{1/(2k_*)}.$$

All comparisons here use the same graph measure of the individual solution; its constants are uniform by the preceding lemmas.

The starting exponent $2k_*$ can exceed the exponent in (27.23). A second, elementary nested-patch argument closes this point. Write $M(s) = \sup_{Q_s} U$. Interpolation gives

$$\left(\int_{\Gamma_f|Q_s} U^{2k_*} \right)^{1/(2k_*)} \leq M(s)^{1-1/k_*} \left(\int_{\Gamma_f|Q_s} U^2 \right)^{1/(2k_*)}.$$

The last integral is uniformly bounded by (27.23). If $k_* = 1$ this already suffices. Otherwise Young's inequality applied to (27.28) gives, for any $q \in (0, 1)$,

$$(27.29) \quad M(r) \leq qM(s) + C(n, q)(s - r)^{-2}.$$

Choose increasing radii r_j tending to a fixed larger radius, with $r_{j+1} - r_j$ proportional to 2^{-j} , and take $q < 1/4$. Iteration sums a geometric series with ratio $4q < 1$. The remaining $q^j M(r_j)$ tends to zero: for each individual smooth solution all these patches lie in one compact neighborhood with finite supremum. Thus $M(r_0) \leq C(n)$ uniformly. A finite cover gives a uniform Z bound on a compact set containing B and the support of K .

Outside this set $K = 0$. At an interior maximum with $Z > Z_0(n)$, $\mathrm{d}Z = 0$ and

$$\mathrm{div}(3uA_\chi \mathrm{grad} Z) = 3u \mathrm{tr}_{A_\chi} \mathrm{Hess} Z \leq 0,$$

whereas $u\Xi > 0$ by (27.19). The outer value is $Z = 0$, so this maximum argument extends the compact bound through the end. This proves the stage-one Z estimate.

In stages two and three $\lambda = 0$, so the same maximum argument works everywhere in the interior. On B the already established face-gradient bounds show that Z can be high only if $t \geq 1$: if $t < 1$, then

$$Z = t + \frac{1}{6} \log(1 + l^2 |\mathrm{d}f|^2) \leq 1 + \frac{1}{6} \log(1 + e^2 C_4^2 \eta_n^{-2}).$$

For $t \geq 1$ the boundary switch satisfies $j_0 = 1$ and the prescribed conormal flux in (26.21) is zero. A high boundary maximum would contradict the boundary point principle applied locally on the high-level set to $\mathrm{div}(3uA_\chi \mathrm{grad} Z) = u\Xi > 0$. For this application the individual smooth solution gives a smooth positive elliptic matrix; no uniform ellipticity constant is being assumed before the bound. Equivalently, the outward derivative at a nonconstant boundary maximum is strictly positive, whereas the zero conormal flux forces it to vanish. This bounds Z in these two stages.

In stage four $f = 0$, hence $t = Z$. At a positive scale multiplying the source and boundary data, the same high-level interior and boundary arguments apply. At zero scale the homogeneous mixed problem gives $Z = 0$. The resulting bound is uniform in the scale.

Finally $Z - t = \frac{1}{6} \log D \geq 0$, so the floor and the upper bound give $D \leq e^{6(C(n)+\epsilon)}$. Since $l \geq \ell > 0$, $|\mathrm{d}f| \leq \ell^{-1} \sqrt{D - 1} \leq C(n)$. The coarse bound for $h = \eta_n f$ gives $|f| \leq C(n)$. This proves all of (27.25). $\square$

The bounded range now permits fixed- n regularity and exhaustion. The final mass estimate will use only the earlier polynomial bounds in Lemma 27.3 and (27.14); the graph Sobolev and Moser constants are not needed in that limit.

28. AXIAL REGULARITY AND THE COUPLED ESTIMATES

The second equation has quadratic growth in both $\text{grad } Z$ and $\text{Hess } f$. A bound for its ellipticity constants therefore does not close the regularity argument. We first prove an independent scalar estimate which gives a positive Morrey exponent for $|\text{Hess } f|^2$. That exponent permits a measure correction in the second equation. All constants in this section may depend arbitrarily on a fixed n ; none of them is subsequently used as a polynomial bound Π_n . In coordinate calculations D_x , or D when there is no ambiguity, denotes ordinary differentiation, whereas $D = 1 + l^2 |\text{grad } f|^2$ in the system retains its earlier meaning.

28.1. An independent scalar estimate.

Theorem 28.1 (Measurable axial coefficient). *Let U be a four-dimensional coordinate ball B_2 , or a half-ball $B_2^+ = B_2 \cap \{x_4 > 0\}$. Assume that the metric is smooth, $C_g^{-1} \text{Id} \leq (g_{ij}) \leq C_g \text{Id}$, and that its coordinate C^2 norm is at most C_g . Assume also uniformly bounded C^3 norms for the charts and inverse charts. On a half-ball use boundary normal coordinates, so $g_{44} = 1$ and $g_{a4} = 0$ for $a < 4$. Let z be C^3 , up to the flat face when present, and constant on that face. Suppose, almost everywhere, that*

$$(28.1) \quad \begin{aligned} A_z : \text{Hess}_g z &= G, & A_z &= \text{Id} - (1 - \tilde{\chi}) e_z \otimes e_z, \\ e_z &= \frac{\text{grad } z}{|\text{grad } z|}, & 0 < \chi_0 \leq \tilde{\chi} &\leq 1. \end{aligned}$$

Here $\tilde{\chi}$ and G are measurable. At $\text{grad } z = 0$, any measurable unit axis making the equation true is allowed. If $|\text{grad } z| \leq M_1$ and $|G| \leq N_1$, then there are $\alpha \in (0, 1)$, $r_0 > 0$ and $C < \infty$ such that

$$(28.2) \quad \|\text{grad } z\|_{C^\alpha(U_1)} \leq C(M_1 + N_1), \quad \int_{B_r(x) \cap U} |\text{Hess}_g z|^2 \, dV_g \leq C(M_1 + N_1)^2 r^{2+2\alpha}$$

for $x \in \overline{U_1}$ and $0 < r \leq r_0$, where $U_1 = B_1$ or B_1^+ . The constants depend only on χ_0 , the stated geometry bounds and the separation from the curved patch edges. The C^α norm may equivalently be taken for the coordinate gradient components. In particular, no modulus of continuity of G or $\tilde{\chi}$, and no bound for a derivative of either, is assumed.

Proof. We prove the assertion on uniformly small full balls, using a reflection at the flat face. Rescaling back and a finite local cover give the stated form. The argument first establishes a local Hessian estimate and a dichotomy: on a smaller ball, either the gradient bound decreases or the solution is close to a nonconstant affine function. A fixed-axis estimate then improves the affine approximation. Iterating these alternatives gives Hölder gradient control, from which the local Hessian estimate yields the Morrey bound.

Doubling and a quantitative Hessian estimate. Subtract the constant boundary value. With $R_4 = \text{diag}(1, 1, 1, -1)$, extend the tangential metric block evenly and set $\tilde{z}(x) = -z(R_4 x)$ below the plane. Tangential derivatives of z vanish on the face and the two normal derivatives agree. Thus $\tilde{z}$ is C^1 ; its distributional coordinate Hessian has no surface measure. The doubled metric is Lipschitz and has the prescribed smooth bounds on each closed side. On the reflected side the gradient and covariant Hessian are $-R_4 \text{grad } z$ and $-R_4 (\text{Hess}_g z) R_4$, respectively. Extend $\tilde{\chi}$ evenly and G oddly. Equation (28.1) then holds almost everywhere on the double. Each individual doubled function is locally $W^{2,p}$ for every finite p , although these norms are not yet uniformly bounded.

Make a constant linear coordinate change so that the metric at the center of a ball is the identity. A plane in the double remains a plane under this change. If the ball is sufficiently small, the coordinate principal matrix $a = (a^{ij})$ in (28.1) satisfies

$$(28.3) \quad |\text{Id} - a(x)|_F \leq 1 - \chi_0/2 =: \kappa < 1.$$

The reason for the strict gap is the exact identity $|\text{Id} - A_z|_F = 1 - \tilde{\chi}$ in an orthonormal frame; the small metric oscillation consumes at most half the gap.

Let $T = D^2\Delta^{-1}$ on $\mathbb{R}^4$, with its values measured in the Frobenius norm. Its L^2 operator norm is exactly one, since its Fourier multiplier satisfies

$$\sum_{i,j} \left| \frac{\xi_i \xi_j}{|\xi|^2} \right|^2 = 1.$$

The Calderón–Zygmund bound gives a finite L^4 operator norm C_4 . Interpolation between 2 and 4 therefore gives $\|T\|_p \leq C_4^{\vartheta(p)}$, where $1/p = (1 - \vartheta(p))/2 + \vartheta(p)/4$ and $\vartheta(p) \downarrow 0$ as $p \downarrow 2$. Choose once and for all $2 < p_0 < 4$ such that $C_4^{\vartheta(p_0)} \kappa < 1$. For a compactly supported y , write $\Delta y = a : D^2 y + (\text{Id} - a) : D^2 y$ and absorb to obtain

$$\|D^2 y\|_p \leq \frac{\|T\|_p}{1 - \kappa \|T\|_p} \|a : D^2 y\|_p, \quad p = 2, p_0.$$

This proves the needed Cordes estimate without differentiating a .

For later use its local, scaled form is

$$(28.4) \quad \|D^2 y\|_{L^p(B_{r/2})} \leq C \left(\|a : D^2 y\|_{L^p(B_r)} + r^{-2} \|y\|_{L^p(B_r)} \right), \quad p = 2, p_0.$$

Here is the localization. A cutoff across a gap h gives the additional terms $Ch^{-1} \|Dy\|_p + Ch^{-2} \|y\|_p$. On a slightly larger ball, derivative interpolation bounds the first by $\varepsilon \|D^2 y\|_p + C_\varepsilon h^{-2} \|y\|_p$. Use concentric gaps decreasing geometrically and choose ε so that the resulting geometric series absorbs the larger-ball Hessian norms. For each individual function these norms are finite, so the terminal term tends to zero. This proves (28.4); the same argument absorbs bounded first order coefficients. In particular the Christoffel terms in the covariant equation cause no difficulty. The analytic inputs here are the scalar Calderón–Zygmund and interpolation estimates [28].

A divergence inequality for the gradient deficit. Divide by $M_1 + N_1$ unless this number is zero, in which case the claim is immediate. Subtract a constant and rescale space by r and height by r . The new gradient has norm at most one and the new forcing has norm at most r . Metric first derivatives and the second fundamental form of a reflecting plane are $O(r)$; curvature on the smooth sides is $O(r^2)$. After normalizing the metric at the center, all of these errors can be made uniformly small. In this paragraph write $P = \text{grad } z$, $H_z = \text{Hess}_g z$, and $s = 1 - |P|_g^2 \geq 0$.

There is an exact cancellation of the measurable axial coefficient:

$$(28.5) \quad A_z \text{grad}(|P|_g^2/2) - GP = (H_z - \Delta_g z g)P =: Q,$$

$$(28.6) \quad \text{div } Q = |H_z|_g^2 - (\Delta_g z)^2 + \text{Ric}_g(P, P).$$

Indeed $\text{grad}(|P|_g^2/2) = H_z P$ and $\Delta_g z = G + (1 - \tilde{\chi})H_z(e_z, e_z)$. The axial parts in the first formula cancel. In the second formula, $\text{div } H_z = \text{grad } \Delta_g z + \text{Ric}_g(P, \cdot)$ cancels the third derivatives. Both identities hold also at $P = 0$, because the first has both sides zero there. At no point is a derivative of G or $\tilde{\chi}$ taken. Choose $\varepsilon_0 > 0$ with $(1 + \varepsilon_0)(1 - \chi_0)^2 < 1$ when $\chi_0 < 1$. Young's inequality, or directly $\Delta_g z = G$ if $\chi_0 = 1$, gives

$$(28.7) \quad |H_z|_g^2 - (\Delta_g z)^2 \geq c |H_z|_g^2 - C |G|^2, \quad c = c(\chi_0) > 0.$$

We compute the only surface error in this identity. In boundary normal coordinates on the original side, $P = z_4 \partial_4$ at the plane and, for $a, b < 4$,

$$(H_z)_{ab} = -\Gamma_{ab}^4 z_4 = \frac{1}{2}(\partial_4 g_{ab}) z_4.$$

Writing $H_B = \frac{1}{2} g^{ab} \partial_4 g_{ab}$ for the mean curvature toward increasing x_4 , this gives

$$Q_4 = (z_{44} - \Delta_g z) z_4 = -H_B z_4^2.$$

Thus the jump of the volume-weighted normal flux on the double is bounded by $C \|\Pi_B\| |P|^2$. In particular it contains no uncontrolled second derivative of z . A bounded plane density $j(x')$ is the distributional divergence of $j(x') \mathbf{1}_{\{x_4 > 0\}} \partial_4$; only a normal derivative is taken in this assertion. After a linear coordinate change use the corresponding constant normal direction. Subtracting this bounded step field handles the signed jump, without imposing any tangential regularity on j .

More explicitly, put $J_g = \sqrt{\det g}$ and let

$$a_1^{ij} = J_g (g^{ij} - (1 - \tilde{\chi}) e_z^i e_z^j)$$

be the coordinate divergence matrix. Let $j = [J_g Q^4]$ be the upper trace minus the lower trace. Reflection gives $Q^- = R_4 Q^+$, so $j = -2 J_g H_B z_4^2$. Then

$$\operatorname{div}(J_g Q) = J_g (|H_z|^2 - (\Delta_g z)^2 + \operatorname{Ric}_g(P, P)) + j \, d\mathcal{H}^3|_{\{x_4=0\}}.$$

Since $a_1 Ds + 2J_g GP = -2J_g Q$, the error field $2J_g GP + 2j(x') \mathbf{1}_{\{x_4 > 0\}} \partial_4$ cancels the plane contribution exactly.

Consequently (28.5)–(28.7), with volume factors included, imply on a full normalized ball

$$(28.8) \quad \operatorname{div}(a_1 Ds + E_1) \leq -c_1 |H_z|^2 + e_1, \quad \|E_1\|_\infty + \|e_1\|_\infty \rightarrow 0$$

as the normalized source and geometry errors tend to zero. The matrix a_1 is bounded, symmetric and uniformly elliptic, with constants depending only on χ_0 and the fixed comparison bounds. For example, away from the plane one may take the volume-weighted matrix of A_z , $E_1 = 2\sqrt{\det g} GP$, and $e_1 \leq C(|G|^2 + |\operatorname{Ric}_g|)$; the step field just described supplies the remaining term in E_1 .

The gradient drop or affine alternative. Fix $r_* = 1/32$. We claim that, for every $b_0 > 0$, there are $r_* < k_* < 1$ and $\delta_{\text{alt}} > 0$, depending only on the fixed parameters and b_0 , such that a normalized solution on B_1 , with gradient at most one and errors at most δ_{alt} , satisfies one of

$$(28.9) \quad \sup_{B_{r_*}} |\operatorname{grad} z|_g \leq k_*; \quad \|z - L\|_{L^\infty(B_{r_*})} \leq r_* b_0, \quad \frac{1}{2} \leq |DL| \leq 2,$$

where L is a coordinate affine function.

To prove the claim, consider a sequence with errors tending to zero and $\inf_{B_{r_*}} s \rightarrow 0$. Weak Harnack for nonnegative scalar divergence supersolutions, applied to (28.8) with its bounded vector and scalar errors, yields for some $q > 0$

$$\left(|B_{1/2}|^{-1} \int_{B_{1/2}} s^q \right)^{1/q} \leq C \left(\inf_{B_{r_*}} s + \|E_1\|_\infty + \|e_1\|_\infty \right) \rightarrow 0.$$

One obtains these fixed radii from the usual concentric weak Harnack inequality by a finite chain of overlapping balls. Thus $s \rightarrow 0$ in measure. The scalar weak Harnack and energy estimates used here are valid for bounded measurable divergence matrices [28].

There is also $Ds \rightarrow 0$ in L^1 on smaller balls. Indeed, for a fixed cutoff ζ and $b > 0$, test the weak inequality by $\zeta^2(b - s)_+$. Discarding its favorable Hessian term and using Young's inequality gives

$$\int_{\{s < b\}} \zeta^2 |Ds|^2 \leq C b^2 \int |D\zeta|^2 + C \|E_1\|_\infty^2 + C b (\|E_1\|_\infty + \|e_1\|_\infty).$$

The full local L^2 norm of Ds is bounded by (28.4), the gradient bound and the geometry bounds. On $\{s \geq b\}$, Cauchy–Schwarz and convergence in measure give a vanishing L^1 norm. On $\{s < b\}$ the displayed estimate bounds the limiting L^1 norm by Cb . Letting $b \downarrow 0$ proves the claim. Testing (28.8) with a fixed nonnegative cutoff equal to one on $B_{1/4}$ now gives

$$\int_{B_{1/4}} |H_z|^2 \longrightarrow 0.$$

The coordinate Hessian also tends to zero in L^2 , because the connection coefficients tend to zero and Dz is bounded. After subtracting $z(0)$, the functions are equi-Lipschitz. Poincaré gives convergence of their gradients in L^2 to constants, and compactness gives uniform convergence to an affine function on smaller balls. Since $s \rightarrow 0$ in measure and the metrics converge to the identity, its slope has length one. If (28.9) failed for every $k_* \uparrow 1$ and every tolerance tending to zero, this conclusion would contradict failure of its affine alternative. This proves the claim.

All derivatives for a fixed exceptional axis. For a constant Euclidean unit vector e_0 , let

$$(28.10) \quad \mathcal{A}_0 Y := \Delta_{e_0^\perp} Y + \tilde{\chi}(x) \partial_{e_0} Y.$$

If $Y_0 \in W^{2,2}(B_{3/4})$ and $\mathcal{A}_0 Y_0 = 0$ almost everywhere, we claim that, for some $\alpha_1 \in (0, 1)$,

$$(28.11) \quad \|DY_0\|_{C^{\alpha_1}(B_{1/4})} \leq C \|Y_0\|_{W^{2,2}(B_{3/4})}.$$

Set $v_1 = \partial_{e_0} Y_0 \in W^{1,2}$. Differentiating the equation in distributions, with no differentiation of its coefficient as a separate function, gives

$$\Delta_{e_0^\perp} v_1 + \partial_{e_0}(\tilde{\chi} \partial_{e_0} v_1) = 0.$$

This is a scalar uniformly elliptic divergence equation. Its local boundedness and Hölder estimate bound v_1 in C^{α_1} . Caccioppoli, subtracting the value at the center of a ball of radius h , then gives

$$(28.12) \quad \int_{B_h(x)} |Dv_1|^2 \leq Ch^{2+2\alpha_1} \|Y_0\|_{W^{2,2}(B_{3/4})}^2$$

at all centers in a fixed smaller ball. The power follows from h^{-2} for the cutoff, $h^{2\alpha_1}$ for the squared oscillation, and volume h^4 .

Normalize the norm on the right to one. Since $\Delta Y_0 = (1 - \tilde{\chi}) \partial_{e_0} v_1$, it follows that

$$(28.13) \quad \int_{B_h(x)} |\Delta Y_0| \leq Ch^{3+\alpha_1}.$$

Let N be the Newtonian potential of $\zeta \Delta Y_0$, where ζ is supported inside $B_{1/2}$ and equals one on a slightly smaller ball. The remainder $Y_0 - N$ is harmonic there, with bounded local L^2 norm; the potential has such a bound by local integrability of $|x|^{-2}$ and the L^2 bound for ΔY_0 . For the gradient of N , the kernel and its derivative are bounded by $C|x|^{-3}$ and $C|x|^{-4}$. If two points are separated by h , the annuli of radii $s \leq 4h$ give, by (28.13), $C \sum s^{\alpha_1} \leq Ch^{\alpha_1}$. On annuli $s > 4h$ the kernel difference instead gives $Ch \sum s^{\alpha_1-1} \leq Ch^{\alpha_1}$. The sums run over dyadic scales; both converge because $0 < \alpha_1 < 1$. The harmonic remainder has bounded smooth derivatives on the final smaller ball. Thus every component of DY_0 , and not only $\partial_{e_0} Y_0$, satisfies (28.11).

Improvement of a nonzero affine approximation. Suppose on B_1 that

$$(28.14) \quad \|z - L\|_\infty \leq b, \quad 0 < b \leq b_0, \quad \frac{1}{4} \leq |DL| \leq 4, \quad |\text{grad } z|_g \leq 8,$$

and the source, metric oscillation from the identity and metric first derivatives are at most $b\delta_{\text{aff}}$. Put $Y = (z - L)/b$. Its coordinate equation has uniformly bounded right hand side on B_1 : the covariant Hessian of L is a Christoffel symbol times its bounded slope, and division by b leaves a bound $C\delta_{\text{aff}}$. Consequently (28.4) gives a uniform $W^{2,p_0}(B_{7/8})$ bound for Y .

Let $e_0 = DL/|DL|$ and let $a_0 = \text{Id} - (1 - \tilde{\chi})e_0 \otimes e_0$. Since $Dz = DL + bDY$, the set $E_b = \{|bDY| > \sqrt{b}\} \cap B_{3/4}$ has measure at most $Cb^{p_0/2}$. On its complement the actual gradient is bounded away from zero and the coordinate principal matrix differs from a_0 by at most $C(\sqrt{b} + b\delta_{\text{aff}})$. On E_b this difference is bounded. Hölder's inequality, with $1/2 = 1/p_0 + 1/q$, shows explicitly that

$$(28.15) \quad \|a_0 : D^2Y\|_{L^2(B_{3/4})} \leq C \left(\delta_{\text{aff}} + \sqrt{b_0} + b_0^{(p_0-2)/4} \right) =: \varepsilon_1.$$

Zeros of Dz can occur only in the exceptional set when b_0 is small. The arbitrary permitted choice of their axis is therefore covered by the same estimate.

Remove this residual by solving $a_0 : D^2y_1 = a_0 : D^2Y$ on $B_{3/4}$ with zero Dirichlet data. To justify this solve for measurable $\tilde{\chi}$, recall the integration-by-parts identity on a Euclidean ball for zero-trace y :

$$\int ((\Delta y)^2 - |D^2y|^2) = \int_{\partial B} H_{\partial B} (\partial_\nu y)^2 \geq 0.$$

It extends by density to $W^{2,2} \cap W_0^{1,2}$. Hence the Dirichlet inverse of Δ has Hessian norm at most one. On this space the map

$$y \mapsto \Delta_D^{-1}(a_0 : D^2Y + (\text{Id} - a_0) : D^2y)$$

is a contraction in the Hessian L^2 norm with factor at most $1 - \chi_0$. Poincaré and the Dirichlet Laplace estimate identify this as a complete norm and give

$$(28.16) \quad \|y_1\|_{W^{2,2}(B_{3/4})} \leq C\varepsilon_1.$$

The function $Y_0 = Y - y_1$ has uniformly bounded $W^{2,2}$ norm and satisfies the homogeneous fixed-axis equation. Therefore (28.11) applies to it.

The passage to a small supremum norm must use a separate estimate in dimension four. Choose any $0 < \eta < \min\{1, 2 - 4/p_0\}$. The W^{2,p_0} bound gives a uniform $C^{0,\eta}$ modulus for Y on smaller balls. The fixed-axis estimate and the local L^2 norm give a uniform Lipschitz bound for Y_0 . Thus y_1 has a common $C^{0,\eta}$ modulus on $B_{1/4}$. If $m = \|y_1\|_{L^\infty(B_{1/8})}$, a ball of radius $cm^{1/\eta}$ around a point with absolute value at least $m/2$ has absolute value at least $m/4$, provided m is small. Hence

$$\|y_1\|_2^2 \geq cm^{2+4/\eta}, \quad m \leq C\varepsilon_1^{\eta/(\eta+2)}.$$

If m is not small, the same reasoning on a fixed smaller ball first excludes that case when ε_1 tends to zero. This argument uses no embedding of $W^{2,2}(\mathbb{R}^4)$ into C^0 .

Choose $0 < \alpha_2 < \alpha_1$, then $0 < \lambda_0 < 1/8$ so that the Taylor remainder constant in (28.11) satisfies $C\lambda_0^{1+\alpha_1} \leq \frac{1}{2}\lambda_0^{1+\alpha_2}$. Next choose $\delta_{\text{aff}}, b_0 > 0$ so that the correction supremum above is at most $\frac{1}{2}\lambda_0^{1+\alpha_2}$. Taylor approximation of Y_0 at the center gives an affine L_1 such that

$$(28.17) \quad \|z - L_1\|_{L^\infty(B_{\lambda_0})} \leq b\lambda_0^{1+\alpha_2}, \quad |DL_1 - DL| \leq Cb.$$

Reduce b_0 further so that

$$(28.18) \quad Cb_0/(1 - \lambda_0^{\alpha_2}) < 1/4.$$

These estimates concern the almost everywhere coordinate equation on full balls and therefore apply equally to doubled patches.

The complete scaling iteration. The choices are ordered as follows: first p_0, α_1 and their constants; then $\alpha_2, \lambda_0, \delta_{\text{aff}}, b_0$; then k_*, δ_{alt} from (28.9) for that b_0 . Finally shrink the initial spatial scale so that its normalized source and geometry errors are bounded by δ , where

$$(28.19) \quad \delta \leq c \min\{\delta_{\text{alt}}, b_0 \delta_{\text{aff}} / r_*\}.$$

Here the fixed $c > 0$ incorporates all comparison constants in the geometry rescalings. This scale depends only on the theorem's parameters, since division by $M_1 + N_1$ makes the initial source at most one. Write the resulting function as u and center all following balls at zero.

After m successive gradient drops the actual rescaled solution is

$$u_m(x) = \frac{u(r_*^m x) - u(0)}{r_*^m k_*^m}, \quad |\text{grad } u_m|_{g_m} \leq 1, \quad \|G_m\|_\infty \leq \delta(r_*/k_*)^m.$$

The metric remains the identity at the center; its oscillation and first derivative errors are at most $C\delta r_*^m$, with its curvature error shrinking at least as fast. A reflecting plane has the correspondingly scaled second fundamental form. Since $r_* < k_*$, the next application of the alternative is legitimate. If the affine alternative first occurs after these m drops, rescale its ball by setting $v_0(x) = u_m(r_* x)/r_*$. Then v_0 has affine error at most b_0 , slope in $[1/2, 2]$, bounded gradient, and source at most

$$r_* \delta(r_*/k_*)^m \leq b_0 \delta_{\text{aff}}.$$

Its geometry satisfies the same tolerance by (28.19).

In all subsequent affine steps use

$$v_i(x) = \lambda_0^{-i} v_0(\lambda_0^i x), \quad b_i = b_0 \lambda_0^{i\alpha_2}.$$

The derivative $Dv_i(x) = Dv_0(\lambda_0^i x)$ retains the actual gradient bound. The source and first order geometry errors are at most $b_0 \delta_{\text{aff}} \lambda_0^i \leq b_i \delta_{\text{aff}}$, because $\alpha_2 < 1$. The affine function used at stage i is rescaled in the same way, including its constant term. Thus (28.17) applies inductively. Its changes of slope sum to less than $1/4$ by (28.18), so all slopes remain in $[1/4, 4]$. Only the auxiliary residual in the proof of an improvement is divided by b_i ; that residual is never asserted to solve the original nonlinear axial equation.

Choose

$$0 < \alpha \leq \min\{\alpha_2, \log k_*/\log r_*\}.$$

If the drops never end, the gradient on $B_{r_*^m}$ is at most $k_*^m \leq r_*^{m\alpha}$, and subtraction of the central value gives a constant affine approximation with error $C r_*^{m(1+\alpha)}$. If the drops end after m steps, the affine approximation at $R_i = r_*^{m+1} \lambda_0^i$ has error at most

$$b_0 k_*^m r_*^{m+1} \lambda_0^{i(1+\alpha_2)} \leq b_0 r_*^{-\alpha} R_i^{1+\alpha}.$$

The previous drop scales have the same type of estimate. Adjacent radii have ratio r_* or λ_0 , both fixed, so every intermediate radius has an affine approximation with error $C r^{1+\alpha}$. This conclusion holds at every center of the smaller full patches, including centers whose balls cross the plane.

For clarity, the usual comparison of affine approximations gives the gradient conclusion directly. If $L_r, L_{r/2}$ are two such approximations, then $\|L_r - L_{r/2}\|_{L^\infty(B_{r/2})} \leq C r^{1+\alpha}$, whence $|DL_r - DL_{r/2}| \leq C r^\alpha$. Their slopes converge, and the limit is Dz since z is C^1 . At points separated by h , compare the two approximations on overlapping balls of radius $4h$; the same affine norm estimate gives a slope difference Ch^α . Their limits differ from those slopes by Ch^α as well. This proves the first estimate in (28.2).

Finally take an affine approximation L on B_r with error $C r^{1+\alpha}$. Its L^2 error is at most $C r^{3+\alpha}$. The coordinate equation for $z - L$ has right hand side equal to G plus a bounded Christoffel

coefficient times Dz , whose L^2 norm is at most Cr^2 before these iterative rescalings. Equation (28.4) with $p = 2$ therefore gives

$$\|D^2z\|_{L^2(B_{r/2})} \leq C(r^{1+\alpha} + r^2).$$

The difference $\text{Hess}_g z - D^2z$ has L^2 norm at most Cr^2 . Since $\alpha < 1$, squaring and restoring $M_1 + N_1$ proves the second estimate in (28.2). $\square$

28.2. A scalar equation with a measure source. The next lemma describes the second, scalar regularity mechanism. Its hypotheses permit the surface measure produced by a prescribed conormal flux, as well as the squared Hessian supplied by Theorem 28.1.

Lemma 28.2. *On a ball in $\mathbb{R}^4$ let Z be continuous, bounded by M_Z , and locally $W^{1,2}$. Suppose*

$$(28.20) \quad \text{div}(\mathcal{A}DZ + B_1) = E,$$

where $\mathcal{A}$ is a bounded measurable symmetric matrix with ellipticity constants $0 < \lambda \leq \Lambda$, B_1 is bounded, and E is a signed measure satisfying

$$(28.21) \quad |E| \leq C_0(1 + |DZ|^2) \, dx + \mu_1, \quad \mu_1(B_h(x)) \leq C_\mu h^{2+\beta}, \quad 0 < \beta \leq 1.$$

The growth bound is required at every center and sufficiently small radius in a larger patch. Assume the weak product rule used below is valid; in particular it is valid if Z is smooth on the two closed sides of a flat interface, continuous across it, and the equation includes the normal-flux jump. Then Z has a uniform Hölder modulus on every fixed smaller ball. Its constants depend only on $M_Z, \lambda, \Lambda, C_0, C_\mu, \beta, \|B_1\|_\infty$ and the separation of the patches.

Proof. Put $\mathcal{L} = -\text{div}(\mathcal{A}D)$, and let $q_s = e^{skZ}$ for $s \in \{1, -1\}$. The product rule gives the exact distributional identity

$$(28.22) \quad \mathcal{L}q_s = \text{div}(skq_s B_1) - skq_s E - k^2 q_s (\mathcal{A}DZ \cdot DZ + B_1 \cdot DZ).$$

For verification, $\text{div}(\mathcal{A}DZ) = E - \text{div} B_1$ and $skq_s \text{div} B_1 = \text{div}(skq_s B_1) - k^2 q_s B_1 \cdot DZ$. This also checks the sign of the last drift term for both choices of s . In the piecewise smooth case, integration on the two sides gives this rule including the plane measure, since q_s is continuous and multiplies the flux jump by its common trace.

By ellipticity and Young's inequality,

$$-k^2 q_s \mathcal{A}DZ \cdot DZ - k^2 q_s B_1 \cdot DZ \leq -\frac{1}{2} k^2 \lambda q_s |DZ|^2 + \frac{k^2 q_s}{2\lambda} |B_1|^2.$$

Choose $k \geq \max\{1, 4C_0/\lambda\}$. The term $kC_0 q_s |DZ|^2$ coming from $-skq_s E$ is then absorbed. The bounded range of Z therefore gives, for both signs,

$$(28.23) \quad \mathcal{L}q_s \leq \text{div} F_s + \mu_2, \quad \|F_s\|_\infty \leq C, \quad \mu_2 = C(dx + \mu_1) \geq 0.$$

In particular the gradient-square term has been eliminated before any estimate for its small-ball integral is used.

On B_r solve with zero Dirichlet boundary value

$$\mathcal{L}h_s = \text{div} F_s + \mu_2.$$

The correction exists in $W_0^{1,2}(B_r)$ and satisfies

$$(28.24) \quad \|h_s\|_\infty \leq C(r + r^\beta) =: \delta_r.$$

Here are the details of the measure part of this assertion. Extend $\mathcal{A}$ measurably with the same ellipticity to B_{2r} . The Green upper bound for symmetric uniformly elliptic divergence operators with bounded measurable coefficients, together with domain monotonicity, gives

$$0 \leq G_{B_r}(x, y) \leq G_{B_{2r}}(x, y) \leq C|x - y|^{-2}, \quad x, y \in B_r.$$

This is the scalar Green comparison theorem of Littman–Stampacchia–Weinberger [38, Theorem 7.1 and Remark 2, p. 66]; using the larger ball keeps the two points uniformly away from its boundary. Dyadic annuli and (28.21) give

$$\begin{aligned} \sup_{x \in B_r} \int_{B_r} G_{B_r}(x, y) \, d\mu_2(y) &\leq C \sum_{j \geq 0} (2^{-j}r)^{-2} \mu_2(B_{2^{1-j}r}(x)) \\ &\leq C \sum_{j \geq 0} ((2^{-j}r)^2 + (2^{-j}r)^\beta) \leq C(r^2 + r^\beta). \end{aligned}$$

The finitely many largest annuli are absorbed in the same bound. All ball-growth assertions needed here hold on the fixed larger patch.

For completeness this potential also defines an energy solution. Mollify the restriction of μ_2 to B_r , and restrict the resulting smooth nonnegative density again to B_r . The growth bound is uniform under mollification: for radii larger than the mollifying radius use an enlarged ball, and for smaller radii use the bound for the mollified density. The variational Dirichlet solutions have the preceding uniform supremum bound and satisfy

$$\lambda \int_{B_r} |Dh_j|^2 \leq \int_{B_r} h_j \, d\mu_{2,j} \leq \|h_j\|_\infty \mu_{2,j}(B_r).$$

Weak compactness in $W_0^{1,2}$ and distributional convergence give the desired solution with measure source and the same bound. The bounded divergence source has a variational solution and, for any fixed $p > 4$, the scalar bounded-solution estimate gives

$$\|h_{F_s}\|_\infty \leq Cr^{1-4/p} \|F_s\|_{L^p(B_r)} \leq Cr.$$

This follows as well by scaling the zero-boundary divergence-source estimate in [38, Theorem 2.6, p. 50]; see also [28]. Adding these two solutions proves (28.24).

Set $M_s = \sup_{B_r} q_s$. By (28.23),

$$W_s = M_s - q_s + h_s + \delta_r \geq 0, \quad \mathcal{L}W_s \geq 0.$$

Write $M = \sup_{B_r} Z$, $m = \inf_{B_r} Z$ and $\omega = M - m$. At least half of $B_{r/2}$ satisfies either $Z \leq (M + m)/2$ or $Z \geq (M + m)/2$. In the first case select $s = 1$; in the second select $s = -1$. On that set $M_s - q_s \geq c\omega$, because the magnitudes of both exponential derivatives are bounded above and away from zero on $[-M_Z, M_Z]$. Since $h_s + \delta_r \geq 0$, weak Harnack for W_s gives

$$\inf_{B_{r/4}} (M_s - q_s) \geq c'\omega - 2\delta_r.$$

In the first case this decreases the supremum of Z ; in the second it increases the infimum. Converting back with the same exponential derivative bounds yields

$$(28.25) \quad \text{osc}_{B_{r/4}} Z \leq (1 - c'') \text{osc}_{B_r} Z + C(r + r^\beta), \quad c'' > 0.$$

Choose $\gamma > 0$ with $\gamma < \beta$ and $1 - c'' < 4^{-\gamma}$. Iterating (28.25) on $r, r/4, r/16, \dots$ gives a convergent geometric series after division by the corresponding γ power of the radius. Consequently $\text{osc}_{B_h(x)} Z \leq Ch^\gamma$ at all centers in smaller balls. This proves the lemma. $\square$

28.3. Application to the homotopy and the boundary estimates.

Proposition 28.3. *Fix the prepared data, ϵ , a sufficiently large n and any stage of the homotopy. Every smooth solution on an allowed finite truncation Ω_R with $t \geq -\epsilon$ has uniform local estimates for every finite number of derivatives of f and Z , up to both the inner and outer boundary faces. The constants on the fixed compact part and on uniform unit patches of the end are independent of R and of the homotopy parameters. They may depend arbitrarily on n . On each fixed finite truncation they therefore bound $\|Z\|_{C^{1,\alpha_0}}$ for every prescribed $0 < \alpha_0 < 1$.*

Proof. We use the uniform family of interior and boundary patches described in the statement, shrinking them by fixed factors when necessary.

The first Hölder estimates and the reflected equation. The bounded range (27.25) keeps $f, Z, t, \text{grad } f$ bounded and l, d, χ, u above positive constants depending on n . Divide the trace equation by $l/\sqrt{D}$ to obtain

$$(28.26) \quad \begin{aligned} A_\chi : \text{Hess}_g f &= G_f, & G_f &= \frac{\sqrt{D}}{l} (F_{\text{presc}} - \text{tr}_{A_\chi} K), \\ |G_f| &\leq C(n), & 0 < \chi_0(n) &\leq \chi \leq 1. \end{aligned}$$

All prescribed trace expressions have bounded values on this range. Every boundary value of f is constant on its face. Thus Theorem 28.1 gives a Hölder bound for Df and

$$(28.27) \quad \int_{B_r(x) \cap \Omega_R} |D_x^2 f|^2 \, dx \leq C(n) r^{2+2\alpha}.$$

The coordinate and covariant Hessians differ by a bounded Christoffel term times Df , so they have the same bound at these radii. The bound also holds for the odd double at a face.

In coordinates set

$$\mathcal{A}^{ij} = 3\sqrt{\det g} \, u \, A_\chi^{ij}, \quad B_1^i = \lambda\sqrt{\det g} \, u \, (A_\chi K(w, \cdot))^i,$$

using $\lambda = 0$ in the later stages. The coordinate Z equation is (28.20), with these bounded coefficients and the appropriate scalar homotopy factor in E . Its matrix is symmetric and uniformly elliptic at fixed n . Differentiating the implicit relation for t gives the exact covector identity

$$(28.28) \quad 3 \, dZ = (3 + pv) \, dt + H^f(w, \cdot).$$

Since $3 + pv \geq 3$ and all its coefficients are bounded on the range, $|D_x t| \leq C(n)(1 + |D_x Z| + |D_x^2 f|)$ in coordinates. The quadratic formula (26.6) and the bounded lower order terms in the source therefore give

$$(28.29) \quad |E| \leq C(n)(1 + |D_x Z|^2 + |D_x^2 f|^2)$$

in the interior. No derivative of the principal matrix has been used to obtain this bound.

At a face take the domain to be $x_4 > 0$ and write $Q = \mathcal{A}DZ + B_1$. For $x_4 < 0$ define, with $s = 1$ or -1 ,

$$\tilde{Z}(x) = sZ(R_4 x), \quad \tilde{\mathcal{A}}(x) = R_4 \mathcal{A}(R_4 x) R_4, \quad \tilde{B}_1(x) = sR_4 B_1(R_4 x).$$

The reflected flux is $\tilde{Q}(x) = sR_4 Q(R_4 x)$. Integration by parts on the two sides shows that its divergence has the reflected bulk source $sE(R_4 x)$ and the plane term

$$(28.30) \quad (1 + s)Q_4^+ \, d\mathcal{H}^3|_{\{x_4=0\}}.$$

Indeed the upper normal component is Q_4^+ and the lower one is $-sQ_4^+$, so their difference is $(1 + s)Q_4^+$. For an inner face use even reflection, $s = 1$. The total conormal flux is prescribed by (26.21); its coordinate density Q_4^+ is bounded on the established range. For an outer face use odd reflection, $s = -1$. The zero Dirichlet condition makes Z continuous across the face, and the plane term vanishes. Even reflection at the inner face is also continuous. Each individual reflected function belongs to $W^{1,2}$ and is smooth on the two closed sides, which verifies the weak chain rule required in Lemma 28.2.

In the full patch, (28.29) and (28.30) give a signed measure dominated by

$$C(n)(1 + |D_x Z|^2) \, dx + \mu_1, \quad \mu_1 = C(n)|D_x^2 f|^2 \, dx + 2|Q_4^+| \, d\mathcal{H}^3|_{\{x_4=0\}},$$

where the Hessian density is reflected and the plane term is omitted for an odd face or an interior patch. By (28.27), this positive measure satisfies, at every center,

$$\mu_1(B_h(x)) \leq C(n)h^{2+\beta}, \quad \beta = \min\{2\alpha, 1\} > 0.$$

For balls crossing the plane this follows either from the doubled estimate or by enlarging a ball centered on the plane by a fixed factor. Lemma 28.2 now supplies a uniform Hölder modulus for Z , including at both boundary types.

Schauder for f and the ordinary normal condition for Z . The actual coefficients in (28.26) are smooth functions of (x, Z, Df) on the bounded range. This remains true at $Df = 0$, as is manifest from

$$A_\chi = \text{Id} - \frac{3+p}{3+pv} w \otimes w.$$

The source is smooth in these variables and f , uniformly over the four parameter ranges. The Hölder bounds just proved therefore make the coefficients and source uniformly $C^{0,\theta}$ for some $\theta > 0$. The scalar Dirichlet Schauder estimate, with the constant face data, gives

$$(28.31) \quad \|f\|_{C^{2,\theta}(U')} \leq C(n)$$

on every smaller interior or boundary patch [28]. This application has Hölder principal coefficients, a Hölder right hand side, smooth boundary, and uniform ellipticity; it does not apply a system estimate.

Now expand the divergence equation for Z . Derivatives of its coefficients, which depend smoothly on (x, Z, Df) , are sums of bounded terms and bounded multiples of DZ, D^2f . The latter Hessian is bounded by (28.31). Thus

$$(28.32) \quad a_2^{ij}(x)D_{ij}Z = R_2, \quad |R_2| \leq C(n)(1 + |DZ|^2),$$

with bounded $C^{0,\theta}$ uniformly elliptic principal coefficients. The source contains no second derivative of Z .

On an inner face Df is normal and hence $A_\chi\nu = \chi\nu$. Dividing the prescribed flux by $3u\chi > 0$ gives

$$(28.33) \quad \partial_\nu Z = b_2(x, Z, f, Df).$$

The function b_2 is smooth on the bounded range, separately on each homotopy stage, with uniform bounds through their matching endpoints. Extend its displayed expression smoothly in the spatial variable into the collar. Equation (28.31) then gives

$$|D_x b_2(x, Z, f, Df)| \leq C(n)(1 + |D_x Z|).$$

Consequently the Sobolev trace theorem bounds the $W^{1-1/p,p}$ norm of this normal datum by $C(n)(1 + \|Z\|_{W^{1,p}})$ on a larger patch, for each finite $p > 1$. This derivative count costs DZ and D^2f , never D^2Z .

Linear estimates and absorption of the quadratic term. Fix $p > 4$. On a ball or smooth half-patch of size h , the local linear $W^{2,p}$ estimate for (28.32), with zero outer Dirichlet data or (28.33), has the form

$$(28.34) \quad \|D^2Z\|_{L^p(Q_h)} \leq C_* \|DZ\|_{L^{2p}(Q_{2h})}^2 + C_h(1 + \|Z\|_{W^{1,p}(Q_{2h})}).$$

Here C_* is independent of sufficiently small h ; C_h may contain inverse powers of h . The constants are uniform over the locations and the truncations in question.

We specify the linear estimate underlying this assertion. Freeze the continuous principal matrix at the patch center. The interior constant-coefficient Hessian estimate follows from the Euclidean one by a fixed linear change of coordinates. At the boundary, normal coordinates give no mixed normal–tangential entries in the frozen matrix, because its exceptional axis is normal there. A tangential linear transformation and a normal dilation reduce the frozen principal part to the

Laplacian and preserve the plane. Subtract a Sobolev extension of the normal datum, or of the Dirichlet datum, to obtain homogeneous boundary data. Even reflection for homogeneous normal data and odd reflection for homogeneous Dirichlet data give the constant-coefficient half-space estimate. The extension costs precisely the $W^{1-1/p,p}$ norm for the normal datum. Coefficient oscillation contributes $\sup |a_2 - a_2(x_0)| \|D^2 Z\|_p$ and is absorbed on small patches, uniformly by the established Hölder bound. Cutoffs and interpolation handle the first order terms. Scaling shows that the coefficient of the interior L^p source norm is the constant C_* , independent of h ; the lower order and trace terms account for C_h . This proves (28.34) by the scalar local estimates [28]. The construction requires no meeting of different boundary types, and the inner and outer faces here are disjoint.

For use in its nonlinear right hand side, the scaled scalar interpolation inequality on Q_h is

$$(28.35) \quad \|DZ\|_{L^{2p}(Q_h)}^2 \leq C \operatorname{osc}_{Q_h} Z \|D^2 Z\|_{L^p(Q_h)} + Ch^{4/p-2} (\operatorname{osc}_{Q_h} Z)^2.$$

One can obtain it on the unit patch by subtracting a constant, using a bounded Sobolev extension on a smooth ball or half-patch, and applying the Gagliardo–Nirenberg inequality with derivative orders 1, 2, exponents $2p, p, \infty$ and parameter $1/2$ [37, Theorem 1.3]. To see the interpolation mechanism directly for a compactly supported function v , integration by parts gives

$$\int |Dv|^{2p} \leq C_p \|v\|_\infty \int |Dv|^{2p-2} |D^2 v| \leq C_p \|v\|_\infty \left(\int |Dv|^{2p} \right)^{1-1/p} \|D^2 v\|_p.$$

Division gives the required product estimate. Extension and cutoff produce the additional $\|v\|_\infty^2$ term on a bounded unit patch. Finally rescaling $x = hy$ multiplies a squared L^{2p} gradient norm and the Hessian product by $h^{4/p-2}$, which proves precisely the last power in (28.35).

Here is a cover argument that absorbs larger-patch norms without assuming their uniform boundedness in advance. Use uniformly comparable patches of radius h covering the finite truncation, and their fixed-factor enlargements. Each enlargement is covered by at most N smaller patches, where N is independent of h , R and the total number of patches. Let

$$S = \sup_j \|D^2 Z\|_{L^p(Q_h^j)}.$$

For each individual smooth solution on the finite truncation, S is finite. The Hölder estimate for Z gives $\operatorname{osc}_{Q_{2h}^j} Z \leq C(n)h^\gamma$. Apply (28.35) in the enlarged patches and then cover them by smaller ones. Its contribution to (28.34) is at most $C_* C(n)h^\gamma N^{1/p} S + C(n, h)$. Choose h small enough, also satisfying the linear freezing condition, that this coefficient is at most $1/4$. For the remaining first order term, local interpolation gives, for any $\varepsilon > 0$,

$$\|DZ\|_{L^p(Q_{2h}^j)} \leq \varepsilon \|D^2 Z\|_{L^p(Q_{2h}^j)} + C(h, \varepsilon) \|Z\|_{L^p(Q_{2h}^j)}.$$

Choose ε so that $C_h \varepsilon N^{1/p} \leq 1/4$. The bounded range controls the last term. Taking the supremum in (28.34) now gives $S \leq C(n, h) + S/2$, and hence a uniform bound for S . First order interpolation supplies the corresponding local $W^{2,p}$ bounds for Z . All constants may depend on the now fixed h and on n , but not on the total number of patches or R .

Alternating Schauder estimates and uniform geometry. Since $p > 4$, Sobolev embedding gives $Z \in C^{1,\theta'}$ for some $\theta' > 0$. In conjunction with (28.31), the trace equation now has $C^{1,\theta''}$ coefficients and source for $0 < \theta'' \leq \min\{\theta, \theta'\}$. One differentiation and the scalar Dirichlet Schauder estimate give $f \in C^{3,\theta''}$. The equation for Z then has $C^{0,\theta''}$ source, and its normal datum (28.33) is $C^{1,\theta''}$. The scalar normal-derivative or Dirichlet Schauder estimate gives $Z \in C^{2,\theta''}$. The boundary estimate is justified by the same frozen half-space problem used above: its normal derivative does not annihilate a nonzero decaying solution of the homogeneous principal

equation. Smooth normal coordinates and smooth normal field therefore satisfy its obliqueness condition.

Repeating in the order f then Z increases the number of derivatives by one each time. The trace equation uses Z without derivatives; the expanded second equation uses only D^2f and DZ in its lower terms; and its boundary expression uses only Df . These are exactly the derivative counts needed for the iteration. The smooth fixed coefficients give all finite orders.

The compact inner faces have uniform smooth collars. On the prepared end, the metric and all fixed coefficient fields have uniform bounds on unit patches. Large coordinate spheres have boundary normal charts with uniform bounds of each finite order, and are separated from the inner faces. Thus every radius, ellipticity bound, chart constant and local covering number used above is independent of the sufficiently large truncation radius. Finally a uniform local C^2 bound controls any prescribed C^{1,α_0} norm, $\alpha_0 < 1$, on a fixed finite truncation. This proves the proposition. $\square$

Remark 28.4. Theorem 28.1 also applies in the scalar trial problem below. Its separate height and gradient bounds give bounded source and a positive axial eigenvalue without the floor. Interpolation of the principal matrix with Id preserves its three unit transverse eigenvalues. A trial $Z \in C^{1,\alpha_0}$ then supplies the coefficient regularity for scalar Schauder estimates through $f \in C^{3,\alpha_0}$; this requires DZ , but no second derivative of the trial input.

29. EXISTENCE AND THE ENERGY COMPARISON

We now construct solutions of the system, exhaust the asymptotically flat end, and compare the resulting Riemannian energy with the energy of the prepared data. The distinction between two kinds of constants is essential. Constants in the scalar solution map and in local regularity may depend arbitrarily on a fixed n ; constants used in the final flux estimate are the polynomial bounds Π_n established earlier.

29.1. A scalar solution map on the whole trial space. Fix $0 < \alpha < 1$, a sufficiently large integer n , and an allowed truncation Ω_R . In this subsection constants may depend on R , n , and a bound for the trial input in

$$X_R = \{Z \in C^{1,\alpha}(\overline{\Omega}_R) : Z|_{S_R} = 0\}.$$

The four stages of the homotopy are parametrized consecutively by $s \in [0, 4]$, with $s = 0$ the desired system and $s = 4$ its zero terminal scale. Denote their trace prescriptions by $F_s^{\text{presc}}(x, w, h, t)$. All dependence suppressed in this notation is the smooth dependence specified in the homotopy.

Lemma 29.1 (Scalar trial problem). *For every $Z \in X_R$ and $s \in [0, 4]$, the prescribed trace equation with the assigned constant Dirichlet value on every boundary component has a unique solution $f_s[Z] \in C^{3,\alpha}(\overline{\Omega}_R)$. The map $(s, Z) \mapsto f_s[Z]$ is continuous into $C^{3,\alpha}$ and maps bounded trial sets to bounded sets, uniformly in s . No lower bound for the implicitly defined t is imposed on the trial inputs.*

Proof. The normalized equation is

$$(29.1) \quad A_\chi : \text{Hess}_g f = G_s(x, Z, f, df), \quad G_s = \frac{\sqrt{D}}{l} (F_s^{\text{presc}} - \text{tr}_{A_\chi} K).$$

The implicit relation defining t is smoothly solvable for every (Z, df) . Neither t , the matrix, nor the positive prefactor $\sqrt{D}/l$ depends on the value of f . The prescriptions satisfy

$$|F_s^{\text{presc}} - h| \leq C, \quad \partial_h F_s^{\text{presc}} \geq 1, \quad h = \eta_n f,$$

with the other arguments held fixed. For a second parameter $\beta \in [0, 1]$, use the scalar interpolation

$$(29.2) \quad A_\beta : \text{Hess}_g f = G_\beta, \quad A_\beta = (1 - \beta) \text{Id} + \beta A_\chi, \quad G_\beta = (1 - \beta) \eta_n f + \beta G_s,$$

with the same Dirichlet values. At a positive interior maximum above a fixed height threshold, both source summands are positive, whereas the left side is nonpositive. The negative minimum argument is identical with signs reversed. Including the assigned boundary values gives $|h| \leq C$, uniformly in both parameters and the bounded trial set.

We next check the degeneracy at large gradient, before invoking any regularity theorem. Put $\sigma = |df|_g$. Uniformly for bounded Z , $\sigma \rightarrow \infty$ forces $t \rightarrow -\infty$: a lower bound on t would bound $e^{6t}l(t)^2$ away from zero, contradicting $e^{6Z} = e^{6t}(1 + l(t)^2\sigma^2)$. Thus eventually $t < 0$ and $l(t) = e^{nt}$, giving the exact equation

$$(29.3) \quad e^{6Z} = e^{6t} + e^{(2n+6)t}\sigma^2.$$

Writing $\gamma_n = 6/(n+3)$, its consequences, uniformly for Z in a fixed bounded interval, are

$$(29.4) \quad \begin{aligned} t &= -\frac{\log \sigma}{n+3} + \frac{3Z}{n+3} + o(1), \\ d &= e^{-6nZ/(n+3)}\sigma^{-\gamma_n}(1 + o(1)), & \chi &= \frac{3}{n+3}d(1 + o(1)), \\ \frac{\sqrt{D}}{l} &= \sigma(1 + o(1)). \end{aligned}$$

For example, the first line follows by taking logarithms of $e^{(2n+6)t}\sigma^2 = e^{6Z}(1-d)$; the last follows directly from $\sqrt{D}/l = (\sigma^2 + l^{-2})^{1/2}$. Since $n \geq 4$, one has $\gamma_n < 1$. Compactness of the remaining bounded-gradient range therefore gives constants $c, C > 0$ such that

$$(29.5) \quad \chi_\beta := 1 - \beta + \beta\chi \geq \frac{c}{1+\sigma}, \quad |G_\beta| \leq C(1+\sigma).$$

The estimate for the source uses the height bound just proved. Notice also the exact structural identity

$$A_\beta = \text{Id} - (1 - \chi_\beta)e \otimes e, \quad e = \text{grad } f / |\text{grad } f|, \quad 0 < \chi_\beta \leq 1.$$

In particular, all three transverse eigenvalues remain exactly one throughout the scalar interpolation.

Here is a gradient estimate using only (29.5). Extend g smoothly across the boundary to a compact enlargement. All distances below are distances of this extension, at scales below its injectivity radius. If H_0 is the uniform bound for $|f|$, set

$$\psi(r) = \frac{1}{k} \log(1 + B'r), \quad q_1(r) = \psi'(r), \quad \psi''(r) = -kq_1(r)^2.$$

First choose k large relative to the constants in (29.5) and the bounded geometry. Next choose $h_0 < 1/(4k)$ below all collar, injectivity, and distinct-boundary-component separation scales. Finally choose B' so large that

$$(29.6) \quad q_1 \geq 1 \quad (0 \leq r \leq h_0), \quad \psi(h_0) > 2H_0.$$

These requirements are compatible: $q_1(h_0)$ tends to $1/(kh_0) > 4$ and $\psi(h_0)$ tends to infinity as $B' \rightarrow \infty$.

On the inward collar of a boundary face with constant value f_0 , let r be its distance function. The functions $f_0 \pm \psi(r)$ are upper and lower barriers. To check the upper barrier at a hypothetical positive maximum of $f - f_0 - \psi(r)$, the common gradient is $q_1 \text{grad } r$. The tangential Hessian of the barrier contributes at most Cq_1 , and its axial contribution is at most $-ckq_1^2/(1+q_1)$. By enlarging k this is strictly less than $-C(1+q_1)$, the lower bound for the equation's source at the solution height. Ellipticity and the second derivative test contradict the equation. For the lower

barrier both inequalities reverse. At the outer edge of the collar the comparisons follow from (29.6). Consequently

$$(29.7) \quad |f(x) - f_0| \leq \psi(\text{dist}(x, \text{face})) \quad \text{in its } h_0 \text{ collar.}$$

For completeness, the same modulus controls interior pairs. Suppose that

$$f(x) - f(y) - \psi(\text{dist}(x, y))$$

has a positive maximum on pairs of distance at most h_0 . Neither the diagonal nor the distance- h_0 set contains such a maximum. If an endpoint lies on a face, the other endpoint lies in that face's collar; (29.7) and monotonicity of ψ exclude this as well. Thus both endpoints are interior. Let $r > 0$ be their distance and orient their unique short geodesic from y to x . The first derivative test says that their gradients are $q_1(r)$ times the corresponding parallel unit tangent vectors.

Vary both endpoints in each of three parallel transverse directions. The index-form estimate for the second variation of distance, using the parallel field along the geodesic, is bounded above by Cr . The second derivative test for the two-point maximum therefore gives

$$\text{tr}_{e^\perp} \text{Hess } f(x) - \text{tr}_{e^\perp} \text{Hess } f(y) \leq Cq_1r.$$

Separate axial variations at the two endpoints give

$$\text{Hess } f(x)(e, e) \leq \psi'', \quad \text{Hess } f(y)(e, e) \geq -\psi''.$$

Because the transverse coefficients are exactly one at both endpoints, while the axial coefficients satisfy (29.5), the difference of the two left sides of (29.2) is at most

$$Cq_1r - \frac{2ckq_1^2}{1 + q_1}.$$

Their source difference is at least $-2C(1 + q_1)$. Since $q_1 \geq 1$, these inequalities contradict each other for the already sufficiently large k . No comparison of the two axial coefficients was used. Only variations of the interior endpoints were required, so the short geodesic need not remain inside Ω_R . We have proved the two-point modulus; letting $y \rightarrow x$ gives $|df| \leq \psi'(0) = B'/k$ everywhere, including the boundary by (29.7).

The scalar interpolation is now uniformly elliptic. On the given smooth interior and constant-Dirichlet boundary patches, its equation has the exact form required by Theorem 28.1: the gradient and source are bounded, and $0 < \chi_0 \leq \chi_\beta \leq 1$. That theorem first gives a uniform Hölder bound for df , at some positive exponent. The coefficients and source in (29.2) are smooth compositions of x, Z, f, df , so the scalar Dirichlet Schauder estimate gives $C^{2, \alpha'}$ bounds for some $\alpha' > 0$. These bounds make f, df Lipschitz. Together with $Z \in C^{1, \alpha}$, this yields $C^{0, \alpha}$ coefficients and source, hence $C^{2, \alpha}$ bounds. The compositions are now $C^{1, \alpha}$ and the next Dirichlet estimate gives a $C^{3, \alpha}$ bound.

Subtract a smooth extension of the assigned boundary values and work in the affine space with model $Y_R = C_0^{3, \alpha}(\overline{\Omega}_R)$, where the zero trace is on the entire boundary. The equation defines a continuously differentiable map into $C^{1, \alpha}(\overline{\Omega}_R)$. Its derivative in f is

$$L\varphi = A_\beta^{ij} \nabla_i \nabla_j \varphi + b^i \nabla_i \varphi - c(x)\varphi, \quad c(x) = \eta_n \left[(1 - \beta) + \beta \frac{\sqrt{D}}{l} \partial_h F_s^{\text{presc}} \right] > 0.$$

Derivatives of A_β and t with respect to df contribute only first derivative terms in φ . Its coefficients are $C^{1, \alpha}$ and the ellipticity and positivity are uniform on the bounded range. The maximum principle removes the homogeneous Dirichlet kernel. Linear Dirichlet solvability and Schauder estimates therefore make $L : Y_R \rightarrow C^{1, \alpha}$ an isomorphism [28]. The implicit function theorem proves openness in β . The bounds above prove closedness: subsequences converge in lower Hölder norms, their limits solve the equation, and the uniform $C^{3, \alpha}$ estimates retain that

regularity. At $\beta = 0$ the equation is $\Delta_g f = \eta_n f$, with ordinary Dirichlet solvability. Thus the scalar equation is solvable at $\beta = 1$.

For uniqueness, at a positive maximum of the difference of two solutions their gradients agree. Their matrices and positive prefactors therefore agree, whereas strict height monotonicity makes the source difference positive, contradicting the Hessian comparison. The reverse comparison completes uniqueness. Applying the same implicit function theorem with parameters (s, Z) proves continuous dependence into $C^{3,\alpha}$. At stage junctions use one-sided parameter neighborhoods and the agreement of the prescriptions. The estimates were uniform on bounded trial sets, which proves the last assertion. $\square$

29.2. The frozen return map and degree. For a trial pair $(f_s[Z], Z)$, evaluate all expressions in the homotopy at that pair. Write $A = uA_\chi$, let $B_s = \lambda uA_\chi K(w, \cdot)$ be its drift, and write Q_s and q_s for the prescribed interior source and inner flux, including the common terminal scale. Define $T_s Z = Y$ by the linear problem

$$(29.8) \quad \begin{aligned} \operatorname{div}_g(3A \operatorname{grad} Y + B_s) &= Q_s && \text{in } \Omega_R, \\ (3A \operatorname{grad} Y + B_s) \cdot \nu &= q_s && \text{on } B, \\ Y &= 0 && \text{on } S_R. \end{aligned}$$

Thus only the occurrence of $3 \operatorname{grad} Z$ in the flux is replaced by $3 \operatorname{grad} Y$; the coefficients, source, and boundary expression are frozen. In q_s the value of F is always its trace prescription.

The matrix A is symmetric positive definite and $C^{1,\alpha}$. On bounded trial sets its ellipticity constants, $C^{1,\alpha}$ norm, and those of B_s, q_s are bounded; Q_s is bounded in $C^{0,\alpha}$. In particular the source uses dZ and $\operatorname{Hess} f$, whose regularity is already available. With

$$\mathcal{H}_R = \{Y \in H^1(\Omega_R) : Y|_{S_R} = 0\},$$

the weak problem has coercive bilinear form $3 \int_{\Omega_R} A \operatorname{grad} Y \cdot \operatorname{grad} \varphi$. Indeed Ω_R is connected and S_R is nonempty, so Poincaré's inequality holds on $\mathcal{H}_R$. The prescribed flux is a bounded linear functional by the trace inequality. Lax–Milgram gives a unique weak solution. At an inner face f is constant, so $A_\chi \nu = \chi \nu$, and its flux condition is equivalently

$$\partial_\nu Y = \frac{q_s - B_s \cdot \nu}{3u\chi}.$$

This is a $C^{1,\alpha}$ ordinary normal datum. The disjoint Dirichlet and normal faces have no corners. Linear boundary and interior estimates give $Y \in C^{2,\alpha}$ with bounds uniform on bounded trial sets; the weak energy estimate supplies the lower norm in these estimates. Coefficient dependence and uniqueness, or the same linear estimates applied to differences, give continuous dependence. Consequently

$$(29.9) \quad T : [0, 4] \times X_R \longrightarrow X_R \quad \text{is continuous and compact on bounded sets.}$$

Here compactness uses the compact embedding $C^{2,\alpha} \hookrightarrow C^{1,\alpha}$ on the fixed truncation.

Every fixed point is smooth. Initially it has $Z \in C^{2,\alpha}$ and $f \in C^{3,\alpha}$. The trace equation raises the regularity of f , and then expansion of the divergence equation, with its normal boundary datum, raises the regularity of Z . Repetition gives all orders. The smooth a priori estimates from the preceding sections therefore apply to every fixed point.

Proposition 29.2 (Existence and exhaustion). *Fix the prepared geometry and $0 < \epsilon < 1$. For every sufficiently large n and every allowed R , system (26.20) has a smooth solution on $\bar{\Omega}_R$ with $t > -\epsilon$. For each such fixed n there is a sequence $R \rightarrow \infty$ whose solutions converge smoothly on compact subsets up to B to a solution on $\bar{\Omega}$ with $t \geq -\epsilon$. This solution satisfies the prescribed inner boundary conditions, the height interval (27.11), and the weighted trace and flux estimates of Lemma 27.3.*

Proof. Choose n large enough for Proposition 27.1, the collar estimates, and the polynomial absorption conditions used in Proposition 27.6. Fix an allowed R . By (27.25) and Proposition 28.3, choose M_0 strictly larger than every $C^{1,\alpha}$ norm of an above-floor fixed point on this truncation. Define

$$(29.10) \quad \mathcal{O}_s = \left\{ Z \in X_R : \|Z\|_{C^{1,\alpha}} < M_0, \quad \min_{\Omega_R} t(Z, df_s[Z]) > -\epsilon \right\}.$$

Lemma 29.1 and smooth implicit dependence of t make their union relatively open in $[0, 4] \times X_R$.

For clarity, the varying domain causes no extra degree assumption. Let $\mathcal{K}$ consist of fixed points in the closure of this union with norm at most M_0 . A sequence in $\mathcal{K}$ has a convergent subsequence because of (29.9); continuity makes its limit a fixed point. The limit has $\min t \geq -\epsilon$. Proposition 26.7 excludes equality, and the choice of M_0 excludes contact with the norm boundary. Thus $\mathcal{K}$ is compact and lies in the open union of the sections.

Fix a parameter s_0 . Cover its compact fixed point slice by finitely many small open balls in X_R whose closed balls remain in all $\mathcal{O}_s$ for s close to s_0 . Let U be their union. All fixed points in the closed sections for sufficiently close parameters lie in U : otherwise a sequence of omitted fixed points converging to the slice would contradict compactness and its containment in the open set U . Excision identifies the degree on $\mathcal{O}_s$ with that on U , and fixed-domain homotopy invariance makes the latter constant. If the slice is empty, compactness gives no fixed points nearby and the same conclusion with degree zero. Hence $\deg(\text{Id} - T_s, \mathcal{O}_s, 0)$ is locally constant, and therefore constant, on $[0, 4]$ [24].

At $s = 4$, the scalar prescription is $\text{tr}_{A_X} H^f = h + D_1(x, w)$ with zero boundary heights. At an interior extremum $w = 0$ and $D_1 = 0$, so the maximum principle gives $f = 0$ for every trial Z , and then $t = Z$. The terminal scale sets both frozen right sides to zero and the drift is absent. Testing the frozen equation with Y gives $Y = 0$; thus T_4 is the zero map, even though its positive coefficients may depend on the input. Its sole fixed point, $Z = 0$, belongs to $\mathcal{O}_4$ and has degree one. The degree at $s = 0$ is consequently one, proving existence there.

Now keep n fixed. The preceding choices of sufficiently large n used only the already proved uniform height and collar results and polynomial losses such as $\Pi_n \eta_n / \ell \rightarrow 0$. The arbitrary fixed- n constants in the scalar construction impose no subsequent condition on n . The bounds in Proposition 28.3 are uniform in allowed R on each compact set, including its portions on B . A diagonal Arzelà–Ascoli argument along $R \rightarrow \infty$ gives smooth convergence on all these sets. The equation and inner data pass to the limit, as do $t \geq -\epsilon$, the height interval, and the collar estimates. Smooth convergence on the fixed collars passes the weighted trace and flux inequalities as well. Strictness of the floor in this exhaustion limit is unnecessary below. $\square$

29.3. Decay on the end.

Lemma 29.3 (End estimates). *For every solution obtained in Proposition 29.2, with n fixed, there is $a_n > 0$ such that, for every integer $j \geq 0$,*

$$(29.11) \quad |\nabla^j f| + |\nabla^j(t - Z)| \leq C_j(n) e^{-a_n r} \quad \text{on the distant end.}$$

Moreover

$$(29.12) \quad t, Z = O_2(r^{-2}), \quad \Delta_g t = O(r^{-4-\delta_1}).$$

The constants in these assertions may depend arbitrarily on n . The zeroth-order estimates and (29.11) hold uniformly for the finite-truncation solutions up to their outer faces.

Proof. Choose a fixed sphere beyond the supports of K, C , and all collar terms. Along each finite-truncation solution the trace equation is the homogeneous linear equation

$$(29.13) \quad A_\chi : \text{Hess}_g f - c(x)f = 0, \quad c(x) = \eta_n \frac{\sqrt{D}}{l} \geq c_n > 0.$$

By (27.25) the coefficients are uniformly elliptic; by Proposition 28.3 all their derivatives are bounded on uniform unit patches, independently of R . On the end,

$$(A_\chi : \text{Hess}_g - c)e^{-ar} = e^{-ar}(a^2 A_\chi(\text{dr}, \text{dr}) - a A_\chi : \text{Hess}_g r - c).$$

Choose $a = a_n > 0$ so small that the first term is less than $c_n/4$, and then enlarge the fixed sphere so the second term has magnitude less than $c_n/4$. Thus $e^{-a_n r}$ is a strict supersolution. A fixed multiple dominates both signs of f on the initial sphere and dominates the zero Dirichlet data on S_R . Linear comparison gives $|f| \leq C(n)e^{-a_n r}$ uniformly in R . Local estimates for (29.13), including the zero-Dirichlet estimates on the uniform outer-sphere charts, give the same exponential decay for every derivative, after harmless reduction of a_n if necessary. The identity

$$Z - t = \frac{1}{6} \log(1 + l(t)^2 |df|^2)$$

and the uniform smooth local bounds then give (29.11). Every graph contribution to the metric and to the equations is exponentially small with unit-patch derivatives.

At zero graph and $K = 0$, the exact formulas give

$$u = e^{2Z} l(Z), \quad V = 3u \text{grad } Z, \quad \mathcal{T} = 3(p(Z) + 1) |dZ|^2.$$

Hence the actual divergence equation, divided by u , becomes

$$(29.14) \quad 3\Delta_g Z + 3\mathbf{b}_n(Z) |dZ|_g^2 = \rho - m_0(Z)C_n\rho_0 + \mathcal{E}_n, \quad \mathbf{b}_n(z) = p(z) + 2 - \delta_0(p(z) + 1),$$

where $\mathcal{E}_n$ and its derivatives on unit patches are $O_j(e^{-a_n r})$. Define the strictly increasing function

$$(29.15) \quad \Phi(0) = 0, \quad \Phi'(z) = \exp\left(\int_0^z \mathbf{b}_n(s) \, ds\right).$$

Its derivative is bounded above and away from zero on the fixed bounded range of Z . The chain rule cancels the quadratic term:

$$(29.16) \quad \Delta_g \Phi(Z) = \frac{\Phi'(Z)}{3}(\rho - m_0(Z)C_n\rho_0 + \mathcal{E}_n), \quad |\Delta_g \Phi(Z)| \leq C(n)r^{-4-\delta_1}.$$

Put $\beta_1 = \delta_1/2$ and $W(r) = r^{-2}(1 - r^{-\beta_1})$. In dimension four,

$$\Delta_\delta W = -\beta_1(2 + \beta_1)r^{-4-\beta_1}, \quad \Delta_g W = -\beta_1(2 + \beta_1)r^{-4-\beta_1} + O(r^{-6}).$$

Since $0 < \beta_1 < \delta_1 < 1$, a sufficiently large multiple of W is positive and has negative Laplacian dominating the absolute source in (29.16) on a sufficiently distant end. Enlarge the multiple to dominate $|\Phi(Z)|$ on its initial sphere. Its value is positive on S_R , whereas $\Phi(Z) = 0$ there. Applying the maximum principle to both $\Phi(Z) - CW$ and $-\Phi(Z) - CW$ gives

$$|\Phi(Z)| \leq C(n)r^{-2}, \quad |Z| \leq C(n)r^{-2},$$

uniformly on truncations and then in the exhaustion limit.

Here are the differentiated estimates, which do not follow from the pointwise barrier alone. On an annulus of radius r_1 , rescale by $x = r_1 y$ and set $U(y) = r_1^2 \Phi(Z(r_1 y))$. Its supremum and its Poisson source in the rescaled equation are bounded: the source is r_1^4 times the right side of (29.16), of size $O_n(r_1^{-\delta_1})$. The rescaled background coefficients have uniform smooth bounds. Interior $W^{2,p}$ estimates, for a fixed $p > 4$, give bounded $C^{1,\theta}$ norms of U on a smaller annulus. The weights have their differentiated power decay, the compositions with Z now have bounded scaled Hölder norms, and the exponential errors remain bounded with all scaled derivatives. The right side therefore has a bounded $C^{0,\theta}$ norm. Schauder estimates give a bounded $C^{2,\theta}$ norm of

U . The inverse of Φ has bounded smooth derivatives on the relevant range. Rescaling back gives $Z = O_2(r^{-2})$, and (29.11) gives the same statement for t . Finally $|\mathrm{d}Z|^2 = O(r^{-6})$ in (29.14); since $\delta_1 < 1$, this is lower order than $r^{-4-\delta_1}$. Equation (29.11) now gives the asserted bound for $\Delta_g t$. $\square$

29.4. Scalar curvature, boundary sign, and finite flux. For the desired system $F = h + C$ and $w(h) = \eta_n a \sigma$. Substitute these into (26.5) and use $\operatorname{div} V = u\Xi$ to obtain

$$(29.17) \quad \begin{aligned} \frac{1}{2}e^{2t} \operatorname{Scal}_{\widehat{g}} &= \mu + J(w) + w(C) - \rho - (h + C)\tau \\ &\quad + (1 - \delta_0)(\mathcal{T} + \eta_n a \sigma) + m_0 \mathcal{P} \geq 0. \end{aligned}$$

Indeed $|w| \leq 1$, and on $\operatorname{supp} K$ the height interval (27.11) permits use of (25.16):

$$\mu + J(w) + w(C) - \rho - (h + C)\tau \geq \mu - |J| - |\mathrm{d}C| - \rho - |(h + C)\tau| \geq 0.$$

Outside $\operatorname{supp} K$ the same slack inequality needs no height restriction. The other terms are nonnegative by (26.14) and the definitions of the penalty and δ_0 . Thus the nonmaximal trace term has been retained and controlled at this final step.

The inner flux condition and (26.18) give exactly

$$(29.18) \quad H + 3\partial_\nu t = -N_B, \quad \widehat{H} = -e^{-t}\sqrt{d}N_B < 0 \quad \text{on every component of } B.$$

All quantities here are finite and $d > 0$ for each fixed n . Furthermore (29.11) and (29.12) imply

$$(29.19) \quad \widehat{g} - \delta = O_2(r^{-2}), \quad \operatorname{Scal}_{\widehat{g}} = O(r^{-4-\delta_1}).$$

For the scalar estimate one may use the conformal scalar formula with $\bar{g} = g + O_j(e^{-a_n r})$: $e^{2t} \operatorname{Scal}_{\widehat{g}} = \operatorname{Scal}_{\bar{g}} - 6|\mathrm{d}t|_{\bar{g}}^2$. The prepared scalar decay and (29.12) give precisely (29.19). The nonnegative scalar curvature is therefore integrable. The inequality

$$(29.20) \quad \widehat{g} = e^{2t}(g + l^2 \mathrm{d}f \otimes \mathrm{d}f) \geq e^{-2\epsilon} g$$

also proves completeness with B included: a $\widehat{g}$ -Cauchy sequence is g -Cauchy, its limit belongs to the closed complete exterior, and smooth local metric equivalence gives convergence in $\widehat{g}$.

Proposition 29.4 (Energy comparison). *For each fixed $\epsilon > 0$ and all sufficiently large n , the limiting metric has finite ADM energy and*

$$(29.21) \quad E_{\widehat{g}} \leq E_g + \frac{\Pi_n}{3\omega_3} \left(\ell + \frac{\ell^2}{\eta_n} \right).$$

The polynomial Π_n depends only on the prepared data, ϵ , and the fixed geometric choices. In particular the error tends to zero as $n \rightarrow \infty$ with ϵ fixed.

Proof. We first establish the exact sign and normalization at fixed n . Lemma 29.3 gives $u = 1 + O_n(r^{-2})$ and

$$(29.22) \quad V = 3 \operatorname{grad}_g t + O_n(r^{-5}) + O_n(e^{-a_n r}).$$

Also $u\Xi \in L^1(\Omega, \mathrm{d}V_g)$: on the end the quadratic term is $O_n(r^{-6})$, the graph and height-gradient terms are exponentially small, and the remaining weights are $O_n(r^{-4-\delta_1})$. The divergence theorem on a large truncation consequently gives a finite limit

$$(29.23) \quad \mathfrak{F} := \lim_{r \rightarrow \infty} \int_{S_r} V \cdot \nu_g \, \mathrm{d}A_g = \int_{\Omega} u\Xi \, \mathrm{d}V_g + \int_B V_\nu \, \mathrm{d}A_g.$$

The plus sign before the inner flux arises because the domain-outward normal on B is $-\nu$.

To compute the metric flux, write $\widehat{g} = e^{2t}g$ up to exponential errors. The leading perturbation relative to g is $2t\delta_{ij}$, so in four dimensions the change of its energy numerator is

$$\partial_j(2t\delta_{ij}) - \partial_i(2t\delta_{jj}) = -6\partial_i t.$$

All omitted derivatives are $O_n(r^{-5})$ or exponentially small. Their sphere integrals tend to zero. Since $dt = O_n(r^{-3})$ and $g - \delta = O_2(r^{-2})$, changing the normal and measure in its flux from Euclidean to background ones also has vanishing error. Equation (29.22) thus proves both existence of the new energy and the exact relation

$$(29.24) \quad E_{\hat{g}} - E_g = -\frac{1}{\omega_3} \lim_{r \rightarrow \infty} \int_{S_r} \partial_{n_\delta} t \, dA_\delta = -\frac{\mathfrak{F}}{3\omega_3}.$$

We now use polynomial constants only. Apply Lemma 27.3 with $\varphi = 1$ and use (27.14). On the fixed compact collars $\mathcal{C}$ one has $\min_{\mathcal{C}} \rho > 0$, and therefore

$$\begin{aligned} \int_B |V_\nu| \, dA_g &\leq \Pi_n \frac{\eta_n}{\ell} \int_{\mathcal{C}} u(1 + \sqrt{\mathcal{T}}) \, dV_g \\ &\leq \Pi_n \frac{\eta_n}{\ell} \int_{\mathcal{C}} u(\rho + \delta_0 \mathcal{T}) \, dV_g. \end{aligned}$$

The polynomial absorbs the fixed comparison constant in the second line. Because $\eta_n/\ell = e^{-\epsilon n/2}$, for sufficiently large n this is at most one half of $\int_{\Omega} u(\rho + \delta_0 \mathcal{T})$.

It remains to estimate the penalty. Its support in an above-floor solution lies in $-\epsilon \leq t \leq -\epsilon + 1/n$. For $n > 1/\epsilon$ this interval is negative, and uniformly there

$$(29.25) \quad l \asymp \ell, \quad L_0 \asymp \ell, \quad u \leq C(\ell + \ell^2 \sigma), \quad uv \leq C\ell^2 \sigma, \quad u a \sigma = L_0 l \sigma^2 \geq c\ell^2 \sigma^2.$$

For example $uv = L_0 l^2 \sigma^2 / \sqrt{1 + l^2 \sigma^2} \leq L_0 l \sigma$. Thus on that band

$$um_0 \mathcal{P} \leq \Pi_n [\ell \rho_0 + \ell^2 \sigma (\rho_0 + \rho_1)].$$

Young's inequality applied to $\sqrt{\eta_n} \ell \sigma$ and $\Pi_n \ell (\rho_0 + \rho_1) / \sqrt{\eta_n}$, together with (29.25), yields everywhere

$$(29.26) \quad um_0 \mathcal{P} \leq \frac{1}{2} \delta_0 u \eta_n a \sigma + \Pi_n \left[\ell \rho_0 + \frac{\ell^2}{\eta_n} (\rho_0^2 + \rho_1^2) \right].$$

The enlarged Π_n is still polynomial. The three residual weights are integrable against dV_g in dimension four: $\rho_0 = r^{-4-\delta_1}$, $\rho_0^2 = r^{-8-2\delta_1}$, and $\rho_1^2 = r^{-6}$ on the end, and they are bounded on the compact part. Combining (29.23), the boundary absorption, and (29.26) gives

$$\mathfrak{F} \geq -\Pi_n \left(\ell + \frac{\ell^2}{\eta_n} \right).$$

This and (29.24) prove (29.21). Finally $\ell = e^{-\epsilon n}$ and $\ell^2/\eta_n = e^{-\epsilon n/2}$; both dominate every polynomial loss. No constant from the fixed- n decay or Schauder estimates appears in this bound. $\square$

29.5. A minimal enclosing hypersurface. The boundary B has strictly negative mean curvature in $\hat{g}$. The full-perimeter enclosure of Proposition 23.1 therefore gives the smooth minimal boundary needed for the Riemannian comparison. We relate its total three-volume to the original enclosing quantity A_* .

Lemma 29.5 (Minimal enclosure). *Let Ω and B be the exterior and its boundary from Proposition 25.6, and let $\epsilon > 0$. Suppose $\hat{g}$ is a smooth metric on Ω , complete with B included, such that $\hat{g} - \delta = O_2(r^{-2})$ on its end, $\hat{H}_B < 0$ for the normal into Ω , and $\hat{g} \geq e^{-2\epsilon} g$. Then there is a nonempty compact smooth embedded minimal hypersurface Σ enclosing B . Its exterior is connected, complete with boundary, and has the same single asymptotically flat end. The hypersurface is outer area-minimizing, with disconnected competitors permitted, and its area $\hat{A} = |\Sigma|_{\hat{g}}$ satisfies*

$$(29.27) \quad \hat{A} \geq e^{-3\epsilon} A_*.$$

Proof. Proposition 25.6 gives a smooth connected orientable four-dimensional exterior Ω with nonempty compact smooth boundary B and exactly one asymptotically flat end. By hypothesis, $\hat{g}$ is smooth, complete with B included, and satisfies $\hat{g} - \delta = O_2(r^{-2})$. Thus $(X, g) = (\Omega, \hat{g})$ satisfies the geometric hypotheses of Proposition 23.1, with asymptotic exponent $q = 2 > 1$.

Apply that Proposition to obtain a full-perimeter minimizing enclosure of B , and denote its frontier by Σ . Since $\hat{H}_B < 0$ for the normal into Ω , the strict inner-barrier conclusion places Σ in $\text{int } \Omega$ and makes it a nonempty compact smooth embedded minimal hypersurface. Its closed exterior is smooth, connected, complete with Σ included, and has the same single end. The Proposition also gives outer area-minimization there, with all boundary components counted and disconnected enclosing competitors permitted.

For every smooth enclosing cut Γ of B , its closed connected outer domain is also an admissible outer domain in the original prepared manifold, by Proposition 25.6. This includes frontier coincident with B . Restricting $\hat{g} \geq e^{-2\epsilon}g$ to its three-dimensional tangent planes gives

$$(29.28) \quad |\Gamma|_{\hat{g}} \geq e^{-3\epsilon}|\Gamma|_g \geq e^{-3\epsilon}A_*.$$

Apply this to Σ to obtain (29.27). $\square$

29.6. The Riemannian comparison. We use the following dimension-four instance of the boundary Riemannian Penrose inequality [18, Theorem 1.4, p. 84]. Let (N, g_N) be a smooth connected four-dimensional Riemannian manifold, complete with its nonempty compact boundary included, having exactly one end, which is asymptotically flat. Suppose in that end

$$(g_N)_{ij} - \delta_{ij} = O_2(r^{-p_0}), \quad p_0 > 1, \quad R_{g_N} = O(r^{-s_0}), \quad s_0 > 4,$$

and $R_{g_N} \geq 0$ everywhere. If its boundary is minimal and outer area-minimizing, then

$$(29.29) \quad E_{g_N} \geq \frac{1}{2} \left(\frac{|\partial N|_{g_N}}{\omega_3} \right)^{2/3}.$$

The boundary may have one or more components, and the ADM normalization is $1/(6\omega_3)$. The numerical inequality does not require a spin assumption; the additional spin hypothesis in the published theorem concerns its equality conclusion. We use only (29.29).

29.7. The prepared-data energy theorem.

Theorem 29.6 (Energy inequality for prepared data). *Let (M^4, g, K) be smooth, connected, and orientable, complete with a nonempty compact smooth boundary S included, and with exactly one end, which is asymptotically flat. Let K be a smooth compactly supported symmetric covariant tensor. Suppose, with a smooth positive function r equal to the coordinate radius on the end, constants $c_0 > 0$ and $0 < \delta_1 < 1$, that*

$$\mu - |J|_g \geq c_0 r^{-4-\delta_1} \quad \text{on } M, \quad H + \text{tr}_S K < 0 \quad \text{on every component of } S,$$

where the boundary normal points into M . Suppose on the end that

$$g - \delta = O_k(r^{-2}) \quad \text{for every integer } k \geq 0, \quad \text{Scal}_g = O(r^{-4-\delta_1}),$$

and that the ADM energy E_g is finite in the stated normalization. Let A_* be the infimum of the total g -three-volume of the entire intrinsic boundaries of connected closed smooth outer domains containing the distant end, with interiors disjoint from S , including disconnected cuts and frontier coincident with S . Then

$$(29.30) \quad E_g \geq \frac{1}{2} \left(\frac{A_*}{\omega_3} \right)^{2/3}.$$

Proof. Proposition 25.6 supplies the closed exterior Ω with its nonempty compact smooth boundary B and the data used in this section. It is connected, orientable, and one-ended, and its background energy is still E_g . Fix $0 < \epsilon < 1$. For every sufficiently large n , Proposition 29.2 constructs a smooth metric $\hat{g}$ on this exterior. By (29.17) and (29.19), it has nonnegative scalar curvature and the pointwise end bounds

$$\hat{g} - \delta = O_2(r^{-2}), \quad R_{\hat{g}} = O(r^{-4-\delta_1}), \quad 0 < \delta_1 < 1.$$

It is complete by (29.20), and every component of B has strictly negative mean curvature by (29.18).

Lemma 29.5 supplies a nonempty compact smooth minimal boundary Σ , with connected one-ended exterior and outer area-minimization against disconnected competitors. Restricting the metric to this exterior preserves scalar curvature, asymptotic decay and ADM energy. Thus the metric and scalar exponents in (29.29) are $p_0 = 2 > 1$ and $s_0 = 4 + \delta_1 > 4$, and all hypotheses of that theorem hold. Proposition 29.4 gives its finite energy. The enclosure is an admissible smooth cut, so (29.27) compares its three-volume with A_* .

Combining the Riemannian inequality with (29.21) gives

$$E_g + \frac{\Pi_n}{3\omega_3} \left(\ell + \frac{\ell^2}{\eta_n} \right) \geq E_{\hat{g}} \geq \frac{1}{2} \left(\frac{|\Sigma|_{\hat{g}}}{\omega_3} \right)^{2/3} \geq \frac{1}{2} e^{-2\epsilon} \left(\frac{A_*}{\omega_3} \right)^{2/3}.$$

For each fixed ϵ let $n \rightarrow \infty$, after the fixed- n exhaustion already carried out. Then let $\epsilon \downarrow 0$. This proves (29.30). $\square$

Theorem 29.6 supplies the energy estimate used in Proposition 24.8. Removing strictification and then the rest-end replacement, applying the positive-shell argument, and truncating the unwanted ends complete Theorem 22.2.

29.8. The general maximal DEC energy inequality. The maximal special case requires no assumption that a momentum limit exists. The following end calculation supplies that limit from the decay and integrability already imposed on the data.

Lemma 29.7 (Existence of the momentum flux). *On a smooth four-dimensional asymptotically flat end, suppose $g - \delta = O_2(r^{-q})$ and $K = O_1(r^{-1-q})$ for some $q > 1$. If $|J|_g$ is integrable, where $J_i = \nabla^j(K_{ij} - (\text{tr}_g K)g_{ij})$, then each ADM momentum limit in (22.2) exists and is finite.*

Proof. In the given coordinates set

$$\pi_{ij}^g = K_{ij} - (\text{tr}_g K)g_{ij}, \quad \pi_{ij}^\delta = K_{ij} - (\text{tr}_\delta K)\delta_{ij}.$$

The differentiated decay assumptions give

$$(29.31) \quad \pi^g - \pi^\delta = O_1(r^{-1-2q}), \quad \partial_j \pi_{ij}^\delta - J_i = O(r^{-2-2q}).$$

To verify the second bound, expand the covariant divergence as

$$J_i = g^{jk}(\partial_k \pi_{ij}^g - \Gamma_{ki}^a \pi_{aj}^g - \Gamma_{kj}^a \pi_{ia}^g).$$

Replacing g^{jk} by δ^{jk} costs $O(r^{-q})O(r^{-2-q})$; replacing π^g by π^δ costs the derivative of the first bound in (29.31). Each connection term has size $O(r^{-1-q})O(r^{-1-q})$. These are exactly the asserted errors.

Their Euclidean radial volume bound is Cr^{1-2q} , integrable because $q > 1$. The asymptotic metric and Euclidean volume and covector norms are uniformly comparable, so $\partial_j \pi_{ij}^\delta \in L^1$. The Euclidean divergence theorem between two large spheres makes the flux $\int_{S_R} \pi_{ij}^\delta n_\delta^j dA_\delta$ a Cauchy function of R , with finite limit. By the first bound in (29.31), its difference from the flux of π^g is $O(R^{2-2q}) \rightarrow 0$. This is the tensor appearing in (22.2), which proves the lemma. $\square$

Corollary 29.8 (Maximal DEC energy inequality). *Let (X^4, g_X, K_X) be smooth and connected, with X orientable, g_X complete as a metric space with its nonempty compact smooth boundary S_X included, and with exactly one end, which is asymptotically flat and diffeomorphic to the complement of a closed ball in $\mathbb{R}^4$. Let K_X be a smooth symmetric covariant two-tensor. Suppose*

$$\begin{aligned} \operatorname{tr}_{g_X} K_X &= 0, & \mu_X &= \frac{1}{2}(R_{g_X} - |K_X|_{g_X}^2) \geq |\operatorname{div}_{g_X} K_X|_{g_X}, \\ \mu_X, |\operatorname{div}_{g_X} K_X|_{g_X} &\in L^1(X, dV_{g_X}). \end{aligned}$$

Assume for some $q > 1$ that

$$g_X - \delta = O_2(r^{-q}), \quad K_X = O_1(r^{-1-q}),$$

and that the ADM energy E_X in (22.1) is finite. On every component of S_X , require $H + \operatorname{tr}_{S_X} K_X \leq 0$ with the normal into X . For the one-ended enclosing-cut infimum of Definition 22.1, denoted here by $a_{g_X}(S_X)$, one has

$$E_X \geq \frac{1}{2} \left(\frac{a_{g_X}(S_X)}{2\pi^2} \right)^{2/3}.$$

The boundary may have any topology and finitely many components. No symmetry, vacuum, spin, outermostness, outer area-minimization or momentum-flux assumption is imposed.

Proof. Maximality identifies the momentum density with $J_X = \operatorname{div}_{g_X} K_X$. Lemma 29.7 supplies the finite ADM momentum. All hypotheses of Theorem 22.2 therefore hold for this one-ended exterior. Its invariant-mass inequality in particular gives the displayed energy bound. The theorem has already been proved using the Riemannian inequality (29.29), so this corollary is a consequence of the proof, not an input to it. $\square$

30. VARIATION AT EQUALITY AND LOCALIZATION OF THE AREA TERM

Throughout this section, (Ω, g, K) satisfies the hypotheses of Definition 22.3, and equality holds in Theorem 22.2. Thus Ω has one end and its only boundary S is a connected, outermost, outer area-minimizing future MOTS. We use the full enclosing-cut convention of Definition 22.1. Set

$$(30.1) \quad \begin{aligned} A &= \operatorname{Area}_g(S), & m &= \sqrt{E^2 - |P|^2}, & r_0 &= (A/\omega)^{1/3} = \sqrt{2m}, \\ b^0 &= E/m, & b^i &= -P_i/m, & \mathbf{m}(a) &= \frac{1}{2}(a/\omega)^{2/3}, & c &= 6\omega \mathbf{m}'(A) = 2/r_0. \end{aligned}$$

The strict future timelikeness needed here is already part of Theorem 22.2. For nearby data with charges $(\tilde{E}, \tilde{P})$, define the fixed linear charge functional

$$\mathcal{E}(\tilde{g}, \tilde{K}) = b^0 \tilde{E} + \sum_{i=1}^4 b^i \tilde{P}_i.$$

The reverse Cauchy–Schwarz inequality on the future timelike cone gives

$$(30.2) \quad \mathcal{E}(\tilde{g}, \tilde{K}) \geq \sqrt{\tilde{E}^2 - |\tilde{P}|^2}.$$

Equality holds at the original data. In particular, whenever the varied data satisfy the numerical theorem, their enclosing infimum $a(\tilde{g})$ satisfies

$$(30.3) \quad \mathcal{E}(\tilde{g}, \tilde{K}) - \mathbf{m}(a(\tilde{g})) \geq 0.$$

The varied data need not preserve outermostness or outer area minimization: those assumptions are absent from the numerical theorem.

The right-derivative formula in Proposition 23.2 accounts for all minimizing frontiers without choosing a differentiable family. It gives the positive-measure identity of Proposition 30.6. Compact normal tests and outermostness then localize its area term to S in Proposition 30.8, supplying the identity used to construct the causal adjoint field in Theorem 31.1.

30.1. Constraint derivatives and a strict direction. On the closed unit ball bundle of g , introduce

$$(30.4) \quad \begin{aligned} C(x, v) &= 2(\mu(x) + J_x(v)) = R_g + \tau^2 - |K|_g^2 + 2J_x(v), \quad |v|_g \leq 1, \\ \mathcal{A} &= \{(x, v) : C(x, v) = 0\}. \end{aligned}$$

DEC is equivalent to $C \geq 0$ on this bundle. When the metric varies, we identify its unit ball with the original one by the positive symmetric inverse square root. More precisely, if $g_s(V, W) = g(G_s V, W)$, then $I_s = G_s^{-1/2}$ sends the g unit ball isometrically to the g_s unit ball. For a variation $H = (h, p) = (\dot{g}_0, \dot{K}_0)$, all derivatives C'_H below include this identification; in particular,

$$\dot{I}_0 v = -\frac{1}{2} h^\# v.$$

Lemma 30.1 (Conformal constraint identities). *For a positive smooth function f , let $g_f = f^2 g$ and $K_f = fK$. Using $f^{-1}v$ in the varied unit ball, one has*

$$(30.5) \quad f^2 C_f(x, f^{-1}v) = C(x, v) + 6f^{-1}[-\Delta_g f + K(\nabla f, v)],$$

$$(30.6) \quad f\theta_{+,f} = \theta_+ + 3\partial_\nu \log f.$$

Here ν points into the exterior and $\Delta_g = \operatorname{div}_g \nabla$.

Proof. Apply Lemma 24.2. Its density and momentum identities give

$$f^2 \mu_f = \mu - 3f^{-1} \Delta_g f, \quad f^2 J_f(f^{-1}v) = J(v) + 3f^{-1} K(\nabla f, v).$$

Adding twice these expressions proves (30.5). Equation (30.6) is the expansion formula in that same lemma, with the stated orientation. $\square$

Lemma 30.2 (Strict conformal direction). *Choose*

$$1 < q_0 < \min(q, 2), \quad 0 < \delta < \min(q_0, 1), \quad w = r^{-4-\delta},$$

where the end radius is extended to a positive smooth function on Ω . There is a smooth nonnegative function ϕ , smooth up to S , such that

$$(30.7) \quad \partial_\nu \phi = -1 \quad \text{on } S, \quad -\Delta_g \phi - |K|_g |\mathrm{d}\phi|_g \geq w, \quad \phi = O_2(r^{-2}), \quad -\Delta_g \phi / w \rightarrow 1.$$

Its limiting Euclidean sphere gradient flux exists and is finite. For

$$Q = (2\phi g, \phi K)$$

one consequently has

$$(30.8) \quad C'_Q \geq 6w \quad \text{on } \mathcal{A}, \quad -\theta'_Q = 3 \quad \text{on } S.$$

Moreover, uniformly for all $|v|_g \leq 1$ on the end, $C'_Q/w \rightarrow 6$.

Proof. Choose a smooth majorant $d_0 \geq |K|_g$ which on the end is a sufficiently large radial multiple of r^{-1-q_0} . It can be chosen with all symbol bounds there. We solve

$$(30.9) \quad -\Delta_g \phi = d_0 \sqrt{|\mathrm{d}\phi|_g^2 + r^{-6}} + w, \quad \partial_\nu \phi = -1 \quad \text{on } S, \quad \phi \rightarrow 0 \quad \text{at infinity}.$$

First truncate at a large coordinate sphere S_R , imposing $\phi = 0$ on S_R , and replace d_0 in the equation by td_0 , $0 \leq t \leq 1$. The inner and outer boundary components are disjoint smooth hypersurfaces; there is no corner where the two boundary conditions meet. The $t = 0$ problem is the invertible mixed Laplace problem. Every linearization in ϕ adds to $-\Delta_g$ a smooth drift of norm at most d_0 , with homogeneous Neumann data at S and homogeneous Dirichlet data at S_R . The maximum and boundary point principles give uniqueness, and the mixed elliptic Fredholm theory then gives invertibility. These are the usual local elliptic and boundary estimates on a fixed smooth truncated domain; see [28].

Solutions are nonnegative. Indeed a negative minimum cannot occur in the interior because the right side of the equation is positive. At S , the outward normal of the truncated domain is $-\nu$, so the prescribed outward derivative is $+1$, also excluding such a minimum by the boundary point principle. The outer Dirichlet value is zero.

For completeness, the continuation bounds do not require a monotonicity assumption in the unknown. Fix $p > 4$. The mixed $W^{2,p}$ estimate, the linear growth of the right side in $d\phi$, and first-derivative interpolation give, on this fixed truncation,

$$\|\phi\|_{W^{2,p}} \leq C(1 + \|\phi\|_{L^\infty}),$$

uniformly in t . If the supremum were unbounded, division by that supremum and compactness in C^1 would produce a nonnegative function z with supremum one satisfying

$$-\Delta_g z = t_* d_0 |dz|_g, \quad \partial_\nu z = 0 \text{ on } S, \quad z = 0 \text{ on } S_R$$

for some $t_* \in [0, 1]$. At points where $dz \neq 0$ the right side is a bounded drift applied to dz , and it is zero otherwise. The strong maximum and boundary point principles for this bounded-drift equation contradict the positive maximum. The resulting $W^{2,p}$ and Schauder bounds close the continuation argument and give a smooth solution at $t = 1$ on every sufficiently large truncation.

We next make the bounds independent of R . For $W = r^{-2} - r^{-2-\delta}$, the Euclidean identity

$$-\Delta_\delta W = (2 + \delta)\delta r^{-4-\delta}$$

and the AF estimates imply, beyond a fixed radius R_1 ,

$$(30.10) \quad -\Delta_g W - d_0 |dW|_g \geq c_1 w > 0.$$

The metric errors and the drift term are smaller than w because $\delta < q_0 < q$. Also $d_0 r^{-3} = o(w)$. If M_R is the supremum of the truncated solution on the fixed core bounded by S_{R_1} , a sufficiently large fixed multiple of $(1 + M_R)W$ is a supersolution on $R_1 \leq r \leq R$. To see this, use $\sqrt{a^2 + b^2} \leq a + b$ and (30.10); choose the multiple also to dominate M_R on S_{R_1} . It dominates the zero outer value. Comparison is valid by writing the difference of the two gradient nonlinearities as a bounded drift. Therefore

$$(30.11) \quad 0 \leq \phi_R \leq C(1 + M_R)W \quad (R_1 \leq r \leq R).$$

If M_R diverged along a sequence, normalize by M_R . Local mixed estimates near S and interior estimates elsewhere give a subsequential limit z on every fixed compact set. More precisely, (30.11) bounds ϕ_R/M_R on a neighborhood of the fixed closed core. Local Neumann $W^{2,p}$ estimates at S , and interior $W^{2,p}$ estimates away from it, give uniform bounds there for $p > 4$. Compactness in C^1 on the closed core retains both $\sup_{\text{core}} z = 1$ and $\partial_\nu z = 0$. The limit satisfies $-\Delta_g z = d_0 |dz|_g$, homogeneous Neumann data on S , and, by (30.11), $z \rightarrow 0$ at infinity. Its positive global maximum is attained in a compact set. The same strong maximum and boundary point principles again give a contradiction. Thus M_R is bounded, and exhaustion produces a smooth solution of (30.9) with $\phi = O(r^{-2})$.

Here is the derivative control needed later, using only the stated derivatives of the original data. On a fixed annulus in the y variables put $g_\rho(y) = (g_{ij}(\rho y))$, $U_\rho(y) = \rho^2 \phi(\rho y)$, and $a_\rho(y) = \rho d_0(\rho y)$. The exact rescaled equation is

$$(30.12) \quad -\Delta_{g_\rho} U_\rho = a_\rho \sqrt{|dU_\rho|_{g_\rho}^2 + |y|^{-6}} + \rho^{-\delta} |y|^{-4-\delta}.$$

The metric components have uniform $C^{1,1}$ bounds from $g - \delta = O_2$. Consequently both the principal coefficients of this Laplacian and its first-order coefficients have uniform $C^{0,\alpha}$ bounds for every fixed $\alpha < 1$. No third derivative of g is needed. The chosen radial d_0 gives $a_\rho = O(\rho^{-q_0})$ with uniform symbol bounds, independently of derivatives of K . First-derivative interpolation in the local $W^{2,p}$ estimates gives a uniform $W^{2,p}$ bound on a smaller annulus. Taking $p > 4$

controls dU_ρ in $C^{0,\alpha}$ for some $\alpha > 0$. The positive $|y|^{-6}$ term is bounded away from zero there, so the entire right side of (30.12) is uniformly $C^{0,\alpha}$. The usual Schauder estimate, including the first-order Laplacian coefficients with these $C^{0,\alpha}$ bounds, now gives a uniform $C^{2,\alpha}$ bound on a further smaller annulus. Scaling back proves $\phi = O_2(r^{-2})$ without an additional metric or tensor falloff derivative. The equation now yields

$$d_0 \sqrt{|d\phi|_g^2 + r^{-6}} = O(r^{-4-q_0}) = o(w),$$

which proves both the differential inequality and the last limit in (30.7). The right side of (30.9) is integrable. The divergence theorem consequently gives a finite limiting physical sphere flux. Its difference from the Euclidean sphere flux tends to zero, since $g - \delta = O(r^{-q})$ and $d\phi = O(r^{-3})$.

Finally, differentiating Lemma 30.1 at $f = 1 + s\phi$ gives

$$C'_Q = -2\phi C + 6[-\Delta_g \phi + K(\nabla \phi, v)], \quad \theta'_Q = -\phi \theta_+ + 3\partial_\nu \phi.$$

On $\mathcal{A}$ the first term is zero, and on S we have $\theta_+ = 0$. This proves (30.8). On the full end ball bundle, $C = O(r^{-2-q})$, $\phi C = O(r^{-4-q}) = o(w)$, and $K(\nabla \phi, v) = O(r^{-4-q}) = o(w)$. Thus the normalized derivative tends uniformly to six. $\square$

30.2. Feasible variations and a measure identity.

Definition 30.3 (Variation space). Let $\mathcal{V}_0$ be the real vector space generated by all smooth compactly supported pairs (h, p) up to S and the following five pairs, cut off smoothly to the end. The scalar pair is

$$h_{ij} = r^{-2}\delta_{ij}, \quad p = 0.$$

For each vector e in a fixed basis of $\mathbb{R}^4$, the momentum pair has $h = 0$ and is defined by

$$(30.13) \quad p - (\text{tr}_\delta p)\delta = r^{-3}(e \otimes \hat{x} + \hat{x} \otimes e - (e \cdot \hat{x})\delta), \quad \hat{x} = x/r.$$

Set $\mathcal{V} = \mathbb{R}Q + \mathcal{V}_0$. We write its elements as $H = aQ + H_0$, where $H_0 = (h_0, p_0) \in \mathcal{V}_0$.

Lemma 30.4 (End behavior of the variation space). *The five prototypes have independent charge derivatives. For every fixed $H_0 \in \mathcal{V}_0$,*

$$(30.14) \quad C'_{H_0} = O(r^{-4-q}) = o(w)$$

uniformly on the full end ball bundle. The coefficient a in Definition 30.3 is uniquely determined, and

$$(30.15) \quad C'_H/w \longrightarrow 6a$$

uniformly there. All elements of $\mathcal{V}$ have well-defined charge derivatives.

Proof. Trace reversal on symmetric tensors in dimension four is invertible. The right side of (30.13), written as

$$D_{ij} = r^{-4}(e_i x_j + x_i e_j - (e \cdot x)\delta_{ij}),$$

satisfies $\partial_j D_{ij} = 0$ and $D_{ij}\hat{x}^j = r^{-3}e_i$. The scalar prototype satisfies the Euclidean linearized scalar constraint because $\Delta_\delta r^{-2} = 0$ off the origin. With the ADM normalizations (22.1)–(22.2), the scalar prototype has $E' = 1$, $P' = 0$, and the momentum prototype for e has $E' = 0$, $P' = e/3$. Background corrections to these fluxes vanish by the AF estimates, so the derivatives are independent.

On the end, $h_0 = O_2(r^{-2})$ and $p_0 = O_1(r^{-3})$. Their Euclidean linearized scalar and momentum constraints vanish. Each remaining linear constraint term has a factor from the decaying background: for example, $(g - \delta)\partial^2 h_0$, $\partial g \partial h_0$, $\partial^2 g h_0$, Kp_0 , or $(g - \delta)\partial p_0$ has order at most r^{-4-q} . The variation of the vector identification contributes $-J(h_0^\sharp v)$, of the same order. This proves

(30.14). Lemma 30.2 then proves (30.15) and the uniqueness of a : an element of $\mathcal{V}_0$ cannot have the normalized end derivative of Q .

Compact variations have zero charge derivatives. The prototypes have the fluxes just computed. For Q , the momentum derivative is zero and

$$E'_Q = -\frac{1}{\omega} \lim_{R \rightarrow \infty} \int_{|x|=R} \partial_r \phi \, dA_\delta,$$

which is finite by Lemma 30.2. Products with background errors have zero limiting flux. Linearity completes the proof. $\square$

Let $\mathcal{T}$ be the compact space of minimizing frontiers for g , with their filled sets, from Proposition 23.2. For $T \in \mathcal{T}$, write

$$(30.16) \quad a'_T(h) = \frac{1}{2} \int_T \text{tr}_T h \, dA_g.$$

This is a continuous function of T for each fixed variation h .

Lemma 30.5 (Strict linear inequalities give feasible nonlinear data). *Suppose $H = aQ + H_0 \in \mathcal{V}$ has $a > 0$ and, for some $\varepsilon > 0$,*

$$C'_H \geq \varepsilon w \text{ on } \mathcal{A}, \quad -\theta'_H > 0 \text{ on } S.$$

Then, for every sufficiently small $s > 0$,

$$(30.17) \quad g_s = e^{2as\phi}(g + sh_0), \quad K_s = e^{as\phi}(K + sp_0)$$

satisfies every hypothesis of Theorem 22.2. Its boundary has strictly negative future expansion. The path has derivative H , is uniformly comparable to g , has uniform large-sphere barriers, and its charges are differentiable at $s = 0$.

Proof. Positivity of the metric and uniform comparability hold for small s , since h_0 and ϕ are bounded relative to g . They also give completeness with the boundary included. The topology, smoothness, end count and orientability remain those of Ω .

We check DEC on the whole ball of test vectors. First use the intermediate metric and tensor $\bar{g}_s = g + sh_0$, $\bar{K}_s = K + sp_0$, identifying its vector ball with that of g by $\bar{I}_s$. For sufficiently large r , Taylor expansion and the Euclidean cancellations in Lemma 30.4 give, uniformly in $|v|_g \leq 1$,

$$\bar{C}_s(x, \bar{I}_s v) = C(x, v) + O(sr^{-4-q}) + O(s^2 r^{-6}).$$

Here the remainder is uniform in r , s in a fixed small interval, and the whole vector ball. To check all its components, the coordinate constraint expressions have the schematic forms

$$\begin{aligned} R_g &= g^{-1} \partial^2 g + \text{smooth}(g^{-1})(\partial g)^2, \\ J &= \text{smooth}(g^{-1}) \partial K + \text{smooth}(g^{-1})(\partial g) K, \end{aligned}$$

and $2\mu - R_g$ is quadratic in K , with smooth coefficients in g^{-1} . Each second s derivative on the intermediate path is $O(r^{-6})$: the largest terms are $h_0 \partial^2 h_0$, $(\partial h_0)^2$, p_0^2 , $h_0 \partial p_0$, and $(\partial h_0) p_0$. Terms containing an original K , ∂K , or background metric derivative decay faster. This calculation includes all occurrences of $\tau = g^{ij} K_{ij}$, without a separate trace estimate. The vector identification also has the uniform matrix expansion

$$\bar{I}_s = (\text{Id} + sh_0^\sharp)^{-1/2} = \text{Id} - \frac{s}{2} h_0^\sharp + O(s^2 r^{-4});$$

the eigenvalues lie in a fixed positive interval, so the matrix-function Taylor bound is uniform. Multiplication by the original or varied momentum covector keeps the same remainder order. In

fact the preceding calculation gives the componentwise estimates

$$(30.18) \quad \begin{aligned} \bar{\mu}_s &= \mu + O(sr^{-4-q}) + O(s^2r^{-6}), \\ \bar{J}_s &= J + O(sr^{-4-q}) + O(s^2r^{-6}). \end{aligned}$$

Apply Lemma 30.1 to these intermediate data with $f_s = e^{as\phi}$. After multiplication by the positive factor f_s^2 , the constraint for the final data is

$$(30.19) \quad C(x, v) + s(6aw + o(w)) + O(s^2r^{-6}),$$

again uniformly for the full vector ball. Indeed, $-\Delta_g\phi/w \rightarrow 1$ and $K(\nabla\phi, v) = o(w)$. Changing the metric, tensor, and vector identification in these conformal differential terms adds only the above decay orders, while the extra exponential term uses $|\mathrm{d}\phi|^2 = O(r^{-6})$. Since $a > 0$ and $r^{-6} = O(w)$, one can first choose a fixed large radius so the coefficient of s in (30.19) is positive, and then choose s small so its remainder is absorbed. DEC follows on this fixed far end, using $C \geq 0$.

The remaining base region and its closed unit ball bundle are compact. The derivative of the constraint is strictly positive on its original zero set. By continuity it is positive on a neighborhood of that zero set; on the complement the original constraint has a positive minimum. A uniform Taylor expansion therefore gives $C_s \geq 0$ everywhere in the compact region for small positive s . The same compactness argument on S , where $\theta_+ = 0$, gives $\theta_{+,s} < 0$.

The new densities are integrable componentwise, not merely after contraction with active vectors. The exact conformal formulas give

$$\begin{aligned} f_s J_s &= \bar{J}_s + 3as\bar{K}_s(\nabla_{\bar{g}_s}\phi, \cdot), \\ f_s^2 \mu_s &= \bar{\mu}_s - 3as\Delta_{\bar{g}_s}\phi - 3a^2s^2|\mathrm{d}\phi|_{\bar{g}_s}^2. \end{aligned}$$

The extra covector in the first line is $O(sr^{-4-q}) + O(s^2r^{-6})$. Also $\Delta_{\bar{g}_s}\phi = \Delta_g\phi + O(sr^{-6})$, with $\Delta_g\phi = O(w)$. Together with (30.18), these identities express the new densities as bounded smooth scaling factors times the original integrable densities, plus integrable errors of orders $r^{-4-\delta}$, r^{-4-q} , and r^{-6} . Uniform metric comparability controls both norms and volume elements. The differentiated falloff has exponent $\min(q, 2) > 1$, uniformly for s in a small fixed interval, so large coordinate spheres remain mean convex uniformly.

The charge limits can also be checked before differentiation. Write $\pi = K - \tau g$. Since conformal trace reversal gives $\pi_s = f_s(\bar{K}_s - \bar{\tau}_s\bar{g}_s)$, expansion on the end yields

$$\pi_s = \pi + s(p_0 - (\mathrm{tr}_\delta p_0)\delta) + O(sr^{-3-q}) + O(s^2r^{-5}).$$

The terms with the original trace are of size $h_0K = O(r^{-3-q})$; multiplication by $f_s - 1 = O(sr^{-2})$ adds only the displayed error orders. All errors have zero limiting sphere flux. Therefore $P_s = P + sP'_{H_0}$ in the prescribed coordinates, without an extra decay assumption on τ . Similarly,

$$g_s - g - s(h_0 + 2a\phi g) = O_1(s^2r^{-4}),$$

whose energy flux vanishes. The finite flux of ϕ and the scalar prototype thus give $E_s = E + sE'_H$. In particular both charge functions exist and are differentiable with the derivatives computed in Lemma 30.4. The path and its first two parameter derivatives are uniformly bounded relative to g , as required by Proposition 23.2. This verifies all claims and all hypotheses of the numerical theorem. $\square$

Proposition 30.6 (Multiplier identity at equality). *There exist a probability measure ϖ on $\mathcal{T}$, a nonnegative finite measure η on S , a nonnegative locally finite measure Λ on $\mathcal{A}$, and $z \geq 0$, with $\int_{\mathcal{A}} w \, \mathrm{d}\Lambda < \infty$, such that for every $H = aQ + H_0 = (h, p) \in \mathcal{V}$,*

$$(30.20) \quad 6\omega\mathcal{E}'(H) - c \int_{\mathcal{T}} a'_T(h) \, \mathrm{d}\varpi(T) = \langle C'_H, \Lambda \rangle - \int_S \theta'_H \, \mathrm{d}\eta + za.$$

The constraint pairing is absolutely defined, since C'_H/w is bounded on the compactified active set.

Proof. Compactify $\mathcal{A}$ by one end point. If $\mathcal{A}$ is already compact, adjoin an isolated point; if it is empty, use just that point. The ball fibers are compact and the base has one end, so this is a compact Hausdorff space, denoted $\bar{\mathcal{A}}$. By Lemma 30.4, C'_H/w extends continuously to it with value $6a$ at the added point. On the disjoint compact union

$$\mathcal{K} = \bar{\mathcal{A}} \sqcup S \sqcup \mathcal{T}$$

consider the linear map to $C(\mathcal{K})$ whose three components are

$$(30.21) \quad H \mapsto \left(\frac{C'_H}{w}, -\theta'_H, ca'_T(h) - 6\omega\mathcal{E}'(H) \right).$$

Its image does not meet the open cone of functions strictly positive everywhere. Otherwise its value at the added point gives $a > 0$, and Lemma 30.5 supplies actual feasible data (30.17). Let $a(s)$ be their enclosing infimum. The area derivative formula in Proposition 23.2 gives

$$a'(0+) = \min_{T \in \mathcal{T}} a'_T(h).$$

Strict positivity of the third component in (30.21), and compactness of $\mathcal{T}$, imply

$$\left. \frac{d}{ds} \right|_{0+} 6\omega(\mathcal{E}(g_s, K_s) - \mathbf{m}(a(s))) = 6\omega\mathcal{E}'(H) - c \min_{T \in \mathcal{T}} a'_T(h) < 0.$$

The expression starts at zero and is nonnegative by (30.3), a contradiction.

Hahn–Banach separation of a linear subspace from this open convex cone gives a nonzero positive continuous functional on $C(\mathcal{K})$ annihilating the image. The subspace need not be closed: its closure is still disjoint from the open cone. The Riesz representation theorem identifies the functional with a nonzero finite positive measure on $\mathcal{K}$.

Its mass on $\mathcal{T}$ is positive. If that mass were zero, testing Q would give a strictly positive value on every remaining component: $C'_Q/w \geq 6$ on $\mathcal{A}$, the added-point value is six, and $-\theta'_Q = 3$ on S . This would contradict annihilation. Normalize the mass on $\mathcal{T}$ to one and call the resulting measure there ϖ . Call its restriction to S η . On $\mathcal{A}$, divide the remaining measure by the positive function w to obtain Λ ; this is locally finite and has finite weighted mass. If the normalized added-point mass is λ_∞ , put $z = 6\lambda_\infty$. Rearranging annihilation of (30.21) is exactly (30.20). $\square$

30.3. Normal graph tests. The measure ϖ may initially be supported on several enclosures of the same area. The next argument uses variations of the original data to show that it is supported on the original horizon. It is local in the open exterior and assumes no complete ambient development.

Lemma 30.7 (Compact normal tests annihilating active constraints). *For every $\ell \in C_c^\infty(\text{int } \Omega)$, there are smooth variations $H_j = (h_j, p_j) \in \mathcal{V}_0$, all supported in a fixed compact subset of the open exterior, such that*

$$h_j = 2\ell K, \quad C'_{H_j} \rightarrow 0$$

uniformly on the active rays over that compact set.

Proof. Use local Lorentzian metrics in Gaussian time, $\mathbf{g}_j = -dt^2 + g_j(t)$, near a compact neighborhood of $\text{supp } \ell$, and prescribe $g_j(0) = g$, $\partial_t g_j(0) = 2K$. At $t = 0$, prescribe their spatial Einstein tensors to be

$$(30.22) \quad (\mathbf{G}_j)_{ik} = (D_j)_{ik}, \quad D_j = \frac{J \otimes J}{\mu + 1/j}.$$

This is possible by choosing the second time derivative of g_j . Indeed its coefficient in the spatial Einstein equation is a nonzero multiple of trace reversal on symmetric spatial tensors, which is invertible in dimension four. The normal components are fixed by the Gauss–Codazzi constraints:

$$\mathbf{G}_j(n, n) = \mu, \quad \mathbf{G}_j(n, e_i) = J_i.$$

A smooth choice of the second time jet realizes the prescribed tensor in a sufficiently small time neighborhood. No equation or energy condition away from the initial slice is required.

We first check the convergence of these tensors along the slice, within the chosen open interior neighborhood. Put $Z = \{\mu = 0\}$ there. At every point of Z , smoothness and $\mu \geq 0$ give $d\mu = 0$. DEC implies $|J| \leq \mu$, so $J = 0$ and $\nabla J = 0$ there as well; for example, along a smooth curve through the point, $\mu = O(t^2)$ and thus $J = O(t^2)$. The bounds

$$(30.23) \quad |D_j| \leq \mu, \quad |\nabla D_j| \leq 2|\nabla J| + |d\mu|$$

follow by differentiating the quotient and using $|J| \leq \mu$. On every compact subset of this interior neighborhood, these imply convergence in C^1 to

$$(30.24) \quad D = \begin{cases} J \otimes J / \mu, & \mu > 0, \\ 0, & \mu = 0. \end{cases}$$

To justify the assertion at the zero set, choose a small neighborhood of its intersection with the compact set. The right sides of (30.23) are uniformly small there by continuity and their vanishing on Z . On the remaining compact part, μ is bounded away from zero and the usual quotient derivatives converge uniformly. The zero extension in (30.24) is differentiable on Z , with derivative zero, because $|D| \leq \mu$ and $d\mu = 0$ there. The same bounds give continuity of that derivative.

The full tensors $\mathbf{G}_j$ restricted to the slice, and their covariant derivatives in spatial directions, consequently converge in C^0 . The spacetime connection in a spatial direction along $t = 0$ depends only on the common g, K , not on the chosen second time jets.

Consider a positive-density active ray. Its activity and DEC imply $|v| = 1$, $|J| = \mu$, and $v = -J^\sharp / \mu$. Thus $\zeta = n + v$ is future null. On a neighborhood in the slice where $\mu > 0$, define a covector α by

$$\alpha(n) = \mu, \quad \alpha|_{T\Omega} = J.$$

It is causal and nonnegative on the future cone, and the limiting Einstein tensor along that neighborhood is $\mathbf{G}_0 = \alpha \otimes \alpha / \mu$. At the active point,

$$\alpha(\zeta) = 0, \quad \alpha = -\mu \zeta^\flat.$$

Parallel transport ζ along any spatial curve. It remains future null, so its pairing with α is nonnegative and has value zero at the active point. Its derivative there is zero. Consequently

$$(30.25) \quad (\nabla_{e_i} \alpha)(\zeta) = 0, \quad (\nabla_{e_i} \mathbf{G}_0)(e_i, \zeta) = 0$$

at that point, for every spatial orthonormal vector e_i .

Contracted Bianchi, with signature $(-, +, +, +, +)$, gives the exact contraction needed for the graph variation:

$$(30.26) \quad (\nabla_n \mathbf{G}_j)(n, \zeta) = \sum_{i=1}^4 (\nabla_{e_i} \mathbf{G}_j)(e_i, \zeta) \longrightarrow 0.$$

There is no assertion of convergence for every normal derivative of $\mathbf{G}_j$; only (30.26) is used.

Now vary the hypersurface to the graph $t = s\ell(x)$ in $\mathbf{g}_j$. Its induced data have metric derivative $h_j = 2\ell K$ and some smooth tensor derivative p_j . They give compactly supported variations on Ω : outside a fixed neighborhood of $\text{supp } \ell$ the graph is the original slice, so extend the variations by

zero. The same first jets of $g_j(t)$ along the initial slice imply uniform compact bounds, independent of j , for the derivatives of the unit normal n_s and of the isometrically identified vector v_s .

These first derivatives can be written explicitly. The graph velocity is ℓn . Orthogonality to its tangent vectors and the unit-normal condition give $\nabla_s n_s|_0 = \nabla \ell$. The material derivative of a graph-pushed vector v is $d\ell(v)n + \ell K^\sharp v$. Since $h_j = 2\ell K$, its isometric identification has derivative $I'_0 v = -\ell K^\sharp v$, canceling the last spatial term. Hence

$$(30.27) \quad \nabla_s n_s|_0 = \nabla \ell, \quad \nabla_s v_s|_0 = d\ell(v)n.$$

Differentiating the constraint $2\mathbf{G}_j(n_s, n_s + v_s)$ and using Bianchi gives the exact formula

$$(30.28) \quad \begin{aligned} \frac{1}{2}C'_{H_j}(x, v) &= \ell \sum_i (\nabla_{e_i} \mathbf{G}_j)(e_i, \zeta) + \mathbf{G}_j(\nabla \ell, \zeta) \\ &+ \mathbf{G}_j(n, \nabla \ell + d\ell(v)n). \end{aligned}$$

Thus only spatial first derivatives of the Einstein tensor occur in the final expression. At a positive-density active ray the first limiting term is zero by (30.25), the second is zero because $\mathbf{G}_0(\cdot, \zeta) = 0$, and the last is zero because

$$\alpha(\nabla \ell + d\ell(v)n) = J(\nabla \ell) + \mu d\ell(v) = 0.$$

Equivalently, $\zeta_s = n_s + v_s$ remains null at a unit active vector, and its derivative pairs to zero with the proportional null covector α .

If $\mu = 0$, the full tensor $\mathbf{G}_j$ and all its spatial covariant derivatives are zero at the point, by (30.22), (30.23), and the vanishing first jets of μ, J . Bianchi therefore proves the same cancellation for every $|v| \leq 1$, including non-unit active vectors. The argument-derivative terms vanish there as well.

Let $L \Subset \text{int } \Omega$ contain the fixed support. The limiting version of the right side of (30.28) is zero at every active ray, by the positive- and zero-density arguments. All its coefficients are bounded on L , uniformly for $|v| \leq 1$. Consequently

$$\sup_{\mathcal{A}|_L} |C'_{H_j}| \leq C_\ell \|\mathbf{G}_j - \mathbf{G}_0\|_{C^1_{\text{sp}}(L)} \longrightarrow 0,$$

where the norm uses the common connection in spatial directions along the initial slice. This estimate contains no division by μ and includes sequences of active rays approaching Z . The support remains strictly inside the open exterior throughout; no vanishing normal derivative of the densities at the original boundary is asserted. This proves the required uniform convergence with the derivative convention (30.4). $\square$

Proposition 30.8 (Localization of the area multiplier). *For ϖ -almost every element of $\mathcal{T}$, its frontier equals S , where ϖ is the measure in Proposition 30.6. Consequently, for every variation $H = (h, p)$ in $\mathcal{V}$,*

$$(30.29) \quad \int_{\mathcal{T}} a'_T(h) \, d\varpi(T) = a'_S(h) = \frac{1}{2} \int_S \text{tr}_S h \, dA_g.$$

In particular the multiplier identity becomes

$$(30.30) \quad 6\omega \mathcal{E}'(H) - c a'_S(h) = \langle C'_H, \Lambda \rangle - \int_S \theta'_H \, d\eta + za.$$

Proof. Apply (30.20) to the variations in Lemma 30.7. Their charge, boundary, and end-coefficient terms vanish, and Λ is finite over their fixed compact support. The uniform constraint-derivative convergence therefore gives

$$(30.31) \quad \int_{\mathcal{T}} \int_T \ell \, \text{tr}_T K \, dA_g \, d\varpi(T) = 0 \quad \text{for every } \ell \in C_c^\infty(\text{int } \Omega).$$

Define the finite aggregate area measure ρ by

$$\int f \, d\rho = \int_{\mathcal{T}} \int_T f \, dA_g \, d\varpi(T).$$

All frontiers lie in one compact set. By Proposition 23.3, off S the value $\text{tr}_T K$ is a single continuous function $k(x)$ on their union. Thus (30.31) says $k\rho = 0$ as a signed measure in the open exterior. It follows that $k = 0$ ρ -almost everywhere. Fubini, followed by smoothness of each free part of a minimizing frontier, shows that for ϖ -almost every T ,

$$(30.32) \quad \text{tr}_T K = 0 \quad \text{everywhere on } T \setminus S.$$

Indeed a nonzero value on a smooth free part would persist on an open piece of positive area. Such a T is minimal away from S by Proposition 23.1, and hence is a MOTS there.

Fix one of these frontiers. At any contact point with S , it has the tangent plane and orientation of S , because its filled set contains the entire obstacle. In a local graph chart, let f be its graph and ψ the smooth obstacle graph. The regularity in Proposition 23.1 gives $f \in C^{1,\alpha} \cap W^{2,2}$. On the coincidence set $f = \psi$, one has $Df = D\psi$. Locality of weak Sobolev derivatives, applied to the components of $Df - D\psi$, gives $D^2 f = D^2 \psi$ almost everywhere there. Since S is a MOTS, the same MOTS equation holds almost everywhere on the contact set. By (30.32), it holds off that set as well.

We spell out why no singular contact term remains. In these charts the MOTS equation has the form

$$(30.33) \quad \partial_i \mathcal{F}^i(x, f, Df) = \mathcal{B}(x, f, Df),$$

where the coefficients are smooth after extending the one-sided smooth background across S , and the matrix $\mathcal{F}_{p_j}^i$ is uniformly elliptic on the local bounded gradient range. As $f \in W^{2,2}$, its left side is an L^2 function and the almost-everywhere equation is also the weak divergence-form equation. For a local coordinate derivative f_k , distributional differentiation of (30.33) yields

$$\partial_i (a^{ij} \partial_j f_k) = \partial_i (\delta_{ik} \mathcal{B} - \mathcal{F}_{x_k}^i - \mathcal{F}_z^i f_k), \quad a^{ij} = \mathcal{F}_{p_j}^i(x, f, Df).$$

All coefficients and the displayed right-hand vector field are $C^{0,\alpha}$. The interior divergence-form Schauder estimate gives $f_k \in C^{1,\alpha}$ locally, hence $f \in C^{2,\alpha}$. Further differentiation bootstraps to smoothness [28]. Thus T is a smooth MOTS through contact.

If T touches S , strong comparison gives local coincidence of the two ordered smooth MOTS with the same orientation. To check the applicable maximum principle, their nonnegative graph difference solves a linear uniformly elliptic equation with bounded lower-order coefficients. Its zeroth-order coefficient can be replaced by a nonpositive one to obtain the inequality for the strong minimum principle; an interior zero then forces local vanishing. Therefore the intersection with S is open and closed in S . Connectedness of S implies that S is a component of T . Since every minimizing frontier has total area $A = \text{Area}_g(S)$, there can be no additional component, and $T = S$.

If T does not touch S , it is a compact smooth embedded two-sided MOTS entirely in $\text{int } \Omega$. Its connected exterior side and full enclosure of the obstacle were proved in Proposition 23.1. With all components and the orientation toward the end counted, it is exactly an enclosing hypersurface excluded by the outermostness hypothesis. This alternative is impossible. Thus $T = S$ for ϖ -almost every T , proving (30.29) and (30.30). $\square$

31. A CAUSAL STATIONARY FIELD ON THE ORIGINAL EXTERIOR

We continue with the original equality data (Ω, g, K) of Definition 22.3. In particular, $S = \partial\Omega$ is connected, $\theta_+ = 0$ on S , and S is outer area-minimizing. The constants fixed in Section 30 are

$$r_0 = \sqrt{2m}, \quad b^0 = \frac{E}{m}, \quad b^i = -\frac{P_i}{m}, \quad c = \frac{2}{r_0}, \quad \kappa = \frac{c}{2} = \frac{1}{r_0}.$$

Thus $(b^0)^2 - |b|^2 = 1$ and $b^0 > 0$. We fix the same exponent $1 < q_0 < \min\{q, 2\}$ as in the strict variation constructed there. The purpose of this section is to extract a smooth field from the multiplier and to retain all information about the original second fundamental form.

Theorem 31.1 (Causal stationary field). *There are a smooth function u and a smooth vector field X on Ω , smooth up to its one-sided boundary, with the following properties.*

(i) *They satisfy $u \geq |X|_g$, $u > 0$ in $\text{int } \Omega$, and*

$$(31.1) \quad u\mu + J(X) = 0.$$

With symmetric indices denoting averaging, their equations are

$$(31.2) \quad \nabla_{(i} X_{j)} = -u K_{ij},$$

$$(31.3) \quad \Delta u = -\text{div}_g(K(X, \cdot)^\sharp),$$

$$(31.4) \quad \nabla^2 u = u(\text{Ric}_g + \tau K - 2K^2) - \mathcal{L}_X K + X_{(i} J_{j)}.$$

Here $X_i = g_{ij} X^j$ and $(K^2)_{ij} = K_{ik} K^k_{j}$.

(ii) *On the chosen asymptotically flat end,*

$$(31.5) \quad (u, X) = (b^0, b) + O_2(r^{-q_0}).$$

(iii) *At the boundary, for the unit normal ν pointing into Ω ,*

$$(31.6) \quad X = u\nu, \quad \partial_\nu u + K(X, \nu) = \kappa.$$

Either $u > 0$ everywhere on S or $u = 0$ everywhere on S .

(iv) *On $\mathbb{R} \times \text{int } \Omega$, set*

$$(31.7) \quad \mathbf{g} = -u^2 dt^2 + g_{ij}(dx^i + X^i dt)(dx^j + X^j dt), \quad \xi = \partial_t, \quad N = u^2 - |X|_g^2.$$

For the future normal $n = u^{-1}(\xi - X)$, the slice $t = 0$ has exactly the original induced metric g and second fundamental form K , with the convention of Theorem 22.4. Moreover,

$$(31.8) \quad \text{Ric}_{\mathbf{g}}(\xi, \cdot) = 0, \quad \text{Ric}_{\mathbf{g}} = 0 \quad \text{on } \{N > 0\}.$$

(v) *We have $N \geq 0$, $N \rightarrow 1$ at infinity, and $N > 0$ in a full collar just outside S . More precisely, in the positive boundary-lapse case,*

$$(31.9) \quad N|_S = 0, \quad dN|_S = 2u\kappa\nu^\flat = 2\kappa X^\flat.$$

In the zero boundary-lapse case, if $s \geq 0$ is the original g -normal distance from S , there are smooth one-sided fields d, Y_2 such that

$$(31.10) \quad u = sd, \quad X = s^2 Y_2, \quad d|_S = \kappa, \quad N = s^2(d^2 - s^2|Y_2|_g^2).$$

Consequently all zeros of N in $\text{int } \Omega$ lie in a compact subset of that interior.

The proof occupies the remainder of this section. At no point do we assume that the original data are vacuum or that they lie in a prescribed stationary development. The metric in (31.7) is constructed from the multiplier.

31.1. From a positive measure to the interior adjoint equations. By Proposition 30.6 and the area localization in Proposition 30.8, the multiplier identity is

$$(31.11) \quad 6\omega \mathcal{E}'(H) - c a'_S(h) = \langle C'_H, \Lambda \rangle - \int_S \theta'_H d\eta + za, \quad H = aQ + H_0.$$

The coefficient a is the coefficient of the strict direction Q in the variation space. We recall that

$$C(x, v) = R_g + \tau^2 - |K|^2 + 2J(v), \quad |v|_g \leq 1,$$

that Λ is a positive locally finite measure on the active set $C = 0$, and that the test-vector identification for a metric variation h has $\dot{v} = -\frac{1}{2}h^\sharp v$.

Let u be the scalar pushforward of Λ to Ω , and let X be its vector first moment. Initially these symbols denote measures: for a compactly supported scalar f and covector α ,

$$\langle u, f \rangle = \int f(x) d\Lambda(x, v), \quad \langle X, \alpha \rangle = \int \alpha_x(v) d\Lambda(x, v).$$

The unit-ball condition and positivity give $|X|_g \leq u$ as measures, while activity gives $\mu u + J(X) = 0$ as a measure. We use dV_g as the reference density when expressing the resulting distributional equations. After establishing smoothness, we use the same letters for the smooth densities of these measures.

Lemma 31.2 (Interior adjoint equations). *The moment measures have smooth densities on $\text{int } \Omega$. They satisfy (31.1)–(31.4) there, and $u \geq |X|_g$ there.*

Proof. For a variation supported in the open exterior, every term of (31.11) except the constraint pairing vanishes. For a pure tensor variation $p = \dot{K}$, that pairing is

$$2\langle u, \tau \text{tr}_g p - K : p \rangle + 2\langle X, \nabla^j(p_{ij} - (\text{tr}_g p)g_{ij}) \rangle.$$

Its distributional adjoint equation is

$$(31.12) \quad u(\tau g - K) - \nabla_{(i} X_{j)} + (\text{div } X)g = 0.$$

Taking the trace in four dimensions gives $3u\tau + 3 \text{div } X = 0$. Substitution proves (31.2).

For the conformal variation $(h, p) = (2\psi g, \psi K)$, the constraint transformation already established in Section 30 gives, on active rays,

$$C'_H(x, v) = 6\{-\Delta\psi + K(\nabla\psi, v)\}.$$

Its adjoint is (31.3). This calculation is valid for measure moments, so both equations hold in distributions before any regularity assertion.

Diverge (31.2) and substitute $\text{div } X = -u\tau$. Commuting covariant derivatives gives a Laplace equation for X whose right side consists of smooth coefficients times $u, X, \nabla u$. In (31.3), the right side consists of smooth coefficients times $X, \nabla X$. Thus the coupled system has diagonal Laplace principal part on the scalar and vector bundles; all couplings have order at most one. Distributional interior elliptic regularity, iterated with the smooth coefficients, makes both densities smooth. The measure inequalities and complementarity now give the asserted pointwise statements.

It remains to verify the full metric equation, including its matter term. We may now work with smooth u, X . For a pure metric variation h , keep covariant K , scalar u , and contravariant X fixed. The constraint pairing is the variation of

$$(31.13) \quad \int [u(R + \tau^2 - |K|^2) + 2X^i \nabla^j (K_{ij} - \tau g_{ij})] dV_g$$

minus $\int h(X, J^\sharp) dV_g$. The latter is precisely the contribution of $\dot{v} = -\frac{1}{2}h^\sharp v$. In using the varying volume form in (31.13), no term has been added: its original integrand is $2u\mu + 2J(X) = 0$ pointwise.

For completeness, put $\pi^{ij} = K^{ij} - \tau g^{ij}$. Integrating the shift term once by parts gives $-\int \pi^{ij}(\mathcal{L}_X g)_{ij} dV_g$, with unchanged boundary terms outside the variation support. At fixed covariant K ,

$$\delta \pi^{ij} = -h^i_a K^{aj} - h^j_a K^{ia} + (K : h)g^{ij} + \tau h^{ij}.$$

Variation of the last integral, followed by integration of the term $-\pi^{ij}(\mathcal{L}_X h)_{ij}$, has coefficient

$$(31.14) \quad \mathcal{L}_X K - (\operatorname{div} X)K - \{X(\tau) + K^{ab}\nabla_a X_b\}g.$$

One can see the cancellations directly: if $B = \mathcal{L}_X g$, the variation of π contributes $2\operatorname{Sym}(KB) - (\operatorname{tr} B)K - \tau B$; integration of $\mathcal{L}_X h$ contributes $(\mathcal{L}_X \pi)^{ij} + (\operatorname{div} X)\pi^{ij}$ with indices lowered; and the volume variation contributes $-\frac{1}{2}(\pi : B)g$. Together they are (31.14).

The scalar and quadratic terms in (31.13) have coefficient

$$\nabla^2 u - (\Delta u)g - u \operatorname{Ric}_g + 2u(K^2 - \tau K) + u\mu g.$$

Adding (31.14) and the test-vector correction therefore gives

$$(31.15) \quad \begin{aligned} 0 = & \nabla^2 u - (\Delta u)g - u \operatorname{Ric}_g + 2u(K^2 - \tau K) + u\mu g \\ & + \mathcal{L}_X K - (\operatorname{div} X)K - \{X(\tau) + K^{ab}\nabla_a X_b\}g - X_{(i}J_{j)}. \end{aligned}$$

Equation (31.2) gives $K^{ab}\nabla_a X_b = -u|K|^2$. Moreover, using $\nabla^j K_{ij} = J_i + \nabla_i \tau$ in (31.3) gives

$$(31.16) \quad \Delta u + X(\tau) = u|K|^2 - J(X) = u(|K|^2 + \mu).$$

Together with $\operatorname{div} X = -u\tau$, these identities cancel the scalar multiples of g in (31.15) and give (31.4). $\square$

31.2. Boundary regularity and the asymptotic constants.

Lemma 31.3 (Prolongation and absence of boundary atoms). *The fields u, X extend smoothly to the one-sided boundary S . Their first jet at one interior point determines them on the connected interior. The original moment measures have no boundary-supported part, and hence are exactly $u dV_g$ and $X dV_g$ on all of Ω .*

Proof. Put $B_{ij} = -uK_{ij}$. Differentiate $\nabla_{(i}X_{j)} = B_{ij}$ in three arrangements, add two of the resulting equations, and subtract the third. Commuting derivatives gives

$$(31.17) \quad \begin{aligned} \nabla_i \nabla_j X_k &= \nabla_i B_{jk} + \nabla_j B_{ik} - \nabla_k B_{ij} + \mathcal{R}_{ijk}(X), \\ \mathcal{R}_{ijk}(X) &= -\frac{1}{2}([\nabla_j, \nabla_i]X)_k + ([\nabla_i, \nabla_k]X)_j + ([\nabla_j, \nabla_k]X)_i. \end{aligned}$$

Every commutator is a curvature contraction with X . Thus this equation uses only $K\nabla u$, $u\nabla K$, and curvature times X . Expanding the Lie derivative in (31.4) gives

$$(31.18) \quad \begin{aligned} \nabla_i \nabla_j u &= u(\operatorname{Ric}_{g,ij} + \tau K_{ij} - 2(K^2)_{ij}) - X^a \nabla_a K_{ij} \\ &\quad - K_{aj} \nabla_i X^a - K_{ia} \nabla_j X^a + X_{(i}J_{j)}. \end{aligned}$$

Consequently

$$\mathcal{J} = (u, X, \nabla u, \nabla X)$$

obeys a homogeneous linear first-order system, with smooth coefficients built from g, K , their required derivatives, and J . Along any smooth curve, this is a linear ODE for $\mathcal{J}$. Uniqueness along curves proves the first-jet assertion.

In a compact collar of S , start on an interior collar section and solve this ODE along the normal segments down to S . Its coefficients are smooth up to the one-sided boundary. Smooth dependence on the initial point and on the normal parameter provides the smooth extension, agreeing with the original solution for positive collar parameter. The pointwise inequalities and complementarity extend by continuity.

The original moments could still have parts supported on S ; the interior regularity alone does not exclude them. Let s be geodesic collar distance, let $f \in C^\infty(S)$ be arbitrary and extended constantly along the normal segments, and take a collar cutoff χ equal to one near S . Choose the compact pure metric variation

$$(31.19) \quad h = \frac{s^2 \chi(s) f}{6} (g - ds \otimes ds), \quad p = 0.$$

This is one of the allowed smooth compact variations, with strict-direction coefficient $a = 0$. Its value and first jet vanish on S . The trace along the three-dimensional collar slices is $s^2 \chi f / 2$, so its second normal derivative at S is f . At S , the scalar variation

$$R'_g(h) = \nabla^i \nabla^j h_{ij} - \Delta(\text{tr}_g h) - \text{Ric}_g : h$$

therefore equals $-f$: the only nonzero second jets are normal-normal derivatives of tangential components, whereas $h_{ss} = h_{sA} = 0$. The variation of $\tau^2 - |K|^2$ uses only h , that of J at fixed covariant K uses only $h, \nabla h$, and the test-vector correction uses only h . Consequently $C'_H|_S = -f$, independently of the active ray v .

The area derivative vanishes because $h|_S = 0$; the expansion derivative vanishes because $h|_S = \nabla h|_S = p|_S = 0$; and the charge derivative vanishes by compact support. Integrating the smooth interior densities by parts gives zero as well: the adjoint vanishes, and its scalar boundary terms involve only $h, \nabla h$, while its momentum boundary terms involve only h . This is ordinary integration up to S using the one-sided smooth fields, not distributional differentiation of a zero extension across S . Thus (31.11) says that the scalar boundary measure annihilates every $f \in C^\infty(S)$, and that measure is zero. The inequality $|X|_g \leq u$ as measures eliminates the vector boundary part. Moreover, since u is the pushforward of the positive measure Λ , the entire measure Λ has zero mass over S , not merely zero first moment there. $\square$

We first obtain constant limits from a Hessian estimate and causality. Related radial estimates for asymptotic Killing fields appear in [12, Proposition 2.1 and Appendix C]. The normalization of the limits will then follow from the end variations.

Lemma 31.4 (Asymptotics of causal fields). *Let g be a smooth metric uniformly comparable to the Euclidean metric on an end $\mathbb{R}^4 \setminus \overline{B}_R$. Let u be a smooth function and X a smooth vector field there with $u \geq |X|_g$. Write Y for all coordinate components of (u, X) . Suppose that, for some $q_0 > 1$ and a constant C , the coordinate derivatives satisfy*

$$(31.20) \quad |\partial^2 Y| \leq C r^{-1-q_0} |\partial Y| + C r^{-2-q_0} |Y|.$$

Then there are constants u_∞ and $X_\infty \in \mathbb{R}^4$ such that

$$(u, X) = (u_\infty, X_\infty) + O_2(r^{-q_0}).$$

Proof. Fix a sufficiently large coordinate sphere of radius R and put

$$D(r) = \sup_{\omega \in S^3} |\partial Y(r\omega)|, \quad \mathcal{M}(r) = \sup_{R \leq t \leq r} D(t).$$

Along each ray, $|Y(t\omega)| \leq C + \int_R^t D(a) da \leq C + t\mathcal{M}(t)$, uniformly in ω . Integrating $\partial_r(\partial Y)$ using (31.20), and then taking the supremum over directions and radii at most r , gives

$$\mathcal{M}(r) \leq C + C \int_R^r t^{-1-q_0} \mathcal{M}(t) dt.$$

The kernel is integrable. Gronwall therefore bounds $\mathcal{M}$ uniformly, so $\partial Y = O(1)$ and $Y = O(r)$. Equation (31.20) now gives $\partial^2 Y = O(r^{-1-q_0})$. Each coordinate derivative has a radial limit $L(\omega)$, with error $O(r^{-q_0})$. Any two points on the sphere of radius r can be joined by a spherical arc of length at most πr ; integrating the Hessian along this arc gives an $O(r^{-q_0})$ difference between

their gradients. Letting $r \rightarrow \infty$ shows that $L(\omega)$ is one constant matrix L . Radial integration of $\partial Y - L$ then gives

$$\partial Y = L + O(r^{-q_0}), \quad Y = Lx + O(1),$$

where boundedness of the second error uses $q_0 > 1$. Since $u \geq 0$ in every direction, the linear part of u vanishes. Uniform comparability of g with the Euclidean metric and $|X|_g \leq u$ then force the linear part of X to vanish as well. Thus $Y = O(1)$ and $\partial Y = O(r^{-q_0})$. Substitution in (31.20), using $q_0 > 1$, improves the Hessian bound to $O(r^{-2-q_0})$. Integrating each gradient component radially toward its zero limit gives $\partial Y = O(r^{-1-q_0})$. A further radial integration gives $Y = B(\omega) + O(r^{-q_0})$; comparison of values along the same spherical arcs now gives an $O(r^{-q_0})$ difference, so B too is independent of direction. This proves the asserted constant limit and both differentiated estimates. $\square$

Lemma 31.5 (Asymptotic normalization and positive lapse). *Equation (31.5) holds, and $u > 0$ on $\text{int } \Omega$.*

Proof. Let Y denote all coordinate components of the pair (u, X) . The prolonged system and the original O_2 metric and O_1 tensor decay imply Equation (31.20). Indeed curvature, ∇K , and J have order $-2 - q_0$, while K and the Christoffel symbols have order $-1 - q_0$. Equations (31.17) and (31.18) use no higher background derivatives. Converting their covariant Hessians to coordinate Hessians adds terms with coefficients Γ , $\partial \Gamma$, and Γ^2 , controlled by the stated O_2 metric decay.

Lemma 31.4 now gives constants with

$$(31.21) \quad (u, X) = (u_\infty, X_\infty) + O_2(r^{-q_0}).$$

Use the five end prototypes of Definition 30.3, whose charge derivatives are computed in Lemma 30.4, in (31.11). Their strict-direction coefficient is $a = 0$, so the term za is absent; their boundary and area terms vanish because the prototypes are supported away from S . Integration by parts using the interior adjoint equations leaves only the outer boundary flux. By (31.21), its limit is

$$(31.22) \quad 6\omega(u_\infty E' + X_\infty^i P'_i).$$

For clarity, the leading integrand is

$$u_\infty(\partial_j h_{ij} - \partial_i h_{jj}) + 2X_\infty^j(p_{ij} - (\text{tr}_\delta p)\delta_{ij}).$$

This is exactly (31.22) with the energy and momentum normalizations of Equations (22.1)–(22.2). Every other boundary term contains a decaying background or adjoint coefficient, or its derivative, paired with $h = O_2(r^{-2})$ or $p = O_1(r^{-3})$ at the corresponding order. Multiplication by the sphere area $O(r^3)$ makes each such flux tend to zero. The volume pairing converges as well: the prototype constraint derivatives are $O(r^{-4-q})$ and the moment densities are bounded.

The scalar prototype $h = r^{-2}\delta$, $p = 0$, has $(E', P') = (1, 0)$. For the tensor prototype indexed by a constant vector e , its Euclidean trace reversal contracts with the sphere normal to $r^{-3}e$, and thus has $(E', P') = (0, e/3)$. These independent charges identify $(u_\infty, X_\infty) = (b^0, b)$ from (31.11) and (31.22). This proves (31.5).

If u vanished at an interior point, its nonnegativity would give $du = 0$ there. Taylor's theorem and $|X| \leq u$ would give $X = 0$ and $\nabla X = 0$ at the same point. The whole first jet would vanish, contradicting Lemma 31.3 and $u_\infty = b^0 > 0$. $\square$

31.3. Boundary fluxes and the two lapse cases.

Lemma 31.6 (Boundary equations). *The multiplier measure at S is $d\eta = 2u dA_g$, and (31.6) holds.*

Proof. Take arbitrary compact pure tensor variations up to S . The actual outward normal of the exterior domain at its inner boundary is $-\nu$. Integration by parts in (31.11) therefore gives

$$-2 \int_S [p(X, \nu) - (\text{tr}_g p) X_\nu] dA_g - \int_S (\text{tr}_S p) d\eta = 0.$$

Arbitrary mixed components $p_{\nu A}$ imply $X_{TS} = 0$, and arbitrary tangential trace implies $d\eta = 2X_\nu dA_g$.

Now take $(h, p) = (2\psi g, \psi K)$ with arbitrary compact support up to S . At a MOTS, its expansion derivative is $3\partial_\nu \psi$, while its area derivative is $3 \int_S \psi dA_g$. Integrating its constraint pairing with (31.3) gives

$$-3c \int_S \psi dA_g = 6 \int_S [u \partial_\nu \psi - \{\partial_\nu u + K(X, \nu)\} \psi] dA_g - 3 \int_S \partial_\nu \psi d\eta.$$

The boundary value and normal derivative of ψ can be prescribed independently. Their coefficients give $d\eta = 2u dA_g$ and $\partial_\nu u + K(X, \nu) = c/2$. Together with the tensor-variation conclusions these are exactly (31.6). $\square$

Lemma 31.7 (Boundary lapse dichotomy). *Either $u > 0$ at every point of S or $u = 0$ identically on S .*

Proof. Let L_S be the expansion variation at S under normal motion with speed $s\nu$. Its principal part is $-\Delta_S$. More explicitly, in a geodesic normal extension of ν , if II is the second fundamental form of S in (Ω, g) ,

$$L_S s = -\Delta_S s + 2K(\nu, \nabla_S s) + \left[-|\text{II}|^2 - \text{Ric}_g(\nu, \nu) + \text{tr}_{TS}(\nabla_\nu K) \right] s.$$

Only the smoothness of the lower coefficients and the displayed principal part will be needed.

Extend $s\nu$ to a smooth compactly supported vector field Y_1 up to the boundary, and use $H = (\mathcal{L}_{Y_1} g, \mathcal{L}_{Y_1} K)$ in (31.11). These are legitimate one-sided tensor variations; one may compute their jets using any smooth extension across S . Their area and expansion derivatives are respectively $\int_S H s dA_g$ and $L_S s$.

We check the active-ray derivative with the specified vector identification. At a fixed interior point put $\mathcal{A}(v) = \nabla_v Y_1$, so $h^\sharp = \mathcal{A} + \mathcal{A}^*$, where the adjoint uses g . Let ϕ_t be the local flow of Y_1 and let v_t be the isometrically identified test vector for $g_t = \phi_t^* g$. Naturality gives

$$C_{\phi_t^* g, \phi_t^* K}(x, v_t) = C_{g, K}(\phi_t x, d\phi_t(v_t)).$$

At $t = 0$, the covariant velocity of the vector on the right, relative to parallel transport along $\phi_t x$, is

$$\mathcal{A}(v) + \dot{v} = \mathcal{A}(v) - \frac{1}{2}(\mathcal{A} + \mathcal{A}^*)v = \frac{1}{2}(\mathcal{A} - \mathcal{A}^*)v.$$

It is perpendicular to v . The base derivative of C , with v parallel transported, vanishes at an interior active ray because it differentiates a nonnegative function at a two-sided interior minimum. The remaining vector derivative is $2J(\frac{1}{2}(\mathcal{A} - \mathcal{A}^*)v)$, also zero: at an active unit ray $J = -\mu v^\flat$, and at an active ray with $|v| < 1$ we have $\mu = J = 0$. Therefore $C'_H = 0$ at every interior active ray.

The vector field Y_1 need not preserve S : its compact Lie derivatives are permitted tensor variations, independently of feasibility of a flow on the whole exterior. No normal derivative at a one-sided boundary minimum is being set equal to zero. Such boundary rays contribute

no integral because Λ has zero mass over S by Lemma 31.3. Thus the entire constraint pairing vanishes. The multiplier identity and $d\eta = 2u dA_g$ now read

$$-c \int_S H s dA_g = -2 \int_S u L_{SS} dA_g,$$

and hence give

$$(31.23) \quad 2L_S^* u = cH.$$

Here the adjoint is with respect to dA_g on the closed manifold S .

Outer area minimization, tested against every nonnegative outward normal speed, gives $H \geq 0$ pointwise. Hence $L_S^* u \geq 0$ and $u \geq 0$ on S . Increase the zeroth coefficient of L_S^* by a constant until that coefficient is nonnegative. The inequality remains valid because $u \geq 0$. The strong minimum principle for this operator with principal part $-\Delta_S$ shows that a zero of u forces u to vanish on the connected surface S . Otherwise it is everywhere positive. $\square$

31.4. The constructed spacetime and its null stress.

Lemma 31.8 (Killing development and timelike collars). *The metric in (31.7) has all the properties in parts (iv) and (v) of Theorem 31.1.*

Proof. The interior positivity of u makes (31.7) a smooth Lorentzian metric, with future normal $n = u^{-1}(\partial_t - X)$ after choosing the indicated time orientation. In the convention of the problem, the second fundamental form of a lapse-shift metric is

$$\frac{1}{2u}(\partial_t g - \mathcal{L}_X g).$$

Its coefficients here are independent of t , and (31.2) identifies this tensor with the original K . The vector field $\xi = \partial_t$ is Killing by construction.

We record the curvature computation to fix both the matter term and the sign convention. The Gauss and normal-variation identities for a general lapse-shift metric, with positive second fundamental form convention, are

$$\begin{aligned} \text{Ric}_{\mathbf{g},ij} &= \text{Ric}_{g,ij} + \tau K_{ij} - 2(K^2)_{ij} + u^{-1}(\partial_t K_{ij} - (\mathcal{L}_X K)_{ij} - (\nabla^2 u)_{ij}), \\ \text{Ric}_{\mathbf{g}}(n, n) &= -u^{-1}(\partial_t \tau - X\tau) - |K|^2 + u^{-1}\Delta u. \end{aligned}$$

Tracing these equations uses

$$g^{ij}(\partial_t K_{ij} - (\mathcal{L}_X K)_{ij}) = \partial_t \tau - X\tau + 2u|K|^2.$$

In the stationary case this yields

$$(31.24) \quad \text{Ric}_{\mathbf{g},ij} = \text{Ric}_{g,ij} + \tau K_{ij} - 2(K^2)_{ij} - u^{-1}((\mathcal{L}_X K)_{ij} + (\nabla^2 u)_{ij}),$$

$$(31.25) \quad R_{\mathbf{g}} = R_g + \tau^2 + |K|^2 - 2u^{-1}(\Delta u + X\tau).$$

Equation (31.16) and complementarity show that

$$R_{\mathbf{g}} = 2\mu + \frac{2J(X)}{u} = 0.$$

Writing $\mathbf{G}$ for the Einstein tensor, the constraints and (31.4) consequently give

$$(31.26) \quad \mathbf{G}(n, n) = \mu, \quad \mathbf{G}(n, e_i) = J_i, \quad u\mathbf{G}_{ij} = -X_{(i}J_{j)}.$$

These identities can also be compared with the transversal Killing construction in [13]; its momentum symbol has the opposite sign to the one used here. The null-fluid structure of a modified constraint adjoint is discussed in [30, Section 6.1]. We use the displayed direct calculation, in four spatial dimensions, rather than importing a stationarity or dimensional-transfer theorem.

If $\mu = 0$, the dominant energy condition gives $J = 0$, so all entries in (31.26) vanish. If $\mu > 0$, the inequalities

$$0 = u\mu + J(X) \geq u\mu - |J||X| \geq 0$$

force $|J| = \mu$, $|X| = u$, and $J = -\mu X^\flat/u$. Since $\mathbf{g}(\xi, n) = -u$ and $\mathbf{g}(\xi, e_i) = X_i$, the tensor with components (31.26) is exactly

$$(31.27) \quad \mathbf{G} = \frac{\mu}{u^2} \xi^\flat \otimes \xi^\flat \quad \text{where } \mu > 0.$$

The Killing one-form in this equation is the spacetime one-form. It is null wherever the displayed tensor is nonzero. Thus $\mathbf{G}(\xi, \cdot) = 0$ everywhere. Since $R_{\mathbf{g}} = 0$, this is the first assertion of (31.8). If $N > 0$, then $|X| < u$, so complementarity forces $\mu = J = 0$; all the Ricci components vanish. This proves the second assertion. An internal null region has not been excluded or assumed vacuum.

It remains to examine the original boundary and the end. Equation (31.5) gives $N \rightarrow (b^0)^2 - |b|^2 = 1$. At S , Equation (31.6) gives $X = u\nu$, so $N = 0$ and all tangential derivatives of N vanish. If $u > 0$ on S , Equation (31.2) gives $\langle \nabla_\nu X, \nu \rangle = -uK(\nu, \nu)$. Hence

$$\partial_\nu N = 2u(\partial_\nu u + uK(\nu, \nu)) = 2u\kappa,$$

which is (31.9). Compactness of S supplies a full collar on which $N > 0$.

In the other case, $u|_S = 0$ and $X|_S = 0$. Tangential derivatives of X vanish. The normal-normal and normal-tangential components of (31.2) then give $\nabla_\nu X = 0$ on S , so every first derivative of X vanishes there. Smooth one-sided division by normal distance gives $u = sd$ and $X = s^2 Y_2$. The boundary condition gives $d|_S = \partial_\nu u|_S = \kappa > 0$. This is (31.10), and once again compactness gives a full collar with $N > 0$. Together with positivity far out on the unique end, these collars place all interior zeros of N in an interior compact set. $\square$

Proof of Theorem 31.1. Combine Lemmas 31.2, 31.3, 31.5, 31.6, 31.7, and 31.8. $\square$

32. VANISHING TWIST AND THE COMPLETE STATIC BASE

We continue with the equality data and the fields supplied by Theorem 31.1. The stationary metric

$$\mathbf{g} = -u^2 dt^2 + g_{ij}(dx^i + X^i dt)(dx^j + X^j dt) \quad \text{on } \mathbb{R} \times \text{int } \Omega$$

is smooth because $u > 0$ in the interior. Its Killing field is $\xi = \partial_t$. Recall that

$$N = u^2 - |X|_g^2 \geq 0, \quad \text{Ric}_{\mathbf{g}}(\xi, \cdot) = 0, \quad \text{Ric}_{\mathbf{g}} = 0 \quad \text{where } N > 0.$$

Moreover, $N \rightarrow 1$ at the end, and $N > 0$ on a full interior collar of S . Thus the interior zero set

$$Z = \{x \in \text{int } \Omega : N(x) = 0\}$$

is compactly contained in $\text{int } \Omega$. No regularity of Z as a subset is assumed. In particular, it may have interior or infinitely many components.

32.1. The Killing current near a null set. On the open set $\mathcal{P} = \{N > 0\} \subset \text{int } \Omega$, put

$$(32.1) \quad \begin{aligned} h_b &= g + N^{-1} X^\flat \otimes X^\flat, & C_b &= g^{-1} - u^{-2} X \otimes X = h_b^{-1}, \\ A_b &= N^{-1} X^\flat, & F &= dA_b, & \lambda &= \sqrt{N}. \end{aligned}$$

Here $X^\flat$ is formed with g . Direct substitution gives

$$(32.2) \quad \mathbf{g} = -N(dt - A_b)^2 + h_b.$$

The tensor C_b extends smoothly and nonnegatively through Z , although h_b and A_b need not extend there. For a covector α write $|\alpha|_{C_b}^2 = C_b^{ij} \alpha_i \alpha_j$.

Lemma 32.1 (Killing-current identities). *Let $\mathcal{G}_{\alpha\beta} = \nabla_\alpha^{\mathbf{g}} \xi_\beta$. On the interior of Ω one has*

$$(32.3) \quad u^{-1} \operatorname{div}_g(u C_b dN) = -2|\mathcal{G}|_{\mathbf{g}}^2.$$

On $\mathcal{P}$ the antisymmetric tensor

$$(32.4) \quad Q^{ij} = u N C_b^{ik} C_b^{j\ell} F_{k\ell} \quad \text{satisfies} \quad \nabla_i^g Q^{ij} = 0.$$

The norm in (32.3) is the Lorentzian contraction of both tensor indices; it need not be nonnegative.

Proof. The Killing identities and $\operatorname{Ric}_{\mathbf{g}}(\xi, \cdot) = 0$ give

$$\nabla^{\mathbf{g}, \alpha} \mathcal{G}_{\alpha\beta} = 0, \quad \square_{\mathbf{g}}(\mathbf{g}(\xi, \xi)) = 2|\mathcal{G}|_{\mathbf{g}}^2.$$

For a stationary scalar function the wave operator is $u^{-1} \operatorname{div}_g(u C_b d\cdot)$: the spacetime volume density is $u\sqrt{\det g}$ and its spatial inverse-metric block is C_b . Since $\mathbf{g}(\xi, \xi) = -N$, this proves (32.3), including at Z where all its entries are smooth.

For the second identity set $\theta = dt - A_b$. On $\mathcal{P}$,

$$\xi^\flat = -N\theta, \quad \mathcal{G} = \frac{1}{2} d\xi^\flat = \frac{1}{2} (-dN \wedge \theta + NF).$$

The inverse-metric dual of θ is proportional to ∂_t , so the term containing θ has zero spatial-spatial component after both indices are raised. Consequently

$$\mathcal{G}^{ij} = \frac{N}{2} C_b^{ik} C_b^{j\ell} F_{k\ell}.$$

The stationary divergence identity, with density $u\sqrt{\det g}$, now gives (32.4). In converting coordinate divergence to g -covariant divergence, the additional Christoffel term vanishes by antisymmetry. $\square$

Lemma 32.2 (A transverse logarithmic estimate). *Suppose $Z \neq \emptyset$. On a neighborhood of Z with compact closure in the interior, let*

$$B = |X|_g, \quad e = X/B, \quad D = e^\perp,$$

and denote by $d_D N$ the g -orthogonal projection of dN to D . This neighborhood can be chosen so that u and B are bounded below by positive constants. For every smooth cutoff ζ compactly supported there,

$$(32.5) \quad \sup_{\varepsilon > 0} \int u \zeta^2 \frac{|dN|_{C_b}^2}{(N + \varepsilon)^2} dV_g < \infty.$$

In particular, $d_D \log N$ is square-integrable on the positive set in every smaller neighborhood of Z .

Proof. At a point of Z , $B = u > 0$. Compactness provides the positive lower bounds after shrinking a neighborhood. On that neighborhood the eigenvalues of C_b relative to g are 1 on D and N/u^2 in the e direction. Thus

$$(32.6) \quad |dN|_{C_b}^2 = |d_D N|_g^2 + \frac{N}{u^2} (eN)^2.$$

Smoothness and nonnegativity of N also give

$$(32.7) \quad |dN|_g^2 \leq CN.$$

For completeness, work in a slightly larger compact interior neighborhood, where the Hessian is bounded. Taylor's inequality along a short geodesic in the direction $-\nabla N$ gives $0 \leq N - t|dN| + Ct^2/2$. Increasing C so that $t = |dN|/C$ stays in this neighborhood proves (32.7).

Let $n = u^{-1}(\xi - X)$ be the future unit normal to the original slice. In a local orthonormal frame n, e, e_a , with the e_a in D , differentiating $\mathbf{g}(\xi, \xi) = -N$ gives

$$\mathcal{G}_{in} = -\frac{N_i/2 + B\mathcal{G}_{ie}}{u} \quad \text{for each spatial index } i.$$

As $|\mathcal{G}|_{\mathbf{g}}^2 = 2 \sum_{i < j} \mathcal{G}_{ij}^2 - 2 \sum_i \mathcal{G}_{in}^2$, we obtain the exact expansion

$$(32.8) \quad \begin{aligned} -2|\mathcal{G}|_{\mathbf{g}}^2 &= \frac{|dN|_g^2}{u^2} + \frac{4B}{u^2} \sum_a N_a \mathcal{G}_{ae} - \frac{4N}{u^2} \sum_a \mathcal{G}_{ae}^2 - 4 \sum_{a < b} \mathcal{G}_{ab}^2 \\ &\leq C(N + |d_D N|_g). \end{aligned}$$

All coefficients of $\mathcal{G}$ are bounded here. This estimate is pointwise and does not require a global frame for D .

Write $f = -2|\mathcal{G}|_{\mathbf{g}}^2$ and test $\operatorname{div}_g(uC_b dN) = uf$ against $\zeta^2/(N + \varepsilon)$. There is no singular boundary in this test: Equation (32.3) is smooth across Z , and $\varepsilon > 0$. Integration by parts gives

$$(32.9) \quad \begin{aligned} \int u \zeta^2 \frac{|dN|_{C_b}^2}{(N + \varepsilon)^2} dV_g &= \int \frac{u \zeta^2 f}{N + \varepsilon} dV_g \\ &\quad + 2 \int \frac{u \zeta \langle d\zeta, dN \rangle_{C_b}}{N + \varepsilon} dV_g. \end{aligned}$$

In the first term use (32.8), $N/(N + \varepsilon) \leq 1$, and $|d_D N| \leq |dN|_{C_b}$. Young's inequality absorbs its derivative term into, say, one quarter of the left side. The second integral is handled by Cauchy's inequality for the nonnegative tensor C_b and another quarter of that side. The remaining terms are bounded by a constant times $\int u(\zeta^2 + |d\zeta|_{C_b}^2) dV_g$, uniformly in ε . This proves (32.5). Fatou's lemma and (32.6) give the last assertion. $\square$

32.2. Removal of all twist boundary terms.

Proposition 32.3 (Vanishing of the twist). *On all of $\mathcal{P}$, including every component, $F = dA_b = 0$.*

Proof. On the asymptotic end the constant limit of A_b is the covector $a_\infty = b^i dx^i$, because $N \rightarrow 1$ and $g \rightarrow \delta$. Choose a smooth function f on Ω which equals $b^i x^i$ sufficiently far out and is zero off an end neighborhood disjoint from S and Z . Then

$$\beta = A_b - df, \quad d\beta = F, \quad \beta = O(r^{-q_0}) \quad \text{at infinity.}$$

For any nonnegative smooth function χ compactly supported in $\mathcal{P}$, multiply (32.4) by $\chi\beta_j$ and integrate. Antisymmetry gives

$$(32.10) \quad \frac{1}{2} \int_{\mathcal{P}} \chi u N C_b^{ik} C_b^{j\ell} F_{ij} F_{k\ell} dV_g = - \int_{\mathcal{P}} Q^{ij} (\partial_i \chi) \beta_j dV_g.$$

The integrand on the left is nonnegative, and its quadratic form in F is positive definite at each point of $\mathcal{P}$.

The contraction in the right side has a cancellation that is needed at the null loci. Lowering its remaining index with g , one finds

$$(32.11) \quad \begin{aligned} g_{ik} Q^{kj} (A_b)_j &= \frac{N}{u} F_{ij} X^j \\ &= \frac{1}{u} \left[X^j (dX^b)_{ij} - \frac{|X|^2 N_i - X(N) X_i}{N} \right]. \end{aligned}$$

Indeed $C_b A_b = X/u^2$, and the projection in the other raised index does not change $F_{ij} X^j$. This covector is orthogonal to X . Where $X \neq 0$, its last numerator is $B^2(d_D N)_i$. The coordinate-free numerator in (32.11) will also be used where $X = 0$.

We choose cutoffs near the three possible escape loci with disjoint transition neighborhoods. If $Z = \emptyset$, the first cutoff below is identically one.

The interior null set. Choose nested relatively compact interior neighborhoods of Z on which u and B are bounded below and Lemma 32.2 applies. A smooth cutoff η is one on the smaller neighborhood and supported in the larger. Let $\vartheta : [0, \infty) \rightarrow [0, 1]$ be smooth, zero on $[0, 1]$ and one on $[2, \infty)$, with bounded derivative, and set

$$\chi_{Z,\varepsilon} = 1 - \eta + \eta\vartheta(N/\varepsilon).$$

For all sufficiently small ε , N has a positive lower bound on the support of $d\eta$, so differentiation of η adds no term. The only transition is $\varepsilon < N < 2\varepsilon$, and its derivative is bounded by $C|dN|/\varepsilon$. Here $df = 0$. Orthogonality in (32.11) replaces dN by $d_D N$ in the contraction. Since $N \asymp \varepsilon$ on the band and the smooth term $X \lrcorner dX^\flat$ is bounded, the absolute contribution to the right side of (32.10) is at most

$$(32.12) \quad C \int_{\{\varepsilon < N < 2\varepsilon\} \cap \text{supp } \eta} (1 + |d_D \log N|_g^2) dV_g = o(1).$$

The integrable majorant comes from Lemma 32.2. The characteristic functions of these bands tend pointwise to zero even at points of Z . Thus dominated convergence proves the last assertion without an assumption on the measure or structure of Z .

The asymptotic end. Let $\chi_{\infty,R}$ be one for $r \leq R$, zero for $r \geq 2R$, with derivative bounded by C/R , and extend it by one over the compact part. From the adjoint end estimates,

$$Q = O(r^{-1-q_0}), \quad \beta = O(r^{-q_0}).$$

The transition annulus has volume $O(R^4)$, so its contribution is

$$(32.13) \quad O(R^{2-2q_0}) = o(1),$$

since $q_0 > 1$.

The original boundary. Let s be original g -normal distance from S , increasing into Ω , and let $\chi_{S,\sigma}$ change from zero to one on $\sigma < s < 2\sigma$, with derivative bounded by C/σ . Again $df = 0$ near this collar. If $u|_S > 0$, the boundary conclusions of Theorem 31.1 give

$$dN = 2\kappa X^\flat \quad \text{on } S, \quad X = u\nu \quad \text{on } S, \quad N \asymp s.$$

In particular $d_D N = O(s)$, so (32.11) is bounded. Its contraction with the normal ∇s is $O(s)$: it annihilates X , and $\nabla s - X/u = O(s)$. Multiplication by $1/\sigma$ and integration over a collar of volume $O(\sigma)$ therefore gives $O(\sigma)$.

If $u|_S = 0$, the same theorem supplies smooth one-sided fields d, Y_2 with

$$(32.14) \quad u = sd, \quad X = s^2 Y_2, \quad d|_S = \kappa, \quad N = s^2(d^2 - s^2|Y_2|^2).$$

Thus $u \asymp s$, $N \asymp s^2$, $dX^\flat = O(s)$, and $dN = O(s)$. Both terms inside the brackets of (32.11) are $O(s^3)$; its entire value is $O(s^2)$. This gives a boundary cutoff cost $O(\sigma^2)$.

Now take $\chi = \chi_{Z,\varepsilon} \chi_{S,\sigma} \chi_{\infty,R}$. For positive parameters with sufficiently small ε, σ and large R , it is a compactly supported smooth test on $\mathcal{P}$. Multiplication by the other two factors does not increase any of the three estimates. Therefore the absolute value of the right side of (32.10) tends to zero as $\varepsilon, \sigma \downarrow 0$ and $R \rightarrow \infty$. Choose a sequence of such parameters; its cutoffs tend pointwise to one on $\mathcal{P}$. Fatou's lemma gives zero for the integral of the nonnegative density on the left. Its continuity and pointwise positive definiteness force $F = 0$ everywhere on $\mathcal{P}$. This integration is on an open manifold with compactly supported tests and remains valid if $\mathcal{P}$ has infinitely many components. $\square$

32.3. Static equations and the only finite-distance attachment. Let $\mathcal{U}$ be the component of $\mathcal{P}$ containing the distant asymptotic end. Since $F = 0$ on all of $\mathcal{P}$, the Poincaré lemma supplies a local primitive of A_b on every sufficiently small ball in any positive component. Replacing t locally by $T = t - a$ with $da = A_b$ turns (32.2) into $-\lambda^2 dT^2 + h_b$. Its Ricci components are

$$(\text{Ric}_{\mathbf{g}})_{TT} = \lambda \Delta_{h_b} \lambda, \quad (\text{Ric}_{\mathbf{g}})_{ij} = (\text{Ric}_{h_b})_{ij} - \lambda^{-1} (\text{Hess}_{h_b} \lambda)_{ij}.$$

These follow directly from the nonzero mixed Christoffel symbols $\Gamma_{Ti}^T = \partial_i \log \lambda$ and $\Gamma_{TT}^i = \lambda h_b^{ij} \partial_j \lambda$. Vacuum on $\mathcal{P}$ therefore gives, on every positive component,

$$(32.15) \quad \lambda \text{Ric}_{h_b} = \text{Hess}_{h_b} \lambda, \quad \Delta_{h_b} \lambda = 0, \quad \text{Scal}_{h_b} = 0.$$

This conclusion uses local primitives only; no global time primitive or topological conclusion has yet been used.

Lemma 32.4 (Connectedness of the positive set). *The set $\mathcal{P}$ is connected and equals $\mathcal{U}$. In particular, the positive collar of S belongs to $\mathcal{U}$.*

Proof. The unique asymptotic end lies in $\mathcal{U}$ outside a compact set. Any other component V of $\mathcal{P}$ therefore has compact closure in the original exterior Ω , with its boundary included. Its frontier lies in $S \cup Z$: an interior frontier point with $N > 0$ would have a connected positive neighborhood meeting V and hence belong to V . The frontier is nonempty, since otherwise V would be both open and closed in the connected interior of Ω , which also contains $\mathcal{U}$.

The continuous function $\lambda = \sqrt{N}$ vanishes on this frontier and is positive in V . It therefore attains a positive maximum at an interior point of V . The strong maximum principle for $\Delta_{h_b} \lambda = 0$ makes λ constant on V , contradicting its zero frontier values. Thus there is no such component V . The collar supplied by Theorem 31.1 lies in $\mathcal{P} = \mathcal{U}$, as claimed. $\square$

Proposition 32.5 (Completeness and regular horizon attachment). *The component $\mathcal{U}$ satisfies $0 < \lambda < 1$. Every approach in $\mathcal{U}$ to an interior zero of N has infinite h_b -length. Adjoining S gives a complete manifold with smooth one-sided metric h_b and lapse λ , in the boundary structure specified below, and*

$$(32.16) \quad h_b|_{TS} = g|_{TS}, \quad \lambda|_S = 0, \quad \partial_{\nu_{h_b}} \lambda = \kappa, \quad \Pi_{h_b} = 0.$$

Here ν_{h_b} points into the base exterior. The metric has an even reflection and the lapse an odd reflection of class at least C^2 . More precisely:

- (i) If $u|_S > 0$, then N is a defining function in the original smooth structure and $dN = 2\kappa X^\flat$ on S . The base boundary structure uses $\lambda = \sqrt{N}$ in place of N . In that structure the reflected metric and signed lapse are smooth.
- (ii) If $u|_S = 0$, the one-sided base metric and lapse are smooth in the original boundary structure. Their even and odd reflections in base normal distance are C^2 .

In either case the base attachment is homeomorphic to the original attachment of S . There are no other finite-distance boundary attachments.

Proof. First consider an interior approach to Z . With the notation of Lemma 32.2, Equation (32.11) and $F = 0$ give

$$B^2 d_D \log N = X^j (dX^\flat)_{ij} dx^i, \quad |d_D \log N|_g \leq C.$$

Together with (32.7) this yields

$$(32.17) \quad |d \log N|_{h_b}^2 = |d_D \log N|_g^2 + \frac{(eN)^2}{u^2 N} \leq C$$

on the positive set near Z . Along any curve approaching Z , $\log N \rightarrow -\infty$, which is incompatible with finite length and (32.17).

The original space (Ω, g) , with its boundary included, is a complete locally compact length space and hence proper. Since $h_b \geq g$ on $\mathcal{U}$, a finite-length base curve is g -Cauchy and has a limit in Ω . A limit in the interior with $N > 0$ lies in $\mathcal{U}$ and is a regular base point. A limit in Z is excluded by the preceding bound. Escape to the asymptotic end also has infinite length. Thus the only possible finite-distance attachment is S , whose positive collar belongs to $\mathcal{U}$ by Lemma 32.4.

Next, λ tends to one at infinity, and it tends to zero at any finite limiting boundary of $\mathcal{U}$ in the original manifold. If $\lambda > 1$ somewhere, continuity on the compact part of the original space and its end limit give an attained interior maximum. The strong maximum principle in (32.15) would make λ constant on $\mathcal{U}$, contradicting its limit one. Hence $\lambda \leq 1$. If equality held at an interior point, the same principle would give $N \equiv 1$ on $\mathcal{U}$. Such a component has no interior frontier, because continuity would give $N = 1 > 0$ at each frontier point. It is therefore both open and closed in the connected $\text{int } \Omega$, so it is the entire interior. This contradicts $N \rightarrow 0$ at S . Thus $0 < \lambda < 1$.

It remains to construct and check the attachment of S .

Positive boundary lapse. Here $dN = 2u\kappa\nu^b$ on S , with $u > 0$, so (N, y^A) are original local boundary coordinates. The identity $dN = 2\kappa X^b$ at $N = 0$ and smooth division give

$$X^b = a dN + N\beta_A dy^A, \quad a(0, y) = \frac{1}{2\kappa},$$

where a and β_A are smooth in (N, y) . Passing to (λ, y) , with $N = \lambda^2$, the base metric components are

$$\begin{aligned} (h_b)_{\lambda\lambda} &= 4\lambda^2 g_{NN} + 4a^2, \\ (h_b)_{\lambda A} &= 2\lambda(g_{NA} + a\beta_A), \\ (h_b)_{AB} &= g_{AB} + \lambda^2\beta_A\beta_B. \end{aligned} \tag{32.18}$$

All original coefficients on the right are evaluated at (λ^2, y) . The first and third expressions are smooth even functions of signed λ , and the middle one is smooth odd. They therefore define a smooth metric invariant under $(\lambda, y) \mapsto (-\lambda, y)$. At the boundary the cross terms vanish and $(h_b)_{\lambda\lambda} = \kappa^{-2}$; the tangential metric is $g|_{TS}$. Nondegeneracy holds on both sides after restricting the collar. The reflection fixes S , so its second fundamental form vanishes, and its inward unit normal is $\kappa\partial_\lambda$ there. This proves (32.16). The lapse is the signed coordinate λ , hence is smooth and odd. Changes between these boundary charts are smooth after the substitution $N = \lambda^2$, so the construction is compatible on overlapping collars.

Zero boundary lapse. Equation (32.14) gives

$$h_b = g + \frac{s^2 Y_2^b \otimes Y_2^b}{d^2 - s^2 |Y_2|^2}, \quad \lambda = s \sqrt{d^2 - s^2 |Y_2|^2}.$$

Both are smooth one-sided in the original boundary structure, $h_b = g$ at S , and $\partial_{\nu_{h_b}} \lambda = \kappa$. Since Ric_{h_b} is smooth up to S , the first equation in (32.15) extends continuously and gives $\text{Hess}_{h_b} \lambda = 0$ there. Its tangential part equals $\kappa \Pi_{h_b}$, proving that the second form vanishes. Use base Gaussian coordinates (ρ, y) , so $h_b = d\rho^2 + h_{AB}(\rho, y) dy^A dy^B$. At $\rho = 0$ we have

$$\partial_\rho h_{AB} = 0, \quad \lambda = 0, \quad \partial_\rho^2 \lambda = 0.$$

Even reflection of h_{AB} and odd reflection of λ thus match derivatives through order two. This proves the claimed C^2 reflection and all of (32.16).

These constructions give a positive one-sided smooth base metric up to S . In the first case replacing $N \geq 0$ by $\lambda = \sqrt{N} \geq 0$ is a homeomorphism; in the second case the original structure is retained. The attachment therefore has the same underlying topology as the original one. If a base Cauchy sequence approaches S in the original topology, the corresponding regular

base coordinates converge, so it converges in the attached base metric as well. Combining this observation with the finite-length argument above proves completeness with S included. $\square$

33. CLASSIFICATION OF THE FOUR-DIMENSIONAL STATIC BASE

We now classify the static base obtained in Proposition 32.5. The positive set $\mathcal{U} = \{N > 0\}$ is connected and contains the boundary collar. On $\mathcal{U}$ the metric h_b and the lapse $\lambda = \sqrt{N}$ satisfy

$$(33.1) \quad \lambda \operatorname{Ric}_{h_b} = \operatorname{Hess}_{h_b} \lambda, \quad \Delta_{h_b} \lambda = 0, \quad \operatorname{Scal}_{h_b} = 0, \quad 0 < \lambda < 1.$$

The base is complete with S attached; approaches to interior zeros of N have infinite base length. Its boundary data are

$$(33.2) \quad h_b|_{TS} = g|_{TS}, \quad \lambda|_S = 0, \quad \partial_{\nu_{h_b}} \lambda = \kappa, \quad \Pi_{h_b} = 0, \quad \kappa = \frac{1}{\sqrt{2m}}.$$

These are conclusions of the preceding argument, not hypotheses on the original data. In particular, neither simple connectivity nor absence of additional complete ends of the base is available yet.

Our goal is both the explicit static metric and the identity $\mathcal{U} = \operatorname{int} \Omega$ in Proposition 33.5. The argument allows additional complete ends until Euclidean rigidity of the conformal double excludes interior null regions and identifies the whole exterior.

33.1. Improvement of the static asymptotics. After a constant linear change of the coordinates at infinity, the positive limiting matrix of h_b becomes the identity. For a fixed $q_0 \in (1, \min\{q, 2\})$ we then have $h_b - \delta = O_2(r^{-q_0})$ and $\lambda - 1 = O_2(r^{-q_0})$. Introduce

$$(33.3) \quad U = \log \lambda, \quad \gamma = \lambda h_b.$$

The conformal length factor from h_b to γ is $e^{U/2}$. The four-dimensional Ricci transformation and (33.1) give

$$(33.4) \quad \operatorname{Ric}_\gamma = \frac{3}{2} dU \otimes dU, \quad \Delta_\gamma U = 0.$$

Indeed, $\operatorname{Ric}_{h_b} = \operatorname{Hess}_{h_b} U + dU \otimes dU$ and $\Delta_{h_b} U = -|dU|_{h_b}^2$, which yield both identities directly.

Lemma 33.1 (Static end expansion). *There are coordinates harmonic for γ on a sufficiently distant part of the end, a number $\epsilon \in (0, 1)$, a constant $M_1 > 0$, and homogeneous harmonic polynomials H_1, H_2 of degrees one and two such that, for every fixed nonnegative integer k ,*

$$(33.5) \quad \gamma - \delta = O_k(r^{-2-\epsilon}),$$

$$(33.6) \quad U = -\frac{M_1}{r^2} + \frac{H_1(x)}{r^4} + \frac{H_2(x)}{r^6} + O_k(r^{-4-\epsilon}).$$

In addition $\gamma - \delta = O_k(r^{-3})$.

Proof. We give the coordinate and expansion arguments, since the initial falloff alone would not justify the compactification below.

Extend γ from a sufficiently distant region to a smooth metric on $\mathbb{R}^4$, using a cutoff interpolation with δ . Write its Laplacian as

$$\Delta_\gamma = \Delta_\delta + D^{ij} \partial_i \partial_j + D^i \partial_i.$$

By making the interpolation sufficiently far out, the scaled $C^{0,\alpha}$ norms of D^{ij} and $(1+r)D^i$ are as small as desired, for any fixed $\alpha \in (0, 1)$. Here a scaled norm means the ordinary norm after rescaling each annulus to fixed size. The initial differentiated falloff gives the needed Hölder estimates: second derivatives bound the scaled Hölder seminorm of first derivatives. Moreover $D^i = O(r^{-1-q_0})$ in these scaled norms.

Put $a = 1 - q_0 \in (-1, 0)$. On all of $\mathbb{R}^4$, with weight $1 + r$, the Euclidean Newton operator maps scaled $C^{0,\alpha}$ functions of order $a - 2$ to scaled $C^{2,\alpha}$ functions of order a . For completeness, the zeroth-order estimate follows from the kernel $C|x - y|^{-2}$ by splitting the integral into $|y| < r/2$, a comparable annulus, and $|y| > 2r$; convergence uses $-2 < a < 0$. Interior elliptic estimates after rescaling give the two derivative and Hölder bounds. The small coefficient norms therefore make the map obtained from

$$\Delta_\delta v^i = -D^i - D^{ab}\partial_a\partial_b v^i - D^a\partial_a v^i$$

a contraction in that weighted space. Its solution satisfies $v = O(r^{1-q_0})$ with the stated scaled $C^{2,\alpha}$ bounds, and $x^i + v^i$ are γ -harmonic coordinates on the far end. They are indeed coordinates there: their first derivative tends to the identity, and a cutoff of the correction can be chosen to have globally small first derivative. Local elliptic regularity makes the coordinates smooth wherever the original metric is smooth.

In the harmonic chart, $\gamma - \delta$ and U initially have order $-q_0$ in scaled $C^{1,\alpha}$: one derivative of the transformed metric uses two derivatives of the coordinate correction. We do not assume a weighted second-derivative estimate at this stage. The chart and the fields are nevertheless smooth locally, so Equations (33.4) take the coupled elliptic form

$$(33.7) \quad -\frac{1}{2}\gamma^{ab}\partial_a\partial_b\gamma_{ij} + Q_{ij}(\gamma, \partial\gamma) = \frac{3}{2}U_iU_j, \quad \gamma^{ab}\partial_a\partial_b U = 0,$$

where Q is quadratic in first derivatives, with smooth coefficients depending on γ . On an annulus rescaled to unit size, the initial $C^{1,\alpha}$ norms of $\gamma - \delta$ and U are $O(R^{-q_0})$. Both quadratic right sides have $C^{0,\alpha}$ norm $O(R^{-2q_0})$, and the principal coefficients are uniformly elliptic with bounded $C^{1,\alpha}$ norm. Interior Schauder estimates on a smaller annulus therefore restore $C^{2,\alpha}$ bounds of order R^{-q_0} for both unknowns. Differentiating this coupled system now inductively gives $\gamma - \delta, U = O_k(r^{-q_0})$ for every fixed k . Thus no higher derivative falloff is being assumed in the original coordinates. It follows in particular that

$$(33.8) \quad \Delta_\delta(\gamma - \delta), \Delta_\delta U = O_k(r^{-2-2q_0}).$$

We use the following elementary multipole fact in dimension four. If a decaying smooth function w on an end satisfies

$$\Delta_\delta w = O_k(r^{-4-j-\eta}), \quad j \in \{0, 1, 2, \dots\}, \quad 0 < \eta < 1,$$

then

$$(33.9) \quad w = \sum_{d=0}^j \frac{P_d(x)}{r^{2+2d}} + O_k(r^{-2-j-\eta}),$$

with P_d homogeneous harmonic of degree d . To prove this, cut w off to the end and represent the result by the whole-space Newton potential of its Laplacian. The difference is an entire harmonic function tending to zero, hence vanishes. The source has convergent moments through degree j . Taylor expansion of the kernel in the region $|y| < r/2$ has remainder bounded by $Cr^{-3-j}|y|^{j+1}$; integration, and estimation of the complementary region after subtraction of the same moments, gives the error in (33.9). The local singularity of the Newton kernel is integrable. Rescaled elliptic estimates supply the differentiated error bounds. This also proves the statement simultaneously for each fixed finite number of derivatives.

Choose $\epsilon \in (0, 1)$ with $4 + \epsilon < 2 + 2q_0$. Applying (33.9) first with $j = 0$ gives monopole expansions for $\gamma - \delta$ and U . Write the metric monopole as D_{ij}/r^2 . The harmonic-coordinate condition is

$$\partial_j(\sqrt{\det \gamma} \gamma^{ji}) = 0.$$

Its leading homogeneous term implies

$$(D - \tfrac{1}{2}(\text{tr } D) \text{Id})x = 0 \quad \text{for every } x.$$

Taking the trace in dimension four gives $D = 0$. This proves (33.5). Since $U = O_k(r^{-2})$, its equation now improves to $\Delta_\delta U = O_k(r^{-6-\epsilon})$. The case $j = 2$ of (33.9) gives (33.6), with the sign of its constant temporarily undetermined. In particular, the strict exponent in this source estimate excludes a logarithmic term at quadrupole order. The first use of (33.9) likewise accounts for the entire order r^{-2} tensor term, rather than only its spherical average. The metric equation in (33.7) has Euclidean source $O_k(r^{-6})$ at this stage. The case $j = 1$ of (33.9), with any smaller positive exponent if necessary, gives $\gamma - \delta = O_k(r^{-3})$. No expansion of the metric beyond this order is asserted or needed.

Finally $-U$ is a positive γ -harmonic function. On a sufficiently distant region the function $r^{-2} + r^{-2-\epsilon}$ is positive and subharmonic for γ ; its favorable Euclidean Laplacian dominates the metric error. A small positive multiple lies below $-U$ on a fixed large sphere, and both tend to zero at infinity. The maximum principle therefore keeps it below $-U$ throughout the region. Taking the limit of $r^2(-U)$ proves $M_1 > 0$. $\square$

Remark 33.2. The coefficient in Lemma 33.1 has the expected four-dimensional mass normalization:

$$h_b = e^{-U} \gamma = (1 + M_1/r^2) \delta + O_k(r^{-3}), \quad \lambda = 1 - M_1/r^2 + O_k(r^{-3}).$$

Direct substitution in the energy flux with factor $1/(6\omega_3)$ gives base energy M_1 . We have not identified that chart with the original ADM frame. The equality $M_1 = m$ will instead follow from the boundary constant κ .

33.2. The complete conformal double. Define on $\mathcal{U}$

$$(33.10) \quad k_\pm = \left(\frac{1 \pm \lambda}{2} \right)^2 h_b.$$

The four-dimensional scalar conformal formula is

$$(33.11) \quad \text{Scal}_{f^2 h} = f^{-3}(-6\Delta_h f + \text{Scal}_h f).$$

Thus both k_+ and k_- are scalar flat. In the variables of (33.3), the useful exact identities are

$$(33.12) \quad k_+ = \cosh^2(U/2) \gamma, \quad k_- = \sinh^2(U/2) \gamma.$$

Lemma 33.3 (A zero-mass complete space). *Glue a plus and a minus copy along their common attached boundary S , and add one point at the minus asymptotic end. The resulting space W is a connected orientable smooth boundaryless four-manifold with a complete C^2 scalar-flat metric k_0 . This metric is smooth except possibly at the compact gluing hypersurface and the added point, and has a distinguished end satisfying*

$$(33.13) \quad k_0 - \delta = O_k(r^{-3}) \quad \text{for every fixed } k.$$

No assertion about the other ends of W is needed here.

Proof. The smooth manifold structure is fixed before the metric is smoothed. At the seam use product charts from the regular base collar, with a signed normal coordinate on the double. In the positive boundary-lapse case this coordinate is signed λ ; in the zero boundary-lapse case one can use signed base normal distance. Tangential chart changes come from S , so these charts are smoothly compatible across the seam. Their overlaps with the original open base are smooth away from the seam. The added-point chart below is also smoothly compatible on every punctured overlap. Smoothness of this atlas does not assert smoothness of the reflected metric.

By (33.12) and Lemma 33.1, the plus metric satisfies $k_+ - \delta = O_k(r^{-3})$; in particular its energy is zero.

For the minus end use inversion $x = y/|y|^2$ and put $s = |y|$. Equation (33.6) gives

$$(33.14) \quad \frac{U(x(y))}{s^2} = -M_1 + H_1(y) + H_2(y) + O_k(s^{2+\epsilon}).$$

Differentiating the transformed remainder costs the corresponding powers of s^{-1} . The Jacobian of inversion is $s^{-2}A(y)$, where $A(y) = \text{Id} - 2yy^t/|y|^2$ is orthogonal. Consequently the compactifying expression for k_- is

$$(33.15) \quad \left(\frac{\sinh(U/2)}{s^2} \right)^2 [\text{Id} + A(y)(\gamma(x(y)) - \text{Id})A(y)].$$

The first factor is the product

$$\frac{1}{4} \left(\frac{U}{s^2} \right)^2 \left(\frac{\sinh(U/2)}{U/2} \right)^2.$$

It has a C^2 extension with positive value $M_1^2/4$ at the origin. For the second factor, (33.5) gives $\gamma(x(y)) - \text{Id} = O_k(s^{2+\epsilon})$. Each derivative of the angular matrix A costs at most one power of s^{-1} , so its conjugated error also has continuous derivatives through order two vanishing at the origin. This proves a nondegenerate C^2 extension of (33.15) across the added point. Scalar flatness extends there by continuity.

Proposition 32.5 supplies the even C^2 reflection of h_b and the odd C^2 reflection of λ . The two conformal factors in (33.10) are one expression $((1 + \lambda)/2)^2$ in that signed lapse. They therefore glue to a C^2 scalar-flat metric. Orient the second copy oppositely before gluing. The resulting manifold is orientable, and the punctured-ball chart above extends its orientation over the added point.

It remains to check completeness, including possible escapes toward interior zeros of N . The plus length factor is at least $1/2$. On the minus copy its length factor is bounded below away from the chosen end: if a sequence there had $\lambda \rightarrow 1$, compactness in the original exterior would give either a positive-lapse limiting point with value one, contrary to the strict maximum principle, or a limiting boundary of the positive component, where $\lambda \rightarrow 0$. The latter possibility includes all interior zero loci and the attachment to S . Thus the complete base ends remain complete for both conformal metrics. The seam and the added point are regular metric neighborhoods. These observations exhaust the possible finite-distance escapes by Proposition 32.5 and the completeness of the original exterior. $\square$

33.3. Zero-mass rigidity without assumptions on the other ends. We use only the non-negativity part of a smooth positive-mass theorem. The precise statement needed is Theorem 1.2 of Lesourd–Unger–Yau [36]: a complete smooth orientable manifold of dimension $3 \leq n \leq 7$, with nonnegative scalar curvature and a distinguished asymptotically Schwarzschild end, has nonnegative mass at that end; its other ends are unrestricted. Definition 1.9 in that reference requires the remainder from the Schwarzschild metric to have weighted $C^{2,\alpha}$ order $n - 1$. There is no spin hypothesis. The following argument reduces our C^2 zero-mass situation to exactly that smooth nonnegativity statement.

Lemma 33.4 (Zero-mass rigidity for the constructed space). *Let (W, k_0) have the properties in Lemma 33.3. Then (W, k_0) is isometric to Euclidean $\mathbb{R}^4$. The isometry is smooth on every region where k_0 is smooth, including each smooth one-sided structure at the seam.*

Proof. Suppose first that Ric_{k_0} is nonzero somewhere. By continuity it is nonzero at a point in the smooth part of the metric. Choose a smooth compactly supported symmetric tensor p in that part for which

$$(33.16) \quad I := \int_W \text{Scal}'_{k_0}(p) dV_{k_0} = - \int_W \langle \text{Ric}_{k_0}, p \rangle dV_{k_0} > 0.$$

For example, a negative cutoff multiple of Ric_{k_0} works. In this proof only, $s > 0$ denotes a small metric-variation parameter. Smooth the compact nonsmooth loci to obtain smooth metrics ℓ_s equal to k_0 outside one fixed compact set, with

$$\ell_s = k_0 + sp + o(s) \quad \text{in } C^2$$

on that compact set. More explicitly, choose fixed nested relatively compact neighborhoods of the attachment loci, and a smooth cutoff χ equal to one on the smaller neighborhood and supported in the larger one. Finite-chart mollification and a partition of unity give a smooth tensor t_s approximating $k_0 + sp$ in C^2 on the larger neighborhood with error at most s^2 . Set $\ell_s = \chi t_s + (1 - \chi)(k_0 + sp)$ there and retain $k_0 + sp$ elsewhere. Where χ is not one, the old metric is smooth, so this defines a globally smooth metric. The perturbation and all smoothing are confined to a fixed compact set; every other end is unchanged. For small s the metrics are uniformly comparable to k_0 and therefore complete. Their scalar curvatures $R_s = \text{Scal}_{\ell_s}$ have support in a fixed compact set K , satisfy $\|R_s\|_\infty = O(s)$, and obey

$$(33.17) \quad \int_W R_s dV_{\ell_s} = sI + o(s) > 0.$$

We construct a positive smooth conformal factor satisfying

$$(33.18) \quad -6\Delta_{\ell_s} f_s + R_s f_s = 0, \quad f_s = 1 + O(s) \quad \text{uniformly on } W,$$

$$(33.19) \quad f_s - 1 = O_k(r^{-2}) \quad \text{on the distinguished end,}$$

$$(33.20) \quad \int_{S_R} \partial_{\nu_{\ell_s}} f_s dA_{\ell_s} = \frac{1}{6} \int_W R_s f_s dV_{\ell_s} \quad \text{for every sufficiently large } R.$$

In particular, the flux in (33.20) is not being obtained by assuming zero flux at unspecified ends.

Take connected smooth compact exhaustions D_j whose boundary in the distinguished end is a large coordinate sphere S_{R_j} . The remaining boundary components lie outside each prescribed compact subset for sufficiently large j . Such an exhaustion is obtained by exhausting the complement of a fixed end neighborhood, joining to full coordinate annuli, and smoothing the compact joins. Impose $f = 1$ on S_{R_j} and homogeneous Neumann conditions on every other boundary component. These boundary components are disjoint, so the mixed problem has no interface corners.

We first record a uniform estimate for the compactly supported potential. If v has zero Dirichlet value on S_{R_j} , then for every fixed compact set K' in a fixed connected core neighborhood,

$$(33.21) \quad \|v\|_{L^2(K', \ell_s)} \leq C_{K'} \|\nabla v\|_{L^2(D_j, \ell_s)},$$

with a constant independent of sufficiently large j and small s . To see this first on a fixed end annulus, integrate along coordinate rays to the Dirichlet sphere. In Euclidean polar coordinates,

$$|v(t, \theta)|^2 \leq \left(\int_t^{R_j} r^3 |\partial_r v(r, \theta)|^2 dr \right) \left(\int_t^{R_j} r^{-3} dr \right), \quad \int_t^{R_j} r^{-3} dr \leq \frac{1}{2t^2}.$$

Integration over the unit sphere and over the fixed annulus controls its L^2 norm by the full gradient energy. On a fixed connected neighborhood joining that annulus to K' , the Poincaré inequality with the annulus as an anchor propagates the estimate. Uniform metric comparison on that neighborhood and the tail makes the constants uniform in s .

For $v = f - 1$ the variational problem is

$$(33.22) \quad \int_{D_j} \left(\langle \nabla v, \nabla w \rangle + \frac{R_s}{6} vw \right) dV_{\ell_s} = -\frac{1}{6} \int_{D_j} R_s w dV_{\ell_s},$$

for tests w vanishing on the Dirichlet sphere. By (33.21) with a fixed neighborhood of K , the left quadratic form is bounded below by $(1 - Cs)\|\nabla v\|_2^2$, and the right side is bounded in absolute

value by $Cs\|\nabla w\|_2$. On each finite domain the gradient norm is a Hilbert norm on the Dirichlet test space. Lax–Milgram therefore solves (33.22) for small s , with

$$(33.23) \quad \|\nabla v\|_{L^2(D_j)} = O(s), \quad \|v\|_{L^2(K')} = O(s)$$

for every fixed compact K' as above. Smooth elliptic regularity holds for the finite-domain solutions.

These estimates also give a uniform supremum bound near K . First, interior $W^{2,2}$ estimates for $\Delta_{\ell_s} v = (R_s/6)(1+v)$ give $O(s)$ on a slightly smaller fixed neighborhood. In dimension four this implies every finite local L^p bound, though not yet an L^∞ bound. Applying the equation once more gives $W^{2,p} = O(s)$ for $p > 2$, and hence $\|v\|_\infty = O(s)$ there. The constants are uniform because the metrics have uniform C^2 bounds and uniform ellipticity on those neighborhoods, and $R_s = O(s)$ in L^∞ . Derivative bounds on the smoothing error beyond those needed here are unnecessary.

Outside a fixed neighborhood of K , the function v is harmonic. Its values at the inner boundary are bounded by Cs , its outer Dirichlet values are zero, and its other outer normal derivatives are zero. Here is a global weak test that avoids assumptions on the remaining components. Take $(v - Cs)_+$, which vanishes near K , and extend it by zero through that neighborhood. It is an admissible test in (33.22); both terms containing R_s vanish, leaving the integral of its squared gradient equal to zero. Connectedness of D_j and its zero Dirichlet trace imply $(v - Cs)_+ = 0$ everywhere. The negative excess gives the other bound. Thus $|v| \leq Cs$ throughout D_j , with no geometry of the other ends used. On the distinguished end, a multiple of $r^{-2}(1 - r^{-\eta})$, with $0 < \eta < 1$, is a positive strictly superharmonic barrier sufficiently far out for the metric (33.13). Comparison on the truncated annuli gives the uniform bound $|v| \leq C_s r^{-2}$ there.

For each fixed small s , pass to a subsequence converging smoothly on compact subsets as $j \rightarrow \infty$. The limit yields (33.18) and the zeroth-order decay in (33.19). Rescaled end estimates and differentiation give all its fixed differentiated orders. Positivity follows from the uniform bound $f_s = 1 + O(s)$. For a fixed sphere S_R outside the support of R_s , integrate the finite-domain equation on

$$B_{j,R} = D_j \setminus \{x \text{ in the distinguished end} : r > R\}.$$

Its boundary consists of S_R and only artificial boundaries with homogeneous Neumann data; the Dirichlet sphere S_{R_j} has been removed. For large j it contains the whole support of R_s , so

$$\int_{S_R} \partial_{\nu_{\ell_s}} f_{s,j} dA_{\ell_s} = \frac{1}{6} \int_{D_j} R_s f_{s,j} dV_{\ell_s}.$$

The curvature has fixed compact support. Local convergence thus passes the derivative on the fixed sphere and the integral on that fixed support to the limit, proving (33.20) exactly. No integration over a limiting noncompact inner region is involved. The limiting factor may have harmonic behavior at the other ends; neither its limiting values nor individual end fluxes are required.

The metric $f_s^2 \ell_s$ is smooth, complete and scalar flat by (33.11) and the uniform positive bounds for f_s . Its distinguished end is asymptotically Schwarzschild in the precise sense required by the stated positive-mass theorem. Indeed, on that end $\ell_s = k_0 = \delta + O_k(r^{-3})$ and $\Delta_{\ell_s} f_s = 0$. Combining this with (33.19) gives $\Delta_\delta(f_s - 1) = O_k(r^{-7})$. The multipole estimate (33.9) therefore implies

$$(33.24) \quad f_s = 1 + \frac{a_s}{r^2} + O_k(r^{-3}), \quad f_s^2 \ell_s = (1 + a_s/r^2)^2 \delta + O_k(r^{-3}).$$

The differentiated bounds supply the weighted $C^{2,\alpha}$ remainder of order three, required in dimension four.

On the other hand, (33.17) and the uniform bound in (33.18) give

$$\int_W R_s f_s dV_{\ell_s} = sI + o(s) > 0.$$

Thus (33.20) has strictly positive outward flux. Physical and Euclidean fluxes have the same limit by the falloff. From (33.24), that limit is $-2\omega_3 a_s$, so $a_s < 0$. Equivalently, direct computation with the prescribed energy normalization gives

$$(33.25) \quad E(f_s^2 \ell_s) = -\frac{1}{\omega_3} \lim_{R \rightarrow \infty} \int_{S_R} \partial_r f_s dA_\delta = 2a_s < 0.$$

The background end contributes zero energy, and products of its error with the conformal error contribute zero limiting flux. This contradicts the smooth arbitrary-ends positive-mass theorem: the dimension is four, the manifold is orientable and boundaryless, the metric is complete and smooth, its scalar curvature vanishes, and (33.24) has exactly the required distinguished end. Hence $\text{Ric}_{k_0} = 0$.

For clarity concerning regularity, a C^2 Ricci-flat metric becomes smooth in harmonic coordinates by the elliptic Ricci equation; this is the harmonic-coordinate regularity statement of DeTurck–Kazdan [25, Theorem 5.2] for C^2 Einstein metrics in dimension at least three. We may consequently apply Bishop–Gromov comparison and its rigidity statement [46]. The asymptotic volume ratio is at least the Euclidean value using the distinguished end alone: paths from a fixed base point to a fixed end sphere, followed by coordinate rays, have length $r + o(r)$ uniformly in angle, and the end volume form tends to its Euclidean value. Thus large metric balls contain asymptotically Euclidean coordinate annuli of the corresponding radius. Ricci nonnegativity gives the opposite inequality for the volume ratio. Its value is therefore one, and the equality case of Bishop–Gromov implies that W is Euclidean space globally.

The isometry is smooth wherever the original metric is smooth. Its smoothness in each one-sided seam structure can also be seen directly, without asserting that the reflected metric was smooth. The C^2 metric has C^1 connection coefficients in the original seam charts. A flat parallel coframe extends across the seam by parallel transport, with at least C^1 dependence on its base point. In coordinates each member α satisfies

$$\partial_j \alpha_i = \Gamma_{ji}^k \alpha_k.$$

On each closed one-sided chart the connection coefficients are smooth up to the boundary. This identity inductively makes the coframe smooth up to that boundary. Its closed forms integrate to Euclidean coordinates agreeing with the isometry on the open side after a fixed Euclidean motion, hence also at the seam by continuity. This proves the asserted smoothness separately in both one-sided structures. $\square$

33.4. The original positive component and its spherical boundary.

Proposition 33.5 (Global static classification). *The positive component is the whole original open exterior: $\mathcal{U} = \text{int } \Omega$ and $N > 0$ there. Its attached boundary is S . There is a global isometry of the static base onto the Schwarzschild–Tangherlini spatial exterior of mass m , under which*

$$(33.26) \quad h_b = \left(1 + \frac{m}{2\rho^2}\right)^2 (d\rho^2 + \rho^2 g_{\mathbb{S}^3}), \quad \lambda = \frac{1 - m/(2\rho^2)}{1 + m/(2\rho^2)}, \quad \rho \geq \sqrt{m/2}.$$

The isometry is smooth on the open base and in its one-sided regular boundary structure. The attachment is homeomorphic to the original attachment of S in Ω and restricts smoothly on S . In particular the open base is simply connected and S is a round three-sphere in its induced metric. The base reflection is smooth in signed λ in the positive boundary-lapse case; its angular coordinates can be taken constant along base normal geodesics.

Proof. By Lemmas 33.3 and 33.4, the constructed space W is Euclidean. Suppose a sequence in $\mathcal{U}$ approaches an interior zero of N in the original compact part of Ω . It escapes every compact subset of W : the zero locus is not a point of the plus copy, it cannot converge to S , and the only new point lies at the minus end in a neighborhood disjoint from that sequence. Yet the sequence remains outside a fixed distant tail of the distinguished end of W .

This is impossible in Euclidean space. A coordinate cross-sphere in the distinguished end is a compact embedded sphere. Its tail has only that sphere as its frontier and is the unbounded component of its complement. Jordan–Brouwer separation makes the other side bounded; its closure is compact. Thus the complement of the tail is compact. This use of separation does not require a smooth Schoenflies theorem in dimension four.

It follows that $\mathcal{U}$ has no boundary in $\text{int } \Omega$. The latter is connected, so $\mathcal{U} = \text{int } \Omega$. We now identify the plus and minus copies in the Euclidean double explicitly.

At S , under the change $k_+ = f^2 h_b$ with $f = (1 + \lambda)/2$, the shape operator transforms by

$$\mathcal{S}_{k_+} = f^{-1}(\mathcal{S}_{h_b} + \nu_{h_b}(\log f) \text{Id}).$$

Using (33.2), its value is $2\kappa \text{Id}$ for the normal toward the plus side. In Euclidean coordinates write z for position and ν for that normal. With $d_* = (2\kappa)^{-1}$, the tangential derivative of $z - d_*\nu$ vanishes. Connectedness of S makes this vector a constant center. The image of S lies on the sphere of radius d_* about that center, and is both open there by local immersion and closed by compactness. It is the entire round sphere. Embeddedness follows from the global ambient isometry. The plus side, which contains the distinguished infinity, is its Euclidean exterior.

Set $\psi = 2/(1 + \lambda)$. Then $h_b = \psi^2 k_+$. Scalar flatness and (33.11) show that ψ is harmonic on that Euclidean exterior. It is equal to two on the sphere and tends to one at infinity. Uniqueness by the maximum principle therefore gives, in centered Euclidean radius ρ ,

$$(33.27) \quad \psi = 1 + \frac{d_*^2}{\rho^2}, \quad \lambda = \frac{1 - d_*^2/\rho^2}{1 + d_*^2/\rho^2}.$$

Since $\kappa = 1/\sqrt{2m}$, we have $d_*^2 = m/2$, which proves (33.26). For an invariant check on the coefficient from Lemma 33.1, the flux of the γ -harmonic function U is independent of the end cross-section. Its limit is $2\omega_3 M_1$ in the harmonic coordinates and $2\omega_3 m$ in the explicit coordinates of (33.27). The divergence theorem between homologous cross-sections therefore gives $M_1 = m$, without identifying the two asymptotic charts. The area radius

$$r = \rho \left(1 + \frac{m}{2\rho^2}\right)$$

satisfies

$$(33.28) \quad N = 1 - \frac{2m}{r^2}, \quad h_b = N^{-1} dr^2 + r^2 g_{\mathbb{S}^3} \quad \text{on } \text{int } \Omega.$$

The apparent degeneracy of the area-radius expression at the boundary is removed by the isotropic radius or the regular base collar. Its boundary radius is $r_0 = \sqrt{2m}$.

The smooth one-sided boundary assertion follows from Lemma 33.4 and the regular base collar of Proposition 32.5. In the positive boundary-lapse case this collar is expressed in $\lambda = \sqrt{N}$, with a smooth reflection; replacing an original defining coordinate N by $\sqrt{N}$ changes the smooth boundary coordinate but not the underlying topology or the smooth structure on S itself. In the zero boundary-lapse case the base collar is already smooth in the original one-sided structure. The spherical angular coordinates in (33.26) are constant along the base normal geodesics from S . In a reflected collar their normal projection to S is smooth and invariant under reflection. These facts give precisely the regular boundary identification needed to recover the hypersurface in horizon coordinates. Finally the Euclidean exterior of a ball in dimension four is simply connected, proving the topological assertions. $\square$

34. RECOVERY OF THE ORIGINAL HYPERSURFACE AND THE CONVERSE

The static classification identifies a metric on the orbit space. We now recover the original spacelike hypersurface, including its second fundamental form and its smooth structure at the horizon. We then compute the invariant ADM mass of every admissible Tangherlini slice, allowing a nonzero asymptotic boost.

34.1. The original graph in the open exterior. Throughout the forward implication, assume equality and all the horizon hypotheses of Definition 22.3. Retain

$$m = \sqrt{E^2 - |P|^2}, \quad r_0 = \sqrt{2m}, \quad \kappa = r_0^{-1}.$$

Proposition 33.5 identifies the whole open base $\text{int } \Omega$ with the Tangherlini spatial exterior of mass m :

$$(34.1) \quad h_b = N^{-1} dr^2 + r^2 g_{\mathbb{S}^3}, \quad N = 1 - \frac{r_0^2}{r^2}, \quad r > r_0.$$

In particular $N > 0$ throughout the original interior, the open base is simply connected, and its regular boundary attachment is homeomorphic to the original attachment of S . The adjoint field is the original field of Theorem 31.1; all occurrences of $X^\flat$ below use the original metric g .

Lemma 34.1 (Recovery on the open exterior). *There is a smooth function T_0 on $\text{int } \Omega$ such that the map*

$$x \mapsto (T_0(x), r(x), \vartheta(x))$$

into the static Tangherlini exterior induces the original g and K . Here (r, ϑ) are the global base coordinates in Equation (34.1), with $\vartheta \in \mathbb{S}^3$.

Proof. The one-form $A_b = N^{-1} X^\flat$ is closed by the staticity conclusion. Simple connectedness gives a global primitive with

$$(34.2) \quad dT_0 = -A_b.$$

On $\mathbb{R} \times \text{int } \Omega$ put $T = t + T_0(x)$. The definitions of h_b , A_b , and $N = u^2 - |X|_g^2$ give the exact identity

$$(34.3) \quad \begin{aligned} -N dT^2 + h_b &= -N (dt - A_b)^2 + h_b \\ &= -u^2 dt^2 + g_{ij} (dx^i + X^i dt)(dx^j + X^j dt). \end{aligned}$$

Thus the graph $T = T_0$ is exactly the original $t = 0$ slice and induces g . Static time is future increasing, and its future unit normal is $n = u^{-1}(\partial_t - X)$. With the convention of the theorem, the lapse-shift identity is

$$\partial_t g = 2uK + \mathcal{L}_X g.$$

The metric on the right of Equation (34.3) is stationary, and Equation (31.2) therefore gives its second form as $-\mathcal{L}_X g/(2u) = K$. This proves the claim for the original tensor, without a further deformation. These local sign conventions also agree with the Killing initial data identities in [13, Equations (2.6) and (2.14)]. $\square$

34.2. Extension through the horizon in the original smooth structure. The base has two possible regular boundary structures. When $u|_S > 0$, N is an original smooth defining function, whereas the regular base coordinate is $\lambda = \sqrt{N}$. When $u|_S = 0$, the base is smooth in the original collar coordinate. The following lemma treats these cases before passing to regular spacetime coordinates.

Lemma 34.2 (Smooth horizon extension). *The graph in Lemma 34.1 extends smoothly, in the original smooth structure of Ω , to S in the regular maximal Tangherlini extension. Its boundary is a smooth section of the corresponding future horizon if $u|_S > 0$, and is the bifurcation sphere if $u|_S = 0$.*

Proof. Connectedness of S and Theorem 31.1 leave exactly the two cases just described.

Positive boundary lapse. Equation (31.9) states that $dN = 2\kappa X^\flat$ on S and that N is an original defining function. Consequently the smooth covector $X^\flat - (2\kappa)^{-1} dN$ vanishes on S . Dividing its components by N in an original collar gives a smooth one-form β_0 such that

$$(34.4) \quad A_b = \frac{1}{2\kappa} d \log N + \beta_0, \quad T_0 = -\frac{1}{2\kappa} \log N + B_0.$$

Here B_0 is smooth up to S : its differential is $-\beta_0$, and its values on an interior collar section determine its smooth extension by integration along collar segments. The primitive in Equation (34.2) is already single-valued, so this construction has no period ambiguity.

We also need the angular coordinate ϑ to be smooth in the original coordinate N , not only in λ . The specific reflection matters here. In the original-derived coordinates (λ, y) , with $N = \lambda^2$, Equation (32.18) has even $\lambda\lambda$ and AB coefficients and odd λA coefficients, all original functions being evaluated at (λ^2, y) . Thus the smooth isometric reflection is exactly

$$\mathcal{R}(\lambda, y) = (-\lambda, y).$$

Let p be normal projection to S in this reflected base collar. Reflection fixes S pointwise and reverses its unit normals, so $p \circ \mathcal{R} = p$. The smooth one-sided isometry in Proposition 33.5 sends these normal geodesics to the radial normal geodesics of the classified base. If ϑ_S denotes its smooth angular boundary identification, the original angular map on the positive collar is therefore $\vartheta = \vartheta_S \circ p$. This formula extends it smoothly to the reflected collar and makes it invariant under the displayed $\mathcal{R}$. In particular it is even in the specific λ coordinate obtained from original N , with its tangential parameters fixed. A smooth even function of λ is a smooth one-sided function of λ^2 : Taylor expansion has only even powers, and the differentiated remainder gives this assertion to every finite order. Applying this in angular charts proves that ϑ is smooth in the original (N, y) collar.

Zero boundary lapse. Let $s \geq 0$ be original outward collar distance. By Equation (31.10), there are smooth d, Y_2 with

$$(34.5) \quad u = sd, \quad X = s^2 Y_2, \quad d|_S = \kappa, \quad N = s^2(d^2 - s^2|Y_2|_g^2).$$

It follows that A_b , and hence T_0 , extend smoothly and finitely to S . The positive square root on the exterior is

$$\lambda = s\sqrt{d^2 - s^2|Y_2|_g^2},$$

which is smooth one-sided in s . In this case the original and regular one-sided base structures agree. The smooth one-sided base identification therefore makes ϑ smooth in the original collar as well.

Regular spacetime coordinates. Define the tortoise coordinate and Kruskal coordinates by

$$(34.6) \quad \begin{aligned} r_* &= r + \frac{r_0}{2} \log \frac{r - r_0}{r + r_0}, \\ U &= -\exp(-\kappa(T - r_*)), \quad V = \exp(\kappa(T + r_*)). \end{aligned}$$

Differentiation gives $dr_*/dr = N^{-1}$. More precisely,

$$(34.7) \quad e^{\kappa r_*} = \sqrt{N} D_*(N), \quad D_*(N) = e^{r/r_0} \frac{r}{r + r_0}, \quad r = \frac{r_0}{\sqrt{1 - N}}.$$

Thus D_* is smooth positive near $N = 0$, with $D_*(0) = e/2$. The two-dimensional static part becomes

$$-N dT^2 + N^{-1} dr^2 = -\frac{1}{\kappa^2 D_*(N)^2} dU dV.$$

Moreover $-UV = ND_*(N)^2$ has a smooth local inverse for N near zero. These are regular horizon coordinates: the coefficient of $dU dV$ is smooth negative nonzero and r is smooth across the horizon.

In the positive boundary-lapse case, substituting Equation (34.4) into Equation (34.6) on the graph gives

$$(34.8) \quad U = -ND_*(N)e^{-\kappa B_0}, \quad V = D_*(N)e^{\kappa B_0}.$$

Both functions are smooth in the original collar, and S maps to $U = 0, V > 0$. Since its angular map identifies S with $\mathbb{S}^3$, this is a smooth section of the future horizon.

In the zero boundary-lapse case, the same graph formulas are

$$U = -\lambda D_*(N)e^{-\kappa T_0}, \quad V = \lambda D_*(N)e^{\kappa T_0}.$$

Equation (34.5) makes them smooth in s , and both vanish on S . Together with the angular identification, they map S onto the bifurcation sphere. This proves the asserted smooth extension in both cases. $\square$

Proposition 34.3 (Global recovery of the original exterior). *Equality under the horizon hypotheses gives a global smooth spacelike embedding of the original Ω , including S , in the regular maximal Tangherlini extension of mass m . The embedding pulls back the spacetime metric and the future second fundamental form to the original g and K . Its image lies in the closure of the corresponding exterior, its boundary has the form stated in Lemma 34.2, and its end approaches the corresponding spatial infinity.*

Proof. Combine the maps in Lemmas 34.1 and 34.2. Their pullback metric is g on the open exterior and remains g at S by continuity. Since g is positive definite, the extended map is a spacelike immersion there.

It is injective on the interior by the global base identification, and on the boundary by the round-sphere identification. No interior point can have the same image as a boundary point, since their area radii are respectively $r > r_0$ and $r = r_0$. It is also proper. A compact set of the regular spacetime has bounded area radius; its preimage is a closed subset of a bounded annulus in the attached base. Such an annulus is compact, and the base attachment is homeomorphic to the original attachment. A proper injective immersion is an embedding.

To obtain the normal at the boundary, take a smooth future timelike field in regular spacetime coordinates, project it to the normal line of the spacelike image, and normalize. Its normal projection is timelike and nonzero; the result is a smooth future unit normal. It agrees with the interior normal by uniqueness of the future choice. Hence the second-form identity from Lemma 34.1 extends to S . If desired, the smooth collar expressions extend locally across S as a

spacelike immersion, since immersion and spacelikeness are open conditions. No data beyond S are prescribed or asserted.

Finally the inverse-base identity gives

$$(34.9) \quad |dT_0|_{h_b}^2 = \frac{|X|_g^2}{u^2 N} \longrightarrow \frac{|b|^2}{(b^0)^2} < 1,$$

using Equation (31.5) and $N \rightarrow 1$. Choose $a < 1$ slightly larger than the limiting slope. Increasing a fixed base radius if necessary absorbs the radial length factor $N^{-1/2}$, so integration along base radial rays, starting on a compact sphere, gives $|T_0| \leq ar + C$ uniformly on the end. Thus the end remains in a spacelike cone of the static exterior and approaches its spatial infinity. $\square$

34.3. Asymptotic geometry of an admissible Tangherlini slice. For the converse write $M_0 > 0$ for the geometric mass parameter of the ambient Tangherlini spacetime, to distinguish it from the invariant ADM mass that we will compute. In isotropic spatial coordinates y on its chosen exterior, with future static time T , its metric is

$$(34.10) \quad \mathbf{g}_{M_0} = - \left(\frac{1 - M_0/(2|y|^2)}{1 + M_0/(2|y|^2)} \right)^2 dT^2 + \left(1 + \frac{M_0}{2|y|^2} \right)^2 \sum_{i=1}^4 (dy^i)^2.$$

The following argument applies to the end of every slice in the converse class. It derives its asymptotic plane from the prescribed intrinsic decay.

Lemma 34.4 (Asymptotic affine plane). *Let a smooth spacelike hypersurface in the Tangherlini exterior of mass $M_0 > 0$ have an end with coordinates $x \in \mathbb{R}^4 \setminus \overline{B}$ and induced data satisfying*

$$g - \delta = O_2(|x|^{-q}), \quad K = O_1(|x|^{-1-q}), \quad q > 1.$$

Assume the hypersurface is smoothly embedded with a compact remaining part and boundary, if present, in the exterior closure. For any $1 < q_0 < \min(q, 2)$ there are constants $L \in \text{GL}(4, \mathbb{R})$, $a \in \mathbb{R}^4$, $L_0 \in \mathbb{R}^4$, and $a_0 \in \mathbb{R}$ such that on the end

$$(34.11) \quad \begin{aligned} y &= Lx + L_0 + O_2(|x|^{1-q_0}), \\ T &= a \cdot x + a_0 + O_2(|x|^{1-q_0}), \quad L^t L - a \otimes a = I. \end{aligned}$$

In particular the limiting tangent plane is spacelike and x gives Euclidean orthonormal coordinates on it.

Proof. We first control the spatial projection and then obtain its affine asymptotics.

The end leaves every bounded area radius. Gauss and Codazzi express the all-tangential and one-normal spacetime curvature components in a slice orthonormal frame in terms of Rm_g , K^2 , and ∇K . All tend to zero by the stated decay. Vacuum $\text{Ric}_{ij} = 0$ expresses the two-normal curvature components as sums of all-tangential ones. Thus every component tends to zero, and so does the spacetime curvature-square invariant.

In the static orthonormal frame of Tangherlini, curvature components with one time index and three spatial indices vanish. All remaining squared components contribute nonnegatively to that invariant. The angular sectional curvature, computed from the static warped product, is

$$\frac{1 - |\nabla r|_{h_{\text{static}}}^2}{r^2} = \frac{2M_0}{r^4}.$$

The invariant is consequently bounded below by a positive constant times M_0^2/r^8 , including at the horizon by regularity. Hence the area radius r , and therefore $|y|$, tend to infinity as $|x| \rightarrow \infty$.

A finite timelike limiting normal. For the estimates in original end coordinates, write $R = |x|$. Decompose the future static Killing field along the slice as $\partial_T = un + X$. On its exterior tail $u > |X|_g$. Local translates along this transverse field give stationary lapse-shift coordinates. The vacuum Killing initial data equations, with the present second-form convention, are

$$(34.12) \quad \nabla_{(i} X_{j)} = -uK_{ij}, \quad \nabla^2 u = u(\text{Ric}_g + \tau K - 2K^2) - \mathcal{L}_X K.$$

These follow respectively from the Killing equation and its normal derivative using the vacuum spatial Ricci equation. Their local form is dimension independent; we use them here in four spatial dimensions. The same local identities are recorded in [13, Section 2].

Commuting derivatives in the first equation expresses $\nabla^2 X$ as curvature times X and permutations of $\nabla(uK)$. If Y denotes the coordinate components of the pair (u, X) , the prescribed decay therefore gives

$$(34.13) \quad |\partial^2 Y| \leq C|x|^{-1-q_0}|\partial Y| + C|x|^{-2-q_0}|Y|.$$

Here only two derivatives of g and one of K are used.

The metric g is uniformly comparable to the Euclidean metric on the end, the lapse and shift are causal, and $q_0 > 1$. Applying Lemma 31.4 to Equation (34.13) gives finite constant limits with $O_2(R^{-q_0})$ errors. Their normalization follows separately from $u^2 - |X|_g^2 = -\mathbf{g}_{M_0}(\partial_T, \partial_T) \rightarrow 1$, using the already proved $|y| \rightarrow \infty$. Thus

$$(34.14) \quad u = u_\infty + O_2(R^{-q_0}), \quad X = X_\infty + O_2(R^{-q_0}), \quad u_\infty^2 - |X_\infty|_\delta^2 = 1.$$

Proper spatial projection and affine jets. Set $N = u^2 - |X|_g^2$. Pairing ∂_T with tangent vectors to the slice gives the exact identities

$$(34.15) \quad dT = -\frac{X^\flat}{N}, \quad y^* h_{\text{static}} = h_b = g + N^{-1} X^\flat \otimes X^\flat.$$

The spatial projection y is a local diffeomorphism, since the static Killing field is timelike and transverse to the spacelike hypersurface. Equation (34.14) makes h_b asymptotic to a positive constant matrix. Because $|y| \rightarrow \infty$, both its Jacobian and inverse Jacobian are uniformly bounded far out.

We make the target coverage and path lifting explicit. Choose R_* large enough that the preceding Jacobian bounds hold on the original end $E_* := \{|x| > R_*\}$. Choose R_0 larger than the maximum of $|y|$ on ∂E_* , and put

$$V_0 = \{y \in \mathbb{R}^4 : |y| > R_0\}, \quad U_0 = E_* \cap y^{-1}(V_0).$$

The local diffeomorphism $y : U_0 \rightarrow V_0$ is proper. Indeed a sequence whose images remain in a compact subset of V_0 cannot escape to original infinity, by $|y| \rightarrow \infty$, or approach ∂E_* , by the choice of R_0 . It therefore has a convergent subsequence in U_0 . The image is open by local invertibility and closed by properness. It is nonempty, and V_0 is connected, so the image is all of V_0 .

A proper local diffeomorphism is a covering: inverse neighborhoods of its finite fibers give evenly covered neighborhoods, with properness excluding additional sheets arriving from outside those neighborhoods. In particular, radial paths down to a fixed sphere $|y| = R_1 > R_0$ lift. Their endpoints lie in the compact preimage of that sphere. The uniform inverse-Jacobian bound gives a lifted length at most $C(|y| - R_1)$ and hence $|x| \leq C'(1 + |y|)$. Conversely, integrating the Jacobian bound along original coordinate rays gives $|y| \leq C(1 + |x|)$. Hence $|y| \asymp |x|$.

The connection transformation law for Equation (34.15) now yields $\partial^2 y = O(|x|^{-1-q_0})$: the connection of h_b has that order, while the static connection is $O(|y|^{-3})$, and $q_0 < 2$. The first identity in Equation (34.15) similarly gives the required Hessian bound for T . Integration on radial rays and comparison on spheres first yield constant gradient limits and then constant

translation limits. The latter comparison costs $O(R^{1-q_0})$, which tends to zero. This proves the two expansions in Equation (34.11). Finally the limit of the induced metric is $L^t L - a \otimes a = I$. In particular L is invertible and the affine plane is spacelike. $\square$

34.4. The invariant ADM mass of the slice.

Proposition 34.5 (Mass of every admissible Tangherlini slice). *Every hypersurface in the converse class of Theorem 22.4, in Tangherlini spacetime of geometric mass $M_0 > 0$, has invariant ADM mass M_0 . More precisely, in coordinates adapted to its asymptotic boost, its charges are $E = M_0 \Gamma$ and $P = M_0 \Gamma v e_1$, where $0 \leq v < 1$ and $\Gamma = (1 - v^2)^{-1/2}$.*

Proof. We compare the slice with its limiting plane and compute the flux on that plane.

Comparison with the affine plane. Apply Lemma 34.4. Use the same parameter x on the affine plane in Equation (34.11), and denote its induced data by (g^*, K^*) . In affine Minkowski coordinates adapted to this plane, the residual displacement of the actual embedding is a spacetime vector Z , with time component Z^0 , satisfying $Z = O_2(|x|^{1-q_0})$. The ambient metric differs from Minkowski by $O_2(|x|^{-2})$ near both embeddings. Consequently

$$(34.16) \quad \begin{aligned} g_{ij} - g_{ij}^* &= \partial_i Z_j + \partial_j Z_i + O_1(|x|^{-2q_0}), \\ K_{ij} - K_{ij}^* &= \partial_i \partial_j Z^0 + O(|x|^{-1-2q_0}). \end{aligned}$$

Here the indices on the spatial components of Z are Euclidean. To check the error orders, the quadratic Minkowski metric term is $O_1(|x|^{-2q_0})$. Evaluation of the ambient perturbation at the displaced point, or its contraction with a displaced tangent vector, gives $O_1(|x|^{-2-q_0})$, which is smaller since $q_0 < 2$. For the second form use $-\langle n, \nabla_i \partial_j \iota \rangle$. The normal differs from the constant future affine normal by $O(|x|^{-q_0})$; multiplying this by $\partial^2 Z$ gives $O(|x|^{-1-2q_0})$. Ambient connection and evaluation errors have order $O(|x|^{-3-q_0})$, again smaller. Euclidean trace reversal may be used in the difference with errors of the same allowed order.

The only possible additional limiting fluxes in Equation (34.16) come from its displayed linear terms. For the metric term the energy vector is

$$F_i = \Delta Z_i - \partial_i \operatorname{div} Z, \quad \partial_i F_i = 0.$$

For the second-form term the momentum tensor is

$$B_{ij} = \partial_i \partial_j Z^0 - \delta_{ij} \Delta Z^0, \quad \partial_j B_{ij} = 0.$$

Extend Z smoothly by a cutoff through the bounded Euclidean interior. The divergence theorem makes both linear fluxes exactly zero on every sufficiently large sphere. The remaining flux errors are $O(R^{2-2q_0})$, and hence tend to zero because $q_0 > 1$. It is therefore enough to compute the charges of the affine plane in the ambient metric (34.10).

The four-dimensional boost calculation. Translations change the ambient leading mass terms only by $O(|x|^{-3})$ and do not affect the charges. After spatial rotations and an orthogonal choice of x on the plane, write it as

$$T = \Gamma v x_1, \quad y_1 = \Gamma x_1, \quad y_A = x_A \quad (A = 2, 3, 4).$$

Extend these coordinates by $T = \Gamma(t + v x_1)$ and $y_1 = \Gamma(x_1 + vt)$. The leading perturbation of the ambient Minkowski metric is

$$M_0 |y|^{-2} \left(2 dT^2 + \sum_i (dy^i)^2 \right).$$

Set $s_*^2 = \Gamma^2 x_1^2 + |x_\perp|^2$ and $H_* = M_0 s_*^{-2}$. The needed components of its boost are

$$h_{11} = \Gamma^2 (1 + 2v^2) H_*, \quad h_{AB} = \delta_{AB} H_*, \quad h_{t1} = 3\Gamma^2 v H_*, \quad \partial_t H_* = v \partial_1 H_*.$$

Since $\Gamma^2(1 + 2v^2) + 2 = 3\Gamma^2$, differentiation gives

$$(34.17) \quad \partial_j g_{ij}^* - \partial_i g_{jj}^* = 6M_0 \Gamma^2 x_i s_*^{-4} + o(|x|^{-3}).$$

For the stated future convention, the linear second form is $(\partial_t h_{ij} - \partial_i h_{tj} - \partial_j h_{ti})/2$. Its trace reversal is

$$(34.18) \quad K^* - (\text{tr}_{g^*} K^*)g^* = 3M_0 \Gamma^2 v s_*^{-4} \begin{pmatrix} x_1 & x_\perp^t \\ x_\perp & -\Gamma^2 x_1 I_3 \end{pmatrix} + o(|x|^{-3}).$$

This also fixes the sign of P_1 in the chosen boost coordinates.

The ellipsoid $\{\Gamma^2 x_1^2 + |x_\perp|^2 < 1\}$ has Euclidean volume $\omega/(4\Gamma)$. Its radial volume formula gives

$$(34.19) \quad \int_{\mathbb{S}^3} (\Gamma^2 n_1^2 + |n_\perp|^2)^{-2} dA = \frac{\omega}{\Gamma}.$$

Equations (34.17)–(34.19), with the energy factor $1/(6\omega)$ and momentum factor $1/(3\omega)$, give

$$(34.20) \quad E = M_0 \Gamma, \quad P_1 = M_0 \Gamma v, \quad P_A = 0 \quad (A = 2, 3, 4).$$

The transverse momentum integrals vanish by oddness. Thus $\sqrt{E^2 - |P|^2} = M_0$, as asserted. $\square$

Lemma 34.6 (Volume of a horizon section). *A smooth spacelike section of a future Tangherlini horizon of geometric mass $M_0 > 0$, including its bifurcation sphere, has induced metric $(2M_0)g_{\mathbb{S}^3}$ and three-volume $\omega(2M_0)^{3/2}$.*

Proof. In the regular coordinates of Equation (34.6), the future horizon is $U = 0$ with area radius $r = \sqrt{2M_0}$. The generator direction ∂_V is the kernel of its induced form. Thus pullback to any section leaves precisely $r^2 g_{\mathbb{S}^3}$, regardless of the value of V along the section. A compact connected spacelike section has an angular projection which is a local diffeomorphism and hence a covering of $\mathbb{S}^3$; compactness gives surjectivity, and simple connectedness of $\mathbb{S}^3$ makes that covering a diffeomorphism. The same induced metric holds at the bifurcation sphere. Taking its volume proves the formula. $\square$

Completion of the proof of Theorem 22.4. For equality data, Theorem 22.2 gives $m > 0$. The preceding adjoint, staticity, and classification results apply to the original exterior, and Proposition 34.3 supplies the required global smooth embedding with the original g and K , including its normal and its horizon boundary. The ambient mass is the same $m = \sqrt{E^2 - |P|^2}$ by the scale in Equation (34.1).

Conversely, take a Tangherlini hypersurface satisfying every exterior, decay, and horizon hypothesis in the theorem. If its ambient geometric mass is M_0 , Proposition 34.5 gives invariant ADM mass M_0 , and Lemma 34.6 gives $A = \omega(2M_0)^{3/2}$. Outer area minimization, which is part of this converse class, identifies A with the original enclosing infimum. Therefore

$$\sqrt{E^2 - |P|^2} = M_0 = \frac{1}{2} \left(\frac{A}{\omega} \right)^{2/3}.$$

This proves both directions, including slices with nonzero original K or nonzero original ADM momentum. $\square$

Part 5. Local anti-de Sitter perturbations

35. LOCAL CONFORMAL PERTURBATIONS OF SCHWARZSCHILD–ANTI-DE SITTER

For cosmological constant $\Lambda = -3$, the Schwarzschild–anti-de Sitter mass–horizon relation suggests

$$(35.1) \quad m_{\text{AH}} \geq \sqrt{\frac{A}{16\pi}} \left(1 + \frac{A}{4\pi} \right),$$

where m_{AH} is the Lorentz norm of the hyperbolic mass covector. We prove this inequality for sufficiently small maximal vacuum conformal perturbations of any positive-mass Schwarzschild–anti-de Sitter exterior, using the area A of its prescribed future marginally outer trapped boundary. The transverse-traceless (TT) tensor generating the perturbation is fixed before the small parameter interval is chosen. The result concerns each given solution branch; it asserts neither a uniform neighborhood theorem nor existence of a branch for every seed.

Khuri and Kopiński [32, Theorem 2] proved a perturbative inequality in this conformal setting under an additional condition: the static-lapse-weighted integral of the squared seed norm must dominate a multiple of the squared H^1 norm of its normal component on a domain containing the boundary [32, Equation (2.5)]. Both sides scale quadratically with the seed, so shrinking the perturbation parameter alone cannot remove that condition. Our sharp TT boundary identity and treatment of its null directions remove this requirement for each fixed seed and branch.

The geometric ingredients have related precedents. Chruściel and Herzlich [22] develop the hyperbolic mass covector and its covariance. Neves and Tian [42] construct constant mean curvature (CMC) foliations near infinity under positive-mass Schwarzschild–anti-de Sitter asymptotic hypotheses, and Ambrozio [6] proves a local Riemannian Penrose inequality using CMC foliations. Here we derive the mass identities and finite Taylor families of CMC spheres directly. We need neither an actual global CMC foliation at nonzero parameter nor a spacetime extension through the inner boundary.

Proof strategy. Call the left side of (35.1) minus the right side the deficit. The proof separates three cases for the fixed seed. After establishing actual mass and area expansions, we show that the constant and linear coefficients vanish and the quadratic coefficient is nonnegative. Its kernel has one radial direction and three directions with degree-one angular dependence. A static TT identity will identify these four directions explicitly.

The nonradial kernel directions require a fourth-order argument. A change of spacetime slice, made only at the level of finite Taylor expansions, removes their first extrinsic-curvature coefficient. Null transport preserves the boundary area through the required order, and a second TT square gives a nonnegative quartic deficit. Equality would force the changed metric to be a round warped-product jet through second order. Conformality of the original branch then gives a differential identity incompatible with a nonzero decaying degree-one mode. Thus every nonradial null direction has a strictly positive quartic coefficient.

The remaining radial seeds satisfy an exact conserved-mass identity. Combining these alternatives with the actual remainder estimates proves (35.1), with equality precisely for radial seeds at sufficiently small positive parameters. The interval may depend on the fixed background, seed, and branch. The main theorem uses the actual boundary area without outermostness or outer area-minimization; a separate corollary adds those geometric hypotheses and identifies this area with the least enclosing area.

36. GEOMETRIC SETTING AND THE THEOREM

Fix $m_0 > 0$, and let $a > 0$ be the unique positive root of $1 + a^2 - 2m_0/a = 0$. On

$$M = [0, \infty) \times S^2, \quad S = \{0\} \times S^2,$$

let

$$(36.1) \quad g_0 = ds^2 + r(s)^2 \sigma, \quad r(0) = a, \quad r'(s) > 0 \quad (s > 0), \quad (r')^2 = 1 + r^2 - \frac{2m_0}{r},$$

where σ is the unit round metric. The coordinate s is smooth at the boundary, and $n = \partial_s$ points from S toward infinity. On the end, $r = r(s)$ is a coordinate and

$$(36.2) \quad b = \frac{dr^2}{1+r^2} + r^2\sigma = d\eta^2 + \sinh^2 \eta \sigma, \quad \eta = \operatorname{arsinh} r.$$

In particular, r is asymptotic to a positive constant times e^s .

Our Laplacian is $\Delta = \operatorname{div} \operatorname{grad}$, with nonpositive eigenvalues on a closed manifold. The spaces $C_\tau^{k,\alpha}$ use the hyperbolic metric b in the end: the tensor and its covariant derivatives through order k are $O(e^{-\tau s})$, and the correspondingly weighted α -Hölder seminorms on unit background balls are bounded. On a compact set we use ordinary $C^{k,\alpha}$ regularity. This definition is equivalent to one using hyperbolic distance.

Fix $0 < \alpha < 1$, $3/2 < \tau < 3$, and a smooth symmetric tensor $q \in C_\tau^{1,\alpha}$, smooth up to S , satisfying

$$(36.3) \quad \operatorname{tr}_{g_0} q = 0, \quad \operatorname{div}_{g_0} q = 0.$$

Throughout this part the seed q is independent of ϵ . Suppose a smooth branch of positive solutions, defined for $0 \leq \epsilon < \epsilon_*$, satisfies

$$(36.4) \quad \Delta_{g_0} \phi_\epsilon = \frac{3}{4}(\phi_\epsilon^5 - \phi_\epsilon) - \frac{\epsilon^2}{8}|q|_{g_0}^2 \phi_\epsilon^{-7} \quad \text{on } M,$$

$$(36.5) \quad \partial_s \phi_\epsilon = \frac{\epsilon}{4} q_{ss} \phi_\epsilon^{-3} \quad \text{on } S,$$

$$(36.6) \quad \phi_0 = 1, \quad \phi_\epsilon - 1 \in C_\tau^{2,\alpha}, \quad \|\phi_\epsilon - 1\|_{C_\tau^{2,\alpha}} \rightarrow 0 \quad \text{as } \epsilon \rightarrow 0.$$

No differentiability in stronger weighted spaces is assumed. Set

$$(36.7) \quad g_\epsilon = \phi_\epsilon^4 g_0, \quad K_\epsilon = \epsilon \phi_\epsilon^{-2} q, \quad A_\epsilon = |S|_{g_\epsilon}.$$

36.1. Mass and boundary conventions. Let x_i , $1 \leq i \leq 3$, be the coordinate functions on the unit sphere. Define $V_0 = \sqrt{1+r^2}$, $V_i = r x_i$, and $e_\epsilon = g_\epsilon - b$. We use the flux normalization

$$(36.8) \quad p_\mu(\epsilon) = \frac{1}{16\pi} \lim_{R \rightarrow \infty} \int_{\{r=R\}} [V_\mu(\operatorname{div}_b e_\epsilon - d \operatorname{tr}_b e_\epsilon) + (\operatorname{tr}_b e_\epsilon) dV_\mu - e_\epsilon(\operatorname{grad}_b V_\mu, \cdot)](\nu_b) dA_b,$$

where ν_b points toward increasing r . All operators and the area form in this formula are those of b . Assume that these limits are finite and that the covector is future timelike for all sufficiently small positive ϵ . Thus

$$m_{\text{AH}}(\epsilon) = \sqrt{p_0(\epsilon)^2 - \sum_{i=1}^3 p_i(\epsilon)^2}$$

is the positive mass norm. This normalization gives $p_0(0) = m_0$ and $p_i(0) = 0$. The p_i are spatial components of the hyperbolic mass covector, not angular momentum charges.

For a two-sided surface with unit normal ν toward the chosen end, put

$$H = \operatorname{div}_\Sigma \nu, \quad \theta_+ = H + \operatorname{tr}_\Sigma K.$$

For a spacetime realization our convention is $K(X, Y) = \bar{g}(\bar{\nabla}_X n_{\text{future}}, Y)$; in Gaussian normal coordinates this means $\partial_t g = 2K$. A future marginally outer trapped surface, or MOTS, has $\theta_+ = 0$.

The standard conformal identities give

$$(36.9) \quad \operatorname{tr}_{g_\epsilon} K_\epsilon = 0, \quad \operatorname{div}_{g_\epsilon} K_\epsilon = 0, \quad R_{g_\epsilon} + 6 = |K_\epsilon|_{g_\epsilon}^2.$$

Indeed, for a tracefree q the divergence identity is $\operatorname{div}_{\phi^4 g_0}(\phi^{-2} q) = \phi^{-6} \operatorname{div}_{g_0} q$. Also

$$R_{\phi^4 g_0} = \phi^{-5}(-8\Delta_{g_0}\phi - 6\phi) = -6 + \epsilon^2 \phi^{-12} |q|_{g_0}^2.$$

The boundary S is totally geodesic for g_0 . Its normal in g_ϵ is $\phi_\epsilon^{-2} \partial_s$, and

$$(36.10) \quad H_{g_\epsilon}(S) = 4\phi_\epsilon^{-3} \partial_s \phi_\epsilon = \epsilon \phi_\epsilon^{-6} q_{ss}, \quad \operatorname{tr}_S K_\epsilon = -\epsilon \phi_\epsilon^{-6} q_{ss}.$$

Consequently it is a future MOTS. Its mean curvature need not vanish.

36.2. Statement. Define

$$(36.11) \quad b_*(A) = \sqrt{\frac{A}{16\pi}} \left(1 + \frac{A}{4\pi}\right), \quad \mathfrak{D}_q(\epsilon) = m_{\text{AH}}(\epsilon) - b_*(A_\epsilon).$$

A tensor is called *radial* if it is invariant under every rotation of the S^2 factor.

Theorem 36.1. *For every background, fixed TT seed, and solution branch satisfying (36.1)–(36.6) and the mass assumptions above, there is $0 < \epsilon_0 \leq \epsilon_*$ such that*

$$m_{\text{AH}}(\epsilon) \geq \sqrt{\frac{A_\epsilon}{16\pi}} \left(1 + \frac{A_\epsilon}{4\pi}\right) \quad (0 \leq \epsilon < \epsilon_0).$$

For all sufficiently small positive ϵ , equality holds if the seed is radial, and the inequality is strict otherwise.

The theorem makes no sign assumption on $q_{ss}|_S$. It also does not require outermostness or outer area-minimization. In particular it implies the following formulation with the usual geometric boundary hypotheses. Here an enclosing surface separates S from the end and bounds with S a compact region. Write $A_{\min}(S)$ for the infimum of areas of smooth enclosing surfaces, allowing disconnected competitors and S itself.

Corollary 36.2. *In addition to the hypotheses of Theorem 36.1, suppose $q_{ss} \geq 0$ on S . Assume, for every sufficiently small positive ϵ , that no compact smooth embedded two-sided surface in the interior enclosing S has $\theta_+ \leq 0$ everywhere, and that every smooth enclosing surface has area at least A_ϵ . Disconnected enclosing competitors are allowed, and S itself is allowed in the area infimum. Then the exact local Penrose inequality holds with $A_{\min}(S) = A_\epsilon$.*

The outermostness condition excludes all weakly future outer trapped enclosing surfaces, not only additional MOTS. These additional assumptions are relevant to the geometric formulation of the conjecture but are not needed in the coefficient argument below.

36.3. Static operators and Taylor notation. We use

$$(36.12) \quad \begin{aligned} N &= r', & \kappa &= N'(0) = \frac{1+3a^2}{2a}, & B &= \operatorname{Ric}_{g_0} + 3g_0, \\ L &= \Delta_{g_0} - 3, & D &= -\Delta_{a^2\sigma} + \frac{2\kappa}{a}. \end{aligned}$$

The letter D denotes the boundary operator, while $\mathfrak{D}_q$ denotes the mass deficit.

Lemma 36.3. *The tensor B and the static lapse satisfy*

$$(36.13) \quad B = B_s \, ds^2 + B_t r^2 \sigma, \quad B_s = 1 - \frac{2m_0}{r^3}, \quad B_t = 1 + \frac{m_0}{r^3},$$

$$(36.14) \quad \nabla^2 N = NB, \quad \operatorname{tr}_{g_0} B = 3, \quad \operatorname{div}_{g_0} B = 0, \quad LN = 0.$$

Moreover D is positive on all spherical harmonics and

$$(36.15) \quad b_*(4\pi a^2) = m_0, \quad b'_*(4\pi a^2) = \frac{\kappa}{8\pi}.$$

Proof. The radial and tangential sectional curvatures of $ds^2 + r^2\sigma$ are $-r''/r$ and $(1 - (r')^2)/r^2$. Differentiating (36.1) gives $N' = r + m_0/r^2$ and $N'' = N(1 - 2m_0/r^3)$. The Ricci eigenvalues are therefore $-2 - 2m_0/r^3$ and $-2 + m_0/r^3$, giving (36.13) and $R_{g_0} = -6$. The radial and tangential Hessian eigenvalues of N are N'' and $N'N/r$, respectively, proving the first identity in (36.14). Its trace gives $LN = 0$. The divergence identity follows from the contracted Bianchi identity and the constant scalar curvature. At S , $B_t = \kappa/a$ and $2\kappa/a > 0$, so $D > 0$. Finally $2m_0 = a(1 + a^2)$ gives (36.15) by direct differentiation. $\square$

We write $[\mathcal{F}]_j$ for the coefficient of ϵ^j in a Taylor expansion; coefficients do not include an extra factorial. Thus $u_j[q] = [\phi_\epsilon - 1]_j$ and $c_j[q] = [\mathfrak{D}_q]_j$ when the indicated actual expansions have been established. For a tensor T we also use the weighted function norm

$$\|T\|_\beta = \sup_M e^{\beta s} |T|_{g_0}.$$

On the end this is equivalent to using b . Finite angular type means that the span of all rotational pullbacks of the function or tensor is finite-dimensional. For functions this is precisely a finite sum of spherical-harmonic spaces, with arbitrary smooth radial coefficients.

Some geometric arguments use Taylor families of metrics and surfaces only to order p , with identities read in the finite Taylor algebra $\mathbb{R}[\epsilon]/(\epsilon^{p+1})$. On any compact radial interval, their coefficients can be represented by a smooth family and all differential-geometric identities then hold to the retained order. At S , a formal displacement may have negative first coefficient. It is evaluated using the given smooth one-sided jets: extend sufficiently many coefficients smoothly across $s = 0$, perform the finite Taylor calculation, and retain only the prescribed order. Every boundary derivative of an identity holding on $s \geq 0$ is fixed by those jets. This convention does not assume a constrained extension at a fixed point with $s < 0$. The time-recursion argument below will specify how the constraint identities are used after such a displacement.

37. WEIGHTED COEFFICIENTS AND ACTUAL EXPANSIONS

We first justify the passage between finite Taylor calculations and the given solution branch. Throughout this section, fix

$$(37.1) \quad \frac{3}{2} < \beta < \min\{\tau, \sqrt{3}\}.$$

We use the weighted norms and Taylor conventions of Section 36.3. Boundary norms without a weight are uniform norms on S .

37.1. A scalar inverse and its asymptotics. Equivariant linear differential operations preserve finite angular type. Products and contractions preserve it after enlarging the angular space by a finite tensor product. On functions, the round Laplacian preserves these spaces: it is the sum of squares of the three infinitesimal rotation fields. Its restriction is self-adjoint, so we may split into finitely many eigenspaces with eigenvalues $\lambda \geq 0$ for $-\Delta_\sigma$.

Lemma 37.1 (Comparison and finite angular inverse). *There are constants $c_\beta, C_\beta > 0$ with the following properties. Suppose $z = o(e^{-\beta s})$ uniformly at infinity and*

$$(L - V)z = F, \quad \|V\|_{C^0(M)} \leq c_\beta.$$

If either $z|_S = b$ or $\partial_s z|_S = b$, then

$$(37.2) \quad \|z\|_\beta \leq C_\beta (\|F\|_\beta + \|b\|_{C^0(S)}).$$

The constants do not depend on an angular cutoff. The same estimate holds for $\partial_s z - a_ z = b$ if $\|a_*\|_{C^0(S)} \leq \beta/2$.*

If F and b are smooth and of finite angular type, and $F = O(e^{-\gamma s})$ for some $\gamma > 3$, there is a unique decaying finite angular type solution of $Lu = F$ with either of the above Dirichlet or Neumann data. It satisfies

$$(37.3) \quad u = r^{-3}\zeta(x) + o(r^{-3}), \quad \partial_s(u - r^{-3}\zeta(x)) = o(r^{-3})$$

for a smooth finite angular type function ζ . These statements hold after any fixed number of angular differentiations. If $\partial_s^j F = O(e^{-\gamma s})$ for $0 \leq j \leq k$, the remainder in (37.3) can also be differentiated in s through order $k + 2$, with each resulting remainder $o(r^{-3})$.

Proof. Let $h = e^{-\beta s}$. Since $N/r \geq 0$,

$$Lh = (\beta^2 - 2\beta N/r - 3)h \leq -(3 - \beta^2)h.$$

Choose $c_\beta < (3 - \beta^2)/2$. A sufficiently large multiple Ah satisfies $(L - V)(Ah) \leq -|F|$. For Dirichlet data, enlarge A to dominate $|b|$. For Neumann data, enlarge it so that $-\beta A - b < 0$ and $-\beta A + b < 0$ on S . The functions $Ah - z$ and $Ah + z$ are positive sufficiently far out. A negative interior minimum contradicts their differential inequalities, because the zeroth-order coefficient of $L - V$ is negative. A negative minimum on S is impossible in the Neumann case: the inward derivative at such a minimum is nonnegative, whereas the chosen derivative is negative. This proves (37.2). For the Robin condition, apply the maximum argument to z/h . At a positive boundary maximum its inward derivative is nonpositive, whereas $\partial_s(z/h) = (\beta + a_*)(z/h) + b$ on S . Since $\beta + a_* \geq \beta/2$, a sufficiently large positive maximum is impossible; apply the same argument to $-z/h$.

For the existence assertion, work in each angular eigenspace. The radial equation is

$$(37.4) \quad y'' + 2\frac{N}{r}y' - \left(3 + \frac{\lambda}{r^2}\right)y = F_\lambda.$$

First impose the required condition at 0 and $y(T) = 0$ on a finite interval. The homogeneous problem has trivial kernel: multiplication by $r^2 y$ and integration gives

$$0 = -\int_0^T r^2 (y')^2 ds - \int_0^T (3r^2 + \lambda)y^2 ds,$$

with zero boundary terms. Thus the finite-interval linear problem is solvable. The barrier estimate is independent of T . On each compact interval, the equation bounds y' and then y'' : the mean value theorem supplies a point with bounded y' , and the first-order equation for y' propagates that bound. A subsequence therefore converges on compact intervals to a solution with $y = O(e^{-\beta s})$. The same local argument on unit intervals gives $y' = O(e^{-\beta s})$.

Now $N/r = 1 + O(e^{-2s})$ and $r^{-2} = O(e^{-2s})$. Hence

$$y'' + 2y' - 3y = G, \quad G = O(e^{-\gamma' s}), \quad \gamma' = \min\{\gamma, \beta + 2\} > 3.$$

Variation of constants for the roots 1, -3 eliminates the growing homogeneous term and gives

$$y = ce^{-3s} + O(e^{-\gamma' s}), \quad y' = -3ce^{-3s} + O(e^{-\gamma' s}).$$

For example the particular solution is a linear combination of

$$e^s \int_s^\infty e^{-t} G(t) dt \quad \text{and} \quad e^{-3s} \int_s^\infty e^{3t} G(t) dt;$$

both integrals converge with the claimed bounds. Since $r = Ce^s(1 + O(e^{-2s}))$, this is (37.3). The equation gives the second differentiated remainder, and differentiated equations give the rest. Angular differentiations are bounded operations on the fixed finite spaces. Any other solution tending uniformly to zero also satisfies the initial weighted bound: compare on sufficiently long intervals with $Ah + \delta$, where $\delta > 0$ dominates the far boundary value, and then let $\delta \downarrow 0$. The

same ODE improvement applies. The improved decay permits comparison for a difference of two solutions, proving uniqueness. $\square$

37.2. Coefficient equations and the required spatial derivatives. Put $\psi = \phi - 1$, $h_q = q_{ss}|_S$, and $Q_q = |q|_{g_0}^2$. Define

$$\begin{aligned} \mathcal{A}(z) &= \frac{15}{2}z^2 + \frac{15}{2}z^3 + \frac{15}{4}z^4 + \frac{3}{4}z^5, \\ \mathcal{N}_{\epsilon,q}(z) &= \mathcal{A}(z) - \frac{\epsilon^2}{8}Q_q(1+z)^{-7}, \\ (37.5) \quad \mathcal{B}_{\epsilon,q}(z) &= \frac{\epsilon}{4}h_q(1+z)^{-3}. \end{aligned}$$

The actual equations become

$$(37.6) \quad L\psi = \mathcal{N}_{\epsilon,q}(\psi), \quad \partial_s \psi|_S = \mathcal{B}_{\epsilon,q}(\psi|_S).$$

For a finite angular type seed, define $u_j = u_j[q]$ recursively. Set $P_{j-1} = \sum_{i < j} \epsilon^i u_i$, and let

$$\begin{aligned} (37.7) \quad F_j &= [\mathcal{N}_{\epsilon,q}(P_{j-1})]_j, & B_j &= [\mathcal{B}_{\epsilon,q}(P_{j-1}|_S)]_j, \\ Lu_j &= F_j, & \partial_s u_j|_S &= B_j, & u_j &\longrightarrow 0. \end{aligned}$$

Here $[\cdot]_j$ extracts the coefficient of ϵ^j ; inverse powers are Taylor expanded at 1. The first two equations are explicitly

$$\begin{aligned} (37.8) \quad F_1 &= 0, & B_1 &= \frac{1}{4}h_q, \\ F_2 &= \frac{15}{2}u_1^2 - \frac{1}{8}Q_q, & B_2 &= -\frac{3}{4}h_q u_1|_S. \end{aligned}$$

Only preceding coefficients occur on the right of (37.7). The construction does not require an actual solution branch for the seed.

Lemma 37.2 (Finite coefficient asymptotics). *For every smooth finite angular type seed in $C_\tau^{1,\alpha}$, the coefficients through degree two are well defined and*

$$u_j = r^{-3}\zeta_j(x) + o(r^{-3}), \quad j = 1, 2.$$

The first coefficient has this asymptotic with arbitrarily many spatial derivatives. The second has it with the first spatial derivatives, in particular with every derivative needed at degree two below.

For the fourth-order construction, suppose specifically that

$$(37.9) \quad q = \text{Hess } v - Bv, \quad Lv = 0, \quad v \longrightarrow 0, \\ v|_S \text{ has only constant and degree-one spherical modes.}$$

Corollary 39.3 will show that every seed with zero quadratic deficit has this form, so fourth-order estimates are needed only in this class. Then q and all its derivatives have $O(r^{-3})$ scaled components. The coefficients $u_1, \dots, u_4$ exist and have the above asymptotic with arbitrarily many spatial derivatives.

Let $p = 2$ in the first case and $p = 4$ in the second. In the coordinate $\eta = \text{arsinh } r$, the conformal metric jet $G = (1 + \sum_{j=1}^p \epsilon^j u_j)^4 g_0$, taken through degree p , satisfies

$$\begin{aligned} (37.10) \quad G - b &= r^{-3}(C \, d\eta^2 + E r^2 \sigma) + o(r^{-3}), \\ C &= 2m_0 + 4 \sum_{j=1}^p \epsilon^j \zeta_j, & E &= 4 \sum_{j=1}^p \epsilon^j \zeta_j. \end{aligned}$$

For coefficient j , the remainder has this decay after angular and η derivatives through order $p - j + 1$, in scaled coframes $d\eta, r \, dx$.

Proof. Lemma 37.1 constructs u_1 with homogeneous interior equation. Its derivatives have the asserted asymptotics by (37.4). In degree two the interior source is a sum of $u_1^2 = O(r^{-6})$ and $|q|^2 = O(r^{-2\tau})$, with $\min\{6, 2\tau\} > 3$. Its first radial derivative has the same bound because $q \in C_\tau^{1,\alpha}$. Finite angular type supplies any fixed number of angular derivatives. The inverse lemma therefore gives the claimed second coefficient and, in particular, its first differentiated asymptotic. No bound on higher radial derivatives of a general seed is asserted here.

For (37.9), solve the homogeneous Dirichlet problem separately in its four boundary modes. The homogeneous ODE gives $v = O(r^{-3})$ with all differentiated versions. The scaled components of the connection, B , and all their derivatives are bounded at infinity. It follows that $\text{Hess } v - Bv$ has $O(r^{-3})$ components with all derivatives. Inductively, every interior source in (37.7) is a sum of products containing at least two factors among lower solution coefficients and q . It therefore has $O(r^{-6})$ decay with all derivatives. Another application of the inverse lemma completes the induction through degree four.

Finally $g_0 - b = 2m_0 r^{-3} d\eta^2 + O(r^{-5})$ in scaled components, with all derivatives. In each positive degree the only r^{-3} term in $(1 + \sum \epsilon^j u_j)^4 - 1$ is $4r^{-3} \zeta_j$; products of two solution coefficients are $O(r^{-6})$. This proves (37.10). At $p = 2$, the derivative count is two for coefficient one and one for coefficient two, both already established. At $p = 4$ all the needed derivatives are available from the stronger case. $\square$

37.3. An exact Green formula for the mass. Define the test functions

$$(37.11) \quad W_0 = N, \quad W_i = r x_i \quad (1 \leq i \leq 3).$$

The background identities and the radial equation give

$$(37.12) \quad L W_0 = 0, \quad L W_i = -3m_0 r^{-2} x_i.$$

Lemma 37.3 (Green representation of the mass). *For every sufficiently small member of the actual branch,*

$$(37.13) \quad p_\mu(\epsilon) - p_\mu(0) = \frac{1}{2\pi} \left\{ \int_S (\psi \partial_s W_\mu - W_\mu \partial_s \psi) \, dA_{g_0} - \int_M (W_\mu L\psi - \psi L W_\mu) \, dV_{g_0} \right\}.$$

In particular the right-hand side is an absolutely convergent integral expression. More generally the linear functional

$$(37.14) \quad \mathcal{G}_\mu(u, F, B) = \frac{1}{2\pi} \left\{ \int_S (u \partial_s W_\mu - W_\mu B) \, dA_{g_0} - \int_M (W_\mu F - u L W_\mu) \, dV_{g_0} \right\}$$

is bounded in the norms $\|u\|_\beta + \|F\|_{2\beta} + \|B\|_{C^0(S)}$. For finite angular type coefficient jets, this same functional computes the coefficient of their metric mass.

Proof. First fix ϵ ; no parameter limit is taken in this argument. The assumed weighted regularity gives $\psi, \nabla^b \psi = O(r^{-\tau})$. The difference between the actual metric perturbation $\phi^4 g_0 - g_0$ and $4\psi b$ consists of terms with scaled components and first derivatives

$$O(r^{-2\tau}) + O(r^{-\tau-3}).$$

Indeed $\phi^4 - 1 - 4\psi = O(\psi^2)$ and $g_0 - b = O_1(r^{-3})$. Since a mass test function and its derivative have size $O(r)$ and the sphere area has size $O(r^2)$, these terms give flux errors

$$(37.15) \quad O(r^{3-2\tau}) + O(r^{-\tau}) \longrightarrow 0.$$

For the tensor $4\psi b$ in dimension three,

$$\text{div}_b(4\psi b) - d \, \text{tr}_b(4\psi b) = -8 \, d\psi,$$

and the remaining two terms of the prescribed mass integrand add $8\psi \, dV_\mu$. Thus the flux difference is $1/(2\pi)$ times the flux of $\psi \, \text{grad}_b V_\mu - V_\mu \, \text{grad}_b \psi$. Replacing the hyperbolic normal

by ∂_s produces a vanishing error since the relative normal change is $O(r^{-3})$. For the time test, replacing $\sqrt{1+r^2}$ by N also produces a vanishing error, because their difference and its first radial derivative are $O(r^{-2})$. The sphere area forms are the same.

The g_0 divergence of $\psi \operatorname{grad} W_\mu - W_\mu \operatorname{grad} \psi$ is $\psi L W_\mu - W_\mu L \psi$. Apply the divergence theorem to $[0, T] \times S^2$. The outward normal of this integration domain at its inner boundary is $-\partial_s$, yielding exactly the sign of the boundary term in (37.13). Finally $L\psi = O(r^{-2\tau})$ by (37.6); hence the bulk integrals converge, and $T \rightarrow \infty$ proves the identity.

For the asserted boundedness, $W_\mu = O(e^s)$ and $dV_{g_0} = O(e^{2s}) ds d\omega$. The $W_\mu F$ term is dominated by a constant times

$$\|F\|_{2\beta} e^{(3-2\beta)s},$$

which is integrable by (37.1). The $u L W_i$ term is dominated by $C\|u\|_\beta e^{-\beta s}$, and $L W_0 = 0$. The boundary terms are bounded by the stated norms. For finite coefficient jets, apply the same finite-radius identity coefficientwise. Lemma 37.2 supplies their differentiated asymptotics, and all nonlinear coefficient products discarded from the flux are $O(r^{-6})$ with first derivatives. Thus their coefficient flux is also (37.14). $\square$

37.4. Approximation on the full space of seeds. Let $\mathcal{T}_{\tau,\alpha}$ denote the real vector space of smooth TT tensors in $C_\tau^{1,\alpha}$. No boundary sign is imposed on this space. We shall construct coefficient functionals on it even when no actual branch is being considered.

Lemma 37.4 (Rotation approximation and coefficient continuity). *Every $q \in \mathcal{T}_{\tau,\alpha}$ is a limit, in $\|\cdot\|_\beta$, of finite angular type tensors $q_\nu \in \mathcal{T}_{\tau,\alpha}$. The approximants may be chosen as averages of rotated copies of q . Their weighted norms are uniformly bounded, and the construction preserves a nonnegative boundary value $q_{ss}|_S$ when that condition holds.*

The maps $q \mapsto u_1[q]$ and $q \mapsto u_2[q]$ defined for finite angular type seeds extend uniquely by these approximations to continuous, respectively linear and quadratic, maps on $\mathcal{T}_{\tau,\alpha}$ with values in the weighted uniform norm. The corresponding pairs (F_j, B_j) in (37.8) extend continuously in the norms $\|\cdot\|_{2\beta}$ and $\|\cdot\|_{C^0(S)}$.

Proof. Let $U(R)q$ denote the pullback by a rotation $R \in SO(3)$, and let dR be normalized invariant volume. Define

$$k_\nu(R) = c_\nu \left(\frac{1 + \operatorname{tr} R}{4} \right)^\nu, \quad \int_{SO(3)} k_\nu dR = 1, \quad q_\nu = \int_{SO(3)} k_\nu(R) U(R)q dR.$$

The expression raised to the power ν lies in $[0, 1]$ and attains one only at the identity. The normalized densities therefore concentrate there. To verify this directly, outside any fixed neighborhood their base is at most $1 - \delta$, whereas on a smaller neighborhood of positive volume it is at least $1 - \delta/2$; the ratio of the corresponding integrals tends to zero.

The map $R \mapsto U(R)q$ is continuous in $\|\cdot\|_\beta$. On a compact part of M this follows from smoothness. On the tail, the bound $Ce^{-(\tau-\beta)s}$ is uniform in R , so the strict weight margin controls it. Concentration gives $\|q_\nu - q\|_\beta \rightarrow 0$. Rotations preserve g_0 , the end background, and s . They commute with trace and divergence and preserve the weighted regularity. Consequently each average is smooth and TT, with the asserted uniform bounds. They also preserve the radial normal, so positivity of $q_{ss}|_S$ is retained by the nonnegative average.

For fixed ν , the density is a polynomial in matrix entries. After a change of integration variable, the rotation orbit of q_ν depends on translates of this polynomial. These translates lie in a finite-dimensional polynomial space; their coefficients multiply finitely many fixed averaged tensors. Thus q_ν has finite angular type.

It remains to check that the coefficient limits are independent of the approximation. On a bounded set in $\|q\|_\beta$, comparison applied to (37.8) gives, for finite type seeds $q, \hat{q}$,

$$\begin{aligned} \|u_1[q] - u_1[\hat{q}]\|_\beta &\leq C\|q - \hat{q}\|_\beta, \\ \|F_2[q] - F_2[\hat{q}]\|_{2\beta} + \|B_2[q] - B_2[\hat{q}]\|_{C^0(S)} &\leq C\|q - \hat{q}\|_\beta, \\ \|u_2[q] - u_2[\hat{q}]\|_\beta &\leq C\|q - \hat{q}\|_\beta. \end{aligned}$$

Here the elementary product estimate is

$$\| |q|^2 - |\hat{q}|^2 \|_{2\beta} \leq (\|q\|_\beta + \|\hat{q}\|_\beta)\|q - \hat{q}\|_\beta.$$

The comparison constant has no angular-cutoff dependence. These estimates prove existence, uniqueness, and continuity of the limits in the complete weighted uniform space. Formula (37.8) preserves their linear and quadratic dependence. No derivative convergence of the limiting coefficients is needed in this construction or in the mass functional. $\square$

37.5. Actual expansions and mass coefficients.

Proposition 37.5 (Expansions of the given branch). *For every prescribed seed and branch in the theorem, set $p = 2$ and take u_j, F_j, B_j from Lemma 37.4. If the seed has the form (37.9), we may instead take $p = 4$ with the finite coefficient construction. In both cases,*

$$\begin{aligned} \left\| \psi - \sum_{j=1}^p \epsilon^j u_j \right\|_\beta &= O(\epsilon^{p+1}), \\ \left\| L\psi - \sum_{j=1}^p \epsilon^j F_j \right\|_{2\beta} &= O(\epsilon^{p+1}), \\ (37.16) \quad \left\| \partial_s \psi|_S - \sum_{j=1}^p \epsilon^j B_j \right\|_{C^0(S)} &= O(\epsilon^{p+1}). \end{aligned}$$

The area and all four mass components have actual expansions to the same order. Their mass coefficients are

$$(37.17) \quad a_{\mu j}[q] = \mathcal{G}_\mu(u_j, F_j, B_j), \quad p_\mu(\epsilon) = p_\mu(0) + \sum_{j=1}^p \epsilon^j a_{\mu j}[q] + O(\epsilon^{p+1}).$$

In particular, the actual deficit satisfies

$$(37.18) \quad \mathfrak{D}_q(\epsilon) = c_2[q]\epsilon^2 + O(\epsilon^3)$$

for every seed under consideration. The coefficient c_2 is a continuous quadratic form on $\mathcal{T}_{\tau, \alpha}$ in the $\|\cdot\|_\beta$ topology, invariant under rotations. For seeds of the form (37.9) there is additionally the actual expansion

$$(37.19) \quad \mathfrak{D}_q(\epsilon) = c_2[q]\epsilon^2 + c_3[q]\epsilon^3 + c_4[q]\epsilon^4 + O(\epsilon^5).$$

The subsequent geometric arguments determine the signs and the vanishing of these coefficients.

Proof. Initial estimate. The given convergence of ψ in $C_\tau^{2, \alpha}$ implies that $\|\psi\|_{C^0}$ is small. Write $\mathcal{A}(\psi) = V_\psi \psi$, extending the quotient by zero at $\psi = 0$. Then $\|V_\psi\|_{C^0} = o(1)$. Moving this term to the potential in (37.6) leaves an interior source bounded by $C\epsilon^2 e^{-2\tau s}$ and boundary derivative bounded by $C\epsilon$. Since $\psi = o(e^{-\beta s})$, Lemma 37.1 gives

$$(37.20) \quad \|\psi\|_\beta = O(\epsilon).$$

Finite angular type seeds. Suppose first that the needed coefficients have been constructed by (37.7). The first bound (37.20) starts an induction. If

$$\|\psi - P_{j-1}\|_\beta = O(\epsilon^j),$$

then ψ and P_{j-1} are both $O(\epsilon e^{-\beta s})$. For such arguments, the mean value theorem applied to (37.5) gives

$$|\mathcal{N}_{\epsilon,q}(\psi) - \mathcal{N}_{\epsilon,q}(P_{j-1})| \leq C\epsilon e^{-\beta s} |\psi - P_{j-1}| = O(\epsilon^{j+1} e^{-2\beta s}).$$

Taylor substitution into the finite polynomial P_{j-1} leaves an error of this same order after its coefficients through degree j are extracted. Every interior term contains at least two decaying factors, counting the factor Q_q as two. On S ,

$$|\mathcal{B}_{\epsilon,q}(\psi) - \mathcal{B}_{\epsilon,q}(P_{j-1})| \leq C\epsilon |\psi - P_{j-1}| = O(\epsilon^{j+1}),$$

and the boundary Taylor error has the same order. Thus $\psi - P_j$ has interior source $O(\epsilon^{j+1} e^{-2\beta s})$ and boundary derivative $O(\epsilon^{j+1})$. It is $o(e^{-\beta s})$ at infinity, so comparison improves its weighted norm to $O(\epsilon^{j+1})$. This proves all three assertions in (37.16) inductively through the required degree. In particular, it proves the fourth-order assertions in the special case (37.9).

Arbitrary seeds through degree two. Let q_ν be the approximants of Lemma 37.4, and put $d_\nu = \|q - q_\nu\|_\beta$. Write $u_{j,\nu}, F_{j,\nu}, B_{j,\nu}$ for their coefficients. These have uniform bounds in the norms of that lemma. We compare them directly with the given branch for q ; no branch for q_ν is introduced. Using (37.20), the equation for $\psi - \epsilon u_{1,\nu}$ has source $O(\epsilon^2 e^{-2\beta s})$ and boundary derivative $O(\epsilon d_\nu + \epsilon^2)$. Hence

$$(37.21) \quad \|\psi - \epsilon u_{1,\nu}\|_\beta \leq C(\epsilon d_\nu + \epsilon^2).$$

Expanding the nonlinearities once more, formula (37.8) and (37.21) imply

$$(37.22) \quad \begin{aligned} \|L\psi - \epsilon^2 F_{2,\nu}\|_{2\beta} &\leq C(\epsilon^2 d_\nu + \epsilon^3), \\ \|\partial_s \psi|_S - \epsilon B_{1,\nu} - \epsilon^2 B_{2,\nu}\|_{C^0(S)} &\leq C((\epsilon + \epsilon^2)d_\nu + \epsilon^3). \end{aligned}$$

For clarity, the only quadratic interior discrepancies are $\psi^2 - \epsilon^2 u_{1,\nu}^2$ and $\epsilon^2(Q_q - Q_{q_\nu})$; their weighted norms are bounded by the first right-hand side. At the boundary the linear discrepancy is $\epsilon(h_q - h_{q_\nu})/4$, and the next one contains $\epsilon h_q \psi - \epsilon^2 h_{q_\nu} u_{1,\nu}$, producing exactly the second bound. All remaining terms are cubic with the stated weights. Comparison applied to $\psi - \epsilon u_{1,\nu} - \epsilon^2 u_{2,\nu}$ now gives

$$\|\psi - \epsilon u_{1,\nu} - \epsilon^2 u_{2,\nu}\|_\beta \leq C((\epsilon + \epsilon^2)d_\nu + \epsilon^3).$$

The constants are independent of ν . Letting $\nu \rightarrow \infty$ and using coefficient continuity proves the full three estimates (37.16) for $p = 2$ and the original arbitrary seed.

Mass, area, and the deficit. Apply the bounded functional in Lemma 37.3 to the three remainders. This proves (37.17) directly. The area is the compact-boundary integral

$$A_\epsilon = \int_S (1 + \psi)^4 dA_{g_0},$$

so its expansion follows from the first remainder alone. If $A_\epsilon = A_0 + \epsilon A_1 + \epsilon^2 A_2 + O(\epsilon^3)$, then

$$A_1 = 4 \int_S u_1 dA_{g_0}, \quad A_2 = \int_S (4u_2 + 6u_1^2) dA_{g_0}.$$

The mass norm is smooth near $(m_0, 0, 0, 0)$ and b_* is smooth near $A_0 = 4\pi a^2$. Their expansions therefore have the same controlled remainders. The constant deficit is zero. Also $F_1 = 0$, $N|_S = 0$, $\partial_s N|_S = \kappa$, and $LN = 0$, so

$$a_{01} = \frac{\kappa}{2\pi} \int_S u_1 dA_{g_0} = b'_*(A_0)A_1.$$

The spatial mass components have zero base value and first enter the mass norm quadratically. This proves that the linear deficit coefficient vanishes.

In degree two the scalar expression is explicitly

$$(37.23) \quad c_2[q] = a_{02} - \frac{1}{2m_0} \sum_{i=1}^3 a_{i1}^2 - b'_*(A_0)A_2 - \frac{1}{2}b''_*(A_0)A_1^2.$$

Lemma 37.4 and the bounded Green functional show that this is a continuous quadratic form on the entire formal seed space $\mathcal{T}_{\tau,\alpha}$. Formula (37.23) uses only the coefficient functions and absolutely convergent Green integrals; it does not assume a geometric integral formula for arbitrary seeds. Rotations leave the area invariant, fix the time mass component, and rotate the three spatial components. Thus c_2 is rotation invariant. The identical finite coefficient argument through degree four proves (37.19). $\square$

Remark 37.6. The derivative control used in (37.15) comes from each actual member's assumed $C_{\tau}^{2,\alpha}$ decay. It is not a consequence of the weighted uniform Taylor estimates. The radius limit is taken for each fixed member to obtain the exact identity (37.13); only then are its integrals expanded. Similarly, each finite angular type coefficient is identified with its own flux before passing to an approximant limit in the bounded functional (37.14). No uniform convergence of discarded derivative flux errors in either parameter or angular cutoff is used.

38. CMC SPHERE JETS AND THEIR HAWKING MASS

We use the finite Taylor conventions of Section 36.3. Taylor expansions of inverses, compositions, and other smooth geometric operations at a rotation-invariant background preserve finite angular type at each fixed order: each coefficient uses only finitely many linear differential operations and tensor products.

38.1. The sphere construction. In the end put $\eta = \operatorname{arsinh} r$, so that

$$b = d\eta^2 + \sinh^2 \eta \sigma.$$

An estimate for a tensor below refers to its components in scaled coframes $d\eta, r \, dx^A$, in fixed smooth angular coordinate charts. A differentiated little-oh estimate includes all mixed radial and angular derivatives of the stated total order.

Proposition 38.1 (CMC sphere jets). *Let $p \geq 1$, and let G be a smooth Riemannian metric jet of order p , with base $G_0 = g_0$, whose coefficients have finite angular type. Suppose that there are angular scalar jets C, E , with $C_0 = 2m_0$ and $E_0 = 0$, such that*

$$(38.1) \quad G - b = r^{-3}(C \, d\eta^2 + E r^2 \sigma) + o(r^{-3}).$$

For the coefficient of degree j , $1 \leq j \leq p$, suppose that the remainder satisfies this estimate after every mixed radial and angular derivative of total order at most $p - j + 1$.

There is a smooth family of sphere jets Σ_s , $0 \leq s < \infty$, given by graphs over $\{s\} \times S^2$, with the following properties:

- (1) Σ_s has constant outward mean curvature through order p , and Σ_0 is minimal through that order.
- (2) The graph coefficients have finite angular type. The normal lapse f for variation in s has base value $f_0 = 1$.
- (3) If $A(s)$ and $H(s)$ are the area and mean curvature jets, then the CMC Hawking mass

$$(38.2) \quad m_H(\Sigma_s) = \sqrt{\frac{A(s)}{16\pi}} \left(1 - \frac{(H(s)^2 - 4)A(s)}{16\pi} \right)$$

satisfies, coefficientwise,

$$(38.3) \quad \lim_{s \rightarrow \infty} m_H(\Sigma_s) = m_{\text{AH}}(G) := \sqrt{p_0(G)^2 - \sum_{i=1}^3 p_i(G)^2}.$$

Here $p_\mu(G)$ is the coefficientwise flux (36.8), and the square root is its formal branch with base value $m_0 > 0$.

The expression (38.2) is the CMC specialization of the asymptotically hyperbolic Hawking mass used in [6, Equation (2)]. The proof below checks its limiting normalization against (36.8) directly.

Proof. We first solve the Jacobi equations on finite radial intervals. At infinity, hyperbolic boosts account for the order-one degree-one displacement. The asymptotic CMC equation selects a frame in which the spatial mass components vanish. We then choose the free graph means to join those spheres to the minimal first leaf.

The graph equation on finite radial intervals. For a normal variation with speed w , the mean-curvature variation is

$$(38.4) \quad Jw = -\Delta_\Sigma w - (\text{Ric}_G(\nu, \nu) + |\Pi|^2)w.$$

Indeed, the induced metric varies by $2w\Pi$, the variation of the unit normal is $-\text{grad}_\Sigma w$, and differentiation of the second form gives the Hessian term, the ambient curvature term, and the quadratic second-form term. Tracing, including the variation of the inverse induced metric, gives (38.4). A tangential velocity adds the directional derivative of H .

For a background coordinate sphere, $H_0 = 2N/r$ and

$$\text{Ric}_{g_0}(\partial_s, \partial_s) = -2 - \frac{2m_0}{r^3}, \quad |\Pi_0|^2 = \frac{2N^2}{r^2}.$$

Thus its Jacobi operator is

$$(38.5) \quad J_s^0 = -\Delta_{r^2\sigma} - \frac{2}{r^2} + \frac{6m_0}{r^3}.$$

On degree- ℓ spherical harmonics its eigenvalue is

$$(38.6) \quad \lambda_\ell(s) = \frac{\ell(\ell+1) - 2}{r^2} + \frac{6m_0}{r^3}.$$

It is strictly positive on every nonconstant mode. In particular, the degree-one eigenvalue is $6m_0/r^3 > 0$. At $s = 0$, using $2m_0 = a(1 + a^2)$, the constant eigenvalue is

$$-\frac{2}{a^2} + \frac{6m_0}{a^3} = 3 + \frac{1}{a^2} = \frac{2\kappa}{a},$$

and hence $J_0^0 = D$, positive on all modes.

Write a graph as $s + \rho(s, x; \epsilon)$, with $\rho_0 = 0$. At degree j , its mean-curvature coefficient is

$$J_s^0 \rho_j + \mathcal{R}_j(s, x),$$

where $\mathcal{R}_j$ is known from the metric coefficients and the lower graph coefficients. To make the first sphere minimal, solve this equation at $s = 0$ with right-hand side zero, using D^{-1} . At general s , prescribe any smooth round mean for ρ_j that agrees with this value at zero, and solve the projection of the equation onto the nonconstant modes. The operator (38.5) preserves the splitting into constants and nonconstants, so the latter equation has a unique solution. At each order it is a finite-dimensional linear equation with a smooth invertible coefficient matrix at every finite radius. This produces smooth coefficients on each finite interval. Induction also shows that the resulting graph at zero is exactly the minimal graph jet already constructed there. The constant eigenvalue need not be inverted away from zero.

The flux aspect. For an angular scalar F , write $\langle F \rangle_\sigma = (4\pi)^{-1} \int_{S^2} F \, dA_\sigma$. The asymptotic form (38.1) gives

$$(38.7) \quad (p_0(G), p_1(G), p_2(G), p_3(G)) = \langle \mu(x)(1, x) \rangle_\sigma, \quad \mu = \frac{C + 3E}{2}.$$

Here μ denotes only a flux aspect. To verify the formula directly, let $e = G - b$. Its leading trace is $(C + 2E)r^{-3}$, and its radial divergence-minus-trace derivative is

$$\begin{aligned} (\operatorname{div}_b e - \operatorname{d} \operatorname{tr}_b e)(\partial_\eta) &= 2 \coth \eta (C - E)r^{-3} - 2\partial_\eta (Er^{-3}) + o(r^{-3}) \\ &= (2C + 4E)r^{-3} + o(r^{-3}). \end{aligned}$$

For $V_0 = \sqrt{1 + r^2}$ or $V_i = rx_i$, let $v = 1$ or x_i , respectively. Both V and $\partial_\eta V$ have leading term rv . The terms $(\operatorname{tr}_b e) \, dV - e(\operatorname{grad}_b V, \cdot)$ add $2Evr^{-2} + o(r^{-2})$ to the normal flux integrand. The total is therefore $(2C + 6E)vr^{-2} + o(r^{-2})$. Multiplying by $r^2 dA_\sigma / (16\pi)$ proves (38.7). The first differentiated remainder assumed in (38.1) makes every omitted flux integral tend to zero.

Boosted coordinates. Realize hyperbolic space as

$$(\cosh \eta, \sinh \eta x) \subset \mathbb{R}^{3,1}.$$

Let $\mathcal{B}(d)$ be the exponential of the Lorentz Lie-algebra matrix with time-space entries $d \in \mathbb{R}^3$ and zero spatial rotation part. Thus $\mathcal{B}(0)$ is the identity. Write its action on future null vectors as

$$(38.8) \quad \mathcal{B}(d)(1, x) = k(x)(1, y(x)).$$

For small d , $k > 0$. In pullback coordinates the old radius is $kr + O(r^{-1})$, the old radial distance is $\eta + \log k + O(r^{-2})$, and the old radial unit covector agrees with the new one up to $O(r^{-1})$ in hyperbolic norm. These statements, including all their needed derivatives, follow by applying the matrix $\mathcal{B}(d)$ to the displayed hyperboloid coordinates. Since $\mathcal{B}(d)$ preserves b , the leading coefficients of its pullback of G are

$$(38.9) \quad C_d = k^{-3}C \circ y, \quad E_d = k^{-3}E \circ y.$$

For example, the angular part of $d \log k$ has hyperbolic norm $O(r^{-1})$; it therefore changes neither leading scaled diagonal coefficient in (38.1).

Differentiating (38.8) tangentially and taking Minkowski inner products gives $y^* \sigma = k^{-2} \sigma$. Consequently $dA_\sigma(y) = k^{-2} dA_\sigma(x)$, and $(1, x) = k\mathcal{B}(d)^{-1}(1, y)$. Changing angular variables proves the exact leading-flux transformation

$$(38.10) \quad \left\langle k^{-3} \mu(y)(1, x) \right\rangle_\sigma = \mathcal{B}(d)^{-1} \langle \mu(x)(1, x) \rangle_\sigma.$$

This establishes the needed covariance directly for the specified flux.

The asymptotic CMC equation. For large s , put $R = r(s)$ and $\eta_R = \operatorname{arsinh} R$. In the coordinates obtained from $\mathcal{B}(d)$, seek a graph

$$(38.11) \quad \eta = \eta_R + \frac{h(x)}{R},$$

where d and h are jets with zero base values, and every coefficient of h has zero constant and degree-one components. The parameters d, h are held fixed when computing an individual leaf. We claim, coefficientwise for bounded input jets in fixed finite angular spaces, that

$$(38.12) \quad R^3(H - 2 \coth \eta_R) = -(\Delta_\sigma + 2)h - (C_d + 3E_d) + o(1).$$

Here are the estimates underlying this claim. In b , the unit normal to (38.11) is

$$\frac{\partial_\eta - r^{-2} \operatorname{grad}_\sigma(h/R)}{\sqrt{1 + r^{-2} |\operatorname{d}(h/R)|_\sigma^2}}.$$

Its divergence gives

$$H_b = 2 \coth \eta_R - R^{-3}(\Delta_\sigma + 2)h + O(R^{-4}).$$

Indeed the slope has hyperbolic norm $O(R^{-2})$, its normalization differs from one by $O(R^{-4})$, and the second Taylor term in $2 \coth(\eta_R + h/R)$ is $O(R^{-4})$. These estimates hold for every coefficient of the finite Taylor expansion when the input coefficients are bounded.

On an unshifted sphere the perturbation in (38.1), after boosting, changes the radial lapse by $C_d R^{-3}/2 + o(R^{-3})$. This contributes $-C_d R^{-3} + o(R^{-3})$ to H . The relative area density changes by $E_d R^{-3} + o(R^{-3})$; its radial logarithmic derivative contributes $-3E_d R^{-3} + o(R^{-3})$. The mixed components in the remainder contribute $o(R^{-3})$, including their tangential-divergence term. For completeness, on a fixed surface the comparison between two metrics b and $\hat{g}$ is

$$\Pi_{\hat{g}}(X, Y) = \hat{g}(\nu_{\hat{g}}, \nu_b) \Pi_b(X, Y) - \hat{g}(\nu_{\hat{g}}, (\nabla^{\hat{g}} - \nabla^b)_X Y).$$

It follows by splitting $\nabla_X^b Y$ into its tangential and normal parts and using normal orthogonality. After tracing, the linear metric correction depends smoothly on the tangent plane, the background shape operator, and the scaled components of $\hat{g} - b, \nabla^b(\hat{g} - b)$. The slope of (38.11) tends to zero, and its background shape operator differs by $o(1)$ from that of a coordinate sphere. Its displacement is $O(R^{-1})$ in radial distance. The leading metric correction on that graph is consequently still $-(C_d + 3E_d)R^{-3}$. Quadratic metric corrections are $O(R^{-6})$. This proves (38.12).

The estimate just proved also justifies its use at every Taylor order through p . A coefficient of G of degree j can undergo at most $p - j$ spatial differentiations from Taylor composition with the graph and the boost. Mean curvature requires one additional derivative of the metric. Thus the mixed derivative allowance $p - j + 1$ in (38.1) controls the coefficient of every remainder after multiplication by R^3 . The base g_0 has the required bounds to all orders. Angular differential operators cause no further restriction within a fixed finite-dimensional angular space. More explicitly, denote the degree- j residual in (38.1) by $\mathcal{E}_j$, and set

$$\omega_j(R) = \max_{a+b \leq p-j+1} \sup_{r \geq R/2} \left\| \partial_\eta^a \nabla_\sigma^b (r^3 \mathcal{E}_j^{\text{scaled}}) \right\|_\infty.$$

Then $\omega_j(R) \rightarrow 0$. For every $n \leq p$, bounded coefficients of d, h in fixed finite angular spaces give the estimate

$$\left\| [R^3(H - 2 \coth \eta_R) + (\Delta_\sigma + 2)h + C_d + 3E_d]_n \right\|_\infty \leq C_{p, \mathcal{V}, M} \left(R^{-1} + \sum_{j=1}^n \omega_j(R) \right).$$

Here $\mathcal{V}$ denotes the finitely many angular spaces and M a bound for the input coefficients; the fixed metric coefficients may also enter the constant. The explicit base residual is $O(r^{-5})$ with all derivatives and is included in the R^{-1} bound. The preceding graph estimates, the product rule, and the derivative count prove this estimate term by term. It is an estimate for Taylor coefficients, exactly as needed for the recursion.

Solving the rescaled equation. Project $R^3(H - 2 \coth \eta_R)$ onto the nonconstant modes and set the projection equal to zero, through order p . At the background, the finite-radius linearization preserves the splitting between the h -modes, of degrees at least two, and the three boost modes. Set

$$\alpha_R = \frac{\sqrt{1 + R^2}}{N(R)}.$$

A graph variation h/R in η has physical normal displacement $\alpha_R h/R$. Equations (38.5)–(38.6) give its exact degree- ℓ multiplier in the rescaled equation:

$$(38.13) \quad \alpha_R(\ell(\ell + 1) - 2 + 6m_0/R), \quad \ell \geq 2.$$

An infinitesimal boost has η -displacement $d \cdot x$. Its physical normal displacement is $\alpha_R d \cdot x$, so its multiplier is $6m_0\alpha_R$. These blocks converge respectively to $\ell(\ell+1) - 2$ and $6m_0$, both invertible. Equivalently, the latter sign and coefficient follow by differentiating $2m_0k^{-3}$ in (38.12), since $k = 1 + d \cdot x + O(|d|^2)$.

At each Taylor order only fixed finite angular spaces are involved. In particular, the Taylor coefficients at $d = 0$ of the boost map and its compositions are finite sums of coordinate functions and angular derivatives of the given profiles. This assertion concerns the finite Taylor expansion: it does not require a finite harmonic cutoff to be preserved by a finite nonzero boost. One can see the closure directly from the infinitesimal boost field

$$\mathcal{K}_a = (a \cdot x)\partial_\eta + \coth \eta \operatorname{grad}_\sigma(a \cdot x).$$

It follows by differentiating the hyperboloid coordinates. Its Lie derivative, repeated finitely many times, uses only radial derivatives, angular derivatives, and multiplication by degree-one profiles, also for tensor fields. It follows that the inverses of the finite-dimensional blocks above are bounded for all sufficiently large R and converge as $R \rightarrow \infty$.

Now solve successively for the coefficients of h, d . At each order their linear operator is the displayed background linearization. The remaining source depends only on lower solved coefficients and the prescribed metric coefficients. Equation (38.12), with its uniform coefficientwise remainder estimate, shows inductively that this source converges once the lower coefficients do. The converging inverse then gives bounded coefficients of h, d with limits. This starts at degree one and proves the assertion through order p . Smooth dependence on s follows at every finite radius from the smooth finite-dimensional equations.

In original coordinates these spheres are graph jets over the background sphere: the angular map has identity as base and can be inverted formally. Their round mean graph heights are smooth scalar functions of s . Choose the free means in the finite-radius recursion to agree with these functions for all sufficiently large s , and to agree with the minimal graph's means at zero, using a smooth interpolation. Inductive uniqueness of the nonconstant graph coefficients makes the resulting recursion coincide with the far construction. This gives the family on the whole half-line. Its base is the coordinate-sphere family, so its lapse has base one.

The Hawking mass limit. Let d_∞ be the coefficientwise limiting boost. The degree-one projection of (38.12) gives

$$\langle \mu_{d_\infty} x_i \rangle_\sigma = 0, \quad \mu_d = (C_d + 3E_d)/2.$$

Writing

$$H_b(R) = 2 \coth \eta_R, \quad B_R = R^3(H - H_b(R)),$$

the round average of that same equation gives

$$B_R = -2\langle \mu_d \rangle_\sigma + o(1).$$

The area of (38.11) satisfies

$$(38.14) \quad \frac{A}{4\pi R^2} = 1 + \delta_R, \quad \delta_R = O(R^{-2})$$

coefficientwise. To see the relevant cancellation, the relative hyperbolic area density is

$$1 + 2 \coth \eta_R \frac{h}{R} + O(R^{-2}),$$

and the metric correction is $O(R^{-3})$. The sole possible relative term of order R^{-1} integrates to zero because every coefficient of h has zero round mean.

Since $H_b(R)^2 - 4 = 4/R^2$, direct substitution into (38.2) gives

$$m_H = -\frac{R\delta_R}{2}\sqrt{1+\delta_R} - \frac{H_b(R)B_R}{4}(1+\delta_R)^{3/2} - \frac{B_R^2}{8R^3}(1+\delta_R)^{3/2}.$$

Therefore

$$\lim_{s \rightarrow \infty} m_H(\Sigma_s) = \langle \mu_{d_\infty} \rangle_\sigma.$$

By (38.10), this last scalar is the time component of $\mathcal{B}(d_\infty)^{-1}p(G)$, whose spatial components vanish. Lorentz invariance and its positive base value m_0 identify it with the formal square root in (38.3). $\square$

Remark 38.2. Lemma 37.2 supplies the derivative budget used here: for a general finite-angular-type seed, the homogeneous first coefficient has all differentiated asymptotics and the second has the first differentiated asymptotic required at order two. The fourth-order construction is used only after reduction to $q = \text{Hess } v - Bv$, with $Lv = 0$ and v consisting of constant and degree-one angular modes. All required differentiated decay is available in this class.

38.2. The Hawking variation identity. On the sphere jets of Proposition 38.1, write

$$\langle u \rangle = \frac{1}{A} \int_{\Sigma_s} u \, dA_G, \quad \bar{f} = \langle f \rangle, \quad \delta f = f - \bar{f}.$$

Here averages are with respect to the induced metric; the round average retains the subscript σ . Let J be the full Jacobi operator (38.4), and let II° be the tracefree second fundamental form.

Lemma 38.3 (CMC Hawking variation). *The following identity holds through the order of the sphere and metric jets:*

$$(38.15) \quad \frac{d}{ds} m_H = \sqrt{\frac{A}{16\pi}} \frac{HA}{8\pi} \left[\bar{f} \left\langle \frac{\text{Scal}_G + 6 + |\text{II}^\circ|^2}{2} \right\rangle + \frac{1}{\bar{f}} \langle \delta f \, J \delta f \rangle \right].$$

Proof. The area and mean-curvature variation formulas give

$$A' = AH\bar{f}, \quad H' = Jf.$$

The latter is a constant on each leaf; any tangential transport term vanishes since H is constant there. For brevity put

$$c = \frac{4\pi}{A} + 3 - \frac{3H^2}{4}, \quad Q_H = \frac{\text{Scal}_G + 6 + |\text{II}^\circ|^2}{2}.$$

Differentiation of (38.2) gives

$$(38.16) \quad m'_H = \sqrt{\frac{A}{16\pi}} \frac{HA}{8\pi} (c\bar{f} - H').$$

The Gauss equation and $|\text{II}|^2 = H^2/2 + |\text{II}^\circ|^2$ imply

$$J1 = K_\Sigma + 3 - \frac{3H^2}{4} - Q_H,$$

where K_Σ is intrinsic Gauss curvature. The sphere jets have $\int K_\Sigma \, dA_G = 4\pi$ coefficientwise. One can verify this without a limiting argument: the variation of integrated scalar curvature on a closed two-dimensional metric is the integral of a divergence minus its variation tensor paired with $\text{Ric} - \frac{1}{2} \text{Scal } g$. The latter tensor is zero in dimension two, so that integral is constant under

every variation and equals its round value. Applying the same identity to the Taylor coefficients proves the assertion for jets. Consequently

$$\langle J1 \rangle = c - \langle Q_H \rangle.$$

The operator J is self-adjoint on the closed leaf. Since $\langle \delta f \rangle = 0$ and $Jf = H'$ is constant,

$$\begin{aligned} H' &= \bar{f}(c - \langle Q_H \rangle) + \langle \delta f J1 \rangle, \\ 0 &= \langle \delta f Jf \rangle = \bar{f} \langle \delta f J1 \rangle + \langle \delta f J \delta f \rangle. \end{aligned}$$

The scalar jet $\bar{f}$ is invertible because its base value is one. Eliminating the common term gives

$$c\bar{f} - H' = \bar{f} \langle Q_H \rangle + \bar{f}^{-1} \langle \delta f J \delta f \rangle.$$

Substitution into (38.16) proves the identity. All calculations may first be made on a compact radial interval. Smooth representatives of the metric and graph jets realize the ordinary geometric variation identities there, and the imposed CMC identities hold to the required Taylor order. $\square$

38.3. The first nonzero coefficient.

Proposition 38.4 (Leading coefficient as a positive integral). *Use the construction of Proposition 38.1 through order $2l$, where $l \geq 1$, and let $A_{\text{leaf}} = A(0)$ be the area jet of its minimal first leaf. Suppose, as ambient and leafwise coefficient identities, respectively, that*

$$(38.17) \quad \text{Scal}_G + 6 = \epsilon^{2l} |k|_{g_0}^2 + O(\epsilon^{2l+1}), \quad \text{II}^\circ = \epsilon^l U + O(\epsilon^{l+1}), \quad \delta f = \epsilon^l F + O(\epsilon^{l+1}).$$

Here k is a smooth tensor on M ; the fields U, F on the background spheres are identified with fields on M . Then every coefficient below degree $2l$ of $m_{\text{AH}}(G) - b_*(A_{\text{leaf}})$ vanishes, and

$$(38.18) \quad \begin{aligned} &[m_{\text{AH}}(G) - b_*(A_{\text{leaf}})]_{2l} \\ &= \frac{1}{16\pi} \int_0^\infty N(s) \int_{\{s\} \times S^2} (|k|^2 + |U|^2 + 2F J_s^0 F) \, dA_{g_0} \, ds \geq 0. \end{aligned}$$

All norms in this formula use the background metrics. The displayed iterated integral is finite. If it vanishes, then k, U, F vanish for $s > 0$, and also at the boundary by continuity.

Proof. The first nonzero coefficient F has zero round mean on each background sphere: extract degree l from $\int_{\Sigma_s} \delta f \, dA_G = 0$, using the vanishing of all lower coefficients of δf . Every term in the bracket in (38.15) starts at degree $2l$. At that degree, all exterior factors can therefore be evaluated at the background:

$$A_0 = 4\pi r^2, \quad H_0 = 2N/r, \quad \bar{f}_0 = 1, \quad \sqrt{\frac{A_0}{16\pi}} H_0 = N.$$

The same reasoning evaluates the norms, the induced measure, and the Jacobi operator inside the leading terms at the background. Composing the ambient scalar coefficient with a graph displacement changes only higher degrees. It follows that

$$(38.19) \quad [m'_H]_{2l} = \frac{N(s)}{16\pi} \int_{\{s\} \times S^2} (|k|^2 + |U|^2 + 2F J_s^0 F) \, dA_{g_0},$$

and all lower coefficient derivatives vanish.

The integrand in (38.19) is nonnegative after sphere integration. Indeed, on the mean-zero field F ,

$$\int_{\{s\} \times S^2} F J_s^0 F \, dA_{g_0} = \sum_{\ell \geq 1} \left(\frac{\ell(\ell+1)-2}{r^2} + \frac{6m_0}{r^3} \right) \|F_\ell\|_{L^2(r^2\sigma)}^2.$$

Only finitely many modes occur, and every displayed eigenvalue is strictly positive at a finite radius. Also $N(s) > 0$ for $s > 0$.

For any finite T , integrate the coefficient identities on $[0, T]$. The first leaf is minimal to the required order, so $m_H(\Sigma_0) = b_*(A_{\text{leaf}})$ through that order. The coefficientwise endpoint limits exist by (38.3). For degree $2l$, the right-hand side of the integrated (38.19) is an increasing function of T and its limit equals the finite limiting endpoint coefficient. This proves both its convergence and (38.18). For lower degrees the integrated derivative is zero, proving their claimed vanishing.

This order of operations uses only smooth jets on finite radial intervals and coefficientwise endpoint limits. In particular, convergence of the derivatives of the asymptotic graph coefficients or of their lapse is not needed to obtain the integral's convergence. If the integral is zero, each of its continuous nonnegative sphere-integrated summands is zero for every $s > 0$. The positivity just established forces $k = U = F = 0$ there. Smoothness gives their boundary values. $\square$

39. THE QUADRATIC ESTIMATE AND ITS COMPLETE KERNEL

All contractions in this section, unless a different metric is indicated, use g_0 . We write dV_0 and dA_0 for its volume form and its induced area form. A function appearing in an integral over S is understood to be restricted to S . Introduce

$$Q(v) = \text{Hess}_{g_0} v - vB.$$

The boundary normal $n = \partial_s$ points into M ; the outward normal of the domain M along its inner boundary is therefore $-n$.

Lemma 39.1 (The weighted TT square identity). *Let k be a smooth TT tensor of finite angular type, with decay $k = O(r^{-\gamma})$ for some $\gamma > 3/2$. Let v be the decaying solution of*

$$(39.1) \quad Lv = 0, \quad v|_S = D^{-1}(k_{ss}|_S).$$

Then $Q(v)$ is TT, $Q(v)_{ss}|_S = k_{ss}|_S$, and

$$(39.2) \quad \int_M N|k|^2 dV_0 - \kappa \int_S v Dv dA_0 = \int_M N|k - Q(v)|^2 dV_0.$$

All integrals in this identity converge. In particular, no sign assumption on $k_{ss}|_S$ is needed.

Proof. The operator $D = -\Delta_S + 2\kappa/a$ is strictly positive on all spherical harmonics. Lemma 37.1, applied to the finitely many modes of the boundary datum, gives the unique solution of (39.1); it has $v = O(r^{-3})$ with the differentiated asymptotics needed below.

The background identities $\text{tr } B = 3$, $\text{div } B = 0$, and $B = \text{Ric}_{g_0} + 3g_0$ imply, by commuting the derivatives of a function,

$$\text{tr } Q(v) = \Delta v - 3v = Lv, \quad (\text{div } Q(v))_i = \nabla_i \Delta v + (\text{Ric}_{ij} - B_{ij}) \nabla^j v = \nabla_i Lv.$$

Thus $Q(v)$ is TT. Since S is totally geodesic, the tangential trace of $\text{Hess } v$ on S is $\Delta_S v$. Moreover $B_t(0) = \kappa/a$. Taking the tangential trace of $Q(v)$ and using its vanishing full trace therefore yields

$$(39.3) \quad Q(v)_{ss}|_S = -\Delta_S v + 2(\kappa/a)v = Dv.$$

Set $\omega = N dv - v dN$. With the averaged convention for the symmetric gradient, $\text{sym } \nabla \omega = (\nabla_i \omega_j + \nabla_j \omega_i)/2$, the mixed products cancel and the static identity $\text{Hess } N = NB$ gives

$$\text{sym } \nabla \omega = N \text{Hess } v - v \text{Hess } N = NQ(v).$$

Symmetry and vanishing divergence of k consequently give, on $M_R = \{0 \leq s \leq s(R)\}$,

$$\int_{M_R} N \langle k, Q(v) \rangle dV_0 = \int_{\partial M_R} k(\nu, \omega^\sharp) dA_0.$$

On S , $N = 0$ and $dN = \kappa ds$, so that $\omega^\sharp = -\kappa\nu n$. Since $\nu = -n$ there, the inner contribution is $+\kappa \int_S vk_{ss} dA_0$. At infinity, $|\omega| = O(r^{-2})$, and hence the outer contribution is $O(r^{-\gamma})$ and tends to zero. We obtain

$$(39.4) \quad \int_M N \langle k, Q(v) \rangle dV_0 = \kappa \int_S vk_{ss} dA_0 = \kappa \int_S vDv dA_0.$$

The same calculation with k replaced by $Q(v)$ gives $\int_M N |Q(v)|^2 dV_0 = \kappa \int_S vDv dA_0$. Expanding the square proves (39.2). Finally, $N dV_0$ has radial size $O(r^3) ds dA_\sigma$, so the integral of $N|k|^2$ converges because $2\gamma > 3$. The stronger decay of $Q(v)$ gives all remaining convergence assertions. $\square$

39.1. The boundary area correction. For the moment let q have finite angular type, and use the conformal metric jet $g = \phi^4 g_0$ through order two. Write $u = u_1[q]$, so that $Lu = 0$ and $u_s|_S = q_{ss}/4$. Apply Proposition 38.1, and let A_{leaf} denote the area of its first, minimal Taylor leaf. This notation concerns that formal leaf alone; it is not an enclosing-area infimum. Let

$$U = [\Pi^\circ]_1, \quad F = [f - \bar{f}]_1,$$

where f is the lapse and $\bar{f}$ its area average on each leaf. Thus F has zero round mean on each background sphere.

Let v solve (39.1) with $k = q$. At the fixed boundary, the conformal boundary equation gives

$$H_g(S) = \epsilon \phi^{-6} q_{ss}.$$

If h_1 is the first normal height of the minimal Taylor leaf, its mean-curvature equation is

$$Dh_1 + q_{ss} = 0, \quad h_1 = -v|_S.$$

The height difference from that leaf to S is therefore $\epsilon v|_S + O(\epsilon^2)$. Area has zero first variation at a minimal leaf, and its Hessian at the base sphere is D . Taylor expansion about the minimal leaf gives

$$(39.5) \quad A(S) - A_{\text{leaf}} = \frac{\epsilon^2}{2} \int_S vDv dA_0 + O(\epsilon^3) \quad \text{as an identity of jets.}$$

Indeed, a first-order perturbation of the area Hessian would multiply two first-order displacements and first enter order three. The first variation at the minimal Taylor leaf vanishes to the order needed. This reasoning also applies when its leading displacement points across the boundary: it uses only the smooth boundary jets and their Taylor extension, as in the construction of the formal leaves.

Proposition 39.2 (The quadratic deficit). *On the real linear space of all smooth TT seeds with the prescribed decay, the first deficit coefficient is zero and the second coefficient $c_2[q]$ is a continuous, rotation-invariant, nonnegative quadratic form. For a finite-angular-type seed it is given by*

$$(39.6) \quad c_2[q] = \frac{1}{16\pi} \int_M N \left(|q - Q(v)|^2 + |U|^2 + 2FJ_s^0 F \right) dV_0,$$

$$J_s^0 = -\Delta_{r^2\sigma} - \frac{2}{r^2} + \frac{6m_0}{r^3}.$$

The last term is integrated on the spheres before radial integration. For every actual branch in Theorem 36.1,

$$(39.7) \quad \mathfrak{D}_q(\epsilon) = c_2[q]\epsilon^2 + o(\epsilon^2).$$

In particular, $c_2[q] > 0$ implies a strictly positive deficit for all sufficiently small positive ϵ .

Proof. For a finite-angular-type seed the scalar constraint gives $\text{Scal}_g + 6 = \epsilon^2 |q|^2 + O(\epsilon^3)$. Proposition 38.4, with $l = 1$, shows that the constant and first coefficients of the mass norm minus $b_*(A_{\text{leaf}})$ vanish, and that its second coefficient is

$$\frac{1}{16\pi} \int_M N(|q|^2 + |U|^2 + 2FJ_s^0 F) \, dV_0.$$

The areas in (39.5) agree through order one, so the first coefficient of the original deficit is also zero. Since $b'_*(4\pi a^2) = \kappa/(8\pi)$, the contribution to its second coefficient from that area difference is $-\kappa(16\pi)^{-1} \int_S v Dv \, dA_0$. Lemma 39.1 now proves (39.6) with precisely the displayed factor and sign.

On a spherical harmonic with $-\Delta_\sigma$ eigenvalue $\lambda \geq 2$, the eigenvalue of J_s^0 is

$$\frac{\lambda - 2}{r^2} + \frac{6m_0}{r^3} > 0.$$

Thus its quadratic form is strictly positive on mean-zero functions at every finite s , and $N > 0$ for $s > 0$. This proves nonnegativity for finite-angular-type seeds without any boundary sign restriction.

Proposition 37.5 defines c_2 on the full real space $\mathcal{T}_{\tau,\alpha}$ of smooth TT seeds as a continuous, rotation-invariant quadratic form, independently of branch existence and boundary sign, and proves $c_1 = 0$. For an arbitrary seed q , take the finite-angular-type TT approximants q_ν from Lemma 37.4. The formula proved above gives $c_2[q_\nu] \geq 0$, and continuity in $\|\cdot\|_\beta$ gives $c_2[q] \geq 0$. Proposition 37.5 also supplies (39.7) for every prescribed actual branch. $\square$

39.2. All null directions. Define the fixed vector space

$$(39.8) \quad \mathcal{K} = \{Q(v) : Lv = 0, \, v \text{ decays}, \, v|_S \in \text{span}_{\mathbb{R}}\{1, x_1, x_2, x_3\}\}.$$

The decaying solutions in this definition have finite angular type and $O(r^{-3})$ decay with all the differentiated bounds obtained from their homogeneous radial equations.

Corollary 39.3 (The complete quadratic kernel). *The kernel of c_2 is exactly $\mathcal{K}$, a four-dimensional real vector space. In particular, the nonzero degree-one directions are genuine quadratic null directions.*

Proof. First suppose q has finite angular type and $c_2[q] = 0$. Equation (39.6) and the strict positivity just proved imply

$$(39.9) \quad q = Q(v), \quad U = 0, \quad F = 0 \quad \text{on } s > 0.$$

Smoothness extends these equalities to the boundary. Coordinate spheres are umbilic under conformal changes of g_0 . At the base minimal sphere, the tracefree second-form variation due to a normal height h_1 is $-(\text{Hess}_S h_1)^\circ$: the radial curvature term is pure trace. Since $h_1 = -v|_S$, the vanishing of $U|_S$ implies $(\text{Hess}_\sigma(v|_S))^\circ = 0$. For completeness, the divergence of this equation on the unit sphere gives $d(\Delta_\sigma v + 2v) = 0$. Subtracting the mean reduces it to $\text{Hess}_\sigma(v - \bar{v}) = -(v - \bar{v})\sigma$. This equation determines its solution along every geodesic from its value and differential at one point, and its solutions are precisely the linear coordinate functions. Thus $v|_S$ is a constant plus a linear harmonic.

Uniqueness of the decaying Dirichlet problem for L then excludes every other angular mode throughout M . Each degree-one radial coefficient is a multiple of the same decaying solution of

$$(39.10) \quad V'' + 2\frac{N}{r}V' - \left(3 + \frac{2}{r^2}\right)V = 0.$$

Hence $q \in \mathcal{K}$. Conversely, every element of $\mathcal{K}$ is null, as we now check; the conditions $U = F = 0$ do not impose a further restriction on its degree-one part.

Take $q = Q(v) \in \mathcal{K}$. The first conformal coefficient u also has only constant and linear modes, by its Neumann equation and decaying uniqueness. Let $h(s, x)$ be the first height coefficient of the CMC spheres, and put $p = N/r$. Conformal variation of mean curvature and first variation of the lapse give

$$(39.11) \quad [H]_1 = J_s^0 h + 4(u_s - pu), \quad [f]_1 = h_s + 2u.$$

Here the second identity follows by taking the normal component of the velocity of the graph $s + \epsilon h(s, x)$ in $g_0 + 4\epsilon u g_0$. On the degree-one part, the CMC equation therefore fixes

$$(39.12) \quad h = -\frac{2r^3}{3m_0}(u_s - pu).$$

All formulas in the next calculation are componentwise on that part. The background equations and (39.10) give

$$p' = \frac{3m_0}{r^3} - \frac{1}{r^2}, \quad p^2 = 1 + \frac{1}{r^2} - \frac{2m_0}{r^3}, \quad u'' = -2pu' + \left(3 + \frac{2}{r^2}\right)u.$$

Consequently,

$$(39.13) \quad \begin{aligned} \frac{1}{r^3} \partial_s [r^3(u' - pu)] &= 3p(u' - pu) + u'' - p'u - pu' \\ &= \left(3 + \frac{2}{r^2} - p' - 3p^2\right)u = \frac{3m_0}{r^3}u. \end{aligned}$$

It follows that $h_s = -2u$ on the degree-one part, so its lapse fluctuation vanishes. The freely chosen radial mean of h changes $[f]_1$ only by a radial function, which disappears upon subtracting the mean. Thus $F = 0$ independently of the radial labeling.

At every background sphere, the tracefree second-form variation of these graphs is $-(\text{Hess}_{r^2\sigma} h)^\circ$; the conformal variation and radial curvature terms are pure trace. The constant and linear modes of h have vanishing tracefree spherical Hessian, so $U = 0$ as well. At S there is also exact agreement with the prescribed first minimal height: on degree one,

$$D_1 = \frac{6m_0}{a^3}, \quad h(0) = -\frac{4u_s(0)}{D_1} = -v(0).$$

We already have $q = Q(v)$, and hence (39.6) proves $c_2[q] = 0$. Existence and uniqueness for each of the four boundary modes show that $\mathcal{K}$ has dimension at most four; it has dimension exactly four because (39.3) and invertibility of D make the map from these boundary data to $Q(v)$ injective.

It remains to rule out additional null directions of arbitrary angular type. Let $\mathcal{Z} = \{q : c_2[q] = 0\}$ in the full smooth TT space. A nonnegative quadratic form has a linear kernel: if $c_2[q] = 0$, nonnegativity of $c_2[q + th]$ for all real t forces its polarized bilinear form to vanish on (q, h) for every h . By Proposition 39.2, $\mathcal{Z}$ is also closed in $\|\cdot\|_\beta$ and invariant under rotations. For $q \in \mathcal{Z}$, let q_j be its polynomial rotational averages from Lemma 37.4. Each such average is a norm limit of finite linear combinations of rotations of q ; therefore $q_j \in \mathcal{Z}$. Each q_j has finite angular type, so the preceding argument puts it in $\mathcal{K}$. Since $q_j \rightarrow q$ in norm and the fixed finite-dimensional space $\mathcal{K}$ is closed in that norm, $q \in \mathcal{K}$. This proves the full assertion without applying a pointwise geometric limit to the CMC fields U and F . $\square$

Remark 39.4 (The boundary sign and the null space). If $v|_S = c + d \cdot x$, then

$$Q(v)_{ss}|_S = D_0 c + D_1 d \cdot x, \quad D_0 = \frac{2\kappa}{a}, \quad D_1 = \frac{2}{a^2} + \frac{2\kappa}{a} = \frac{6m_0}{a^3}.$$

Thus $Q(v)_{ss}|_S \geq 0$ is equivalent to $D_0 c \geq D_1 |d|$. A nonzero pure dipole fails that condition, but a sufficiently large positive radial component makes a mixed radial-dipole seed satisfy it. This is a

compatibility statement about seeds, not an existence claim for branches satisfying outermostness and outer area-minimization. In particular, the boundary sign cannot be used to discard the degree-one null directions from the proof.

40. FOURTH ORDER IN THE QUADRATIC KERNEL

We now treat the kernel left by Proposition 39.2. A statement about a fourth-order jet is an identity modulo ϵ^5 , in the conventions of Section 36.3; it does not assert the existence of the corresponding geometric object at a nonzero parameter.

Proposition 40.1 (The quartic coefficient). *Suppose that $c_2[q] = 0$. Write the representation furnished by Corollary 39.3 as*

$$q = Q(v) := \text{Hess}_{g_0} v - Bv, \quad Lv = 0, \quad v = v_0(s) + v_1(s, x),$$

where v_1 has only degree-one spherical modes and v decays at infinity. Then

$$c_0[q] = c_1[q] = c_2[q] = c_3[q] = 0, \quad c_4[q] \geq 0.$$

If $v_1 \not\equiv 0$, then $c_4[q] > 0$. In particular, in that case the actual deficit $\mathfrak{D}_q(\epsilon)$ is strictly positive for all sufficiently small positive ϵ .

The proof has three steps. Lemmas 40.2 and 40.3 change the slice and compare its boundary area with a minimal leaf, giving a quartic TT square. If that square vanishes, Sections 40.4–40.5 force the degree-one profile to vanish. The actual expansion from Proposition 37.5 then proves the assertion for the given branch.

Throughout the proof we use the stronger decay supplied by the kernel representation. The homogeneous separated equations for v , and Lemma 37.1, give $O(r^{-3})$ decay with every required derivative for v and $q = Q(v)$. The conformal coefficients through degree four consequently have differentiated $O(r^{-3})$ decay and finite angular type. Put

$$u = [\phi_\epsilon]_1, \quad Lu = 0.$$

We use the fourth-order jets of the original data $g_\epsilon = \phi_\epsilon^4 g_0$ and $K_\epsilon = \epsilon \phi_\epsilon^{-2} q$. Their initial constraints and their boundary expansion equation hold as Taylor identities.

40.1. A change of slice at the level of jets.

Lemma 40.2 (The corrected slice). *There is a decaying finite-angular-type function w , with $w|_S = 0$, for which the following construction is defined through degree four. Formally evolve the initial jets in Gaussian time and take the graph*

$$(40.1) \quad T = -\epsilon v + \epsilon^2 w.$$

The induced metric and future second fundamental form, denoted by $\tilde{g}$ and $\tilde{K}$, satisfy

$$(40.2) \quad \tilde{K} = \epsilon^2 k + O(\epsilon^3), \quad \text{tr}_{g_0} k = 0, \quad \text{div}_{g_0} k = 0,$$

$$(40.3) \quad \text{Scal}_{\tilde{g}} + 6 = \epsilon^4 |k|_{g_0}^2 + O(\epsilon^5).$$

Here $k = O(r^{-3})$, with all derivatives needed for the fourth-order CMC construction. Coefficient-wise through degree four,

$$(40.4) \quad \tilde{g} - g_\epsilon = O(r^{-6})$$

with the required differentiated bounds. Thus the mass covectors and their Lorentz norms agree through that degree. Moreover,

$$(40.5) \quad [\tilde{g} - g_\epsilon]_2 = -2vq - Bv^2 - dv \otimes dv = -\text{Hess}_{g_0}(v^2) + dv \otimes dv + Bv^2.$$

Proof. We give the formal evolution and its constraint justification before performing the graph calculation. Write

$$\bar{g} = -dt^2 + \gamma(t), \quad \gamma(0) = g_\epsilon, \quad \mathcal{K}(0) = K_\epsilon,$$

and determine a full formal time series recursively from

$$(40.6) \quad \begin{aligned} \partial_t \gamma &= 2\mathcal{K}, \\ \partial_t \mathcal{K} &= -\text{Ric}_\gamma - 3\gamma - (\text{tr}_\gamma \mathcal{K})\mathcal{K} + 2\mathcal{K} \circ_\gamma \mathcal{K}. \end{aligned}$$

The coefficients take values in the ring of smooth spatial jets modulo ϵ^5 . At each time order the right side is a known spatial differential expression in previously determined coefficients, so the recursion is well defined. A substitution $t = T = O(\epsilon)$ uses only finitely many of these coefficients. Keeping a full time series here ensures that taking time derivatives never treats an omitted time coefficient as zero.

Let $H = \text{tr}_\gamma \mathcal{K}$. The time-index Christoffel symbols are

$$\bar{\Gamma}_{ij}^t = \mathcal{K}_{ij}, \quad \bar{\Gamma}_{tj}^i = \mathcal{K}_j^i,$$

and direct contraction of the curvature gives

$$(40.7) \quad \begin{aligned} \bar{\text{Ric}}_{ij} &= \text{Ric}_{\gamma,ij} + \partial_t \mathcal{K}_{ij} + H\mathcal{K}_{ij} - 2(\mathcal{K} \circ_\gamma \mathcal{K})_{ij}, \\ \bar{\text{Ric}}_{tt} &= -\text{tr}_\gamma(\partial_t \mathcal{K}) + |\mathcal{K}|_\gamma^2, & \bar{\text{Ric}}_{ti} &= (\text{div}_\gamma \mathcal{K})_i - \partial_i H. \end{aligned}$$

Consequently $E := \bar{\text{Ric}} + 3\bar{g}$ has $E_{ij} = 0$. The initial Hamiltonian and momentum constraints give $E_{tt} = E_{ti} = 0$ at $t = 0$. For completeness, the Hamiltonian constraint is $\text{Scal}_\gamma + H^2 - |\mathcal{K}|_\gamma^2 = -6$; tracing (40.6) at $t = 0$ therefore gives $\text{tr}_\gamma \partial_t \mathcal{K} = -3 + |\mathcal{K}|_\gamma^2$, which proves the asserted tt equation with the required sign.

The contracted Bianchi identity propagates these remaining equations formally. Indeed, if $e = E_{tt}$ and $j_i = E_{ti}$, the trace reversal of E has components

$$\hat{E}_{tt} = e/2, \quad \hat{E}_{ti} = j_i, \quad \hat{E}_{ij} = e\gamma_{ij}/2.$$

Its vanishing divergence, evaluated with the displayed Christoffel symbols, is

$$(40.8) \quad \partial_t e = 2 \text{div}_\gamma j - 2He, \quad \partial_t j_i = \frac{1}{2} \partial_i e - H j_i.$$

This homogeneous system determines each next time coefficient of (e, j) from the preceding ones. All coefficients vanish by induction. Thus

$$(40.9) \quad \bar{\text{Ric}} = -3\bar{g}$$

as a formal time series modulo ϵ^5 .

There is no extension hypothesis hidden in this construction. The original coefficient fields are smooth up to S , and their constraint residuals and all one-sided spatial derivatives vanish there. The recursion and (40.8) therefore also hold in their spatial Taylor jets at S . Each finite calculation below involves only finitely many such derivatives. They may, if desired, be realized in a collar across S by their Taylor polynomials in s , to an order larger than the differential order of that calculation. Curvature and its residual are formed before substituting a displacement of order ϵ . Their retained boundary jets vanish, so the substitution is valid even when the leading displacement is negative. This neither constructs nor uses an actual Einstein extension on the other side of S .

Gauss–Codazzi applied as a differential identity to any spacelike graph now gives

$$(40.10) \quad \text{Scal}_{\text{slice}} + 6 = |K_{\text{slice}}|^2 - (\text{tr } K_{\text{slice}})^2, \quad \text{div}(K_{\text{slice}} - (\text{tr } K_{\text{slice}})g_{\text{slice}}) = 0.$$

Equivalently one can form Gaussian coordinates about that graph by the Taylor recursion for its normal geodesics and use (40.7). Either interpretation needs only differential identities of finite jets.

We next compute the graph second form explicitly. With $T_i = \partial_i T$, its tangent fields are $X_i = \partial_i + T_i \partial_t$, and its future unit normal is

$$n_T = \frac{\partial_t + \text{grad}_{\gamma(t)} T}{\sqrt{1 - |\text{dT}|_{\gamma(t)}^2}} \Big|_{t=T}.$$

Using $-\bar{g}(n_T, \bar{\nabla}_{X_i} X_j)$ gives

$$(40.11) \quad \widetilde{K}_{ij} = \frac{\mathcal{K}_{ij}(t) + (\text{Hess}_{\gamma(t)} T)_{ij} - T_k (T_i \mathcal{K}^k_j(t) + T_j \mathcal{K}^k_i(t))}{\sqrt{1 - |\text{dT}|_{\gamma(t)}^2}} \Big|_{t=T}.$$

The Hessian connection in this formula is taken at fixed time before evaluation at $t = T$. The induced metric is $\tilde{g} = \gamma(T) - \text{dT} \otimes \text{dT}$.

At the background initial slice, $\mathcal{K} = 0$ and $\partial_t \mathcal{K} = -B$. A first-order time displacement b therefore changes the second form by $\text{Hess } b - Bb = Q(b)$. Equation (40.1) gives

$$[\widetilde{K}]_1 = q - Q(v) = 0.$$

We choose w to remove the trace at the next degree. This can be done with an explicit scalar source. Since $[\gamma(0)]_1 = 4ug_0$, the conformal Ricci variation and $Lu = 0$ give

$$[\partial_t \mathcal{K}(0)]_1 = 2 \text{Hess } u - 6ug_0, \quad [\mathcal{K}(0)]_2 = -2uq.$$

At the background $\partial_t^2 \mathcal{K}(0) = 0$: differentiating (40.6) there uses $\partial_t \gamma(0) = 0$ and $\mathcal{K}(0) = 0$. Expanding (40.11) to degree two thus gives

$$(40.12) \quad \begin{aligned} k &= Q(w) - 2uq - 2v \text{Hess } u + 6uv g_0 \\ &\quad + 2 \text{d}u \otimes \text{d}v + 2 \text{d}v \otimes \text{d}u - 2 \langle \text{d}u, \text{d}v \rangle_{g_0} g_0. \end{aligned}$$

In particular,

$$\text{tr}_{g_0} k = Lw + 12uv - 2 \langle \text{d}u, \text{d}v \rangle_{g_0}.$$

Choose the unique decaying solution of

$$(40.13) \quad Lw = 2 \langle \text{d}u, \text{d}v \rangle_{g_0} - 12uv, \quad w|_S = 0.$$

Its right side is $O(r^{-6})$, with differentiated bounds, so Lemma 37.1 applies and gives $w = O(r^{-3})$ with the corresponding derivatives. Formula (40.12) gives the asserted decay of k . The degree-two momentum equation in (40.10), together with $\text{tr}_{g_0} k = 0$, gives $\text{div}_{g_0} k = 0$.

The full transformed trace is $O(\epsilon^3)$. Its square consequently begins in degree six, whereas

$$|\widetilde{K}|_g^2 = \epsilon^4 |k|_{g_0}^2 + O(\epsilon^5).$$

This proves (40.3); maximality at higher orders on the changed slice is not needed.

Finally,

$$\gamma(T) - \gamma(0) = 2\mathcal{K}(0)T + \frac{1}{2}\partial_t^2 \gamma(0)T^2 + \dots.$$

In every retained coefficient the first term is $O(r^{-6})$, and every subsequent term has at least two decaying time-displacement factors multiplying bounded background time coefficients. The recursion preserves these bounded differentiated estimates: in the scaled coframe $\text{d}s, r \text{d}x$, the background connection and curvature, with their required derivatives, are bounded; each coefficient is obtained by finitely many covariant differentiations, contractions, and inverse-metric Taylor operations from the smooth initial coefficients. The same bounds hold for $\text{dT} \otimes \text{dT}$, proving (40.4). An $O(r^{-6})$ metric coefficient with its first covariant derivative contributes $O(r^{-3})$ to the

integrated mass flux, since the test potentials and their gradients are $O(r)$ and the sphere area is $O(r^2)$. Thus the leading asymptotic data, every mass component, and the mass norm are unchanged through degree four.

Since $\partial_t^2 \gamma(0) = -2B$ at the background, the degree-two difference is $-2vq - Bv^2 - dv \otimes dv$. Substituting $q = \text{Hess } v - Bv$ and using $\text{Hess}(v^2) = 2v \text{Hess } v + 2dv \otimes dv$ proves (40.5). $\square$

40.2. Transport of the boundary and its area.

Lemma 40.3 (The null cut). *For the changed slice of Lemma 40.2 there is a sphere jet $S^\sharp$, based at S , such that*

$$(40.14) \quad |S^\sharp|_{\tilde{g}} = |S|_{g_\epsilon} + O(\epsilon^5),$$

$$(40.15) \quad H_{\tilde{g}}(S^\sharp) = \epsilon^2 k_{ss}|_S + O(\epsilon^3).$$

Let $S_{\min}$ be the minimal Taylor leaf of the CMC construction for $\tilde{g}$. Their height jets agree through degree one. If

$$(40.16) \quad z = D^{-1}(k_{ss}|_S),$$

then the height of $S^\sharp$ minus that of $S_{\min}$ is $\epsilon^2 z + O(\epsilon^3)$, and

$$(40.17) \quad |S^\sharp|_{\tilde{g}} - |S_{\min}|_{\tilde{g}} = \frac{1}{2} \epsilon^4 \int_S z D z \, dA_{g_0} + O(\epsilon^5).$$

These statements remain valid when some leading boundary displacements are negative.

Proof. On the original initial slice take the future outward null normal $\ell = n_{\text{future}} + \nu$ to S . Its affinely parametrized null geodesics determine a null hypersurface jet, with parameter λ zero at S . These are Taylor solutions of the geodesic and Jacobi equations for the formal metric $\bar{g}$ constructed above. At the initial sphere the expansion is identically zero by the original boundary equation.

Write χ for the null second form and $\theta = \text{tr } \chi$ for its expansion. The initial sphere is umbilic in $\phi_\epsilon^4 g_0$. Moreover the coefficient of ϵ in the tracefree tangential part of K_ϵ equals the tracefree tangential part of $Q(v)$. Since S is totally geodesic in g_0 , that part is $(\text{Hess}_{a^2\sigma}(v|_S))^\circ$. It vanishes for the constant and degree-one modes of $v|_S$. The vanishing initial expansion and this shear calculation together give

$$(40.18) \quad \chi(0, \epsilon) = O(\epsilon^2).$$

The screen space is the positive-definite quotient $\ell^\perp / \text{span}\{\ell\}$. In a parallel orthonormal frame of this space the null shape operator obeys the optical Riccati equation

$$\partial_\lambda \chi = -\chi^2 - \mathcal{R}_\ell, \quad \partial_\lambda \theta = -|\chi|^2 - \overline{\text{Ric}}(\ell, \ell).$$

These equations follow directly from the Jacobi equation: angular Jacobi fields remain orthogonal to ℓ , and their derivative matrix times their inverse is the shape operator. At $(\lambda, \epsilon) = (0, 0)$, spherical symmetry makes $\mathcal{R}_\ell$ a scalar multiple of the screen identity. Its trace is $\overline{\text{Ric}}(\ell, \ell) = 0$ by (40.9) and nullness, so this curvature endomorphism is zero. Thus $\partial_\lambda \chi(0, 0) = 0$.

It follows from (40.18) that every retained monomial of χ has total degree at least two in (λ, ϵ) . Raychaudhuri and the identically zero initial expansion then imply

$$(40.19) \quad \chi = O_{\text{tot}}(2), \quad \theta = O_{\text{tot}}(5).$$

Here $O_{\text{tot}}(j)$ means that each monomial $\lambda^a \epsilon^b$ has $a + b \geq j$. The order statements are made in the formal series modulo ϵ^5 . Equivalently one can choose any sufficiently high finite Taylor lift: an Einstein residual of order ϵ^5 integrated along a segment $\lambda = O(\epsilon)$ contributes only beyond every degree used here. Since the logarithmic screen area density has derivative θ , its change along such a segment vanishes through degree four.

Intersect this null hypersurface jet with $t = T$. At the background initial point, the derivative of $t - T$ along its generator is one. Successive Taylor substitution therefore solves for the intersection parameter, which is $O(\epsilon)$. The angular projection has identity background differential and likewise has a formal inverse. The resulting cut $S^\sharp$ is a graph over S in the changed slice.

The area conclusion also holds for this variable-parameter cut. If J_A are the transported angular Jacobi fields, its tangent fields are $J_A + (\partial_A \lambda)\ell$. Nullness and $\bar{g}(J_A, \ell) = 0$ show that the additional terms do not change the induced metric. Its area density is just the transported screen density evaluated at the variable parameter. This proves (40.14).

The outward future null normal obtained from the new slice differs from ℓ by a scalar with positive, nonzero background value. Expansion is multiplied by that scalar, so it still vanishes through the needed orders. At leading order

$$\mathrm{tr}_{S^\sharp, \tilde{g}} \tilde{K} = -\epsilon^2 k_{ss}|_S + O(\epsilon^3),$$

because $\mathrm{tr}_{g_0} k = 0$. Its zero null expansion gives (40.15).

Both $S^\sharp$ and the minimal Taylor leaf have zero mean-curvature coefficients through degree one. The difference of their first height coefficients is therefore annihilated by the background operator D , which is invertible. At degree two, their already equal first coefficients make all common nonlinear terms cancel, leaving $D(h_2^\sharp - h_2^{\min}) = k_{ss}|_S$. This proves (40.16). The first variation of area at the minimal Taylor leaf vanishes to the retained order. A height difference beginning with $\epsilon^2 z$ consequently has, at degree four, only the base second-variation contribution $\frac{1}{2} \int_S z D z \, dA_{g_0}$. This is (40.17).

All of these assertions are identities in the joint spatial, λ , and ϵ Taylor jets at the original boundary. The boundary-jet justification in Lemma 40.2 permits their evaluation at negative leading displacements. Neither the sign of an affine segment nor that of a height coefficient changes the order calculation or the area Hessian. No area inequality for an actually extended manifold is being applied. $\square$

40.3. The quartic square and its vanishing consequences. Null transport preserves the boundary area through degree four. The subsequent minimal-graph correction contributes the area Hessian in Lemma 40.3; the TT identity absorbs this boundary term.

$$S \text{ in } g_\epsilon \xrightarrow{\text{null transport}} S^\sharp \text{ in } \tilde{g} \xrightarrow{\text{minimal graph}} S_{\min} \text{ in } \tilde{g}.$$

FIGURE 4. Finite Taylor boundary comparisons. The first arrow preserves area through degree four. The signed height difference for the second is $h^\sharp - h^{\min} = \epsilon^2 z + O(\epsilon^3)$, producing the quartic area Hessian (40.17). Negative leading displacements are included.

Apply Proposition 38.1 to $\tilde{g}$ through degree four. Its hypotheses follow from Lemma 40.2, including the unchanged leading asymptotic data. Denote by f the lapse, by $\bar{f}$ its area average on each leaf, and set

$$\delta f = f - \bar{f}, \quad U_j = [\mathrm{II}^\circ]_j, \quad \mathcal{F}_j = [\delta f]_j.$$

At degree two, the scalar contribution is zero by (40.3). Equations (40.14) and (40.17), and the preserved mass, identify the quadratic coefficient of mass minus b_* of the minimal-leaf area with the original zero coefficient $c_2[q]$. The leading-energy identity, Proposition 38.4, with its scalar tensor equal to zero, gives

$$0 = \frac{1}{16\pi} \int_M N(|U_1|^2 + 2\mathcal{F}_1 J_s^0 \mathcal{F}_1) \, dV_{g_0}.$$

The lapse coefficient is mean free on each background sphere, and J_s^0 is strictly positive on that space. Since $N > 0$ for $s > 0$, sphere integration and smoothness imply

$$(40.20) \quad U_1 = 0, \quad \mathcal{F}_1 = 0.$$

We can now use Proposition 38.4 at order four: the scalar-curvature defect starts with $\epsilon^4|k|^2$, and both geometric square terms start in degree four. It follows first that mass minus b_* of the minimal-leaf area has no coefficients below degree four. The area correction (40.17) also starts in that degree. Thus $c_3[q] = 0$, as well as the previously known lower vanishings.

Let v_k be the decaying solution

$$(40.21) \quad Lv_k = 0, \quad v_k|_S = z = D^{-1}(k_{ss}|_S).$$

It is supplied by Lemma 37.1. The TT tensor k has the decay and finite angular type required in Lemma 39.1. Since $b'_*(4\pi a^2) = \kappa/(8\pi)$, subtracting the area correction from the leading-energy identity and applying that lemma gives

$$(40.22) \quad c_4[q] = \frac{1}{16\pi} \int_M N \left(|k - Q(v_k)|_{g_0}^2 + |U_2|^2 + 2\mathcal{F}_2 J_s^0 \mathcal{F}_2 \right) dV_{g_0}.$$

Indeed the subtracted boundary term is exactly $\kappa(16\pi)^{-1} \int_S z D z dA_{g_0}$. The integrals are understood with sphere integration first, as in Proposition 38.4; no pointwise sign is asserted for $\mathcal{F}_2 J_s^0 \mathcal{F}_2$. Its sphere integral is nonnegative. This proves $c_4[q] \geq 0$, and equality implies

$$(40.23) \quad \delta f = O(\epsilon^3), \quad \Pi^\circ = O(\epsilon^3)$$

on every leaf. Boundary values follow by smoothness. We show next that (40.23) excludes a nonzero degree-one part of v .

40.4. Round angular jets and conformal coordinates. We now assume equality in the quartic square. The vanishing in (40.23) makes the changed metric a warped product through degree two. The following lemma will round its angular metric before we compare it with the conformal class of g_0 .

Lemma 40.4 (Rounding a two-jet on the sphere). *Let $\gamma_\epsilon = \sigma + \epsilon P_1 + \epsilon^2 P_2$ be a smooth metric two-jet on S^2 of finite angular type. Suppose its scalar curvature is angularly constant through degree two. Then there are a positive scalar two-jet a_ϵ , with $a_0 = 1$, and a diffeomorphism two-jet Φ_ϵ , based at the identity, such that*

$$\gamma_\epsilon = a_\epsilon \Phi_\epsilon^* \sigma + O(\epsilon^3).$$

All coefficients can be chosen of finite angular type.

Proof. We first prove the infinitesimal decomposition used twice below. For a symmetric tensor P on the round sphere, there are a vector field Y and a function h such that

$$(40.24) \quad P = \mathcal{L}_Y \sigma + h\sigma.$$

For finite-angular-type P , the choices may retain that property.

Let $P^\circ = P - \frac{1}{2}(\text{tr}_\sigma P)\sigma$, and identify vector fields and one-forms using σ . Define

$$\mathcal{C}(Y)_{ij} = \nabla_i Y_j + \nabla_j Y_i - (\text{div } Y)\sigma_{ij}.$$

On the curvature-one sphere, commuting covariant derivatives gives

$$(40.25) \quad \text{div } \mathcal{C}(Y) = (\Delta_\sigma + 1)Y, \quad (\Delta_\sigma + 1)df = d(\Delta_\sigma + 2)f.$$

Here the Laplacian on one-forms is the rough Laplacian $\nabla^i \nabla_i$. The second identity also holds after applying the parallel Hodge star $*$ to one-forms.

Put $\alpha = \text{div } P^\circ$. To decompose it, solve the scalar equations

$$\Delta_\sigma \varphi = \text{div } \alpha, \quad \Delta_\sigma \psi = -\text{div}(*\alpha)$$

with zero means. Both right sides have zero integral. In finite angular spaces these equations are solved by inversion on the nonconstant spherical modes. The residual $\omega = \alpha - d\varphi - *d\psi$ is closed and co-closed. Commuting derivatives of a closed one-form of zero divergence gives $\nabla^i \nabla_i \omega = \omega$; integration against ω forces $\omega = 0$. We have therefore obtained

$$\alpha = d\varphi + *d\psi.$$

The one-forms dx_i and $*dx_i$ correspond to conformal Killing vector fields. Integration by parts against the tracefree tensor P° consequently shows that α is orthogonal to all of them. It follows that both φ and ψ have zero degree-one components. Thus we may solve

$$(\Delta_\sigma + 2)A = \varphi, \quad (\Delta_\sigma + 2)C = \psi$$

on their angular modes, and set $Y^\flat = dA + *dC$. Equation (40.25) gives $\operatorname{div} \mathcal{C}(Y) = \operatorname{div} P^\circ$.

The tracefree tensor $P_* = P^\circ - \mathcal{C}(Y)$ is consequently divergence free. Such a tensor vanishes on the round sphere; here is a direct verification. In an orthonormal frame write its components as $\begin{pmatrix} a & b \\ b & -a \end{pmatrix}$. Its two divergence equations are precisely the two independent Codazzi equations $\nabla_i (P_*)_{jk} = \nabla_j (P_*)_{ik}$. Contracting and commuting these equations on the curvature-one sphere gives

$$\nabla^i \nabla_i (P_*)_{jk} = 2(P_*)_{jk} - \sigma_{jk} \operatorname{tr}_\sigma P_* = 2(P_*)_{jk}.$$

Integration yields $-\int |\nabla P_*|^2 = 2 \int |P_*|^2$, hence $P_* = 0$. Taking $h = \frac{1}{2} \operatorname{tr}_\sigma P - \operatorname{div} Y$ proves (40.24). Every operation used above commutes with rotations or inverts a scalar operator within finitely many spherical eigenspaces, so it preserves finite angular type.

The scalar-curvature derivative in the direction $h\sigma$ is

$$(D \operatorname{Scal})_\sigma(h\sigma) = -\Delta_\sigma h - 2h.$$

The derivative in a Lie direction is zero because the background scalar curvature is constant. If the scalar-curvature variation of P is constant, (40.24) therefore implies that h consists only of a constant and degree-one modes. A degree-one function h_1 satisfies $\operatorname{Hess}_\sigma h_1 = -h_1 \sigma$, so

$$h_1 \sigma = \mathcal{L}_{-\frac{1}{2} \operatorname{grad}_\sigma h_1} \sigma.$$

Thus P is a Lie derivative plus a constant multiple of σ .

Apply this conclusion to P_1 . A formal inverse flow and a scalar scaling remove its first coefficient. Both operations preserve the property that curvature is angularly constant. The resulting metric two-jet has the form $\sigma + \epsilon^2 \hat{P}_2$, so its degree-two curvature equation is the same linearized equation, with no quadratic contribution from a first coefficient. The argument just given removes $\hat{P}_2$ by a degree-two flow and scaling. Undoing the two operations proves the stated rounding. This proves only an identity of two-jets, which is exactly what is required. $\square$

Local conformal tangency forced by equality. Suppose now that $c_4[q] = 0$. Fix a compact interval $I \Subset (0, \infty)$. In coordinates of the CMC sphere jets we can remove tangential shift through degree two by integrating its tangential reparametrization equation on I . Its background value is zero, so the coordinate change is based at the identity. In these coordinates (40.23) says that the lapse is a function of the leaf parameter only and that $\Pi = (H/2)g_{\text{leaf}}$ through degree two. Since H is constant on each leaf, the normal metric variation is

$$\partial_s g_{\text{leaf}} = 2f\Pi = fHg_{\text{leaf}}.$$

Consequently on $I \times S^2$ the metric has the form

$$(40.26) \quad f_*(s)^2 ds^2 + b(s)^2 \gamma_\epsilon(x) + O(\epsilon^3),$$

where all quantities are two-jets, with backgrounds $f_* = 1$, $b = r$, and $\gamma_0 = \sigma$. Angular finiteness is preserved by these Taylor coordinate changes and integrations.

For this warped metric the scalar curvature through degree two is

$$(40.27) \quad b^{-2} \text{Scal}_{\gamma_\epsilon} - \frac{4}{f_* b} \frac{d}{ds} \left(\frac{b'}{f_*} \right) - 2 \left(\frac{b'}{f_* b} \right)^2.$$

Equation (40.3) makes it the constant -6 through degree two. The last two terms in (40.27) are radial, so $\text{Scal}_{\gamma_\epsilon}$ is angularly constant to that order. Lemma 40.4 makes γ_ϵ round up to diffeomorphism and scaling. Absorbing the scaling into b , we obtain $f_*^2 ds^2 + b^2 \sigma$.

This last two-jet is locally conformal to a pullback of g_0 . Indeed solve, coefficientwise on I ,

$$(40.28) \quad \frac{y'}{r(y)} = \frac{f_*}{b},$$

with background $y = s$ and a fixed identity background initial value at an interior point of I . This is a nonsingular scalar ODE, and it gives

$$f_*^2 ds^2 + b^2 \sigma = \frac{b^2}{r(y)^2} (dy^2 + r(y)^2 \sigma) \mod \epsilon^3.$$

Composing the coordinate changes already made shows, in the original coordinates on the interval, that

$$(40.29) \quad \tilde{g} = \omega_\epsilon \Psi_\epsilon^* g_0 + O(\epsilon^3), \quad \omega_0 = 1, \quad \Psi_0 = \text{id}.$$

Only this local statement is used.

There is a useful consequence of the known first coefficient in (40.29). Write $\omega_\epsilon = 1 + \epsilon a_1 + \epsilon^2 a_2$ and express the diffeomorphism two-jet as

$$\Psi_\epsilon^* g_0 = g_0 + \epsilon \mathcal{L}_X g_0 + \epsilon^2 (\mathcal{L}_Z g_0 + \tfrac{1}{2} \mathcal{L}_X^2 g_0).$$

The graph calculation shows that $\tilde{g} - g_\epsilon$ starts in degree two, so $[\tilde{g}]_1 = 4u g_0$. Hence $\mathcal{L}_X g_0 = \lambda g_0$ for some function λ . It follows that

$$\mathcal{L}_X^2 g_0 = (X\lambda + \lambda^2) g_0,$$

and the mixed scalar term $a_1 \mathcal{L}_X g_0$ is pure trace as well. Therefore

$$(40.30) \quad [\tilde{g}]_2 = \mathcal{L}_Z g_0 + c g_0$$

for a function c on $I \times S^2$. The original metric coefficient $[g_\epsilon]_2$ is itself pure trace. Thus the same Lie-derivative-plus-trace conclusion holds for their difference. Since $\text{Hess}(v^2) = \frac{1}{2} \mathcal{L}_{\text{grad}(v^2)} g_0$, (40.5) gives the necessary condition

$$(40.31) \quad dv \otimes dv + Bv^2 = \mathcal{L}_Y g_0 + c g_0 \quad \text{on } I \times S^2$$

for some local Y, c . Their choices may depend on I .

40.5. The degree-one obstruction.

Lemma 40.5 (A nonzero degree-one mode is obstructed). *Let $v = v_0(s) + V(s)x$ solve $Lv = 0$, where x is one fixed unit-axis linear coordinate on S^2 and $V = O(r^{-3})$. If (40.31) holds on every compact radial interval in the interior, with possibly different local vector fields and functions, then $V \equiv 0$.*

Proof. Set $Y_2 = x^2 - 1/3$, a degree-two spherical eigenfunction. Project (40.31) onto its scalar, gradient, and tangential Hessian modes. Write the contributing radial vector-field coefficient as $p(s)Y_2\partial_s$, its tangential coefficient as $d(s)\text{grad}_\sigma Y_2$, and the scalar-multiple coefficient as $c(s)Y_2$. These are the only generator components contributing to the tested modes. This follows either from the one-form decomposition in Lemma 40.4 or by integration by parts, using $\Delta_\sigma Y_2 = -6Y_2$. In particular, the coexact component is orthogonal to both the gradient test and the electric Hessian test, and the other spherical eigenspaces are orthogonal as well.

For clarity, the angular identity

$$dx \otimes dx = \frac{1}{2} \text{Hess}_\sigma Y_2 + x^2 \sigma$$

shows all terms of the left side in these modes. With $B = B_s ds^2 + B_t r^2 \sigma$, equality of the radial, mixed, Hessian, and tangential scalar coefficients respectively gives

$$(40.32) \quad \begin{aligned} V'^2 + B_s V^2 &= 2p' + c, & \frac{VV'}{2} &= p + r^2 d', \\ \frac{V^2}{2} &= 2r^2 d, & V^2 + r^2 B_t V^2 &= 2rNp + r^2 c. \end{aligned}$$

Terms involving v_0 have only degrees zero or one and do not enter these equations. Solving the last three equations yields

$$d = \frac{V^2}{4r^2}, \quad p = \frac{NV^2}{2r}, \quad c = \frac{3m_0 V^2}{r^3}.$$

Substitute these expressions in the first equation and use

$$B_s = 1 - \frac{2m_0}{r^3}, \quad B_t = 1 + \frac{m_0}{r^3}, \quad N' = r + \frac{m_0}{r^2}, \quad N^2 = 1 + r^2 - \frac{2m_0}{r}.$$

After cancellation one obtains

$$(40.33) \quad \left(V' - \frac{N}{r} V \right)^2 = \frac{6m_0}{r^3} V^2.$$

The local generators have disappeared from this relation. Because the radial interval was arbitrary, it holds throughout the end even though their choices need not agree on overlapping intervals.

For $z = V/r$, (40.33) reads

$$|z'| = \sqrt{6m_0} r^{-3/2} |z|.$$

The coefficient on the right is integrable toward infinity, and $z \rightarrow 0$. The integral differential inequality, applied backwards between s and $S > s$, gives

$$|z(s)| \leq |z(S)| \exp \left(\int_s^S \sqrt{6m_0} r(t)^{-3/2} dt \right).$$

Letting $S \rightarrow \infty$ proves $z = 0$ on the end. The homogeneous degree-one ODE obtained from $Lv = 0$ then gives $V \equiv 0$ by uniqueness. $\square$

Completion of the proof of Proposition 40.1. The coefficient vanishings and nonnegativity were proved in (40.20)–(40.22). Suppose that the degree-one part of v is nonzero and that $c_4[q] = 0$. All three degree-one radial coefficients solve the same decaying homogeneous ODE. Decaying uniqueness makes them proportional, so after a fixed rotation their sum is $V(s)x$ for one axis and a nonzero radial function V . The vanishing consequences of (40.22) give (40.31) on each interior radial interval. Lemma 40.5 forces $V = 0$, a contradiction. Thus $c_4[q] > 0$.

Finally this is a statement about the actual deficit, not only a formal inequality. The kernel representation gives precisely the finite angular type and differentiated decay needed for the fourth-order expansion in Proposition 37.5. Lemma 37.3 converts the weighted function and equation remainders to the mass remainder. Consequently

$$\mathfrak{D}_q(\epsilon) = c_4[q]\epsilon^4 + O(\epsilon^5)$$

in the case under consideration. Its positive coefficient gives strict positivity for all sufficiently small positive ϵ , with the interval allowed to depend on the fixed data. The radial kernel, including the zero seed, is handled by the exact argument in the next section. $\square$

41. RADIAL EQUALITY AND THE NONLINEAR CONCLUSION

Proposition 41.1 (Exact radial equality). *Suppose the fixed TT seed q is invariant under rotations of S^2 . Every branch in Theorem 36.1 is radial for all sufficiently small ϵ , and on that parameter interval*

$$m_{\text{AH}}(\epsilon) = b_*(A_\epsilon).$$

This includes the zero seed, and requires no sign for $q_{ss}|_S$.

Proof. We first prove radially of the actual solution, rather than merely of its Taylor coefficients. Fix a rotation $\mathcal{R}$, and set $\hat{\phi} = \phi_\epsilon \circ \mathcal{R}$ and $w = \hat{\phi} - \phi_\epsilon$. Because q is radial, $\hat{\phi}$ satisfies the same equation and boundary condition as ϕ_ϵ . Their difference satisfies

$$(41.1) \quad (\Delta_{g_0} - c_\epsilon)w = 0, \quad w_s|_S = c_\epsilon^* w|_S,$$

where divided differences of the nonlinearities give, uniformly on M and uniformly in $\mathcal{R}$,

$$c_\epsilon = 3 + o(1), \quad c_\epsilon^* = O(\epsilon).$$

Indeed, the derivatives of the interior and boundary nonlinearities with respect to their positive scalar argument z are respectively

$$\frac{3}{4}(5z^4 - 1) + \frac{7\epsilon^2}{8}|q|^2 z^{-8}, \quad -\frac{3\epsilon}{4}q_{ss}z^{-4}.$$

The asserted bounds follow from boundedness of q and the given uniform convergence $\phi_\epsilon \rightarrow 1$.

Choose $3/2 < \beta < \min(\tau, \sqrt{3})$. Then $w = o(e^{-\beta s})$, and (41.1) has the form of Lemma 37.1 with $V = c_\epsilon - 3$, Robin coefficient $a_* = c_\epsilon^*$, and zero interior and boundary data. For sufficiently small ϵ , uniformly in the rotation, $\|V\|_\infty \leq c\beta$ and $\|a_*\|_\infty \leq \beta/2$. The comparison estimate gives $w = 0$, so the actual ϕ_ϵ is radial.

Write the invariant seed as $q = A_q(s) ds^2 + B_q(s)r^2\sigma$. Its trace and divergence equations give

$$B_q = -A_q/2, \quad A'_q + 3(N/r)A_q = 0.$$

Consequently, for a constant $C_q \in \mathbb{R}$,

$$(41.2) \quad q = C_q r^{-3} \left(ds^2 - \frac{1}{2} r^2 \sigma \right).$$

Fix a sufficiently small parameter. Define proper radial distance ℓ and the actual area radius R by

$$d\ell = \phi_\epsilon^2 ds, \quad R = r\phi_\epsilon^2.$$

Then the actual metric and second-form data are

$$g_\epsilon = d\ell^2 + R^2\sigma, \quad K_\epsilon = -2k d\ell^2 + kR^2\sigma, \quad k = -\frac{\epsilon C_q}{2R^3}.$$

A dot denotes an ℓ -derivative. The momentum and scalar constraints are therefore

$$(41.3) \quad \dot{k} = -3\frac{\dot{R}}{R}k, \quad -\frac{4\ddot{R}}{R} + \frac{2(1 - \dot{R}^2)}{R^2} + 6 = 6k^2.$$

The scalar-curvature formula here follows from the sectional curvatures $-\ddot{R}/R$ and $(1 - \dot{R}^2)/R^2$ of this warped product.

Define

$$(41.4) \quad \mathcal{M} = \frac{R}{2} (1 + R^2 - \dot{R}^2 + R^2 k^2).$$

Direct differentiation, using the momentum equation, gives

$$\dot{\mathcal{M}} = \frac{\dot{R}}{2} (1 + 3R^2 - \dot{R}^2 - 2R\ddot{R} - 3R^2 k^2) = 0$$

by the scalar equation in (41.3). No division by $\dot{R}$ has been used, so conservation is valid at turning points of R as well. At the boundary the MOTS equation reads

$$\frac{2\dot{R}}{R} + 2k = 0.$$

Substitution in (41.4) yields

$$(41.5) \quad \mathcal{M}|_S = \frac{R|_S}{2}(1 + (R|_S)^2) = b_*(4\pi(R|_S)^2) = b_*(A_\epsilon).$$

We identify the same constant with the specified mass flux at infinity. Put $\psi = \phi_\epsilon - 1$. Its actual radial equation gives

$$\psi'' + 2\frac{N}{r}\psi' - 3\psi = O(\psi^2) + O(r^{-6}),$$

where (41.2) was used in the source estimate. The assumed weighted decay gives $\psi, \psi' = O(e^{-\tau s})$, and $N/r = 1 + O(e^{-2s})$. It follows that

$$\psi'' + 2\psi' - 3\psi = O(e^{-\gamma s}), \quad \gamma = \min(2\tau, \tau + 2, 6) > 3.$$

Variation of constants for the roots $1, -3$ excludes the growing solution by the prescribed decay and yields, as in Lemma 37.1, a constant $c = c(\epsilon)$ with

$$(41.6) \quad \phi_\epsilon = 1 + cr^{-3} + o(r^{-3}), \quad \partial_s \phi_\epsilon = -3cNr^{-4} + o(r^{-3}).$$

In particular,

$$R = r + 2cr^{-2} + o(r^{-2}), \quad \dot{R} = N + 2r\frac{\partial_s \phi_\epsilon}{\phi_\epsilon} = N - 6cNr^{-3} + o(r^{-2}).$$

Using $N^2 = 1 + r^2 - 2m_0/r$, we obtain

$$1 + R^2 - \dot{R}^2 = \frac{2m_0 + 16c}{r} + o(r^{-1}).$$

Also $R^3 k^2 = O(R^{-3})$, so

$$(41.7) \quad \lim_{\ell \rightarrow \infty} \mathcal{M} = m_0 + 8c.$$

For a direct flux normalization check, rotational invariance implies $p_i = 0$ for $i = 1, 2, 3$. Equation (41.6) implies

$$g_\epsilon - g_0 = 4cr^{-3}b + o(r^{-3})$$

in scaled components, with its first differentiated asymptotic. For a perturbation fb in dimension three, the mass-flux covector integrand with test function V simplifies to $2(f \, dV - V \, df)$. With $f = 4cr^{-3}$, $V = V_0 = \sqrt{1 + r^2}$, and $\nu_b = \sqrt{1 + r^2} \partial_r$, its normal component is

$$2(f \partial_{\nu_b} V_0 - V_0 \partial_{\nu_b} f) = 32cr^{-2} + 24cr^{-4}.$$

The error terms give zero limiting flux. Integrating over a coordinate sphere of area $4\pi r^2$ and dividing by 16π gives $p_0 - m_0 = 8c$. Since $p_0 \rightarrow m_0 > 0$ as $\epsilon \rightarrow 0$,

$$m_{\text{AH}} = p_0 = m_0 + 8c = \mathcal{M} = b_*(A_\epsilon)$$

for sufficiently small ϵ , by (41.5)–(41.7). This argument includes $C_q = 0$. In that case one can also apply the comparison estimate of Lemma 37.1 to $\phi_\epsilon - 1$ and conclude $\phi_\epsilon \equiv 1$ directly. $\square$

Completion of the proof of Theorem 36.1. Fix the seed and the given solution branch. If q is radial, Proposition 41.1 gives exact equality on a sufficiently small parameter interval. Suppose henceforth that q is not radial. Proposition 39.2 gives the actual expansion

$$\mathfrak{D}_q(\epsilon) = c_2[q]\epsilon^2 + o(\epsilon^2), \quad c_2[q] \geq 0.$$

If $c_2[q] > 0$, this proves strict positivity of the deficit for every sufficiently small positive ϵ .

If $c_2[q] = 0$, Corollary 39.3 gives $q = Q(v)$ with $Lv = 0$ and with only constant and linear angular modes. Its linear part is nonzero, for otherwise q would be radial. These homogeneous modes have the stronger differentiated decay needed for the actual fourth-order expansion in Proposition 37.5. Proposition 40.1 then gives

$$\mathfrak{D}_q(\epsilon) = c_4[q]\epsilon^4 + o(\epsilon^4), \quad c_4[q] > 0,$$

with all coefficients below order four equal to zero. Again the actual deficit is strictly positive for every sufficiently small positive ϵ .

At $\epsilon = 0$ the background identity $m_0 = b_*(4\pi a^2)$ gives equality. The preceding alternatives exhaust all fixed seeds. Choose $\epsilon_0 > 0$ small enough for the applicable alternative and the given branch, with $\epsilon_0 \leq \epsilon_*$. This proves the exact inequality on $0 \leq \epsilon < \epsilon_0$, equality for radial seeds, and strict inequality for nonradial seeds at positive parameters in that interval. The size of the interval is allowed to depend on the fixed seed and branch; no uniform positive lower bound for c_2 or c_4 is asserted or required. $\square$

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